

SIXTH EDITION

BIOLOGY



RAVEN
JOHNSON

The Origin of Living Things

Unraveling the Mystery of How Geckos Defy Gravity

Science is most fun when it tickles your imagination. This is particularly true when you see something your common sense tells you just *can't* be true. Imagine, for example, you are lying on a bed in a tropical hotel room. A little lizard, a blue gecko about the size of a toothbrush, walks up the wall beside you and upside down across the ceiling, stopping for a few moments over your head to look down at you, and then trots over to the far wall and down.

There is nothing at all unusual in what you have just imagined. Geckos are famous for strolling up walls in this fashion. How do geckos perform this gripping feat? Investigators have puzzled over the adhesive properties of geckos for decades. What force prevents gravity from dropping the gecko on your nose?

The most reasonable hypothesis seemed suction—salamanders' feet form suction cups that let them climb walls, so maybe geckos' do too. The way to test this is to see if the feet adhere in a vacuum, with no air to create suction. Salamander feet don't, but gecko feet do. It's not suction.

How about friction? Cockroaches climb using tiny hooks that grapple onto irregularities in the surface, much as rock-climbers use crampons. Geckos, however, happily run up walls of smooth polished glass that no cockroach can climb. It's not friction.

Electrostatic attraction? Clothes in a dryer stick together because of electrical charges created by their rubbing together. You can stop this by adding a "static remover" like a Cling-free sheet that is heavily ionized. But a gecko's feet still adhere in ionized air. It's not electrostatic attraction.

Could it be glue? Many insects use adhesive secretions from glands in their feet to aid climbing. But there are no glands cells in the feet of a gecko, no secreted chemicals, no footprints left behind. It's not glue.

There is one tantalizing clue, however, the kind that experimenters love. Gecko feet seem to get stickier on some surfaces than others. They are less sticky on low-energy surfaces like Teflon, and more sticky on surfaces made of



Defying gravity. This gecko lizard is able to climb walls and walk upside down across ceilings. Learning how geckos do this is a fascinating bit of experimental science.

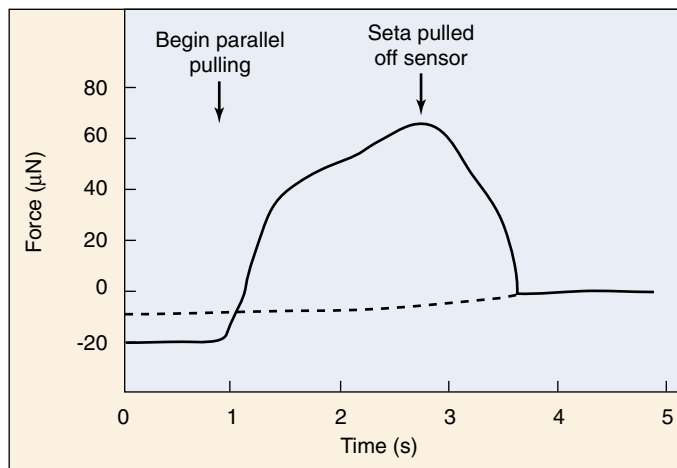
polar molecules. This suggests that geckos are tapping directly into the molecular structure of the surfaces they walk on!

Tracking down this clue, Kellar Autumn of Lewis & Clark College in Portland, Oregon, and Robert Full of the University of California, Berkeley, took a closer look at gecko feet. Geckos have rows of tiny hairs called setae on the bottoms of their feet, like the bristles of some trendy toothbrush. When you look at these hairs under the microscope, the end of each seta is divided into 400 to 1000 fine projections called spatulae. There are about half a million of these setae on each foot, each only one-tenth the diameter of a human hair.

Autumn and Full put together an interdisciplinary team of scientists and set out to measure the force produced by a single seta. To do this, they had to overcome two significant experimental challenges:

Isolating a single seta. No one had ever isolated a single seta before. They succeeded in doing this by surgically plucking a hair from a gecko foot under a microscope and bonding the hair onto a microprobe. The microprobe was fitted into a specially designed micromanipulator that can move the mounted hair in various ways.

Measuring a very small force. Previous research had shown that if you pull on a whole gecko, the adhesive force sticking each of the gecko's feet to the wall is about 10 Newtons (N), which is like supporting 1 kg. Because each foot has half a million setae, this predicts that a single seta would produce about 20 microNewtons of force. That's a very tiny amount to measure. To attempt the measurement, Autumn and Full recruited a mechanical engineer from Stanford, Thomas Kenny. Kenny is an expert at building instruments that can measure forces at the atomic level.



The sliding step experiment. The adhesive force of a single seta was measured. An initial push perpendicularly put the seta in contact with the sensor. Then, with parallel pulling, the force continued to increase over time to a value of 60 microNewtons (after this, the seta began to slide and pulled off the sensor). In a large number of similar experiments, adhesion forces typically approach 200 microNewtons.

The Experiment

Once this team had isolated a seta and placed it in Kenny's device, "We had a real nasty surprise," says Autumn. For two months, pushing individual seta against a surface, they couldn't get the isolated hair to stick at all!

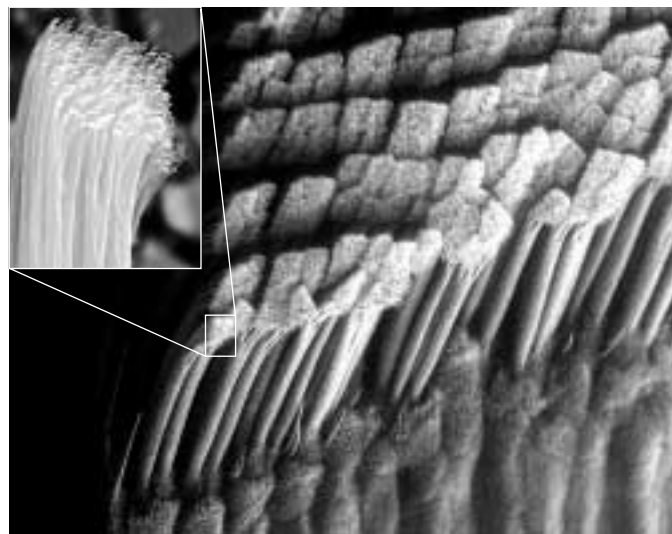
This forced the research team to stand back and think a bit. Finally it hit them. Geckos don't walk by pushing their feet down, like we do. Instead, when a gecko takes a step, it pushes the palm of the foot into the surface, then uncurls its toes, sliding them backwards onto the surface. This shoves the forest of tips *sideways* against the surface.

Going back to their instruments, they repeated their experiment, but this time they oriented the seta to approach the surface from the side rather than head-on. This had the effect of bringing the many spatulae on the tip of the seta into direct contact with the surface.

To measure these forces on the seta from the side, as well as the perpendicular forces they had already been measuring, the researchers constructed a micro-electromechanical cantilever. The apparatus consisted of two piezoresistive layers deposited on a silicon cantilever to detect force in both parallel and perpendicular angles.

The Results

With the seta oriented properly, the experiment yielded results. Fantastic results. The attachment force measured by the machine went up 600-fold from what the team had been measuring before. A single seta produced not the 20 microNewtons of force predicted by the whole-foot measurements, but up to an astonishing 200 microNewtons (see graph above)! Measuring many individual seta, adhesive forces averaged 194 ± 25 microNewtons.



Closeup look at a gecko's foot. The setae on a gecko's foot are arranged in rows, and point backwards, away from the toenail. Each seta branches into several hundred spatulae (inset photo).

Two hundred microNewtons is a tiny force, but stupendous for a single hair only 100 microns long. Enough to hold up an ant. A million hairs could support a small child. A little gecko, ceiling walking with 2 million of them (see photos above), could theoretically carry a 90-pound backpack—talk about being over-engineered.

If a gecko's feet stick *that* good, how do geckos ever become unstuck? The research team experimented with unattaching individual seta; they used yet another micro-instrument, this one designed by engineer Ronald Fearing also from U.C. Berkeley, to twist the hair in various ways. They found that tipped past a critical angle, 30 degrees, the attractive forces between hair and surface atoms weaken to nothing. The trick is to tip a foot hair until its projections let go. Geckos release their feet by curling up each toe and peeling it off, just the way we remove tape.

What is the source of the powerful adhesion of gecko feet? The experiments do not reveal exactly what the attractive force is, but it seems almost certain to involve interactions at the atomic level. For a gecko's foot to stick, the hundreds of spatulae at the tip of each seta must butt up squarely against the surface, so the individual atoms of each spatula can come into play. When two atoms approach each other very closely—closer than the diameter of an atom—a subtle nuclear attraction called Van der Waals forces comes into play. These forces are individually very weak, but when lots of them add their little bits, the sum can add up to quite a lot.

Might robots be devised with feet tipped with artificial setae, able to walk up walls? Autumn and Full are working with a robotics company to find out. Sometimes science is not only fun, but can lead to surprising advances.



To explore this experiment further, go to the Virtual Lab at www.mhhe.com/raven6/vlab1.mhtml

1

The Science of Biology

Concept Outline

1.1 Biology is the science of life.

Properties of Life. Biology is the science that studies living organisms and how they interact with one another and their environment.

1.2 Scientists form generalizations from observations.

The Nature of Science. Science employs both deductive reasoning and inductive reasoning.

How Science Is Done. Scientists construct hypotheses from systematically collected objective data. They then perform experiments designed to disprove the hypotheses.

1.3 Darwin's theory of evolution illustrates how science works.

Darwin's Theory of Evolution. On a round-the-world voyage Darwin made observations that eventually led him to formulate the hypothesis of evolution by natural selection.

Darwin's Evidence. The fossil and geographic patterns of life he observed convinced Darwin that a process of evolution had occurred.

Inventing the Theory of Natural Selection. The Malthus idea that populations cannot grow unchecked led Darwin, and another naturalist named Wallace, to propose the hypothesis of natural selection.

Evolution After Darwin: More Evidence. In the century since Darwin, a mass of experimental evidence has supported his theory of evolution, now accepted by practically all practicing biologists.

1.4 This book is organized to help you learn biology.

Core Principles of Biology. The first half of this text is devoted to general principles that apply to all organisms, the second half to an examination of particular organisms.



FIGURE 1.1

A replica of the *Beagle*, off the southern coast of South America. The famous English naturalist, Charles Darwin, set forth on H.M.S. *Beagle* in 1831, at the age of 22.

You are about to embark on a journey—a journey of discovery about the nature of life. Nearly 180 years ago, a young English naturalist named Charles Darwin set sail on a similar journey on board H.M.S. *Beagle* (figure 1.1 shows a replica of the *Beagle*). What Darwin learned on his five-year voyage led directly to his development of the theory of evolution by natural selection, a theory that has become the core of the science of biology. Darwin's voyage seems a fitting place to begin our exploration of biology, the scientific study of living organisms and how they have evolved. Before we begin, however, let's take a moment to think about what biology is and why it's important.

1.1 Biology is the science of life.

Properties of Life

In its broadest sense, *biology is the study of living things—the science of life*. Living things come in an astounding variety of shapes and forms, and biologists study life in many different ways. They live with gorillas, collect fossils, and listen to whales. They isolate viruses, grow mushrooms, and examine the structure of fruit flies. They read the messages encoded in the long molecules of heredity and count how many times a hummingbird's wings beat each second.

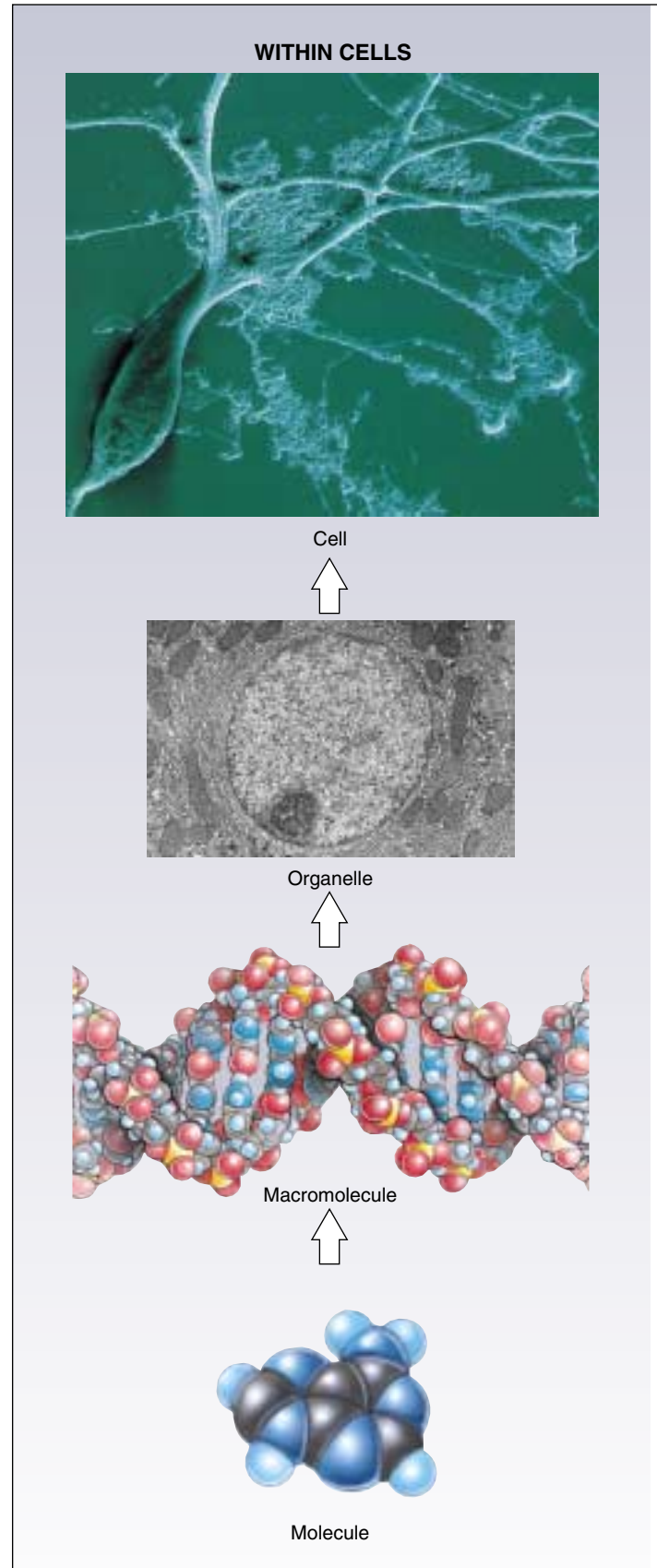
What makes something “alive”? Anyone could deduce that a galloping horse is alive and a car is not, but *why*? We cannot say, “If it moves, it’s alive,” because a car can move, and gelatin can wiggle in a bowl. They certainly are not alive. What characteristics *do* define life? All living organisms share five basic characteristics:

- 1. Order.** All organisms consist of one or more cells with highly ordered structures: atoms make up molecules, which construct cellular organelles, which are contained within cells. This hierarchical organization continues at higher levels in multicellular organisms and among organisms (figure 1.2).
- 2. Sensitivity.** All organisms respond to stimuli. Plants grow toward a source of light, and your pupils dilate when you walk into a dark room.
- 3. Growth, development, and reproduction.** All organisms are capable of growing and reproducing, and they all possess hereditary molecules that are passed to their offspring, ensuring that the offspring are of the same species. Although crystals also “grow,” their growth does not involve hereditary molecules.
- 4. Regulation.** All organisms have regulatory mechanisms that coordinate the organism’s internal functions. These functions include supplying cells with nutrients, transporting substances through the organism, and many others.
- 5. Homeostasis.** All organisms maintain relatively constant internal conditions, different from their environment, a process called homeostasis.

All living things share certain key characteristics: order, sensitivity, growth, development and reproduction, regulation, and homeostasis.

FIGURE 1.2

Hierarchical organization of living things. Life is highly organized—from small and simple to large and complex, within cells, within multicellular organisms, and among organisms.



WITHIN MULTICELLULAR ORGANISMS



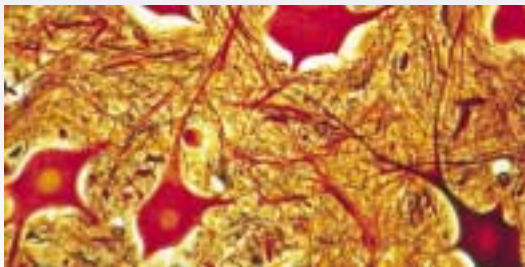
Organism



Organ system



Organ



Tissue

AMONG ORGANISMS



Ecosystem



Community



Species



Population

1.2 Scientists form generalizations from observations.

The Nature of Science

Biology is a fascinating and important subject, because it dramatically affects our daily lives and our futures. Many biologists are working on problems that critically affect our lives, such as the world's rapidly expanding population and diseases like cancer and AIDS. The knowledge these biologists gain will be fundamental to our ability to manage the world's resources in a suitable manner, to prevent or cure diseases, and to improve the quality of our lives and those of our children and grandchildren.

Biology is one of the most successful of the "natural sciences," explaining what our world is like. To understand biology, you must first understand the nature of science. The basic tool a scientist uses is thought. To understand the nature of science, it is useful to focus for a moment on how scientists think. They reason in two ways: deductively and inductively.

Deductive Reasoning

Deductive reasoning applies general principles to predict specific results. Over 2200 years ago, the Greek Eratosthenes used deductive reasoning to accurately estimate the circumference of the earth. At high noon on the longest day of the year, when the sun's rays hit the bottom of a deep well in the city of Syene, Egypt, Eratosthenes measured the length of the shadow cast by a tall obelisk in Alexandria, about 800 kilometers to the north. Because he knew the distance between the two cities and the height of the obelisk, he was able to employ the principles of Euclidean geometry to correctly deduce the circumference of the earth (figure 1.3). This sort of analysis of specific cases using general principles is an example of deductive reasoning. It is the reasoning of mathematics and philosophy and is used to test the validity of general ideas in all branches of knowledge. General principles are constructed and then used as the basis for examining specific cases.

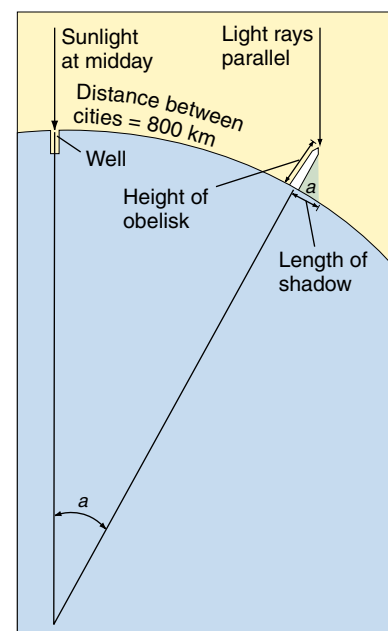
Inductive Reasoning

Inductive reasoning uses specific observations to construct general scientific principles. *Webster's Dictionary* defines science as systematized knowledge derived from observation and experiment carried on to determine the principles underlying what is being studied. In other words, a scientist determines principles from observations, discovering general principles by careful examination of specific cases. Inductive reasoning first became important to science in the 1600s in Europe, when Francis Bacon, Isaac Newton, and others began to use the results of particular experiments to infer general principles about how the world operates. If

FIGURE 1.3

Deductive reasoning: How Eratosthenes estimated the circumference of the earth using deductive reasoning. 1. On a day when sunlight shone straight down a deep well at Syene in Egypt, Eratosthenes measured the length of the shadow cast by a tall obelisk in the city of Alexandria, about 800 kilometers away. 2. The shadow's length and the obelisk's height formed two sides of a triangle. Using the recently developed principles of Euclidean geometry, he calculated the angle, a , to be 7° and $12'$, exactly $\frac{1}{50}$ of a circle (360°). 3. If angle $a = \frac{1}{50}$ of a circle, then the distance between the obelisk (in Alexandria) and the well (in Syene) must equal $\frac{1}{50}$ of the circumference of the earth. 4. Eratosthenes had heard that it was a 50-day camel trip from Alexandria to Syene. Assuming that a camel travels about 18.5 kilometers per day, he estimated the distance between obelisk and well as 925 kilometers (using different units of measure, of course).

5. Eratosthenes thus deduced the circumference of the earth to be $50 \times 925 = 46,250$ kilometers. Modern measurements put the distance from the well to the obelisk at just over 800 kilometers. Employing a distance of 800 kilometers, Eratosthenes's value would have been $50 \times 800 = 40,000$ kilometers. The actual circumference is 40,075 kilometers.



you release an apple from your hand, what happens? The apple falls to the ground. From a host of simple, specific observations like this, Newton inferred a general principle: all objects fall toward the center of the earth. What Newton did was construct a mental model of how the world works, a family of general principles consistent with what he could see and learn. Scientists do the same today. They use specific observations to build general models, and then test the models to see how well they work.

Science is a way of viewing the world that focuses on objective information, putting that information to work to build understanding.

How Science Is Done

How do scientists establish which general principles are true from among the many that might be true? They do this by systematically testing alternative proposals. If these proposals prove inconsistent with experimental observations, they are rejected as untrue. After making careful observations concerning a particular area of science, scientists construct a **hypothesis**, which is a suggested explanation that accounts for those observations. A hypothesis is a proposition that might be true. Those hypotheses that have not yet been disproved are retained. They are useful because they fit the known facts, but they are always subject to future rejection if, in the light of new information, they are found to be incorrect.

Testing Hypotheses

We call the test of a hypothesis an **experiment** (figure 1.4). Suppose that a room appears dark to you. To understand why it appears dark, you propose several hypotheses. The first might be, “There is no light in the room because

the light switch is turned off.” An alternative hypothesis might be, “There is no light in the room because the light-bulb is burned out.” And yet another alternative hypothesis might be, “I am going blind.” To evaluate these hypotheses, you would conduct an experiment designed to eliminate one or more of the hypotheses. For example, you might test your hypotheses by reversing the position of the light switch. If you do so and the light does not come on, you have disproved the first hypothesis. Something other than the setting of the light switch must be the reason for the darkness. Note that a test such as this does not prove that any of the other hypotheses are true; it merely demonstrates that one of them is not. A successful experiment is one in which one or more of the alternative hypotheses is demonstrated to be inconsistent with the results and is thus rejected.

As you proceed through this text, you will encounter many hypotheses that have withstood the test of experiment. Many will continue to do so; others will be revised as new observations are made by biologists. Biology, like all science, is in a constant state of change, with new ideas appearing and replacing old ones.

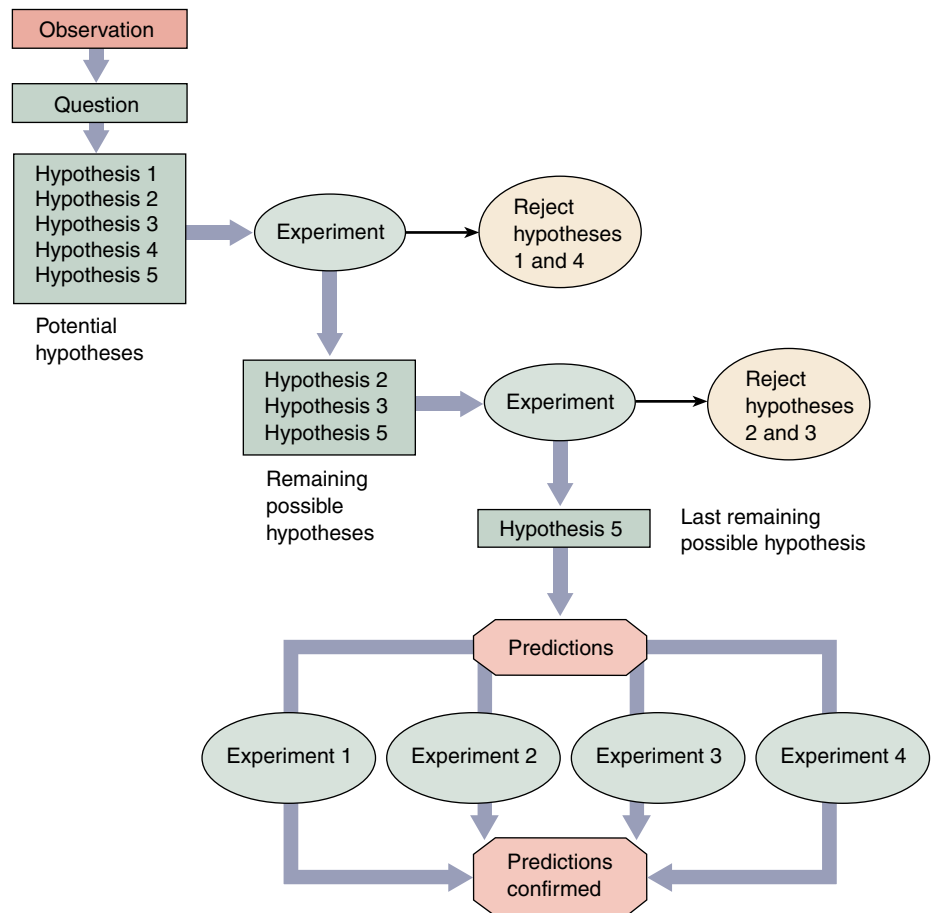


FIGURE 1.4
How science is done. This diagram illustrates the way in which scientific investigations proceed. First, scientists make observations that raise a particular question. They develop a number of potential explanations (hypotheses) to answer the question. Next, they carry out experiments in an attempt to eliminate one or more of these hypotheses. Then, predictions are made based on the remaining hypotheses, and further experiments are carried out to test these predictions. As a result of this process, the least unlikely hypothesis is selected.

Establishing Controls

Often we are interested in learning about processes that are influenced by many factors, or **variables**. To evaluate alternative hypotheses about one variable, all other variables must be kept constant. This is done by carrying out two experiments in parallel: in the first experiment, one variable is altered in a specific way to test a particular hypothesis; in the second experiment, called the **control experiment**, that variable is left unaltered. In all other respects the two experiments are identical, so any difference in the outcomes of the two experiments must result from the influence of the variable that was changed. Much of the challenge of experimental science lies in designing control experiments that isolate a particular variable from other factors that might influence a process.

Using Predictions

A successful scientific hypothesis needs to be not only valid but useful—it needs to tell you something you want to know. A hypothesis is most useful when it makes predictions, because those predictions provide a way to test the validity of the hypothesis. If an experiment produces results inconsistent with the predictions, the hypothesis must be rejected. On the other hand, if the predictions are supported by experimental testing, the hypothesis is supported. The more experimentally supported predictions a hypothesis makes, the more valid the hypothesis is. For example, Einstein's hypothesis of relativity was at first provisionally accepted because no one could devise an experiment that invalidated it. The hypothesis made a clear prediction: that the sun would bend the path of light passing by it. When this prediction was tested in a total eclipse, the light from background stars was indeed bent. Because this result was unknown when the hypothesis was being formulated, it provided strong support for the hypothesis, which was then accepted with more confidence.

Developing Theories

Scientists use the word **theory** in two main ways. A “theory” is a proposed explanation for some natural phenomenon, often based on some general principle. Thus one speaks of the principle first proposed by Newton as the “theory of gravity.” Such theories often bring together concepts that were previously thought to be unrelated, and offer unified explanations of different phenomena. Newton's theory of gravity provided a single explanation for objects falling to the ground and the orbits of planets around the sun. “Theory” is also used to mean the body of interconnected concepts, supported by scientific reasoning and experimental evidence, that explains the facts in some area of study. Such a theory provides an indispensable framework for organizing a body of knowledge. For example, quantum theory in physics brings together a

set of ideas about the nature of the universe, explains experimental facts, and serves as a guide to further questions and experiments.

To a scientist, such theories are the solid ground of science, that of which we are most certain. In contrast, to the general public, *theory* implies just the opposite—a *lack* of knowledge, or a guess. Not surprisingly, this difference often results in confusion. In this text, theory will always be used in its scientific sense, in reference to an accepted general principle or body of knowledge.

To suggest, as many critics outside of science do, that evolution is “just a theory” is misleading. The hypothesis that evolution has occurred is an accepted scientific fact; it is supported by overwhelming evidence. Modern evolutionary theory is a complex body of ideas whose importance spreads far beyond explaining evolution; its ramifications permeate all areas of biology, and it provides the conceptual framework that unifies biology as a science.

Research and the Scientific Method

It used to be fashionable to speak of the “scientific method” as consisting of an orderly sequence of logical “either/or” steps. Each step would reject one of two mutually incompatible alternatives, as if trial-and-error testing would inevitably lead one through the maze of uncertainty that always impedes scientific progress. If this were indeed so, a computer would make a good scientist. But science is not done this way. As British philosopher Karl Popper has pointed out, successful scientists without exception design their experiments with a pretty fair idea of how the results are going to come out. They have what Popper calls an “imaginative preconception” of what the truth might be. A hypothesis that a successful scientist tests is not just any hypothesis; rather, it is an educated guess or a hunch, in which the scientist integrates all that he or she knows and allows his or her imagination full play, in an attempt to get a sense of what *might* be true (see Box: How Biologists Do Their Work). It is because insight and imagination play such a large role in scientific progress that some scientists are so much better at science than others, just as Beethoven and Mozart stand out among most other composers.

Some scientists perform what is called basic research, which is intended to extend the boundaries of what we know. These individuals typically work at universities, and their research is usually financially supported by their institutions and by external sources, such as the government, industry, and private foundations. Basic research is as diverse as its name implies. Some basic scientists attempt to find out how certain cells take up specific chemicals, while others count the number of dents in tiger teeth. The information generated by basic research contributes to the growing body of scientific knowledge, and it provides the scientific foundation utilized by applied research. Scientists who conduct applied research are often employed in

How Biologists Do Their Work

The Consent

*Late in November, on a single night
Not even near to freezing, the ginkgo trees
That stand along the walk drop all their leaves
In one consent, and neither to rain nor to wind
But as though to time alone: the golden and
green*

*Leaves litter the lawn today, that yesterday
Had spread aloft their fluttering fans of light.
What signal from the stars? What senses took it
in?*

*What in those wooden motives so decided
To strike their leaves, to down their leaves,
Rebellion or surrender? And if this
Can happen thus, what race shall be exempt?
What use to learn the lessons taught by time,
If a star at any time may tell us: Now.*

Howard Nemerov

What is bothering the poet Howard Nemerov is that life is influenced by forces he cannot control or even identify. It is the job of biologists to solve puzzles such as the one he poses, to identify and try to understand those things that influence life.

Nemerov asks why ginkgo trees (figure 1.A) drop all their leaves at once. To find an answer to questions such as this, biologists and other scientists pose *possible* answers and then try to determine which answers are false. Tests of alternative possibilities are called experiments. To



FIGURE 1.A
A ginkgo tree.

learn why the ginkgo trees drop all their leaves simultaneously, a scientist would first formulate several possible answers, called hypotheses:

Hypothesis 1: Ginkgo trees possess an internal clock that times the release of leaves to match the season. On the day Nemerov describes, this clock sends a “drop” signal (perhaps a chemical) to all the leaves at the same time.

Hypothesis 2: The individual leaves of ginkgo trees are each able to sense day length, and

when the days get short enough in the fall, each leaf responds independently by falling.

Hypothesis 3: A strong wind arose the night before Nemerov made his observation, blowing all the leaves off the ginkgo trees.

Next, the scientist attempts to eliminate one or more of the hypotheses by conducting an experiment. In this case, one might cover some of the leaves so that they cannot use light to sense day length. If hypothesis 2 is true, then the covered leaves should not fall when the others do, because they are not receiving the same information. Suppose, however, that despite the covering of some of the leaves, all the leaves still fall together. This result would eliminate hypothesis 2 as a possibility. Either of the other hypotheses, and many others, remain possibilities.

This simple experiment with ginkgoes points out the essence of scientific progress: science does not prove that certain explanations are true; rather, it proves that others are not. Hypotheses that are inconsistent with experimental results are rejected, while hypotheses that are not proven false by an experiment are provisionally accepted. However, hypotheses may be rejected in the future when more information becomes available, if they are inconsistent with the new information. Just as finding the correct path through a maze by trying and eliminating false paths, scientists work to find the correct explanations of natural phenomena by eliminating false possibilities.

some kind of industry. Their work may involve the manufacturing of food additives, creating of new drugs, or testing the quality of the environment.

After developing a hypothesis and performing a series of experiments, a scientist writes a paper carefully describing the experiment and its results. He or she then submits the paper for publication in a scientific journal, but before it is published, it must be reviewed and accepted by other scientists who are familiar with that particular field of research. This process of careful evaluation, called peer review, lies at the heart of modern science, fostering careful work, precise description, and thoughtful analysis. When an important discovery is announced in a paper, other scientists attempt to reproduce the result, providing a check on accuracy and honesty. Nonreproducible results are not taken seriously for long.

The explosive growth in scientific research during the second half of the twentieth century is reflected in the enormous number of scientific journals now in existence. Although some, such as *Science* and *Nature*, are devoted to a wide range of scientific disciplines, most are extremely specialized: *Cell Motility and the Cytoskeleton*, *Glycoconjugate Journal*, *Mutation Research*, and *Synapse* are just a few examples.

The scientific process involves the rejection of hypotheses that are inconsistent with experimental results or observations. Hypotheses that are consistent with available data are conditionally accepted. The formulation of the hypothesis often involves creative insight.

1.3 Darwin's theory of evolution illustrates how science works.

Darwin's Theory of Evolution

Darwin's theory of evolution explains and describes how organisms on earth have changed over time and acquired a diversity of new forms. This famous theory provides a good example of how a scientist develops a hypothesis and how a scientific theory grows and wins acceptance.

Charles Robert Darwin (1809–1882; figure 1.5) was an English naturalist who, after 30 years of study and observation, wrote one of the most famous and influential books of all time. This book, *On the Origin of Species by Means of Natural Selection, or The Preservation of Favoured Races in the Struggle for Life*, created a sensation when it was published, and the ideas Darwin expressed in it have played a central role in the development of human thought ever since.

In Darwin's time, most people believed that the various kinds of organisms and their individual structures resulted from direct actions of the Creator (and to this day many people still believe this to be true). Species were thought to be specially created and unchangeable, or immutable, over the course of time. In contrast to these views, a number of earlier philosophers had presented the view that living things must have changed during the history of life on earth. Darwin proposed a concept he called natural selection as a coherent, logical explanation for this process, and he brought his ideas to wide public attention. His book, as its title indicates, presented a conclusion that differed sharply from conventional wisdom. Although his theory did not challenge the existence of a Divine Creator, Darwin argued that this Creator did not simply create things and then leave them forever unchanged. Instead, Darwin's God expressed Himself through the operation of natural laws that produced change over time, or **evolution**. These views put Darwin at odds with most people of his time, who believed in a literal interpretation of the Bible and accepted the idea of a fixed and constant world. His revolu-

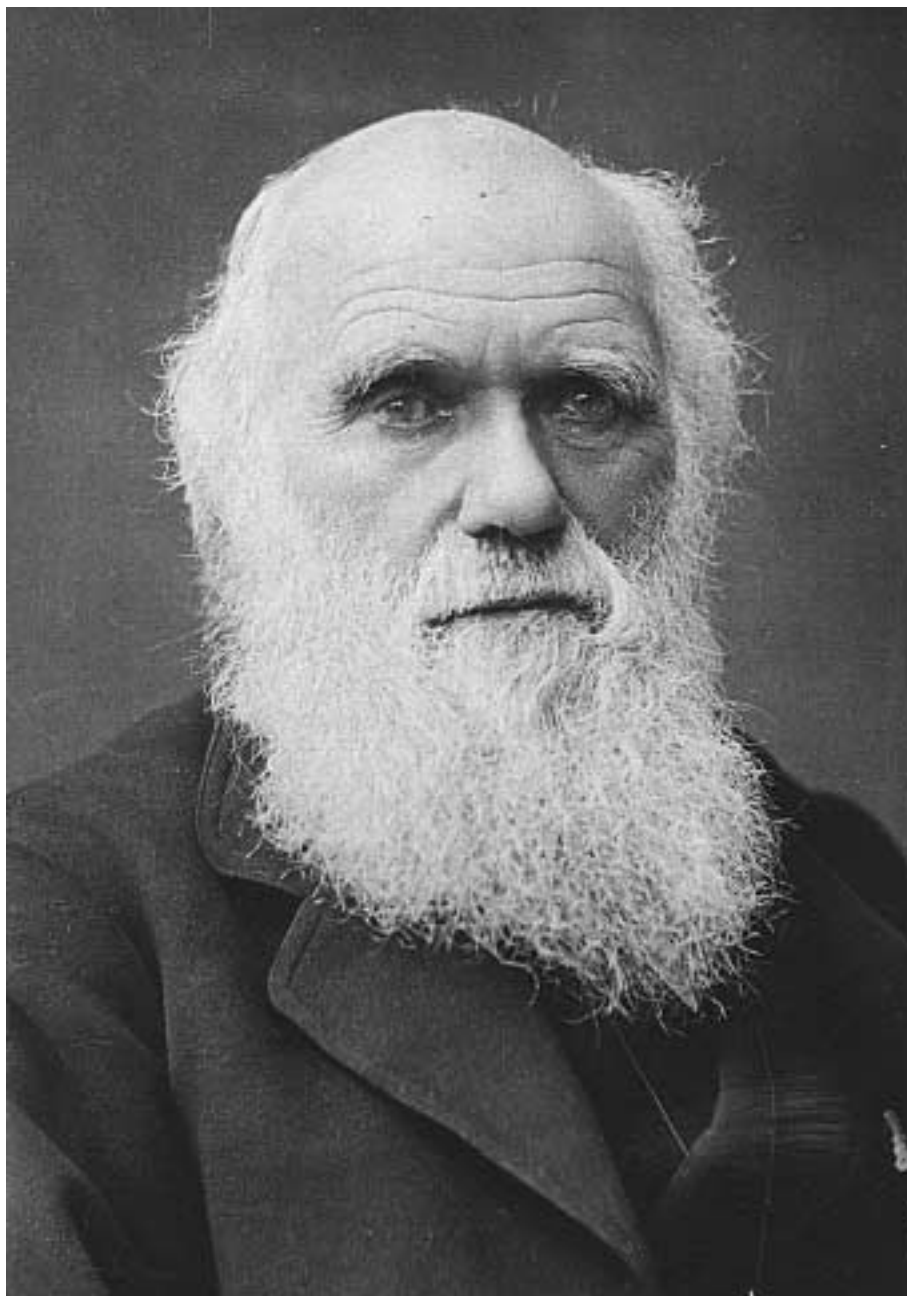


FIGURE 1.5

Charles Darwin. This newly rediscovered photograph taken in 1881, the year before Darwin died, appears to be the last ever taken of the great biologist.

tionary theory deeply troubled not only many of his contemporaries but Darwin himself.

The story of Darwin and his theory begins in 1831, when he was 22 years old. On the recommendation of one of his professors at Cambridge University, he was selected to serve

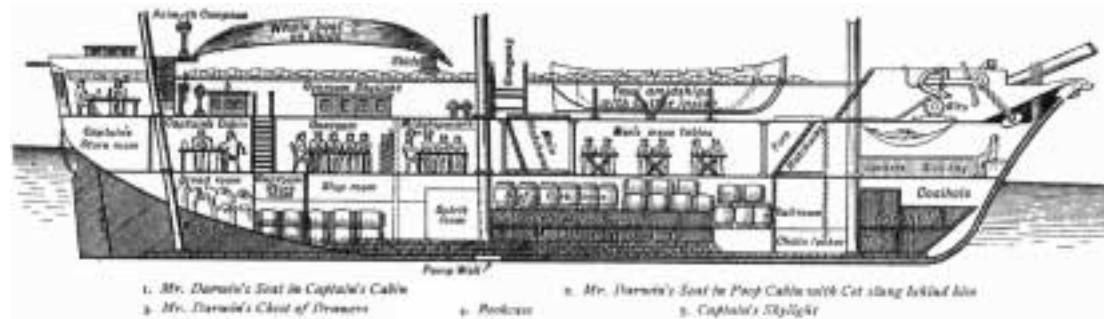


FIGURE 1.6

The five-year voyage of H.M.S. *Beagle*. Most of the time was spent exploring the coasts and coastal islands of South America, such as the Galápagos Islands. Darwin's studies of the animals of the Galápagos Islands played a key role in his eventual development of the theory of evolution by means of natural selection.

FIGURE 1.7

Cross section of the *Beagle*. A 10-gun brig of 242 tons, only 90 feet in length, the *Beagle* had a crew of 74 people! After he first saw the ship, Darwin wrote to his college professor Henslow: "The absolute want of room is an evil that nothing can surmount."



as naturalist on a five-year navigational mapping expedition around the coasts of South America (figure 1.6), aboard H.M.S. *Beagle* (figure 1.7). During this long voyage, Darwin had the chance to study a wide variety of plants and animals on continents and islands and in distant seas. He was able to explore the biological richness of the tropical forests, examine the extraordinary fossils of huge extinct mammals in Patagonia at the southern tip of South America, and observe the remarkable series of related but distinct forms of life on the Galápagos Islands, off the west coast of South America. Such an opportunity clearly played an important role in the development of his thoughts about the nature of life on earth.

When Darwin returned from the voyage at the age of 27, he began a long period of study and contemplation. During the next 10 years, he published important books on several

different subjects, including the formation of oceanic islands from coral reefs and the geology of South America. He also devoted eight years of study to barnacles, a group of small marine animals with shells that inhabit rocks and pilings, eventually writing a four-volume work on their classification and natural history. In 1842, Darwin and his family moved out of London to a country home at Down, in the county of Kent. In these pleasant surroundings, Darwin lived, studied, and wrote for the next 40 years.

Darwin was the first to propose natural selection as an explanation for the mechanism of evolution that produced the diversity of life on earth. His hypothesis grew from his observations on a five-year voyage around the world.

Darwin's Evidence

One of the obstacles that had blocked the acceptance of any theory of evolution in Darwin's day was the incorrect notion, widely believed at that time, that the earth was only a few thousand years old. Evidence discovered during Darwin's time made this assertion seem less and less likely. The great geologist Charles Lyell (1797–1875), whose *Principles of Geology* (1830) Darwin read eagerly as he sailed on the *Beagle*, outlined for the first time the story of an ancient world of plants and animals in flux. In this world, species were constantly becoming extinct while others were emerging. It was this world that Darwin sought to explain.

What Darwin Saw

When the *Beagle* set sail, Darwin was fully convinced that species were immutable. Indeed, it was not until two or three years after his return that he began to consider seriously the possibility that they could change. Nevertheless, during his five years on the ship, Darwin observed a number of phenomena that were of central importance to him in reaching his ultimate conclusion (table 1.1). For example, in the rich fossil beds of southern South America, he observed fossils of extinct armadillos similar to the armadillos that still lived in the same area (figure 1.8). Why would similar living and fossil organisms be in the same area unless the earlier form had given rise to the other?

Repeatedly, Darwin saw that the characteristics of similar species varied somewhat from place to place. These geographical patterns suggested to him that organismal lineages change gradually as species migrate from one area to another. On the Galápagos Islands, off the coast of Ecuador, Darwin encountered giant land tortoises. Surprisingly, these tortoises were not all identical. In fact, local residents and the sailors who captured the tortoises for food could tell which island a particular tortoise had come from just by looking at its shell. This distribution of physical variation suggested that all of the tortoises were related, but that they had changed slightly in appearance after becoming isolated on different islands.

In a more general sense, Darwin was struck by the fact that the plants and animals on these relatively young volcanic islands resembled those on the nearby coast of South America. If each one of these plants and animals had been created independently and simply placed on the Galápagos Islands, why didn't they resemble the plants and animals of islands with similar climates, such as those off the coast of Africa, for example? Why did they resemble those of the adjacent South American coast instead?

The fossils and patterns of life that Darwin observed on the voyage of the *Beagle* eventually convinced him that evolution had taken place.

Table 1.1 Darwin's Evidence that Evolution Occurs

FOSSILS

- 1. Extinct species, such as the fossil armadillo in figure 1.8, most closely resemble living ones in the same area, suggesting that one had given rise to the other.
- 2. In rock strata (layers), progressive changes in characteristics can be seen in fossils from earlier and earlier layers.

GEOGRAPHICAL DISTRIBUTION

- 3. Lands with similar climates, such as Australia, South Africa, California, and Chile, have unrelated plants and animals, indicating that diversity is not entirely influenced by climate and environment.
- 4. The plants and animals of each continent are distinctive; all South American rodents belong to a single group, structurally similar to the guinea pigs, for example, while most of the rodents found elsewhere belong to other groups.

OCEANIC ISLANDS

- 5. Although oceanic islands have few species, those they do have are often unique (endemic) and show relatedness to one another, such as the Galápagos tortoises. This suggests that the tortoises and other groups of endemic species developed after their mainland ancestors reached the islands and are, therefore, more closely related to one another.
- 6. Species on oceanic islands show strong affinities to those on the nearest mainland. Thus, the finches of the Galápagos Islands closely resemble a finch seen on the western coast of South America. The Galápagos finches do *not* resemble the birds on the Cape Verde Islands, islands in the Atlantic Ocean off the coast of Africa that are similar to the Galápagos. Darwin visited the Cape Verde Islands and many other island groups personally and was able to make such comparisons on the basis of his own observations.

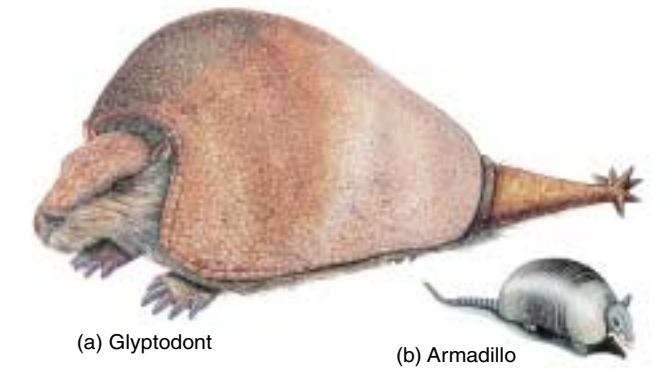


FIGURE 1.8 Fossil evidence of evolution. The now-extinct glyptodont (a) was a 2000-kilogram South American armadillo, much larger than the modern armadillo (b), which weighs an average of about 4.5 kilograms. (Drawings are not to scale.)

Inventing the Theory of Natural Selection

It is one thing to observe the results of evolution, but quite another to understand how it happens. Darwin's great achievement lies in his formulation of the hypothesis that evolution occurs because of natural selection.

Darwin and Malthus

Of key importance to the development of Darwin's insight was his study of Thomas Malthus's *Essay on the Principle of Population* (1798). In his book, Malthus pointed out that populations of plants and animals (including human beings) tend to increase geometrically, while the ability of humans to increase their food supply increases only arithmetically. A *geometric progression* is one in which the elements increase by a constant *factor*; for example, in the progression 2, 6, 18, 54, . . . , each number is three times the preceding one. An *arithmetic progression*, in contrast, is one in which the elements increase by a constant *difference*; in the progression 2, 6, 10, 14, . . . , each number is four greater than the preceding one (figure 1.9).

Because populations increase geometrically, virtually any kind of animal or plant, if it could reproduce unchecked, would cover the entire surface of the world within a surprisingly short time. Instead, populations of species remain fairly constant year after year, because death limits population numbers. Malthus's conclusion provided the key ingredient that was necessary for Darwin to develop the hypothesis that evolution occurs by natural selection.

Sparked by Malthus's ideas, Darwin saw that although every organism has the potential to produce more offspring than can survive, only a limited number actually do survive and produce further offspring. Combining this observation with what he had seen on the voyage of the *Beagle*, as well as with his own experiences in breeding domestic animals, Darwin made an important association (figure 1.10): Those individuals that possess superior physical, behavioral, or other attributes are more likely to survive than those that are not so well endowed. By surviving, they gain the opportunity to pass on their favorable characteristics to their offspring. As the frequency of these characteristics increases in the population, the nature of the population as a whole will gradually change. Darwin called this process selection. The driving force he identified has often been referred to as survival of the fittest.

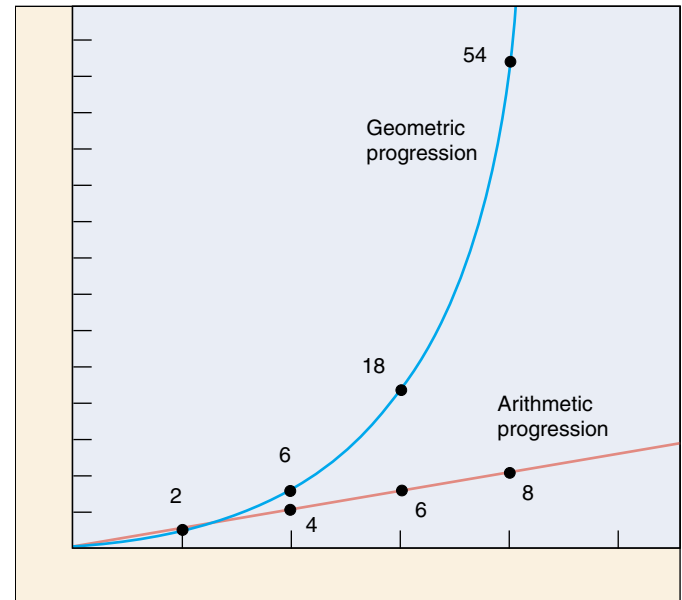


FIGURE 1.9

Geometric and arithmetic progressions. A geometric progression increases by a constant factor (e.g., $\times 2$ or $\times 3$ or $\times 4$), while an arithmetic progression increases by a constant difference (e.g., units of 1 or 2 or 3). Malthus contended that the human growth curve was geometric, but the human food production curve was only arithmetic. Can you see the problems this difference would cause?

"Can we doubt . . . that individuals having any advantage, however slight, over others, would have the best chance of surviving and procreating their kind? On the other hand, we may feel sure that any variation in the least degree injurious would be rigidly destroyed. This preservation of favorable variations, I call Natural Selection."

FIGURE 1.10

An excerpt from Charles Darwin's *On the Origin of Species*.

Natural Selection

Darwin was thoroughly familiar with variation in domesticated animals and began *On the Origin of Species* with a detailed discussion of pigeon breeding. He knew that breeders selected certain varieties of pigeons and other animals, such as dogs, to produce certain characteristics, a process Darwin called **artificial selection**. Once this had been done, the animals would breed true for the characteristics that had been selected. Darwin had also observed that the differences purposely developed between domesticated races or breeds were often greater than those that separated wild species. Domestic pigeon breeds, for example, show much greater variety than all of the hundreds of wild species of pigeons found throughout the world. Such relationships suggested to Darwin that evolutionary change could occur in nature too. Surely if pigeon breeders could foster such variation by “artificial selection,” nature could do the same, playing the breeder’s role in selecting the next generation—a process Darwin called **natural selection**.

Darwin’s theory thus incorporates the hypothesis of evolution, the process of natural selection, and the mass of new evidence for both evolution and natural selection that Darwin compiled. Thus, Darwin’s theory provides a simple and direct explanation of biological diversity, or why animals are different in different places: because habitats differ in their requirements and opportunities, the organisms with characteristics favored locally by natural selection will tend to vary in different places.

Darwin Drafts His Argument

Darwin drafted the overall argument for evolution by natural selection in a preliminary manuscript in 1842. After showing the manuscript to a few of his closest scientific friends, however, Darwin put it in a drawer, and for 16 years turned to other research. No one knows for sure why Darwin did not publish his initial manuscript—it is very thorough and outlines his ideas in detail. Some historians have suggested that Darwin was shy of igniting public criticism of his evolutionary ideas—there could have been little doubt in his mind that his theory of evolution by natural selection would spark controversy. Others have pro-



FIGURE 1.11
Darwin greets his monkey ancestor. In his time, Darwin was often portrayed unsympathetically, as in this drawing from an 1874 publication.

posed that Darwin was simply refining his theory all those years, although there is little evidence he altered his initial manuscript in all that time.

Wallace Has the Same Idea

The stimulus that finally brought Darwin’s theory into print was an essay he received in 1858. A young English naturalist named Alfred Russel Wallace (1823–1913) sent the essay to Darwin from Malaysia; it concisely set forth the theory of evolution by means of natural selection, a theory Wallace had developed independently of Darwin. Like Darwin, Wallace had been greatly influenced by Malthus’s 1798 essay. Colleagues of Wallace, knowing of Darwin’s work, encouraged him to communicate with Darwin. After receiving Wallace’s essay, Darwin arranged for a joint presentation of their ideas at a seminar in London. Darwin then completed his own book, expanding the 1842 manuscript which he had written so long ago, and submitted it for publication.

Publication of Darwin’s Theory

Darwin’s book appeared in November 1859 and caused an immediate sensation. Many people were deeply disturbed by the suggestion that human beings were descended from the same ancestor as apes (figure 1.11). Darwin did not actually discuss this idea in his book, but it followed directly from the principles he outlined. In a subsequent book, *The Descent of Man*, Darwin presented the argument directly, building a powerful case that humans and living apes have common ancestors. Although people had long accepted that humans closely resembled apes in many characteristics, the possibility that there might be a direct evolutionary relationship was unacceptable to many. Darwin’s arguments for the theory of evolution by natural selection were so compelling, however, that his views were almost completely accepted within the intellectual community of Great Britain after the 1860s.

The fact that populations do not really expand geometrically implies that nature acts to limit population numbers. The traits of organisms that survive to produce more offspring will be more common in future generations—a process Darwin called natural selection.

Evolution After Darwin: More Evidence

More than a century has elapsed since Darwin's death in 1882. During this period, the evidence supporting his theory has grown progressively stronger. There have also been many significant advances in our understanding of how evolution works. Although these advances have not altered the basic structure of Darwin's theory, they have taught us a great deal more about the mechanisms by which evolution occurs. We will briefly explore some of this evidence here; in chapter 21 we will return to the theory of evolution and examine the evidence in more detail.

The Fossil Record

Darwin predicted that the fossil record would yield intermediate links between the great groups of organisms, for example, between fishes and the amphibians thought to have arisen from them, and between reptiles and birds. We now know the fossil record to a degree that was unthinkable in the nineteenth century. Recent discoveries of microscopic fossils have extended the known history of life on earth back to about 3.5 billion years ago. The discovery of other fossils has supported Darwin's predictions and has shed light on how organisms have, over this enormous time span, evolved from the simple to the complex. For vertebrate animals especially, the fossil record is rich and exhibits a graded series of changes in form, with the evolutionary parade visible for all to see (see Box: Why Study Fossils?).

The Age of the Earth

In Darwin's day, some physicists argued that the earth was only a few thousand years old. This bothered Darwin, because the evolution of all living things from some single

original ancestor would have required a great deal more time. Using evidence obtained by studying the rates of radioactive decay, we now know that the physicists of Darwin's time were wrong, very wrong: the earth was formed about 4.5 billion years ago.

The Mechanism of Heredity

Darwin received some of his sharpest criticism in the area of heredity. At that time, no one had any concept of genes or of how heredity works, so it was not possible for Darwin to explain completely how evolution occurs. Theories of heredity in Darwin's day seemed to rule out the possibility of genetic variation in nature, a critical requirement of Darwin's theory. Genetics was established as a science only at the start of the twentieth century, 40 years after the publication of Darwin's *On the Origin of Species*. When scientists began to understand the laws of inheritance (discussed in chapter 13), the heredity problem with Darwin's theory vanished. Genetics accounts in a neat and orderly way for the production of new variations in organisms.

Comparative Anatomy

Comparative studies of animals have provided strong evidence for Darwin's theory. In many different types of vertebrates, for example, the same bones are present, indicating their evolutionary past. Thus, the forelimbs shown in figure 1.12 are all constructed from the same basic array of bones, modified in one way in the wing of a bat, in another way in the fin of a porpoise, and in yet another way in the leg of a horse. The bones are said to be **homologous** in the different vertebrates; that is, they have the same evolutionary origin, but they now differ in structure and function. This contrasts with **analogous** structures, such as the wings of birds and butterflies, which have similar structure and function but different evolutionary origins.

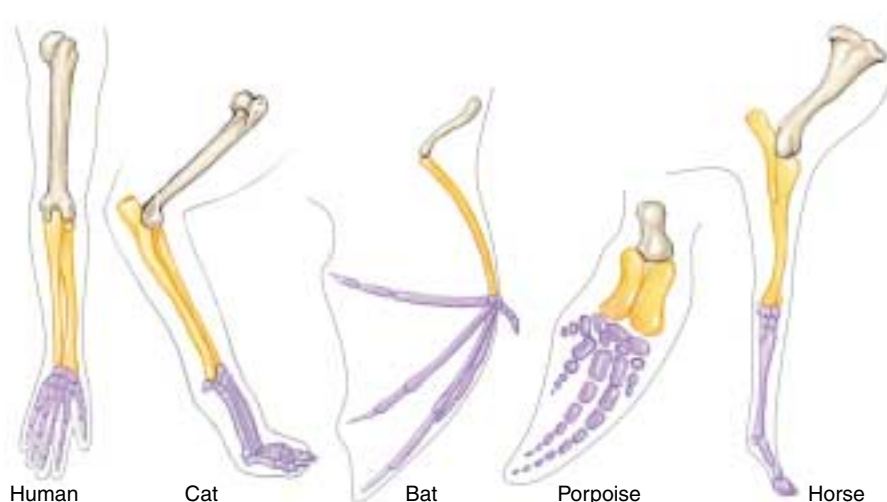


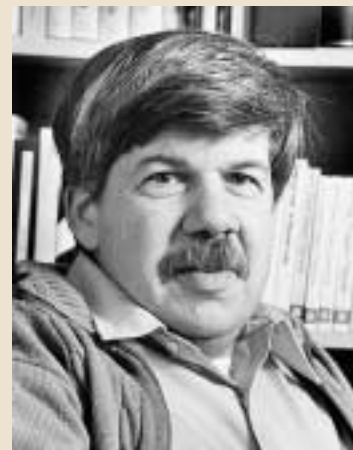
FIGURE 1.12
Homology among vertebrate limbs. The forelimbs of these five vertebrates show the ways in which the relative proportions of the forelimb bones have changed in relation to the particular way of life of each organism.

Why Study Fossils?

I grew up on the streets of New York City, in a family of modest means and little formal education, but with a deep love of learning. Like many urban kids who become naturalists, my inspiration came from a great museum—in particular, from the magnificent dinosaurs on display at the American Museum of Natural History. As we all know from *Jurassic Park* and other sources, dinomania in young children (I was five when I saw my first dinosaur) is not rare—but nearly all children lose the passion, and the desire to become a paleontologist becomes a transient moment between policeman and fireman in a chronology of intended professions. But I persisted and became a professional paleontologist, a student of life's history as revealed by the evidence of fossils (though I ended up working on snails rather than dinosaurs!). Why?

I remained committed to paleontology because I discovered, still as a child, the wonder of one of the greatest transforming ideas ever discovered by science: evolution. I learned that those dinosaurs, and all creatures that have ever lived, are bound together in a grand family tree of physical relationships, and that the rich and fascinating changes of life, through billions of years in

Flight has evolved three separate times among vertebrates. Birds and bats are still with us, but pterosaurs, such as the one pictured, became extinct with the dinosaurs about 65 million years ago.



Stephen Jay Gould
Harvard University

geological time, occur by a natural process of evolutionary transformation—"descent with modification," in Darwin's words. I was thrilled to learn that humans had arisen from apelike ancestors, who had themselves evolved from the tiny mouselike mammals that had lived in the time of dinosaurs and seemed then so inconspicuous, so unsuccessful, and so unpromising.

Now, at mid-career (I was born in 1941) I remain convinced that I made the right choice, and committed to learn and convey, as much as I can as long as I can, about evolution and the history of life. We can learn a great deal about the *process* of evolution

by studying modern organisms. But history is complex and unpredictable—and principles of evolution (like natural selection) cannot specify the *pathway* that life's history has actually followed. Paleontology holds the archives of the pathway—the fossil record of past life, with its fascinating history of mass extinctions, periods of rapid change, long episodes of stability, and constantly changing patterns of dominance and diversity. Humans represent just one tiny, largely fortuitous, and late-arising twig on the enormously arborescent bush of life. Paleontology is the study of this grandest of all bushes.

Molecular Biology

Biochemical tools are now of major importance in efforts to reach a better understanding of how evolution occurs. Within the last few years, for example, evolutionary biologists have begun to "read" genes, much as you are reading this page. They have learned to recognize the order of the "letters" of the long DNA molecules, which are present in every living cell and which provide the genetic information for that organism. By comparing the sequences of "letters" in the DNA of different groups of animals or plants, we can specify the degree of relationship among the groups more precisely than by any other means. In many cases, detailed family trees can then be constructed. The consistent pattern emerging from a growing mountain of data is one of progressive change over time, with more distantly related species showing more differences in their DNA than closely related ones, just as Darwin's theory predicts. By measuring the degree of difference in the genetic coding, and by interpreting the information available from the fossil record, we

can even estimate the *rates* at which evolution is occurring in different groups of organisms.

Development

Twentieth-century knowledge about growth and development further supports Darwin's theory of evolution. Striking similarities are seen in the developmental stages of many organisms of different species. Human embryos, for example, go through a stage in which they possess the same structures that give rise to the gills in fish, a tail, and even a stage when the embryo has fur! Thus, the development of an organism (its ontogeny) often yields information about the evolutionary history of the species as a whole (its phylogeny).

Since Darwin's time, new discoveries of the fossil record, genetics, anatomy, and development all support Darwin's theory.

1.4 This book is organized to help you learn biology.

Core Principles of Biology

From centuries of biological observation and inquiry, one organizing principle has emerged: biological diversity reflects history, a record of success, failure, and change extending back to a period soon after the formation of the earth. The explanation for this diversity, the theory of evolution by natural selection, will form the backbone of your study of biological science, just as the theory of the covalent bond is the backbone of chemistry, or the theory of quantum mechanics is that of physics. Evolution by natural selection is a thread that runs through everything you will learn in this book.

Basic Principles

The first half of this book is devoted to a description of the basic principles of biology, introduced through a levels-of-organization framework (see figure 1.2). At the molecular, organellar, and cellular levels of organization, you will be introduced to *cell biology*. You will learn how cells are constructed and how they grow, divide, and communicate. At the organismal level, you will learn the principles of *genetics*, which deal with the way that individual traits are transmitted from one generation to the next. At the population level,

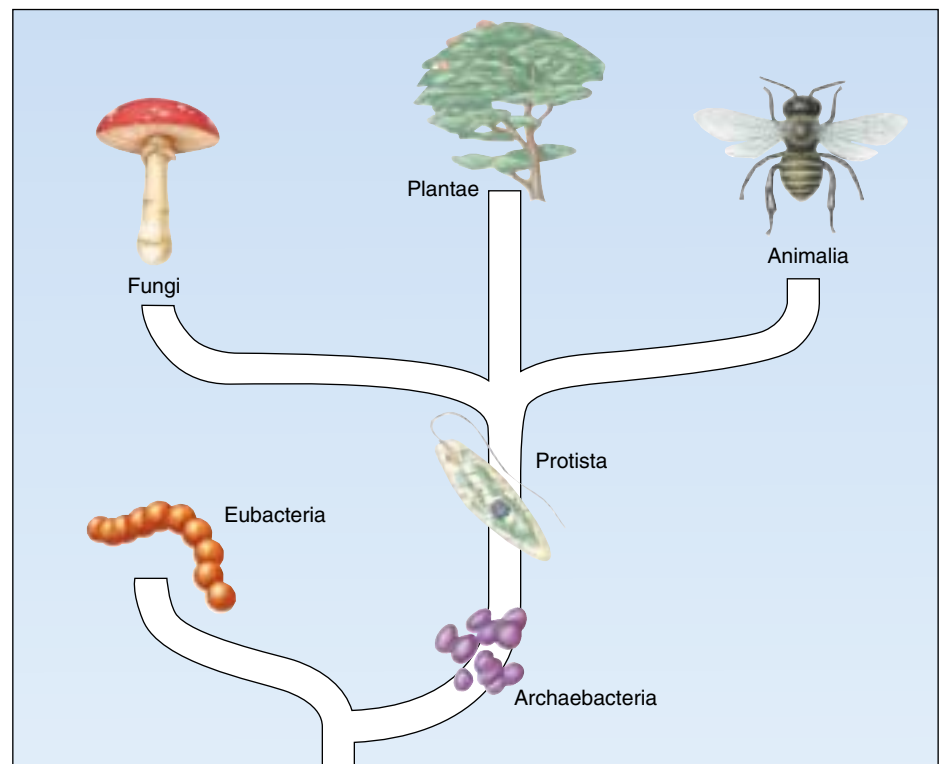
you will examine *evolution*, the gradual change in populations from one generation to the next, which has led through natural selection to the biological diversity we see around us. Finally, at the community and ecosystem levels, you will study *ecology*, which deals with how organisms interact with their environments and with one another to produce the complex communities characteristic of life on earth.

Organisms

The second half of the book is devoted to an examination of organisms, the products of evolution. It is estimated that at least 5 million different kinds of plants, animals, and microorganisms exist, and their diversity is *incredible* (figure 1.13). Later in the book, we will take a particularly detailed look at the vertebrates, the group of animals of which we are members. We will consider the vertebrate body and how it functions, as this information is of greatest interest and importance to most students.

As you proceed through this book, what you learn at one stage will give you the tools to understand the next. The core principle of biology is that biological diversity is the result of a long evolutionary journey.

FIGURE 1.13
The diversity of life. Biologists categorize all living things into six major groups called kingdoms: archaeobacteria, eubacteria, protists, fungi, plants, and animals.





Summary

Questions

Media Resources

1.1 Biology is the science of life.

- Living things are highly organized, whether as single cells or as multicellular organisms, with several hierarchical levels.

1. What are the characteristics of living things?



- Art Activity: Biological organization

1.2 Scientists form generalizations from observations.

- Science is the determination of general principles from observation and experimentation.
- Scientists select the best hypotheses by using controlled experiments to eliminate alternative hypotheses that are inconsistent with observations.
- A group of related hypotheses supported by a large body of evidence is called a theory. In science, a theory represents what we are most sure about. However, there are no absolute truths in science, and even theories are accepted only conditionally.
- Scientists conduct basic research, designed to gain information about natural phenomena in order to contribute to our overall body of knowledge, and applied research, devoted to solving specific problems with practical applications.

2. What is the difference between deductive and inductive reasoning? What is a hypothesis?
3. What are variables? How are control experiments used in testing hypotheses?
4. How does a hypothesis become a theory? At what point does a theory become accepted as an absolute truth, no longer subject to any uncertainty?
5. What is the difference between basic and applied research?



- Scientists on Science: Why Paleontology?
- Experiments: Probability and Hypothesis Testing in Biology

1.3 Darwin's theory of evolution illustrates how science works.

- One of the central theories of biology is Darwin's theory that evolution occurs by natural selection. It states that certain individuals have heritable traits that allow them to produce more offspring in a given kind of environment than other individuals lacking those traits. Consequently, those traits will increase in frequency through time.
- Because environments differ in their requirements and opportunities, the traits favored by natural selection will vary in different environments.
- This theory is supported by a wealth of evidence acquired over more than a century of testing and questioning.

6. Describe the evidence that led Darwin to propose that evolution occurs by means of natural selection. What evidence gathered since the publication of Darwin's theory has lent further support to the theory?
7. What is the difference between homologous and analogous structures? Give an example of each.



- Introduction to Evolution
- Before Darwin
- Voyage of the Beagle
- Natural Selection
- The Process of Natural Selection
- Evidence for Evolution



- Student Research: The Search for Medicinal Plants on Science Articles
- 140 Years Without Darwin Are Enough
- Bird-Killing Cats: Nature's Way of Making Better Bids

1.4 This book is organized to help you learn biology.

- Biological diversity is the result of a long history of evolutionary change. For this reason evolution is the core of the science of biology.
- Considered in terms of levels-of-organization, the science of biology can be said to consist of subdisciplines focusing on particular levels. Thus one speaks of molecular biology, cell biology, organismal biology, population biology, and community biology.

8. Can you think of any alternatives to levels-of-organization as ways of organizing the mass of information in biology?

2

The Nature of Molecules

Concept Outline

2.1 Atoms are nature's building material.

Atoms. All substances are composed of tiny particles called atoms, each a positively charged nucleus around which orbit negative electrons.

Electrons Determine the Chemical Behavior of Atoms.

Electrons orbit the nucleus of an atom; the closer an electron's orbit to the nucleus, the lower its energy level.

2.2 The atoms of living things are among the smallest.

Kinds of Atoms. Of the 92 naturally occurring elements, only 11 occur in organisms in significant amounts.

2.3 Chemical bonds hold molecules together.

Ionic Bonds Form Crystals. Atoms are linked together into molecules, joined by chemical bonds that result from forces like the attraction of opposite charges or the sharing of electrons.

Covalent Bonds Build Stable Molecules. Chemical bonds formed by the sharing of electrons can be very strong, and require much energy to break.

2.4 Water is the cradle of life.

Chemistry of Water. Water forms weak chemical associations that are responsible for much of the organization of living chemistry.

Water Atoms Act Like Tiny Magnets. Because electrons are shared unequally by the hydrogen and oxygen atoms of water, a partial charge separation occurs. Each water atom acquires a positive and negative pole and is said to be "polar."

Water Clings to Polar Molecules. Because the opposite partial charges of polar molecules attract one another, water tends to cling to itself and other polar molecules and to exclude nonpolar molecules.

Water Ionizes. Because its covalent bonds occasionally break, water contains a low concentration of hydrogen (H^+) and hydroxide (OH^-) ions, the fragments of broken water molecules.

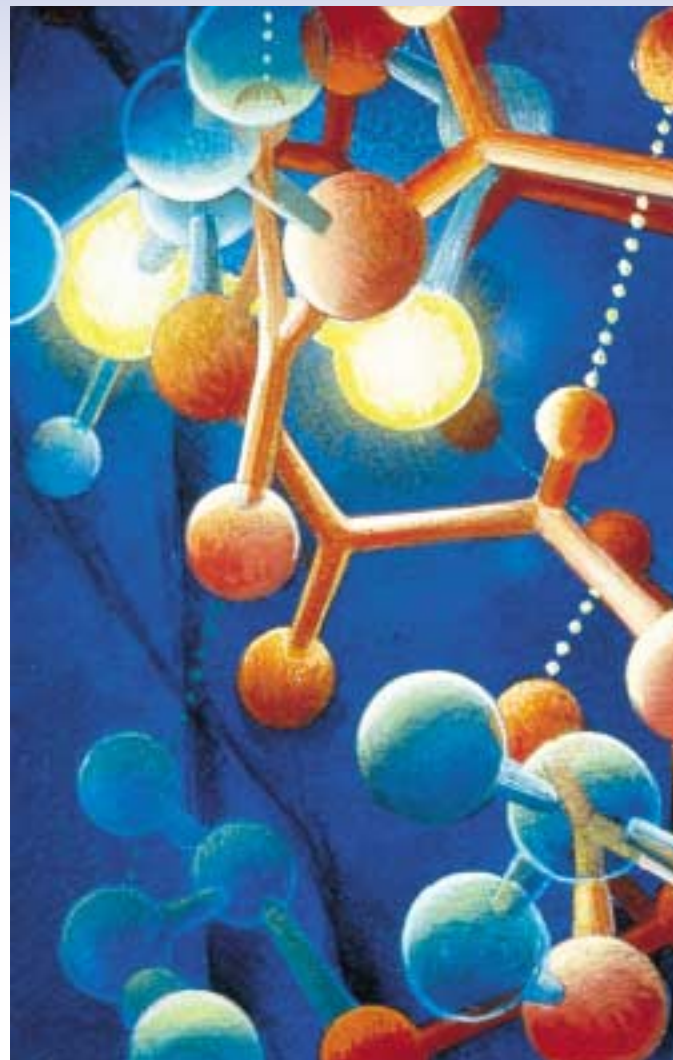


FIGURE 2.1

Cells are made of molecules. Specific, often simple, combinations of atoms yield an astonishing diversity of molecules within the cell, each with unique functional characteristics.

About 10 to 20 billion years ago, an enormous explosion likely marked the beginning of the universe. With this explosion began the process of evolution, which eventually led to the origin and diversification of life on earth. When viewed from the perspective of 20 billion years, life within our solar system is a recent development, but to understand the origin of life, we need to consider events that took place much earlier. The same processes that led to the evolution of life were responsible for the evolution of molecules (figure 2.1). Thus, our study of life on earth begins with physics and chemistry. As chemical machines ourselves, we must understand chemistry to begin to understand our origins.

2.1 Atoms are nature's building material.

Atoms

Any substance in the universe that has mass (see below) and occupies space is defined as **matter**. All matter is composed of extremely small particles called **atoms**. Because of their size, atoms are difficult to study. Not until early in this century did scientists carry out the first experiments suggesting what an atom is like.

The Structure of Atoms

Objects as small as atoms can be “seen” only indirectly, by using very complex technology such as tunneling microscopy. We now know a great deal about the complexities of atomic structure, but the simple view put forth in 1913 by the Danish physicist Niels Bohr provides a good starting point. Bohr proposed that every atom possesses an orbiting cloud of tiny subatomic particles called **electrons** whizzing around a core like the planets of a miniature solar system. At the center of each atom is a small, very dense nucleus formed of two other kinds of subatomic particles, **protons** and **neutrons** (figure 2.2).

Within the nucleus, the cluster of protons and neutrons is held together by a force that works only over short subatomic distances. Each proton carries a positive (+) charge, and each electron carries a negative (–) charge. Typically an atom has one electron for each proton. The number of protons (the atom's **atomic number**) determines the chemical character of the atom, because it dictates the number of electrons orbiting the nucleus which are available for chemical activity. Neutrons, as their name implies, possess no charge.

Atomic Mass

The terms *mass* and *weight* are often used interchangeably, but they have slightly different meanings. *Mass* refers to the amount of a substance, while *weight* refers to the force gravity exerts on a substance. Hence, an object has the same *mass* whether it is on the earth or the moon, but its

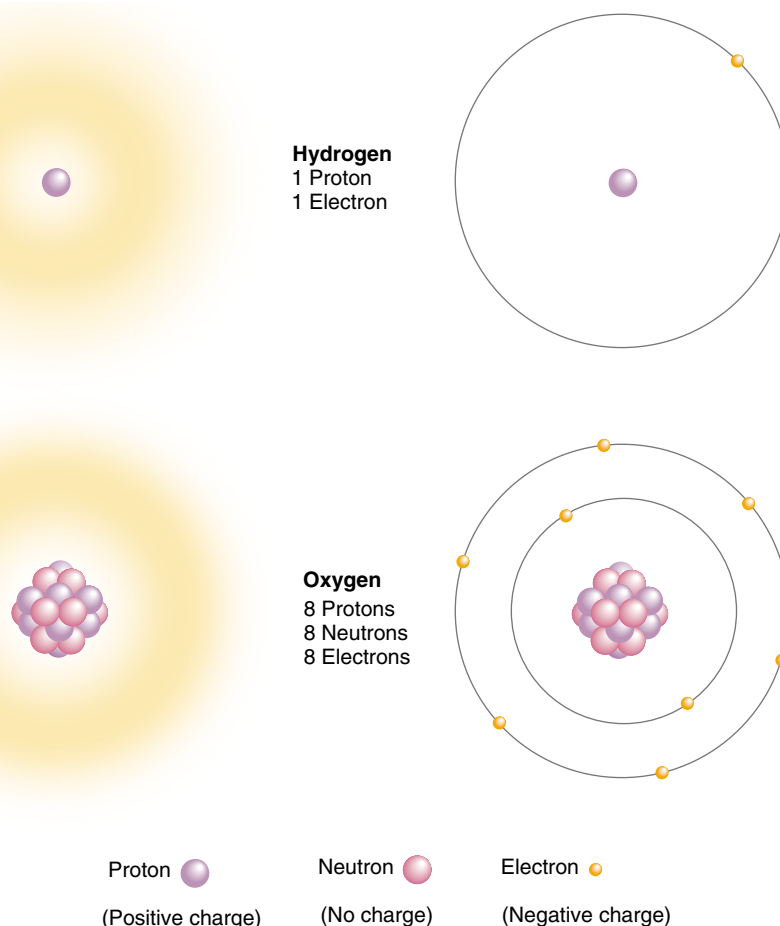
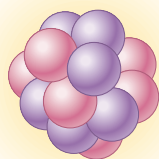


FIGURE 2.2

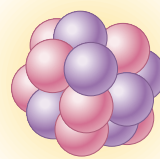
Basic structure of atoms. All atoms have a nucleus consisting of protons and neutrons, except hydrogen, the smallest atom, which has only one proton and no neutrons in its nucleus. Oxygen, for example, has eight protons and eight neutrons in its nucleus. Electrons spin around the nucleus a far distance away from the nucleus.

weight will be greater on the earth because the earth's gravitational force is greater than the moon's. The **atomic mass** of an atom is equal to the sum of the masses of its protons and neutrons. Atoms that occur naturally on earth contain from 1 to 92 protons and up to 146 neutrons.

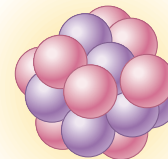
The mass of atoms and subatomic particles is measured in units called *daltons*. To give you an idea of just how small these units are, note that it takes 602 million million billion (6.02×10^{23}) daltons to make 1 gram! A proton weighs approximately 1 dalton (actually 1.009 daltons), as does a neutron (1.007 daltons). In contrast, electrons weigh only $\frac{1}{1840}$ of a dalton, so their contribution to the overall mass of an atom is negligible.



Carbon-12
6 Protons
6 Neutrons
6 Electrons



Carbon-13
6 Protons
7 Neutrons
6 Electrons



Carbon-14
6 Protons
8 Neutrons
6 Electrons

FIGURE 2.3

The three most abundant isotopes of carbon. Isotopes of a particular atom have different numbers of neutrons.

Isotopes

Atoms with the same atomic number (that is, the same number of protons) have the same chemical properties and are said to belong to the same **element**. Formally speaking, an element is any substance that cannot be broken down to any other substance by ordinary chemical means. However, while all atoms of an element have the same number of protons, they may not all have the same number of neutrons. Atoms of an element that possess different numbers of neutrons are called **isotopes** of that element. Most elements in nature exist as mixtures of different isotopes. Carbon (C), for example, has three isotopes, all containing six protons (figure 2.3). Over 99% of the carbon found in nature exists as an isotope with six neutrons. Because its total mass is 12 daltons (6 from protons plus 6 from neutrons), this isotope is referred to as carbon-12, and symbolized ^{12}C . Most of the rest of the naturally occurring carbon is carbon-13, an isotope with seven neutrons. The rarest carbon isotope is carbon-14, with eight neutrons. Unlike the other two isotopes, carbon-14 is unstable: its nucleus tends to break up into elements with lower atomic numbers. This nuclear breakup, which emits a significant amount of energy, is called radioactive decay, and isotopes that decay in this fashion are **radioactive isotopes**.

Some radioactive isotopes are more unstable than others and therefore decay more readily. For any given isotope, however, the rate of decay is constant. This rate is usually expressed as the **half-life**, the time it takes for one half of the atoms in a sample to decay. Carbon-14, for example, has a half-life of about 5600 years. A sample of carbon containing 1 gram of carbon-14 today would contain 0.5 gram of carbon-14 after 5600 years, 0.25 gram 11,200 years from now, 0.125 gram 16,800 years from now, and so on. By determining the ratios of the different isotopes of carbon and other elements in biological samples and in rocks, scientists are able to accurately determine when these materials formed.

While there are many useful applications of radioactivity, there are also harmful side effects that must be considered in any planned use of radioactive substances. Radioactive substances emit energetic subatomic particles that have the po-

tential to severely damage living cells, producing mutations in their genes, and, at high doses, cell death. Consequently, exposure to radiation is now very carefully controlled and regulated. Scientists who work with radioactivity (basic researchers as well as applied scientists such as X-ray technologists) wear radiation-sensitive badges to monitor the total amount of radioactivity to which they are exposed. Each month the badges are collected and scrutinized. Thus, employees whose work places them in danger of excessive radioactive exposure are equipped with an “early warning system.”

Electrons

The positive charges in the nucleus of an atom are counterbalanced by negatively charged electrons orbiting at varying distances around the nucleus. Thus, atoms with the same number of protons and electrons are electrically neutral, having no net charge.

Electrons are maintained in their orbits by their attraction to the positively charged nucleus. Sometimes other forces overcome this attraction and an atom loses one or more electrons. In other cases, atoms may gain additional electrons. Atoms in which the number of electrons does not equal the number of protons are known as **ions**, and they carry a net electrical charge. An atom that has more protons than electrons has a net positive charge and is called a **cation**. For example, an atom of sodium (Na) that has lost one electron becomes a sodium ion (Na^+), with a charge of +1. An atom that has fewer protons than electrons carries a net negative charge and is called an **anion**. A chlorine atom (Cl) that has gained one electron becomes a chloride ion (Cl^-), with a charge of -1.

An atom consists of a nucleus of protons and neutrons surrounded by a cloud of electrons. The number of its electrons largely determines the chemical properties of an atom. Atoms that have the same number of protons but different numbers of neutrons are called isotopes. Isotopes of an atom differ in atomic mass but have similar chemical properties.

Electrons Determine the Chemical Behavior of Atoms

The key to the chemical behavior of an atom lies in the arrangement of its electrons in their orbits. It is convenient to visualize individual electrons as following discrete circular orbits around a central nucleus, as in the Bohr model of the atom. However, such a simple picture is not realistic. It is not possible to precisely locate the position of any individual electron precisely at any given time. In fact, a particular electron can be anywhere at a given instant, from close to the nucleus to infinitely far away from it.

However, a particular electron is more likely to be located in some positions than in others. The area around a nucleus where an electron is most likely to be found is called the orbital of that electron (figure 2.4). Some electron orbitals near the nucleus are spherical (*s* orbitals), while others are dumbbell-shaped (*p* orbitals). Still other orbitals, more distant from the nucleus, may have different shapes. Regardless of its shape, no orbital may contain more than two electrons.

Almost all of the volume of an atom is empty space, because the electrons are quite far from the nucleus relative to its size. If the nucleus of an atom were the size of an apple, the orbit of the nearest electron would be more than 1600 meters away. Consequently, the nuclei of two atoms never come close enough in nature to interact with each other. It is for this reason that an atom's electrons, not its protons or neutrons, determine its chemical behavior. This

also explains why the isotopes of an element, all of which have the same arrangement of electrons, behave the same way chemically.

Energy within the Atom

All atoms possess energy, defined as the ability to do work. Because electrons are attracted to the positively charged nucleus, it takes work to keep them in orbit, just as it takes work to hold a grapefruit in your hand against the pull of gravity. The grapefruit is said to possess *potential energy*, the ability to do work, because of its position; if you were to release it, the grapefruit would fall and its energy would be reduced. Conversely, if you were to move the grapefruit to the top of a building, you would increase its potential energy. Similarly, electrons have potential energy of position. To oppose the attraction of the nucleus and move the electron to a more distant orbital requires an input of energy and results in an electron with greater potential energy. This is how chlorophyll captures energy from light during photosynthesis (chapter 10)—the light excites electrons in the chlorophyll. Moving an electron closer to the nucleus has the opposite effect: energy is released, usually as heat, and the electron ends up with less potential energy (figure 2.5).

A given atom can possess only certain discrete amounts of energy. Like the potential energy of a grapefruit on a step of a staircase, the potential energy contributed by the position of an electron in an atom can have only certain values.

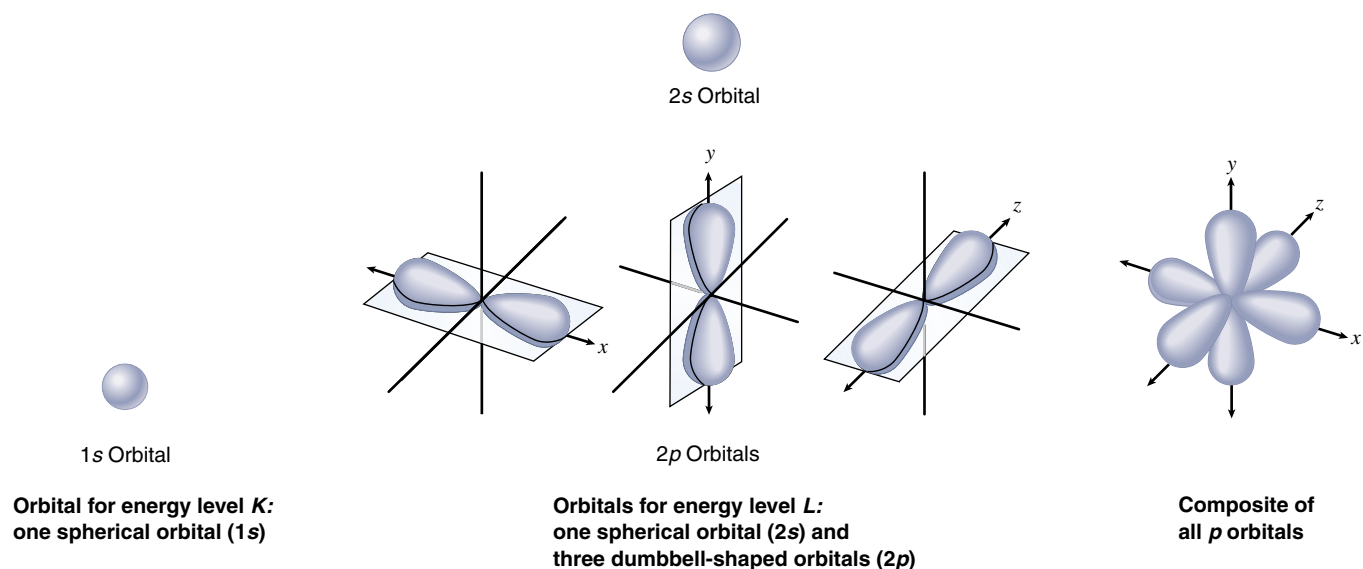
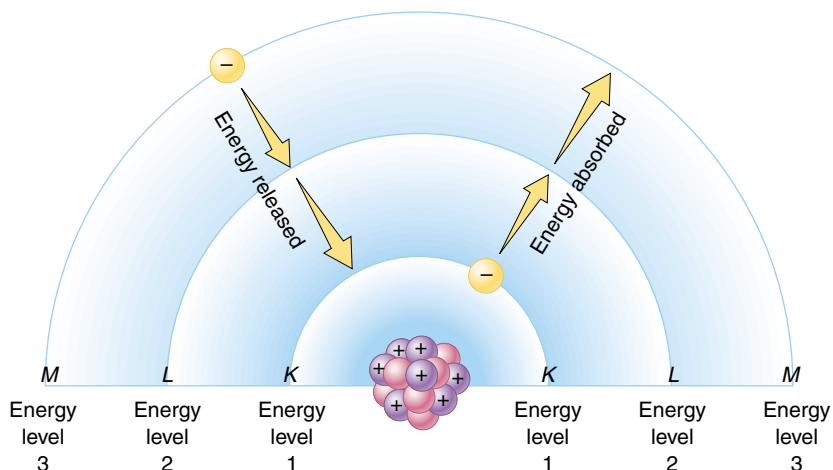


FIGURE 2.4

Electron orbitals. The lowest energy level or electron shell, which is nearest the nucleus, is level *K*. It is occupied by a single *s* orbital, referred to as 1s. The next highest energy level, *L*, is occupied by four orbitals: one *s* orbital (referred to as the 2s orbital) and three *p* orbitals (each referred to as a 2*p* orbital). The four *L*-level orbitals compactly fill the space around the nucleus, like two pyramids set base-to-base.

FIGURE 2.5

Atomic energy levels. When an electron absorbs energy, it moves to higher energy levels farther from the nucleus. When an electron releases energy, it falls to lower energy levels closer to the nucleus.



Every atom exhibits a ladder of potential energy values, rather than a continuous spectrum of possibilities, a discrete set of orbits at particular distances from the nucleus.

During some chemical reactions, electrons are transferred from one atom to another. In such reactions, the loss of an electron is called **oxidation**, and the gain of an electron is called **reduction** (figure 2.6). It is important to realize that when an electron is transferred in this way, it keeps its energy of position. In organisms, chemical energy is stored in high-energy electrons that are transferred from one atom to another in reactions involving oxidation and reduction.

Because the amount of energy an electron possesses is related to its distance from the nucleus, electrons that are the same distance from the nucleus have the same energy, even if they occupy different orbitals. Such electrons are said to occupy the same **energy level**. In a schematic diagram of an atom (figure 2.7), the nucleus is represented as a small circle and the electron energy levels are drawn as concentric rings, with the energy level increasing with distance from the nucleus. Be careful not to confuse energy levels, which are drawn as rings to indicate an electron's *energy*, with orbitals, which have a variety of three-dimensional shapes and indicate an electron's most likely *location*.

Electrons orbit a nucleus in paths called orbitals. No orbital can contain more than two electrons, but many orbitals may be the same distance from the nucleus and, thus, contain electrons of the same energy.

FIGURE 2.7

Electron energy levels for helium and nitrogen. Gold balls represent the electrons. Each concentric circle represents a different distance from the nucleus and, thus, a different electron energy level.

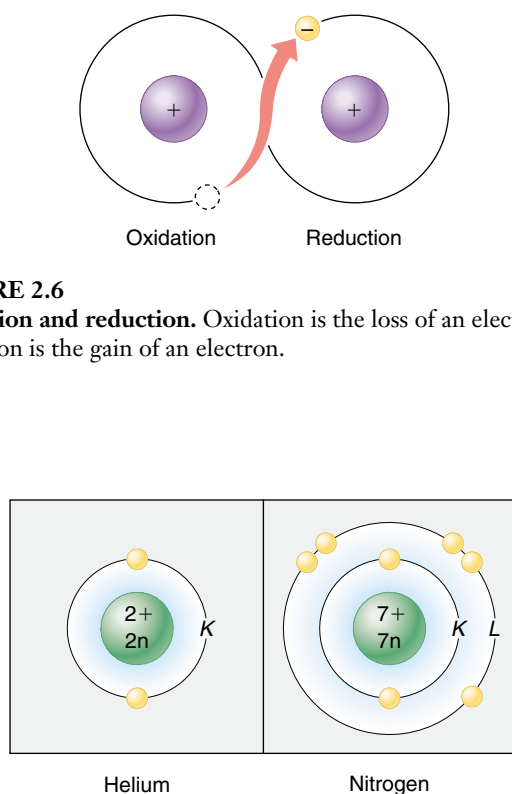
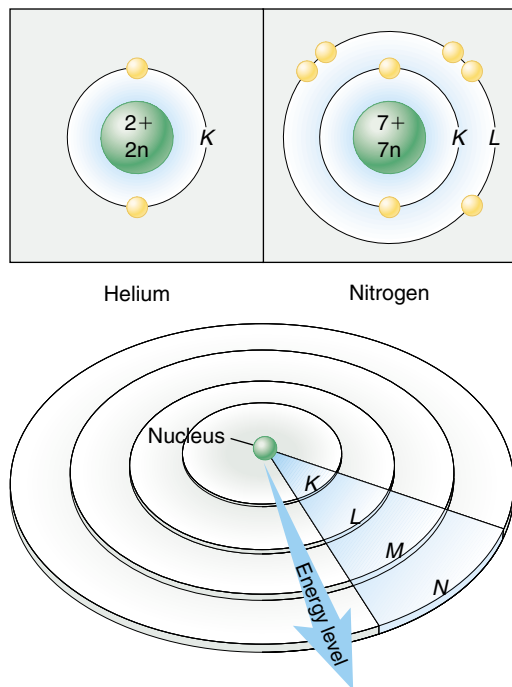


FIGURE 2.6

Oxidation and reduction. Oxidation is the loss of an electron; reduction is the gain of an electron.



2.2 The atoms of living things are among the smallest.

Kinds of Atoms

There are 92 naturally occurring elements, each with a different number of protons and a different arrangement of electrons. When the nineteenth-century Russian chemist Dmitri Mendeleev arranged the known elements in a table according to their atomic mass (figure 2.8), he discovered one of the great generalizations in all of science. Mendeleev found that the elements in the table exhibited a pattern of chemical properties that repeated itself in groups of eight elements. This periodically repeating pattern lent the table its name: the periodic table of elements.

The Periodic Table

The eight-element periodicity that Mendeleev found is based on the interactions of the electrons in the outer energy levels of the different elements. These electrons are called **valence electrons** and their interactions are the basis for the differing chemical properties of the elements. For most of the atoms important to life, an outer energy

level can contain no more than eight electrons; the chemical behavior of an element reflects how many of the eight positions are filled. Elements possessing all eight electrons in their outer energy level (two for helium) are **inert**, or nonreactive; they include helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe), and radon (Rn). In sharp contrast, elements with seven electrons (one fewer than the maximum number of eight) in their outer energy level, such as fluorine (F), chlorine (Cl), and bromine (Br), are highly reactive. They tend to gain the extra electron needed to fill the energy level. Elements with only one electron in their outer energy level, such as lithium (Li), sodium (Na), and potassium (K), are also very reactive; they tend to lose the single electron in their outer level.

Mendeleev's periodic table thus leads to a useful generalization, the **octet rule** (Latin *octo*, "eight") or **rule of eight**: atoms tend to establish completely full outer energy levels. Most chemical behavior can be predicted quite accurately from this simple rule, combined with the tendency of atoms to balance positive and negative charges.

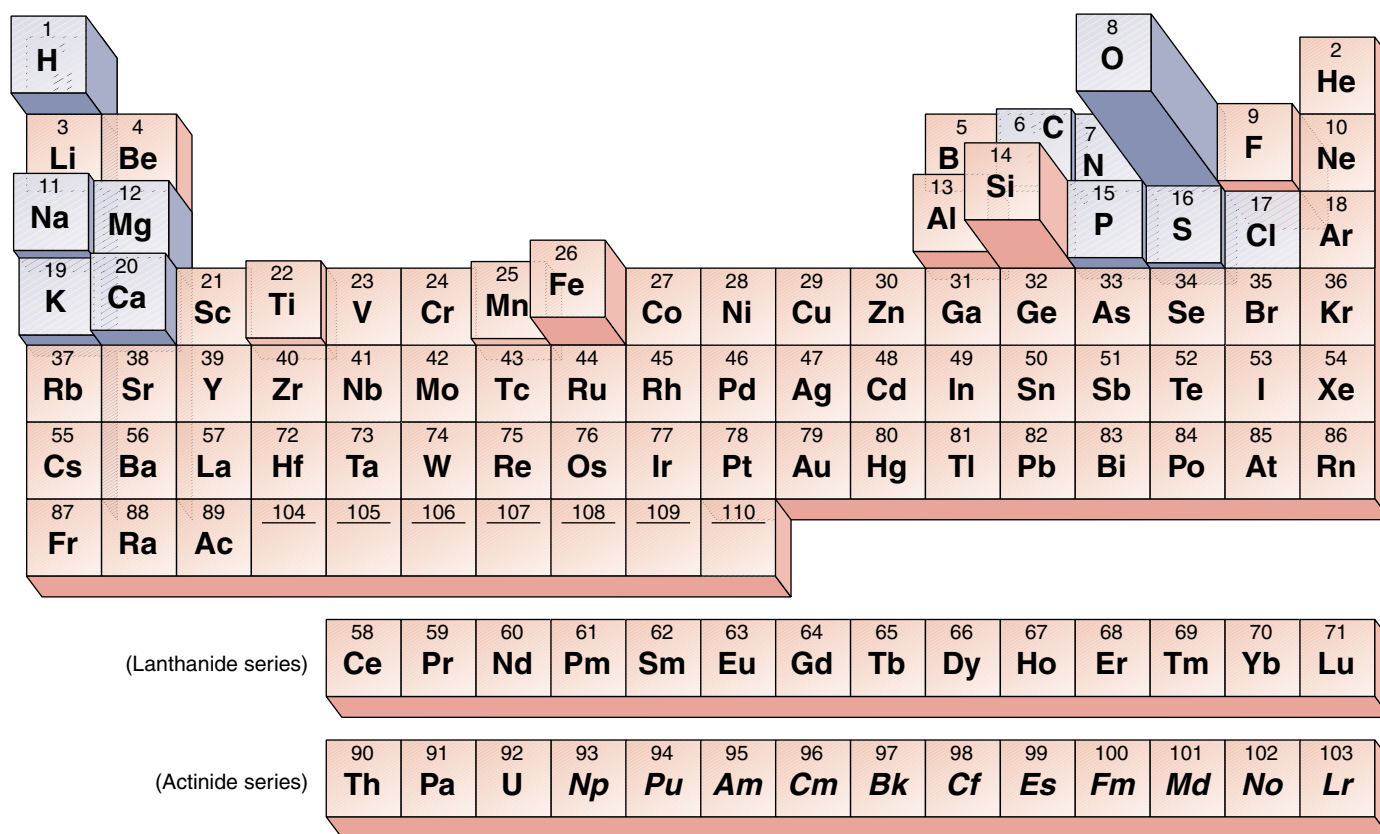


FIGURE 2.8

Periodic table of the elements. In this representation, the frequency of elements that occur in the earth's crust is indicated by the height of the block. Elements found in significant amounts in living organisms are shaded in blue.

Table 2.1 The Most Common Elements on Earth and Their Distribution in the Human Body

Element	Symbol	Atomic Number	Approximate Percent of Earth's Crust by Weight	Percent of Human Body by Weight	Importance or Function
Oxygen	O	8	46.6	65.0	Required for cellular respiration; component of water
Silicon	Si	14	27.7	Trace	
Aluminum	Al	13	6.5	Trace	
Iron	Fe	26	5.0	Trace	Critical component of hemoglobin in the blood
Calcium	Ca	20	3.6	1.5	Component of bones and teeth; triggers muscle contraction
Sodium	Na	11	2.8	0.2	Principal positive ion outside cells; important in nerve function
Potassium	K	19	2.6	0.4	Principal positive ion inside cells; important in nerve function
Magnesium	Mg	12	2.1	0.1	Critical component of many energy-transferring enzymes
Hydrogen	H	1	0.14	9.5	Electron carrier; component of water and most organic molecules
Manganese	Mn	25	0.1	Trace	
Fluorine	F	9	0.07	Trace	
Phosphorus	P	15	0.07	1.0	Backbone of nucleic acids; important in energy transfer
Carbon	C	6	0.03	18.5	Backbone of organic molecules
Sulfur	S	16	0.03	0.3	Component of most proteins
Chlorine	Cl	17	0.01	0.2	Principal negative ion outside cells
Vanadium	V	23	0.01	Trace	
Chromium	Cr	24	0.01	Trace	
Copper	Cu	29	0.01	Trace	Key component of many enzymes
Nitrogen	N	7	Trace	3.3	Component of all proteins and nucleic acids
Boron	B	5	Trace	Trace	
Cobalt	Co	27	Trace	Trace	
Zinc	Zn	30	Trace	Trace	Key component of some enzymes
Selenium	Se	34	Trace	Trace	
Molybdenum	Mo	42	Trace	Trace	Key component of many enzymes
Tin	Sn	50	Trace	Trace	
Iodine	I	53	Trace	Trace	Component of thyroid hormone

Distribution of the Elements

Of the 92 naturally occurring elements on earth, only 11 are found in organisms in more than trace amounts (0.01% or higher). These 11 elements have atomic numbers less than 21 and, thus, have low atomic masses. Table 2.1 lists the levels of various elements in the human body; their levels in other organisms are similar. Inspection of this table suggests that the distribution of elements in living systems is by no means accidental. The most common elements inside organisms are not the elements that are most abundant in the

earth's crust. For example, silicon, aluminum, and iron constitute 39.2% of the earth's crust, but they exist in trace amounts in the human body. On the other hand, carbon atoms make up 18.5% of the human body but only 0.03% of the earth's crust.

Ninety-two elements occur naturally on earth; only eleven of them are found in significant amounts in living organisms. Four of them—oxygen, hydrogen, carbon, nitrogen—constitute 96.3% of the weight of your body.

2.3 Chemical bonds hold molecules together.

Ionic Bonds Form Crystals

A group of atoms held together by energy in a stable association is called a **molecule**. When a molecule contains atoms of more than one element, it is called a **compound**. The atoms in a molecule are joined by **chemical bonds**; these bonds can result when atoms with opposite charges attract (ionic bonds), when two atoms share one or more pairs of electrons (covalent bonds), or when atoms interact in other ways. We will start by examining **ionic bonds**, which form when atoms with opposite electrical charges (ions) attract.

A Closer Look at Table Salt

Common table salt, sodium chloride (NaCl), is a lattice of ions in which the atoms are held together by ionic bonds (figure 2.9). Sodium has 11 electrons: 2 in the inner energy level, 8 in the next level, and 1 in the outer (valence) level. The valence electron is unpaired (free) and has a strong tendency to join with another electron. A stable configuration can be achieved if the valence electron is lost to another atom that also has an unpaired electron. The loss of this electron results in the formation of a positively charged sodium ion, Na^+ .

The chlorine atom has 17 electrons: 2 in the inner energy level, 8 in the next level, and 7 in the outer level. Hence, one of the orbitals in the outer energy level has an unpaired electron. The addition of another electron to the outer level fills that level and causes a negatively charged chloride ion, Cl^- , to form.

When placed together, metallic sodium and gaseous chlorine react swiftly and explosively, as the sodium atoms donate electrons to chlorine, forming Na^+ and Cl^- ions. Because opposite charges attract, the Na^+ and Cl^- remain associated in an **ionic compound**, NaCl , which is electrically neutral. However, the electrical attractive force holding NaCl together is not directed specifically between particular Na^+ and Cl^- ions, and no discrete sodium chloride molecules form. Instead, the force exists between any one ion and all neighboring ions of the opposite charge, and the ions aggregate in a crystal matrix with a precise geometry. Such aggregations are what we know as salt crystals. If a salt such as NaCl is placed in water, the electrical attraction of the water molecules, for reasons we will point out later in this chapter, disrupts the forces holding the ions in their crystal matrix, causing the salt to dissolve into a roughly equal mixture of free Na^+ and Cl^- ions.

An ionic bond is an attraction between ions of opposite charge in an ionic compound. Such bonds are not formed between particular ions in the compound; rather, they exist between an ion and all of the oppositely charged ions in its immediate vicinity.

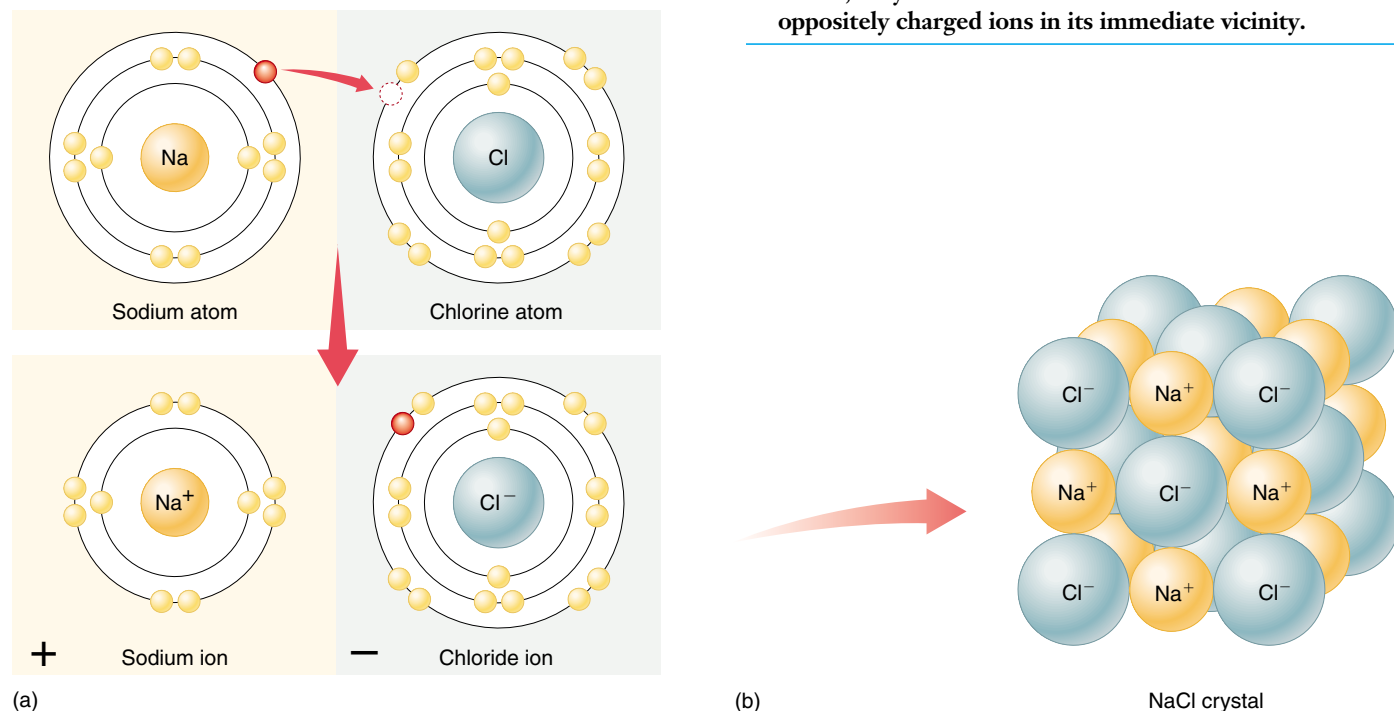


FIGURE 2.9

The formation of ionic bonds by sodium chloride. (a) When a sodium atom donates an electron to a chlorine atom, the sodium atom becomes a positively charged sodium ion, and the chlorine atom becomes a negatively charged chloride ion. (b) Sodium chloride forms a highly regular lattice of alternating sodium ions and chloride ions.

Covalent Bonds Build Stable Molecules

Covalent bonds form when two atoms share one or more pairs of valence electrons. Consider hydrogen (H) as an example. Each hydrogen atom has an unpaired electron and an unfilled outer energy level; for these reasons the hydrogen atom is unstable. When two hydrogen atoms are close to each other, however, each atom's electron can orbit both nuclei. In effect, the nuclei are able to share their electrons. The result is a diatomic (two-atom) molecule of hydrogen gas (figure 2.10).

The molecule formed by the two hydrogen atoms is stable for three reasons:

1. **It has no net charge.** The diatomic molecule formed as a result of this sharing of electrons is not charged, because it still contains two protons and two electrons.
2. **The octet rule is satisfied.** Each of the two hydrogen atoms can be considered to have two orbiting electrons in its outer energy level. This satisfies the octet rule, because each shared electron orbits both nuclei and is included in the outer energy level of *both* atoms.
3. **It has no free electrons.** The bonds between the two atoms also pair the two free electrons.

Unlike ionic bonds, covalent bonds are formed between two specific atoms, giving rise to true, discrete molecules. While ionic bonds can form regular crystals, the more specific associations made possible by covalent bonds allow the formation of complex molecular structures.

Covalent Bonds Can Be Very Strong

The strength of a covalent bond depends on the number of shared electrons. Thus **double bonds**, which satisfy the octet rule by allowing two atoms to share *two* pairs of electrons, are stronger than **single bonds**, in which only one electron pair is shared. This means more chemical energy is required to break a double bond than a single bond. The strongest covalent bonds are **triple bonds**, such as those that link the two nitrogen atoms of nitrogen gas molecules. Covalent bonds are represented in chemical formulations as lines connecting atomic symbols, where each line between two bonded atoms represents the sharing of one pair of electrons. The **structural formulas** of hydrogen gas and oxygen gas are H—H and O=O, respectively, while their **molecular formulas** are H₂ and O₂.

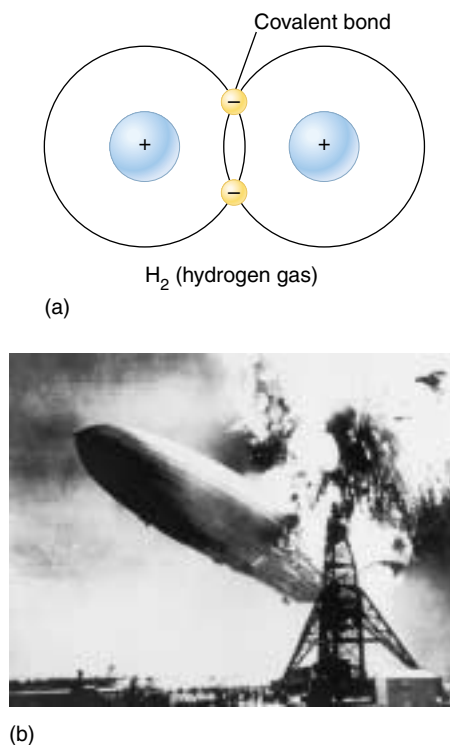


FIGURE 2.10

Hydrogen gas. (a) Hydrogen gas is a diatomic molecule composed of two hydrogen atoms, each sharing its electron with the other. (b) The flash of fire that consumed the *Hindenburg* occurred when the hydrogen gas that was used to inflate the dirigible combined explosively with oxygen gas in the air to form water.

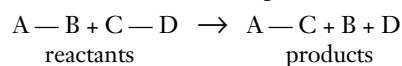
Molecules with Several Covalent Bonds

Molecules often consist of more than two atoms. One reason that larger molecules may be formed is that a given atom is able to share electrons with more than one other atom. An atom that requires two, three, or four additional electrons to fill its outer energy level completely may acquire them by sharing its electrons with two or more other atoms.

For example, the carbon atom (C) contains six electrons, four of which are in its outer energy level. To satisfy the octet rule, a carbon atom must gain access to four additional electrons; that is, it must form four covalent bonds. Because four covalent bonds may form in many ways, carbon atoms are found in many different kinds of molecules.

Chemical Reactions

The formation and breaking of chemical bonds, the essence of chemistry, is called a **chemical reaction**. All chemical reactions involve the shifting of atoms from one molecule or ionic compound to another, without any change in the number or identity of the atoms. For convenience, we refer to the original molecules before the reaction starts as **reactants**, and the molecules resulting from the chemical reaction as **products**. For example:



The extent to which chemical reactions occur is influenced by several important factors:

1. **Temperature.** Heating up the reactants increases the rate of a reaction (as long as the temperature isn't so high as to destroy the molecules).
2. **Concentration of reactants and products.** Reactions proceed more quickly when more reactants are available. An accumulation of products typically speeds reactions in the reverse direction.
3. **Catalysts.** A catalyst is a substance that increases the rate of a reaction. It doesn't alter the reaction's equilibrium between reactants and products, but it does shorten the time needed to reach equilibrium, often dramatically. In organisms, proteins called enzymes catalyze almost every chemical reaction.

A covalent bond is a stable chemical bond formed when two atoms share one or more pairs of electrons.

2.4 Water is the cradle of life.



(a)



(b)



(c)

FIGURE 2.11

Water takes many forms. As a liquid, water fills our rivers and runs down over the land to the sea. (a) The iceberg on which the penguins are holding their meeting was formed in Antarctica from a huge block of ice that broke away into the ocean water. (b) When water cools below 0°C , it forms beautiful crystals, familiar to us as snow and ice. However, water is not always plentiful. (c) At Badwater, in Death Valley, California, there is no hint of water except for the broken patterns of dried mud.

Chemistry of Water

Of all the molecules that are common on earth, only **water** exists as a liquid at the relatively low temperatures that prevail on the earth's surface, three-fourths of which is covered by liquid water (figure 2.11). When life was originating, water provided a medium in which other molecules could move around and interact without being held in place by strong covalent or ionic bonds. Life evolved as a result of these interactions, and it is still inextricably tied to water. Life began in water and evolved there for 3 billion years before spreading to land. About two-thirds of any organism's body is composed of water, and no organism can grow or reproduce in any but a water-rich environment. It is no accident that tropical rain forests are bursting with life, while dry deserts appear almost lifeless except when water becomes temporarily plentiful, such as after a rainstorm.

The Atomic Structure of Water

Water has a simple atomic structure. It consists of an oxygen atom bound to two hydrogen atoms by two single covalent bonds (figure 2.12a). The resulting molecule is stable: it satisfies the octet rule, has no unpaired electrons, and carries no net electrical charge.

The single most outstanding chemical property of water is its ability to form weak chemical associations with only 5 to 10% of the strength of covalent bonds. This

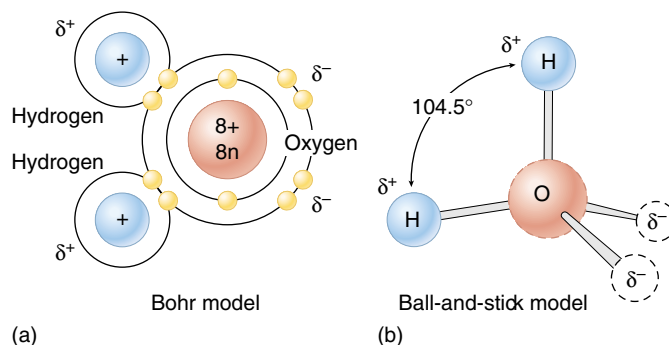


FIGURE 2.12

Water has a simple molecular structure. (a) Each molecule is composed of one oxygen atom and two hydrogen atoms. The oxygen atom shares one electron with each hydrogen atom. (b) The greater electronegativity of the oxygen atom makes the water molecule polar: water carries two partial negative charges (δ^-) near the oxygen atom and two partial positive charges (δ^+), one on each hydrogen atom.

property, which derives directly from the structure of water, is responsible for much of the organization of living chemistry.

The chemistry of life is water chemistry. The way in which life first evolved was determined in large part by the chemical properties of the liquid water in which that evolution occurred.

Water Atoms Act Like Tiny Magnets

Both the oxygen and the hydrogen atoms attract the electrons they share in the covalent bonds of a water molecule; this attraction is called **electronegativity**. However, the oxygen atom is more electronegative than the hydrogen atoms, so it attracts the electrons more strongly than do the hydrogen atoms. As a result, the shared electrons in a water molecule are far more likely to be found near the oxygen nucleus than near the hydrogen nuclei. This stronger attraction for electrons gives the oxygen atom two partial negative charges (δ^-), as though the electron cloud were denser near the oxygen atom than around the hydrogen atoms. Because the water molecule as a whole is electrically neutral, each hydrogen atom carries a partial positive charge (δ^+). The Greek letter delta (δ) signifies a partial charge, much weaker than the full unit charge of an ion.

What would you expect the shape of a water molecule to be? Each of water's two covalent bonds has a partial charge at each end, δ^- at the oxygen end and δ^+ at the hydrogen end. The most stable arrangement of these charges is a *tetrahedron*, in which the two negative and two positive charges are approximately equidistant from one another (figure 2.12*b*). The oxygen atom lies at the center of the tetrahedron, the hydrogen atoms occupy two of the apexes, and the partial negative charges occupy the other two apexes. This results in a bond angle of 104.5° between the two covalent oxygen-hydrogen bonds. (In a regular tetrahedron, the bond angles would be 109.5° ; in water, the partial negative charges occupy more space than the hydrogen atoms, and, therefore, they compress the oxygen-hydrogen bond angle slightly.)

The water molecule, thus, has distinct “ends,” each with a partial charge, like the two poles of a magnet. (These partial

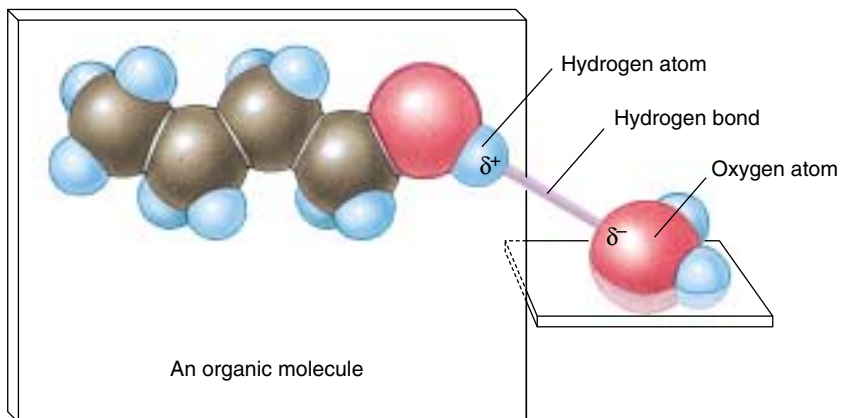


FIGURE 2.13
Structure of a hydrogen bond.

charges are much less than the unit charges of ions, however.) Molecules that exhibit charge separation are called **polar molecules** because of their magnet-like poles, and water is one of the most polar molecules known. *The polarity of water underlies its chemistry and the chemistry of life.*

Polar molecules interact with one another, as the δ^- of one molecule is attracted to the δ^+ of another. Because many of these interactions involve hydrogen atoms, they are called **hydrogen bonds** (figure 2.13). Each hydrogen bond is individually very weak and transient, lasting on average only $\frac{1}{100,000,000,000}$ second (10^{-11} sec). However, the cumulative effects of large numbers of these bonds can be enormous. Water forms an abundance of hydrogen bonds, which are responsible for many of its important physical properties (table 2.2).

The water molecule is very polar, with ends that exhibit partial positive and negative charges. Opposite charges attract, forming weak linkages called hydrogen bonds.

Table 2.2 The Properties of Water

Property	Explanation	Example of Benefit to Life
Cohesion	Hydrogen bonds hold water molecules together	Leaves pull water upward from the roots; seeds swell and germinate
High specific heat	Hydrogen bonds absorb heat when they break, and release heat when they form, minimizing temperature changes	Water stabilizes the temperature of organisms and the environment
High heat of vaporization	Many hydrogen bonds must be broken for water to evaporate	Evaporation of water cools body surfaces
Lower density of ice	Water molecules in an ice crystal are spaced relatively far apart because of hydrogen bonding	Because ice is less dense than water, lakes do not freeze solid
High polarity	Polar water molecules are attracted to ions and polar compounds, making them soluble	Many kinds of molecules can move freely in cells, permitting a diverse array of chemical reactions

Water Clings to Polar Molecules

The polarity of water causes it to be attracted to other polar molecules. When the other molecules are also water, the attraction is referred to as **cohesion**. When the other molecules are of a different substance, the attraction is called **adhesion**. It is because water is cohesive that it is a liquid, and not a gas, at moderate temperatures.

The cohesion of liquid water is also responsible for its **surface tension**. Small insects can walk on water (figure 2.14) because at the air-water interface all of the hydrogen bonds in water face downward, causing the molecules of the water surface to cling together. Water is adhesive to any substance with which it can form hydrogen bonds. That is why substances containing polar molecules get “wet” when they are immersed in water, while those that are composed of nonpolar molecules (such as oils) do not.

The attraction of water to substances like glass with surface electrical charges is responsible for capillary action: if a glass tube with a narrow diameter is lowered into a beaker of water, water will rise in the tube above the level of the water in the beaker, because the adhesion of water to the glass surface, drawing it upward, is stronger than the force of gravity, drawing it down. The narrower the tube, the greater the electrostatic forces between the water and the glass, and the higher the water rises (figure 2.15).

Water Stores Heat

Water moderates temperature through two properties: its high specific heat and its high heat of vaporization. The temperature of any substance is a measure of how rapidly its individual molecules are moving. Because of the many hydrogen bonds that water molecules form with one another, a large input of thermal energy is required to break these bonds before the individual water molecules can begin moving about more freely and so have a higher temperature. Therefore, water is said to have a high **specific heat**, which is defined as the amount of heat that must be absorbed or lost by 1 gram of a substance to change its temperature by 1 degree Celsius ($^{\circ}\text{C}$). Specific heat measures the extent to which a substance resists changing its temperature when it absorbs or loses heat. Because polar substances tend to form hydrogen bonds, and energy is needed to break these bonds, the more polar a substance is, the higher is its specific heat. The specific heat of water (1 calorie/gram/ $^{\circ}\text{C}$) is twice that of most carbon compounds and nine times that of iron. Only ammonia, which is more polar than water and forms very strong hydrogen bonds, has a higher specific heat than water (1.23 calories/gram/ $^{\circ}\text{C}$). Still, only 20% of the hydrogen bonds are broken as water heats from 0° to 100°C .

Because of its high specific heat, water heats up more slowly than almost any other compound and holds its temperature longer when heat is no longer applied. This characteristic enables organisms, which have a high water con-



FIGURE 2.14

Cohesion. Some insects, such as this water strider, literally walk on water. In this photograph you can see the dimpling the insect's feet make on the water as its weight bears down on the surface. Because the surface tension of the water is greater than the force that one foot brings to bear, the strider glides atop the surface of the water rather than sinking.

FIGURE 2.15

Capillary action. Capillary action causes the water within a narrow tube to rise above the surrounding water; the adhesion of the water to the glass surface, which draws water upward, is stronger than the force of gravity, which tends to draw it down. The narrower the tube, the greater the surface area available for adhesion for a given volume of water, and the higher the water rises in the tube.



tent, to maintain a relatively constant internal temperature. The heat generated by the chemical reactions inside cells would destroy the cells, if it were not for the high specific heat of the water within them.

A considerable amount of heat energy (586 calories) is required to change 1 gram of liquid water into a gas. Hence, water also has a high **heat of vaporization**. Because the transition of water from a liquid to a gas requires the input of energy to break its many hydrogen bonds, the evaporation of water from a surface causes cooling of that surface. Many organisms dispose of excess body heat by evaporative cooling; for example, humans and many other vertebrates sweat.

At low temperatures, water molecules are locked into a crystal-like lattice of hydrogen bonds, forming the solid we call ice (figure 2.16). Interestingly, ice is less dense than liquid water because the hydrogen bonds in ice space the water molecules relatively far apart. This unusual feature enables icebergs to float. Were it otherwise, ice would cover nearly all bodies of water, with only shallow surface melting annually.

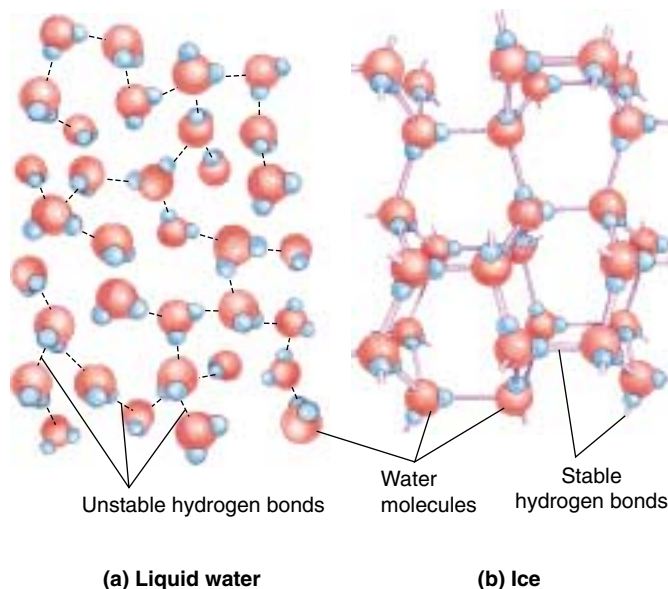


FIGURE 2.16

The role of hydrogen bonds in an ice crystal. (a) In liquid water, hydrogen bonds are not stable and constantly break and reform. (b) When water cools below 0°C, the hydrogen bonds are more stable, and a regular crystalline structure forms in which the four partial charges of one water molecule interact with the opposite charges of other water molecules. Because water forms a crystal latticework, ice is less dense than liquid water and floats. If it did not, inland bodies of water far from the earth's equator might never fully thaw.

Water Is a Powerful Solvent

Water is an effective solvent because of its ability to form hydrogen bonds. Water molecules gather closely around any substance that bears an electrical charge, whether that substance carries a full charge (ion) or a charge separation (polar molecule). For example, sucrose (table sugar) is composed of molecules that contain slightly polar hydroxyl (OH) groups. A sugar crystal dissolves rapidly in water because water molecules can form hydrogen bonds with individual hydroxyl groups of the sucrose molecules. Therefore, sucrose is said to be *soluble* in water. Every time a sucrose molecule dissociates or breaks away from the crystal, water molecules surround it in a cloud, forming a **hydration shell** and preventing it from associating with other sucrose molecules. Hydration shells also form around ions such as Na^+ and Cl^- (figure 2.17).

Water Organizes Nonpolar Molecules

Water molecules always tend to form the maximum possible number of hydrogen bonds. When nonpolar molecules such as oils, which do not form hydrogen bonds, are placed

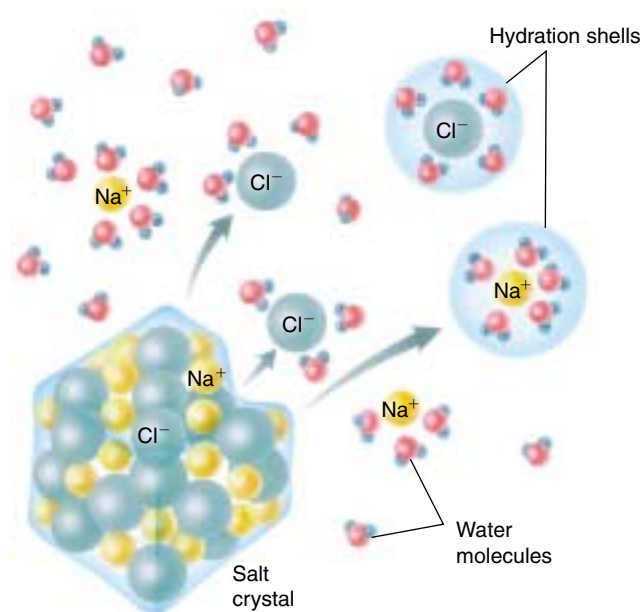


FIGURE 2.17

Why salt dissolves in water. When a crystal of table salt dissolves in water, individual Na^+ and Cl^- ions break away from the salt lattice and become surrounded by water molecules. Water molecules orient around Cl^- ions so that their partial positive poles face toward the negative Cl^- ion; water molecules surrounding Na^+ ions orient in the opposite way, with their partial negative poles facing the positive Na^+ ion. Surrounded by hydration shells, Na^+ and Cl^- ions never reenter the salt lattice.

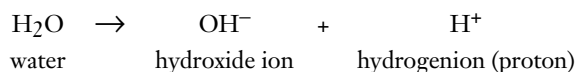
in water, the water molecules act to exclude them. The nonpolar molecules are forced into association with one another, thus minimizing their disruption of the hydrogen bonding of water. In effect, they shrink from contact with water and for this reason they are referred to as **hydrophobic** (Greek *hydros*, “water” and *phobos*, “fearing”). In contrast, polar molecules, which readily form hydrogen bonds with water, are said to be **hydrophilic** (“water-loving”).

The tendency of nonpolar molecules to aggregate in water is known as **hydrophobic exclusion**. By forcing the hydrophobic portions of molecules together, water causes these molecules to assume particular shapes. Different molecular shapes have evolved by alteration of the location and strength of nonpolar regions. As you will see, much of the evolution of life reflects changes in molecular shape that can be induced in just this way.

Water molecules, which are very polar, cling to one another, so that it takes considerable energy to separate them. Water also clings to other polar molecules, causing them to be soluble in water solution, but water tends to exclude nonpolar molecules.

Water Ionizes

The covalent bonds within a water molecule sometimes break spontaneously. In pure water at 25°C, only 1 out of every 550 million water molecules undergoes this process. When it happens, one of the protons (hydrogen atom nuclei) dissociates from the molecule. Because the dissociated proton lacks the negatively charged electron it was sharing in the covalent bond with oxygen, its own positive charge is no longer counterbalanced, and it becomes a positively charged ion, H^+ . The rest of the dissociated water molecule, which has retained the shared electron from the covalent bond, is negatively charged and forms a **hydroxide ion** (OH^-). This process of spontaneous ion formation is called **ionization**:



At 25°C, a liter of water contains $\frac{1}{10,000,000}$ (or 10^{-7}) mole of H^+ ions. (A **mole** is defined as the weight in grams that corresponds to the summed atomic masses of all of the atoms in a molecule. In the case of H^+ , the atomic mass is 1, and a mole of H^+ ions would weigh 1 gram. One mole of any substance always contains 6.02×10^{23} molecules of the substance.) Therefore, the **molar concentration** of hydrogen ions (represented as $[H^+]$) in pure water is 10^{-7} mole/liter. Actually, the hydrogen ion usually associates with another water molecule to form a hydronium (H_3O^+) ion.

pH

A more convenient way to express the hydrogen ion concentration of a solution is to use the **pH scale** (figure 2.18). This scale defines pH as the negative logarithm of the hydrogen ion concentration in the solution:

$$pH = -\log [H^+]$$

Because the logarithm of the hydrogen ion concentration is simply the exponent of the molar concentration of H^+ , the pH equals the exponent times -1 . Thus, pure water, with an $[H^+]$ of 10^{-7} mole/liter, has a pH of 7. Recall that for every H^+ ion formed when water dissociates, an OH^- ion is also formed, meaning that the dissociation of water produces H^+ and OH^- in equal amounts. Therefore, a pH value of 7 indicates neutrality—a balance between H^+ and OH^- —on the pH scale.

Note that the pH scale is *logarithmic*, which means that a difference of 1 on the scale represents a tenfold change in hydrogen ion concentration. This means that a solution with a pH of 4 has *10 times* the concentration of H^+ than is present in one with a pH of 5.

Acids. Any substance that dissociates in water to increase the concentration of H^+ ions is called an acid. Acidic solu-

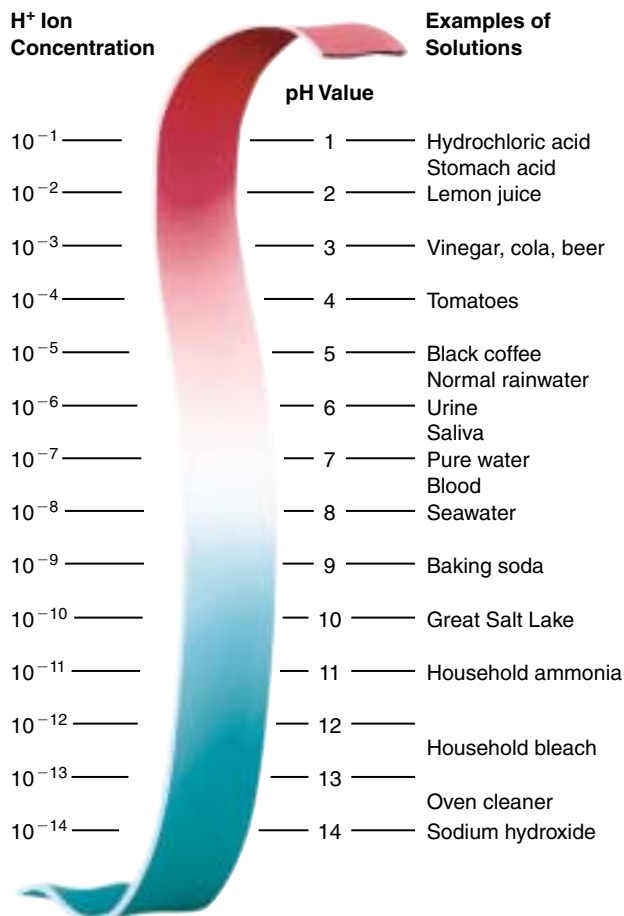


FIGURE 2.18

The pH scale. The pH value of a solution indicates its concentration of hydrogen ions. Solutions with a pH less than 7 are acidic, while those with a pH greater than 7 are basic. The scale is logarithmic, so that a pH change of 1 means a tenfold change in the concentration of hydrogen ions. Thus, lemon juice is 100 times more acidic than tomato juice, and seawater is 10 times more basic than pure water, which has a pH of 7.

tions have pH values below 7. The stronger an acid is, the more H^+ ions it produces and the lower its pH. For example, hydrochloric acid (HCl), which is abundant in your stomach, ionizes completely in water. This means that 10^{-1} mole per liter of HCl will dissociate to form 10^{-1} mole per liter of H^+ ions, giving the solution a pH of 1. The pH of champagne, which bubbles because of the carbonic acid dissolved in it, is about 2.

Bases. A substance that combines with H^+ ions when dissolved in water is called a base. By combining with H^+ ions, a base lowers the H^+ ion concentration in the solution. Basic (or alkaline) solutions, therefore, have pH values above 7. Very strong bases, such as sodium hydroxide (NaOH), have pH values of 12 or more.

Buffers

The pH inside almost all living cells, and in the fluid surrounding cells in multicellular organisms, is fairly close to 7. Most of the biological catalysts (enzymes) in living systems are extremely sensitive to pH; often even a small change in pH will alter their shape, thereby disrupting their activities and rendering them useless. For this reason it is important that a cell maintain a constant pH level.

Yet the chemical reactions of life constantly produce acids and bases within cells. Furthermore, many animals eat substances that are acidic or basic; cola, for example, is a strong (although dilute) acidic solution. Despite such variations in the concentrations of H^+ and OH^- , the pH of an organism is kept at a relatively constant level by buffers (figure 2.19).

A **buffer** is a substance that acts as a reservoir for hydrogen ions, donating them to the solution when their concentration falls and taking them from the solution when their concentration rises. What sort of substance will act in this way? Within organisms, most buffers consist of pairs of substances, one an acid and the other a base. The key buffer in human blood is an acid-base pair consisting of carbonic acid (acid) and bicarbonate (base). These two substances interact in a pair of reversible reactions. First, carbon dioxide (CO_2) and H_2O join to form carbonic acid (H_2CO_3), which in a second reaction dissociates to yield bicarbonate ion (HCO_3^-) and H^+ (figure 2.20). If some acid or other substance adds H^+ ions to the blood, the HCO_3^- ions act as a base and remove the excess H^+ ions by forming H_2CO_3 . Similarly, if a basic substance removes H^+ ions from the blood, H_2CO_3 dissociates, releasing more H^+ ions into the blood. The forward and reverse reactions that interconvert H_2CO_3 and HCO_3^- thus stabilize the blood's pH.

The reaction of carbon dioxide and water to form carbonic acid is important because it permits carbon, essential to life, to enter water from the air. As we will discuss in chapter 4, biologists believe that life first evolved in the early oceans. These oceans were rich in carbon because of the reaction of carbon dioxide with water.

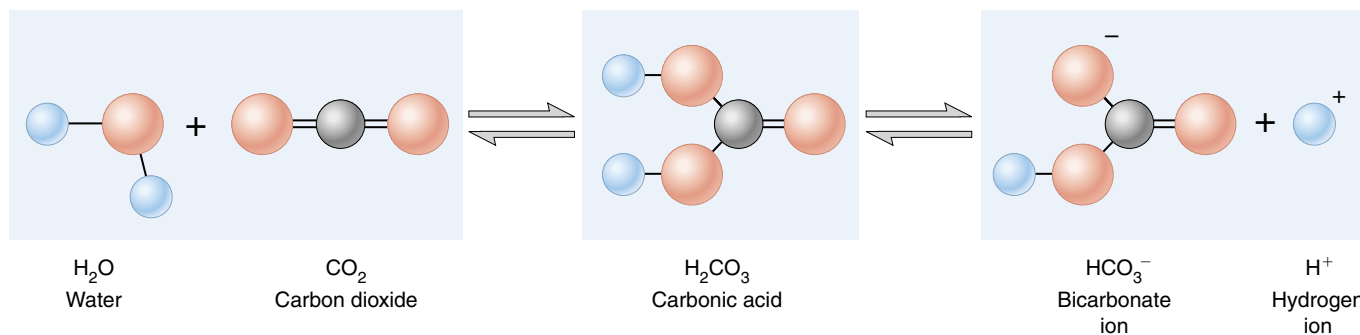


FIGURE 2.20

Buffer formation. Carbon dioxide and water combine chemically to form carbonic acid (H_2CO_3). The acid then dissociates in water, freeing H^+ ions. This reaction makes carbonated beverages acidic, and produced the carbon-rich early oceans that cradled life.

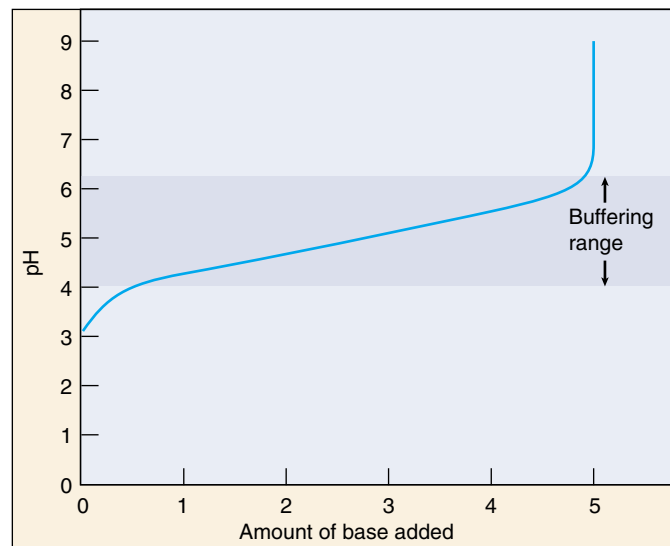


FIGURE 2.19

Buffers minimize changes in pH. Adding a base to a solution neutralizes some of the acid present, and so raises the pH. Thus, as the curve moves to the right, reflecting more and more base, it also rises to higher pH values. What a buffer does is to make the curve rise or fall very slowly over a portion of the pH scale, called the “buffering range” of that buffer.

In a condition called blood acidosis, human blood, which normally has a pH of about 7.4, drops 0.2 to 0.4 points on the pH scale. This condition is fatal if not treated immediately. The reverse condition, blood alkalosis, involves an increase in blood pH of a similar magnitude and is just as serious.

The pH of a solution is the negative logarithm of the H^+ ion concentration in the solution. Thus, low pH values indicate high H^+ concentrations (acidic solutions), and high pH values indicate low H^+ concentrations (basic solutions). Even small changes in pH can be harmful to life.



Summary

Questions

Media Resources

2.1 Atoms are nature's building material.

- The smallest stable particles of matter are protons, neutrons, and electrons, which associate to form atoms.
- The core, or nucleus, of an atom consists of protons and neutrons; the electrons orbit around the nucleus in a cloud. The farther an electron is from the nucleus, the faster it moves and the more energy it possesses.
- The chemical behavior of an atom is largely determined by the distribution of its electrons and in particular by the number of electrons in its outermost (highest) energy level. There is a strong tendency for atoms to have a completely filled outer level; electrons are lost, gained, or shared until this condition is reached.

1. An atom of nitrogen has 7 protons and 7 neutrons. What is its atomic number? What is its atomic mass? How many electrons does it have?
2. How do the isotopes of a single element differ from each other?
3. The half-life of radium-226 is 1620 years. If a sample of material contains 16 milligrams of radium-226, how much will it contain in 1620 years? How much will it contain in 3240 years? How long will it take for the sample to contain 1 milligram of radium-226?



- Atomic Structure



- Basic Chemistry
- Atoms

2.2 The atoms of living things are among the smallest.

- More than 95% of the weight of an organism consists of oxygen, hydrogen, carbon, and nitrogen, all of which form strong covalent bonds with one another.

4. What is the octet rule, and how does it affect the chemical behavior of atoms?

2.3 Chemical bonds hold molecules together.

- Ionic bonds form when electrons transfer from one atom to another, and the resulting oppositely charged ions attract one another.
- Covalent bonds form when two atoms share electrons. They are responsible for the formation of most biologically important molecules.

5. What is the difference between an ionic bond and a covalent bond? Give an example of each.



- Bonds
- Ionic Bonds



- Bonds

2.4 Water is the cradle of life.

- The chemistry of life is the chemistry of water (H_2O). The central oxygen atom in water attracts the electrons it shares with the two hydrogen atoms. This charge separation makes water a polar molecule.
- A hydrogen bond is formed between the partial positive charge of a hydrogen atom in one molecule and the partial negative charge of another atom, either in another molecule or in a different portion of the same molecule.
- Water is cohesive and adhesive, has a great capacity for storing heat, is a good solvent for other polar molecules, and tends to exclude nonpolar molecules.
- The H^+ concentration in a solution is expressed by the pH scale, in which pH equals the negative logarithm of the H^+ concentration.

6. What types of atoms participate in the formation of hydrogen bonds? How do hydrogen bonds contribute to water's high specific heat?
7. What types of molecules are hydrophobic? What types are hydrophilic? Why do these two types of molecules behave differently in water?
8. What is the pH of a solution that has a hydrogen ion concentration of 10^{-3} mole/liter? Would such a solution be acidic or basic?



- Water
- pH Scale

3

The Chemical Building Blocks of Life

Concept Outline

3.1 Molecules are the building blocks of life.

The Chemistry of Carbon. Because individual carbon atoms can form multiple covalent bonds, organic molecules can be quite complex.

3.2 Proteins perform the chemistry of the cell.

The Many Functions of Proteins. Proteins can be catalysts, transporters, supporters, and regulators.

Amino Acids Are the Building Blocks of Proteins. Proteins are long chains of various combinations of amino acids.

A Protein's Function Depends on the Shape of the Molecule. A protein's shape is determined by its amino acid sequence.

How Proteins Fold Into Their Functional Shape. The distribution of nonpolar amino acids along a protein chain largely determines how the protein folds.

How Proteins Unfold. When conditions such as pH or temperature fluctuate, proteins may denature or unfold.

3.3 Nucleic acids store and transfer genetic information.

Information Molecules. Nucleic acids store information in cells. RNA is a single-chain polymer of nucleotides, while DNA possesses two chains twisted around each other.

3.4 Lipids make membranes and store energy.

Phospholipids Form Membranes. The spontaneous aggregation of phospholipids in water is responsible for the formation of biological membranes.

Fats and Other Kinds of Lipids. Organisms utilize a wide variety of water-insoluble molecules.

Fats as Food. Fats are very efficient energy storage molecules because of their high proportion of C—H bonds.

3.5 Carbohydrates store energy and provide building materials.

Simple Carbohydrates. Sugars are simple carbohydrates, often consisting of six-carbon rings.

Linking Sugars Together. Sugars can be linked together to form long polymers, or polysaccharides.

Structural Carbohydrates. Structural carbohydrates like cellulose are chains of sugars linked in a way that enzymes cannot easily attack.

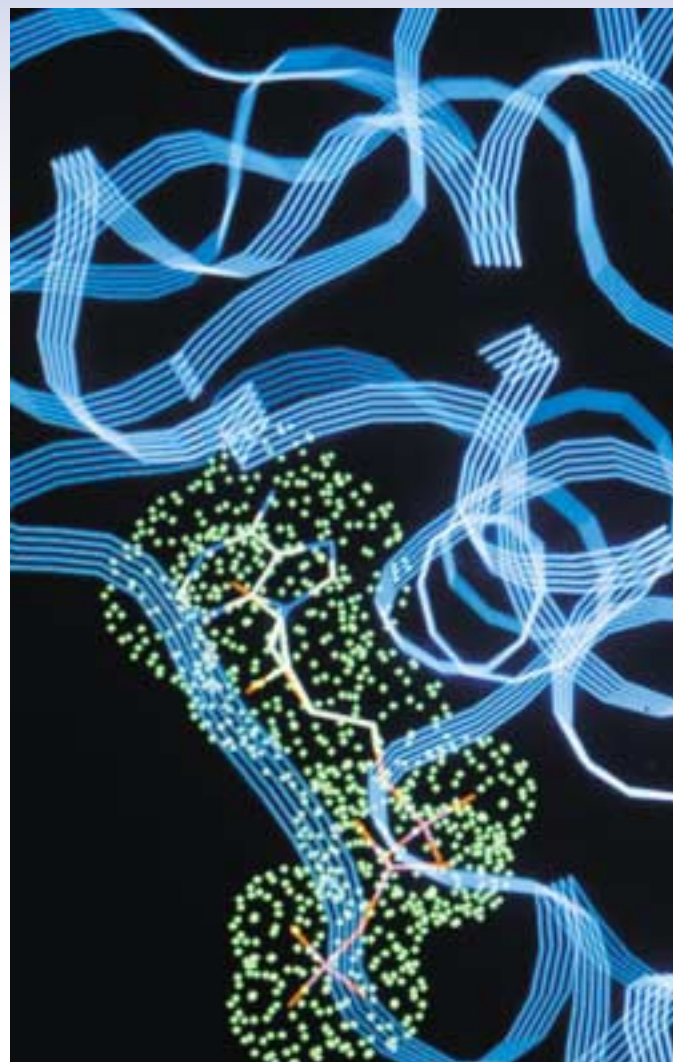


FIGURE 3.1

Computer-generated model of a macromolecule. Pictured is an enzyme responsible for releasing energy from sugar. This complex molecule consists of hundreds of different amino acids linked into chains that form the characteristic coils and folds seen here.

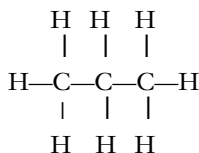
Molecules are extremely small compared with the familiar world we see about us. Imagine: there are more water molecules in a cup than there are stars in the sky. Many other molecules are gigantic, compared with water, consisting of thousands of atoms. These atoms are organized into hundreds of smaller molecules that are linked together into long chains (figure 3.1). These enormous molecules, almost always synthesized by living things, are called macromolecules. As we shall see, there are four general types of macromolecules, the basic chemical building blocks from which all organisms are assembled.

3.1 Molecules are the building blocks of life.

The Chemistry of Carbon

In chapter 2 we discussed how atoms combine to form molecules. In this chapter, we will focus on **organic molecules**, those chemical compounds that contain carbon. The frameworks of biological molecules consist predominantly of carbon atoms bonded to other carbon atoms or to atoms of oxygen, nitrogen, sulfur or hydrogen. Because carbon atoms possess four valence electrons and so can form four covalent bonds, molecules containing carbon can form straight chains, branches, or even rings. As you can imagine, all of these possibilities generate an immense range of molecular structures and shapes.

Organic molecules consisting only of carbon and hydrogen are called **hydrocarbons**. Covalent bonds between carbon and hydrogen are energy-rich. We use hydrocarbons from fossil fuels as a primary source of energy today. Propane gas, for example, is a hydrocarbon consisting of a chain of three carbon atoms, with eight hydrogen atoms bound to it:



Because carbon-hydrogen covalent bonds store considerable energy, hydrocarbons make good fuels. Gasoline, for example, is rich in hydrocarbons.

Functional Groups

Carbon and hydrogen atoms both have very similar electronegativities, so electrons in C—C and C—H bonds are evenly distributed, and there are no significant differences in charge over the molecular surface. For this reason, hydrocarbons are nonpolar. Most organic molecules that are produced by cells, however, also contain other atoms. Because these other atoms often have different electronegativities, molecules containing them exhibit regions of positive or negative charge, and so are polar. These molecules can be thought of as a C—H core to which specific groups of atoms called **functional groups** are attached. For example, a hydrogen atom bonded to an oxygen atom (—OH) is a functional group called a *hydroxyl group*.

Functional groups have definite chemical properties that they retain no matter where they occur. The hydroxyl group, for example, is polar, because its oxygen atom, being very electronegative, draws electrons toward itself (as we saw in chapter 2). Figure 3.2 illustrates the hydroxyl group and other biologically important functional groups. Most chemical reactions that occur within organisms involve the transfer of a functional group as an intact unit from one molecule to another.

Biological Macromolecules

Some organic molecules in organisms are small and simple, containing only one or a few functional groups. Others are large complex assemblies called **macromolecules**. In many cases, these macromolecules are polymers, molecules built by linking together a large number of small, similar chemical subunits, like railroad cars coupled to form a train. For example, complex carbohydrates like starch are polymers of simple ring-shaped sugars, proteins are polymers of amino acids, and nucleic acids (DNA and RNA) are polymers of nucleotides. Biological macromolecules are traditionally grouped into four major categories: proteins, nucleic acids, lipids, and carbohydrates (table 3.1).

Group	Structural Formula	Ball-and-Stick Model	Found In:
Hydroxyl	—OH		Carbohydrates, alcohols
Carbonyl			Formaldehyde
Carboxyl			Amino acids, vinegar
Amino			Ammonia
Sulfhydryl	—S—H		Proteins, rubber
Phosphate			Phospholipids, nucleic acids, ATP
Methyl			Methane gas

FIGURE 3.2

The primary functional chemical groups. These groups tend to act as units during chemical reactions and confer specific chemical properties on the molecules that possess them. Amino groups, for example, make a molecule more basic, while carboxyl groups make a molecule more acidic.

Table 3.1 Macromolecules

Macromolecule	Subunit	Function	Example
PROTEINS			
Globular	Amino acids	Modified glucose	Hemoglobin
Structural	Amino acids	Catalysis; transport	Hair; silk
NUCLEIC ACIDS			
DNA	Nucleotides	Encodes genes	Chromosomes
RNA	Nucleotides	Needed for gene expression	Messenger RNA
LIPIDS			
Fats	Glycerol and three fatty acids	Energy storage	Butter; corn oil; soap
Phospholipids	Glycerol, two fatty acids, phosphate, and polar R groups	Cell membranes	Lecithin
Prostaglandins	Five-carbon rings with two nonpolar tails	Chemical messengers	Prostaglandin E (PGE)
Steroids	Four fused carbon rings	Membranes; hormones	Cholesterol; estrogen
Terpenes	Long carbon chains	Pigments; structural	Carotene; rubber
CARBOHYDRATES			
Starch, glycogen	Glucose	Energy storage	Potatoes
Cellulose	Glucose	Cell walls	Paper; strings of celery
Chitin		Structural support	Crab shells

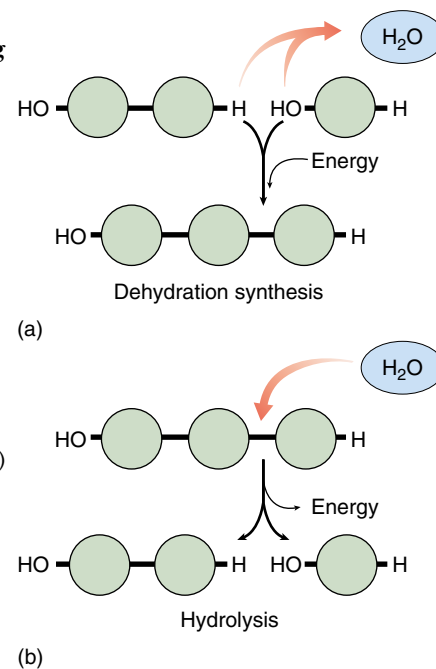
Building Macromolecules

Although the four categories of macromolecules contain different kinds of subunits, they are all assembled in the same fundamental way: to form a covalent bond between two subunit molecules, an —OH group is removed from one subunit and a hydrogen atom (H) is removed from the other (figure 3.3*a*). This condensation reaction is called a **dehydration synthesis**, because the removal of the —OH group and H during the synthesis of a new molecule in effect constitutes the removal of a molecule of water (H_2O). For every subunit that is added to a macromolecule, one water molecule is removed. Energy is required to break the chemical bonds when water is extracted from the subunits, so cells must supply energy to assemble macromolecules. These and other biochemical reactions require that the reacting substances be held close together and that the correct chemical bonds be stressed and broken. This process of positioning and stressing, termed catalysis, is carried out in cells by a special class of proteins known as enzymes.

Cells disassemble macromolecules into their constituent subunits by performing reactions that are essentially the reverse of dehydration—a molecule of water is added instead of removed (figure 3.3*b*). In this process, which is called **hydrolysis** (Greek *hydro*, “water” + *lyse*, “break”), a hydrogen atom is attached to one subunit and a hydroxyl group to the other, breaking a specific covalent bond in the macromolecule. Hydrolytic reactions release the energy that was stored in the bonds that were broken.

FIGURE 3.3
Making and breaking
macromolecules.

(*a*) Biological macromolecules are polymers formed by linking subunits together. The covalent bond between the subunits is formed by dehydration synthesis, an energy-requiring process that creates a water molecule for every bond formed. (*b*) Breaking the bond between subunits requires the returning of a water molecule with a subsequent release of energy, a process called hydrolysis.



Polymers are large molecules consisting of long chains of similar subunits joined by dehydration reactions. In a dehydration reaction, a hydroxyl (—OH) group is removed from one subunit and a hydrogen atom (H) is removed from the other.

3.2 Proteins perform the chemistry of the cell.

The Many Functions of Proteins

We will begin our discussion of macromolecules that make up the bodies of organisms with proteins (see table 3.1). The proteins within living organisms are immensely diverse in structure and function (table 3.2 and figure 3.4).

1. Enzyme catalysis. We have already encountered one class of proteins, enzymes, which are biological catalysts that facilitate specific chemical reactions. Because of this property, the appearance of enzymes was one of the most important events in the evolution of life. Enzymes are globular proteins, with a three-dimensional shape that fits snugly around the chemi-

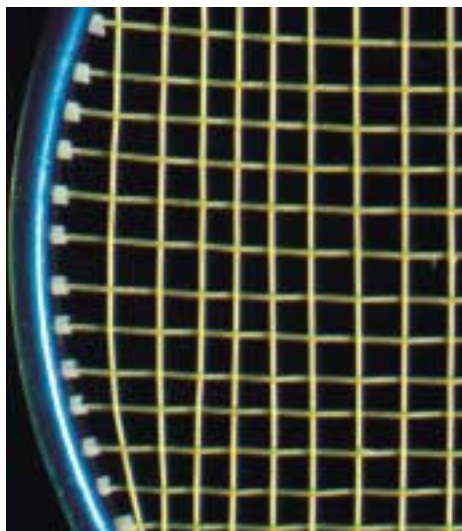
cals they work on, facilitating chemical reactions by stressing particular chemical bonds.

2. Defense. Other globular proteins use their shapes to “recognize” foreign microbes and cancer cells. These cell surface receptors form the core of the body’s hormone and immune systems.

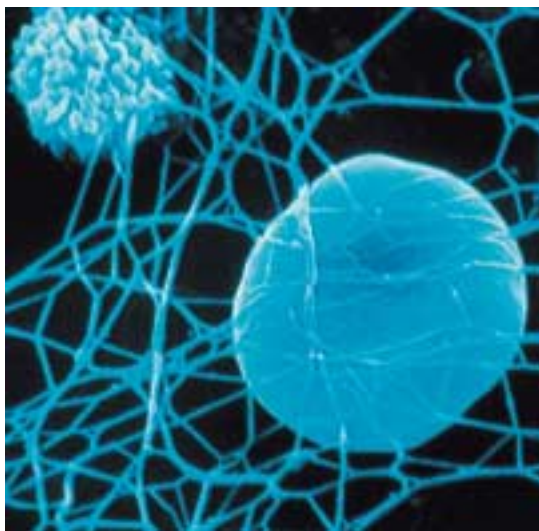
3. Transport. A variety of globular proteins transport specific small molecules and ions. The transport protein hemoglobin, for example, transports oxygen in the blood, and myoglobin, a similar protein, transports oxygen in muscle. Iron is transported in blood by the protein transferrin.

Table 3.2 The Many Functions of Proteins

Function	Class of Protein	Examples	Use
Metabolism (Catalysis)	Enzymes	Hydrolytic enzymes Proteases Polymerases Kinases	Cleave polysaccharides Break down proteins Produce nucleic acids Phosphorylate sugars and proteins
Defense	Immunoglobulins	Antibodies	Mark foreign proteins for elimination
Cell recognition	Toxins	Snake venom	Block nerve function
Transport throughout body	Cell surface antigens Globins	MHC proteins Hemoglobin Myoglobin Cytochromes	“Self” recognition Carries O ₂ and CO ₂ in blood Carries O ₂ and CO ₂ in muscle Electron transport
Membrane transport	Transporters	Sodium-potassium pump Proton pump Anion channels	Excitable membranes Chemiosmosis Transport Cl ⁻ ions
Structure/Support	Fibers	Collagen Keratin Fibrin	Cartilage Hair, nails Blood clot
Motion	Muscle	Actin Myosin	Contraction of muscle fibers Contraction of muscle fibers
Osmotic regulation	Albumin	Serum albumin	Maintains osmotic concentration of blood
Regulation of gene action	Repressors	lac repressor	Regulates transcription
Regulation of body functions	Hormones	Insulin Vasopressin Oxytocin	Controls blood glucose levels Increases water retention by kidneys Regulates uterine contractions and milk production
Storage	Ion binding	Ferritin Casein Calmodulin	Stores iron, especially in spleen Stores ions in milk Binds calcium ions



(a)



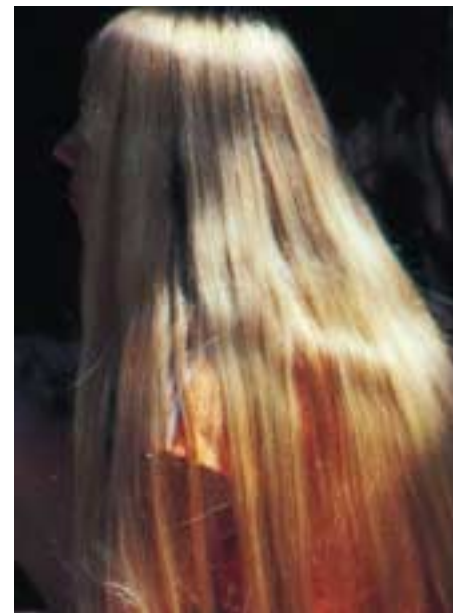
(b)



(c)



(d)



(e)

FIGURE 3.4

Some of the more common structural proteins. (a) Collagen: strings of a tennis racket from gut tissue; (b) fibrin: scanning electron micrograph of a blood clot (3000 $\times$); (c) keratin: a peacock feather; (d) silk: a spider's web; (e) keratin: human hair.

4. Support. Fibrous, or threadlike, proteins play structural roles; these structural proteins (see figure 3.4) include keratin in hair, fibrin in blood clots, and collagen, which forms the matrix of skin, ligaments, tendons, and bones and is the most abundant protein in a vertebrate body.

5. Motion. Muscles contract through the sliding motion of two kinds of protein filament: actin and myosin. **Contractile proteins** also play key roles in the cell's cytoskeleton and in moving materials within cells.

6. Regulation. Small proteins called hormones serve as **intercellular messengers** in animals. Proteins also play many regulatory roles within the cell, turning on and shutting off genes during development, for example. In addition, proteins also receive information, acting as cell surface receptors.

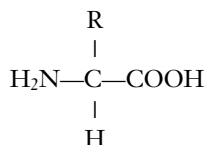
Proteins carry out a diverse array of functions, including catalysis, defense, transport of substances, motion, and regulation of cell and body functions.

Amino Acids Are the Building Blocks of Proteins

Although proteins are complex and versatile molecules, they are all polymers of only 20 amino acids, in a specific order. Many scientists believe amino acids were among the first molecules formed in the early earth. It seems highly likely that the oceans that existed early in the history of the earth contained a wide variety of amino acids.

Amino Acid Structure

An **amino acid** is a molecule containing an amino group (—NH_2), a carboxyl group (—COOH), and a hydrogen atom, all bonded to a central carbon atom:



Each amino acid has unique chemical properties determined by the nature of the side group (indicated by R) covalently bonded to the central carbon atom. For example, when the side group is $\text{—CH}_2\text{OH}$, the amino acid (serine) is polar, but when the side group is —CH_3 , the amino acid (alanine) is nonpolar. The 20 common amino acids are grouped into five chemical classes, based on their side groups:

1. Nonpolar amino acids, such as leucine, often have R groups that contain —CH_2 or —CH_3 .
2. Polar uncharged amino acids, such as threonine, have R groups that contain oxygen (or only —H).
3. Ionizable amino acids, such as glutamic acid, have R groups that contain acids or bases.
4. Aromatic amino acids, such as phenylalanine, have R groups that contain an organic (carbon) ring with alternating single and double bonds.
5. Special-function amino acids have unique individual properties; methionine often is the first amino acid in a chain of amino acids, proline causes kinks in chains, and cysteine links chains together.

Each amino acid affects the shape of a protein differently depending on the chemical nature of its side group. Portions of a protein chain with numerous nonpolar amino acids, for example, tend to fold into the interior of the protein by hydrophobic exclusion.

Proteins Are Polymers of Amino Acids

In addition to its R group, each amino acid, when ionized, has a positive amino (NH_3^+) group at one end and a negative carboxyl (COO^-) group at the other end. The amino and carboxyl groups on a pair of amino acids can undergo a condensation reaction, losing a molecule of water and

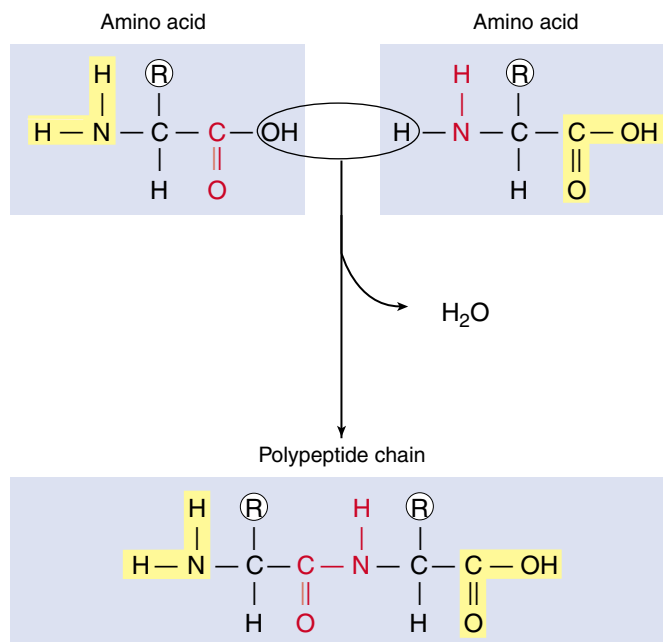


FIGURE 3.5

The peptide bond. A peptide bond forms when the —NH_2 end of one amino acid joins to the —COOH end of another. Because of the partial double-bond nature of peptide bonds, the resulting peptide chain cannot rotate freely around these bonds.

forming a covalent bond. A covalent bond that links two amino acids is called a **peptide bond** (figure 3.5). The two amino acids linked by such a bond are not free to rotate around the N—C linkage because the peptide bond has a partial double-bond character, unlike the N—C and C—C bonds to the central carbon of the amino acid. The stiffness of the peptide bond is one factor that makes it possible for chains of amino acids to form coils and other regular shapes.

A **protein** is composed of one or more long chains, or **polypeptides**, composed of amino acids linked by peptide bonds. It was not until the pioneering work of Frederick Sanger in the early 1950s that it became clear that each kind of protein had a specific amino acid sequence. Sanger succeeded in determining the amino acid sequence of insulin and in so doing demonstrated clearly that this protein had a defined sequence, the same for all insulin molecules in the solution. Although many different amino acids occur in nature, only 20 commonly occur in proteins. Figure 3.6 illustrates these 20 “common” amino acids and their side groups.

A protein is a polymer containing a combination of up to 20 different kinds of amino acids. The amino acids fall into five chemical classes, each with different properties. These properties determine the nature of the resulting protein.

SPECIAL STRUCTURAL PROPERTY

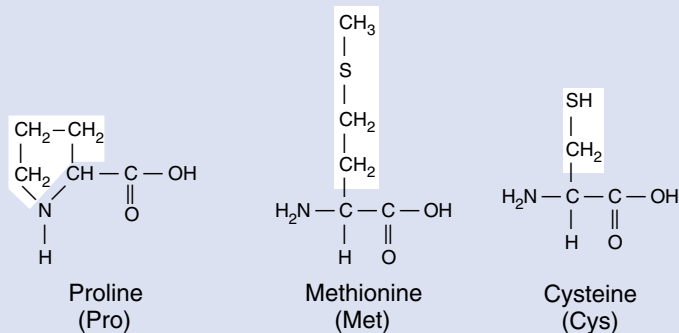
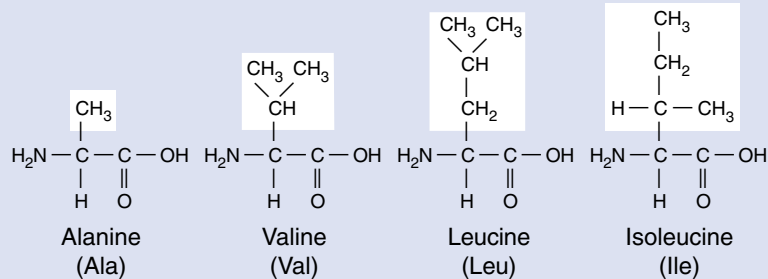


FIGURE 3.6

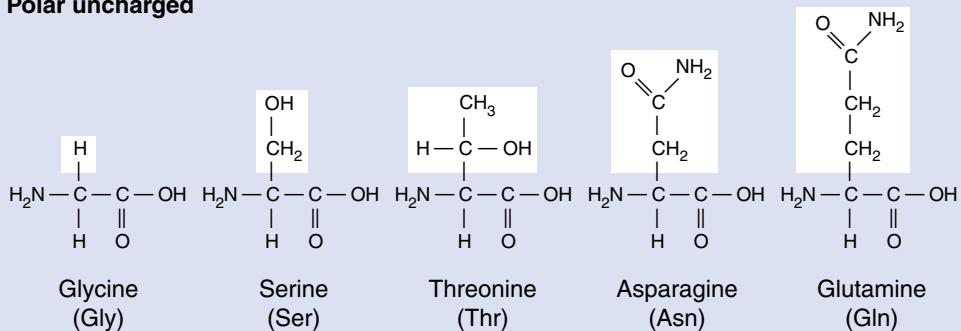
The 20 common amino acids. Each amino acid has the same chemical backbone, but differs in the side, or R, group it possesses. Six of the amino acids are nonpolar because they have —CH_2 or —CH_3 in their R groups. Two of the six are bulkier because they contain ring structures, which classifies them also as aromatic. Another six are polar because they have oxygen or just hydrogen in their R groups; these amino acids, which are uncharged, differ from one another in how polar they are. Five other amino acids are polar and, because they have a terminal acid or base in their R group, are capable of ionizing to a charged form. The remaining three have special chemical properties that allow them to help form links between protein chains or kinks in proteins.

NONAROMATIC

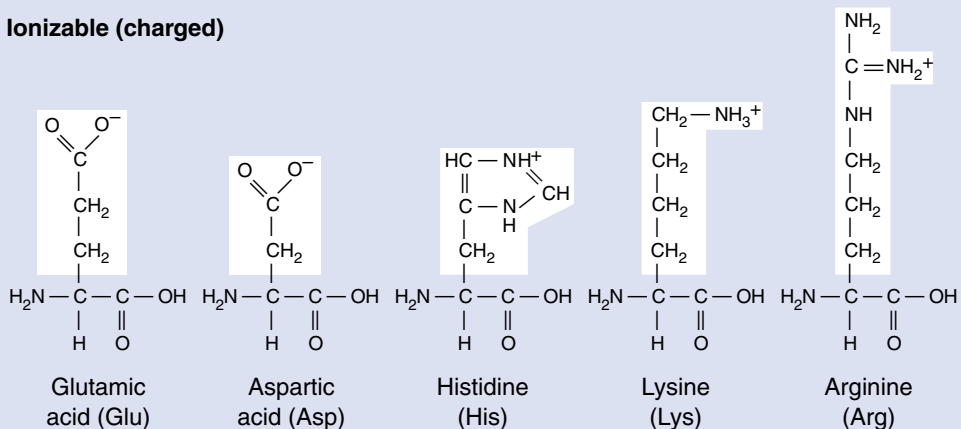
Nonpolar



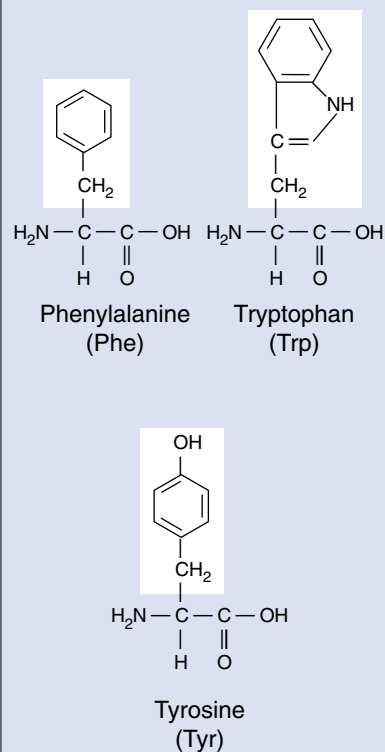
Polar uncharged



Ionizable (charged)



AROMATIC



A Protein's Function Depends on the Shape of the Molecule

The shape of a protein is very important because it determines the protein's function. If we picture a polypeptide as a long strand similar to a reed, a protein might be the basket woven from it.

Overview of Protein Structure

Proteins consist of long amino acid chains folded into complex shapes. What do we know about the shape of these proteins? One way to study the shape of something as small as a protein is to look at it with very short wavelength energy—with X rays. X-ray diffraction is a painstaking procedure that allows the investigator to build up a three-dimensional picture of the position of each atom. The first protein to be analyzed in this way was myoglobin, soon followed by the related protein hemoglobin. As more and more proteins were added to the list, a general principle became evident: in every protein studied, essentially all the internal amino acids are nonpolar ones, amino acids such as leucine, valine, and phenylalanine. Water's tendency to hydrophobically exclude nonpolar molecules literally shoves the nonpolar portions of the amino acid chain into the protein's interior. This positions the nonpolar amino acids in close contact with one another, leaving little empty space inside. Polar and charged amino acids are restricted to the surface of the protein except for the few that play key functional roles.

Levels of Protein Structure

The structure of proteins is traditionally discussed in terms of four levels of structure, as *primary*, *secondary*, *tertiary*, and *quaternary* (figure 3.7). Because of progress in our knowledge of protein structure, two additional levels of structure are increasingly distinguished by molecular biologists: *motifs* and *domains*. Because these latter two elements play important roles in coming chapters, we introduce them here.

Primary Structure. The specific amino acid sequence of a protein is its primary structure. This sequence is determined by the nucleotide sequence of the gene that encodes the protein. Because the R groups that distinguish the various amino acids play no role in the peptide backbone of proteins, a protein can consist of any sequence of amino acids. Thus, a protein containing 100 amino acids could form any of 20^{100} different amino acid sequences (that's the same as 10^{130} , or 1

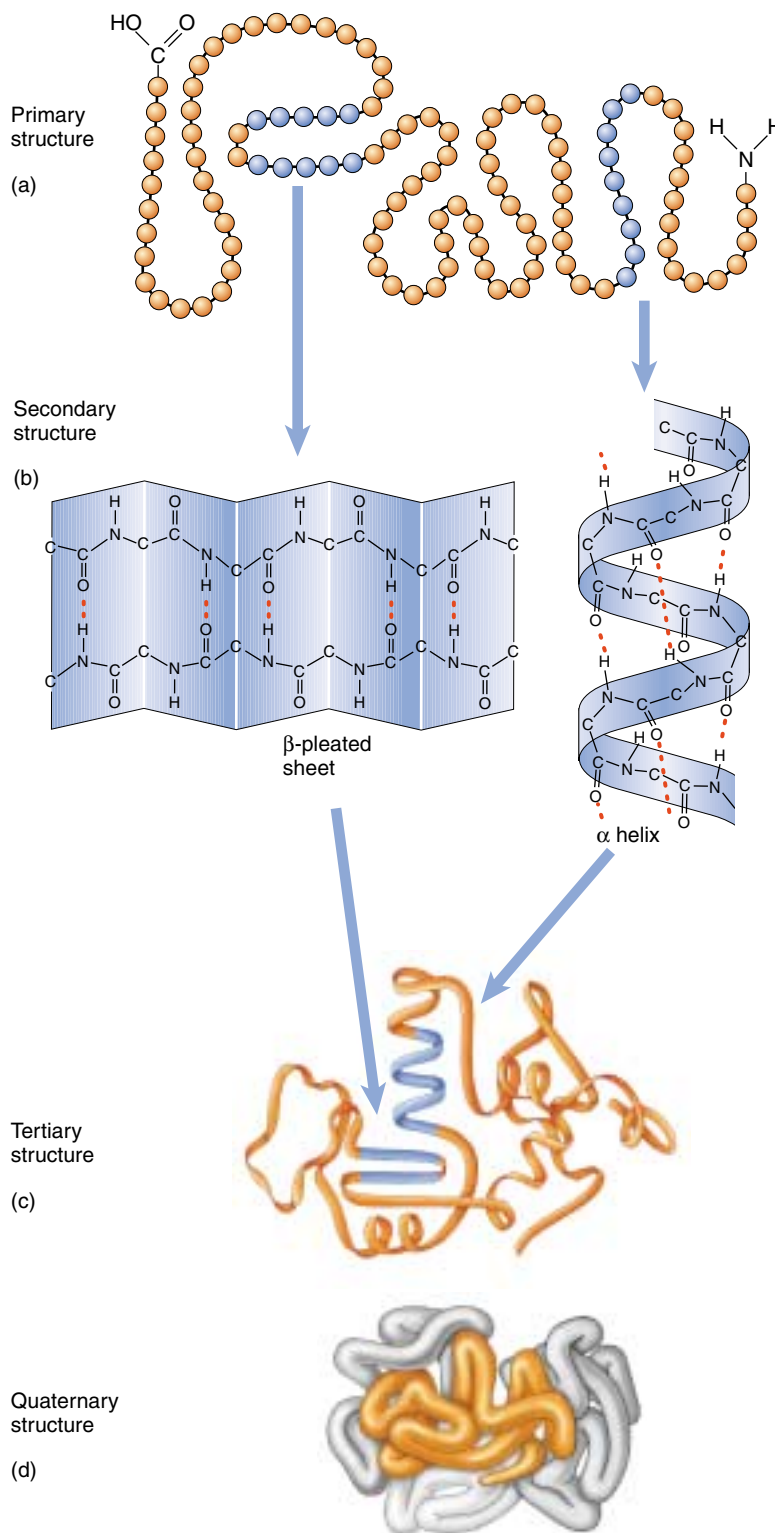


FIGURE 3.7

Levels of protein structure. The amino acid sequence of a protein is called its (a) primary structure. Hydrogen bonds form between nearby amino acids, producing (b) fold-backs called beta-pleated sheets and coils called alpha helices. These fold-backs and coils constitute the protein's secondary structure. A globular protein folds up on itself further to assume a three-dimensional (c) tertiary structure. Many proteins aggregate with other polypeptide chains in clusters; this clustering is called the (d) quaternary structure of the protein.

followed by 130 zeros—more than the number of atoms known in the universe). This is an important property of proteins because it permits such great diversity.

Secondary Structure. The amino acid side groups are not the only portions of proteins that form hydrogen bonds. The —COOH and —NH_2 groups of the main chain also form quite good hydrogen bonds—so good that their interactions with water might be expected to offset the tendency of nonpolar sidegroups to be forced into the protein interior. Inspection of the protein structures determined by X-ray diffraction reveals why they don't—the polar groups of the main chain form hydrogen bonds with each other! Two patterns of H bonding occur. In one, hydrogen bonds form along a single chain, linking one amino acid to another farther down the chain. This tends to pull the chain into a coil called an alpha (α) helix. In the other pattern, hydrogen bonds occur across two chains, linking the amino acids in one chain to those in the other. Often, many parallel chains are linked, forming a pleated, sheet-like structure called a β -pleated sheet. The folding of the amino acid chain by hydrogen bonding into these characteristic coils and pleats is called a protein's secondary structure.

Motifs. The elements of secondary structure can combine in proteins in characteristic ways called motifs, or sometimes “supersecondary structure.” One very common motif is the $\beta \alpha \beta$ motif, which creates a fold or crease; the so-called “Rossmann fold” at the core of nucleotide binding sites in a wide variety of proteins is a $\beta \alpha \beta \alpha \beta$ motif. A second motif that occurs in many proteins is the β barrel, a β sheet folded around to form a tube. A third type of motif, the α turn α motif, is important because many proteins use it to bind the DNA double helix.

Tertiary Structure. The final folded shape of a globular protein, which positions the various motifs and folds nonpolar side groups into the interior, is called a protein's tertiary structure. A protein is driven into its tertiary structure by hydrophobic interactions with water. The final folding of a protein is determined by its primary structure—by the chemical nature of its side groups. Many proteins can be fully unfolded (“denatured”) and will spontaneously refold back into their characteristic shape.

The stability of a protein, once it has folded into its 3-D shape, is strongly influenced by how well its interior fits together. When two nonpolar chains in the interior are in very close proximity, they experience a form of molecular attraction called van der Waal's forces. Individually quite weak, these forces can add up to a strong attraction when many of them come into play, like the combined strength of hundreds of hooks and loops on a strip of Velcro. They are effective forces only over short distances, however; there are no “holes” or cavities in the interior of proteins. That is why there are so many different nonpolar amino acids (alanine, valine, leucine, isoleucine). Each has a dif-

ferent-sized R group, allowing very precise fitting of nonpolar chains within the protein interior. Now you can understand why a mutation that converts one nonpolar amino acid within the protein interior (alanine) into another (leucine) very often disrupts the protein's stability; leucine is a lot bigger than alanine and disrupts the precise way the chains fit together within the protein interior. A change in even a single amino acid can have profound effects on protein shape and can result in loss or altered function of the protein.

Domains. Many proteins in your body are encoded within your genes in functional sections called exons (exons will be discussed in detail in chapter 15). Each exon-encoded section of a protein, typically 100 to 200 amino acids long, folds into a structurally independent functional unit called a **domain**. As the polypeptide chain folds, the domains fold into their proper shape, each more-or-less independent of the others. This can be demonstrated experimentally by artificially producing the fragment of polypeptide that forms the domain in the intact protein, and showing that the fragment folds to form the same structure as it does in the intact protein.

A single polypeptide chain connects the domains of a protein, like a rope tied into several adjacent knots. Often the domains of a protein have quite separate functions—one domain of an enzyme might bind a cofactor, for example, and another the enzyme's substrate.

Quaternary Structure. When two or more polypeptide chains associate to form a functional protein, the individual chains are referred to as subunits of the protein. The subunits need not be the same. Hemoglobin, for example, is a protein composed of two α -chain subunits and two β -chain subunits. A protein's subunit arrangement is called its quaternary structure. In proteins composed of subunits, the interfaces where the subunits contact one another are often nonpolar, and play a key role in transmitting information between the subunits about individual subunit activities.

A change in the identity of one of these amino acids can have profound effects. Sick cell hemoglobin is a mutation that alters the identity of a single amino acid at the corner of the β subunit from polar glutamate to nonpolar valine. Putting a nonpolar amino acid on the surface creates a “sticky patch” that causes one hemoglobin molecule to stick to another, forming long nonfunctional chains and leading to the cell sickling characteristic of this hereditary disorder.

Protein structure can be viewed at six levels: 1. the amino acid sequence, or primary structure; 2. coils and sheets, called secondary structure; 3. folds or creases, called motifs; 4. the three-dimensional shape, called tertiary structure; 5. functional units, called domains; and 6. individual polypeptide subunits associated in a quaternary structure.

How Proteins Fold Into Their Functional Shape

How does a protein fold into a specific shape? Nonpolar amino acids play a key role. Until recently, investigators thought that newly made proteins fold spontaneously as hydrophobic interactions with water shove nonpolar amino acids into the protein interior. We now know this is too simple a view. Protein chains can fold in so many different ways that trial and error would simply take too long. In addition, as the open chain folds its way toward its final form, nonpolar “sticky” interior portions are exposed during intermediate stages. If these intermediate forms are placed in a test tube in the same protein environment that occurs in a cell, they stick to other unwanted protein partners, forming a gluey mess.

Chaperonins

How do cells avoid this? A vital clue came in studies of unusual mutations that prevented viruses from replicating in *E. coli* bacterial cells—it turned out the virus proteins could not fold properly! Further study revealed that normal cells contain special proteins called **chaperonins** that help new proteins fold correctly (figure 3.8). When the *E. coli* gene encoding its chaperone protein is disabled by mutation, the bacteria die, clogged with lumps of incorrectly folded proteins. Fully 30% of the bacteria’s proteins fail to fold to the right shape.

Molecular biologists have now identified more than 17 kinds of proteins that act as molecular chaperones. Many are heat shock proteins, produced in greatly elevated amounts if a cell is exposed to elevated temperature; high temperatures cause proteins to unfold, and heat shock chaperonins help the cell’s proteins refold.

There is considerable controversy about how chaperonins work. It was first thought that they provided a protected environment within which folding could take place unhindered by other proteins, but it now seems more likely that chaperonins rescue proteins that are caught in a wrongly folded state, giving them another chance to fold correctly. When investigators “fed” a deliberately misfolded protein called malate dehydrogenase to chaperonins, the protein was rescued, refolding to its active shape.

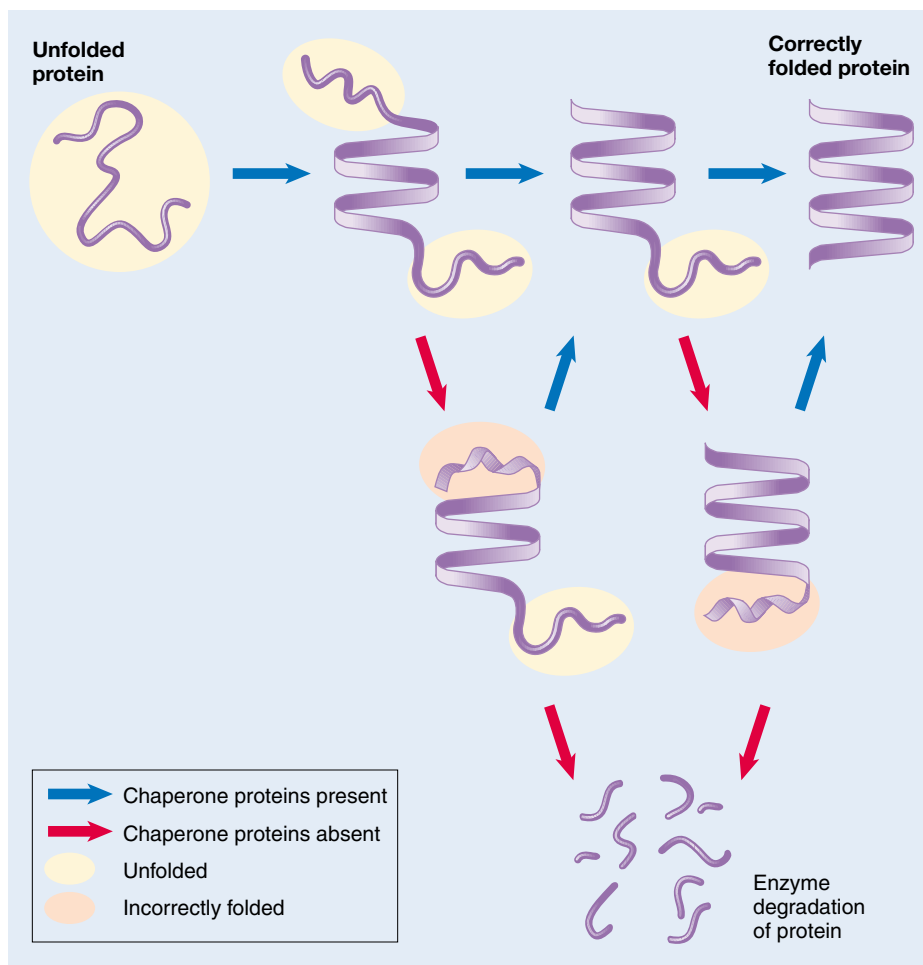


FIGURE 3.8

A current model of protein folding. A newly synthesized protein rapidly folds into characteristic motifs composed of α helices and β sheets, but these elements of structure are only roughly aligned in an open conformation. Subsequent folding occurs more slowly, by trial and error. This process is aided by chaperone proteins, which appear to recognize improperly folded proteins and unfold them, giving them another chance to fold properly. Eventually, if proper folding is not achieved, the misfolded protein is degraded by proteolytic enzymes.

Protein Folding and Disease

There are tantalizing suggestions that chaperone protein deficiencies may play a role in certain diseases by failing to facilitate the intricate folding of key proteins. Cystic fibrosis is a hereditary disorder in which a mutation disables a protein that plays a vital part in moving ions across cell membranes. In at least some cases, the vital membrane protein appears to have the correct amino acid sequence, but fails to fold to its final form. It has also been speculated that chaperone deficiency may be a cause of the protein clumping in brain cells that produces the amyloid plaques characteristic of Alzheimer’s disease.

Proteins called chaperones aid newly produced proteins to fold properly.

How Proteins Unfold

If a protein's environment is altered, the protein may change its shape or even unfold. This process is called **denaturation**. Proteins can be denatured when the pH, temperature, or ionic concentration of the surrounding solution is changed. When proteins are denatured, they are usually rendered biologically inactive. This is particularly significant in the case of enzymes. Because practically every chemical reaction in a living organism is catalyzed by a specific enzyme, it is vital that a cell's enzymes remain functional. That is the rationale behind traditional methods of salt-curing and pickling: prior to the ready availability of refrigerators and freezers, the only practical way to keep microorganisms from growing in food was to keep the food in a solution containing high salt or vinegar concentrations, which denatured the enzymes of microorganisms and kept them from growing on the food.

Most enzymes function within a very narrow range of physical parameters. Blood-borne enzymes that course through a human body at a pH of about 7.4 would rapidly become denatured in the highly acidic environment of the stomach. On the other hand, the protein-degrading enzymes that function at a pH of 2 or less in the stomach would be denatured in the basic pH of the blood. Similarly, organisms that live near oceanic hydrothermal vents have enzymes that work well at the temperature of this extreme environment (over 100°C). They cannot survive in cooler waters, because their enzymes would denature at lower temperatures. Any given organism usually has a "tolerance range" of pH, temperature, and salt concentration. Within that range, its enzymes maintain the proper shape to carry out their biological functions.

When a protein's normal environment is reestablished after denaturation, a small protein may spontaneously refold into its natural shape, driven by the interactions between its nonpolar amino acids and water (figure 3.9). Larger proteins can rarely refold spontaneously because of the complex nature of their final shape. It is important to distinguish denaturation from **dissociation**. The four subunits of hemoglobin (figure 3.10) may dissociate into four individual molecules (two α -globin and two β -globin) without denaturation of the folded globin proteins, and will readily reassume their four-subunit quaternary structure.

Every globular protein has a narrow range of conditions in which it folds properly; outside that range, proteins tend to unfold.

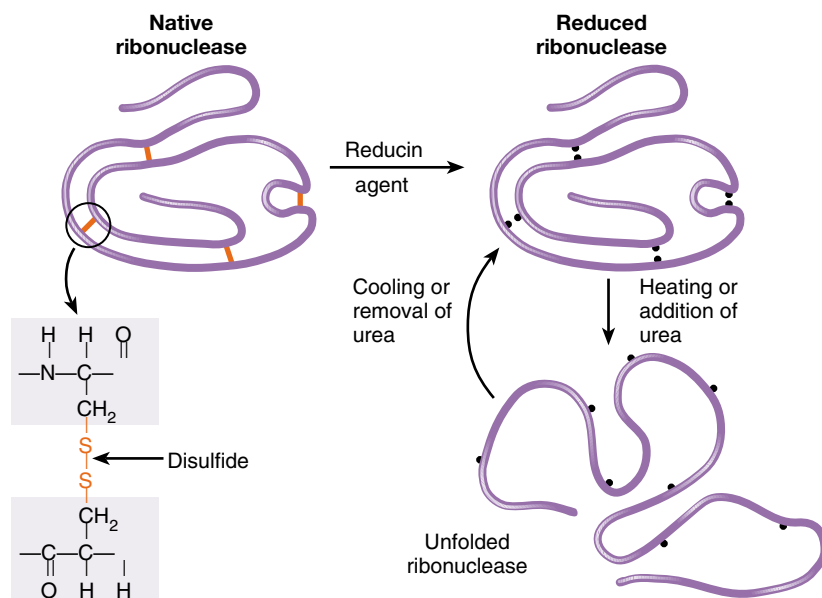


FIGURE 3.9

Primary structure determines tertiary structure. When the protein ribonuclease is treated with reducing agents to break the covalent disulfide bonds that cross-link its chains, and then placed in urea or heated, the protein denatures (unfolds) and loses its enzymatic activity. Upon cooling or removal of urea, it refolds and regains its enzymatic activity. This demonstrates that no information but the amino acid sequence of the protein is required for proper folding: the primary structure of the protein determines its tertiary structure.

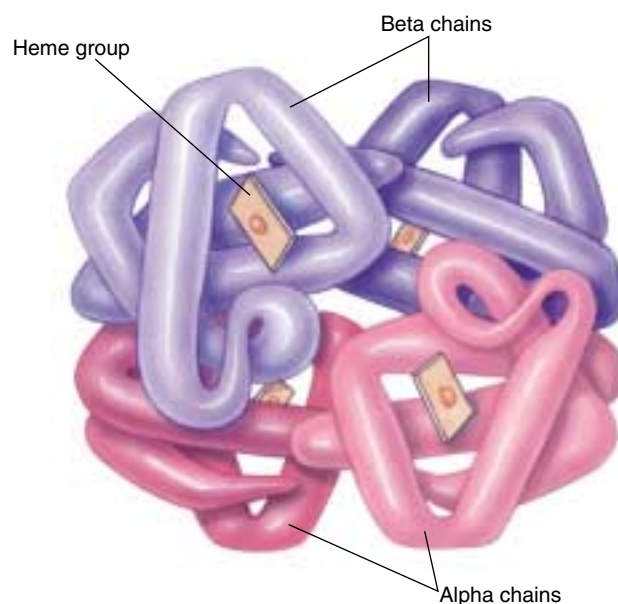


FIGURE 3.10

The four subunits of hemoglobin. The hemoglobin molecule is made of four globin protein subunits, informally referred to as polypeptide chains. The two lower α chains, identical α -globin proteins, are shaded pink; the two upper β chains, identical β -globin proteins, are shaded blue.

3.3 Nucleic acids store and transfer genetic information.

Information Molecules

The biochemical activity of a cell depends on production of a large number of proteins, each with a specific sequence. The ability to produce the correct proteins is passed between generations of organisms, even though the protein molecules themselves are not.

Nucleic acids are the information storage devices of cells, just as disks or tapes store the information that computers use, blueprints store the information that builders use, and road maps store the information that tourists use. There are two varieties of nucleic acids: *deoxyribonucleic acid* (**DNA**; figure 3.11) and *ribonucleic acid* (**RNA**). The way in which DNA encodes the information used to assemble proteins is similar to the way in which the letters on a page encode information (see chapter 14). Unique among macromolecules, nucleic acids are able to serve as templates to produce precise copies of themselves, so that the information that specifies what an organism is can be copied and passed down to its descendants. For this reason, DNA is often referred to as the hereditary material. Cells use the alternative form of nucleic acid, RNA, to read the cell's DNA-encoded information and direct the synthesis of proteins. RNA is similar to DNA in structure and is made as a transcribed copy of portions of the DNA. This transcript passes out into the rest of the cell, where it serves as a blueprint specifying a protein's amino acid sequence. This process will be described in detail in chapter 15.

“Seeing” DNA

DNA molecules cannot be seen with an optical microscope, which is incapable of resolving anything smaller than 1000 atoms across. An electron microscope can image structures as small as a few dozen atoms across, but still cannot resolve the individual atoms of a DNA strand. This limitation was finally overcome in the last decade with the introduction of the scanning-tunneling microscope (figure 3.12).

How do these microscopes work? Imagine you are in a dark room with a chair. To determine the shape of the chair, you could shine a flashlight on it, so that the light bounces off the chair and forms an image on your eye. That's what optical and electron microscopes do; in the latter, the “flashlight” emits a beam of electrons instead of light. You could, however, also reach out and feel the chair's surface with your hand. In effect, you would be putting a probe (your hand) near the chair and measuring how far away the surface is. In a scanning-tunneling microscope, computers advance a probe over the surface of a molecule in steps smaller than the diameter of an atom.

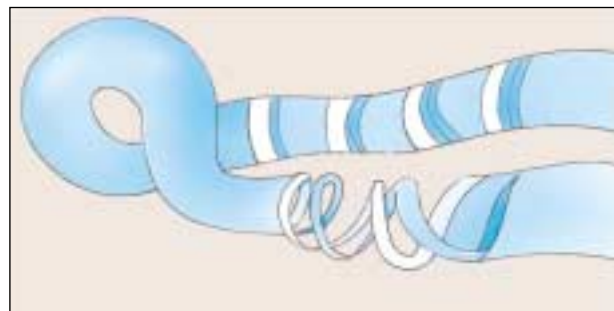
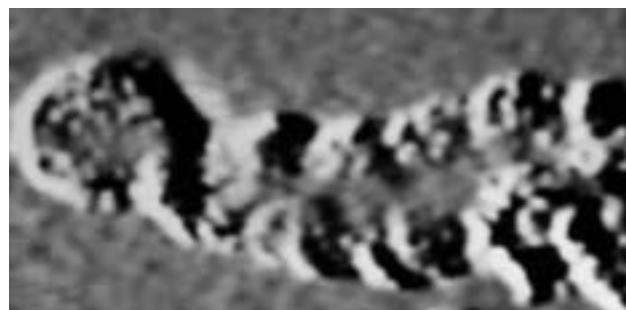


FIGURE 3.11

The first photograph of a DNA molecule. This micrograph, with sketch below, shows a section of DNA magnified a million times! The molecule is so slender that it would take 50,000 of them to equal the diameter of a human hair.

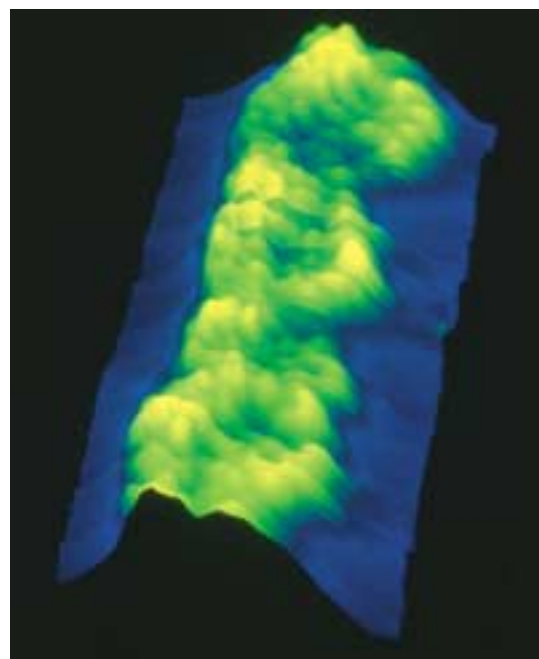


FIGURE 3.12

A scanning tunneling micrograph of DNA (false color; 2,000,000 $\times$). The micrograph shows approximately three turns of the DNA double helix (see figure 3.15).

The Structure of Nucleic Acids

Nucleic acids are long polymers of repeating subunits called **nucleotides**. Each nucleotide consists of three components: a five-carbon sugar (ribose in RNA and deoxyribose in DNA); a phosphate ($-\text{PO}_4$) group; and an organic nitrogen-containing base (figure 3.13). When a nucleic acid polymer forms, the phosphate group of one nucleotide binds to the hydroxyl group of another, releasing water and forming a phosphodiester bond. A **nucleic acid**, then, is simply a chain of five-carbon sugars linked together by phosphodiester bonds with an organic base protruding from each sugar (figure 3.14).

Two types of organic bases occur in nucleotides. The first type, *purines*, are large, double-ring molecules found in both DNA and RNA; they are adenine (A) and guanine (G). The second type, *pyrimidines*, are smaller, single-ring molecules; they include cytosine (C, in both DNA and RNA), thymine (T, in DNA only), and uracil (U, in RNA only).

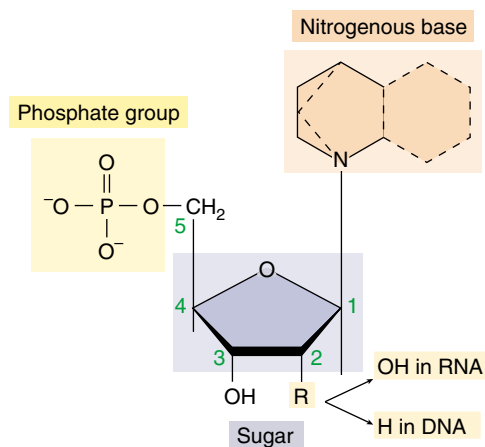


FIGURE 3.13
Structure of a nucleotide. The nucleotide subunits of DNA and RNA are made up of three elements: a five-carbon sugar, an organic nitrogenous base, and a phosphate group.

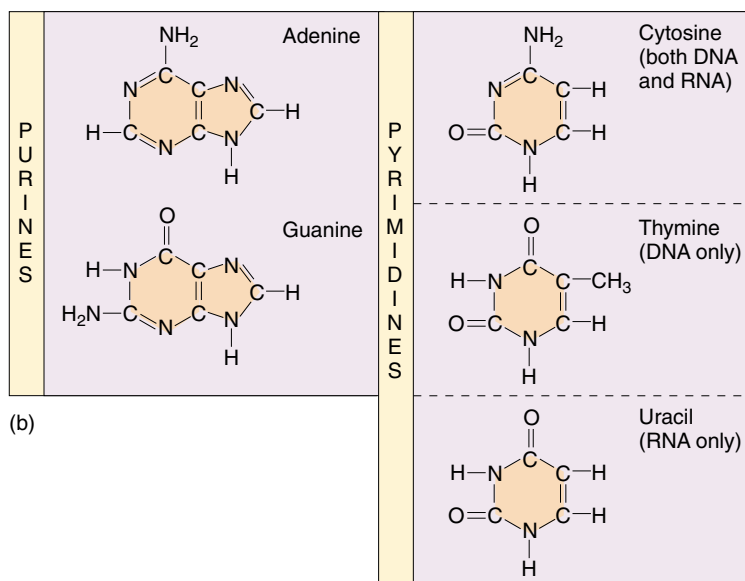
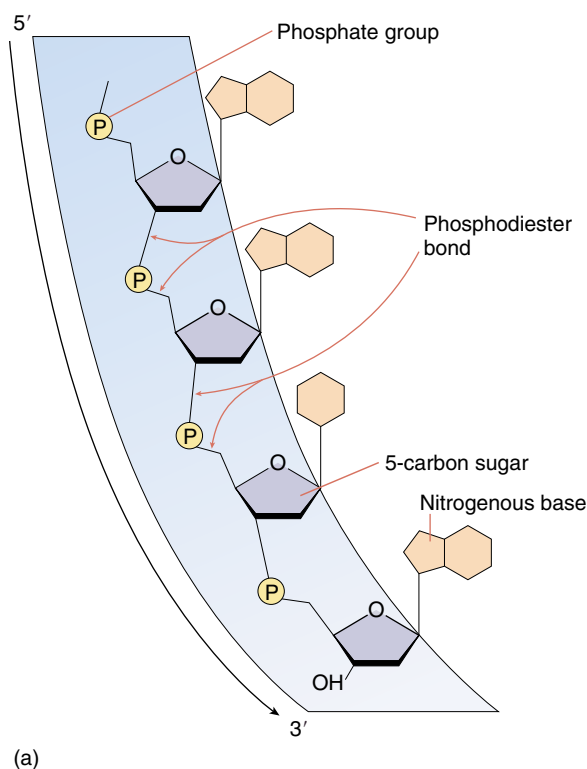


FIGURE 3.14

The structure of a nucleic acid and the organic nitrogen-containing bases. (a) In a nucleic acid, nucleotides are linked to one another via phosphodiester bonds, with organic bases protruding from the chain. (b) The organic nitrogenous bases can be either purines or pyrimidines. In DNA, thymine replaces the uracil found in RNA.

DNA

Organisms encode the information specifying the amino acid sequences of their proteins as sequences of nucleotides in the DNA. This method of encoding information is very similar to that by which the sequences of letters encode information in a sentence. While a sentence written in English consists of a combination of the 26 different letters of the alphabet in a specific order, the code of a DNA molecule consists of different combinations of the four types of nucleotides in specific sequences such as CGCTTACG. The information encoded in DNA is used in the everyday metabolism of the organism and is passed on to the organism's descendants.

DNA molecules in organisms exist not as single chains folded into complex shapes, like proteins, but rather as double chains. Two DNA polymers wind around each other like the outside and inside rails of a circular staircase. Such a winding shape is called a **helix**, and a helix composed of two chains winding about one another, as in DNA, is called a **double helix**. Each step of DNA's helical staircase is a base-pair, consisting of a base in one chain attracted by hydrogen bonds to a base opposite it on the other chain. These hydrogen bonds hold the two chains together as a duplex (figure 3.15). The base-pairing rules are rigid: adenine can pair only with thymine (in DNA) or with uracil (in RNA), and cytosine can pair only with guanine. The bases that participate in base-pairing are said to be **complementary** to each other. Additional details of the structure of DNA and how it interacts with RNA in the production of proteins are presented in chapters 14 and 15.

RNA

RNA is similar to DNA, but with two major chemical differences. First, RNA molecules contain ribose sugars in which the number 2 carbon is bonded to a hydroxyl group. In DNA, this hydroxyl group is replaced by a hydrogen atom. Second, RNA molecules utilize uracil in

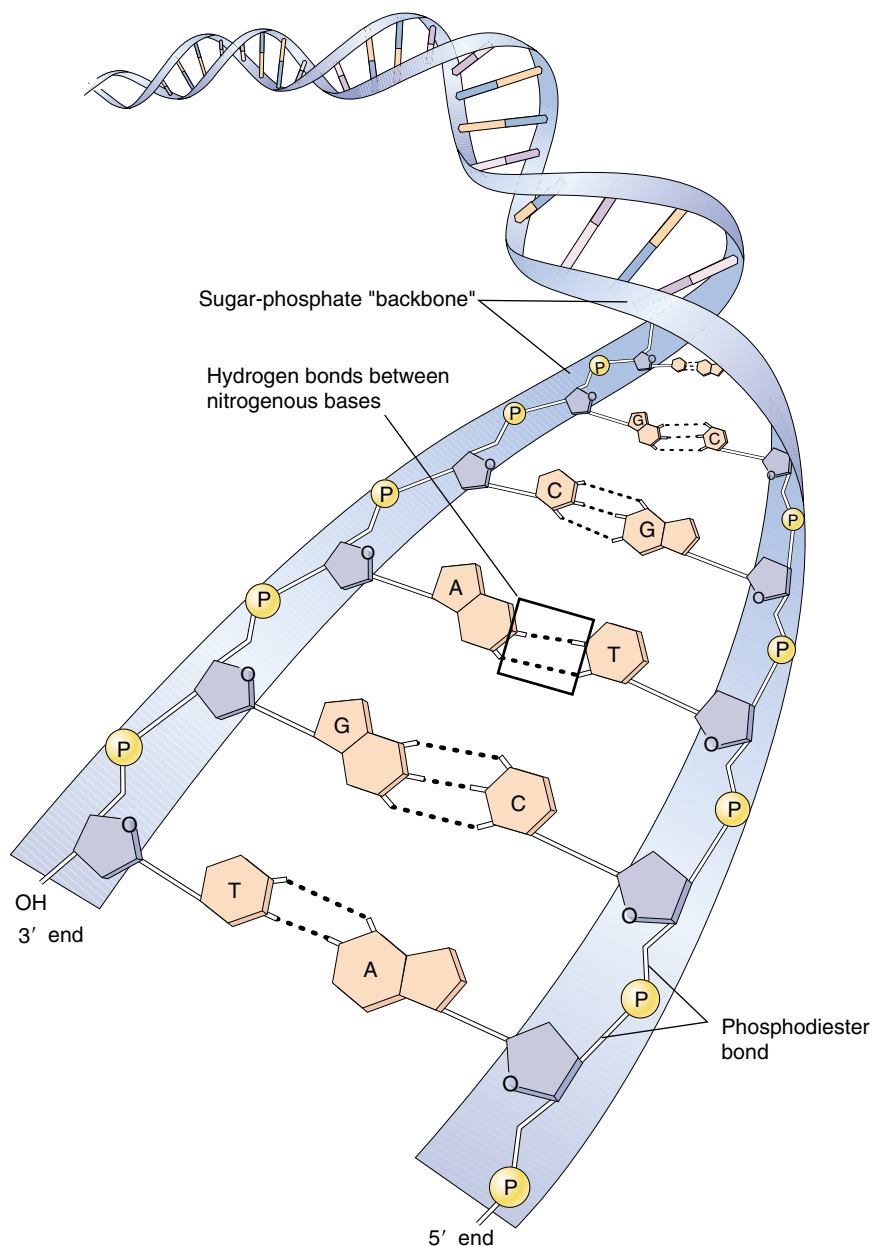


FIGURE 3.15

The structure of DNA. Hydrogen bond formation (dashed lines) between the organic bases, called base-pairing, causes the two chains of a DNA duplex to bind to each other and form a double helix.

place of thymine. Uracil has the same structure as thymine, except that one of its carbons lacks a methyl ($-\text{CH}_3$) group.

Transcribing the DNA message into a chemically different molecule such as RNA allows the cell to tell which is the original information storage molecule and which is the transcript. DNA molecules are always double-stranded (except for a few single-stranded DNA viruses that will be discussed in chapter 33), while the RNA molecules transcribed from DNA are typically single-stranded (figure

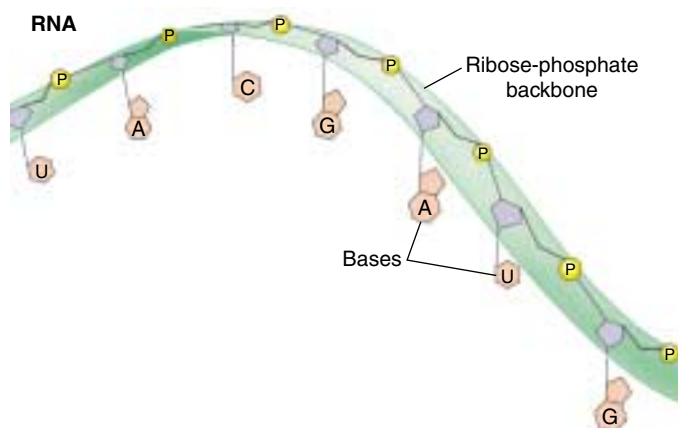
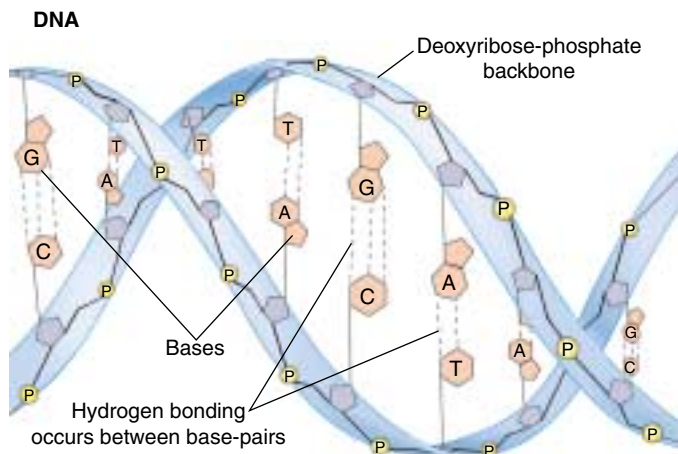


FIGURE 3.16
DNA versus RNA. DNA forms a double helix, uses deoxyribose as the sugar in its sugar-phosphate backbone, and utilizes thymine among its nitrogenous bases. RNA, on the other hand, is usually single-stranded, uses ribose as the sugar in its sugar-phosphate backbone, and utilizes uracil in place of thymine.

3.16). Although there is no chemical reason why RNA cannot form double helices as DNA does, cells do not possess the enzymes necessary to assemble double strands of RNA, as they do for DNA. Using two different molecules, one single-stranded and the other double-stranded, separates the role of DNA in storing hereditary information from the role of RNA in using this information to specify protein structure.

Which Came First, DNA or RNA?

The information necessary for the synthesis of proteins is stored in the cell's double-stranded DNA base sequences. The cell uses this information by first making an RNA transcript of it: RNA nucleotides pair with complementary

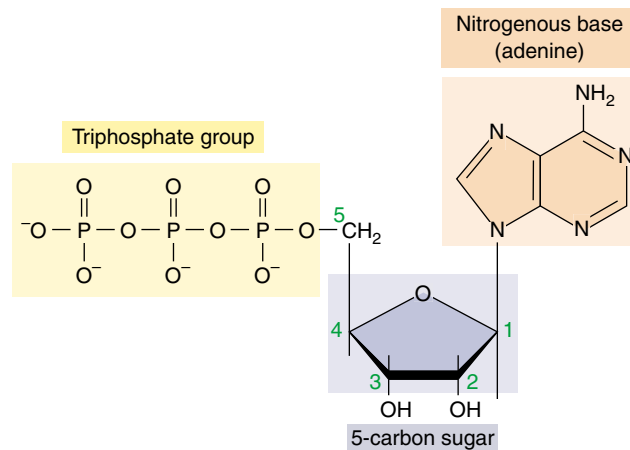


FIGURE 3.17
ATP. Adenosine triphosphate (ATP) contains adenine, a five-carbon sugar, and three phosphate groups. This molecule serves to transfer energy rather than store genetic information.

DNA nucleotides. By storing the information in DNA while using a complementary RNA sequence to actually direct protein synthesis, the cell does not expose the information-encoding DNA chain to the dangers of single-strand cleavage every time the information is used. Therefore, DNA is thought to have evolved from RNA as a means of preserving the genetic information, protecting it from the ongoing wear and tear associated with cellular activity. This genetic system has come down to us from the very beginnings of life.

The cell uses the single-stranded, short-lived RNA transcript to direct the synthesis of a protein with a specific sequence of amino acids. Thus, the information flows from DNA to RNA to protein, a process that has been termed the “central dogma” of molecular biology.

ATP

In addition to serving as subunits of DNA and RNA, nucleotide bases play other critical roles in the life of a cell. For example, adenine is a key component of the molecule *adenosine triphosphate* (ATP; figure 3.17), the energy currency of the cell. It also occurs in the molecules *nicotinamide adenine dinucleotide* (NAD⁺) and *flavin adenine dinucleotide* (FAD⁺), which carry electrons whose energy is used to make ATP.

A nucleic acid is a long chain of five-carbon sugars with an organic base protruding from each sugar. DNA is a double-stranded helix that stores hereditary information as a specific sequence of nucleotide bases. RNA is a single-stranded molecule that transcribes this information to direct protein synthesis.

3.4 Lipids make membranes and store energy.

Lipids are a loosely defined group of molecules with one main characteristic: they are insoluble in water. The most familiar lipids are fats and oils. Lipids have a very high proportion of nonpolar carbon-hydrogen (C—H) bonds, and so long-chain lipids cannot fold up like a protein to sequester their nonpolar portions away from the surrounding aqueous environment. Instead, when placed in water many lipid molecules will spontaneously cluster together and expose what polar groups they have to the surrounding water while sequestering the nonpolar parts of the molecules together within the cluster. This spontaneous assembly of lipids is of paramount importance to cells, as it underlies the structure of cellular membranes.

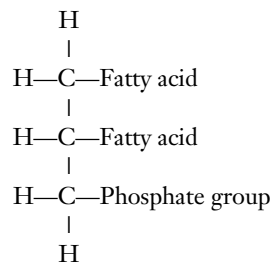
Phospholipids Form Membranes

Phospholipids are among the most important molecules of the cell, as they form the core of all biological membranes. An individual phospholipid is a composite molecule, made up of three kinds of subunits:

1. Glycerol, a three-carbon alcohol, with each carbon bearing a hydroxyl group. Glycerol forms the backbone of the phospholipid molecule.
2. *Fatty acids*, long chains of C—H bonds (hydrocarbon chains) ending in a carboxyl (—COOH) group. Two fatty acids are attached to the glycerol backbone in a phospholipid molecule.

3. Phosphate group, attached to one end of the glycerol. The charged phosphate group usually has a charged organic molecule linked to it, such as choline, ethanolamine, or the amino acid serine.

The phospholipid molecule can be thought of as having a polar “head” at one end (the phosphate group) and two long, very nonpolar “tails” at the other. In water, the nonpolar tails of nearby phospholipids aggregate away from the water, forming two layers of tails pointed toward each other—a lipid bilayer (figure 3.18). Lipid bilayers are the basic framework of biological membranes, discussed in detail in chapter 6.



Because the C—H bonds in lipids are very nonpolar, they are not water-soluble, and aggregate together in water. This kind of aggregation by phospholipids forms biological membranes.

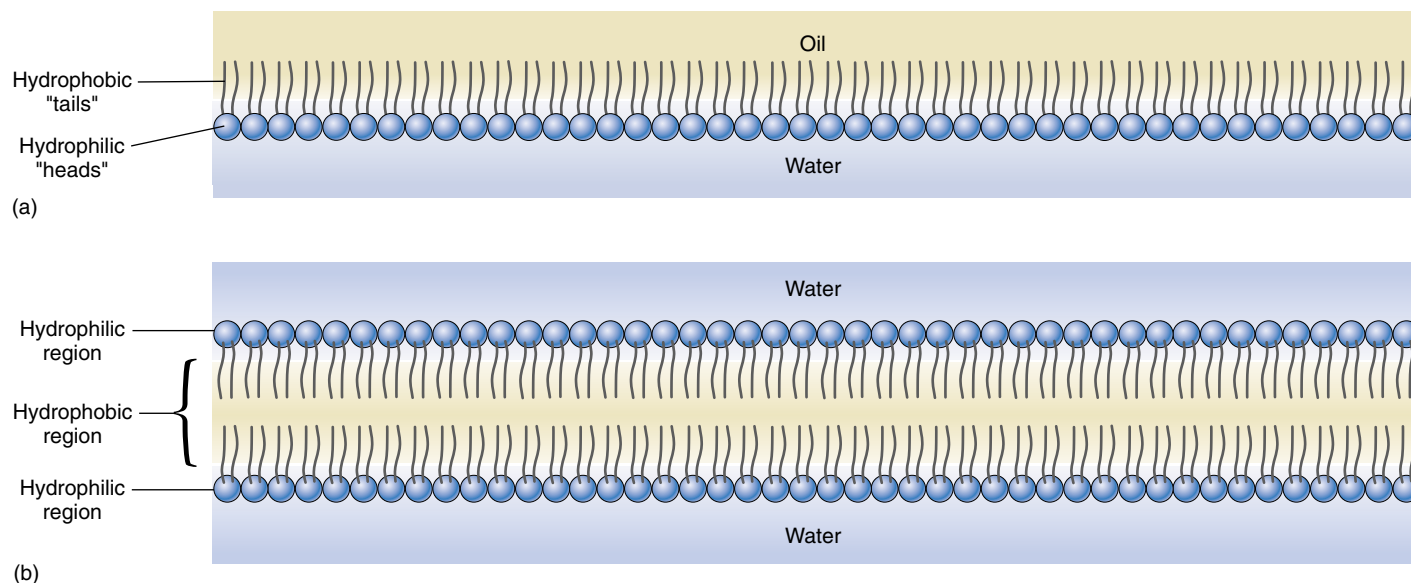


FIGURE 3.18

Phospholipids. (a) At an oil-water interface, phospholipid molecules will orient so that their polar (hydrophilic) heads are in the polar medium, water, and their nonpolar (hydrophobic) tails are in the nonpolar medium, oil. (b) When surrounded by water, phospholipid molecules arrange themselves into two layers with their heads extending outward and their tails inward.

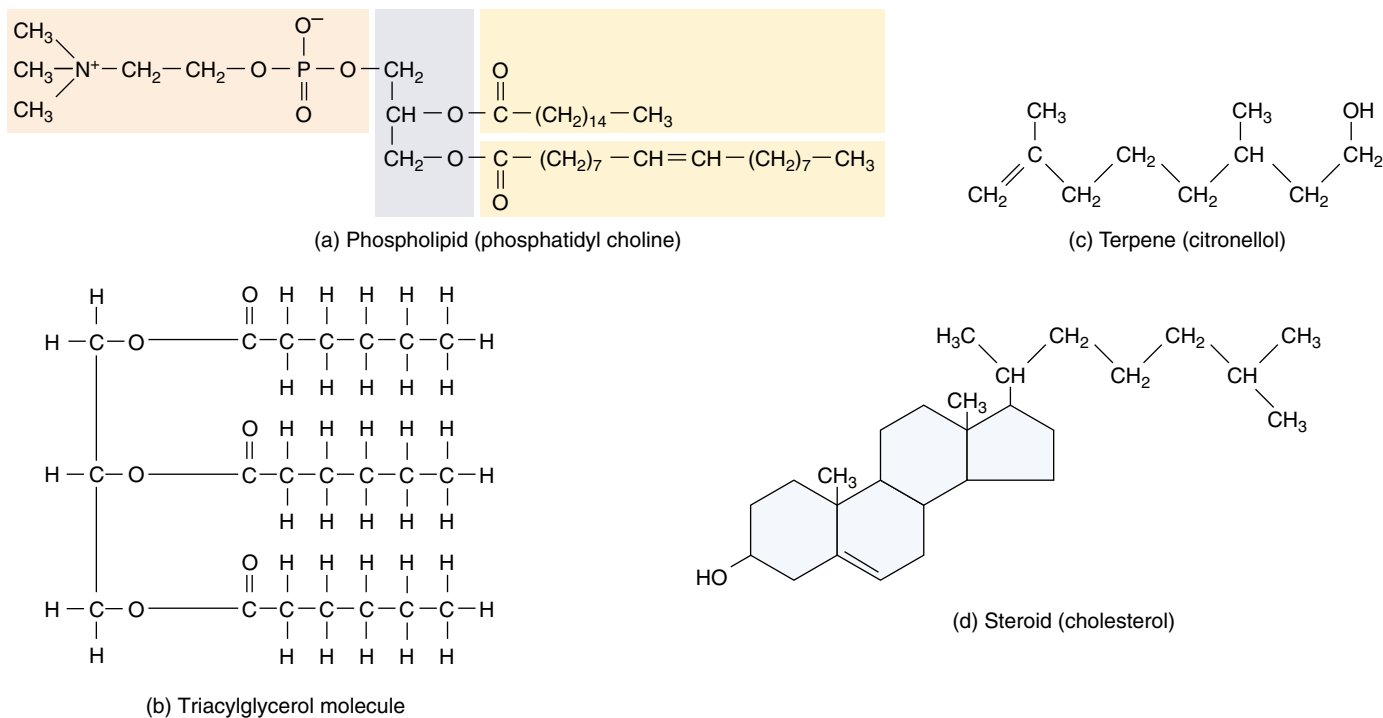


FIGURE 3.19

Lipids. These structures represent four major classes of biologically important lipids: (a) phospholipids, (b) triacylglycerols (triglycerides), (c) terpenes, and (d) steroids.

Fats and Other Kinds of Lipids

Fats are another kind of lipid, but unlike phospholipids, fat molecules do not have a polar end. **Fats** consist of a glycerol molecule to which is attached three fatty acids, one to each carbon of the glycerol backbone. Because it contains three fatty acids, a fat molecule is called a **triglyceride**, or, more properly, a triacylglycerol (figure 3.19). The three fatty acids of a triglyceride need not be identical, and often they differ markedly from one another. Organisms store the energy of certain molecules for long periods in the many C—H bonds of fats.

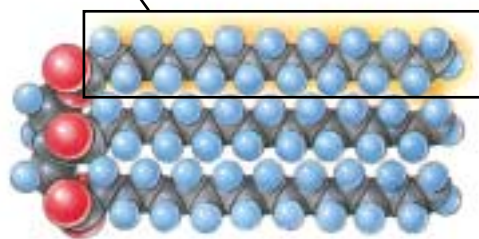
Because triglyceride molecules lack a polar end, they are not soluble in water. Placed in water, they spontaneously clump together, forming fat globules that are very large relative to the size of the individual molecules. Because fats are insoluble, they can be deposited at specific locations within an organism.

Storage fats are one kind of lipid. Oils such as olive oil, corn oil, and coconut oil are also lipids, as are waxes such as beeswax and earwax (see table 3.1). The hydrocarbon chains of fatty acids vary in length; the most common are even-numbered chains of 14 to 20 carbons. If all of the internal carbon atoms in the fatty acid chains are bonded to at least two hydrogen atoms, the fatty acid is said to be **saturated**, because it contains the maximum possible number of hydrogen atoms (figure 3.20). If a fatty acid has double bonds be-

tween one or more pairs of successive carbon atoms, the fatty acid is said to be **unsaturated**. If a given fatty acid has more than one double bond, it is said to be **polyunsaturated**. Fats made from polyunsaturated fatty acids have low melting points because their fatty acid chains bend at the double bonds, preventing the fat molecules from aligning closely with one another. Consequently, a polyunsaturated fat such as corn oil is usually liquid at room temperature and is called an oil. In contrast, most saturated fats such as those in butter are solid at room temperature.

Organisms contain many other kinds of lipids besides fats (see figure 3.19). *Terpenes* are long-chain lipids that are components of many biologically important pigments, such as chlorophyll and the visual pigment retinal. Rubber is also a terpene. *Steroids*, another type of lipid found in membranes, are composed of four carbon rings. Most animal cell membranes contain the steroid cholesterol. Other steroids, such as testosterone and estrogen, function in multicellular organisms as hormones. *Prostaglandins* are a group of about 20 lipids that are modified fatty acids, with two nonpolar “tails” attached to a five-carbon ring. Prostaglandins act as local chemical messengers in many vertebrate tissues.

Cells contain many kinds of molecules in addition to membrane phospholipids that are not soluble in water.



(b) Unsaturated fat

(b) Unsaturated fat

52 **Part I** The Origin of Living Things

3.5 Carbohydrates store energy and provide building materials.

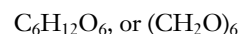
Simple Carbohydrates

Carbohydrates function as energy-storage molecules as well as structural elements. Some are small, simple molecules, while others form long polymers.

Sugars Are Simple Carbohydrates

The **carbohydrates** are a loosely defined group of molecules that contain carbon, hydrogen, and oxygen in the molar ratio 1:2:1. Their empirical formula (which lists the atoms in the molecule with subscripts to indicate how many there are of each) is $(\text{CH}_2\text{O})_n$, where n is the number of carbon atoms. Because they contain many carbon-hydrogen (C—H) bonds, which release energy when they are broken, carbohydrates are well suited for energy storage.

Monosaccharides. The simplest of the carbohydrates are the simple sugars, or **monosaccharides** (Greek *mono*, “single” + Latin *saccharum*, “sugar”). Simple sugars may contain as few as three carbon atoms, but those that play the central role in energy storage have six (figure 3.21). The empirical formula of six-carbon sugars is:



Six-carbon sugars can exist in a straight-chain form, but in an aqueous environment they almost always form rings. The most important of these for energy storage is *glucose* (figure 3.22), a six-carbon sugar which has seven energy-storing C—H bonds.

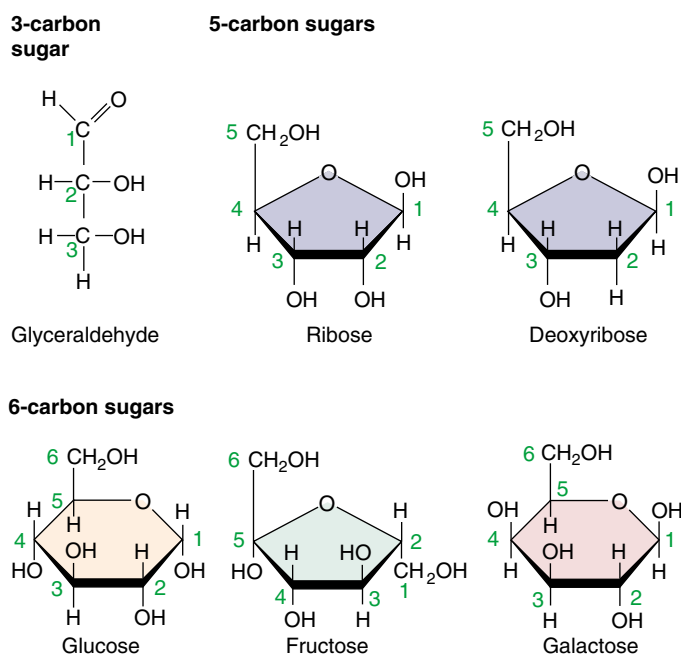


FIGURE 3.21

Monosaccharides. Monosaccharides, or simple sugars, can contain as few as three carbon atoms and are often used as building blocks to form larger molecules. The five-carbon sugars ribose and deoxyribose are components of nucleic acids (see figure 3.15). The six-carbon sugar glucose is a component of large energy-storage molecules.

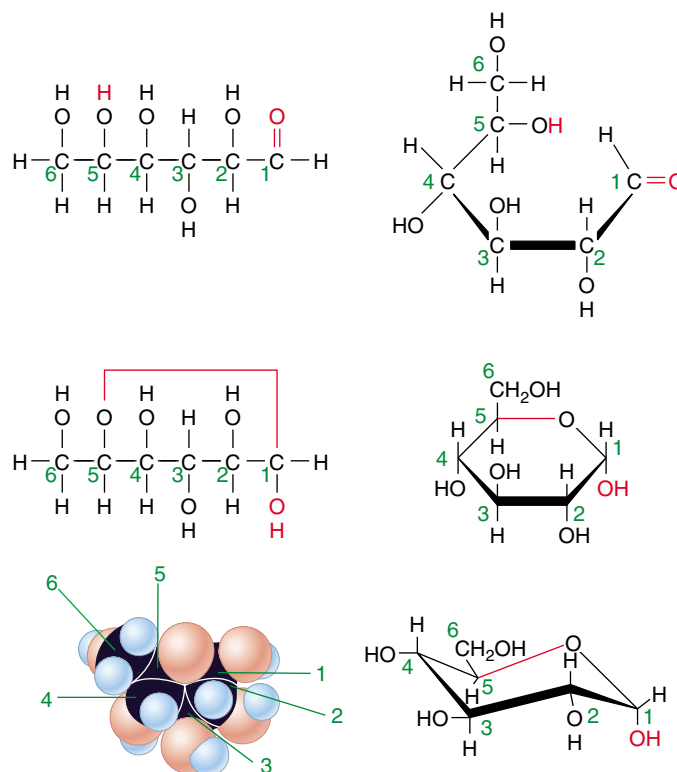


FIGURE 3.22

Structure of the glucose molecule. Glucose is a linear six-carbon molecule that forms a ring shape in solution. The structure of the ring can be represented in many ways; the ones shown here are the most common, with the carbons conventionally numbered (in green) so that the forms can be compared easily. The bold, darker lines represent portions of the molecule that are projecting out of the page toward you—remember, these are three-dimensional molecules!

Disaccharides. Many familiar sugars like sucrose are “double sugars,” two monosaccharides joined by a covalent bond (figure 3.23). Called **disaccharides**, they often play a role in the transport of sugars, as we will discuss shortly.

Polysaccharides. **Polysaccharides** are macromolecules made up of monosaccharide subunits. Starch is a polysaccharide used by plants to store energy. It consists entirely of glucose molecules, linked one after another in long chains. Cellulose is a polysaccharide that serves as a structural building material in plants. It too consists entirely of glucose molecules linked together into chains, and special enzymes are required to break the links.

Sugar Isomers

Glucose is not the only sugar with the formula $C_6H_{12}O_6$. Other common six-carbon sugars such as fructose and galactose also have this same empirical formula (figure 3.24). These sugars are **isomers**, or alternative forms, of glucose. Even though isomers have the same empirical formula, their atoms are arranged in different ways; that is, their three-dimensional structures are different. These structural differences often account for substantial functional differences between the isomers. Glucose and fructose, for example, are *structural isomers*. In fructose, the double-bonded oxygen is attached to an internal carbon rather than to a terminal one. Your taste buds can tell the difference, as fructose tastes much sweeter than glucose, despite the fact that both sugars have the same chemical composition. This structural difference also has an important chemical consequence: the two sugars form different polymers.

Unlike fructose, galactose has the same bond structure as glucose; the only difference between galactose and glucose is the orientation of one hydroxyl group. Because the hydroxyl group positions are mirror images of each other,

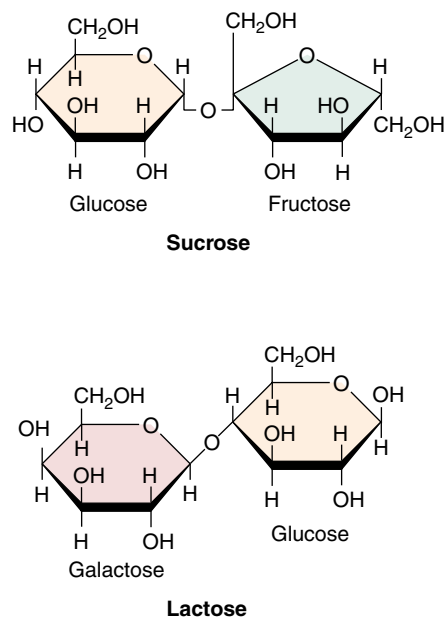


FIGURE 3.23

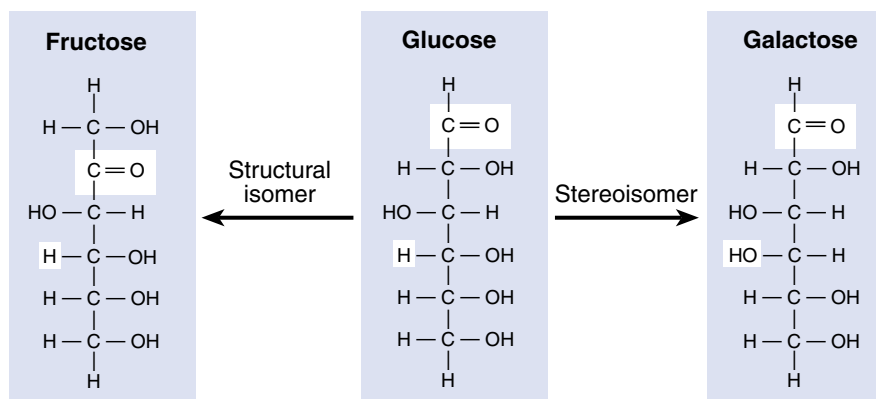
Disaccharides. Sugars like sucrose and lactose are disaccharides, composed of two monosaccharides linked by a covalent bond.

galactose and glucose are called *stereoisomers*. Again, this seemingly slight difference has important consequences, as this hydroxyl group is often involved in creating polymers with distinct functions, such as starch (energy storage) and cellulose (structural support).

Sugars are among the most important energy-storage molecules in organisms, containing many energy-storing C—H bonds. The structural differences among sugar isomers can confer substantial functional differences upon the molecules.

FIGURE 3.24

Isomers and stereoisomers. Glucose, fructose, and galactose are isomers with the empirical formula $C_6H_{12}O_6$. A structural isomer of glucose, such as fructose, has identical chemical groups bonded to different carbon atoms, while a stereoisomer of glucose, such as galactose, has identical chemical groups bonded to the same carbon atoms but in different orientations.



Linking Sugars Together

Transport Disaccharides

Most organisms transport sugars within their bodies. In humans, the glucose that circulates in the blood does so as a simple monosaccharide. In plants and many other organisms, however, glucose is converted into a transport form before it is moved from place to place within the organism. In such a form it is less readily metabolized (used for energy) during transport. Transport forms of sugars are commonly made by linking two monosaccharides together to form a disaccharide (Greek *di*, “two”). Disaccharides serve as effective reservoirs of glucose because the normal glucose-utilizing enzymes of the organism cannot break the bond linking the two monosaccharide subunits. Enzymes that can do so are typically present only in the tissue where the glucose is to be used.

Transport forms differ depending on which monosaccharides link to form the disaccharide. Glucose forms transport disaccharides with itself and many other monosaccharides, including fructose and galactose. When glucose forms a disaccharide with its structural isomer, fructose, the resulting disaccharide is *sucrose*, or table sugar (figure 3.25*a*). Sucrose is the form in which most plants transport glucose and the sugar that most humans (and other animals) eat. Sugarcane is rich in sucrose, and so are sugar beets.

When glucose is linked to its stereoisomer, galactose, the resulting disaccharide is *lactose*, or milk sugar. Many mammals supply energy to their young in the form of lactose. Adults have greatly reduced levels of lactase, the enzyme required to cleave lactose into its two monosaccharide components, and thus cannot metabolize lactose as efficiently. Most of the energy that is channeled into lactose production is therefore reserved for their offspring.

Storage Polysaccharides

Organisms store the metabolic energy contained in monosaccharides by converting them into disaccharides, such as *maltose* (figure 3.25*b*), which are then linked together into insoluble forms that are deposited in specific storage areas in their bodies. These insoluble polysaccharides are long polymers of monosaccharides formed by dehydration synthesis. Plant polysaccharides formed from glucose are called **starches**. Plants store starch as granules within chloroplasts and other organelles. Because glucose is a key metabolic fuel, the stored starch provides a reservoir of energy available for future needs. Energy for cellular work can be retrieved by hydrolyzing the links that bind the glucose subunits together.

The starch with the simplest structure is *amylose*, which is composed of many hundreds of glucose molecules linked together in long, unbranched chains. Each linkage occurs between the number 1 carbon of one glucose molecule and the number 4 carbon of another, so that amylose is, in effect, a longer form of maltose. The long chains of amylose tend to coil up in water (figure 3.26*a*), a property that renders amylose insoluble. Potato starch is about 20% amylose. When amylose is digested by a sprouting potato plant (or by an animal that eats a potato), enzymes first break it into fragments of random length, which are more soluble because they are shorter. Baking or boiling potatoes has the same effect, breaking the chains into fragments. Another enzyme then cuts these fragments into molecules of maltose. Finally, the maltose is cleaved into two glucose molecules, which cells are able to metabolize.

Most plant starch, including the remaining 80% of potato starch, is a somewhat more complicated variant of amylose called *amylopectin* (figure 3.26*b*). Pectins are branched polysaccharides. Amylopectin has short, linear amylose branches consisting of 20 to 30 glucose subunits.

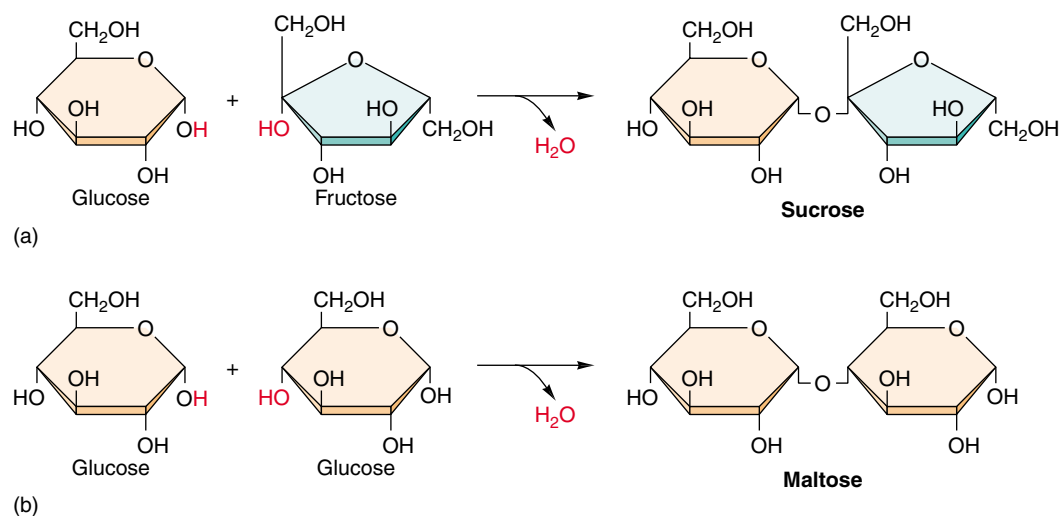


FIGURE 3.25
How disaccharides form.

Some disaccharides are used to transport glucose from one part of an organism's body to another; one example is sucrose (*a*), which is found in sugarcane. Other disaccharides, such as maltose in grain (*b*), are used for storage.

In some plants these chains are cross-linked. The cross-links create an insoluble mesh of glucose, which can be degraded only by another kind of enzyme. The size of the mesh differs from plant to plant; in rice about 100 amylose chains, each with one or two cross-links, forms the mesh.

The animal version of starch is glycogen. Like amylopectin, *glycogen* is an insoluble polysaccharide containing branched amylose chains. In glycogen, the average chain length is much greater and there are more branches than in plant starch (figure 3.26c). Humans and other vertebrates store excess food energy as glycogen in the liver and in muscle cells; when the demand for energy in a tissue increases, glycogen is hydrolyzed to release glucose.

Nonfattening Sweets

Imagine a kind of table sugar that looks, tastes, and cooks like the real thing, but has no calories or harmful side effects. You could eat mountains of candy made from such sweeteners without gaining weight. As Louis Pasteur discovered in the late 1800s, most sugars are “right-handed” molecules, in that the hydroxyl group that binds a critical carbon atom is on the right side. However, “left-handed” sugars, in which the hydroxyl group is on the left side, can be made readily in the laboratory. These synthetic sugars are mirror-image chemical twins of the natural form, but the enzymes that break down sugars in the human digestive system can tell the difference. To digest a sugar molecule, an enzyme must first grasp it, much like a shoe fitting onto a foot, and all of the body’s enzymes are right-handed! A left-handed sugar doesn’t fit, any more than a shoe for the right foot fits onto a left foot.

The Latin word for “left” is *levo*, and left-handed sugars are called **levo-**, or **l-sugars**. They do not occur in nature except for trace amounts in red algae, snail eggs, and seaweed. Because they pass through the body without being used, they can let diet-conscious sweet-lovers have their cake and eat it, too. Nor will they contribute to tooth decay because bacteria cannot metabolize them, either.

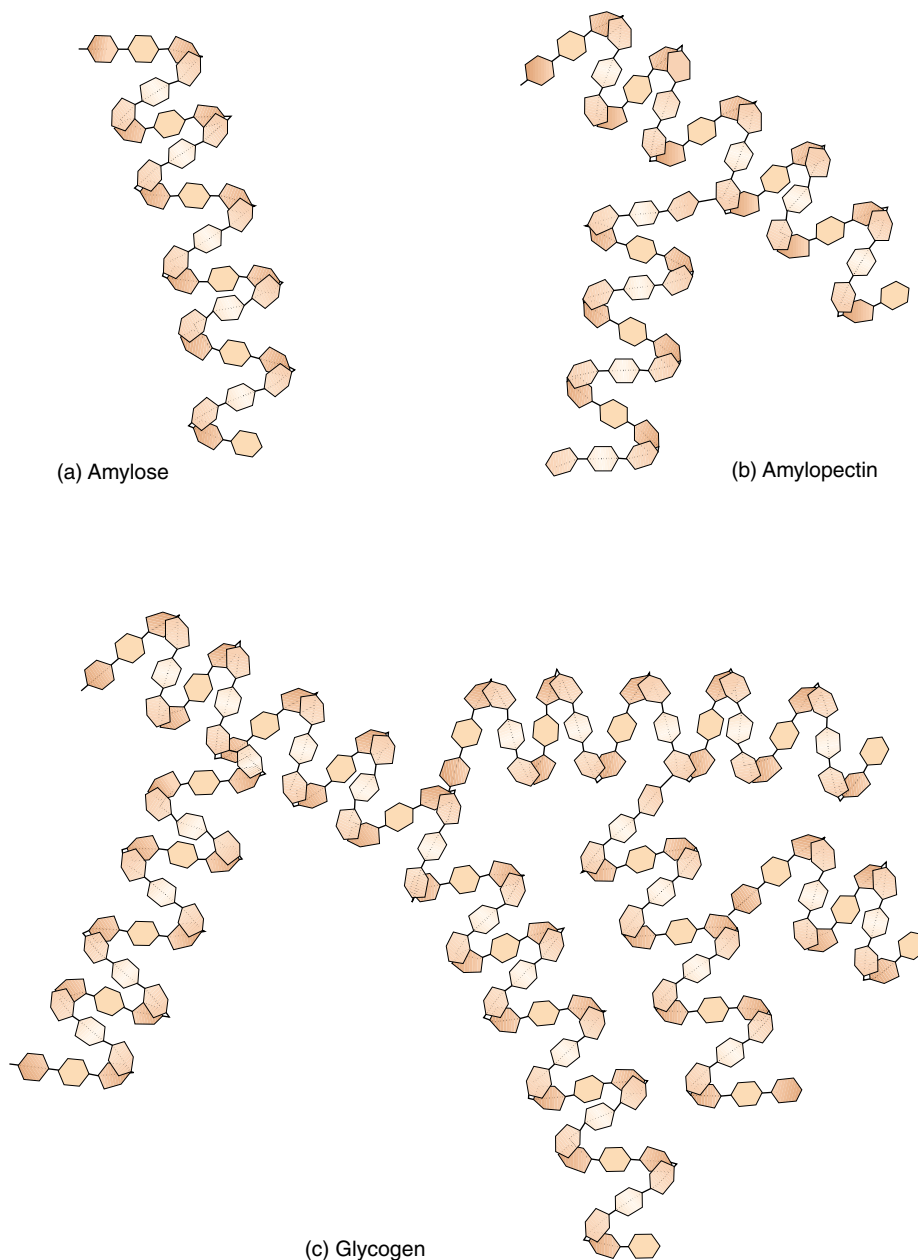


FIGURE 3.26

Storage polysaccharides. Starches are long glucose polymers that store energy in plants. (a) The simplest starches are long chains of maltose called amylose, which tend to coil up in water. (b) Most plants contain more complex starches called amylopectins, which are branched. (c) Animals store glucose in glycogen, which is more extensively branched than amylopectin and contains longer chains of amylose.

Starches are glucose polymers. Most starches are branched and some are cross-linked. The branching and cross-linking render the polymer insoluble and protect it from degradation.

Structural Carbohydrates

While some chains of sugars store energy, others serve as structural material for cells.

Cellulose

For two glucose molecules to link together, the glucose subunits must be the same form. Glucose can form a ring in two ways, with the hydroxyl group attached to the carbon where the ring closes being locked into place either below or above the plane of the ring. If below, it is called the **alpha form**, and if above, the **beta form**. All of the glucose subunits of the starch chain are alpha-glucose. When a chain of glucose molecules consists of all beta-glucose subunits, a polysaccharide with very different properties results. This **structural polysaccharide** is *cellulose*, the chief component of plant cell walls (figure 3.27). Cellulose is chemically similar to amylose, with one important difference: the starch-degrading enzymes that occur in most organisms cannot break the bond between two beta-glucose sugars. This is not because the bond is stronger, but rather because its cleavage requires an enzyme most organisms lack. Because cellulose cannot be broken down readily, it works well as a biological structural material and occurs widely in this role in plants. Those few animals able to break down cellulose find it a rich source of energy. Certain vertebrates, such as cows, can digest cellulose by means of bacteria and protists they harbor in their intestines which provide the necessary enzymes.

Chitin

The structural building material in insects, many fungi, and certain other organisms is called chitin (figure 3.28). *Chitin* is a modified form of cellulose with a nitrogen group added to the glucose units. When cross-linked by proteins, it forms a tough, resistant surface material that serves as the hard exoskeleton of arthropods such as insects and crustaceans (see chapter 46). Few organisms are able to digest chitin.

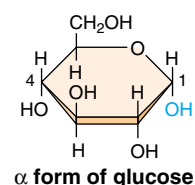
Structural carbohydrates are chains of sugars that are not easily digested. They include cellulose in plants and chitin in arthropods and fungi.



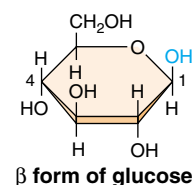
FIGURE 3.27

A journey into wood. This jumble of cellulose fibers (a) is from a yellow pine (*Pinus ponderosa*) (20 $\times$). (b) While starch chains consist of alpha-glucose subunits, (c) cellulose chains consist of beta-glucose subunits. Cellulose fibers can be very strong and are quite resistant to metabolic breakdown, which is one reason why wood is such a good building material.

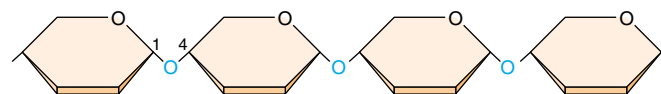
(a)



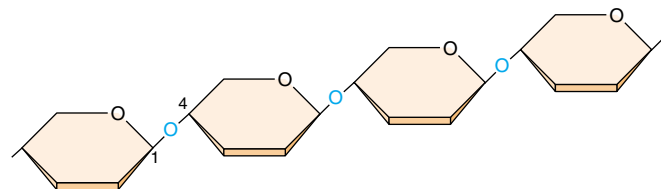
(b)



(c)



Starch: chain of α -glucose subunits



Cellulose: chain of β -glucose subunits



FIGURE 3.28

Chitin. Chitin, which might be considered to be a modified form of cellulose, is the principal structural element in the external skeletons of many invertebrates, such as this lobster.

Chapter 3



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Summary

Questions

Media Resources

3.1 Molecules are the building blocks of life.

- The chemistry of living systems is the chemistry of carbon-containing compounds.
- Carbon's unique chemical properties allow it to polymerize into chains by dehydration synthesis, forming the four key biological macromolecules: carbohydrates, lipids, proteins, and nucleic acids.

1. What types of molecules are formed by dehydration reactions? What types of molecules are formed by hydrolysis?



- Organic Chemistry

3.2 Proteins perform the chemistry of the cell.

- Proteins are polymers of amino acids.
- Because the 20 amino acids that occur in proteins have side groups with different chemical properties, the function and shape of a protein are critically affected by its particular sequence of amino acids.

2. How are amino acids linked to form proteins?

3. Explain what is meant by the primary, secondary, tertiary, and quaternary structure of a protein.



- Explorations: How Proteins Function



- Proteins



- Student Research: A new Protein in Inserts

3.3 Nucleic acids store and transfer genetic information.

- Hereditary information is stored as a sequence of nucleotides in a linear polymer called DNA, which exists in cells as a double helix.
- Cells use the information in DNA by producing a complementary single strand of RNA which directs the synthesis of a protein whose amino acid sequence corresponds to the nucleotide sequence of the DNA from which the RNA was transcribed.

4. What are the three components of a nucleotide? How are nucleotides linked to form nucleic acids?

5. Which of the purines and pyrimidines are capable of forming base-pairs with each other?



- Nucleic Acids

3.4 Lipids make membranes and store energy.

- Fats are one type of water-insoluble molecules called lipids.
- Fats are molecules that contain many energy-rich C—H bonds and, thus, provide an efficient form of long-term energy storage.
- Types of lipids include phospholipids, fats, terpenes, steroids, and prostaglandins.

6. What are the two kinds of subunits that make up a fat molecule, and how are they arranged in the molecule?

7. Describe the differences between a saturated and an unsaturated fat.



- Lipids



- Experiment: Anfinsen: Amino Acid Sequence Determines Protein Shape

3.5 Carbohydrates store energy and provide building materials.

- Carbohydrates store considerable energy in their carbon-hydrogen (C—H) bonds.
- The most metabolically important carbohydrate is glucose, a six-carbon sugar.
- Excess energy resources may be stored in complex sugar polymers called starches (in plants) and glycogen (in animals and fungi).

8. What does it mean to say that glucose, fructose, and galactose are isomers? Which two are structural isomers, and how do they differ from each other? Which two are stereoisomers, and how do they differ from each other?



- Carbohydrates

4

The Origin and Early History of Life

Concept Outline

4.1 All living things share key characteristics.

What Is Life? All known organisms share certain general properties, and to a large degree these properties define what we mean by life.

4.2 There are many ideas about the origin of life.

Theories about the Origin of Life. There are both religious and scientific views about the origin of life. This text treats only the latter—only the scientifically testable.

Scientists Disagree about Where Life Started. The atmosphere of the early earth was rich in hydrogen, providing a ready supply of energetic electrons with which to build organic molecules.

The Miller-Urey Experiment. Experiments attempting to duplicate the conditions of early earth produce many of the key molecules of living organisms.

4.3 The first cells had little internal structure.

Theories about the Origin of Cells. The first cells are thought to have arisen spontaneously, but there is little agreement as to the mechanism.

The Earliest Cells. The earliest fossils are of bacteria too small to see with the unaided eye.

4.4 The first eukaryotic cells were larger and more complex than bacteria.

The First Eukaryotic Cells. Fossils of the first eukaryotic cells do not appear in rocks until 1.5 billion years ago, over 2 billion years after bacteria. Multicellular life is restricted to the four eukaryotic kingdoms of life.

Has Life Evolved Elsewhere? It seems probable that life has evolved on other worlds besides our own. The possible presence of life in the warm waters beneath the surface of Europa, a moon of Jupiter, is a source of current speculation.



FIGURE 4.1

The origin of life. The fortuitous mix of physical events and chemical elements at the right place and time created the first living cells on earth.

There are a great many scientists with intriguing ideas that explain how life may have originated on earth, but there is very little that we know for sure. New hypotheses are being proposed constantly, and old ones reevaluated. By the time this text is published, some of the ideas presented here about the origin of life will surely be obsolete. Thus, the contesting ideas are presented in this chapter in an open-ended format, attempting to make clear that there is as yet no one answer to the question of how life originated on earth. Although recent photographs taken by the Hubble Space Telescope have revived controversy about the age of the universe, it seems clear the earth itself was formed about 4.6 billion years ago. The oldest clear evidence of life—microfossils in ancient rock—are 3.5 billion years old. The origin of life seems to have taken just the right combination of physical events and chemical processes (figure 4.1).

4.1 All living things share key characteristics.

The earth formed as a hot mass of molten rock about 4.6 billion years ago. As the earth cooled, much of the water vapor present in its atmosphere condensed into liquid water, which accumulated on the surface in chemically rich oceans. One scenario for the origin of life is that it originated in this dilute, hot smelly soup of ammonia, formaldehyde, formic acid, cyanide, methane, hydrogen sulfide, and organic hydrocarbons. Whether at the oceans' edge, in hydrothermal deep-sea vents, or elsewhere, the consensus among researchers is that life arose spontaneously from these early waters less than 4 billion years ago. While the way in which this happened remains a puzzle, one cannot escape a certain curiosity about the earliest steps that eventually led to the origin of all living things on earth, including ourselves. How did organisms evolve from the complex molecules that swirled in the early oceans?

What Is Life?

Before we can address this question, we must first consider what qualifies something as "living." What *is* life? This is a difficult question to answer, largely because life itself is not a simple concept. If you try to write a definition of "life," you will find that it is not an easy task, because of the loose manner in which the term is used.

Imagine a situation in which two astronauts encounter a large, amorphous blob on the surface of a planet. How would they determine whether it is alive?

Movement. One of the first things the astronauts might do is observe the blob to see if it moves. Most animals move about (figure 4.2), but movement from one place to another in itself is not diagnostic of life. Most plants and even some animals do not move about, while numerous nonliving objects, such as clouds, do move. The criterion of movement is thus neither *necessary* (possessed by all life) nor *sufficient* (possessed only by life).

Sensitivity. The astronauts might prod the blob to see if it responds. Almost all living things respond to stimuli (figure 4.3). Plants grow toward light, and animals retreat from fire. Not all stimuli produce responses, however. Imagine kicking a redwood tree or singing to a hibernating bear. This criterion, although superior to the first, is still inadequate to define life.

Death. The astronauts might attempt to kill the blob. All living things die, while inanimate objects do not. Death is not easily distinguished from disorder, however; a car that breaks down has not died because it was never alive. Death is simply the loss of life, so this is a circular definition at best. Unless one can detect life, death is a meaningless concept, and hence a very inadequate criterion for defining life.

Complexity. Finally, the astronauts might cut up the blob, to see if it is complexly organized. All living things are complex. Even the simplest bacteria



FIGURE 4.2

Movement. Animals have evolved mechanisms that allow them to move about in their environment. While some animals, like this giraffe, move on land, others move through water or air.



FIGURE 4.3

Sensitivity. This father lion is responding to a stimulus: he has just been bitten on the rump by his cub. As far as we know, all organisms respond to stimuli, although not always to the same ones or in the same way. Had the cub bitten a tree instead of its father, the response would not have been as dramatic.

contain a bewildering array of molecules, organized into many complex structures. However a computer is also complex, but not alive. Complexity is a *necessary* criterion of life, but it is not *sufficient* in itself to identify living things because many complex things are not alive.

To determine whether the blob is alive, the astronauts would have to learn more about it. Probably the best thing they could do would be to examine it more carefully and determine whether it resembles the organisms we are familiar with, and if so, how.

Fundamental Properties of Life

As we discussed in chapter 1, all known organisms share certain general properties. To a large degree, these properties define what we mean by life. The following fundamental properties are shared by all organisms on earth.

Cellular organization. All organisms consist of one or more *cells*—complex, organized assemblages of molecules enclosed within membranes (figure 4.4).

Sensitivity. All organisms respond to stimuli—though not always to the same stimuli in the same ways.

Growth. All living things assimilate energy and use it to grow, a process called *metabolism*. Plants, algae, and some bacteria use sunlight to create covalent carbon-carbon bonds from CO₂ and H₂O through photosynthesis. This transfer of the energy in covalent bonds is essential to all life on earth.

Development. Multicellular organisms undergo systematic gene-directed changes as they grow and mature.

Reproduction. All living things reproduce, passing on traits from one generation to the next. Although some organisms live for a very long time, no organism lives forever, as far as we know. Because all organisms die, ongoing life is impossible without *reproduction*.

Regulation. All organisms have regulatory mechanisms that coordinate internal processes.

Homeostasis. All living things maintain relatively constant internal conditions, different from their environment.

The Key Role of Heredity

Are these properties adequate to define life? Is a membrane-enclosed entity that grows and reproduces alive? Not necessarily. Soap bubbles and proteinoid microspheres spontaneously form hollow bubbles that enclose a small volume of water. These spheres can enclose energy-processing molecules, and they may also grow and subdivide. Despite these features, they are certainly not alive. Therefore, the criteria just listed,

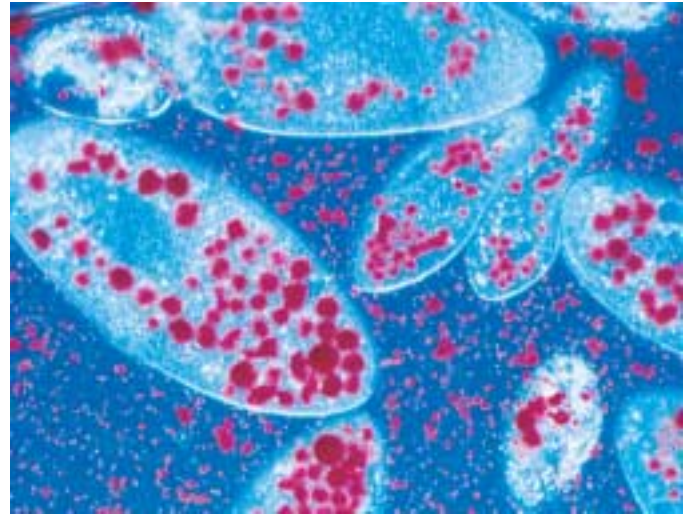


FIGURE 4.4

Cellular organization (150 $\times$). These *Paramecia*, which are complex, single-celled organisms called protists, have just ingested several yeast cells. The yeasts, stained *red* in this photograph, are enclosed within membrane-bounded sacs called digestive vacuoles. A variety of other organelles are also visible.

although necessary for life, are not sufficient to define life. One ingredient is missing—a mechanism for the preservation of improvement.

Heredity. All organisms on earth possess a *genetic system* that is based on the replication of a long, complex molecule called DNA. This mechanism allows for adaptation and evolution over time, also distinguishing characteristics of living things.

To understand the role of heredity in our definition of life, let us return for a moment to proteinoid microspheres. When we examine an individual microsphere, we see it at that precise moment in time but learn nothing of its predecessors. It is likewise impossible to guess what future droplets will be like. The droplets are the passive prisoners of a changing environment, and it is in this sense that they are not alive. The essence of being alive is the ability to encompass change and to reproduce the results of change permanently. Heredity, therefore, provides the basis for the great division between the living and the nonliving. Change does not become evolution unless it is passed on to a new generation. A genetic system is the sufficient condition of life. Some changes are preserved because they increase the chances of survival in a hostile world, while others are lost. Not only did life evolve—evolution is the very essence of life.

All living things on earth are characterized by cellular organization, heredity, and a handful of other characteristics that serve to define the term life.

4.2 There are many ideas about the origin of life.

Theories about the Origin of Life

The question of how life originated is not easy to answer because it is impossible to go back in time and observe life's beginnings; nor are there any witnesses. There is testimony in the rocks of the earth, but it is not easily read, and often it is silent on issues crying out for answers. There are, in principle, at least three possibilities:

1. **Special creation.** Life-forms may have been put on earth by supernatural or divine forces.
2. **Extraterrestrial origin.** Life may not have originated on earth at all; instead, life may have infected earth from some other planet.
3. **Spontaneous origin.** Life may have evolved from inanimate matter, as associations among molecules became more and more complex.

Special Creation. The theory of special creation, that a divine God created life is at the core of most major religions. The oldest hypothesis about life's origins, it is also the most widely accepted. Far more Americans, for example, believe that God created life on earth than believe in the other two hypotheses. Many take a more extreme position, accepting the biblical account of life's creation as factually correct. This viewpoint forms the basis for the very unscientific "scientific creationism" viewpoint discussed in chapter 21.

Extraterrestrial Origin. The theory of **panspermia** proposes that meteors or cosmic dust may have carried significant amounts of complex organic molecules to earth, kicking off the evolution of life. Hundreds of thousands of meteorites and comets are known to have slammed into the early earth, and recent findings suggest that at least some may have carried organic materials. Nor is life on other planets ruled out. For example, the discovery of liquid water under the surface of Jupiter's ice-shrouded moon Europa and suggestions of fossils in



FIGURE 4.5
Lightning. Before life evolved, the simple molecules in the earth's atmosphere combined to form more complex molecules. The energy that drove these chemical reactions may have come from lightning and forms of geothermal energy.

rocks from Mars lend some credence to this idea. The hypothesis that an early source of carbonaceous material is extraterrestrial is testable, although it has not yet been proven. Indeed, NASA is planning to land on Europa, drill through the surface, and send a probe down to see if there is life.

Spontaneous Origin. Most scientists tentatively accept the theory of spontaneous origin, that life evolved from inanimate matter. In this view, the force leading to life was selection. As changes in molecules increased their stability and caused them to persist longer, these molecules could initiate more and more complex associations, culminating in the evolution of cells.

Taking a Scientific Viewpoint

In this book we will focus on the second and third possibilities, attempting to understand whether natural forces could have led to the origin of life and, if so, how the process might have occurred. This is not to say that the first possibility is definitely not the correct one. Any one of the three possibilities might be true. Nor do the second and third possibilities preclude religion (a divine agency might have acted via evolution, for example). However, we are limiting the scope of our inquiry to scientific matters, and only the second and third possibilities permit testable hypotheses to be constructed—that is, explanations that can be tested and potentially disproved.

In our search for understanding, we must look back to the early times. There are fossils of simple living things, bacteria, in rocks 3.5 billion years old. They tell us that life originated during the first billion years of the history of our planet. As we attempt to determine how this process took place, we will first focus on how organic molecules may have originated (figure 4.5), and then we will consider how those molecules might have become organized into living cells.

Panspermia and spontaneous origin are the only testable hypotheses of life's origin currently available.

Scientists Disagree about Where Life Started

While most researchers agree that life first appeared as the primitive earth cooled and its rocky crust formed, there is little agreement as to just where this occurred.

Did Life Originate at the Ocean's Edge?

The more we learn about earth's early history, the more likely it seems that earth's first organisms emerged and lived at very high temperatures. Rubble from the forming solar system slammed into early earth from 4.6 to 3.8 billion years ago, keeping the surface molten hot. As the bombardment slowed down, temperatures dropped. By about 3.8 billion years ago, ocean temperatures are thought to have dropped to a hot 49° to 88°C (120° to 190°F). Between 3.8 and 3.5 billion years ago, life first appeared, promptly after the earth was inhabitable. Thus, as intolerable as early earth's infernal temperatures seem to us today, they gave birth to life.

Very few geochemists agree on the exact composition of the early atmosphere. One popular view is that it contained principally carbon dioxide (CO₂) and nitrogen gas (N₂), along with significant amounts of water vapor (H₂O). It is possible that the early atmosphere also contained hydrogen gas (H₂) and compounds in which hydrogen atoms were bonded to the other light elements (sulfur, nitrogen, and carbon), producing hydrogen sulfide (H₂S), ammonia (NH₃), and methane (CH₄).

We refer to such an atmosphere as a reducing atmosphere because of the ample availability of hydrogen atoms and their electrons. In such a reducing atmosphere it would not take as much energy as it would today to form the carbon-rich molecules from which life evolved.

The key to this reducing atmosphere hypothesis is the assumption that there was very little oxygen around. In an atmosphere with oxygen, amino acids and sugars react spontaneously with the oxygen to form carbon dioxide and water. Therefore, the building blocks of life, the amino acids, would not last long and the spontaneous formation of complex carbon molecules could not occur. Our atmosphere changed once organisms began to carry out photosynthesis, harnessing the energy in sunlight to split water molecules and form complex carbon molecules, giving off gaseous oxygen molecules in the process. The earth's atmosphere is now approximately 21% oxygen.

Critics of the reducing atmosphere hypothesis point out that no carbonates have been found in rocks dating back to the early earth. This suggests that at that time carbon dioxide was locked up in the atmosphere, and if that was the case, then the prebiotic atmosphere would not have been reducing.

Another problem for the reducing atmosphere hypothesis is that because a prebiotic reducing atmosphere would have been oxygen free, there would have been no ozone. Without the protective ozone layer, any organic compounds

that might have formed would have been broken down quickly by ultraviolet radiation.

Other Suggestions

If life did not originate at the ocean's edge under the blanket of a reducing atmosphere, where did it originate?

Under frozen oceans. One hypothesis proposes that life originated under a frozen ocean, not unlike the one that covers Jupiter's moon Europa today. All evidence suggests, however, that the early earth was quite warm and frozen oceans quite unlikely.

Deep in the earth's crust. Another hypothesis is that life originated deep in the earth's crust. In 1988 Gunter Wächtershäuser proposed that life might have formed as a by-product of volcanic activity, with iron and nickel sulfide minerals acting as chemical catalysts to recombine gases spewing from eruptions into the building blocks of life. In later work he and coworkers were able to use this unusual chemistry to build precursors for amino acids (although they did not actually succeed in making amino acids), and to link amino acids together to form peptides. Critics of this hypothesis point out that the concentration of chemicals used in their experiments greatly exceed what is found in nature.

Within clay. Other researchers have proposed the unusual hypothesis that life is the result of silicate surface chemistry. The surface of clays have positive charges to attract organic molecules, and exclude water, providing a potential catalytic surface on which life's early chemistry might have occurred. While interesting conceptually, there is little evidence that this sort of process could actually occur.

At deep-sea vents. Becoming more popular is the hypothesis that life originated at deep-sea hydrothermal vents, with the necessary prebiotic molecules being synthesized on metal sulfides in the vents. The positive charge of the sulfides would have acted as a magnet for negatively charged organic molecules. In part, the current popularity of this hypothesis comes from the new science of genomics, which suggests that the ancestors of today's prokaryotes are most closely related to the bacteria that live on the deep-sea vents.

No one is sure whether life originated at the ocean's edge, under frozen ocean, deep in the earth's crust, within clay, or at deep-sea vents. Perhaps one of these hypotheses will be proven correct. Perhaps the correct theory has not yet been proposed.

When life first appeared on earth, the environment was very hot. All of the spontaneous origin hypotheses assume that the organic chemicals that were the building blocks of life arose spontaneously at that time. How is a matter of considerable disagreement.

The Miller-Urey Experiment

An early attempt to see what kinds of organic molecules might have been produced on the early earth was carried out in 1953 by Stanley L. Miller and Harold C. Urey. In what has become a classic experiment, they attempted to reproduce the conditions at ocean's edge under a reducing atmosphere. Even if this assumption proves incorrect—the jury is still out on this—their experiment is critically important, as it ushered in the whole new field of prebiotic chemistry.

To carry out their experiment, they (1) assembled a reducing atmosphere rich in hydrogen and excluding gaseous oxygen; (2) placed this atmosphere over liquid water, which would have been present at ocean's edge; (3) maintained this mixture at a temperature somewhat below 100°C; and (4) simulated lightning by bombarding it with energy in the form of sparks (figure 4.6).

They found that within a week, 15% of the carbon originally present as methane gas (CH_4) had converted into other simple carbon compounds. Among these compounds were formaldehyde (CH_2O) and hydrogen cyanide (HCN ; figure 4.7). These compounds then combined to form simple molecules, such as formic acid (HCOOH) and urea (NH_2CONH_2), and more complex molecules containing carbon-carbon bonds, including the amino acids glycine and alanine.

As we saw in chapter 3, amino acids are the basic building blocks of proteins, and proteins are one of the major kinds of molecules of which organisms are composed. In similar experiments performed later by other scientists, more than 30 different carbon compounds were identified, including the amino acids glycine, alanine, glutamic acid, valine, proline, and aspartic acid. Other biologically important molecules were also formed in these experiments. For example, hydrogen cyanide contributed to the production of a complex ring-shaped molecule called adenine—one of the bases found in DNA and RNA. Thus, the key molecules of life could have formed in the atmosphere of the early earth.

The Path of Chemical Evolution

A raging debate among biologists who study the origin of life concerns which organic molecules came first, RNA or proteins. Scientists are divided into three camps, those that focus on RNA, protein, or a combination of the two. All three arguments have their strong points. Like the hypotheses that try to account for where life originated, these competing hypotheses are diverse and speculative.

An RNA World. The “RNA world” group feels that without a hereditary molecule, other molecules could not have formed consistently. The “RNA world” argument earned support when Thomas Cech at the University of Colorado discovered ribozymes, RNA molecules that can behave as enzymes, catalyzing their own assembly. Recent work has

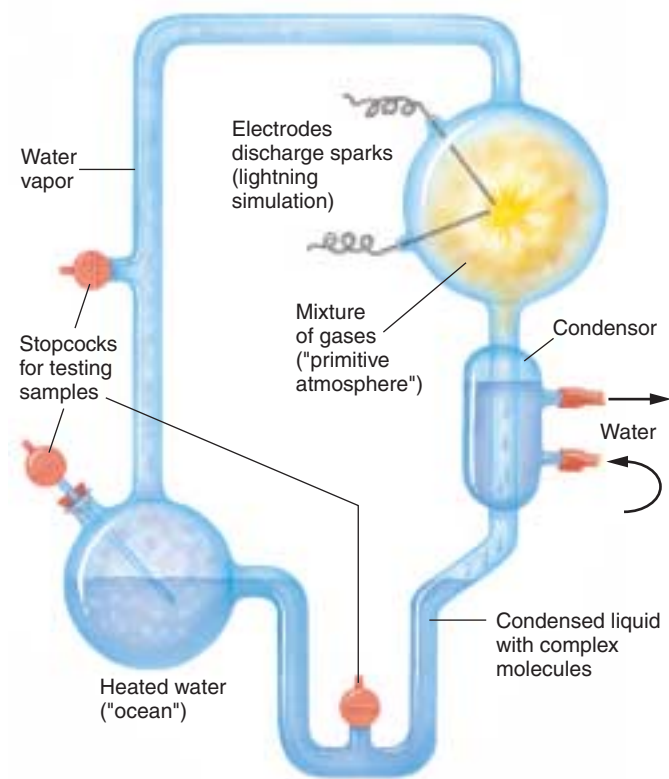


FIGURE 4.6

The Miller-Urey experiment. The apparatus consisted of a closed tube connecting two chambers. The upper chamber contained a mixture of gases thought to resemble the primitive earth's atmosphere. Electrodes discharged sparks through this mixture, simulating lightning. Condensers then cooled the gases, causing water droplets to form, which passed into the second heated chamber, the “ocean.” Any complex molecules formed in the atmosphere chamber would be dissolved in these droplets and carried to the ocean chamber, from which samples were withdrawn for analysis.

shown that the RNA contained in ribosomes (discussed in chapter 5) catalyzes the chemical reaction that links amino acids to form proteins. Therefore, the RNA in ribosomes also functions as an enzyme. If RNA has the ability to pass on inherited information and the capacity to act like an enzyme, were proteins really needed?

A Protein World. The “protein-first” group argues that without enzymes (which are proteins), nothing could replicate at all, heritable or not. The “protein-first” proponents argue that nucleotides, the individual units of nucleic acids such as RNA, are too complex to have formed spontaneously, and certainly too complex to form spontaneously again and again. While there is no doubt that simple proteins are easier to synthesize from abiotic components than nucleotides, both can form in the laboratory under the right conditions. Deciding which came first is a chicken-and-egg paradox. In an effort to shed light on this problem, Julius Rebek and a number of

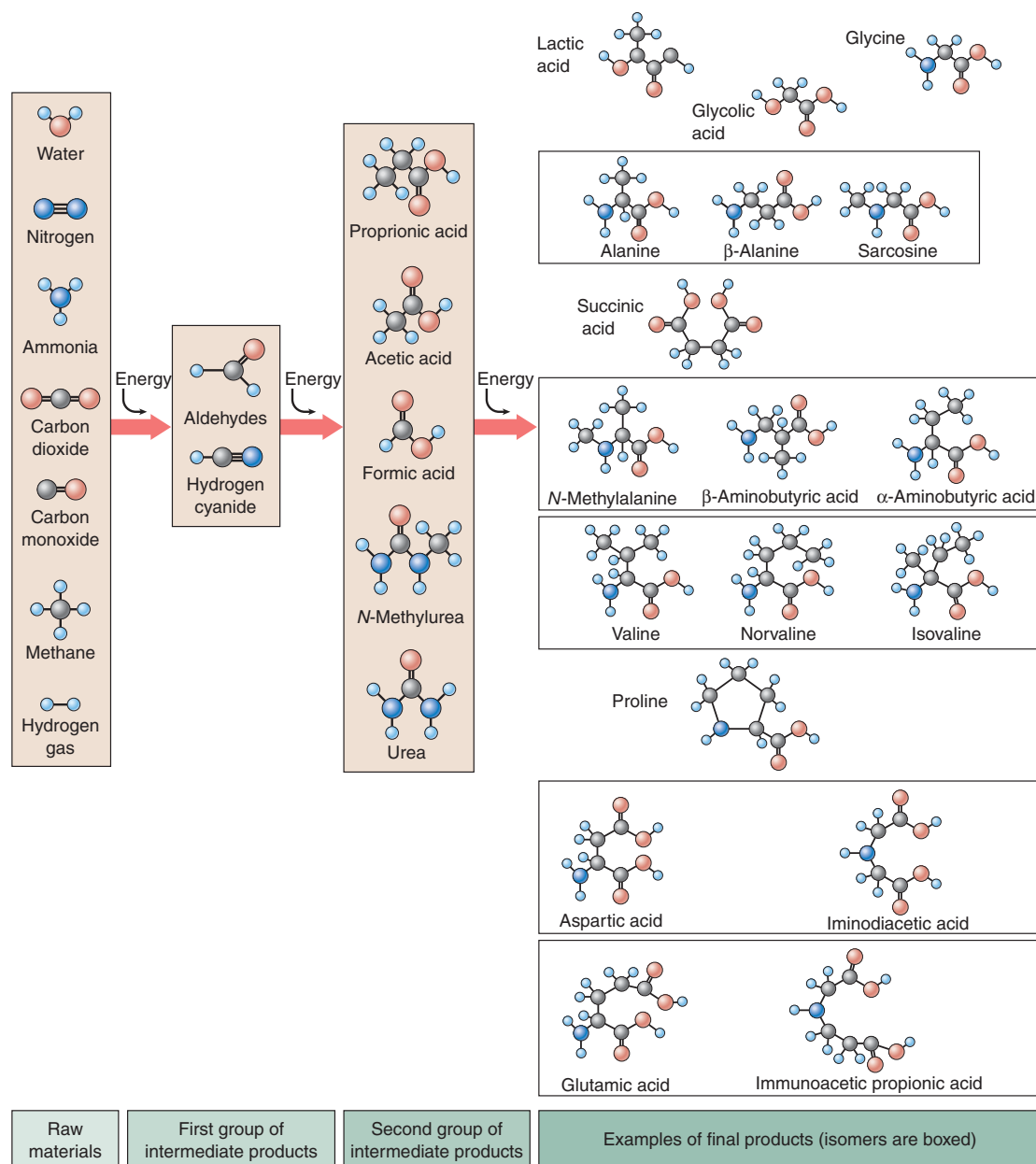


FIGURE 4.7

Results of the Miller-Urey experiment. Seven simple molecules, all gases, were included in the original mixture. Note that oxygen was not among them; instead, the atmosphere was rich in hydrogen. At each stage of the experiment, more complex molecules were formed: first aldehydes, then simple acids, then more complex acids. Among the final products, the molecules that are structural isomers of one another are grouped together in boxes. In most cases only one isomer of a compound is found in living systems today, although many may have been produced in the Miller-Urey experiment.

other chemists have created synthetic nucleotide-like molecules in the laboratory that are able to replicate. Moving even further, Rebek and his colleagues have created synthetic molecules that could replicate and “make mistakes.” This simulates mutation, a necessary ingredient for the process of evolution.

A Peptide-Nucleic Acid World. Another important and popular theory about the first organic molecules assumes key roles for both peptides and nucleic acids. Because RNA

is so complex and unstable, this theory assumes there must have been a pre-RNA world where the peptide-nucleic acid (PNA) was the basis for life. PNA is stable and simple enough to have formed spontaneously, and is also a self-replicator.

Molecules that are the building blocks of living organisms form spontaneously under conditions designed to simulate those of the primitive earth.

4.3 The first cells had little internal structure.

Theories about the Origin of Cells

The evolution of cells required early organic molecules to assemble into a functional, interdependent unit. Cells, discussed in the next chapter, are essentially little bags of fluid. What the fluid contains depends on the individual cell, but every cell's contents differ from the environment outside the cell. Therefore, an early cell may have floated along in a dilute “primordial soup,” but its interior would have had a higher concentration of specific organic molecules.

Cell Origins: The Importance of Bubbles

How did these “bags of fluid” evolve from simple organic molecules? As you can imagine, the answer to this question is a matter for debate. Scientists favoring an “ocean’s edge” scenario for the origin of life have proposed that bubbles may have played a key role in this evolutionary step. A bubble, such as those produced by soap solutions, is a hollow spherical structure. Certain molecules, particularly those with hydrophobic regions, will spontaneously form bubbles in water. The structure of the bubble shields the hydrophobic regions of the molecules from contact with water. If you have ever watched the ocean surge upon the shore, you may have noticed the foamy froth created by the agitated water. The edges of the primitive oceans were more than likely very frothy places bombarded by ultraviolet and other ionizing radiation, and exposed to an atmosphere that may have contained methane and other simple organic molecules.

Oparin’s Bubble Theory

The first bubble theory is attributed to Alexander Oparin, a Russian chemist with extraordinary insight. In the mid-1930s, Oparin suggested that the present-day atmosphere was incompatible with the creation of life. He proposed that life must have arisen from nonliving matter under a set of very different environmental circumstances some time in the distant history of the earth. His was the theory of **primary abiogenesis** (primary because all living cells are now known to come from previously living cells, except in that first case). At the same time, J. B. S. Haldane, a British geneticist, was also independently espousing the same views. Oparin decided that in order for cells to evolve, they must have had some means of developing chemical complexity, separating their contents from their environment by means of a cell membrane, and concentrating materials within themselves. He termed these early, chemical-concentrating bubblelike structures **protobionts**.

Oparin’s theories were published in English in 1938, and for awhile most scientists ignored them. However, Harold Urey, an astronomer at the University of Chicago, was quite taken with Oparin’s ideas. He convinced one of his graduate students, Stanley Miller, to follow Oparin’s rationale and see if he could “create” life. The Urey-Miller experiment has proven to be one of the most significant experiments in the history of science. As a result Oparin’s ideas became better known and more widely accepted.

A Host of Bubble Theories

Different versions of “bubble theories” have been championed by numerous scientists since Oparin. The bubbles they propose go by a variety of names; they may be called *microspheres*, *protocells*, *protobionts*, *micelles*, *liposomes*, or *coacervates*, depending on the composition of the bubbles (lipid or protein) and how they form. In all cases, the bubbles are hollow spheres, and they exhibit a variety of cell-like properties. For example, the lipid bubbles called **coacervates** form an outer boundary with two layers that resembles a biological membrane. They grow by accumulating more subunit lipid molecules from the surrounding medium, and they can form budlike projections and divide by pinching in two, like bacteria. They also can contain amino acids and use them to facilitate various acid-base reactions, including the decomposition of glucose. Although they are not alive, they obviously have many of the characteristics of cells.

A Bubble Scenario

It is not difficult to imagine that a process of chemical evolution involving bubbles or microdrops preceded the origin of life (figure 4.8). The early oceans must have contained untold numbers of these microdrops, billions in a spoonful, each one forming spontaneously, persisting for a while, and then dispersing. Some would, by chance, have contained amino acids with side groups able to catalyze growth-promoting reactions. Those microdrops would have survived longer than ones that lacked those amino acids, because the persistence of both proteinoid microspheres and lipid coacervates is greatly increased when they carry out metabolic reactions such as glucose degradation and when they are actively growing.

Over millions of years, then, the complex bubbles that were better able to incorporate molecules and energy from the lifeless oceans of the early earth would have tended to persist longer than the others. Also favored would have been the microdrops that could use these molecules to expand in size, growing large enough to divide into “daughter”

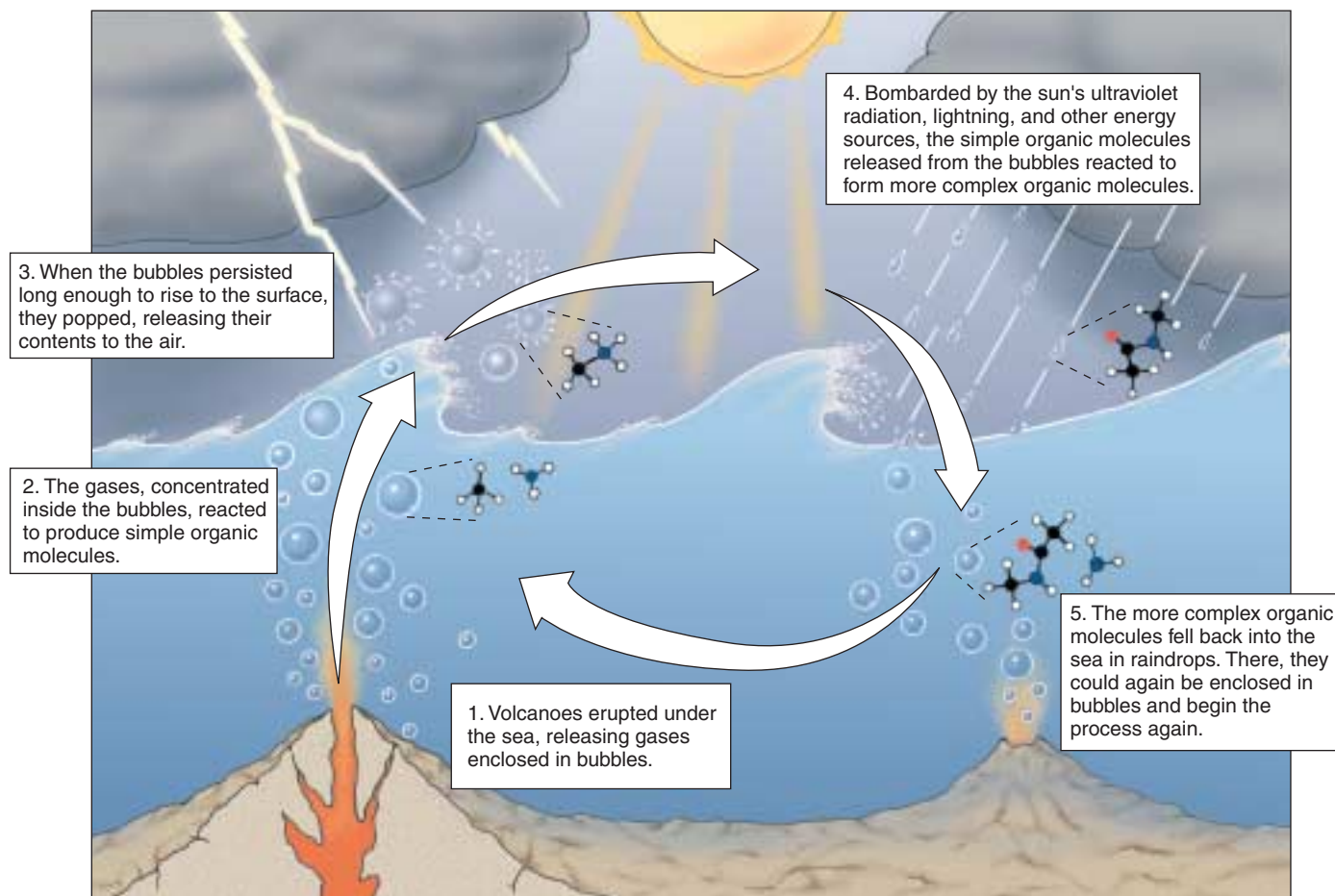


FIGURE 4.8

A current bubble hypothesis. In 1986 geophysicist Louis Lerman proposed that the chemical processes leading to the evolution of life took place within bubbles on the ocean's surface.

microdrops with features similar to those of their “parent” microdrop. The daughter microdrops have the same favorable combination of characteristics as their parent, and would have grown and divided, too. When a way to facilitate the reliable transfer of new ability from parent to offspring developed, heredity—and life—began.

Current Thinking

Whether the early bubbles that gave rise to cells were lipid or protein remains an unresolved argument. While it is true that lipid microspheres (coacervates) will form readily in water, there appears to be no mechanism for their heritable replication. On the other hand, one *can* imagine a heritable mechanism for protein microspheres. Although protein microspheres do not form readily in water, Sidney Fox and his colleagues at the University of Miami have shown that they can form under dry conditions.

The discovery that RNA can act as an enzyme to assemble new RNA molecules on an RNA template has raised

the interesting possibility that neither coacervates nor protein microspheres were the first step in the evolution of life. Perhaps the first components were RNA molecules, and the initial steps on the evolutionary journey led to increasingly complex and stable RNA molecules. Later, stability might have improved further when a lipid (or possibly protein) microsphere surrounded the RNA. At present, those studying this problem have not arrived at a consensus about whether RNA evolved before or after a bubblelike structure that likely preceded cells.

Eventually, DNA took the place of RNA as the replicator in the cell and the storage molecule for genetic information. DNA, because it is a double helix, stores information in a more stable fashion than RNA, which is single-stranded.

Little is known about how the first cells originated. Current hypotheses involve chemical evolution within bubbles, but there is no general agreement about their composition, or about how the process occurred.

The Earliest Cells

What do we know about the earliest life-forms? The fossils found in ancient rocks show an obvious progression from simple to complex organisms, beginning about 3.5 billion years ago. Life may have been present earlier, but rocks of such great antiquity are rare, and fossils have not yet been found in them.

Microfossils

The earliest evidence of life appears in **microfossils**, fossilized forms of microscopic life (figure 4.9). Microfossils were small (1 to 2 micrometers in diameter) and single-celled, lacked external appendages, and had little evidence of internal structure. Thus, they physically resemble present-day **bacteria** (figure 4.10), although some ancient forms cannot be matched exactly. We call organisms with this simple body plan **prokaryotes**, from the Greek words meaning “before” and “kernel,” or “nucleus.” The name reflects their lack of a **nucleus**, a spherical organelle characteristic of the more complex cells of **eukaryotes**.

Judging from the fossil record, eukaryotes did not appear until about 1.5 billion years ago. Therefore, for at least 2 billion years—nearly a half of the age of the earth—bacteria were the only organisms that existed.

Ancient Bacteria: Archaeobacteria

Most organisms living today are adapted to the relatively mild conditions of present-day earth. However, if we look in unusual environments, we encounter organisms that are quite remarkable, differing in form and metabolism from other living things. Sheltered from evolutionary alteration in unchanging habitats that resemble earth’s early environment, these living relics are the surviving representatives of the first ages of life on earth. In places such as the oxygen-free depths of the Black Sea or the boiling waters of hot springs and deep-sea vents, we can find bacteria living at very high temperatures without oxygen.

These unusual bacteria are called **archaeobacteria**, from the Greek word for “ancient ones.” Among the first to be studied in detail have been the **methanogens**, or methane-producing bacteria, among the most primitive bacteria that exist today. These organisms are typically simple in form and are able to grow only in an oxygen-free environment; in fact, oxygen poisons them. For this reason they are said to grow “without air,” or **anaerobically** (Greek *an*, “without” + *aer*, “air” + *bios*, “life”). The methane-producing bacteria convert CO_2 and H_2 into methane gas (CH_4). Although primitive, they resemble all other bacteria in having DNA, a lipid cell membrane, an exterior cell wall, and a metabolism based on an energy-carrying molecule called ATP.

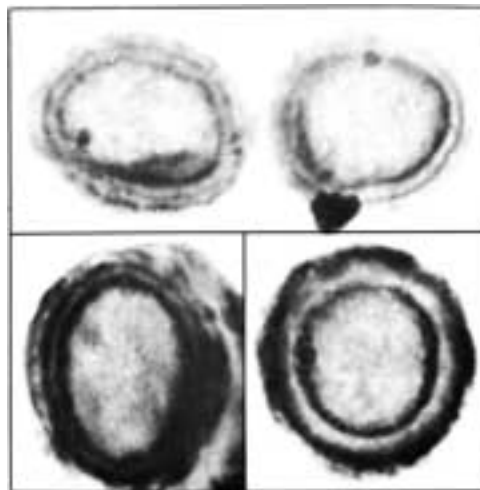


FIGURE 4.9
Cross-sections of fossil bacteria. These microfossils from the Bitter Springs formation of Australia are of ancient cyanobacteria, far too small to be seen with the unaided eye. In this electron micrograph, the cell walls are clearly evident.

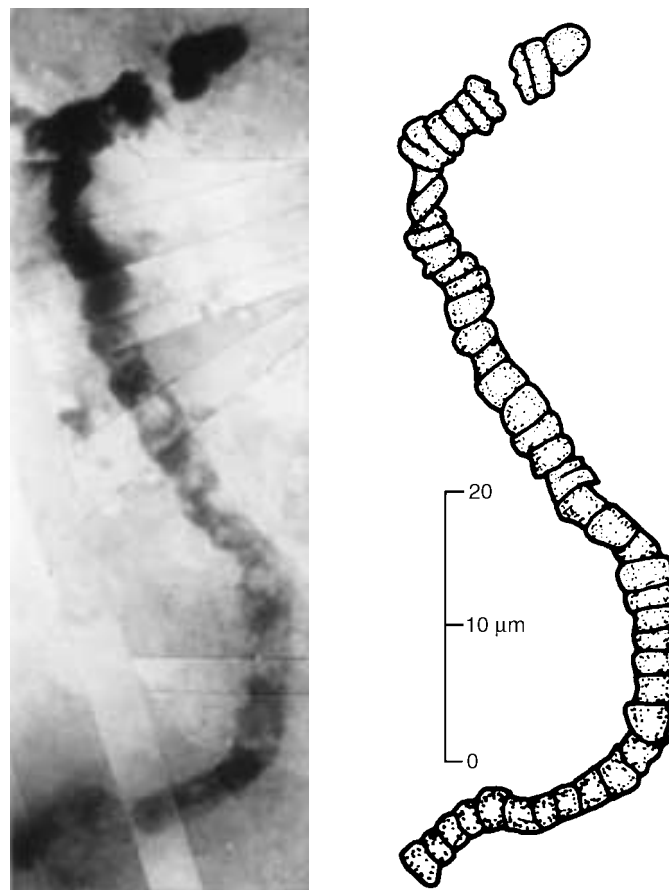


FIGURE 4.10
The oldest microfossil. This ancient bacterial fossil, discovered by J. William Schopf of UCLA in 3.5-billion-year-old rocks in western Australia, is similar to present-day cyanobacteria, as you can see by comparing it to figure 4.11.

Unusual Cell Structures

When the details of cell wall and membrane structure of the methane-producing bacteria were examined, they proved to be different from those of all other bacteria. Archaeobacteria are characterized by a conspicuous lack of a protein cross-linked carbohydrate material called **peptidoglycan** in their cell walls, a key compound in the cell walls of most modern bacteria. Archaeobacteria also have unusual lipids in their cell membranes that are not found in any other group of organisms. There are also major differences in some of the fundamental biochemical processes of metabolism, different from those of all other bacteria. The methane-producing bacteria are survivors from an earlier time when oxygen gas was absent.

Earth's First Organisms?

Other archaeobacteria that fall into this classification are some of those that live in very salty environments like the Dead Sea (**extreme halophiles**—"salt lovers") or very hot environments like hydrothermal volcanic vents under the ocean (**extreme thermophiles**—"heat lovers"). Thermophiles have been found living comfortably in boiling water. Indeed, many kinds of thermophilic archaeobacteria thrive at temperatures of 110°C (230°F). Because these thermophiles live at high temperatures similar to those that may have existed when life first evolved, microbiologists speculate that thermophilic archaeobacteria may be relics of earth's first organisms.

Just how different are extreme thermophiles from other organisms? A methane-producing archaeobacteria called *Methanococcus* isolated from deep-sea vents provides a startling picture. These bacteria thrive at temperatures of 88°C (185°F) and crushing pressures 245 times greater than at sea level. In 1996 molecular biologists announced that they had succeeded in determining the full nucleotide sequence of *Methanococcus*. This was possible because archaeobacterial DNA is relatively small—it has only 1700 genes, coded in a DNA molecule only 1,739,933 nucleotides long (a human cell has 2000 times more!). The thermophile nucleotide sequence proved to be astonishingly different from the DNA sequence of any other organism ever studied; fully two-thirds of its genes are unlike any ever known to science before! Clearly these archaeobacteria separated from other life on earth a long time ago. Preliminary comparisons to the gene sequences of other bacteria suggest that archaeobacteria split from other types of bacteria over 3 billion years ago, soon after life began.

Eubacteria

The second major group of bacteria, the **eubacteria**, have very strong cell walls and a simpler gene architecture. Most bacteria living today are eubacteria. Included in this group are bacteria that have evolved the ability to capture the energy of light and transform it into the energy of chemical



FIGURE 4.11

Living cyanobacteria. Although not multicellular, these bacteria often aggregate into chains such as those seen here.

bonds within cells. These organisms are *photosynthetic*, as are plants and algae.

One type of photosynthetic eubacteria that has been important in the history of life on earth is the cyanobacteria, sometimes called "blue-green algae" (figure 4.11). They have the same kind of chlorophyll pigment that is most abundant in plants and algae, as well as other pigments that are blue or red. Cyanobacteria produce oxygen as a result of their photosynthetic activities, and when they appeared at least 3 billion years ago, they played a decisive role in increasing the concentration of free oxygen in the earth's atmosphere from below 1% to the current level of 21%. As the concentration of oxygen increased, so did the amount of ozone in the upper layers of the atmosphere. The thickening ozone layer afforded protection from most of the ultraviolet radiation from the sun, radiation that is highly destructive to proteins and nucleic acids. Certain cyanobacteria are also responsible for the accumulation of massive limestone deposits.

All bacteria now living are members of either Archaeobacteria or Eubacteria.

4.4 The first eukaryotic cells were larger and more complex than bacteria.

All fossils more than 1.5 billion years old are generally similar to one another structurally. They are small, simple cells; most measure 0.5 to 2 micrometers in diameter, and none are more than about 6 micrometers in diameter. These simple cells eventually evolved into larger, more complex forms—the first eukaryotic cells.

The First Eukaryotic Cells

In rocks about 1.5 billion years old, we begin to see the first microfossils that are noticeably different in appearance from the earlier, simpler forms (figure 4.12). These cells are much larger than bacteria and have internal membranes and thicker walls. Cells more than 10 micrometers in diameter rapidly increased in abundance. Some fossilized cells 1.4 billion years old are as much as 60 micrometers in diameter; others, 1.5 billion years old, contain what appear to be small, membrane-bound structures. Indirect chemical traces hint that eukaryotes may go as far back as 2.7 billion years, although no fossils as yet support such an early appearance of eukaryotes.

These early fossils mark a major event in the evolution of life: a new kind of organism had appeared (figure 4.13). These new cells are called **eukaryotes**, from the Greek words for “true” and “nucleus,” because they possess an internal structure called a nucleus. All organisms other than the bacteria are eukaryotes.

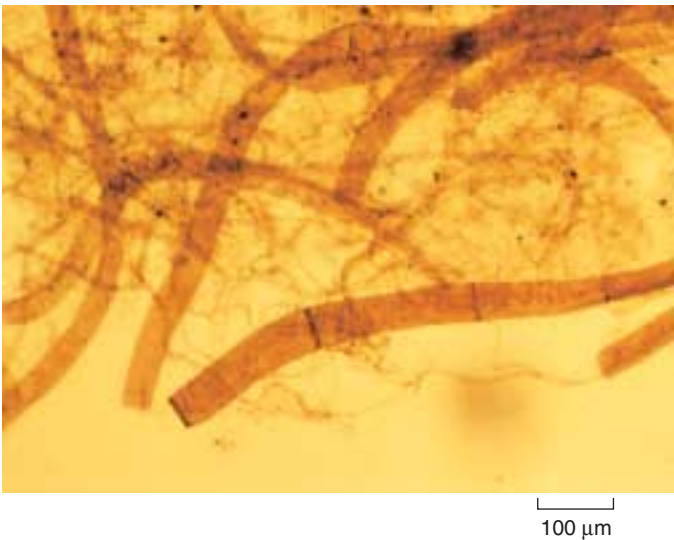


FIGURE 4.12
Microfossil of a primitive eukaryote. This multicellular alga is between 900 million and 1 billion years old.

Origin of the Nucleus and ER

Many bacteria have infoldings of their outer membranes extending into the cytoplasm and serving as passageways to the surface. The network of internal membranes in

CAMBRIAN	PHANEROZOIC	Geological evidence	Millions of years ago	Life forms
		Oldest multicellular fossils	570 600	Appearance of first multicellular organisms
PRECAMBRIAN	PROTEROZOIC	Oldest compartmentalized fossil cells	1500	Appearance of first eukaryotes Appearance of aerobic (oxygen-using) respiration
		Disappearance of iron from oceans and formation of iron oxides	2500	Appearance of oxygen-forming photosynthesis (cyanobacteria)
	ARCHEAN	Oldest definite fossils	3500	Appearance of chemoautotrophs (sulfate respiration) Appearance of life (prokaryotes): anaerobic (methane-producing) bacteria and anaerobic (hydrogen sulfide-forming) photosynthesis
		Oldest dated rocks	4500	Formation of the earth

FIGURE 4.13
The geological timescale. The periods refer to different stages in the evolution of life on earth. The timescale is calibrated by examining rocks containing particular kinds of fossils; the fossils are dated by determining the degree of spontaneous decay of radioactive isotopes locked within rock when it was formed.

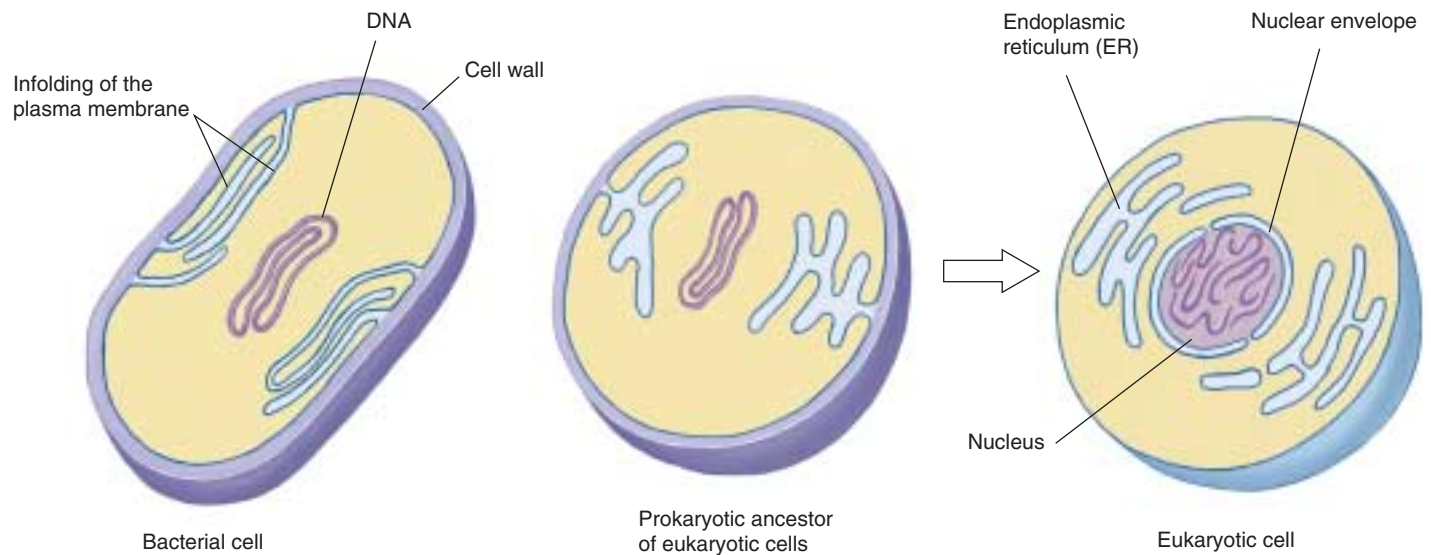


FIGURE 4.14

Origin of the nucleus and endoplasmic reticulum. Many bacteria today have infoldings of the plasma membrane (see also figure 34.7). The eukaryotic internal membrane system called the endoplasmic reticulum (ER) and the nuclear envelope may have evolved from such infoldings of the plasma membrane encasing prokaryotic cells that gave rise to eukaryotic cells.

eukaryotes called endoplasmic reticulum (ER) is thought to have evolved from such infoldings, as is the nuclear envelope, an extension of the ER network that isolates and protects the nucleus (figure 4.14).

Origin of Mitochondria and Chloroplasts

Bacteria that live within other cells and perform specific functions for their host cells are called *endosymbiotic bacteria*. Their widespread presence in nature led Lynn Margulis to champion the **endosymbiotic theory** in the early 1970s. This theory, now widely accepted, suggests that a critical stage in the evolution of eukaryotic cells involved endosymbiotic relationships with prokaryotic organisms. According to this theory, energy-producing bacteria may have come to reside within larger bacteria, eventually evolving into what we now know as mitochondria. Similarly, photosynthetic bacteria may have come to live within other larger bacteria, leading to the evolution of chloroplasts, the photosynthetic organelles of plants and algae. Bacteria with flagella, long whiplike cellular appendages used for propulsion, may have become symbiotically involved with nonflagellated bacteria to produce larger, motile cells. The fact that we now witness so many symbiotic relationships lends general support to this theory. Even stronger support comes from the observation that present-day organelles such as mitochondria, chloroplasts, and centrioles contain their own DNA, which is remarkably similar to the DNA of bacteria in size and character.

Sexual Reproduction

Eukaryotic cells also possess the ability to reproduce sexually, something prokaryotes cannot do effectively. **Sexual reproduction** is the process of producing offspring, with two copies of each chromosome, by fertilization, the union of two cells that each have one copy of each chromosome. The great advantage of sexual reproduction is that it allows for frequent genetic recombination, which generates the variation that is the raw material for evolution. Not all eukaryotes reproduce sexually, but most have the capacity to do so. The evolution of meiosis and sexual reproduction (discussed in chapter 12) led to the tremendous explosion of diversity among the eukaryotes.

Multicellularity

Diversity was also promoted by the development of **multicellularity**. Some single eukaryotic cells began living in association with others, in colonies. Eventually individual members of the colony began to assume different duties, and the colony began to take on the characteristics of a single individual. Multicellularity has arisen many times among the eukaryotes. Practically every organism big enough to see with the unaided eye is multicellular, including all animals and plants. The great advantage of multicellularity is that it fosters specialization; some cells devote all of their energies to one task, other cells to another. Few innovations have had as great an impact on the history of life as the specialization made possible by multicellularity.

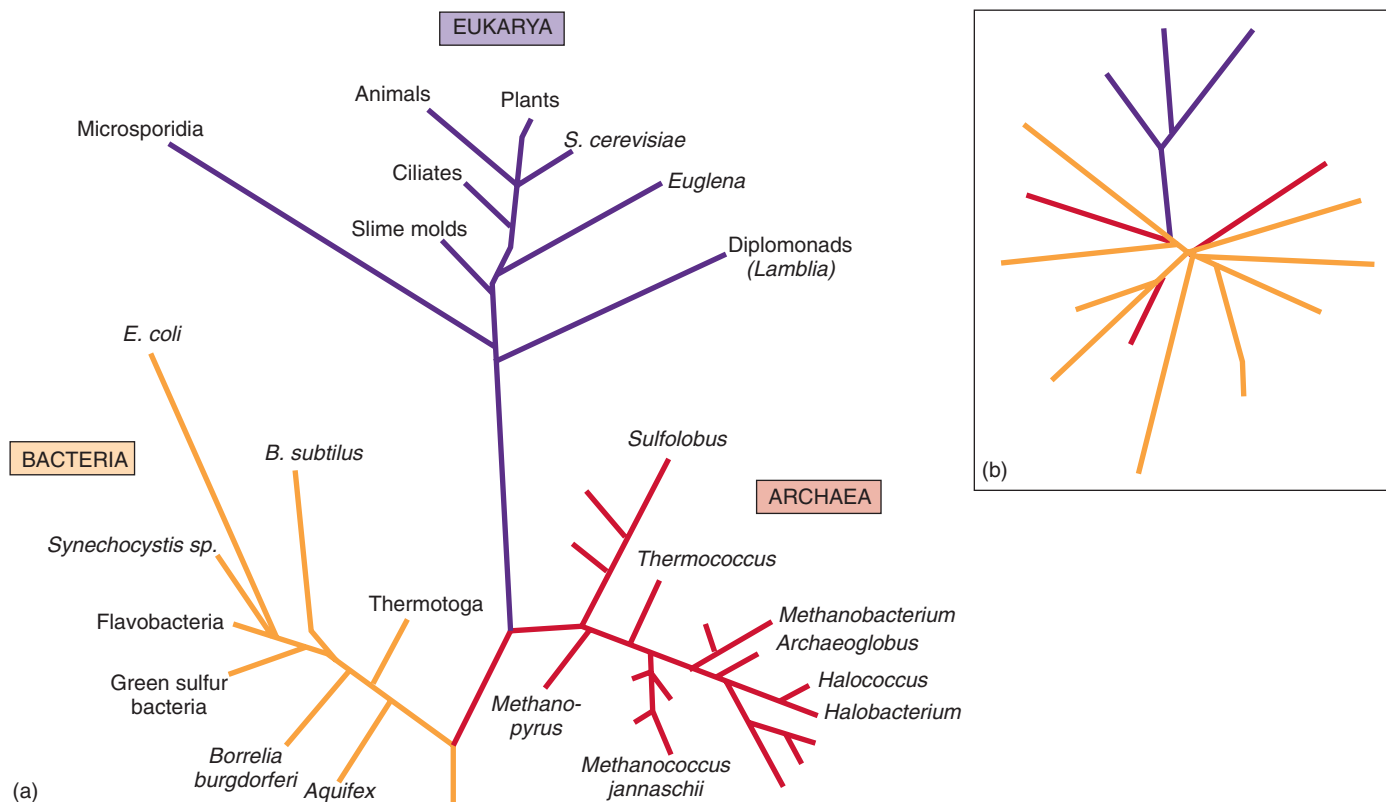


FIGURE 4.15

The three domains of life. The kingdoms Archaeobacteria and Eubacteria are as different from each other as from eukaryotes, so biologists have assigned them a higher category, a “domain.” (a) A three-domain tree of life based on ribosomal RNA consists of the Eukarya, Bacteria, and Archaea. (b) New analyses of complete genome sequences contradict the rRNA tree, and suggest other arrangements such as this one, which splits the Archaea. Apparently genes hopped from branch to branch as early organisms either stole genes from their food or swapped DNA with their neighbors, even distantly related ones.

The Kingdoms of Life

Confronted with the great diversity of life on earth today, biologists have attempted to categorize similar organisms in order to better understand them, giving rise to the science of taxonomy. In later chapters, we will discuss taxonomy and classification in detail, but for now we can generalize that all living things fall into one of three domains which include six kingdoms (figure 4.15):

Kingdom Archaeobacteria: Prokaryotes that lack a peptidoglycan cell wall, including the methanogens and extreme halophiles and thermophiles.

Kingdom Eubacteria: Prokaryotic organisms with a peptidoglycan cell wall, including cyanobacteria, soil bacteria, nitrogen-fixing bacteria, and pathogenic (disease-causing) bacteria.

Kingdom Protista: Eukaryotic, primarily unicellular (although algae are multicellular), photosynthetic or heterotrophic organisms, such as amoebas and paramecia.

Kingdom Fungi: Eukaryotic, mostly multicellular (although yeasts are unicellular), heterotrophic, usually nonmotile organisms, with cell walls of chitin, such as mushrooms.

Kingdom Plantae: Eukaryotic, multicellular, nonmotile, usually terrestrial, photosynthetic organisms, such as trees, grasses, and mosses.

Kingdom Animalia: Eukaryotic, multicellular, motile, heterotrophic organisms, such as sponges, spiders, newts, penguins, and humans.

As more is learned about living things, particularly from the newer evidence that DNA studies provide, scientists will continue to reevaluate the relationships among the kingdoms of life.

For at least the first 1 billion years of life on earth, all organisms were bacteria. About 1.5 billion years ago, the first eukaryotes appeared. Biologists place living organisms into six general categories called kingdoms.

Has Life Evolved Elsewhere?

We should not overlook the possibility that life processes might have evolved in different ways on other planets. A functional genetic system, capable of accumulating and replicating changes and thus of adaptation and evolution, could theoretically evolve from molecules other than carbon, hydrogen, nitrogen, and oxygen in a different environment. Silicon, like carbon, needs four electrons to fill its outer energy level, and ammonia is even more polar than water. Perhaps under radically different temperatures and pressures, these elements might form molecules as diverse and flexible as those carbon has formed on earth.

The universe has 10^{20} (100,000,000,000,000,000,000) stars similar to our sun. We don't know how many of these stars have planets, but it seems increasingly likely that many do. Since 1996, astronomers have been detecting planets orbiting distant stars. At least 10% of stars are thought to have planetary systems. If only 1 in 10,000 of these planets is the right size and at the right distance from its star to duplicate the conditions in which life originated on earth, the "life experiment" will have been repeated 10^{15} times (that is, a million billion times). It does not seem likely that we are alone.

Ancient Bacteria on Mars?

A dull gray chunk of rock collected in 1984 in Antarctica ignited an uproar about ancient life on Mars with the report that the rock contains evidence of possible life. Analysis of gases trapped within small pockets of the rock indicate it is a meteorite from Mars. It is, in fact, the oldest rock known to science—fully 4.5 billion years old. Back then, when this rock formed on Mars, that cold, arid planet was much warmer, flowed with water, and had a carbon dioxide atmosphere—conditions not too different from those that spawned life on earth.

When examined with powerful electron microscopes, carbonate patches within the meteorite exhibit what look like microfossils, some 20 to 100 nanometers in length. One hundred times smaller than any known bacteria, it is not clear they actually are fossils, but the resemblance to bacteria is striking.

Viewed as a whole, the evidence of bacterial life associated with the Mars meteorite is not compelling. Clearly, more painstaking research remains to be done before the discovery can claim a scientific consensus. However, while there is no conclusive evidence of bacterial life associated with this meteorite, it seems very possible that life has evolved on other worlds in addition to our own.

Deep-Sea Vents

The possibility that life on earth actually originated in the vicinity of deep-sea hydrothermal vents is gaining popularity. At the bottom of the ocean, where these vents spewed



FIGURE 4.16

Is there life elsewhere? Currently the most likely candidate for life elsewhere within the solar system is Europa, one of the many moons of the large planet Jupiter.

out a rich froth of molecules, the geological turbulence and radioactive energy battering the land was absent, and things were comparatively calm. The thermophilic archaeobacteria found near these vents today are the most ancient group of organisms living on earth. Perhaps the gentler environment of the ocean depths was the actual cradle of life.

Other Planets

Has life evolved on other worlds within our solar system? There are planets other than ancient Mars with conditions not unlike those on earth. Europa, a large moon of Jupiter, is a promising candidate (figure 4.16). Europa is covered with ice, and photos taken in close orbit in the winter of 1998 reveal seas of liquid water beneath a thin skin of ice. Additional satellite photos taken in 1999 suggest that a few miles under the ice lies a liquid ocean of water larger than earth's, warmed by the push and pull of the gravitational attraction of Jupiter's many large satellite moons. The conditions on Europa now are far less hostile to life than the conditions that existed in the oceans of the primitive earth. In coming decades satellite missions are scheduled to explore this ocean for life.

There are so many stars that life may have evolved many times. Although evidence for life on Mars is not compelling, the seas of Europa offer a promising candidate which scientists are eager to investigate.



Chapter 4

Summary

Questions

Media Resources

4.1 All living things share key characteristics.

- All living things are characterized by cellular organization, growth, reproduction, and heredity.
- Other properties commonly exhibited by living things include movement and sensitivity to stimuli.

1. What characteristics of living things are *necessary* characteristics (possessed by all living things), and which are *sufficient* characteristics (possessed only by living things)?

4.2 There are many ideas about the origin of life.

- Of the many explanations of how life might have originated, only the theories of spontaneous and extraterrestrial origins provide scientifically testable explanations.
- Experiments recreating the atmosphere of primitive earth, with the energy sources and temperatures thought to be prevalent at that time, have led to the spontaneous formation of amino acids and other biologically significant molecules.

2. What molecules are thought to have been present in the atmosphere of the early earth? Which molecule that was notably absent then is now a major component of the atmosphere?



- Origin of Life



- Art Quizzes:
-Miller-Urey Experiment
-Miller-Urey Experiment Results

4.3 The first cells had little internal structure.

- The first cells are thought to have arisen from aggregations of molecules that were more stable and, therefore, persisted longer.
- It has been suggested that RNA may have arisen before cells did, and subsequently became packaged within a membrane.
- Bacteria were the only life-forms on earth for about 1 billion years. At least three kinds of bacteria were present in ancient times: methane utiliziers, anaerobic photosynthesizers, and eventually O₂-forming photosynthesizers.

3. What evidence supports the argument that RNA evolved first on the early earth? What evidence supports the argument that proteins evolved first?



- Art Quiz:
Current Bubble Hypothesis

4. What are coacervates, and what characteristics do they have in common with organisms? Are they alive? Why or why not?

5. What were the earliest known organisms like, and when did they appear? What present-day organisms do they resemble?

4.4 The first eukaryotic cells were larger and more complex than bacteria.

- The first eukaryotes can be seen in the fossil record about 1.5 billion years ago. All organisms other than bacteria are their descendants.
- Biologists group all living organisms into six “kingdoms,” each profoundly different from the others.
- The two most ancient kingdoms contain prokaryotes (bacteria); the other four contain eukaryotes.
- There are approximately 10²⁰ stars in the universe similar to our sun. It is almost certain that life has evolved on planets circling some of them.

6. When did the first eukaryotes appear? By what mechanism are they thought to have evolved from the earlier prokaryotes?

7. What sorts of organisms are contained in each of the six kingdoms of life recognized by biologists?



- Key Events in Earth's History



Biology of the Cell

How Do the Cells of a Growing Plant Know in Which Direction to Elongate?

Sometimes questions that seem simple can be devilishly difficult to answer. Imagine, for example, that you are holding a green blade of grass in your hand. The grass blade has been actively growing, its cells dividing and then stretching and elongating as the blade lengthens. Did you ever wonder how the individual cells within the blade of grass know in what direction to grow?

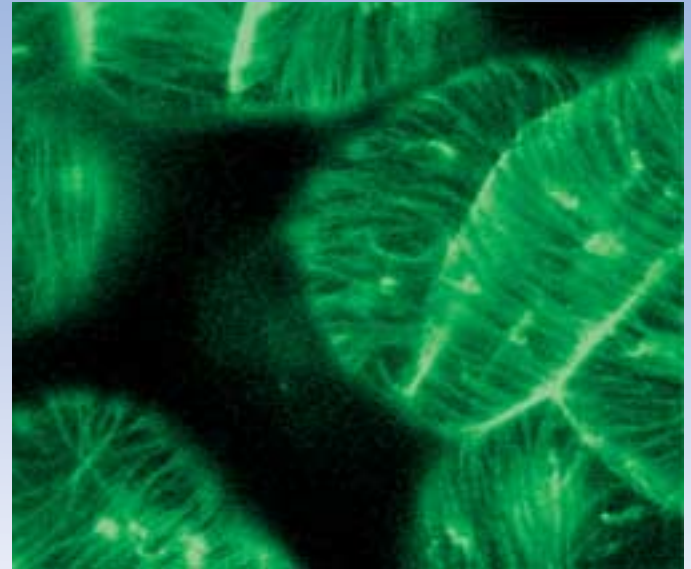
To answer this deceptively simple question, we will first need to provide answers to several others. Like Sherlock Holmes following a trail of clues, we must approach the answer we seek in stages.

Question One. First, we need to ask how a blade of grass is able to grow at all. Plant cells are very different from animal cells in one key respect: every plant cell is encased within a tough cell wall made of cellulose and other tough building materials. This wall provides structural strength and protection to the plant cell, just as armor plate does for a battle tank. But battle tanks can't stretch into longer shapes! How is a plant cell able to elongate?

It works like this. A growing cell first performs a little chemistry to make its wall slightly acidic. The acidity activates enzymes that attack the cell wall from the inside, rearranging cellulose cross-links until the wall loses its rigidity. The cell wall is now able to stretch. The cell then sucks in water, creating pressure. Like blowing up a long balloon, the now-stretchable cell elongates.

Question Two. In a growing plant organ, like the blade of grass, each growing cell balloons out lengthwise. Stating this more formally, a botanist would say the cell elongates parallel to the axis along which the blade of grass is extending. This observation leads to the second question we must answer: How does an individual plant cell control the direction in which it elongates?

It works like this. Before the stretchable cell balloons out, tiny microfibrils of cellulose are laid down along its inside surface. On a per weight basis, these tiny fibrils have the tensile strength of steel! Arrays of these cellulose microfibrils are



Seeing cortical microtubules. Cortical microtubules in epidermal cells of a fava bean are tagged with a fluorescent protein so that their ordered array can be seen.

organized in bands perpendicular to the axis of elongation, like steel belts. These tough bands reinforce the plant cell wall laterally, so that when the cell sucks in water, there is only one way for the cell to expand—lengthwise, along the axis.

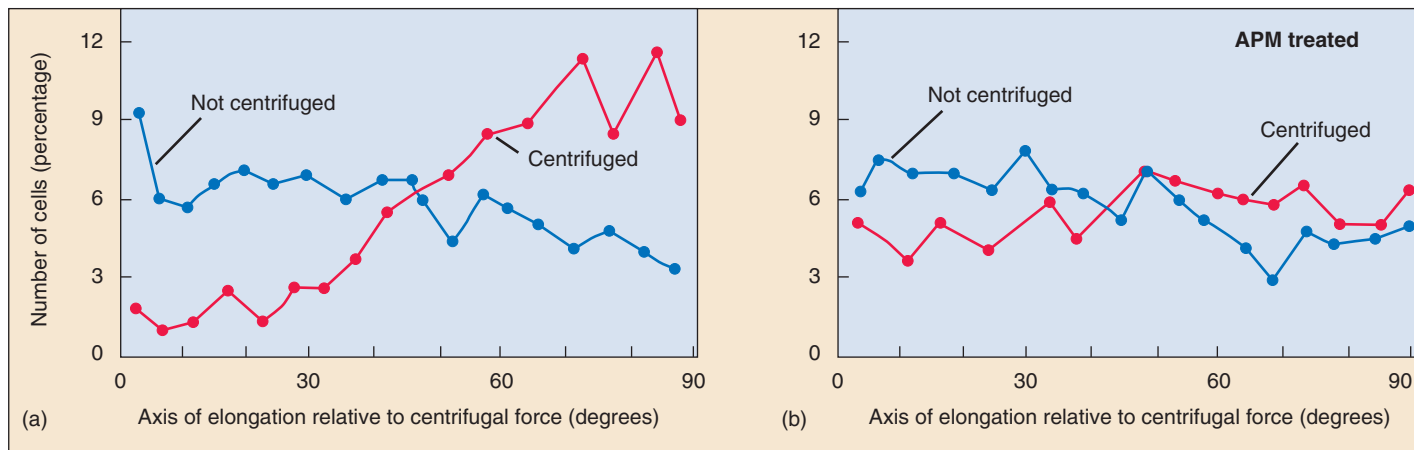
Question Three. Now we're getting somewhere. How are the newly made cellulose microfibrils laid down so that they are oriented correctly, perpendicular to the axis of elongation?

It works like this. The complicated enzymic machine that makes the cellulose microfibrils is guided by special guidereils that run like railroad tracks along the interior surface. The enzyme complex travels along these guidereils, laying down microfibrils as it goes. The guidereils are constructed of chainlike protein molecules called microtubules, assembled into overlapping arrays. Botanists call these arrays of microtubules associated with the interior of the cell surface "cortical microtubules."

Question Four. But we have only traded one puzzle for another. How are the cortical microtubules positioned correctly, perpendicular to the axis of elongation?

It works like this. In newly made cells, the microtubule assemblies are already present, but are not organized. They simply lie about in random disarray. As the cell prepares to elongate by lessening the rigidity of its cell wall, the microtubule assemblies become organized into the orderly transverse arrays we call cortical microtubules.

Question Five. Finally, we arrive at the question we had initially set out to answer. How are microtubule assemblies aligned properly? What sort of signal directs them to orient perpendicular to the axis of elongation? THAT is the question we need to answer.



Effects of centrifugation on cell elongation. (a) Protoplasts (plant cells without cell walls) that were centrifuged showed preferential elongation in a direction approximately perpendicular to the direction of the force. (b) Protoplasts exposed to APM, a microtubule disrupting chemical, exhibited random cell elongation with or without centrifugation.

The Experiment

This question has been addressed experimentally in a simple and direct way in the laboratory of Richard Cyr at Pennsylvania State University. Rigid plant cells conduct mechanical force well from one cell to another, and Carol Wymer (then a graduate student in the Cyr lab) suspected some sort of mechanical force is the signal guiding cortical microtubule alignment.

Wymer set out to test this hypothesis using centrifugation. If cortical microtubules are obtaining their positional information from an applied force, then their alignment should be affected by centrifugal force, and should be impossible if the integrity of the cell wall (which is supposedly transmitting the mechanical force) is perturbed with chemicals that prevent cell wall formation.

Wymer, along with others in the Cyr lab, started out with cells that were not elongated. She isolated protoplasts (cells without walls) from the tobacco plant, *Nicotiana tabacum*, by exposing the plant cells to enzymes that break down the cell wall, creating a spherical plant cell. If allowed to grow in culture, these protoplasts will eventually re-form their cell walls.

In order to examine the effects of directional force on the elongation patterns of plant cells, Wymer and co-workers exposed the tobacco protoplasts to a directional force generated by a centrifuge. Prior experiments had determined that centrifugation at the low speeds used in these experiments does not disrupt the integrity or shape of the protoplasts. The protoplasts were immobilized for centrifugation by embedding them in an agar medium supported in a mold. The embedded protoplasts were spun in a centrifuge at 450 rpm for 15 minutes. Following centrifugation, the embedded cells were cultured for 72 hours, allowing for cell elongation to occur.

Following centrifugation, fluorescently tagged microtubule antibody was applied to the protoplasts, which were then examined with immunofluorescence microscopy for microtubule orientation.

To confirm the involvement of microtubules as sensors of directional force in cell elongation, some protoplasts were incubated prior to centrifugation with a chemical herbicide, APM, which disrupts microtubules.

The Results

The biophysical force of centrifugation had significant effects on the pattern of elongation in the protoplasts following the 72-hour culturing period. The microtubules were randomly arranged in protoplasts that were not centrifuged but were more ordered in protoplasts that had been centrifuged. The microtubules in these cells were oriented parallel to the direction of the force, in a direction approximately perpendicular to the axis of elongation (graph *a* above). These results support the hypothesis that plant cell growth responds to an external biophysical force.

It is true that plant cells are not usually exposed to the type of mechanical force generated by centrifugation but this manipulation demonstrates how a physical force can affect cell growth, assumably by influencing the orientation of cortical microtubules. These could be small, transient biophysical forces acting at the subcellular level.

In preparations exposed to the microtubule disrupting chemical amiprophos-methyl (APM), directed elongation was blocked (graph *b* above). This suggests that reorientation of microtubules is indeed necessary to direct the elongation axis of the plant.

Taken together, these experiments support the hypothesis that the microtubule reorientation that directs cell elongation may be oriented by a mechanical force. Just what the natural force might be is an open question, providing an opportunity for lots of interesting future experiments that are being pursued in the Cyr lab.



To explore this experiment further, go to the Virtual Lab at www.mhhe.com/raven6/vlab2.mhtml

5

Cell Structure

Concept Outline

5.1 All organisms are composed of cells.

Cells. A cell is a membrane-bounded unit that contains DNA and cytoplasm. All organisms are cells or aggregates of cells, descendants of the first cells.

Cells Are Small. The greater relative surface area of small cells enables more rapid communication between the cell interior and the environment.

5.2 Eukaryotic cells are far more complex than bacterial cells.

Bacteria Are Simple Cells. Bacterial cells are small and lack membrane-bounded organelles.

Eukaryotic Cells Have Complex Interiors. Eukaryotic cells are compartmentalized by membranes.

5.3 Take a tour of a eukaryotic cell.

The Nucleus: Information Center for the Cell. The nucleus of a eukaryotic cell isolates the cell's DNA.

The Endoplasmic Reticulum: Compartmentalizing the Cell. An extensive system of membranes subdivides the cell interior.

The Golgi Apparatus: Delivery System of the Cell. A system of membrane channels collects, modifies, packages, and distributes molecules within the cell.

Vesicles: Enzyme Storehouses. Sacs that contain enzymes digest or modify particles in the cell, while other vesicles transport substances in and out of cells.

Ribosomes: Sites of Protein Synthesis. An RNA-protein complex directs the production of proteins.

Organelles That Contain DNA. Some organelles with very different functions contain their own DNA.

The Cytoskeleton: Interior Framework of the Cell. A network of protein fibers supports the shape of the cell and anchors organelles.

Cell Movement. Eukaryotic cell movement utilizes cytoskeletal elements.

Special Things about Plant Cells. Plant cells have a large central vacuole and strong, multilayered cell walls.

5.4 Symbiosis played a key role in the origin of some eukaryotic organelles.

Endosymbiosis. Mitochondria and chloroplasts may have arisen from prokaryotes engulfed by other prokaryotes.

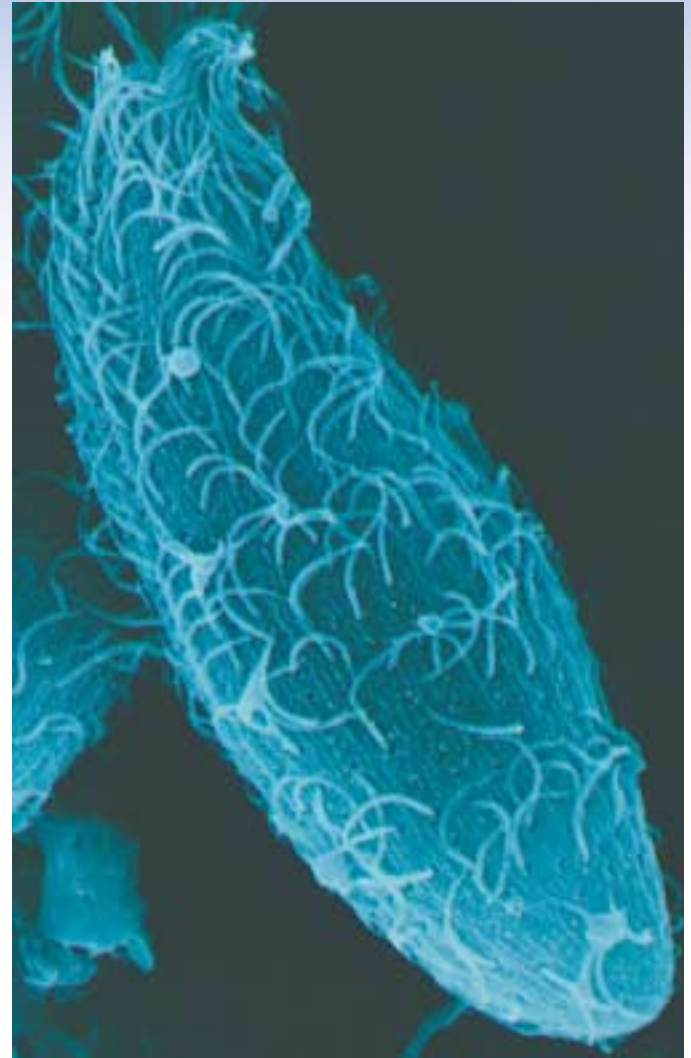


FIGURE. 5.1

The single-celled organism *Dileptus*. The hairlike projections that cover its surface are cilia, which it undulates to propel itself through the water (1000 $\times$).

All organisms are composed of cells. The gossamer wing of a butterfly is a thin sheet of cells, and so is the glistening outer layer of your eyes. The hamburger or tomato you eat is composed of cells, and its contents soon become part of your cells. Some organisms consist of a single cell too small to see with the unaided eye (figure 5.1), while others, like us, are composed of many cells. Cells are so much a part of life as we know it that we cannot imagine an organism that is not cellular in nature. In this chapter we will take a close look at the internal structure of cells. In the following chapters, we will focus on cells in action—on how they communicate with their environment, grow, and reproduce.

5.1 All organisms are composed of cells.

Cells

What is a typical cell like, and what would we find inside it? The general plan of cellular organization varies in the cells of different organisms, but despite these modifications, all cells resemble each other in certain fundamental ways. Before we begin our detailed examination of cell structure, let's first summarize three major features all cells have in common: a plasma membrane, a nucleoid or nucleus, and cytoplasm.

The Plasma Membrane Surrounds the Cell

The **plasma membrane** encloses a cell and separates its contents from its surroundings. The plasma membrane is a phospholipid bilayer about 5 to 10 nanometers (5 to 10 billionths of a meter) thick with proteins embedded in it. Viewed in cross-section with the electron microscope, such membranes appear as two dark lines separated by a lighter area. This distinctive appearance arises from the tail-to-tail packing of the phospholipid molecules that make up the membrane (see figure 3.18). The proteins of a membrane may have large hydrophobic domains, which associate with and become embedded in the phospholipid bilayer.

The proteins of the plasma membrane are in large part responsible for a cell's ability to interact with its environment. *Transport proteins* help molecules and ions move across the plasma membrane, either from the environment to the interior of the cell or vice versa. *Receptor proteins* induce changes within the cell when they come in contact with specific molecules in the environment, such as hormones. *Markers* identify the cell as a particular type. This is especially important in multicellular

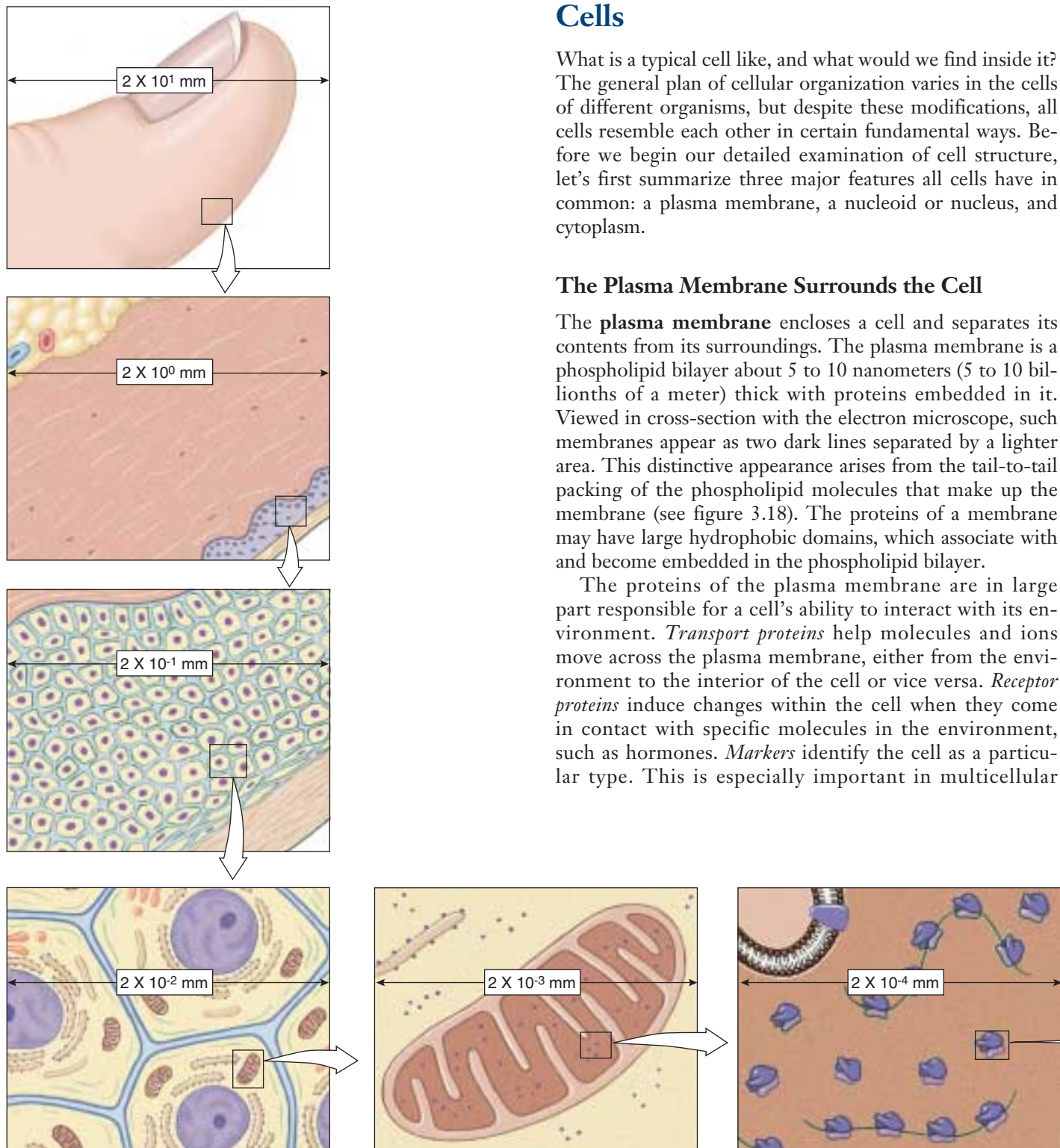


FIGURE 5.2

The size of cells and their contents. This diagram shows the size of human skin cells, organelles, and molecules. In general, the diameter of a human skin cell is 20 micrometers (μm) or 2×10^{-2} mm, of a mitochondrion is 2 μm or 2×10^{-3} mm, of a ribosome is 20 nanometers (nm) or 2×10^{-5} mm, of a protein molecule is 2 nm or 2×10^{-6} mm, and of an atom is 0.2 nm or 2×10^{-7} mm.

organisms, whose cells must be able to recognize each other as they form tissues.

We'll examine the structure and function of cell membranes more thoroughly in chapter 6.

The Central Portion of the Cell Contains the Genetic Material

Every cell contains DNA, the hereditary molecule. In **prokaryotes** (bacteria), most of the genetic material lies in a single circular molecule of DNA. It typically resides near the center of the cell in an area called the **nucleoid**, but this area is not segregated from the rest of the cell's interior by membranes. By contrast, the DNA of **eukaryotes** is contained in the **nucleus**, which is surrounded by two membranes. In both types of organisms, the DNA contains the genes that code for the proteins synthesized by the cell.

The Cytoplasm Comprises the Rest of the Cell's Interior

A semifluid matrix called the **cytoplasm** fills the interior of the cell, exclusive of the nucleus (nucleoid in prokaryotes) lying within it. The cytoplasm contains the chemical wealth of the cell: the sugars, amino acids, and proteins the cell uses to carry out its everyday activities. In eukaryotic cells, the cytoplasm also contains specialized membrane-bounded compartments called **organelles**.

The Cell Theory

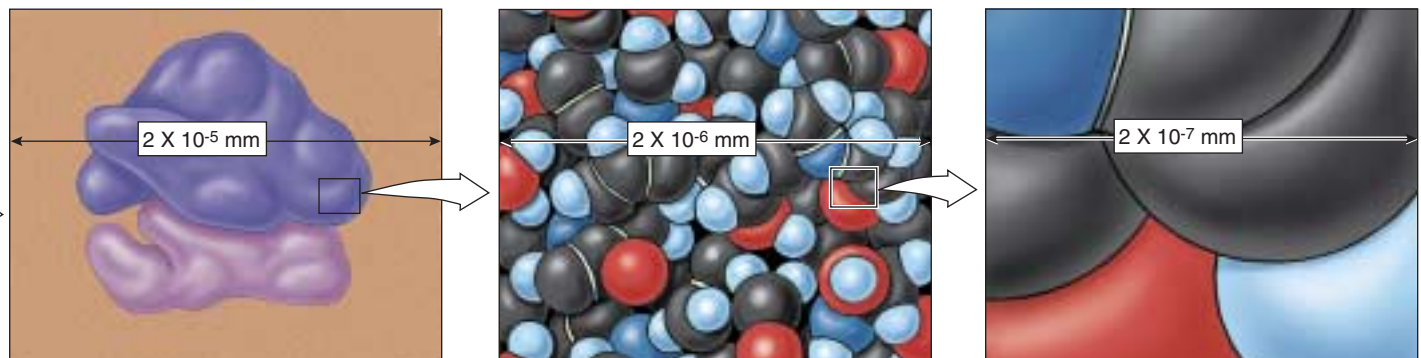
A general characteristic of cells is their microscopic size. While there are a few exceptions—the marine alga *Acetabularia* can be up to 5 centimeters long—a typical eukaryotic cell is 10 to 100 micrometers (10 to 100 millionths of a meter) in diameter (figure 5.2); most bacterial cells are only 1 to 10 micrometers in diameter.

Because cells are so small, no one observed them until microscopes were invented in the mid-seventeenth century. Robert Hooke first described cells in 1665, when he used a microscope he had built to examine a thin slice of cork, a nonliving tissue found in the bark of certain trees. Hooke observed a honeycomb of tiny, empty (because the cells were dead) compartments. He called the compartments in the cork *cellulae* (Latin, “small rooms”), and the term has come down to us as *cells*. The first living cells were observed a few years later by the Dutch naturalist Antonie van Leeuwenhoek, who called the tiny organisms that he observed “animalcules,” meaning little animals. For another century and a half, however, biologists failed to recognize the importance of cells. In 1838, botanist Matthias Schleiden made a careful study of plant tissues and developed the first statement of the cell theory. He stated that all plants “are aggregates of fully individualized, independent, separate beings, namely the cells themselves.” In 1839, Theodor Schwann reported that all animal tissues also consist of individual cells.

The **cell theory**, in its modern form, includes the following three principles:

1. All organisms are composed of one or more cells, and the life processes of metabolism and heredity occur within these cells.
2. Cells are the smallest living things, the basic units of organization of all organisms.
3. Cells arise only by division of a previously existing cell. Although life likely evolved spontaneously in the environment of the early earth, biologists have concluded that no additional cells are originating spontaneously at present. Rather, life on earth represents a continuous line of descent from those early cells.

A cell is a membrane-bounded unit that contains the DNA hereditary machinery and cytoplasm. All organisms are cells or aggregates of cells.



Cells Are Small

How many cells are big enough to see with the unaided eye? Other than egg cells, not many. Most are less than 50 micrometers in diameter, far smaller than the period at the end of this sentence.

The Resolution Problem

How do we study cells if they are too small to see? The key is to understand *why* we can't see them. The reason we can't see such small objects is the limited resolution of the human eye. **Resolution** is defined as the minimum distance two points can be apart and still be distinguished as two separated points. When two objects are closer together than about 100 micrometers, the light reflected from each strikes the same "detector" cell at the rear of the eye. Only when the objects are farther than 100 micrometers apart will the light from each strike different cells, allowing your eye to resolve them as two objects rather than one.

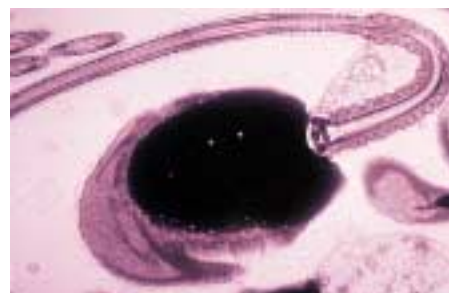
Microscopes

One way to increase resolution is to increase magnification, so that small objects appear larger. Robert Hooke and Antonie van Leeuwenhoek were able to see small cells by magnifying their size, so that the cells appeared larger than the 100-micrometer limit imposed by the human eye. Hooke and van Leeuwenhoek accomplished this feat with **microscopes** that magnified images of cells by bending light through a glass lens. The size of the image that falls on the sheet of detector cells lining the back of your eye depends on how close the object is to your eye—the closer the object, the bigger the image. Your eye, however, is incapable of focusing comfortably on an object closer than about 25 centimeters, because the eye is limited by the size and thickness of its lens. Hooke and van Leeuwenhoek assisted the eye by interposing a glass lens between object and eye. The glass lens adds additional focusing power. Because the glass lens makes the object appear closer, the image on the back of the eye is bigger than it would be without the lens.

Modern light microscopes use two magnifying lenses (and a variety of correcting lenses) that act like back-to-back eyes. The first lens focuses the image of the object on the second lens, which magnifies it again and focuses it on the back of the eye. Microscopes that magnify in stages using several lenses are called **compound microscopes**. They can resolve structures that are separated by more than 200 nm. An image from a compound microscope is shown in figure 5.3*a*.



(a)



(b)



(c)

FIGURE 5.3

Human sperm cells viewed with three different microscopes. (a) Image of sperm taken with a light microscope. (b) Transmission electron micrograph of a sperm cell. (c) Scanning electron micrograph of sperm cells.

Increasing Resolution

Light microscopes, even compound ones, are not powerful enough to resolve many structures within cells. For example, a membrane is only 5 nanometers thick. Why not just add another magnifying stage to the microscope and so increase its resolving power? Because when two objects are closer than a few hundred nanometers, the light beams reflecting from the two images start to overlap. The only way two light beams can get closer together and still be resolved is if their "wavelengths" are shorter.

One way to avoid overlap is by using a beam of electrons rather than a beam of light. Electrons have a much shorter wavelength, and a microscope employing electron beams has 1000 times the resolving power of a light microscope. **Transmission electron microscopes**, so called because the electrons used to visualize the specimens are transmitted through the material, are capable of resolving objects only 0.2 nanometer apart—just twice the diameter of a hydrogen atom! Figure 5.3*b* shows a transmission electron micrograph.

A second kind of electron microscope, the **scanning electron microscope**, beams the electrons onto the surface of the specimen from a fine probe that passes rapidly back and forth. The electrons reflected back from the surface of the specimen, together with other electrons that the specimen itself emits as a result of the bombardment, are amplified and transmitted to a television screen, where the image can be viewed and photographed. Scanning electron microscopy yields striking three-dimensional images and has improved our understanding of many biological and physical phenomena (figure 5.3*c*).

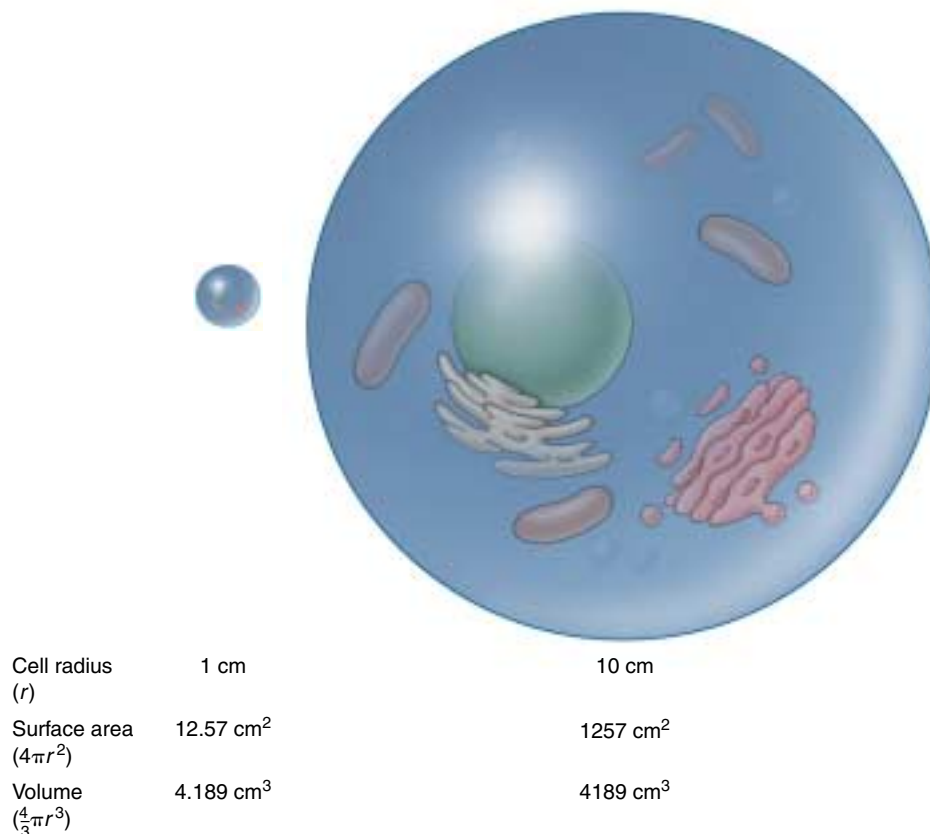


FIGURE 5.4

Surface area-to-volume ratio. As a cell gets larger, its volume increases at a faster rate than its surface area. If the cell radius increases by 10 times, the surface area increases by 100 times, but the volume increases by 1000 times. A cell's surface area must be large enough to meet the needs of its volume.

Why Aren't Cells Larger?

Most cells are not large for practical reasons. The most important of these is communication. The different regions of a cell need to communicate with one another in order for the cell as a whole to function effectively. Proteins and organelles are being synthesized, and materials are continually entering and leaving the cell. All of these processes involve the diffusion of substances at some point, and the larger a cell is, the longer it takes for substances to diffuse from the plasma membrane to the center of the cell. For this reason, an organism made up of many relatively small cells has an advantage over one composed of fewer, larger cells.

The advantage of small cell size is readily visualized in terms of the **surface area-to-volume ratio**. As a cell's size increases, its volume increases much more rapidly than its surface area. For a spherical cell, the increase in surface area is equal to the square of the increase in diameter, while the increase in volume is equal to the cube of the increase in diameter. Thus, if two cells differ by a factor of 10 cm in diameter, the larger cell will have 10², or 100 times, the surface area, but 10³, or 1000 times, the volume, of the smaller cell (figure 5.4). A cell's surface provides its only opportunity for interaction with

the environment, as all substances enter and exit a cell via the plasma membrane. This membrane plays a key role in controlling cell function, and because small cells have more surface area per unit of volume than large ones, the control is more effective when cells are relatively small.

Although most cells are small, some cells are nonetheless quite large and have apparently overcome the surface area-to-volume problem by one or more adaptive mechanisms. For example, some cells have more than one nucleus, allowing genetic information to be spread around a large cell. Also, some large cells actively move material around their cytoplasm so that diffusion is not a limiting factor. Lastly, some large cells, like your own neurons, are long and skinny so that any given point in the cytoplasm is close to the plasma membrane, and thus diffusion between the inside and outside of the cell can still be rapid.

Multicellular organisms usually consist of many small cells rather than a few large ones because small cells function more efficiently. They have a greater relative surface area, enabling more rapid communication between the center of the cell and the environment.

5.2 Eukaryotic cells are far more complex than bacterial cells.

Bacteria Are Simple Cells

Prokaryotes, the bacteria, are the simplest organisms. Prokaryotic cells are small, consisting of cytoplasm surrounded by a plasma membrane and encased within a rigid cell wall, with no distinct interior compartments (figure 5.5). A prokaryotic cell is like a one-room cabin in which eating, sleeping, and watching TV all occur in the same room. Bacteria are very important in the economy of living organisms. They harvest light in photosynthesis, break down dead organisms and recycle their components, cause disease, and are involved in many important industrial processes. Bacteria are the subject of chapter 34.

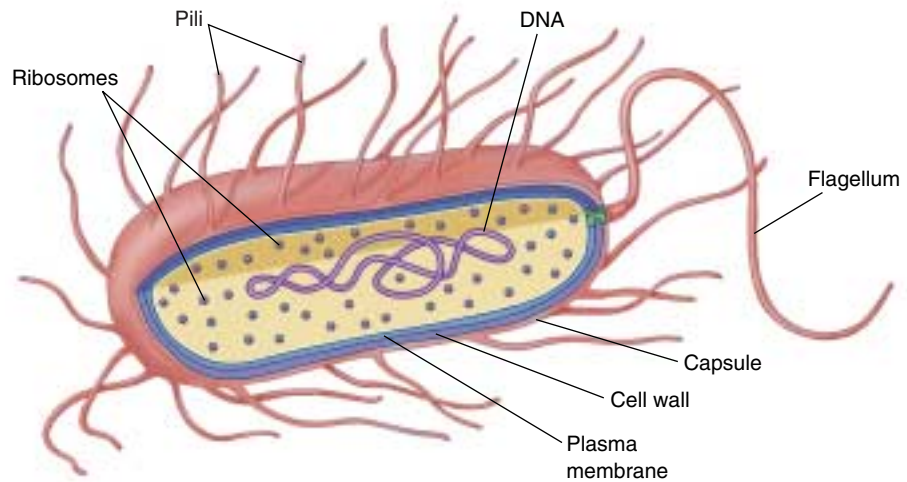


FIGURE 5.5
Structure of a bacterial cell. Generalized cell organization of a bacterium. Some bacteria have hairlike growths on the outside of the cell called pili.

Strong Cell Walls

Most bacteria are encased by a strong **cell wall** composed of *peptidoglycan*, which consists of a carbohydrate matrix (polymers of sugars) that is cross-linked by short polypeptide units. No eukaryotes possess cell walls with this type of chemical composition. With a few exceptions like TB and leprosy-causing bacteria, all bacteria may be classified into two types based on differences in their cell walls detected by the Gram staining procedure. The name refers to the Danish microbiologist Hans Christian Gram, who developed the procedure to detect the presence of certain disease-causing bacteria. **Gram-positive** bacteria have a thick, single-layered cell wall that retains a violet dye from the Gram stain procedure, causing the stained cells to appear purple under a microscope. More complex cell walls have evolved in other groups of bacteria. In them, the wall is multilayered and does not retain the purple dye after Gram staining; such bacteria exhibit the background red dye and are characterized as **gram-negative**.

The susceptibility of bacteria to antibiotics often depends on the structure of their cell walls. Penicillin and vancomycin, for example, interfere with the ability of bacteria to cross-link the peptide units that hold the carbohydrate chains of the wall together. Like removing all the nails from a wooden house, this destroys the integrity of the matrix, which can no longer prevent water from rushing in, swelling the cell to bursting.

Cell walls protect the cell, maintain its shape, and prevent excessive uptake of water. Plants, fungi, and most protists also have cell walls of a different chemical structure, which we will discuss in later chapters.

Long chains of sugars called polysaccharides cover the cell walls of many bacteria. They enable a bacterium to adhere to teeth, skin, food—practically any surface that will support their growth. Many disease-causing bacteria secrete a jellylike protective capsule of polysaccharide around the cell.

Rotating Flagella

Some bacteria use a flagellum (plural, flagella) to move. **Flagella** are long, threadlike structures protruding from the surface of a cell that are used in locomotion and feeding. Bacterial flagella are protein fibers that extend out from a bacterial cell. There may be one or more per cell, or none, depending on the species. Bacteria can swim at speeds up to 20 cell diameters per second by rotating their flagella like screws (figure 5.6). A “motor” unique to bacteria that is embedded within their cell walls and membranes powers the rotation. Only a few eukaryotic cells have structures that truly rotate.

Simple Interior Organization

If you were to look at an electron micrograph of a bacterial cell, you would be struck by the cell’s simple organization. There are few, if any, internal compartments, and while they contain simple structures like ribosomes, most have no membrane-bounded organelles, the kinds so characteristic of eukaryotic cells. Nor do bacteria have a true nucleus. The entire cytoplasm of a bacterial cell is one unit with no internal support structure. Consequently,

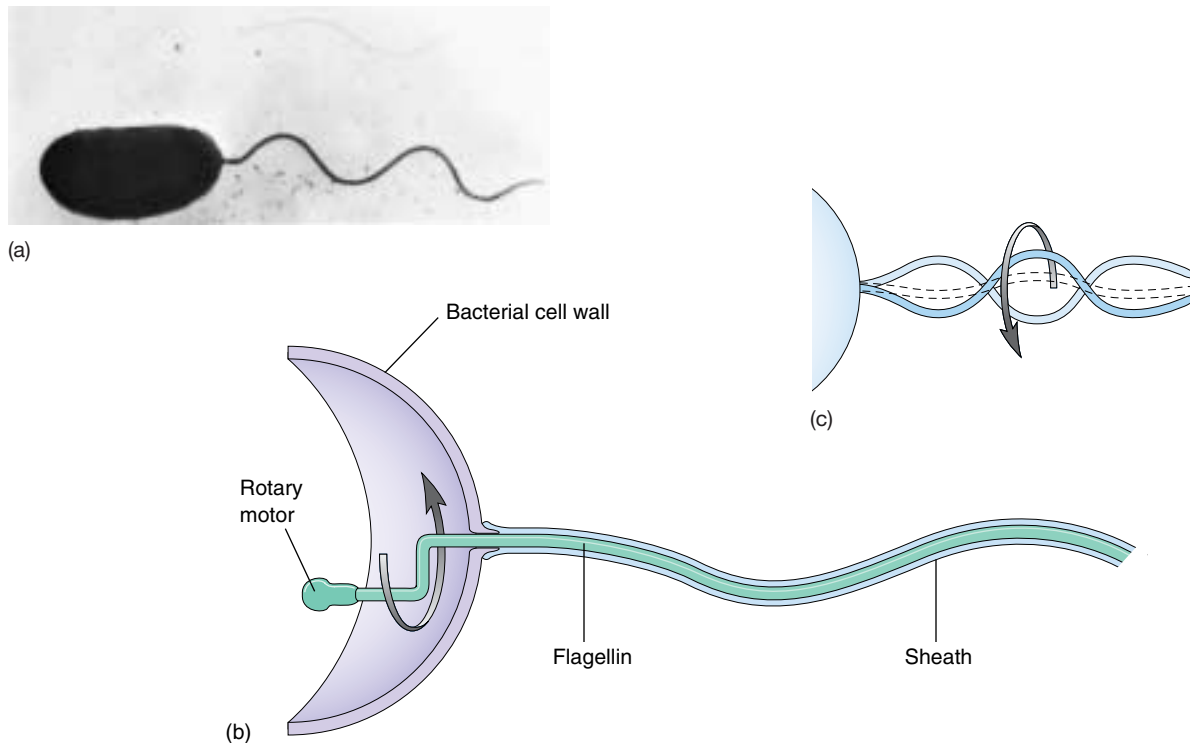


FIGURE 5.6

Bacteria swim by rotating their flagella. (a) The photograph is of *Vibrio cholerae*, the microbe that causes the serious disease cholera. The unsheathed core visible at the top of the photograph is composed of a single crystal of the protein flagellin. (b) In intact flagella, the core is surrounded by a flexible sheath. Imagine that you are standing inside the *Vibrio* cell, turning the flagellum like a crank. (c) You would create a spiral wave that travels down the flagellum, just as if you were turning a wire within a flexible tube. The bacterium creates this kind of rotary motion when it swims.

the strength of the cell comes primarily from its rigid wall (see figure 5.5).

The plasma membrane of a bacterial cell carries out some of the functions organelles perform in eukaryotic cells. When a bacterial cell divides, for example, the bacterial chromosome, a simple circle of DNA, replicates before the cell divides. The two DNA molecules that result from the replication attach to the plasma membrane at different points, ensuring that each daughter cell will contain one of the identical units of DNA. Moreover, some photosynthetic bacteria, such as cyanobacteria and *Prochloron* (figure 5.7), have an extensively folded plasma membrane, with the folds extending into the cell's interior. These membrane folds contain the bacterial pigments connected with photosynthesis.

Because a bacterial cell contains no membrane-bounded organelles, the DNA, enzymes, and other cytoplasmic constituents have access to all parts of the cell. Reactions are not compartmentalized as they are in eukaryotic cells, and the whole bacterium operates as a single unit.

Bacteria are small cells that lack interior organization. They are encased by an exterior wall composed of carbohydrates cross-linked by short polypeptides, and some are propelled by rotating flagella.

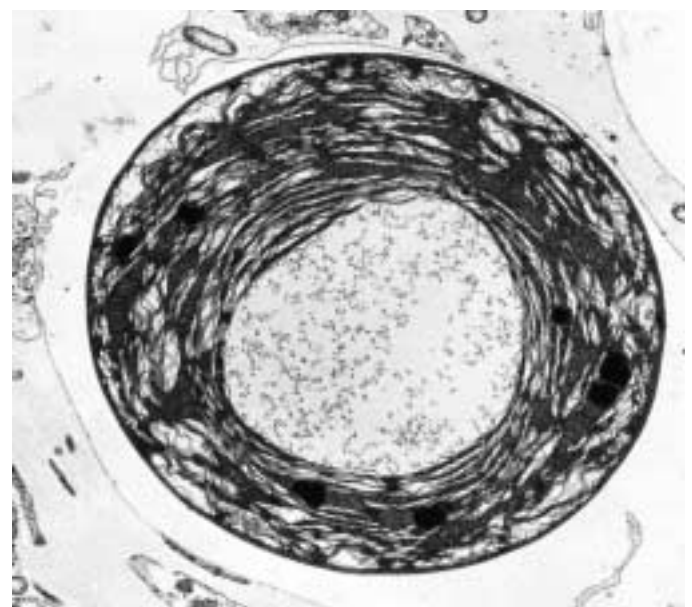


FIGURE 5.7

Electron micrograph of a photosynthetic bacterial cell. Extensive folded photosynthetic membranes are visible in this *Prochloron* cell (14,500 $\times$). The single, circular DNA molecule is located in the clear area in the central region of the cell.

Eukaryotic Cells Have Complex Interiors

Eukaryotic cells (figures 5.8 and 5.9) are far more complex than prokaryotic cells. The hallmark of the eukaryotic cell is compartmentalization. The interiors of eukaryotic cells contain numerous **organelles**, membrane-bounded structures that close off compartments within which multiple biochemical processes can proceed simultaneously and independently. Plant cells often have a large membrane-bounded sac called a **central vacuole**, which stores proteins, pigments, and waste materials. Both plant and animal cells contain **vesicles**, smaller sacs that store and transport a variety of materials. Inside the nucleus, the DNA is

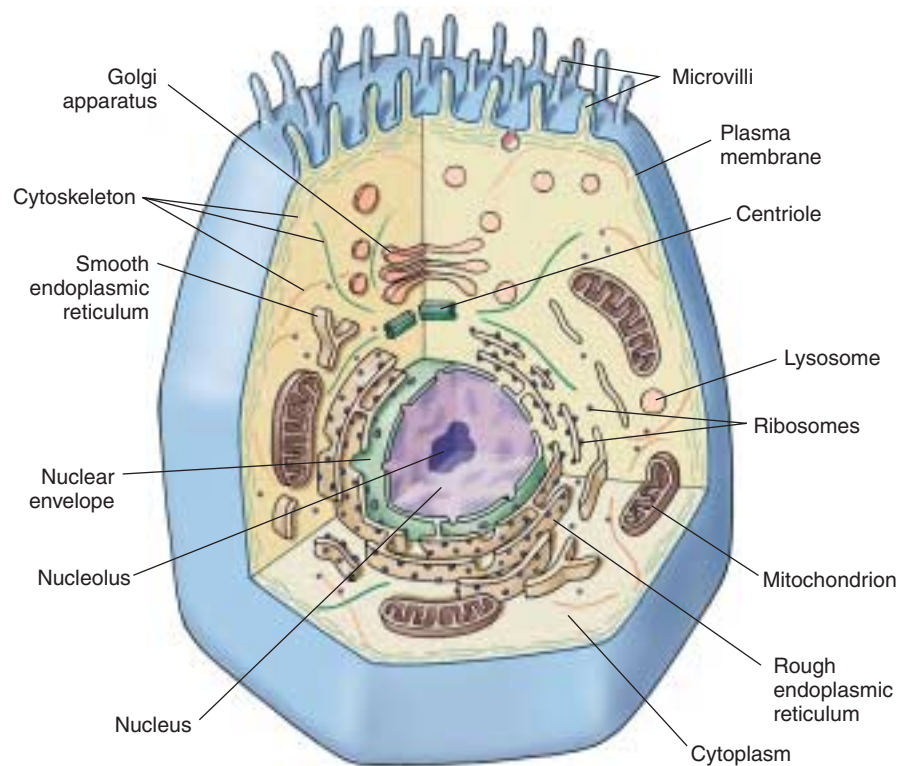
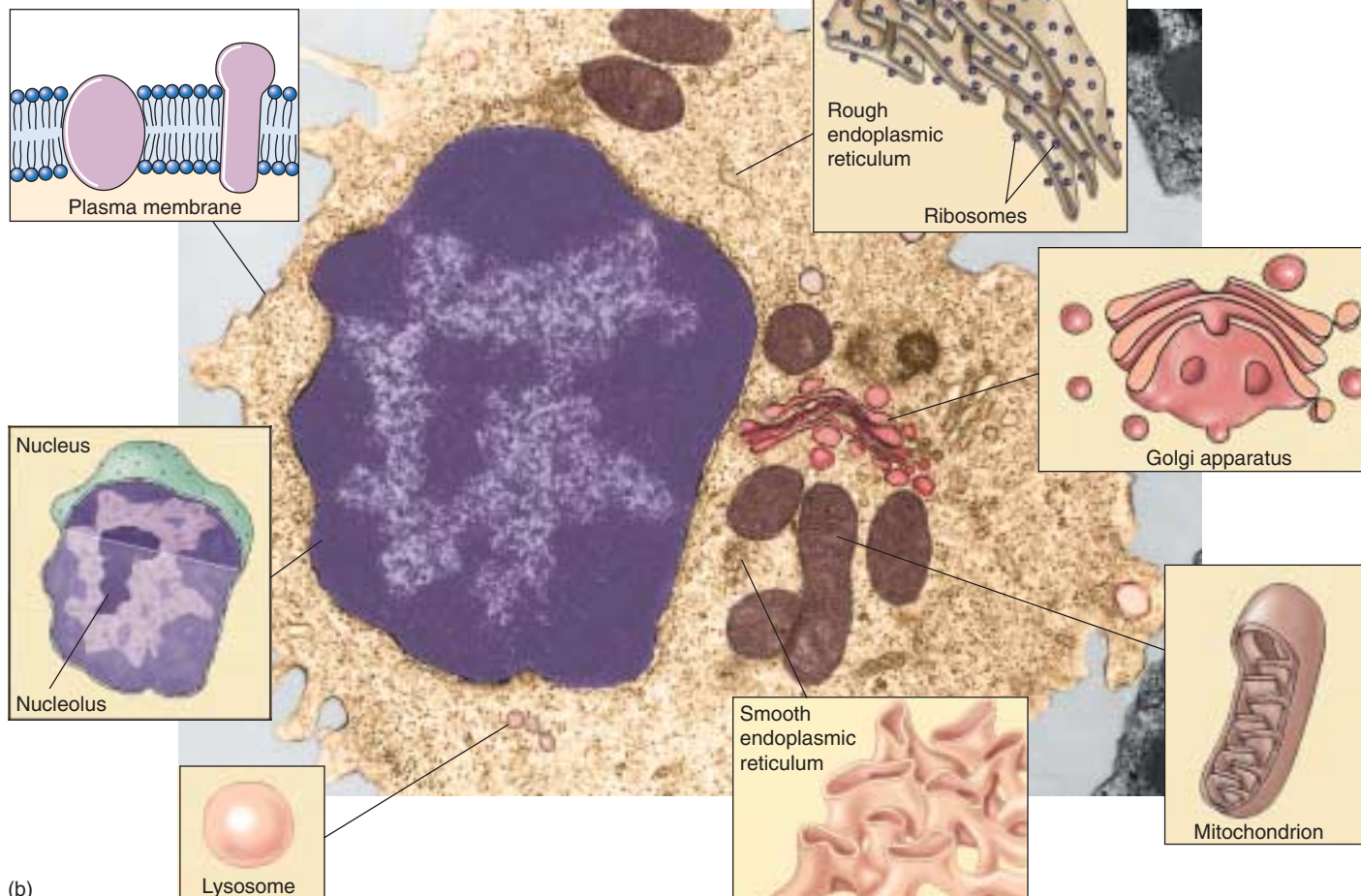


FIGURE 5.8

Structure of an animal cell. (a) A generalized diagram of an animal cell. (b) Micrograph of a human white blood cell (40,500 $\times$) with drawings detailing organelles.

(a)

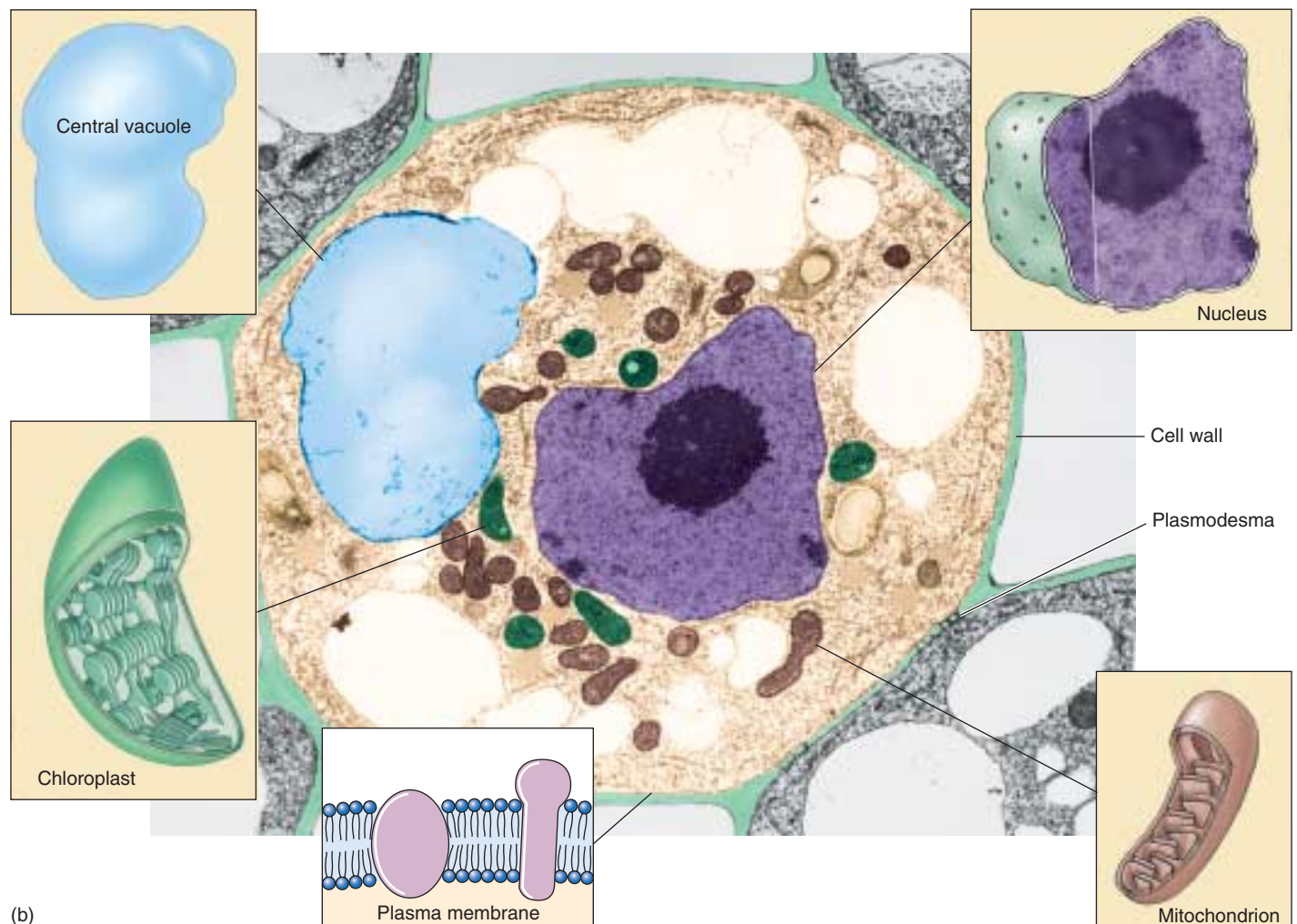
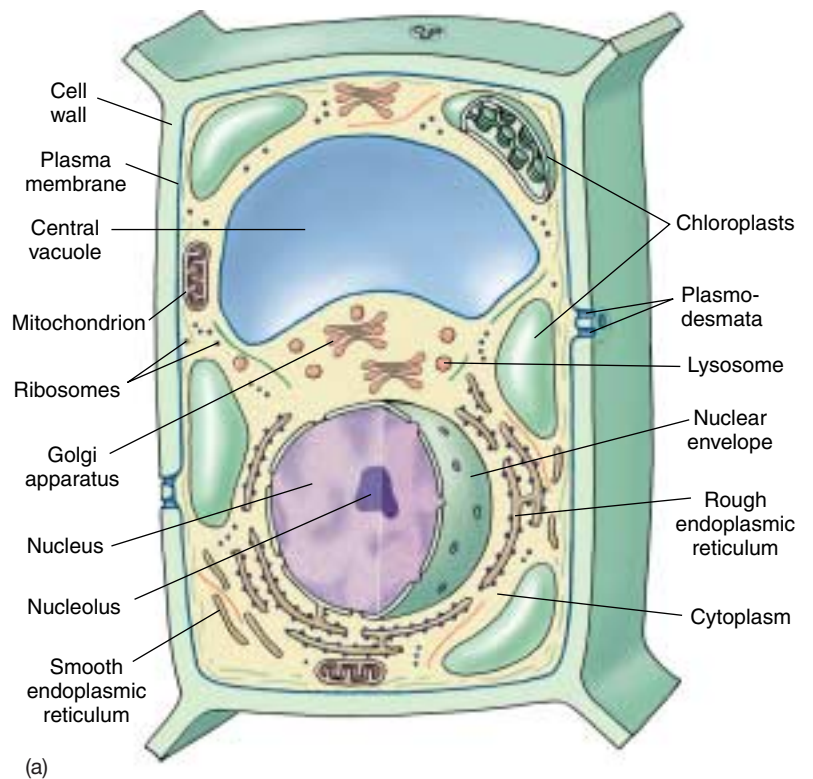


(b)

wound tightly around proteins and packaged into compact units called **chromosomes**. All eukaryotic cells are supported by an internal protein scaffold, the **cytoskeleton**. While the cells of animals and some protists lack cell walls, the cells of fungi, plants, and many protists have strong **cell walls** composed of cellulose or chitin fibers embedded in a matrix of other polysaccharides and proteins. This composition is very different from the peptidoglycan that makes up bacterial cell walls. Let's now examine the structure and function of the internal components of eukaryotic cells in more detail.

Eukaryotic cells contain membrane-bounded organelles that carry out specialized functions.

FIGURE 5.9
Structure of a plant cell. A generalized illustration (a) and micrograph (b) of a plant cell. Most mature plant cells contain large central vacuoles which occupy a major portion of the internal volume of the cell (14,000 $\times$).



5.3 Take a tour of a eukaryotic cell.

The Nucleus: Information Center for the Cell

The largest and most easily seen organelle within a eukaryotic cell is the **nucleus** (Latin, for kernel or nut), first described by the English botanist Robert Brown in 1831. Nuclei are roughly spherical in shape and, in animal cells, they are typically located in the central region of the cell (figure 5.10). In some cells, a network of fine cytoplasmic

filaments seems to cradle the nucleus in this position. The nucleus is the repository of the genetic information that directs all of the activities of a living eukaryotic cell. Most eukaryotic cells possess a single nucleus, although the cells of fungi and some other groups may have several to many nuclei. Mammalian erythrocytes (red blood cells) lose their nuclei when they mature. Many nuclei exhibit a dark-staining zone called the nucleolus, which is a region where intensive synthesis of ribosomal RNA is taking place.

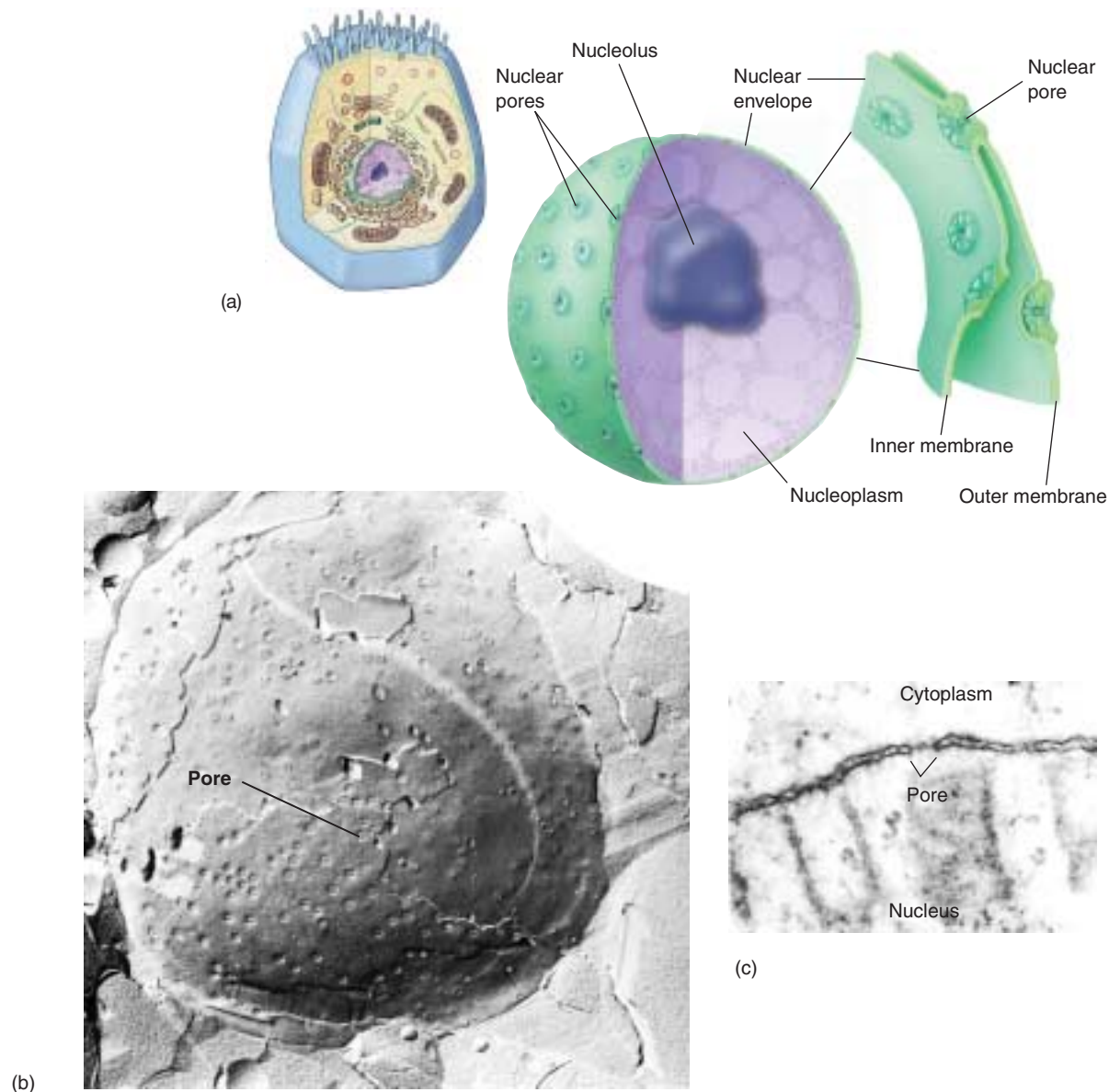


FIGURE 5.10

The nucleus. (a) The nucleus is composed of a double membrane, called a nuclear envelope, enclosing a fluid-filled interior containing the chromosomes. In cross-section, the individual nuclear pores are seen to extend through the two membrane layers of the envelope; the dark material within the pore is protein, which acts to control access through the pore. (b) A freeze-fracture scanning electron micrograph of a cell nucleus showing nuclear pores (9500 $\times$). (c) A transmission electron micrograph (see figure 6.6) of the nuclear membrane showing a nuclear pore.

The Nuclear Envelope: Getting In and Out

The surface of the nucleus is bounded by *two* phospholipid bilayer membranes, which together make up the **nuclear envelope** (see figure 5.10). The outer membrane of the nuclear envelope is continuous with the cytoplasm's interior membrane system, called the endoplasmic reticulum. Scattered over the surface of the nuclear envelope, like craters on the moon, are shallow depressions called **nuclear pores**. These pores form 50 to 80 nanometers apart at locations where the two membrane layers of the nuclear envelope pinch together. Rather than being empty, nuclear pores are filled with proteins that act as molecular channels, permitting certain molecules to pass into and out of the nucleus. Passage is restricted primarily to two kinds of molecules: (1) proteins moving into the nucleus to be incorporated into nuclear structures or to catalyze nuclear activities; and (2) RNA and protein-RNA complexes formed in the nucleus and exported to the cytoplasm.

The Chromosomes: Packaging the DNA

In both bacteria and eukaryotes, DNA contains the hereditary information specifying cell structure and function. However, unlike the circular DNA of bacteria, the DNA of eukaryotes is divided into several linear **chromosomes**. Except when a cell is dividing, its chromosomes are fully extended into threadlike strands, called **chromatin**, of DNA complexed with protein. This open arrangement allows proteins to attach to specific nucleotide sequences along the DNA. Without this access, DNA could not direct the day-to-day activities of the cell. The chromosomes are associated with packaging proteins called **histones**. When a cell prepares to divide, the DNA coils up around the histones into a highly condensed form. In the initial stages of this condensation, units of histone can be seen with DNA wrapped around like a sash. Called **nucleosomes**, these initial aggregations resemble beads on a string (figure 5.11). Coiling continues until the DNA is in a compact mass. Under a light microscope, these fully condensed chromosomes are readily seen in dividing cells as densely staining rods (figure 5.12). After cell division, eukaryotic chromosomes uncoil and can no longer be individually distinguished with a light microscope. Uncoiling the chromosomes into a more extended form permits enzymes to make RNA copies of DNA. Only by means of these RNA copies can the information in the DNA be used to direct the synthesis of proteins.

The nucleus of a eukaryotic cell contains the cell's hereditary apparatus and isolates it from the rest of the cell. A distinctive feature of eukaryotes is the organization of their DNA into complex chromosomes.

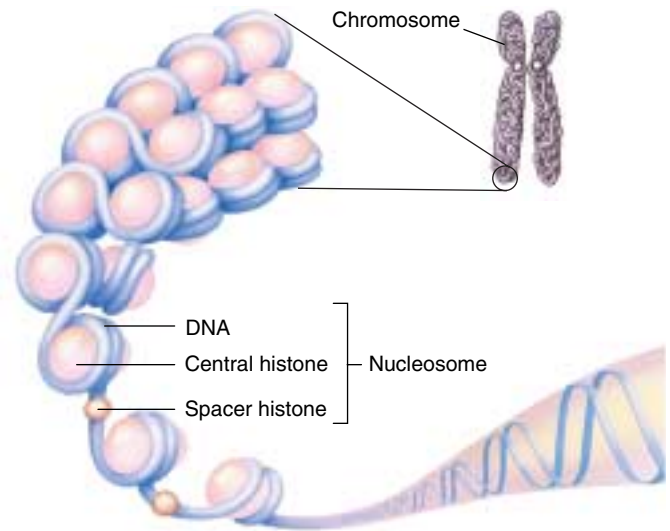


FIGURE 5.11
Nucleosomes. Each nucleosome is a region in which the DNA is wrapped tightly around a cluster of histone proteins.

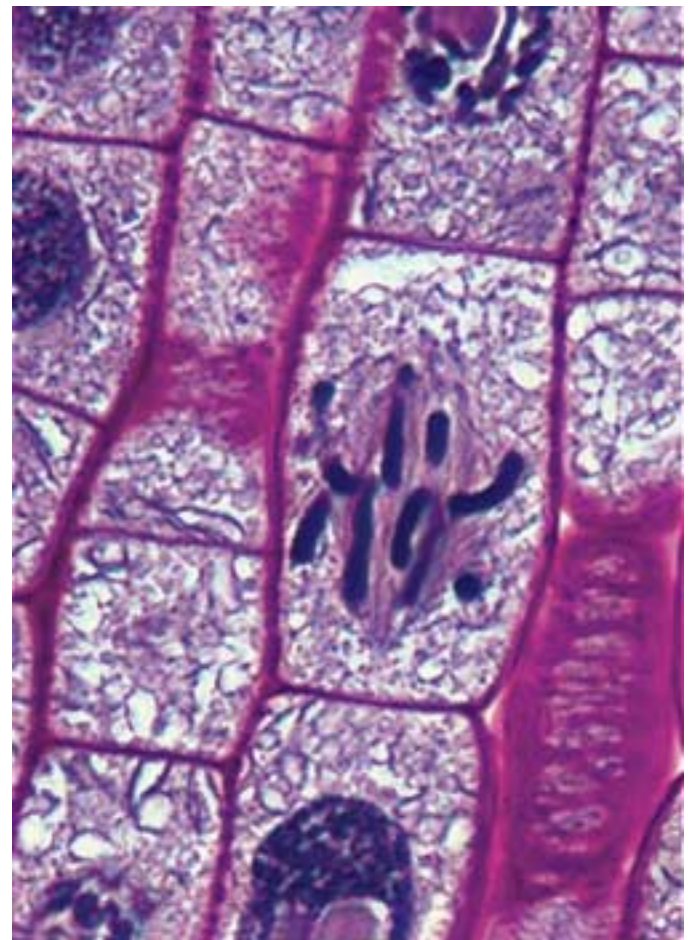


FIGURE 5.12
Eukaryotic chromosomes. These condensed chromosomes within an onion root tip are visible under the light microscope (500 $\times$).

The Endoplasmic Reticulum: Compartmentalizing the Cell

The interior of a eukaryotic cell is packed with membranes (table 5.1). So thin that they are invisible under the low resolving power of light microscopes, this **endomembrane system** fills the cell, dividing it into compartments, channeling the passage of molecules through the interior of the cell, and providing surfaces for the synthesis of lipids and some proteins. The presence of these membranes in eukaryotic cells constitutes one of the most fundamental distinctions between eukaryotes and prokaryotes.

The largest of the internal membranes is called the **endoplasmic reticulum (ER)**. The term *endoplasmic* means “within the cytoplasm,” and the term *reticulum* is Latin for “a little net.” Like the plasma membrane, the ER is composed of a lipid bilayer embedded with proteins. It weaves in sheets through the interior of the cell, creating a series of channels between its folds (figure 5.13). Of the many compartments in eukaryotic cells, the two largest are the inner region of the ER, called the **cisternal space**, and the region exterior to it, the **cytosol**.

Rough ER: Manufacturing Proteins for Export

The ER surface regions that are devoted to protein synthesis are heavily studded with **ribosomes**, large molecular aggregates of protein and ribonucleic acid (RNA) that translate RNA copies of genes into protein (we will examine ribosomes in detail later in this chapter). Through the electron microscope, these ribosome-rich regions of the ER appear pebbly, like the surface of sandpaper, and they are therefore called **rough ER** (see figure 5.13).

The proteins synthesized on the surface of the rough ER are destined to be exported from the cell. Proteins to be exported contain special amino acid sequences called **signal sequences**. As a new protein is made by a free ribosome (one not attached to a membrane), the signal sequence of the growing polypeptide attaches to a recognition factor that carries the ribosome and its partially completed protein to a “docking site” on the surface of the ER. As the protein is assembled it passes through the ER membrane into the interior ER compartment, the cisternal space, from which it is transported by vesicles to the Golgi apparatus (figure 5.14). The protein then travels within vesicles to the inner surface of the plasma membrane, where it is released to the outside of the cell.

Table 5.1 Eukaryotic Cell Structures and Their Functions

Structure	Description	Function
Cell wall	Outer layer of cellulose or chitin; or absent	Protection; support
Cytoskeleton	Network of protein filaments	Structural support; cell movement
Flagella (cilia)	Cellular extensions with 9 + 2 arrangement of pairs of microtubules	Motility or moving fluids over surfaces
Plasma membrane	Lipid bilayer with embedded proteins	Regulates what passes into and out of cell; cell-to-cell recognition
Endoplasmic reticulum	Network of internal membranes	Forms compartments and vesicles; participates in protein and lipid synthesis
Nucleus	Structure (usually spherical) surrounded by double membrane that contains chromosomes	Control center of cell; directs protein synthesis and cell reproduction
Golgi apparatus	Stacks of flattened vesicles	Packages proteins for export from cell; forms secretory vesicles
Lysosomes	Vesicles derived from Golgi apparatus that contain hydrolytic digestive enzymes	Digest worn-out organelles and cell debris; play role in cell death
Microbodies	Vesicles formed from incorporation of lipids and proteins containing oxidative and other enzymes	Isolate particular chemical activities from rest of cell
Mitochondria	Bacteria-like elements with double membrane	“Power plants” of the cell; sites of oxidative metabolism
Chloroplasts	Bacteria-like elements with membranes containing chlorophyll, a photosynthetic pigment	Sites of photosynthesis
Chromosomes	Long threads of DNA that form a complex with protein	Contain hereditary information
Nucleolus	Site of genes for rRNA synthesis	Assembles ribosomes
Ribosomes	Small, complex assemblies of protein and RNA, often bound to endoplasmic reticulum	Sites of protein synthesis

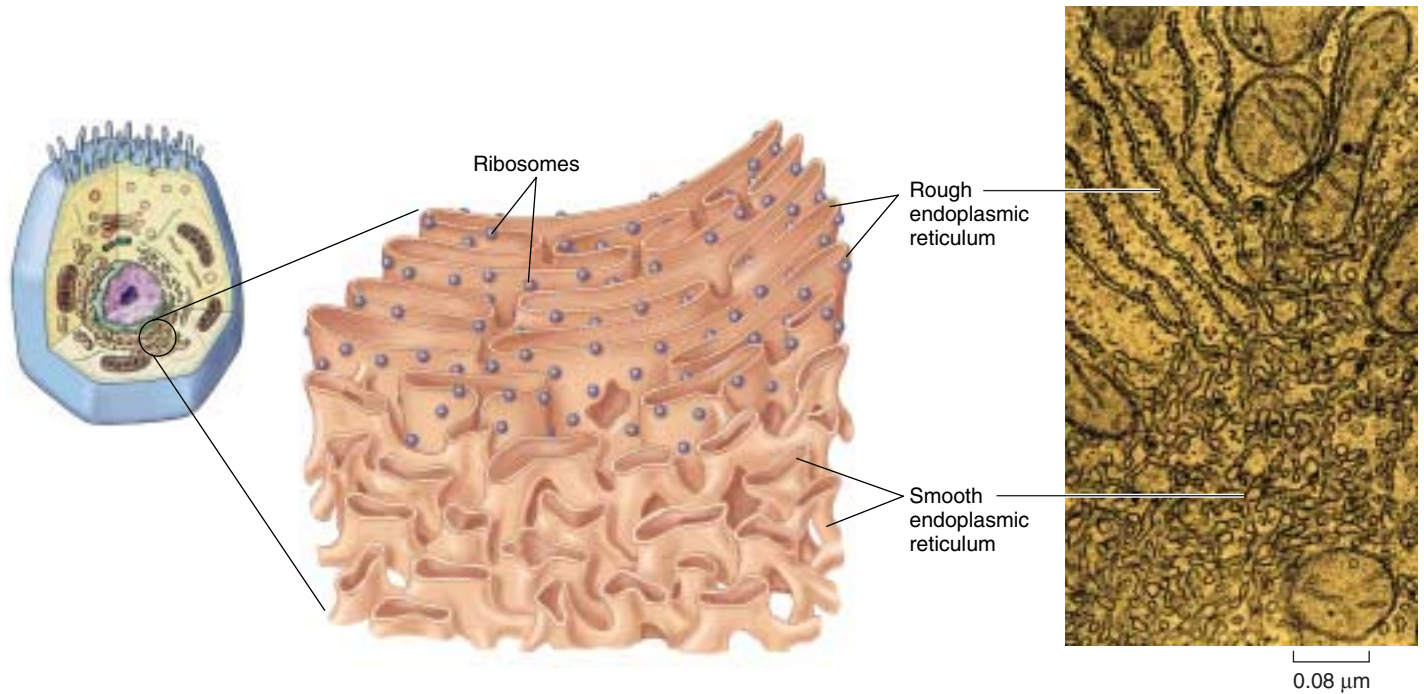


FIGURE 5.13

The endoplasmic reticulum. Ribosomes are associated with only one side of the rough ER; the other side is the boundary of a separate compartment within the cell into which the ribosomes extrude newly made proteins destined for secretion. Smooth endoplasmic reticulum has few to no bound ribosomes.

Smooth ER: Organizing Internal Activities

Regions of the ER with relatively few bound ribosomes are referred to as **smooth ER**. The membranes of the smooth ER contain many embedded enzymes, most of them active only when associated with a membrane. Enzymes anchored within the ER, for example, catalyze the synthesis of a variety of carbohydrates and lipids. In cells that carry out extensive lipid synthesis, such as those in the testes, intestine, and brain, smooth ER is particularly abundant. In the liver, the enzymes of the smooth ER are involved in the detoxification of drugs including amphetamines, morphine, codeine, and phenobarbital.

Some vesicles form at the plasma membrane by budding inward, a process called endocytosis. Some then move into the cytoplasm and fuse with the smooth endoplasmic reticulum. Others form secondary lysosomes or other interior vesicles.

The endoplasmic reticulum (ER) is an extensive system of folded membranes that divides the interior of eukaryotic cells into compartments and channels. Rough ER synthesizes proteins, while smooth ER organizes the synthesis of lipids and other biosynthetic activities.

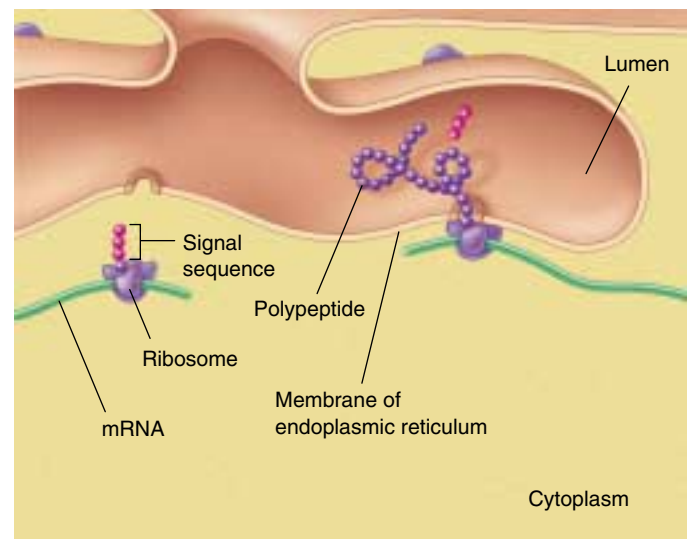


FIGURE 5.14

Signal sequences direct proteins to their destinations in the cell. In this example, a sequence of hydrophobic amino acids (the signal sequence) on a secretory protein attaches them (and the ribosomes making them) to the membrane of the ER. As the protein is synthesized, it passes into the lumen (internal chamber) of the ER. The signal sequence is clipped off after the leading edge of the protein enters the lumen.

The Golgi Apparatus: Delivery System of the Cell

At various locations within the endomembrane system, flattened stacks of membranes called **Golgi bodies** occur, often interconnected with one another. These structures are named for Camillo Golgi, the nineteenth-century Italian physician who first called attention to them. The numbers of Golgi bodies a cell contains ranges from 1 or a few in protists, to 20 or more in animal cells and several hundred in plant cells. They are especially abundant in glandular cells, which manufacture and secrete substances. Collectively the Golgi bodies are referred to as the **Golgi apparatus** (figure 5.15).

The Golgi apparatus functions in the collection, packaging, and distribution of molecules synthesized at one place in the cell and utilized at another location in the cell. A Golgi body has a front and a back, with distinctly different membrane compositions at the opposite ends. The front, or receiving end, is called the *cis* face, and is usually located near ER. Materials move to the *cis* face in transport vesicles that bud off of the ER. These vesicles fuse with the *cis* face, emptying their contents into the interior, or lumen, of the Golgi apparatus. These ER-synthesized molecules then pass through the channels of the Golgi apparatus until they reach the back, or discharging end, called the *trans* face, where they are discharged in secretory vesicles (figure 5.16).

Proteins and lipids manufactured on the rough and smooth ER membranes are transported into the Golgi ap-

paratus and modified as they pass through it. The most common alteration is the addition or modification of short sugar chains, forming a *glycoprotein* when sugars are complexed to a protein and a *glycolipid* when sugars are bound to a lipid. In many instances, enzymes in the Golgi apparatus modify existing glycoproteins and glycolipids made in the ER by cleaving a sugar from their sugar chain or modifying one or more of the sugars.

The newly formed or altered glycoproteins and glycolipids collect at the ends of the Golgi bodies, in flattened stacked membrane folds called **cisternae** (Latin, “collecting vessels”). Periodically, the membranes of the cisternae push together, pinching off small, membrane-bounded secretory vesicles containing the glycoprotein and glycolipid molecules. These vesicles then move to other locations in the cell, distributing the newly synthesized molecules to their appropriate destinations. **Liposomes** are synthetically manufactured vesicles that contain any variety of desirable substances (such as drugs), and can be injected into the body. Because the membrane of liposomes is similar to plasma and organellar membranes, these liposomes serve as an effective and natural delivery system to cells and may prove to be of great therapeutic value.

The Golgi apparatus is the delivery system of the eukaryotic cell. It collects, packages, modifies, and distributes molecules that are synthesized at one location within the cell and used at another.

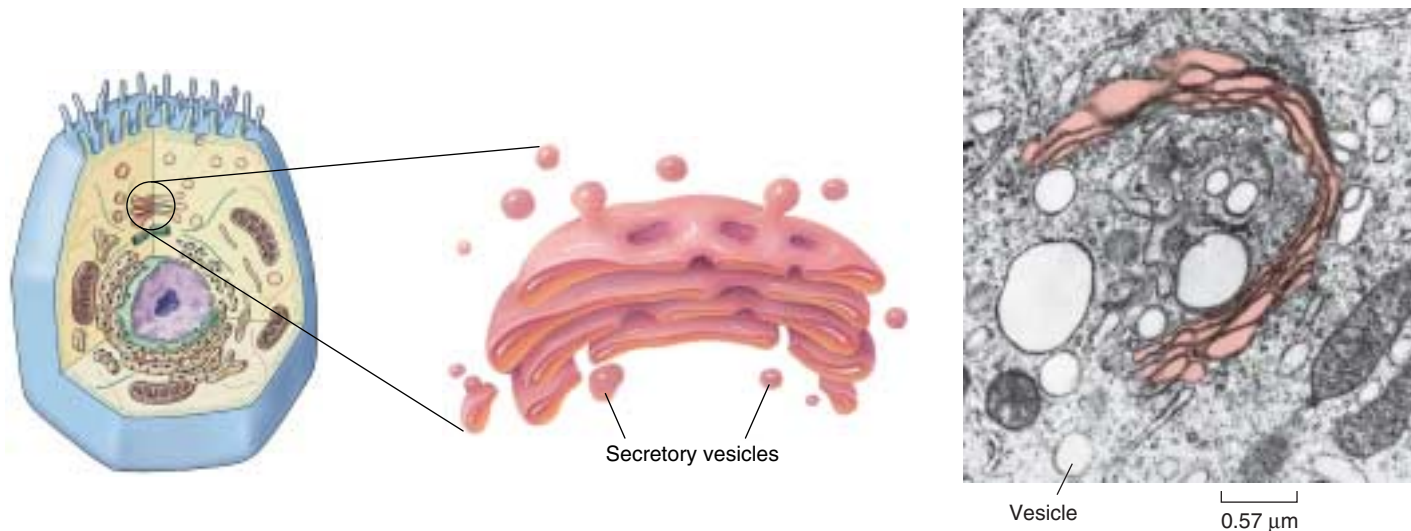


FIGURE 5.15

The Golgi apparatus. The Golgi apparatus is a smooth, concave membranous structure located near the middle of the cell. It receives material for processing on one surface and sends the material packaged in vesicles off the other. The substance in a vesicle could be for export out of the cell or for distribution to another region within the same cell.

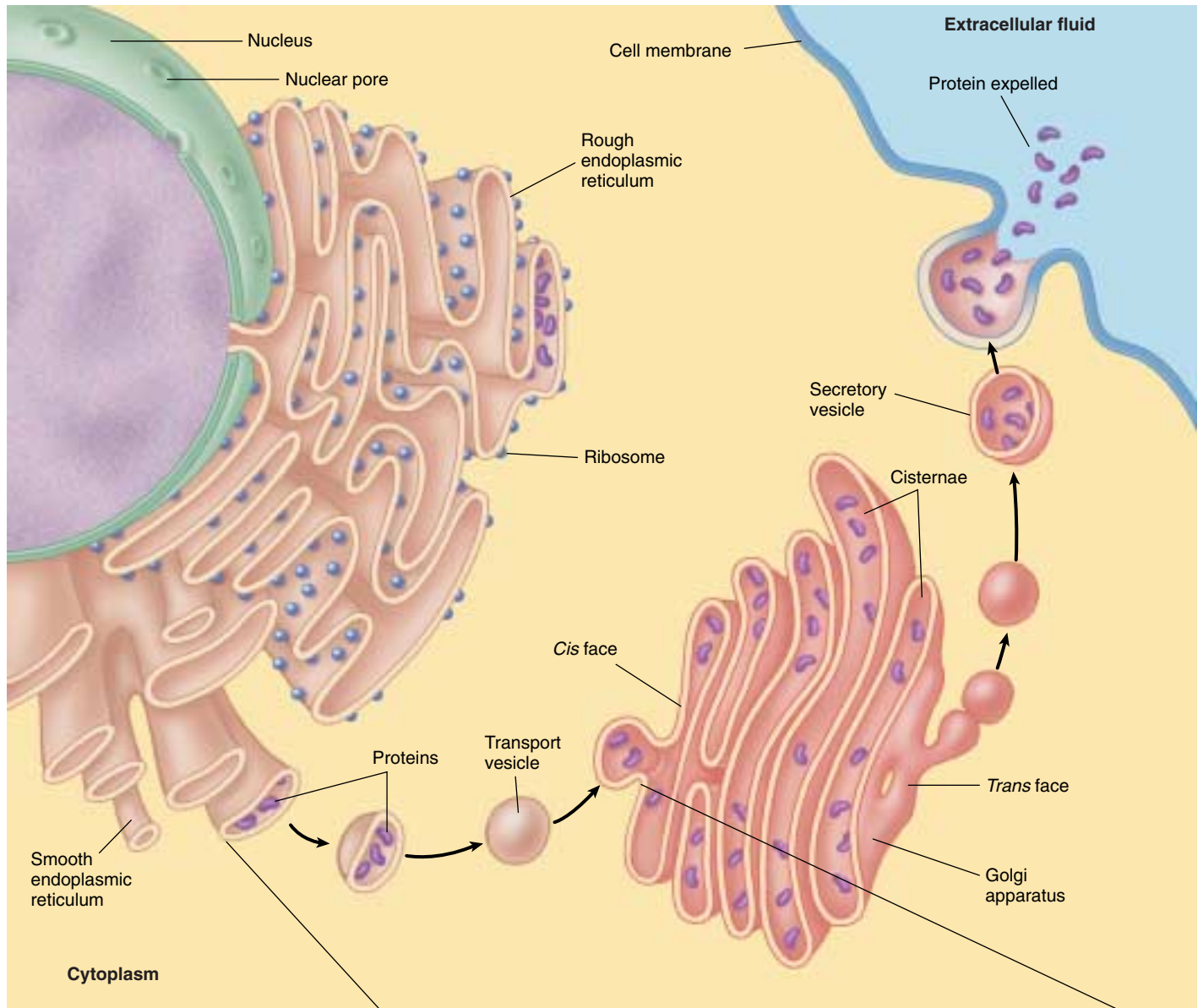
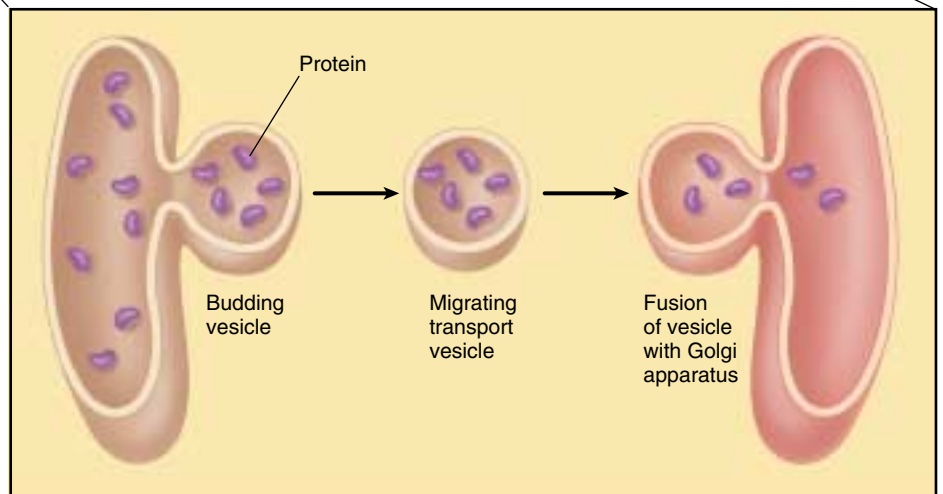


FIGURE 5.16

How proteins are transported within the cell. Proteins are manufactured at the ribosome and then released into the internal compartments of the rough ER. If the newly synthesized proteins are to be used at a distant location in or outside of the cell, they are transported within vesicles that bud off the rough ER and travel to the *cis* face, or receiving end, of the Golgi apparatus. There they are modified and packaged into secretory vesicles. The secretory vesicles then migrate from the *trans* face, or discharging end, of the Golgi apparatus to other locations in the cell, or they fuse with the cell membrane, releasing their contents to the external cellular environment.



Vesicles: Enzyme Storehouses

Lysosomes: Intracellular Digestion Centers

Lysosomes, membrane-bounded digestive vesicles, are also components of the endomembrane system that arise from the Golgi apparatus. They contain high levels of degrading enzymes, which catalyze the rapid breakdown of proteins, nucleic acids, lipids, and carbohydrates. Throughout the lives of eukaryotic cells, lysosomal enzymes break down old organelles, recycling their component molecules and making room for newly formed organelles. For example, mitochondria are replaced in some tissues every 10 days.

The digestive enzymes in lysosomes function best in an acidic environment. Lysosomes actively engaged in digestion keep their battery of hydrolytic enzymes (enzymes that catalyze the hydrolysis of molecules) fully active by pumping protons into their interiors and thereby maintaining a low internal pH. Lysosomes that are not functioning actively do not maintain an acidic internal pH and are called *primary lysosomes*. When a primary lysosome fuses with a food vesicle or other organelle, its pH falls and its arsenal of hydrolytic enzymes is activated; it is then called a *secondary lysosome*.

In addition to breaking down organelles and other structures within cells, lysosomes also eliminate other cells that the cell has engulfed in a process called *phagocytosis*, a specific type of endocytosis (see chapter 6). When a white blood cell, for example, phagocytizes a passing pathogen, lysosomes fuse with the resulting “food vesicle,” releasing their enzymes into the vesicle and degrading the material within (figure 5.17).

Microbodies

Eukaryotic cells contain a variety of enzyme-bearing, membrane-enclosed vesicles called **microbodies**. Microbodies are found in the cells of plants, animals, fungi, and protists. The distribution of enzymes into microbodies is one of the principal ways in which eukaryotic cells organize their metabolism.

While lysosomes bud from the endomembrane system, microbodies grow by incorporating lipids and protein, then dividing. Plant cells have a special type of microbody called a **glyoxysome** that contains enzymes that convert fats into carbohydrates. Another type of microbody, a **peroxisome**, contains enzymes that catalyze the removal of electrons and associated hydrogen atoms (figure 5.18).

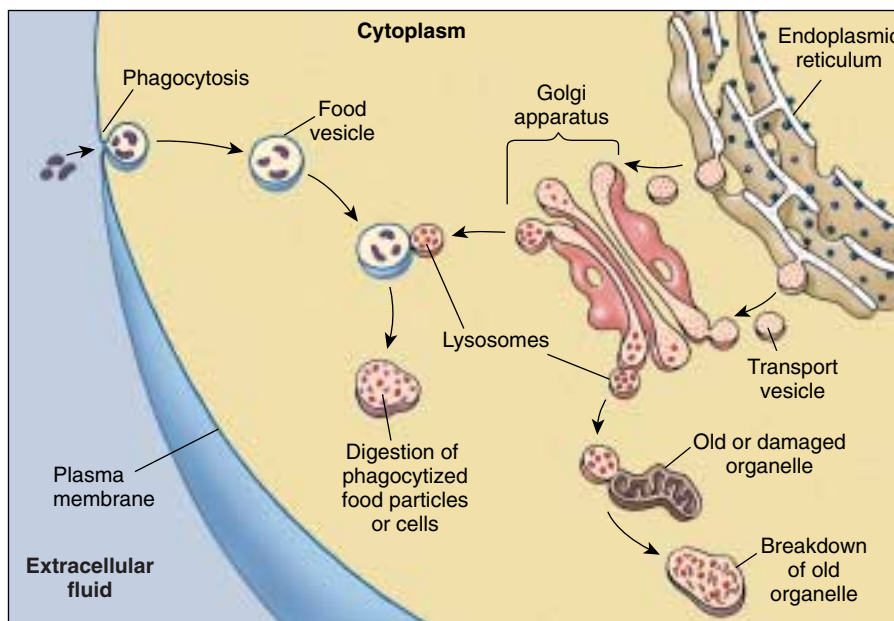


FIGURE 5.17

Lysosomes. Lysosomes contain hydrolytic enzymes that digest particles or cells taken into the cell by phagocytosis and break down old organelles.

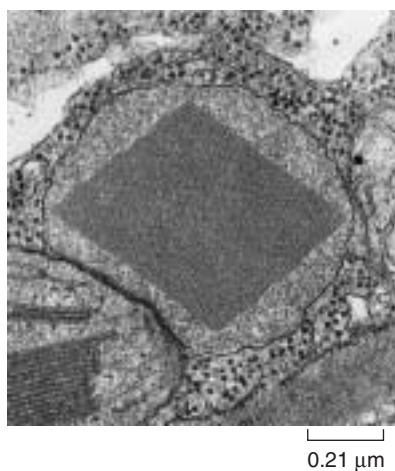


FIGURE 5.18

A peroxisome. Peroxisomes are spherical organelles that may contain a large diamond-shaped crystal composed of protein. Peroxisomes contain digestive and detoxifying enzymes that produce hydrogen peroxide as a by-product.

If these oxidative enzymes were not isolated within microbodies, they would tend to short-circuit the metabolism of the cytoplasm, which often involves adding hydrogen atoms to oxygen. The name *peroxisome* refers to the hydrogen peroxide produced as a by-product of the activities of the oxidative enzymes in the microbody. Hydrogen peroxide is dangerous to cells because of its violent chemical reactivity. However, peroxisomes also contain the enzyme catalase, which breaks down hydrogen peroxide into harmless water and oxygen.

Lysosomes and peroxisomes are vesicles that contain digestive and detoxifying enzymes. The isolation of these enzymes in vesicles protects the rest of the cell from inappropriate digestive activity.

Ribosomes: Sites of Protein Synthesis

Although the DNA in a cell's nucleus encodes the amino acid sequence of each protein in the cell, the proteins are not assembled there. A simple experiment demonstrates this: if a brief pulse of radioactive amino acid is administered to a cell, the radioactivity shows up associated with newly made protein, not in the nucleus, but in the cytoplasm. When investigators first carried out these experiments, they found that protein synthesis was associated with large RNA-protein complexes they called **ribosomes**.

Ribosomes are made up of several molecules of a special form of RNA called ribosomal RNA, or rRNA, bound within a complex of several dozen different proteins. Ribosomes are among the most complex molecular assemblies found in cells. Each ribosome is composed of two subunits (figure 5.19). The subunits join to form a functional ribosome only when they attach to another kind of RNA, called messenger RNA (mRNA) in the cytoplasm. To make proteins, the ribosome attaches to the mRNA, which is a transcribed copy of a portion of DNA, and uses the information to direct the synthesis of a protein.

Bacterial ribosomes are smaller than eukaryotic ribosomes. Also, a bacterial cell typically has only a few thousand ribosomes, while a metabolically active eukaryotic cell, such as a human liver cell, contains several million. Proteins that function in the cytoplasm are made by free ribosomes suspended there, while proteins bound within membranes or destined for export from the cell are assembled by ribosomes bound to rough ER.

The Nucleolus Manufactures Ribosomal Subunits

When cells are synthesizing a large number of proteins, they must first make a large number of ribosomes. To facilitate this, many hundreds of copies of the portion of the DNA encoding the rRNA are clustered together on the chromosome. By transcribing RNA molecules from this cluster, the cell rapidly generates large numbers of the molecules needed to produce ribosomes.

At any given moment, many rRNA molecules dangle from the chromosome at the sites of these clusters of genes that encode rRNA. Proteins associate with the dangling rRNA molecules. These areas where ribosomes are being assembled are easily visible within the nucleus as one or more dark-staining regions, called **nucleoli** (singular, nucleolus; figure 5.20). Nucleoli can be seen under the light microscope even when the chromosomes are extended, unlike the rest of the chromosomes, which are visible only when condensed.

Ribosomes are the sites of protein synthesis in the cytoplasm.

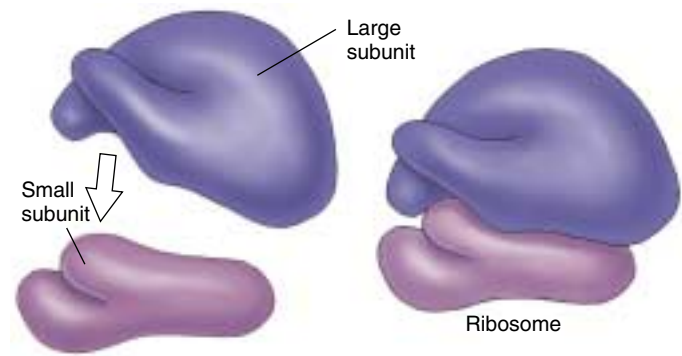


FIGURE 5.19

A ribosome. Ribosomes consist of a large and a small subunit composed of rRNA and protein. The individual subunits are synthesized in the nucleolus and then move through the nuclear pores to the cytoplasm, where they assemble. Ribosomes serve as sites of protein synthesis.

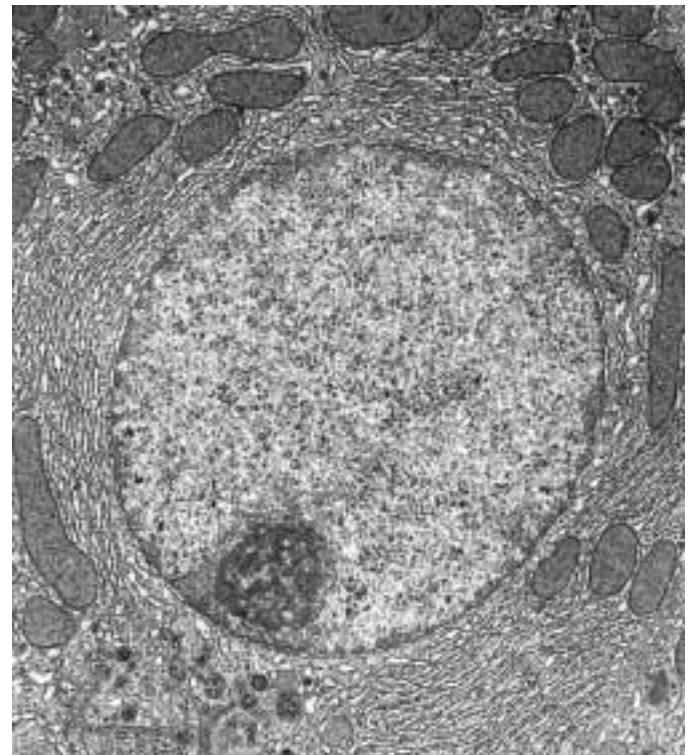


FIGURE 5.20

The nucleolus. This is the interior of a rat liver cell, magnified about 6000 times. A single large nucleus occupies the center of the micrograph. The electron-dense area in the lower left of the nucleus is the nucleolus, the area where the major components of the ribosomes are produced. Partly formed ribosomes can be seen around the nucleolus.

Organelles That Contain DNA

Among the most interesting cell organelles are those in addition to the nucleus that contain DNA.

Mitochondria: The Cell's Chemical Furnaces

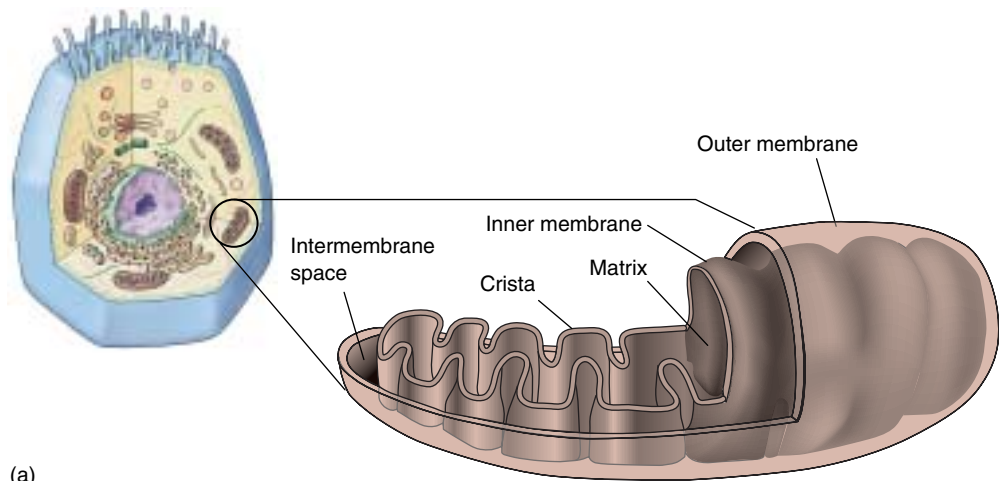
Mitochondria (singular, *mitochondrion*) are typically tubular or sausage-shaped organelles about the size of bacteria and found in all types of eukaryotic cells (figure 5.21). Mitochondria are bounded by two membranes: a smooth outer membrane and an inner one folded into numerous contiguous layers called **cristae** (singular, *crista*). The cristae partition the mitochondrion into two compartments: a **matrix**, lying inside the inner membrane; and an outer compartment, or **intermembrane space**, lying between the two mitochondrial membranes. On the surface of the inner membrane, and also embedded within it, are proteins that carry out oxidative metabolism, the oxygen-requiring process by which energy in macromolecules is stored in ATP. (a)

Mitochondria have their own DNA; this DNA contains several genes that produce proteins essential to the mitochondrion's role in oxidative metabolism. All of these genes are copied into RNA and used to make proteins within the mitochondrion. In this process, the mitochondria employ small RNA molecules and ribosomal components that the mitochondrial DNA also encodes. However, most of the genes that produce the enzymes used in oxidative metabolism are located in the nucleus.

A eukaryotic cell does not produce brand new mitochondria each time the cell divides. Instead, the mitochondria themselves divide in two, doubling in number, and these are partitioned between the new cells. Most of the components required for mitochondrial division are encoded by genes in the nucleus and translated into proteins by cytoplasmic ribosomes. Mitochondrial replication is, therefore, impossible without nuclear participation, and mitochondria thus cannot be grown in a cell-free culture.

Chloroplasts: Where Photosynthesis Takes Place

Plants and other eukaryotic organisms that carry out photosynthesis typically contain from one to several hundred **chloroplasts**. Chloroplasts bestow an obvious advantage on the organisms that possess them: they can manufacture their own food. Chloroplasts contain the photosynthetic pigment chlorophyll that gives most plants their green color.



(b)

FIGURE 5.21

Mitochondria. (a) The inner membrane of a mitochondrion is shaped into folds called cristae, which greatly increase the surface area for oxidative metabolism. (b) Mitochondria in cross-section and cut lengthwise (70,000 $\times$).

The chloroplast body is enclosed, like the mitochondrion, within two membranes that resemble those of mitochondria (figure 5.22). However, chloroplasts are larger and more complex than mitochondria. In addition to the outer and inner membranes, which lie in close association with each other, chloroplasts have a closed compartment of stacked membranes called **grana** (singular, *granum*), which lie internal to the inner membrane. A chloroplast may contain a hundred or more grana, and each granum may contain from a few to several dozen disk-shaped structures called **thylakoids**. On the surface of the thylakoids are the light-capturing photosynthetic pigments, to be discussed in depth in chapter 10. Surrounding the thylakoid is a fluid matrix called the *stroma*.

Like mitochondria, chloroplasts contain DNA, but many of the genes that specify chloroplast components are also located in the nucleus. Some of the elements used in the photosynthetic process, including the specific protein components necessary to accomplish the reaction, are synthesized entirely within the chloroplast.

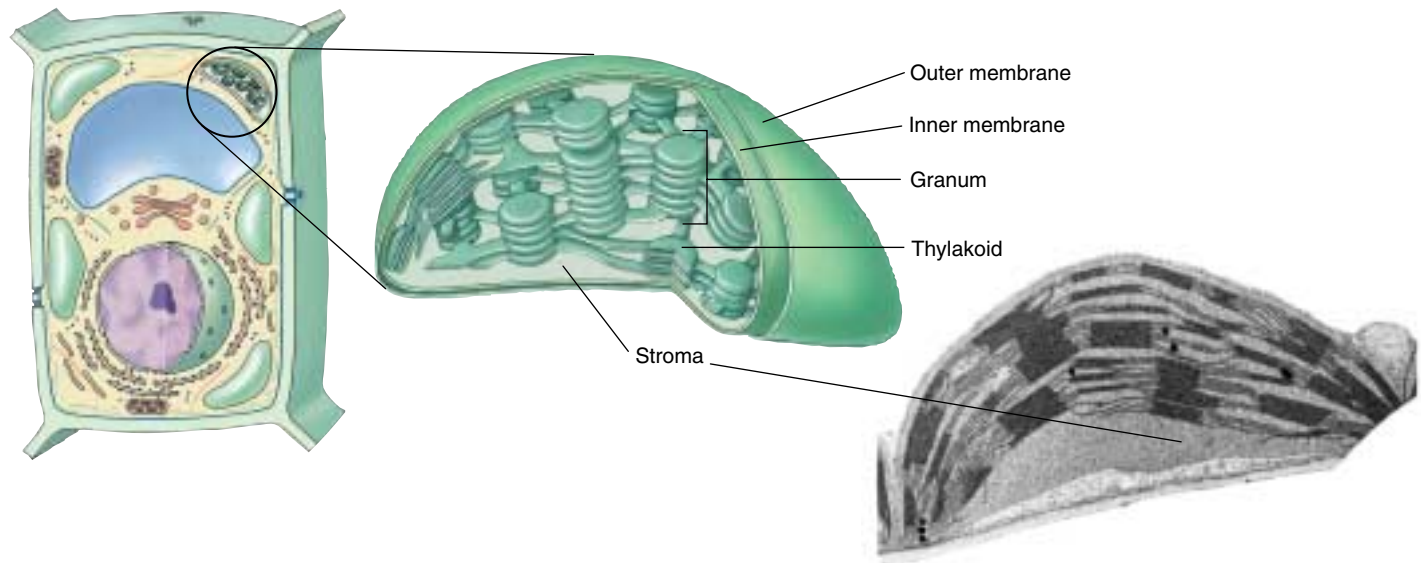


FIGURE 5.22

Chloroplast structure. The inner membrane of a chloroplast is fused to form stacks of closed vesicles called thylakoids. Within these thylakoids, photosynthesis takes place. Thylakoids are typically stacked one on top of the other in columns called grana.

Other DNA-containing organelles in plants are called leucoplasts, which lack pigment and a complex internal structure. In root cells and some other plant cells, leucoplasts may serve as starch storage sites. A leucoplast that stores starch (amylose) is sometimes termed an **amyloplast**. These organelles—chloroplasts, leucoplasts, and amyloplasts—are collectively called **plastids**. All plastids come from the division of existing plastids.

Centrioles: Microtubule Assembly Centers

Centrioles are barrel-shaped organelles found in the cells of animals and most protists. They occur in pairs, usually located at right angles to each other near the nuclear membranes (figure 5.23); the region surrounding the pair in almost all animal cells is referred to as a **centrosome**. Although the matter is in some dispute, at least some centrioles seem to contain DNA, which apparently is involved in producing their structural proteins. Centrioles help to assemble **microtubules**, long, hollow cylinders of the protein tubulin. Microtubules influence cell shape, move the chromosomes in cell division, and provide the functional internal structure of flagella and cilia, as we will discuss later. Centrioles may be contained in areas called **microtubule-organizing centers (MTOCs)**. The cells of plants and fungi lack centrioles, and cell biologists are still in the process of characterizing their MTOCs.

Both mitochondria and chloroplasts contain specific genes related to some of their functions, but both depend on nuclear genes for other functions. Some centrioles also contain DNA, which apparently helps control the synthesis of their structural proteins.

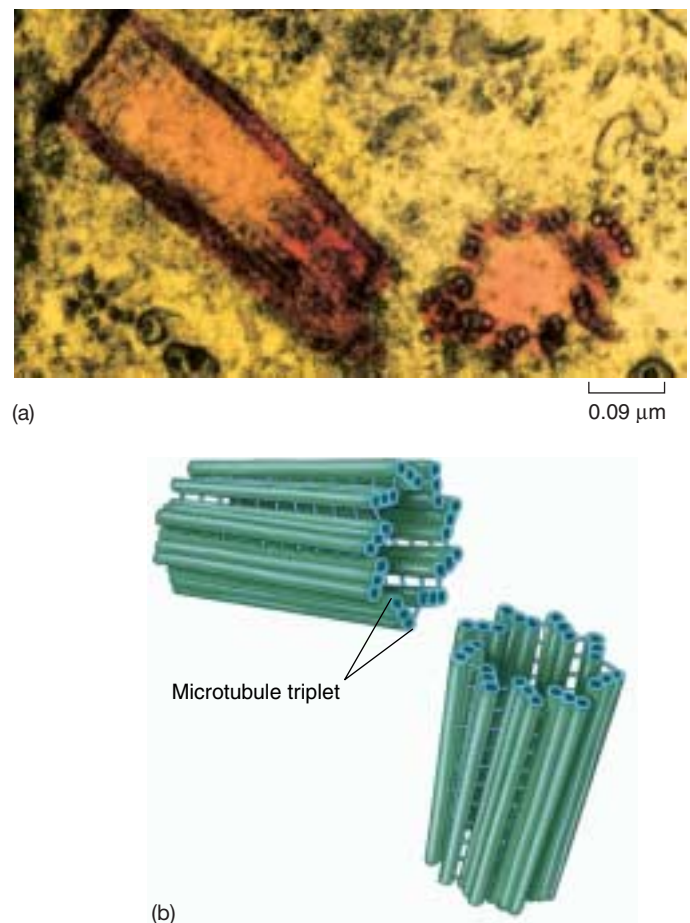


FIGURE 5.23

Centrioles. (a) This electron micrograph shows a pair of centrioles (75,000 $\times$). The round shape is a centriole in cross-section; the rectangular shape is a centriole in longitudinal section. (b) Each centriole is composed of nine triplets of microtubules.

The Cytoskeleton: Interior Framework of the Cell

The cytoplasm of all eukaryotic cells is crisscrossed by a network of protein fibers that supports the shape of the cell and anchors organelles to fixed locations. This network, called the **cytoskeleton** (figure 5.24), is a dynamic system, constantly forming and disassembling. Individual fibers form by **polymerization**, as identical protein subunits attract one another chemically and spontaneously assemble into long chains. Fibers disassemble in the same way, as one subunit after another breaks away from one end of the chain.

Eukaryotic cells may contain three types of cytoskeletal fibers, each formed from a different kind of subunit:

- 1. Actin filaments.** Actin filaments are long fibers about 7 nanometers in diameter. Each filament is composed of two protein chains loosely twined together like two strands of pearls (figure 5.25a). Each “pearl,” or subunit, on the chains is the globular protein **actin**. Actin molecules spontaneously form these filaments, even in a test tube; a cell regulates the rate of their formation through other proteins that act as switches, turning on polymerization when appropriate. Actin filaments are responsible for cellular movements such as contraction, crawling, “pinching” during division, and formation of cellular extensions.
- 2. Microtubules.** Microtubules are hollow tubes about 25 nanometers in diameter, each composed of a ring of 13 protein protofilaments (figure 5.25b). Globular proteins consisting of dimers of alpha and beta *tubulin* subunits polymerize to form the 13 protofilaments. The protofilaments are arrayed side by side around a central core, giving the microtubule its characteristic tube shape. In many cells, microtubules form from MTOC nucleation centers near the center of the cell and radiate toward the periphery. They are in a constant state of flux, continually polymerizing and depolymerizing (the average half-life of a microtubule ranges from 10 minutes in a nondividing animal cell to as short as 20 seconds in a dividing animal cell), unless stabilized by the binding of guanosine triphosphate (GTP) to the ends, which inhibits depolymerization. The ends of the microtubule are designated as “+” (away from the nucleation center) or “-” (toward the nucleation center). Along with allowing for cellular movement, microtubules are responsible for moving materials within the cell itself. Special motor proteins, discussed later in this chapter, move cellular organelles around the cell on microtubular “tracks.” *Kinesin* proteins move organelles toward the “+” end (toward the cell periphery), and *dyneins* move them toward the “-” end.

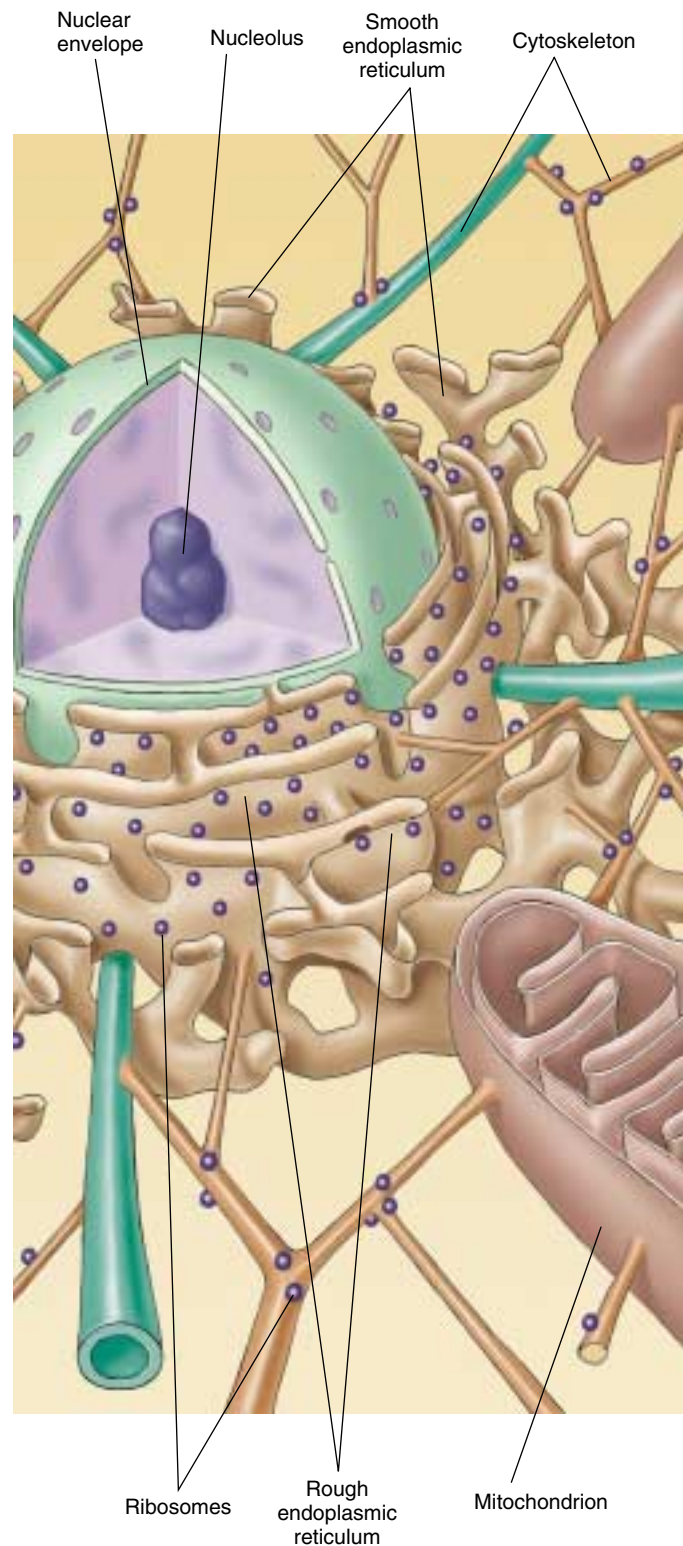


FIGURE 5.24
The cytoskeleton. In this diagrammatic cross-section of a eukaryotic cell, the cytoskeleton, a network of fibers, supports organelles such as mitochondria.

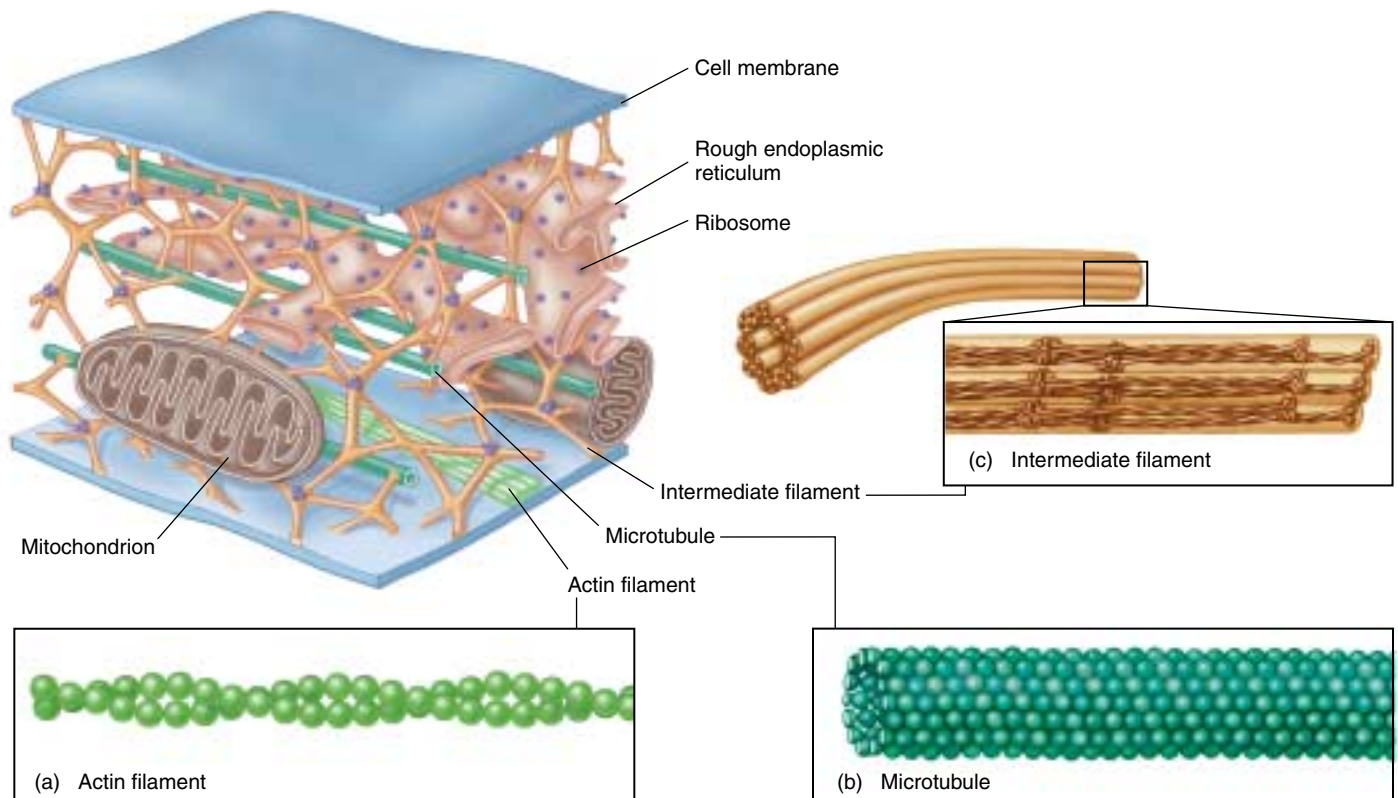


FIGURE 5.25

Molecules that make up the cytoskeleton. (a) Actin filaments. Actin filaments are made of two strands of the fibrous protein actin twisted together and usually occur in bundles. Actin filaments are ubiquitous, although they are concentrated below the plasma membrane in bundles known as stress fibers, which may have a contractile function. (b) Microtubules. Microtubules are composed of 13 stacks of tubulin protein subunits arranged side by side to form a tube. Microtubules are comparatively stiff cytoskeletal elements that serve to organize metabolism and intracellular transport in the nondividing cell. (c) Intermediate filaments. Intermediate filaments are composed of overlapping staggered tetramers of protein. This molecular arrangement allows for a ropelike structure that imparts tremendous mechanical strength to the cell.

3. Intermediate filaments. The most durable element of the cytoskeleton in animal cells is a system of tough, fibrous protein molecules twined together in an overlapping arrangement (figure 5.25c). These fibers are characteristically 8 to 10 nanometers in diameter, intermediate in size between actin filaments and microtubules (which is why they are called intermediate filaments). Once formed, intermediate filaments are stable and usually do not break down. Intermediate filaments constitute a heterogeneous group of cytoskeletal fibers. The most common type, composed of protein subunits called *vimentin*, provides structural stability for many kinds of cells. *Keratin*, another class of intermediate filament, is found in epithelial cells (cells that line organs and body cavities) and associated structures such as hair

and fingernails. The intermediate filaments of nerve cells are called *neurofilaments*.

As we will discuss in the next section, the cytoskeleton provides an interior framework that supports the shape of the cell, stretching the plasma membrane much as the poles of a circus tent. Changing the relative length of cytoskeleton filaments allows cells to rapidly alter their shape, extending projections out or folding inward. Within the cell, the framework of filaments provides a molecular highway along which molecules can be transported.

Elements of the cytoskeleton crisscross the cytoplasm, supporting the cell shape and anchoring organelles in place. There are three principal types of fibers: actin filaments, microtubules, and intermediate filaments.

Cell Movement

Essentially all cell motion is tied to the movement of actin filaments, microtubules, or both. Intermediate filaments act as intracellular tendons, preventing excessive stretching of cells, and actin filaments play a major role in determining the shape of cells. Because actin filaments can form and dissolve so readily, they enable some cells to change shape quickly. If you look at the surfaces of such cells under a microscope, you will find them alive with motion, as projections, called **microvilli** in animal cells, shoot outward from the surface and then retract, only to shoot out elsewhere moments later (figure 5.26).

Some Cells Crawl

It is the arrangement of actin filaments within the cell cytoplasm that allows cells to “crawl,” *literally!* Crawling is a significant cellular phenomenon, essential to inflammation, clotting, wound healing, and the spread of cancer. White blood cells in particular exhibit this ability. Produced in the bone marrow, these cells are released into the circulatory system and then eventually crawl out of capillaries and into the tissues to destroy potential pathogens.

Cells exist in a *gel-sol* state; that is, at any given time, some regions of the cell are rigid (*gel*) and some are more fluid (*sol*). The cell is typically more sol-like in its interior, and more gel-like at its perimeter. To crawl, the cell creates a weak area in the gel perimeter, and then forces the fluid (*sol*) interior through the weak area, forming a **pseudopod** (“false foot”). As a result a large section of cytoplasm oozes off in a different direction, but still remains within the plasma membrane. Once extended, the pseudopod stabilizes into a gel state, assembling actin filaments. Specific membrane proteins in the pseudopod stick to the surface the cell is crawling on, and the rest of the cell is dragged in that direction. The pressure required to force out a developing pseudopod is created when actin filaments in the trailing end of the cell contract, just as squeezing a water balloon at one end forces the balloon to bulge out at the other end.

Moving Material within the Cell

Actin filaments and microtubules often orchestrate their activities to affect cellular processes. For example, during cell reproduction (see chapter 11), newly replicated chromosomes move to opposite sides of a dividing cell because they are attached to shortening microtubules. Then, in animal cells, a belt of actin pinches the cell in two by contracting like a purse string. Muscle cells also use actin filaments to contract their cytoskeletons. The fluttering of an eyelash, the flight of an eagle, and the awkward crawling of a baby all depend on these cytoskeletal movements within muscle cells.



FIGURE 5.26

The surfaces of some cells are in constant motion. This amoeba, a single-celled protist, is advancing toward you, its advancing edges extending projections outward. The moving edges have been said to resemble the ruffled edges of a skirt.

Not only is the cytoskeleton responsible for the cell's shape and movement, but it also provides a scaffold that holds certain enzymes and other macromolecules in defined areas of the cytoplasm. Many of the enzymes involved in cell metabolism, for example, bind to actin filaments; so do ribosomes. By moving and anchoring particular enzymes near one another, the cytoskeleton, like the endoplasmic reticulum, organizes the cell's activities.

Intracellular Molecular Motors

Certain eukaryotic cells must move materials from one place to another in the cytoplasm. Most cells use the endomembrane system as an intracellular highway; the Golgi apparatus packages materials into vesicles that move through the channels of the endoplasmic reticulum to the far reaches of the cell. However, this highway is only effective over short distances. When a cell has to transport materials through long extensions like the axon of a nerve cell, the ER highways are too slow. For these situations, eukaryotic cells have developed high-speed locomotives that run along microtubular tracks.

Four components are required: (1) a **vesicle** or organelle that is to be transported, (2) a **motor molecule** that provides the energy-driven motion, (3) a **connector molecule** that connects the vesicle to the motor molecule, and (4) **microtubules** on which the vesicle will ride like a train on a rail. For example, embedded within the membranes of endoplasmic reticulum is a protein called *kinectin* that bind the ER vesicles to the motor protein *kinesin*. As nature's tiniest motors, these motor proteins literally pull the transport vesicles along the microtubular tracks. Kinesin uses ATP to power its movement toward

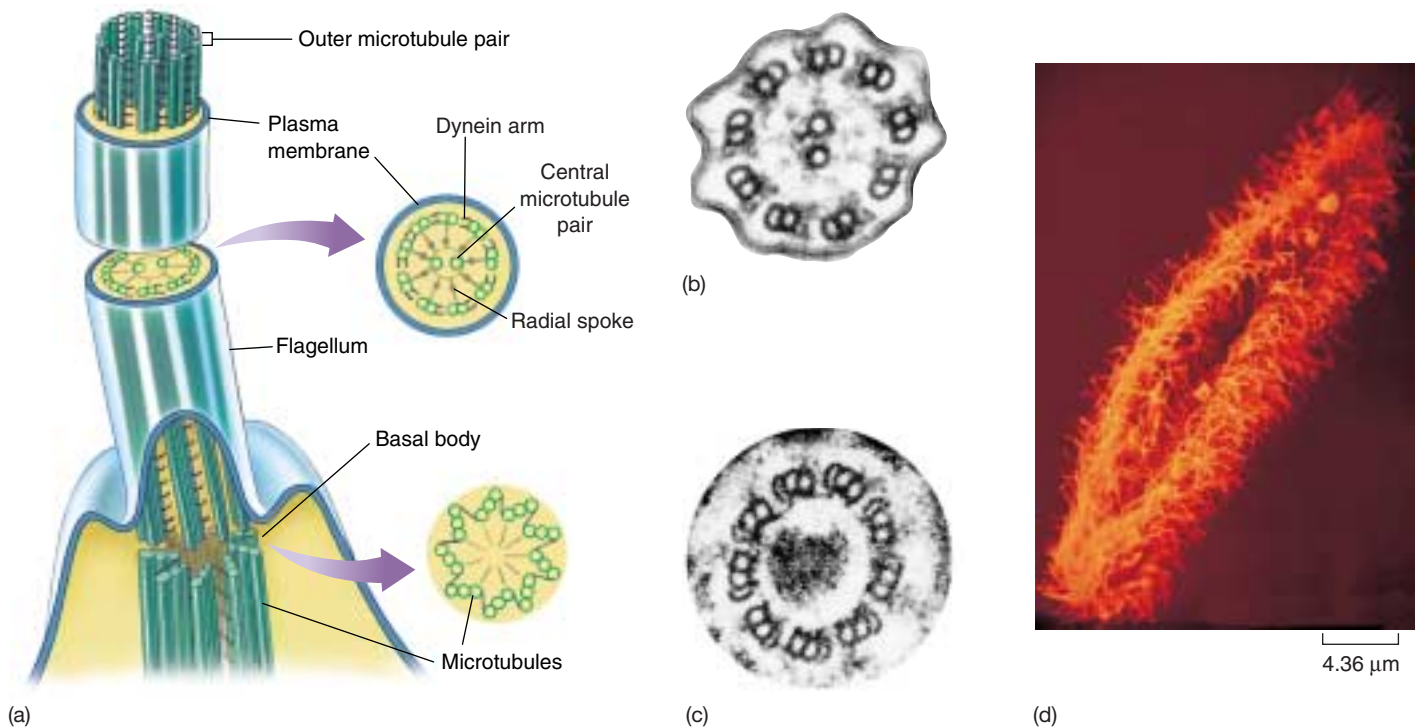


FIGURE 5.27

Flagella and cilia. (a) A eukaryotic flagellum originates directly from a basal body. (b) The flagellum has two microtubules in its core connected by radial spokes to an outer ring of nine paired microtubules with dynein arms. (c) The basal body consists of nine microtubule triplets connected by short protein segments. The structure of cilia is similar to that of flagella, but cilia are usually shorter. (d) The surface of this *Paramecium* is covered with a dense forest of cilia.

the cell periphery, dragging the vesicle with it as it travels along the microtubule. Another vesicle protein (or perhaps a slight modification of kinesin—further research is needed to determine which) binds vesicles to the motor protein dynein, which directs movement in the opposite direction, inward toward the cell's center. (Dynein is also involved in the movement of eukaryotic flagella, as discussed below.) The destination of a particular transport vesicle and its contents is thus determined by the nature of the linking protein embedded within the vesicle's membrane.

Swimming with Flagella and Cilia

Earlier in this chapter, we described the structure of bacterial flagella. Eukaryotic cells have a *completely different* kind of flagellum, consisting of a circle of nine microtubule pairs surrounding two central microtubules; this arrangement is referred to as the **9 + 2 structure** (figure 5.27). As pairs of microtubules move past one another using arms composed of the motor protein dynein, the eukaryotic flagellum undulates rather than rotates. When examined carefully, each flagellum proves to be an outward projection of the cell's interior, containing cytoplasm and enclosed by the plasma membrane. The microtubules of the flagellum are derived

from a **basal body**, situated just below the point where the flagellum protrudes from the surface of the cell.

The flagellum's complex microtubular apparatus evolved early in the history of eukaryotes. Although the cells of many multicellular and some unicellular eukaryotes today no longer possess flagella and are nonmotile, an organization similar to the 9 + 2 arrangement of microtubules can still be found within them, in structures called **cilia** (singular, *cilium*). Cilia are short cellular projections that are often organized in rows (see figure 5.1). They are more numerous than flagella on the cell surface, but have the same internal structure. In many multicellular organisms, cilia carry out tasks far removed from their original function of propelling cells through water. In several kinds of vertebrate tissues, for example, the beating of rows of cilia moves water over the tissue surface. The sensory cells of the vertebrate ear also contain cilia; sound waves bend these cilia, the initial sensory input of hearing. Thus, the 9 + 2 structure of flagella and cilia appears to be a fundamental component of eukaryotic cells.

Some eukaryotic cells use pseudopodia to crawl about within multicellular organisms, while many protists swim using flagella and cilia. Materials are transported within cells by special motor proteins.

Special Things about Plant Cells

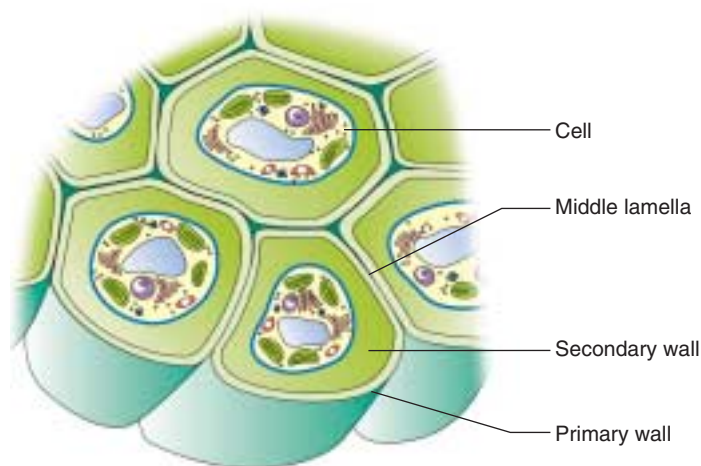
Vacuoles: A Central Storage Compartment

The center of a plant cell usually contains a large, apparently empty space, called the **central vacuole** (figure 5.28). This vacuole is not really empty; it contains large amounts of water and other materials, such as sugars, ions, and pigments. The central vacuole functions as a storage center for these important substances and also helps to increase the surface-to-volume ratio of the plant cell by applying pressure to the cell membrane. The cell membrane expands outward under this pressure, thereby increasing its surface area.

Cell Walls: Protection and Support

Plant cells share a characteristic with bacteria that is not shared with animal cells—that is, plants have **cell walls**, which protect and support the plant cell. Although bacteria also have cell walls, plant cell walls are chemically and structurally different from bacterial cell walls. Cell walls are also present in fungi and some protists. In plants, cell walls are composed of fibers of the polysaccharide cellulose. **Primary walls** are laid down when the cell is still growing, and between the walls of adjacent cells is a sticky substance called the **middle lamella**, which glues the cells together (figure 5.29). Some plant cells produce strong **secondary walls**, which are deposited inside the primary walls of fully expanded cells.

Plant cells store substances in a large central vacuole, and encase themselves within a strong cellulose cell wall.



(a)

FIGURE 5.29

Cell walls in plants. As shown in this drawing (a) and transmission electron micrograph (b), plant cell walls are thicker, stronger, and more rigid than those of bacteria. Primary cell walls are laid down when the cell is young. Thicker secondary cell walls may be added later when the cell is fully grown.

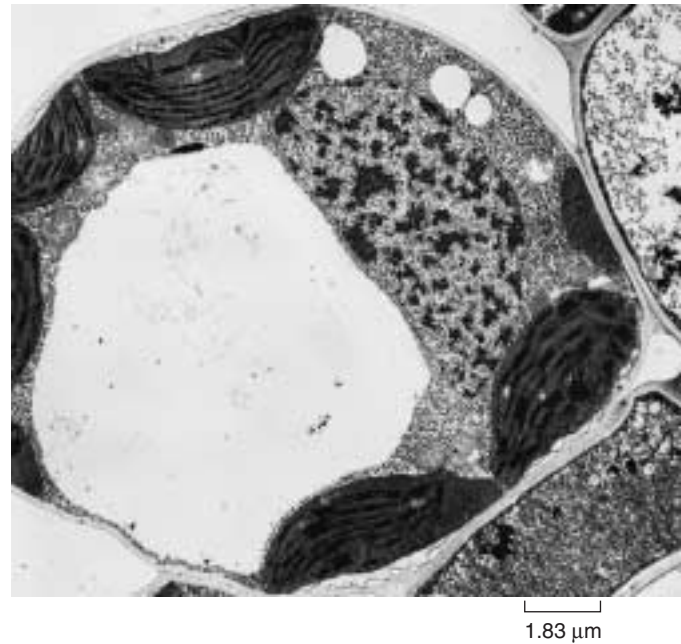
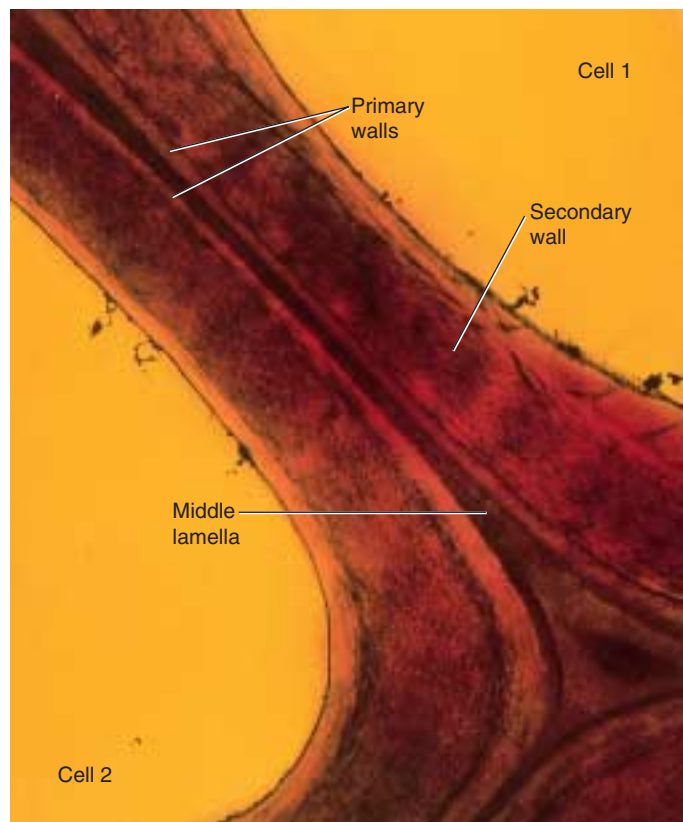


FIGURE 5.28

The central vacuole. A plant's central vacuole stores dissolved substances and can increase in size to increase the surface area of a plant cell.



(b)

5.4 Symbiosis played a key role in the origin of some eukaryotic organelles.

Endosymbiosis

Symbiosis is a close relationship between organisms of different species that live together. The theory of **endosymbiosis** proposes that some of today's eukaryotic organelles evolved by a symbiosis in which one species of prokaryote was engulfed by and lived inside another species of prokaryote that was a precursor to eukaryotes (figure 5.30). According to the endosymbiont theory, the engulfed prokaryotes provided their hosts with certain advantages associated with their special metabolic abilities. Two key eukaryotic organelles are believed to be the descendants of these endosymbiotic prokaryotes: mitochondria, which are thought to have originated as bacteria capable of carrying out oxidative metabolism; and chloroplasts, which apparently arose from photosynthetic bacteria.

The endosymbiont theory is supported by a wealth of evidence. Both mitochondria and chloroplasts are surrounded by two membranes; the inner membrane probably evolved from the plasma membrane of the engulfed bacterium, while the outer membrane is probably derived from the plasma membrane or endoplasmic reticulum of the host cell. Mitochondria are about the same size as most bacteria, and the cristae formed by their inner membranes resemble the folded membranes in various groups of bacteria. Mitochondrial ribosomes are also similar to bacterial ribosomes in size and structure. Both mitochondria and

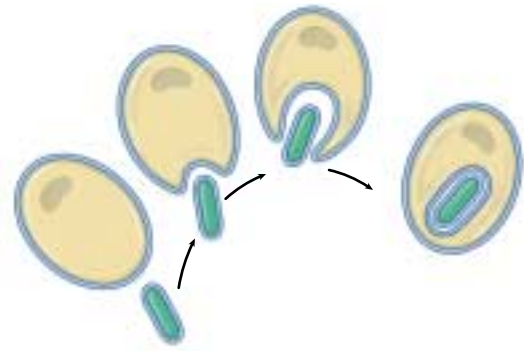


FIGURE 5.30

Endosymbiosis. This figure shows how a double membrane may have been created during the symbiotic origin of mitochondria or chloroplasts.

chloroplasts contain circular molecules of DNA similar to those in bacteria. Finally, mitochondria divide by simple fission, splitting in two just as bacterial cells do, and they apparently replicate and partition their DNA in much the same way as bacteria. Table 5.2 compares and reviews the features of three types of cells.

Some eukaryotic organelles are thought to have arisen by endosymbiosis.

Table 5.2 A Comparison of Bacterial, Animal, and Plant Cells

	Bacterium	Animal	Plant
EXTERIOR STRUCTURES			
Cell wall	Present (protein-polysaccharide)	Absent	Present (cellulose)
Cell membrane	Present	Present	Present
Flagella	May be present (single strand)	May be present	Absent except in sperm of a few species
INTERIOR STRUCTURES			
ER	Absent	Usually present	Usually present
Ribosomes	Present	Present	Present
Microtubules	Absent	Present	Present
Centrioles	Absent	Present	Absent
Golgi apparatus	Absent	Present	Present
Nucleus	Absent	Present	Present
Mitochondria	Absent	Present	Present
Chloroplasts	Absent	Absent	Present
Chromosomes	A single circle of DNA	Multiple; DNA-protein complex	Multiple; DNA-protein complex
Lysosomes	Absent	Usually present	Present
Vacuoles	Absent	Absent or small	Usually a large single vacuole



Chapter 5

Summary

Questions

Media Resources

5.1 All organisms are composed of cells.

- The cell is the smallest unit of life. All living things are made of cells.
- The cell is composed of a nuclear region, which holds the hereditary apparatus, enclosed within the cytoplasm.
- In all cells, the cytoplasm is bounded by a membrane composed of phospholipid and protein.

1. What are the three principles of the cell theory?
2. How does the surface area-to-volume ratio of cells limit the size that cells can attain?



- Exploration: Cell Size



- Surface to Volume

5.2 Eukaryotic cells are far more complex than bacterial cells.

- Bacteria, which have prokaryotic cell structure, do not have membrane-bounded organelles within their cells. Their DNA molecule is circular.
- The eukaryotic cell is larger and more complex, with many internal compartments.

3. How are prokaryotes different from eukaryotes in terms of their cell walls, interior organization, and flagella?



- Art Activities:
 - Animal Cell Structure
 - Plant Cell Structure
 - Nonphotosynthetic Bacterium
 - Cyanobacterium

5.3 Take a tour of a eukaryotic cell.

- A eukaryotic cell is organized into three principal zones: the nucleus, the cytoplasm, and the plasma membrane. Located in the cytoplasm are numerous organelles, which perform specific functions for the cell.
- Many of these organelles, such as the endoplasmic reticulum, Golgi apparatus (which gives rise to lysosomes), and nucleus, are part of a complex endomembrane system.
- Mitochondria and chloroplasts are part of the energy-processing system of the cell.
- The cytoskeleton encompasses a variety of fibrous proteins that provide structural support and perform other functions for the cell.
- Many eukaryotic cells possess flagella or cilia having a 9 + 2 arrangement of microtubules; sliding of the microtubules past one another bends these cellular appendages.
- Cells transport materials long distances within the cytoplasm by packaging them into vesicles that are pulled by motor proteins along microtubule tracks.

4. What is the endoplasmic reticulum? What is its function? How does rough ER differ from smooth ER?



- Art Activities:
 - Anatomy of the Nucleus
 - Golgi Apparatus Structure
 - Mitochondrion Structure
 - Organization of Cristae
 - Chloroplast Structure
 - The Cytoskeleton
 - Plant Cell

5. What is the function of the Golgi apparatus? How do the substances released by the Golgi apparatus make their way to other locations in the cell?

6. What types of eukaryotic cells contain mitochondria? What function do mitochondria perform?



- Endomembrane
- Energy Organelles
- Cytoskeleton

7. What unique metabolic activity occurs in chloroplasts?



- Art Quizzes:
 - Nucleosomes
 - Rough ER and Protein Synthesis
 - Protein Transport

8. What cellular functions do centrioles participate in?

9. What kinds of cytoskeleton fibers are stable and which are changeable?

10. How do cilia compare with eukaryotic flagella?

5.4 Symbiosis played a key role in the origin of some eukaryotic organelles.

- Present-day mitochondria and chloroplasts probably evolved as a consequence of early endosymbiosis: the ancestor of the eukaryotic cell engulfed a bacterium, and the bacterium continued to function within the host cell.

11. What is the endosymbiont theory? What is the evidence supporting this theory?



- Scientists on Science: The Joy of Discovery



6

Membranes

Concept Outline

6.1 Biological membranes are fluid layers of lipid.

The Phospholipid Bilayer. Cells are encased by membranes composed of a bilayer of phospholipid.

The Lipid Bilayer Is Fluid. Because individual phospholipid molecules do not bind to one another, the lipid bilayer of membranes is a fluid.

6.2 Proteins embedded within the plasma membrane determine its character.

The Fluid Mosaic Model. A varied collection of proteins float within the lipid bilayer.

Examining Cell Membranes. Visualizing a plasma membrane requires a powerful electron microscope.

Kinds of Membrane Proteins. The proteins in a membrane function in support, transport, recognition, and reactions.

Structure of Membrane Proteins. Membrane proteins are anchored into the lipid bilayer by their nonpolar regions.

6.3 Passive transport across membranes moves down the concentration gradient.

Diffusion. Random molecular motion results in a net movement of molecules to regions of lower concentration.

Facilitated Diffusion. Passive movement across a membrane is often through specific carrier proteins.

Osmosis. Polar solutes interact with water and can affect the movement of water across semipermeable membranes.

6.4 Bulk transport utilizes endocytosis.

Bulk Passage Into and Out of the Cell. To transport large particles, membranes form vesicles.

6.5 Active transport across membranes is powered by energy from ATP.

Active Transport. Cells transport molecules up a concentration gradient using ATP-powered carrier proteins.

Coupled Transport. Active transport of ions drives coupled uptake of other molecules up their concentration gradients.



FIGURE 6.1

Membranes within a human cell. Sheets of endoplasmic reticulum weave through the cell interior. The large oval is a mitochondrion, itself filled with extensive internal membranes.

Among a cell's most important activities are its interactions with the environment, a give and take that never ceases. Without it, life could not persist. While living cells and eukaryotic organelles (figure 6.1) are encased within a lipid membrane through which few water-soluble substances can pass, the membrane contains protein passageways that permit specific substances to move in and out of the cell and allow the cell to exchange information with its environment. We call this delicate skin of protein molecules embedded in a thin sheet of lipid a **plasma membrane**. This chapter will examine the structure and function of this remarkable membrane.

6.1 Biological membranes are fluid layers of lipid.

The Phospholipid Bilayer

The membranes that encase all living cells are sheets of lipid only two molecules thick; more than 10,000 of these sheets piled on one another would just equal the thickness of this sheet of paper. The lipid layer that forms the foundation of a cell membrane is composed of molecules called **phospholipids** (figure 6.2).

Phospholipids

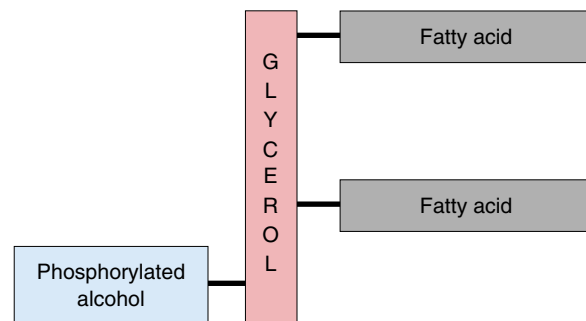
Like the fat molecules you studied in chapter 3, a phospholipid has a backbone derived from a three-carbon molecule called glycerol. Attached to this backbone are fatty acids, long chains of carbon atoms ending in a carboxyl (—COOH) group. A fat molecule has three such chains, one attached to each carbon in the backbone; because these chains are nonpolar, they do not form hydrogen bonds with water, and the fat molecule is not water-soluble. A phospholipid, by contrast, has only two fatty acid chains attached to its backbone. The third carbon on the backbone is attached instead to a highly polar organic alcohol that readily forms hydrogen bonds with water. Because this alcohol is attached by a phosphate group, the molecule is called a *phospholipid*.

One end of a phospholipid molecule is, therefore, strongly nonpolar (water-insoluble), while the other end is strongly polar (water-soluble). The two nonpolar fatty acids extend in one direction, roughly parallel to each other, and the polar alcohol group points in the other direction. Because of this structure, phospholipids are often diagrammed as a polar head with two dangling nonpolar tails (as in figure 6.2*b*).

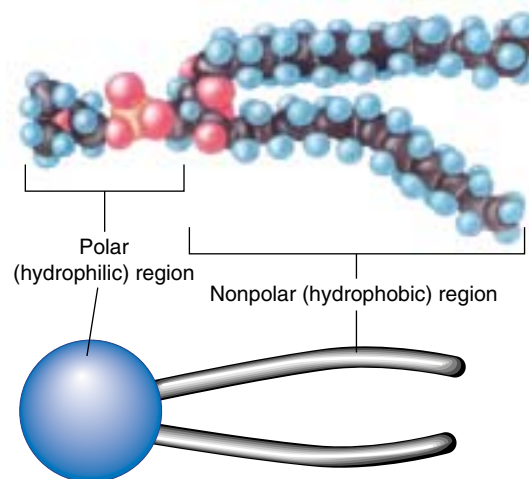
Phospholipids Form Bilayer Sheets

What happens when a collection of phospholipid molecules is placed in water? The polar water molecules repel the long nonpolar tails of the phospholipids as the water molecules seek partners for hydrogen bonding. Due to the polar nature of the water molecules, the nonpolar tails of the phospholipids end up packed closely together, sequestered as far as possible from water. Every phospholipid molecule orients to face its polar head toward water and its nonpolar tails away. When *two* layers form with the tails facing each other, no tails ever come in contact with water. The resulting structure is called a **lipid bilayer** (figure 6.3). Lipid bilayers form spontaneously, driven by the tendency of water molecules to form the maximum number of hydrogen bonds.

The nonpolar interior of a lipid bilayer impedes the passage of any water-soluble substances through the bilayer,



(a)



(b)

FIGURE 6.2

Phospholipid structure. (a) A phospholipid is a composite molecule similar to a triacylglycerol, except that only two fatty acids are bound to the glycerol backbone; a phosphorylated alcohol occupies the third position on the backbone. (b) Because the phosphorylated alcohol usually extends from one end of the molecule and the two fatty acid chains extend from the other, phospholipids are often diagrammed as a polar head with two nonpolar hydrophobic tails.

just as a layer of oil impedes the passage of a drop of water (“oil and water do not mix”). This barrier to the passage of water-soluble substances is the key biological property of the lipid bilayer. In addition to the phospholipid molecules that make up the lipid bilayer, the membranes of every cell also contain proteins that extend through the lipid bilayer, providing passageways across the membrane.

The basic foundation of biological membranes is a lipid bilayer, which forms spontaneously. In such a layer, the nonpolar hydrophobic tails of phospholipid molecules point inward, forming a nonpolar barrier to water-soluble molecules.

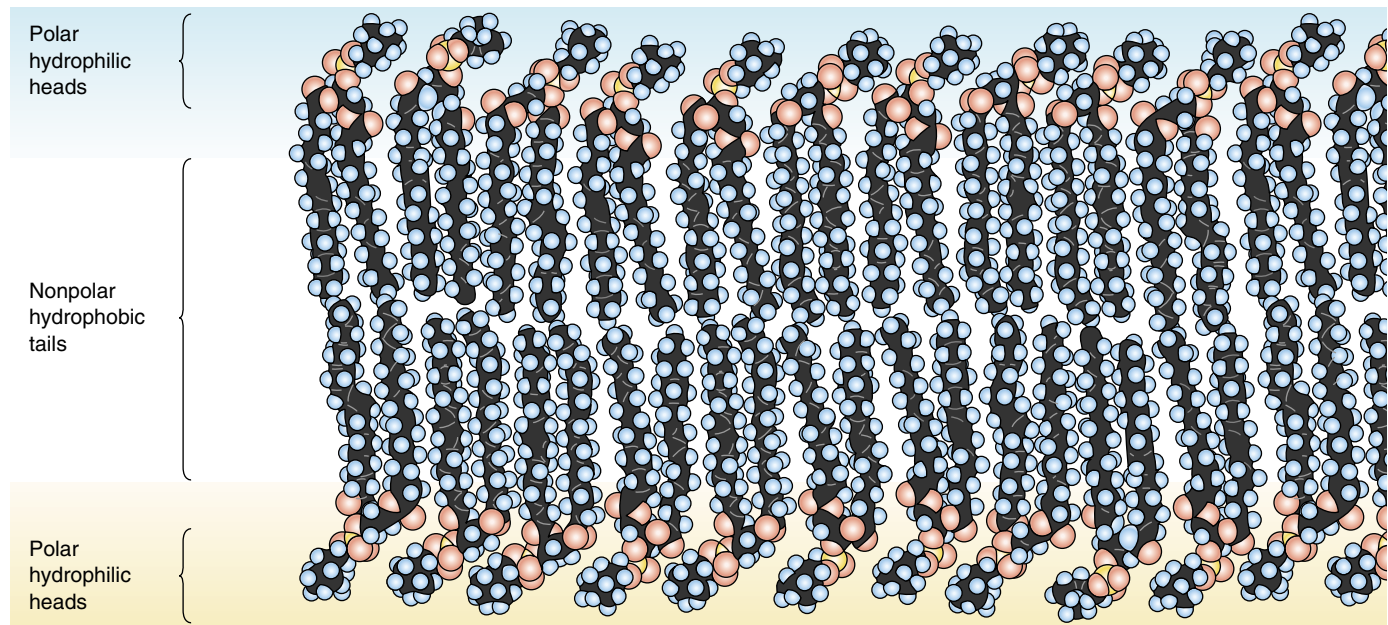


FIGURE 6.3

A phospholipid bilayer. The basic structure of every plasma membrane is a double layer of lipid, in which phospholipids aggregate to form a bilayer with a nonpolar interior. The phospholipid tails do not align perfectly and the membrane is “fluid.” Individual phospholipid molecules can move from one place to another in the membrane.

The Lipid Bilayer Is Fluid

A lipid bilayer is stable because water’s affinity for hydrogen bonding never stops. Just as surface tension holds a soap bubble together, even though it is made of a liquid, so the hydrogen bonding of water holds a membrane together. But while water continually drives phospholipid molecules into this configuration, it does not locate specific phospholipid molecules relative to their neighbors in the bilayer. As a result, individual phospholipids and unanchored proteins are free to move about within the membrane. This can be demonstrated vividly by fusing cells and watching their proteins reassemble (figure 6.4).

Phospholipid bilayers are fluid, with the viscosity of olive oil (and like oil, their viscosity increases as the temperature decreases). Some membranes are more fluid than others, however. The tails of individual phospholipid molecules are attracted to one another when they line up close together. This causes the membrane to become less fluid, because aligned molecules must pull apart from one another before they can move about in the membrane. The greater the degree of alignment, the less fluid the membrane. Some phospholipid tails do not align well because they contain one or more double bonds between carbon atoms, introducing kinks in the tail. Membranes containing such phospholipids are more fluid than membranes that lack them. Most membranes also contain steroid lipids like cholesterol, which can either increase or decrease membrane fluidity, depending on temperature.

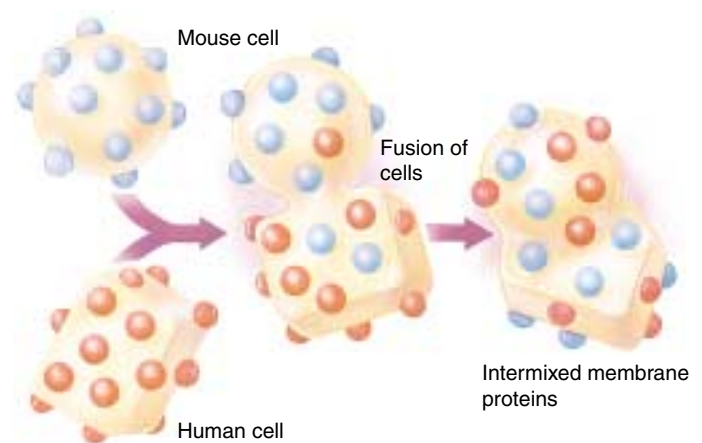


FIGURE 6.4

Proteins move about in membranes. Protein movement within membranes can be demonstrated easily by labeling the plasma membrane proteins of a mouse cell with fluorescent antibodies and then fusing that cell with a human cell. At first, all of the mouse proteins are located on the mouse side of the fused cell and all of the human proteins are located on the human side of the fused cell. However, within an hour, the labeled and unlabeled proteins are intermixed throughout the hybrid cell’s plasma membrane.

The lipid bilayer is liquid like a soap bubble, rather than solid like a rubber balloon.

6.2 Proteins embedded within the plasma membrane determine its character.

The Fluid Mosaic Model

A plasma membrane is composed of both lipids and globular proteins. For many years, biologists thought the protein covered the inner and outer surfaces of the phospholipid bilayer like a coat of paint. The widely accepted Davson-Danielli model, proposed in 1935, portrayed the membrane as a sandwich: a phospholipid bilayer between two layers of globular protein. This model, however, was not consistent with what researchers were learning in the 1960s about the structure of membrane proteins. Unlike most proteins found within cells, membrane proteins are not very soluble in water—they possess long stretches of nonpolar hydrophobic amino acids. If such proteins indeed coated the surface of the lipid bilayer, as the Davson-Danielli model suggests, then their nonpolar portions would separate the polar portions of the phospholipids from water, causing the bilayer to dissolve! Because this doesn't happen, there is clearly something wrong with the model.

In 1972, S. Singer and G. Nicolson revised the model in a simple but profound way: they proposed that the globular proteins are *inserted* into the lipid bilayer, with their nonpolar segments in contact with the nonpolar interior of the bilayer and their polar portions protruding out from the membrane surface. In this model, called the **fluid mosaic model**, a mosaic of proteins float in the fluid lipid bilayer like boats on a pond (figure 6.5).

Components of the Cell Membrane

A eukaryotic cell contains many membranes. While they are not all identical, they share the same fundamental architecture. Cell membranes are assembled from four components (table 6.1):

1. **Lipid bilayer.** Every cell membrane is composed of a phospholipid bilayer. The other components of the membrane are enmeshed within the bilayer, which provides a flexible matrix and, at the same time, imposes a barrier to permeability.

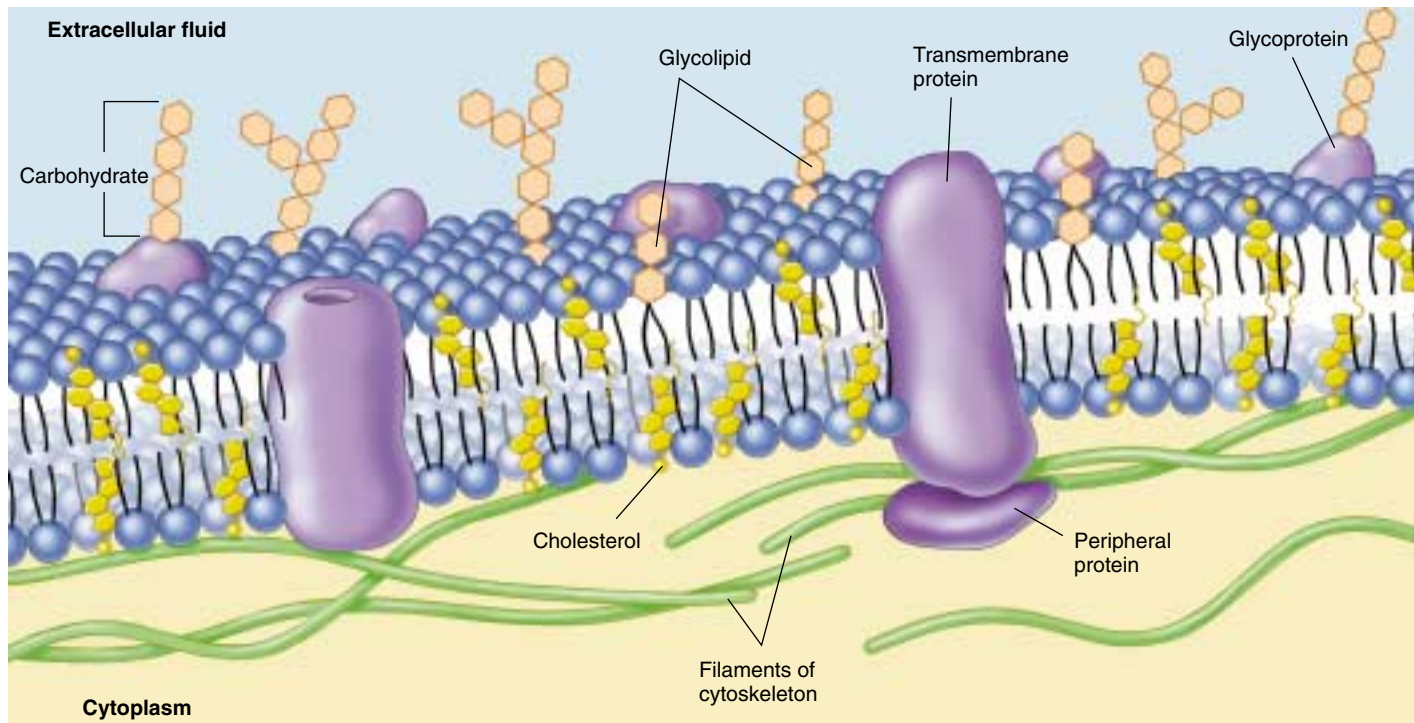


FIGURE 6.5

The fluid mosaic model of the plasma membrane. A variety of proteins protrude through the plasma membrane of animal cells, and nonpolar regions of the proteins tether them to the membrane's nonpolar interior. The three principal classes of membrane proteins are transport proteins, receptors, and cell surface markers. Carbohydrate chains are often bound to the extracellular portion of these proteins, as well as to the membrane phospholipids. These chains serve as distinctive identification tags, unique to particular cells.

Table 6.1 Components of the Cell Membrane				
Component	Composition	Function	How It Works	Example
Phospholipid molecules	Phospholipid bilayer	Provides permeability barrier, matrix for proteins	Excludes water-soluble molecules from nonpolar interior of bilayer	Bilayer of cell is impermeable to water-soluble molecules, like glucose
Transmembrane proteins	Carriers	Transport molecules across membrane against gradient	“Escort” molecules through the membrane in a series of conformational changes	Glycophorin carrier for sugar transport
	Channels	Passively transport molecules across membrane	Create a tunnel that acts as a passage through membrane	Sodium and potassium channels in nerve cells
	Receptors	Transmit information into cell	Signal molecules bind to cell-surface portion of the receptor protein; this alters the portion of the receptor protein within the cell, inducing activity	Specific receptors bind peptide hormones and neurotransmitters
Interior protein network	Spectrins	Determine shape of cell	Form supporting scaffold beneath membrane, anchored to both membrane and cytoskeleton	Red blood cell
	Clathrins	Anchor certain proteins to specific sites, especially on the exterior cell membrane in receptor-mediated endocytosis	Proteins line coated pits and facilitate binding to specific molecules	Localization of low-density lipoprotein receptor within coated pits
Cell surface markers	Glycoproteins	“Self”-recognition	Create a protein/carbohydrate chain shape characteristic of individual	Major histocompatibility complex protein recognized by immune system
	Glycolipid	Tissue recognition	Create a lipid/carbohydrate chain shape characteristic of tissue	A, B, O blood group markers

2. Transmembrane proteins. A major component of every membrane is a collection of proteins that float on or in the lipid bilayer. These proteins provide passageways that allow substances and information to cross the membrane. Many membrane proteins are not fixed in position; they can move about, as the phospholipid molecules do. Some membranes are crowded with proteins, while in others, the proteins are more sparsely distributed.

3. Network of supporting fibers. Membranes are structurally supported by intracellular proteins that reinforce the membrane’s shape. For example, a red blood cell has a characteristic biconcave shape because a scaffold of proteins called spectrin links proteins in the plasma membrane with actin filaments in the cell’s cytoskeleton. Membranes use networks of other proteins to control the lateral movements of some key membrane proteins, anchoring them to specific sites.

4. Exterior proteins and glycolipids. Membrane sections assemble in the endoplasmic reticulum, transfer to the Golgi complex, and then are transported to the plasma membrane. The endoplasmic reticulum adds chains of sugar molecules to membrane proteins and lipids, creating a “sugar coating” called the glycocalyx that extends from the membrane on the outside of the cell only. Different cell types exhibit different varieties of these glycoproteins and glycolipids on their surfaces, which act as cell identity markers.

The fluid mosaic model proposes that membrane proteins are embedded within the lipid bilayer. Membranes are composed of a lipid bilayer within which proteins are anchored. Plasma membranes are supported by a network of fibers and coated on the exterior with cell identity markers.

Examining Cell Membranes

Biologists examine the delicate, filmy structure of a cell membrane using electron microscopes that provide clear magnification to several thousand times. We discussed two types of electron microscopes in chapter 5: the transmission electron microscope (TEM) and the scanning electron microscope (SEM). When examining cell membranes with electron microscopy, specimens must be prepared for viewing.

In one method of preparing a specimen, the tissue of choice is embedded in a hard matrix, usually some sort of epoxy (figure 6.6). The epoxy block is then cut with a microtome, a machine with a very sharp blade that makes incredibly thin slices. The knife moves up and down as the specimen advances toward it, causing transparent “epoxy shavings” less than 1 micrometer thick to peel away from the block of tissue. These shavings are placed on a grid and a beam of electrons is directed through the grid with the TEM. At the high magnification an electron microscope provides, resolution is good enough to reveal the double layers of a membrane.

Freeze-fracturing a specimen is another way to visualize the inside of the membrane. The tissue is embedded in a medium and quick-frozen with liquid nitrogen. The frozen tissue is then “tapped” with a knife, causing a crack between the phospholipid layers of membranes. Proteins, carbohydrates, pits, pores, channels, or any other structure affiliated with the membrane will pull apart (whole, usually) and stick with one side of the split membrane. A very thin coating of platinum is then evaporated onto the fractured surface forming a replica of “cast” of the surface. Once the topography of the membrane has been preserved in the “cast,” the actual tissue is dissolved away, and the “cast” is examined with electron microscopy, creating a strikingly different view of the membrane (see figure 5.10*b*).

Visualizing a plasma membrane requires a very powerful electron microscope. Electrons can either be passed through a sample or bounced off it.

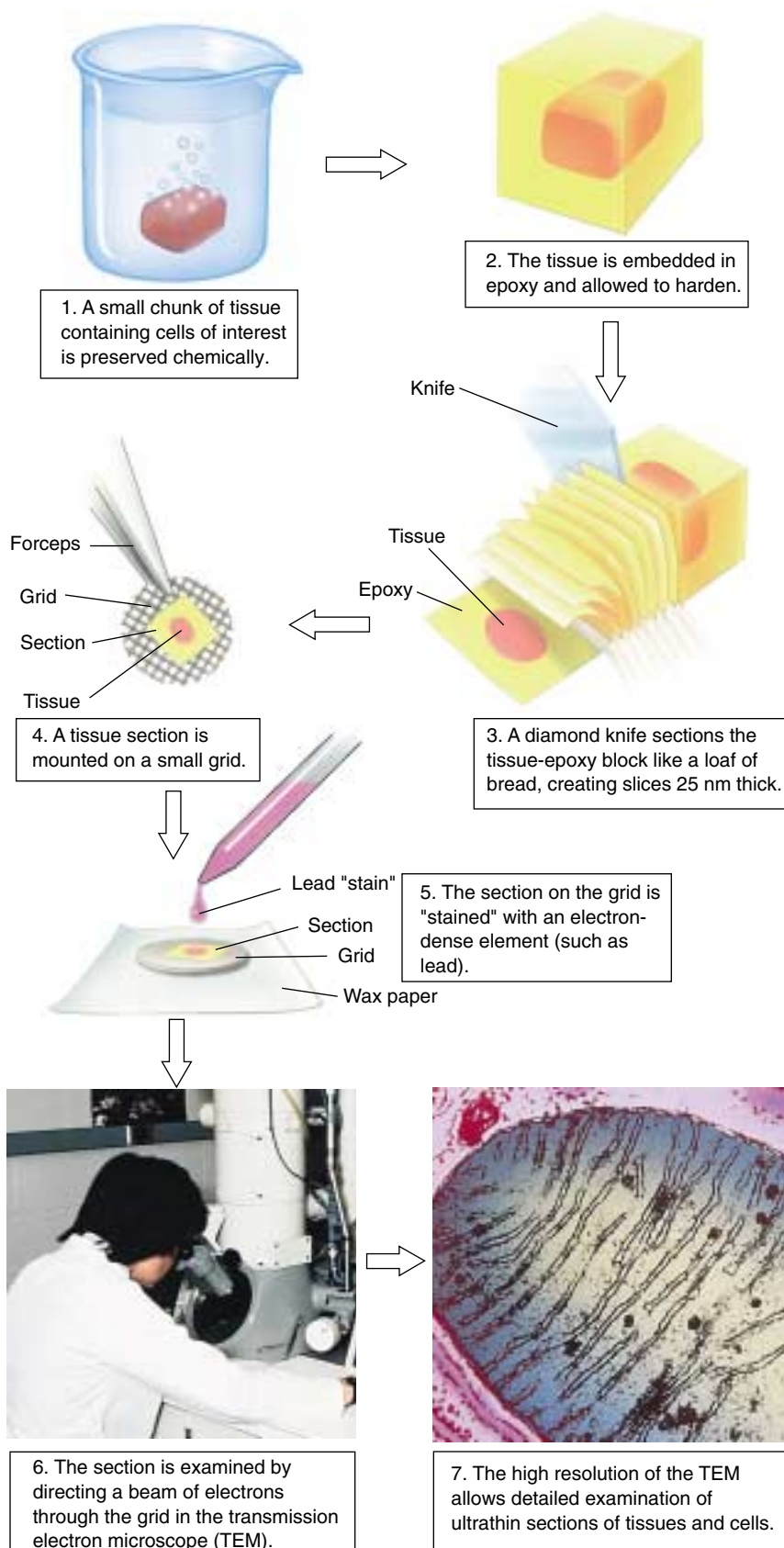


FIGURE 6.6
Thin section preparation for viewing membranes with electron microscopy.

Kinds of Membrane Proteins

As we've seen, the plasma membrane is a complex assembly of proteins enmeshed in a fluid array of phospholipid molecules. This enormously flexible design permits a broad range of interactions with the environment, some directly involving membrane proteins (figure 6.7). Though cells interact with their environment through their plasma membranes in many ways, we will focus on six key classes of membrane protein in this and the following chapter (chapter 7).

- 1. Transporters.** Membranes are very selective, allowing only certain substances to enter or leave the cell, either through channels or carriers. In some instances, they take up molecules already present in the cell in high concentration.
- 2. Enzymes.** Cells carry out many chemical reactions on the interior surface of the plasma membrane, using enzymes attached to the membrane.

- 3. Cell surface receptors.** Membranes are exquisitely sensitive to chemical messages, detecting them with receptor proteins on their surfaces that act as antennae.
- 4. Cell surface identity markers.** Membranes carry cell surface markers that identify them to other cells. Most cell types carry their own ID tags, specific combinations of cell surface proteins characteristic of that cell type.
- 5. Cell adhesion proteins.** Cells use specific proteins to glue themselves to one another. Some act like Velcro, while others form a more permanent bond.
- 6. Attachments to the cytoskeleton.** Surface proteins that interact with other cells are often anchored to the cytoskeleton by linking proteins.

The many proteins embedded within a membrane carry out a host of functions, many of which are associated with transport of materials or information across the membrane.

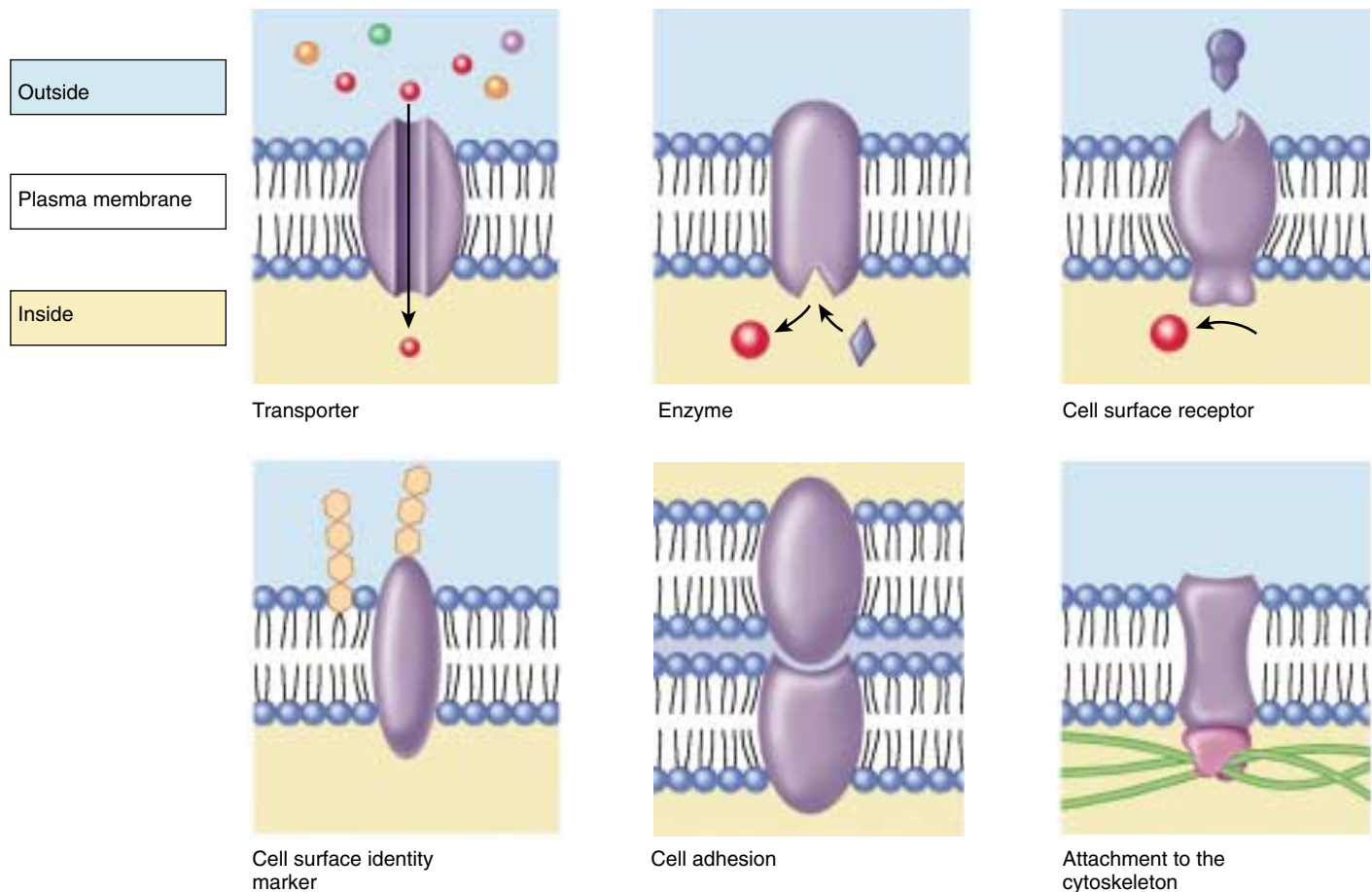


Figure 6.7

Functions of plasma membrane proteins. Membrane proteins act as transporters, enzymes, cell surface receptors, and cell surface markers, as well as aiding in cell-to-cell adhesion and securing the cytoskeleton.

Structure of Membrane Proteins

If proteins float on lipid bilayers like ships on the sea, how do they manage to extend through the membrane to create channels, and how can certain proteins be anchored into particular positions on the cell membrane?

Anchoring Proteins in the Bilayer

Many membrane proteins are attached to the surface of the membrane by special molecules that associate with phospholipids and thereby anchor the protein to the membrane. Like a ship tied up to a floating dock, these proteins are free to move about on the surface of the membrane tethered to a phospholipid.

In contrast, other proteins actually traverse the lipid bilayer. The part of the protein that extends through the lipid bilayer, in contact with the nonpolar interior, consists of one or more nonpolar helices or several β -pleated sheets of nonpolar amino acids (figure 6.8). Because water avoids nonpolar amino acids much as it does nonpolar lipid chains, the nonpolar portions of the protein are held within the interior of the lipid bilayer. Although the polar ends of the protein protrude from both sides of the membrane, the protein itself is locked into the membrane by its nonpolar segments. Any movement of the protein out of the membrane, in either direction, brings the nonpolar regions of the protein into contact with water, which “shoves” the protein back into the interior.

Extending Proteins across the Bilayer

Cells contain a variety of different **transmembrane proteins**, which differ in the way they traverse the bilayer, depending on their functions.

Anchors. A single nonpolar segment is adequate to anchor a protein in the membrane. Anchoring proteins of this sort attach the spectrin network of the cytoskeleton to the interior of the plasma membrane (figure 6.9). Many proteins that function as receptors for extracellular signals are also “single-pass” anchors that pass through the membrane only once. The portion of the receptor that extends out from the cell surface binds to specific hormones or other molecules when the cell encounters them; the binding induces changes at the other end of the protein, in the cell’s interior. In this way, information outside the cell is translated into action within the cell. The mechanisms of cell signaling will be addressed in detail in chapter 7.

Channels. Other proteins have several helical segments that thread their way back and forth through the membrane, forming a channel like the hole in a doughnut. For example, bacteriorhodopsin is one of the key transmembrane proteins that carries out photosynthesis in bacteria. It contains seven nonpolar helical segments that traverse the

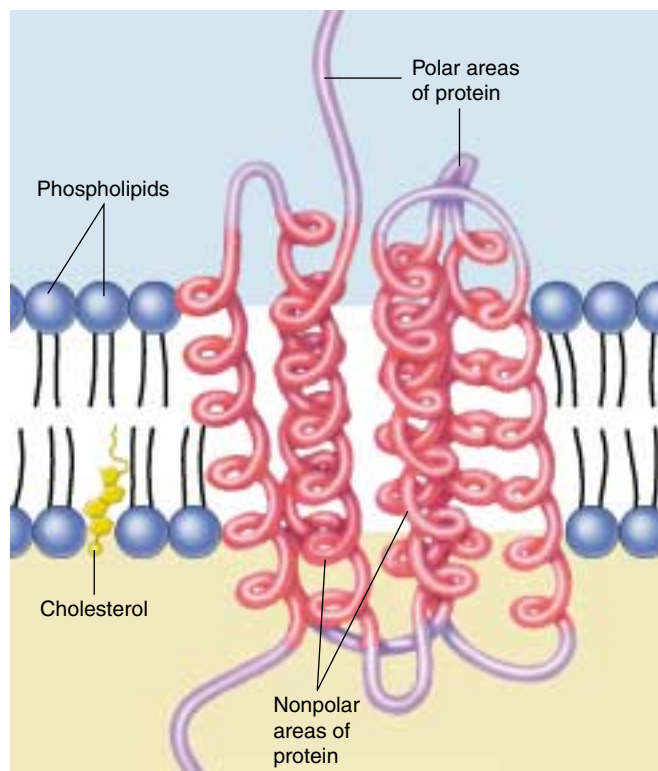


FIGURE 6.8

How nonpolar regions lock proteins into membranes. A spiral helix of nonpolar amino acids (red) extends across the nonpolar lipid interior, while polar (purple) portions of the protein protrude out from the bilayer. The protein cannot move in or out because such a movement would drag nonpolar segments of the protein into contact with water.

membrane, forming a circular pore through which protons pass during the light-driven pumping of protons (figure 6.10). Other transmembrane proteins do not create channels but rather act as carriers to transport molecules across the membrane. All water-soluble molecules or ions that enter or leave the cell are either transported by carriers or pass through channels.

Pores. Some transmembrane proteins have extensive nonpolar regions with secondary configurations of β -pleated sheets instead of α helices. The β sheets form a characteristic motif, folding back and forth in a circle so the sheets come to be arranged like the staves of a barrel. This so-called β barrel, open on both ends, is a common feature of the porin class of proteins that are found within the outer membrane of some bacteria (figure 6.11).

Transmembrane proteins are anchored into the bilayer by their nonpolar segments. While anchor proteins may pass through the bilayer only once, many channels and pores are created by proteins that pass back and forth through the bilayer repeatedly, creating a circular hole in the bilayer.

FIGURE 6.9

Anchoring proteins. Spectrin extends as a mesh anchored to the cytoplasmic side of a red blood cell plasma membrane. The spectrin protein is represented as a twisted dimer, attached to the membrane by special proteins such as junctional complexes and ankyrin; glycoporins can also be involved in attachments. This cytoskeletal protein network confers resiliency to cells like the red blood cell.

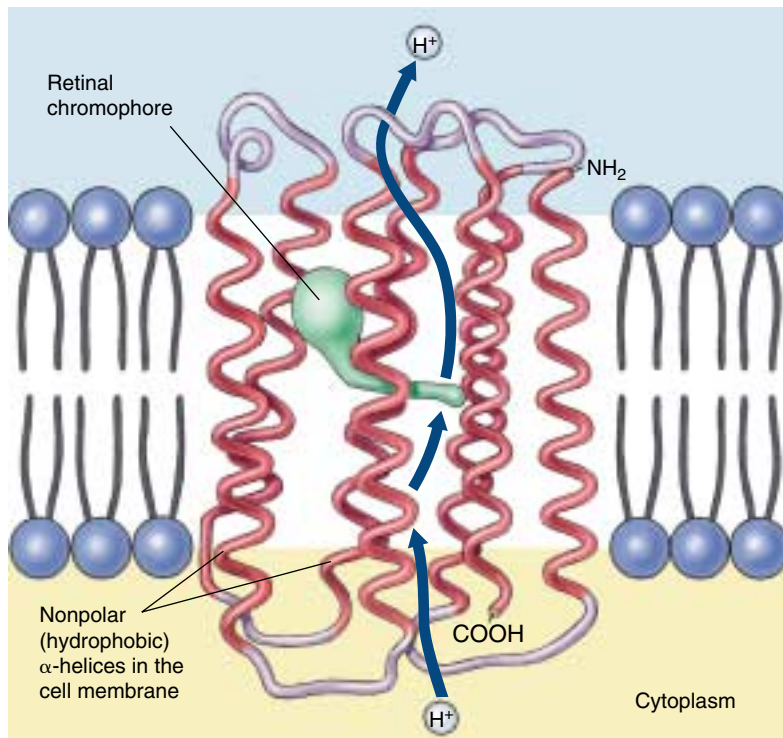
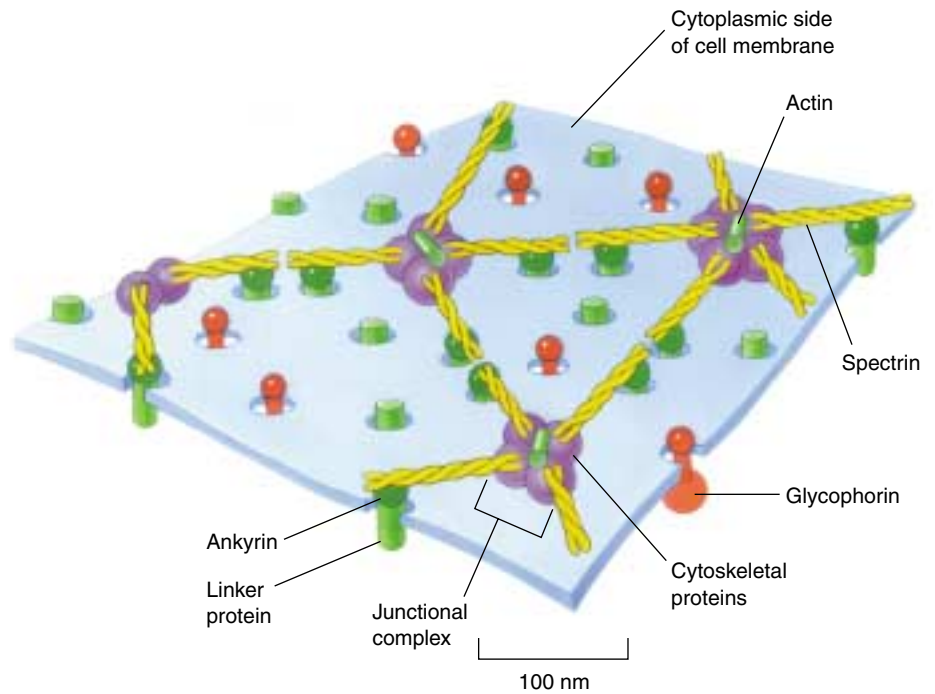


FIGURE 6.10

A channel protein. This transmembrane protein mediates photosynthesis in the bacterium *Halobacterium halobium*. The protein traverses the membrane seven times with hydrophobic helical strands that are within the hydrophobic center of the lipid bilayer. The helical regions form a channel across the bilayer through which protons are pumped by the retinal chromophore (green).

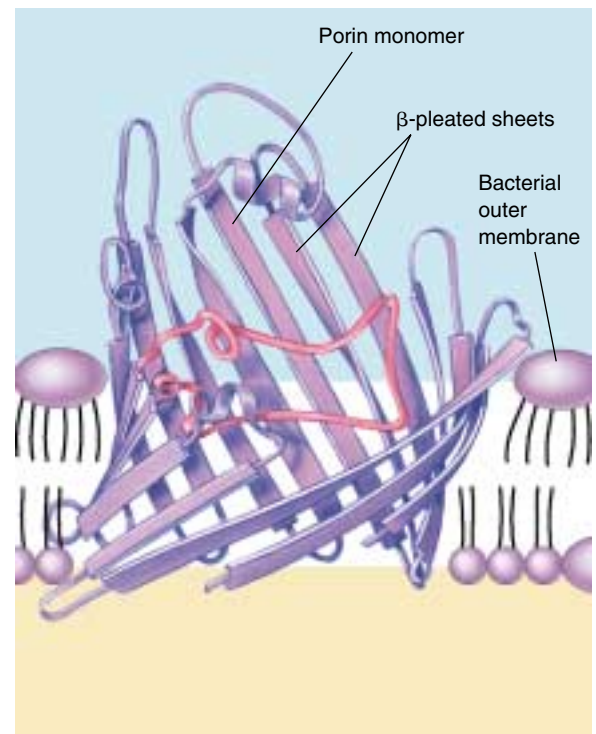


FIGURE 6.11

A pore protein. The bacterial transmembrane protein porin creates large open tunnels called pores in the outer membrane of a bacterium. Sixteen strands of β-pleated sheets run antiparallel to each other, creating a β barrel in the bacterial outer cell membrane. The tunnel allows water and other materials to pass through the membrane.

6.3 Passive transport across membranes moves down the concentration gradient.

Diffusion

Molecules and ions dissolved in water are in constant motion, moving about randomly. This random motion causes a net movement of these substances from regions where their concentration is high to regions where their concentration is lower, a process called **diffusion** (figure 6.12). Net movement driven by diffusion will continue until the concentrations in all regions are the same. You can demonstrate diffusion by filling a jar to the brim with ink, capping it, placing it at the bottom of a bucket of water, and then carefully removing the cap. The ink molecules will slowly diffuse out from the jar until there is a uniform concentration in the bucket and the jar. This uniformity in the concentration of molecules is a type of equilibrium.

Facilitated Transport

Many molecules that cells require, including glucose and other energy sources, are polar and cannot pass through the nonpolar interior of the phospholipid bilayer. These molecules enter the cell through specific channels in the plasma membrane. The inside of the channel is polar and thus “friendly” to the polar molecules, facilitating their transport across the membrane. Each type of biomolecule that is transported across the plasma membrane has its own type of transporter (that is, it has its own channel which fits it like a glove and cannot be used by other molecules). Each channel is said to be selective for that type of molecule, and thus to be **selectively permeable**, as only molecules admitted by the channels it possesses can enter it. The plasma membrane of a cell has many types of channels, each selective for a different type of molecule.

Diffusion of Ions through Channels

One of the simplest ways for a substance to diffuse across a cell membrane is through a channel, as ions do. Ions are solutes (substances dissolved in water) with an unequal number of protons and electrons. Those with an excess of protons are positively charged and called *cations*. Ions with more electrons are negatively charged and called *anions*. Because they are charged, ions interact well with polar molecules like water but are repelled by the nonpolar interior of a phospholipid bilayer. Therefore, ions cannot move between the cytoplasm of a cell and the extracellular fluid without the assistance of membrane transport proteins. **Ion channels** possess a hydrated interior that spans the membrane. Ions can diffuse through the channel in either direction without coming into contact with the hydrophobic tails of the phospholipids in the membrane, and the transported ions do not bind to or otherwise interact with the channel proteins. Two conditions determine the direction of net movement of the ions: their relative concentrations on either side of the membrane, and the voltage across the membrane (a topic we’ll explore in chapter 54). Each type of channel is specific for a particular ion, such as calcium (Ca^{++}) or chloride (Cl^-), or in some cases for a few kinds of ions. Ion channels play an essential role in signaling by the nervous system.

Diffusion is the net movement of substances to regions of lower concentration as a result of random spontaneous motion. It tends to distribute substances uniformly. Membrane transport proteins allow only certain molecules and ions to diffuse through the plasma membrane.

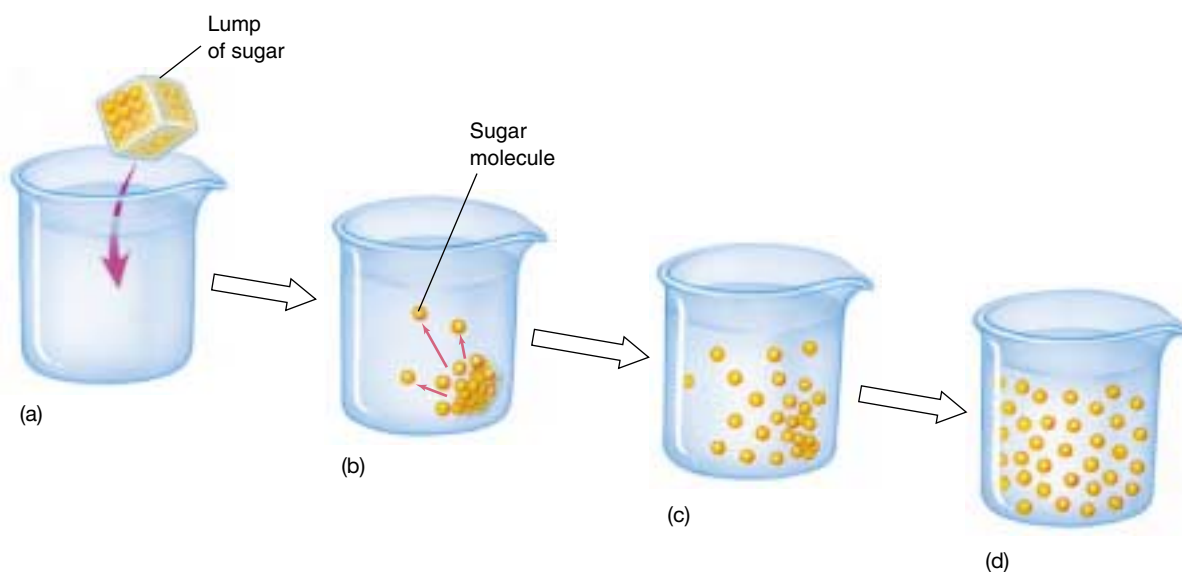


FIGURE 6.12

Diffusion. If a lump of sugar is dropped into a beaker of water (a), its molecules dissolve (b) and diffuse (c). Eventually, diffusion results in an even distribution of sugar molecules throughout the water (d).

Facilitated Diffusion

Carriers, another class of membrane proteins, transport ions as well as other solutes like sugars and amino acids across the membrane. Like channels, carriers are specific for a certain type of solute and can transport substances in either direction across the membrane. Unlike channels, however, they facilitate the movement of solutes across the membrane by physically binding to them on one side of the membrane and releasing them on the other. Again, the direction of the solute's net movement simply depends on its *concentration gradient* across the membrane. If the concentration is greater in the cytoplasm, the solute is more likely to bind to the carrier on the cytoplasmic side of the membrane and be released on the extracellular side. This will cause a net movement from inside to outside. If the concentration is greater in the extracellular fluid, the net movement will be from outside to inside. Thus, the net movement always occurs from areas of high concentration to low, just as it does in simple diffusion, but carriers facilitate the process. For this reason, this mechanism of transport is sometimes called **facilitated diffusion** (figure 6.13).

Facilitated Diffusion in Red Blood Cells

Several examples of facilitated diffusion by carrier proteins can be found in the membranes of vertebrate red blood cells (RBCs). One RBC carrier protein, for example, transports a different molecule in each direction: Cl^- in one direction and bicarbonate ion (HCO_3^-) in the opposite direction. As you will learn in chapter 52, this carrier is important in transporting carbon dioxide in the blood.

A second important facilitated diffusion carrier in RBCs is the glucose transporter. Red blood cells keep their internal concentration of glucose low through a chemical trick: they immediately add a phosphate group to any entering glucose molecule, converting it to a highly charged glucose phosphate that cannot pass back across the membrane. This maintains a steep concentration gradient for glucose, favoring its entry into the cell. The glucose transporter that carries glucose into the cell does not appear to form a channel in the membrane for the glucose to pass through. Instead, the transmembrane protein appears to bind the glucose and then flip its shape, dragging the glucose through the bilayer and releasing it on the inside of the plasma membrane. Once it releases the glucose, the glucose transporter reverts to its original shape. It is then available to bind the next glucose molecule that approaches the outside of the cell.

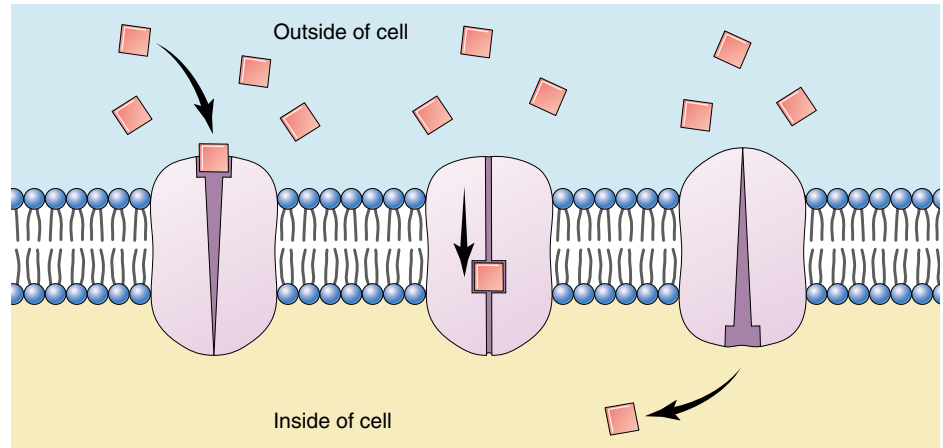


FIGURE 6.13

Facilitated diffusion is a carrier-mediated transport process. Molecules bind to a receptor on the extracellular side of the cell and are conducted through the plasma membrane by a membrane protein.

Transport through Selective Channels Saturates

A characteristic feature of transport through selective channels is that its rate is saturable. In other words, if the concentration gradient of a substance is progressively increased, its rate of transport will also increase to a certain point and then level off. Further increases in the gradient will produce no additional increase in rate. The explanation for this observation is that there are a limited number of carriers in the membrane. When the concentration of the transported substance rises high enough, all of the carriers will be in use and the capacity of the transport system will be saturated. In contrast, substances that move across the membrane by simple diffusion (diffusion through channels in the bilayer without the assistance of carriers) do not show saturation.

Facilitated diffusion provides the cell with a ready way to prevent the buildup of unwanted molecules within the cell or to take up needed molecules, such as sugars, that may be present outside the cell in high concentrations. Facilitated diffusion has three essential characteristics:

1. **It is specific.** Any given carrier transports only certain molecules or ions.
2. **It is passive.** The direction of net movement is determined by the relative concentrations of the transported substance inside and outside the cell.
3. **It saturates.** If all relevant protein carriers are in use, increases in the concentration gradient do not increase the transport rate.

Facilitated diffusion is the transport of molecules and ions across a membrane by specific carriers in the direction of lower concentration of those molecules or ions.

Osmosis

The cytoplasm of a cell contains ions and molecules, such as sugars and amino acids, dissolved in water. The mixture of these substances and water is called an **aqueous solution**. Water, the most common of the molecules in the mixture, is the **solvent**, and the substances dissolved in the water are **solutes**. The ability of water and solutes to diffuse across membranes has important consequences.

Molecules Diffuse down a Concentration Gradient

Both water and solutes diffuse from regions of high concentration to regions of low concentration; that is, they diffuse down their concentration gradients. When two regions are separated by a membrane, what happens depends on whether or not the solutes can pass freely through that membrane. Most solutes, including ions and sugars, are not lipid-soluble and, therefore, are unable to cross the lipid bilayer of the membrane.

Even water molecules, which are very polar, cannot cross a lipid bilayer. Water flows through **aquaporins**, which are specialized channels for water. A simple experiment demonstrates this. If you place an amphibian egg in hypotonic spring water, it does not swell. If you then inject aquaporin mRNA into the egg, the channel proteins are expressed and the egg then swells.

Dissolved solutes interact with water molecules, which form hydration shells about the charged solute. When there is a concentration gradient of solutes, the solutes will move from a high to a low concentration, dragging with them their hydration shells of water molecules. When a membrane separates two solutions, hydration shell water molecules move with the diffusing ions, creating a net movement of water towards the low solute. This net water movement across a membrane by diffusion is called **osmosis** (figure 6.14).

The concentration of *all* solutes in a solution determines the **osmotic concentration** of the solution. If two solutions have unequal osmotic concentrations, the solution with the higher concentration is **hyperosmotic** (Greek *hyper*, “more than”), and the solution with the lower concentration is **hypoosmotic** (Greek *hypo*, “less than”). If the osmotic concentrations of two solutions are equal, the solutions are **isosmotic** (Greek *iso*, “the same”).

In cells, a plasma membrane separates two aqueous solutions, one inside the cell (the cytoplasm) and one outside

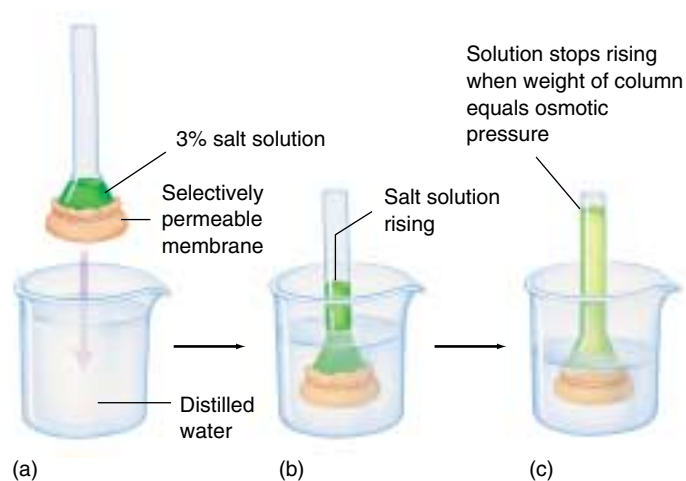
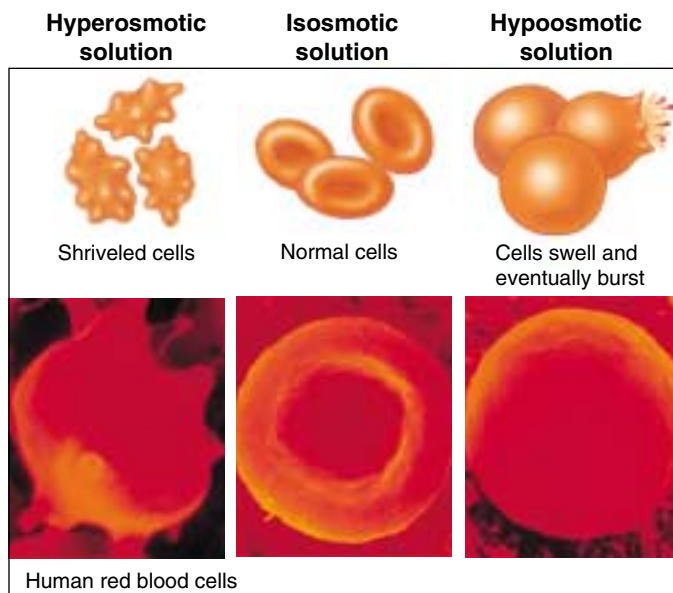
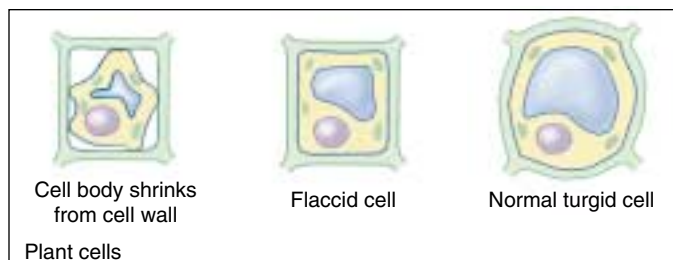


FIGURE 6.14

An experiment demonstrating osmosis. (a) The end of a tube containing a salt solution is closed by stretching a selectively permeable membrane across its face; the membrane allows the passage of water molecules but not salt ions. (b) When this tube is immersed in a beaker of distilled water, the salt cannot cross the membrane, but water can. The water entering the tube causes the salt solution to rise in the tube. (c) Water will continue to enter the tube from the beaker until the weight of the column of water in the tube exerts a downward force equal to the force drawing water molecules upward into the tube. This force is referred to as osmotic pressure.



Human red blood cells



Plant cells

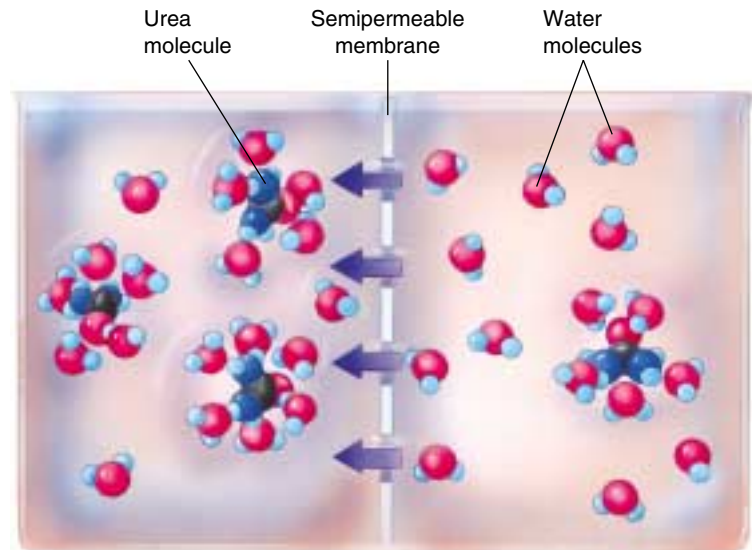
FIGURE 6.15

Osmosis. In a hyperosmotic solution water moves out of the cell toward the higher concentration of solutes, causing the cell to shrivel. In an isosmotic solution, the concentration of solutes on either side of the membrane is the same. Osmosis still occurs, but water diffuses into and out of the cell at the same rate, and the cell doesn't change size. In a hypoosmotic solution the concentration of solutes is higher within the cell than without, so the net movement of water is into the cell.

FIGURE 6.16

How solutes create osmotic pressure.

Charged or polar substances are soluble in water because they form hydrogen bonds with water molecules clustered around them. When a polar solute (illustrated here with urea) is added to the solution on one side of a membrane, the water molecules that gather around each urea molecule are no longer free to diffuse across the membrane; in effect, the polar solute has reduced the number of free water molecules on that side of the membrane, increasing the osmotic pressure. Because the hypoosmotic side of the membrane (on the right, with less solute) has more unbound water molecules than the hyperosmotic side (on the left, with more solute), water moves by diffusion from the right to the left.



(the extracellular fluid). The direction of the net diffusion of water across this membrane is determined by the osmotic concentrations of the solutions on either side (figure 6.15). For example, if the cytoplasm of a cell were *hypoosmotic* to the extracellular fluid, water would diffuse out of the cell, toward the solution with the higher concentration of solutes (and, therefore, the lower concentration of unbound water molecules). This loss of water from the cytoplasm would cause the cell to shrink until the osmotic concentrations of the cytoplasm and the extracellular fluid become equal.

Osmotic Pressure

What would happen if the cell's cytoplasm were *hyperosmotic* to the extracellular fluid? In this situation, water would diffuse into the cell from the extracellular fluid, causing the cell to swell. The pressure of the cytoplasm pushing out against the cell membrane, or **hydrostatic pressure**, would increase. On the other hand, the **osmotic pressure** (figure 6.16), defined as the pressure that must be applied to stop the osmotic movement of water across a membrane, would also be at work. If the membrane were strong enough, the cell would reach an equilibrium, at which the osmotic pressure, which tends to drive water into the cell, is exactly counterbalanced by the hydrostatic pressure, which tends to drive water back out of the cell. However, a plasma membrane by itself cannot withstand large internal pressures, and an isolated cell under such conditions would burst like an overinflated balloon. Accordingly, it is important for animal cells to maintain isosmotic conditions. The cells of bacteria, fungi, plants, and many protists, in contrast, are surrounded by strong cell walls. The cells of these organisms can withstand high internal pressures without bursting.

Maintaining Osmotic Balance

Organisms have developed many solutions to the osmotic dilemma posed by being hyperosmotic to their environment.

Extrusion. Some single-celled eukaryotes like the protist *Paramecium* use organelles called contractile vacuoles to remove water. Each vacuole collects water from various parts of the cytoplasm and transports it to the central part of the vacuole, near the cell surface. The vacuole possesses a small pore that opens to the outside of the cell. By contracting rhythmically, the vacuole pumps the water out of the cell through the pore.

Isosmotic Solutions. Some organisms that live in the ocean adjust their internal concentration of solutes to match that of the surrounding seawater. Isosmotic with respect to their environment, there is no net flow of water into or out of these cells. Many terrestrial animals solve the problem in a similar way, by circulating a fluid through their bodies that bathes cells in an isosmotic solution. The blood in your body, for example, contains a high concentration of the protein albumin, which elevates the solute concentration of the blood to match your cells.

Turgor. Most plant cells are hyperosmotic to their immediate environment, containing a high concentration of solutes in their central vacuoles. The resulting internal hydrostatic pressure, known as **turgor pressure**, presses the plasma membrane firmly against the interior of the cell wall, making the cell rigid. The newer, softer portions of trees and shrubs depend on turgor pressure to maintain their shape, and wilt when they lack sufficient water.

Osmosis is the diffusion of water, but not solutes, across a membrane.

6.4 Bulk transport utilizes endocytosis.

Bulk Passage Into and Out of the Cell

Endocytosis

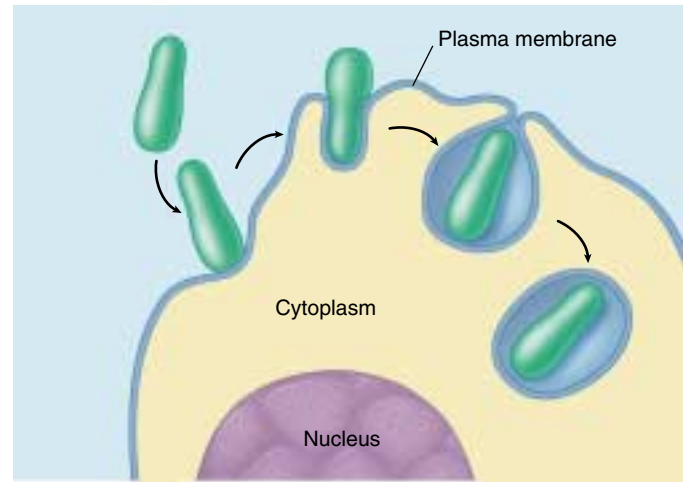
The lipid nature of their biological membranes raises a second problem for cells. The substances cells use as fuel are for the most part large, polar molecules that cannot cross the hydrophobic barrier a lipid bilayer creates. How do organisms get these substances into their cells? One process many single-celled eukaryotes employ is **endocytosis** (figure 6.17). In this process the plasma membrane extends outward and envelops food particles. Cells use three major types of endocytosis: phagocytosis, pinocytosis, and receptor-mediated endocytosis.

Phagocytosis and Pinocytosis. If the material the cell takes in is particulate (made up of discrete particles), such as an organism or some other fragment of organic matter (figure 6.17*a*), the process is called phagocytosis (Greek *phagein*, “to eat” + *cytos*, “cell”). If the material the cell takes in is liquid (figure 6.17*b*), it is called pinocytosis (Greek *pinein*, “to drink”). Pinocytosis is common among animal cells. Mammalian egg cells, for example, “nurse” from surrounding cells; the nearby cells secrete nutrients that the maturing egg cell takes up by pinocytosis. Virtually all eukaryotic cells constantly carry out these kinds of endocytosis, trapping particles and extracellular fluid in vesicles and ingesting them. Endocytosis rates vary from one cell type to another. They can be surprisingly high: some types of white blood cells ingest 25% of their cell volume each hour!

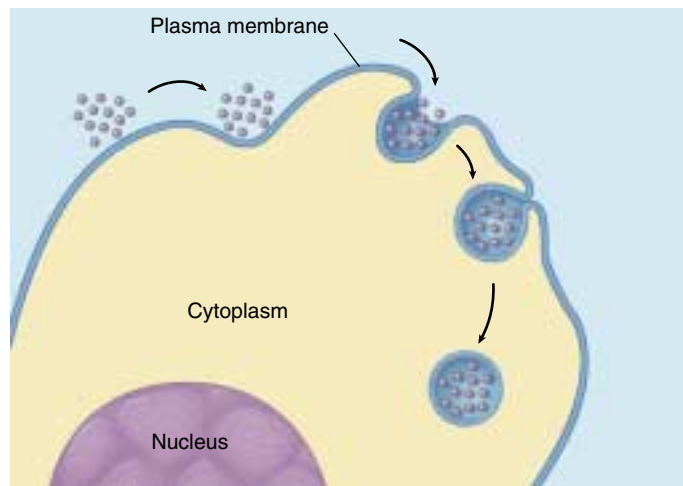
Receptor-Mediated Endocytosis. Specific molecules are often transported into eukaryotic cells through receptor-mediated endocytosis. Molecules to be transported first bind to specific receptors on the plasma membrane. The transport process is specific because only that molecule has a shape that fits snugly into the receptor. The plasma membrane of a particular kind of cell contains a characteristic battery of receptor types, each for a different kind of molecule.

The interior portion of the receptor molecule resembles a hook that is trapped in an indented pit coated with the protein clathrin. The pits act like molecular mousetraps, closing over to form an internal vesicle when the right molecule enters the pit (figure 6.18). The trigger that releases the trap is a receptor protein embedded in the membrane of the pit, which detects the presence of a particular target molecule and reacts by initiating endocytosis. The process is highly specific and very fast.

One type of molecule that is taken up by receptor-mediated endocytosis is called a low density lipoprotein (LDL). The LDL molecules bring cholesterol into the cell



(a) **Phagocytosis**



(b) **Pinocytosis**

FIGURE 6.17

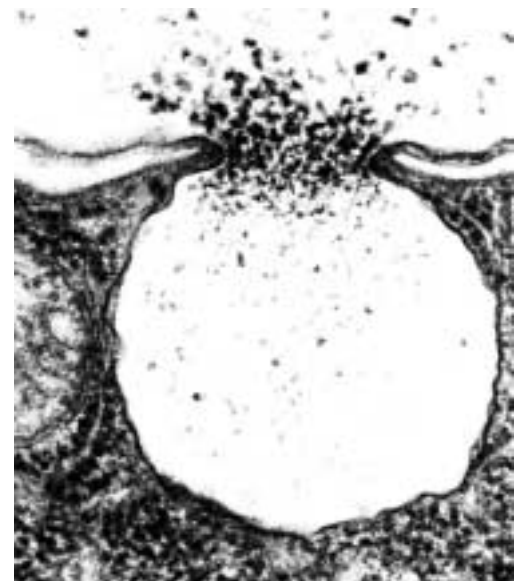
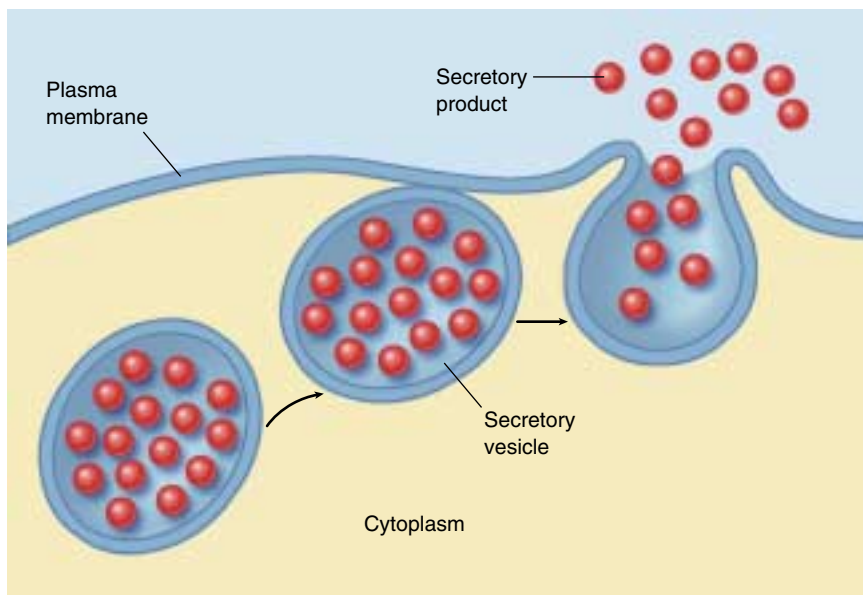
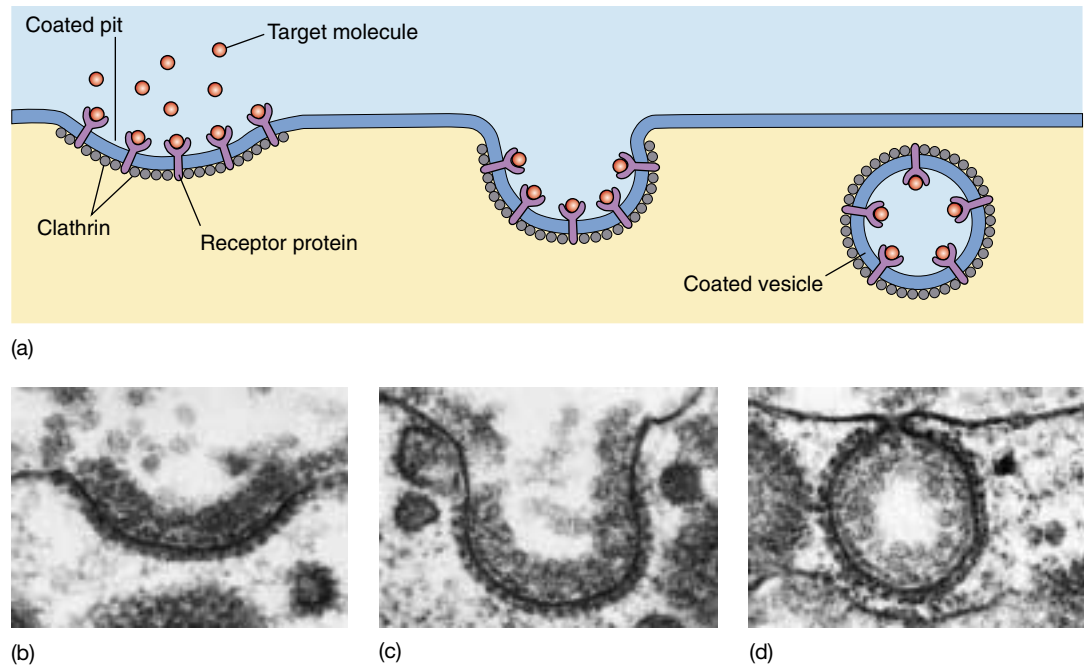
Endocytosis. Both phagocytosis (a) and pinocytosis (b) are forms of endocytosis.

where it can be incorporated into membranes. Cholesterol plays a key role in determining the stiffness of the body's membranes. In the human genetic disease called hypercholesteremia, the receptors lack tails and so are never caught in the clathrin-coated pits and, thus, are never taken up by the cells. The cholesterol stays in the bloodstream of affected individuals, coating their arteries and leading to heart attacks.

Fluid-phase endocytosis is the receptor-mediated pinocytosis of fluids. It is important to understand that endocytosis in itself does not bring substances directly into the cytoplasm of a cell. The material taken in is still separated from the cytoplasm by the membrane of the vesicle.

FIGURE 6.18

Receptor-mediated endocytosis. (a) Cells that undergo receptor-mediated endocytosis have pits coated with the protein clathrin that initiate endocytosis when target molecules bind to receptor proteins in the plasma membrane. (b) A coated pit appears in the plasma membrane of a developing egg cell, covered with a layer of proteins (80,000 $\times$). When an appropriate collection of molecules gathers in the coated pit, the pit deepens (c) and seals off (d) to form a coated vesicle, which carries the molecules into the cell.

**FIGURE 6.19**

Exocytosis. (a) Proteins and other molecules are secreted from cells in small packets called vesicles, whose membranes fuse with the plasma membrane, releasing their contents to the cell surface. (b) A transmission electron micrograph showing exocytosis.

Exocytosis

The reverse of endocytosis is **exocytosis**, the discharge of material from vesicles at the cell surface (figure 6.19). In plant cells, exocytosis is an important means of exporting the materials needed to construct the cell wall through the plasma membrane. Among protists, contractile vacuole discharge is a form of exocytosis. In animal cells, exocytosis

provides a mechanism for secreting many hormones, neurotransmitters, digestive enzymes, and other substances.

Cells import bulk materials by engulfing them with their plasma membranes in a process called endocytosis; similarly, they extrude or secrete material through exocytosis.

6.5 Active transport across membranes is powered by energy from ATP.

Active Transport

While diffusion, facilitated diffusion, and osmosis are passive transport processes that move materials down their concentration gradients, cells can also move substances across the membrane *up* their concentration gradients. This process requires the expenditure of energy, typically ATP, and is therefore called **active transport**. Like facilitated diffusion, active transport involves highly selective protein carriers within the membrane. These carriers bind to the transported substance, which could be an ion or a simple molecule like a sugar (figure 6.20), an amino acid, or a nucleotide to be used in the synthesis of DNA.

Active transport is one of the most important functions of any cell. It enables a cell to take up additional molecules of a substance that is already present in its cytoplasm in concentrations higher than in the extracellular fluid. Without active transport, for example, liver cells would be unable to accumulate glucose molecules from the blood plasma, as the glucose concentration is often higher inside the liver cells than it is in the plasma. Active transport also enables a cell to move substances from its cytoplasm to the extracellular fluid despite higher external concentrations.

The Sodium-Potassium Pump

The use of ATP in active transport may be direct or indirect. Let's first consider how ATP is used directly to move ions against their concentration gradient. More than one-third of all of the energy expended by an animal cell that is not actively dividing is used in the active transport of sodium (Na^+) and potassium (K^+) ions. Most animal cells have a low internal concentration of Na^+ , relative to their surroundings, and a high internal concentration of K^+ . They maintain these concentration differences by actively pumping Na^+ out of the cell and K^+ in. The remarkable protein that transports these two ions across the cell membrane is known as the **sodium-potassium pump** (figure 6.21). The cell obtains the energy it needs to operate the pump from adenosine triphosphate (ATP), a molecule we'll learn more about in chapter 8.

The important characteristic of the sodium-potassium pump is that it is an *active* transport process, transporting Na^+ and K^+ from areas of low concentration to areas of high concentration. This transport up their concentration gradients is the opposite of the passive transport in diffusion; it is achieved only by the constant expenditure of metabolic energy. The sodium-potassium pump works through a series of conformational changes in the transmembrane protein:

Step 1. Three sodium ions bind to the cytoplasmic side of the protein, causing the protein to change its conformation.

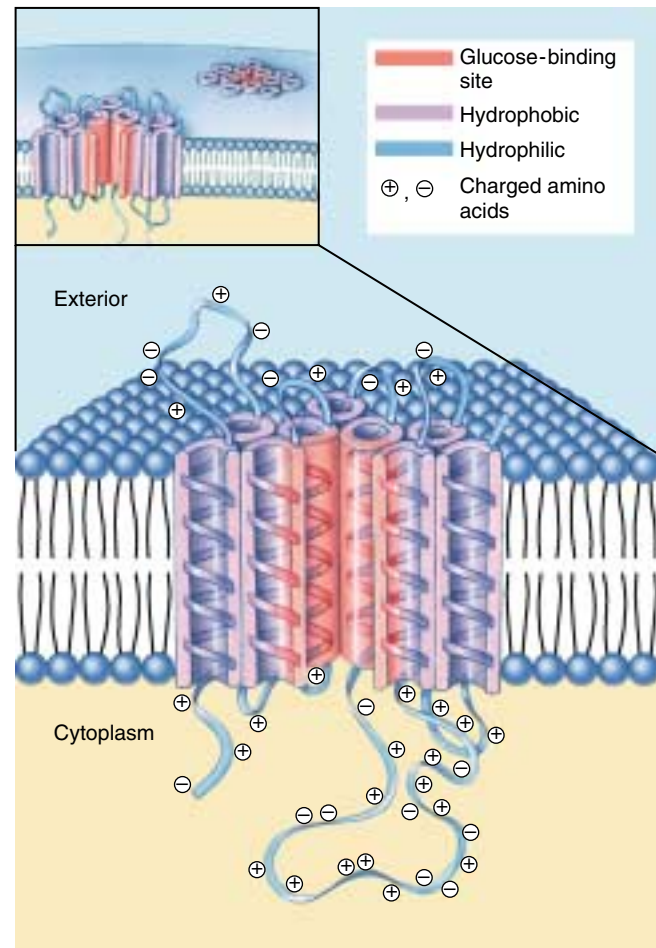


FIGURE 6.20

A glucose transport channel. The molecular structure of this particular glucose transport channel is known in considerable detail. The protein's 492 amino acids form a folded chain that traverses the lipid membrane 12 times. Amino acids with charged groups are less stable in the hydrophobic region of the lipid bilayer and are thus exposed to the cytoplasm or the extracellular fluid. Researchers think the center of the protein consists of five helical segments with glucose-binding sites (in red) facing inward. A conformational change in the protein transports glucose across the membrane by shifting the position of the glucose-binding sites.

Step 2. In its new conformation, the protein binds a molecule of ATP and cleaves it into adenosine diphosphate and phosphate ($\text{ADP} + \text{P}_i$). ADP is released, but the phosphate group remains bound to the protein. The protein is now phosphorylated.

Step 3. The phosphorylation of the protein induces a second conformational change in the protein. This change translocates the three Na^+ across the membrane,

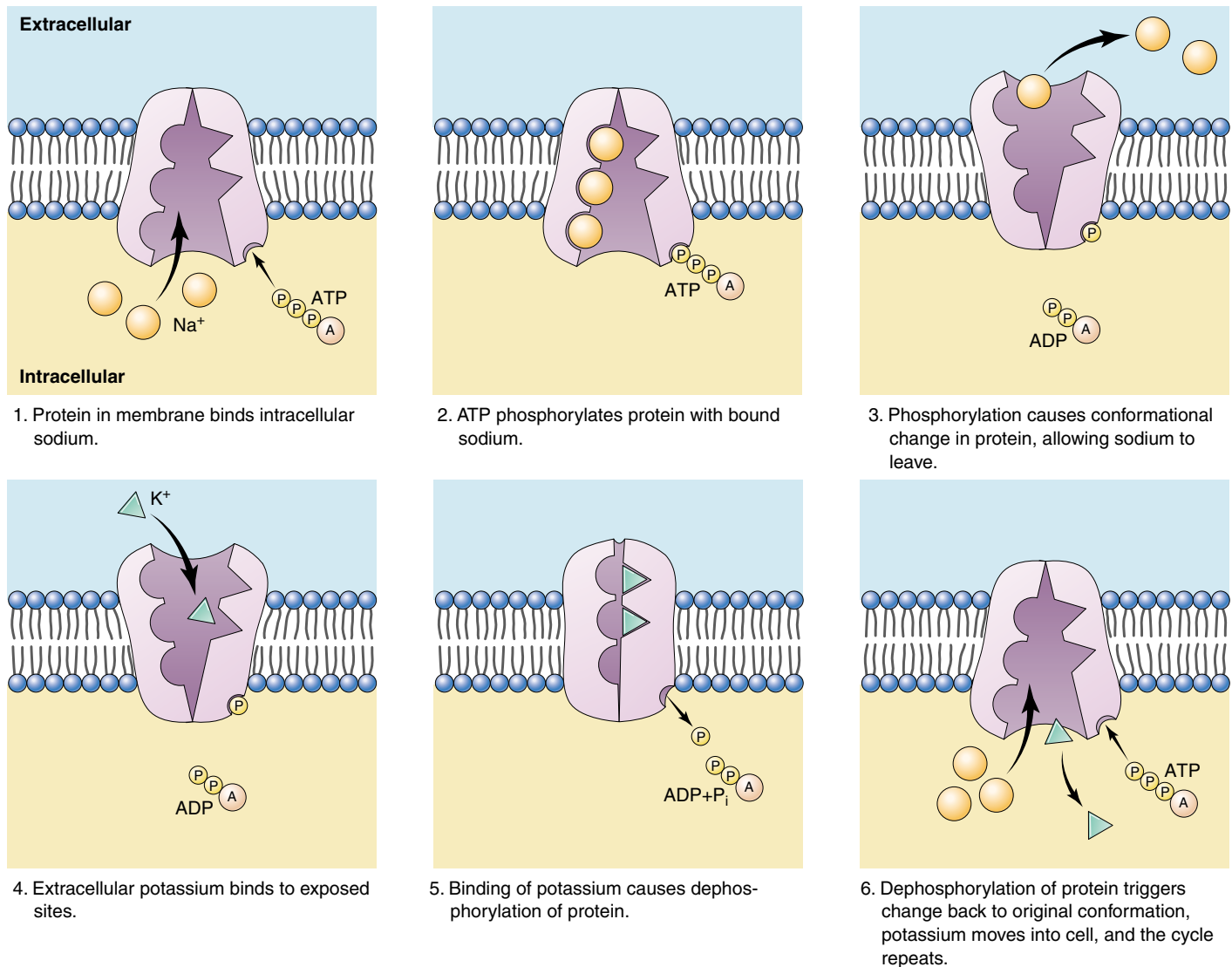


FIGURE 6.21

The sodium-potassium pump. The protein channel known as the sodium-potassium pump transports sodium (Na^+) and potassium (K^+) ions across the cell membrane. For every three Na^+ that are transported out of the cell, two K^+ are transported into the cell. The sodium-potassium pump is fueled by ATP.

so they now face the exterior. In this new conformation, the protein has a low affinity for Na^+ , and the three bound Na^+ dissociate from the protein and diffuse into the extracellular fluid.

Step 4. The new conformation has a high affinity for K^+ , two of which bind to the extracellular side of the protein as soon as it is free of the Na^+ .

Step 5. The binding of the K^+ causes another conformational change in the protein, this time resulting in the dissociation of the bound phosphate group.

Step 6. Freed of the phosphate group, the protein reverts to its original conformation, exposing the two K^+ to the cytoplasm. This conformation has a low affinity for K^+ , so the two bound K^+ dissociate from the protein

and diffuse into the interior of the cell. The original conformation has a high affinity for Na^+ ; when these ions bind, they initiate another cycle.

Three Na^+ leave the cell and two K^+ enter in every cycle. The changes in protein conformation that occur during the cycle are rapid, enabling each carrier to transport as many as 300 Na^+ per second. The sodium-potassium pump appears to be ubiquitous in animal cells, although cells vary widely in the number of pump proteins they contain.

Active transport moves a solute across a membrane up its concentration gradient, using protein carriers driven by the expenditure of chemical energy.

Coupled Transport

Many molecules are transported into cells up a concentration gradient through a process that uses ATP indirectly. The molecules move hand-in-hand with sodium ions or protons that are moving *down* their concentration gradients. This type of active transport, called **cotransport**, has two components:

1. **Establishing the down gradient.** ATP is used to establish the sodium ion or proton *down* gradient, which is greater than the *up* gradient of the molecule to be transported.
2. **Traversing the up gradient.** Cotransport proteins (also called coupled transport proteins) carry the molecule and either a sodium ion or a proton together across the membrane.

Because the *down* gradient of the sodium ion or proton is greater than the *up* gradient of the molecule to be transported, the net movement across the membrane is in the direction of the down gradient, typically into the cell.

Establishing the *Down* Gradient

Either the sodium-potassium pump or the proton pump establishes the down gradient that powers most active transport processes of the cell.

The Sodium-Potassium Pump. The sodium-potassium pump actively pumps sodium ions out of the cell, powered by energy from ATP. This establishes a sodium ion concentration gradient that is lower inside the cell.

The Proton Pump. The **proton pump** pumps protons (H^+ ions) across a membrane using energy derived from energy-rich molecules or from photosynthesis. This creates a proton gradient, in which the concentration of protons is higher on one side of the membrane than the other. Membranes are impermeable to protons, so the only way protons can diffuse back down their concentration gradient is through a second cotransport protein.

Traversing the *Up* Gradient

Animal cells accumulate many amino acids and sugars against a concentration gradient: the molecules are transported into the cell from the extracellular fluid, even though their concentrations are higher inside the cell. These molecules couple with sodium ions to enter the cell down the Na^+ concentration gradient established by the sodium-potassium pump. In this **cotransport process**, Na^+ and a specific sugar or amino acid simultaneously bind to the same transmembrane protein on the outside of the cell, called a **symport** (figure 6.22). Both are then translocated to the inside of the cell, but in the process Na^+ moves *down* its concentration gradient while the sugar or amino acid moves *up* its concentration gradient. In effect, the cell uses some of the energy stored in the Na^+ concentration gradient to accumulate sugars and amino acids.

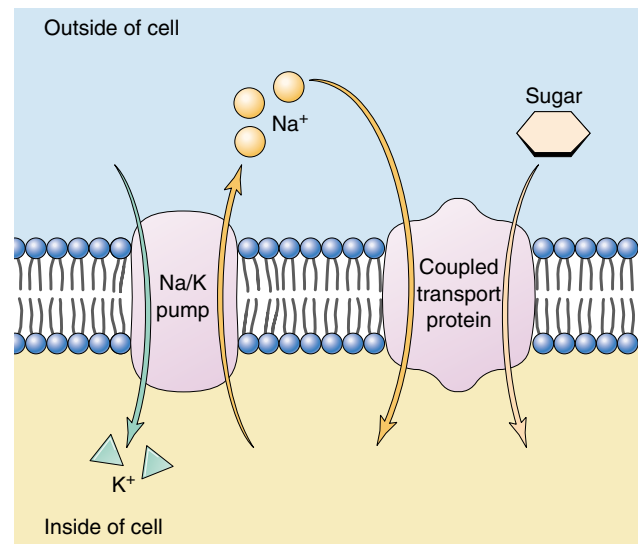


FIGURE 6.22

Cotransport through a coupled transport protein. A membrane protein transports sodium ions into the cell, down their concentration gradient, at the same time it transports a sugar molecule into the cell. The gradient driving the Na^+ entry is so great that sugar molecules can be brought in *against* their concentration gradient.

In a related process, called **countertransport**, the inward movement of Na^+ is coupled with the outward movement of another substance, such as Ca^{++} or H^+ . As in cotransport, both Na^+ and the other substance bind to the same transport protein, in this case called an **antiport**, but in this case they bind on opposite sides of the membrane and are moved in opposite directions. In countertransport, the cell uses the energy released as Na^+ moves down its concentration gradient into the cell to extrude a substance up its concentration gradient.

The cell uses the proton *down* gradient established by the proton pump (figure 6.23) in ATP production. The movement of protons through their cotransport protein is coupled to the production of ATP, the energy-storing molecule we mentioned earlier. Thus, the cell expends energy to produce ATP, which provides it with a convenient energy storage form that it can employ in its many activities. The coupling of the proton pump to ATP synthesis, called **chemiosmosis**, is responsible for almost all of the ATP produced from food (see chapter 9) and all of the ATP produced by photosynthesis (see chapter 10). We know that proton pump proteins are ancient because they are present in bacteria as well as in eukaryotes. The mechanisms for transport across plasma membranes are summarized in table 6.2.

Many molecules are cotransported into cells up their concentration gradients by coupling their movement to that of sodium ions or protons moving down their concentration gradients.

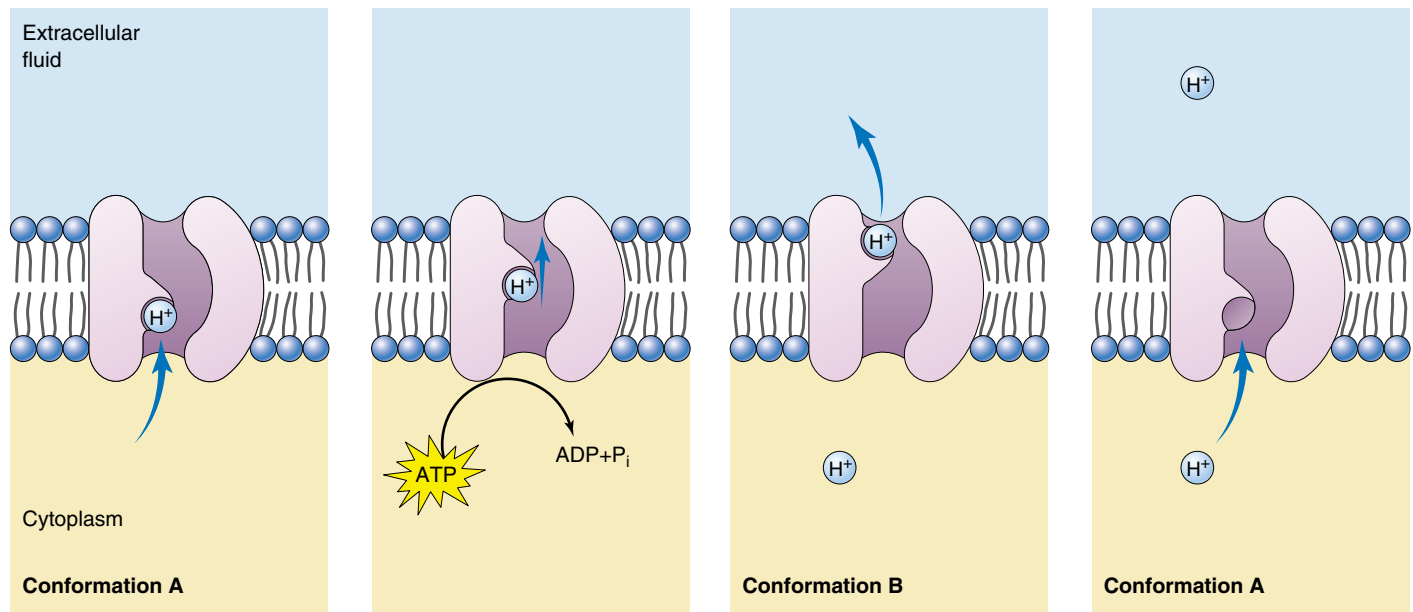


FIGURE 6.23

The proton pump. In this general model of energy-driven proton pumping, the transmembrane protein that acts as a proton pump is driven through a cycle of two conformations: A and B. The cycle A→B→A goes only one way, causing protons to be pumped from the inside to the outside of the membrane. ATP powers the pump.

Table 6.2 Mechanisms for Transport across Cell Membranes

Process	Passage through Membrane	How It Works	Example
PASSIVE PROCESSES			
Diffusion	Direct	Random molecular motion produces net migration of molecules toward region of lower concentration	Movement of oxygen into cells
Facilitated diffusion	Protein carrier	Molecule binds to carrier protein in membrane and is transported across; net movement is toward region of lower concentration	Movement of glucose into cells
Osmosis	Direct	Diffusion of water across differentially permeable membrane	Movement of water into cells placed in a hypotonic solution
ACTIVE PROCESSES			
Endocytosis			
Phagocytosis	Membrane vesicle	Particle is engulfed by membrane, which folds around it and forms a vesicle	Ingestion of bacteria by white blood cells
Pinocytosis	Membrane vesicle	Fluid droplets are engulfed by membrane, which forms vesicles around them	“Nursing” of human egg cells
Carrier-mediated endocytosis	Membrane vesicle	Endocytosis triggered by a specific receptor	Cholesterol uptake
Exocytosis	Membrane vesicle	Vesicles fuse with plasma membrane and eject contents	Secretion of mucus
Active transport			
Na^+/K^+ pump	Protein carrier	Carrier expends energy to export Na^+ against a concentration gradient	Coupled uptake of glucose into cells against its concentration gradient
Proton pump	Protein carrier	Carrier expends energy to export protons against a concentration gradient	Chemiosmotic generation of ATP

Chapter 6



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Summary

Questions

Media Resources

6.1 Biological membranes are fluid layers of lipid.

- Every cell is encased within a fluid bilayer sheet of phospholipid molecules called the plasma membrane.

1. How would increasing the number of phospholipids with double bonds between carbon atoms in their tails affect the fluidity of a membrane?



- Membrane Structure

6.2 Proteins embedded within the plasma membrane determine its character.

- Proteins that are embedded within the plasma membrane have their hydrophobic regions exposed to the hydrophobic interior of the bilayer, and their hydrophilic regions exposed to the cytoplasm or the extracellular fluid.
- Membrane proteins can transport materials into or out of the cell, they can mark the identity of the cell, or they can receive extracellular information.

2. Describe the two basic types of structures that are characteristic of proteins that span membranes.



- Art Activity: Fluid Mosaic Model
- Art Activity: Membrane Protein Diversity

6.3 Passive transport across membranes moves down the concentration gradient.

- Diffusion is the kinetic movement of molecules or ions from an area of high concentration to an area of low concentration.
- Osmosis is the diffusion of water. Because all organisms are composed of mostly water, maintaining osmotic balance is essential to life.

3. If a cell's cytoplasm were hyperosmotic to the extracellular fluid, how would the concentration of solutes in the cytoplasm compare with that in the extracellular fluid?



- Diffusion
- Osmosis



- Diffusion
- Diffusion
- Osmosis



- Student Research: Understanding Membrane Transport

6.4 Bulk transport utilizes endocytosis.

- Materials or volumes of fluid that are too large to pass directly through the cell membrane can move into or out of cells through endocytosis or exocytosis, respectively.
- In these processes, the cell expends energy to change the shape of its plasma membrane, allowing the cell to engulf materials into a temporary vesicle (endocytosis), or eject materials by fusing a filled vesicle with the plasma membrane (exocytosis).

4. How do phagocytosis and pinocytosis differ?

5. Describe the mechanism of receptor-mediated endocytosis.



- Exocytosis/endocytosis



- Exocytosis/endocytosis

6.5 Active transport across membranes is powered by energy from ATP.

- Cells use active transport to move substances across the plasma membrane against their concentration gradients, either accumulating them within the cell or extruding them from the cell. Active transport requires energy from ATP, either directly or indirectly.

6. In what two ways does facilitated diffusion differ from simple diffusion across a membrane?

7. How does active transport differ from facilitated diffusion? How is it similar to facilitated diffusion?



- Exploration: Active Transport



- Active Transport



- Active Transport

7

Cell-Cell Interactions

Concept Outline

7.1 Cells signal one another with chemicals.

Receptor Proteins and Signaling between Cells.

Receptor proteins embedded in the plasma membrane change shape when they bind specific signal molecules, triggering a chain of events within the cell.

Types of Cell Signaling. Cell signaling can occur between adjacent cells, although chemical signals called hormones act over long distances.

7.2 Proteins in the cell and on its surface receive signals from other cells.

Intracellular Receptors. Some receptors are located within the cell cytoplasm. These receptors respond to lipid-soluble signals, such as steroid hormones.

Cell Surface Receptors. Many cell-to-cell signals are water-soluble and cannot penetrate membranes. Instead, the signals are received by transmembrane proteins protruding out from the cell surface.

7.3 Follow the journey of information into the cell.

Initiating the Intracellular Signal. Cell surface receptors often use “second messengers” to transmit a signal to the cytoplasm.

Amplifying the Signal: Protein Kinase Cascades.

Surface receptors and second messengers amplify signals as they travel into the cell, often toward the cell nucleus.

7.4 Cell surface proteins mediate cell-cell interactions.

The Expression of Cell Identity. Cells possess on their surfaces a variety of tissue-specific identity markers that identify both the tissue and the individual.

Intercellular Adhesion. Cells attach themselves to one another with protein links. Some of the links are very strong, others more transient.

Tight Junctions. Adjacent cells form a sheet when connected by tight junctions, and molecules are encouraged to flow through the cells, not between them.

Anchoring Junctions. The cytoskeleton of a cell is connected by an anchoring junction to the cytoskeleton of another cell or to the extracellular matrix.

Communicating Junctions. Many adjacent cells have direct passages that link their cytoplasms, permitting the passage of ions and small molecules.

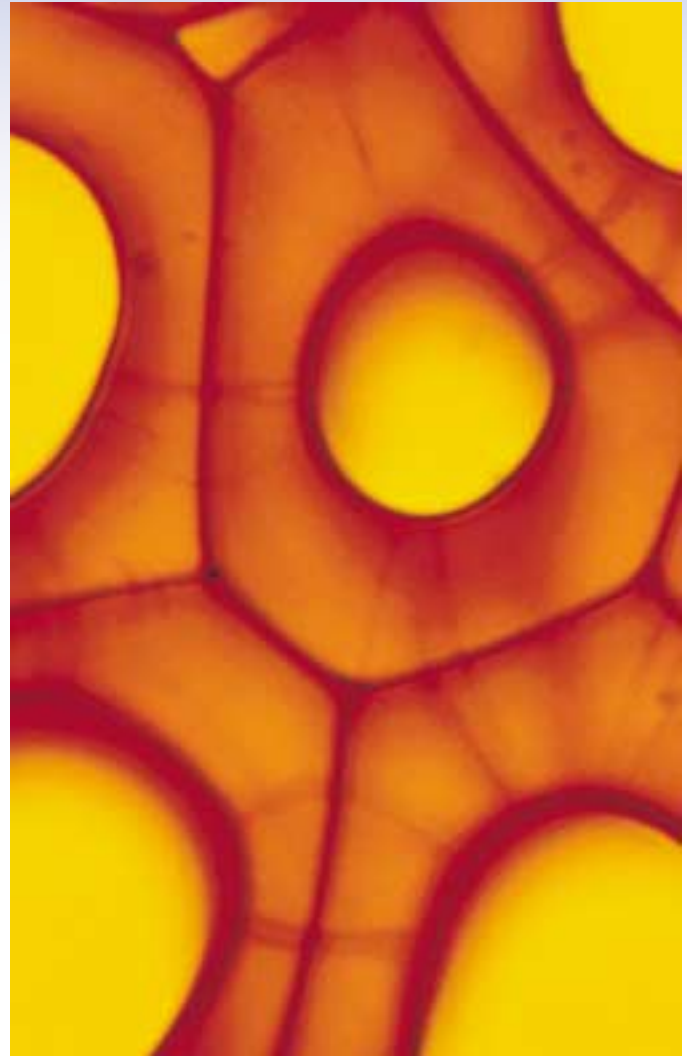


FIGURE 7.1

Persimmon cells in close contact with one another. These plant cells and all cells, no matter what their function, interact with their environment, including the cells around them.

Did you know that each of the 100 trillion cells of your body shares one key feature with the cells of tigers, bumblebees, and persimmons (figure 7.1)—a feature that most bacteria and protists lack? Your cells touch and communicate with one another. Sending and receiving a variety of chemical signals, they coordinate their behavior so that your body functions as an integrated whole, rather than as a massive collection of individual cells acting independently. The ability of cells to communicate with one another is the hallmark of multicellular organisms. In this chapter we will look in detail at how the cells of multicellular organisms interact with one another, first exploring how they signal one another with chemicals and then examining the ways in which their cell surfaces interact to organize tissues and body structures.

7.1 Cells signal one another with chemicals.

Receptor Proteins and Signaling between Cells

Communication between cells is common in nature. Cell signaling occurs in all multicellular organisms, providing an indispensable mechanism for cells to influence one another. The cells of multicellular organisms use a variety of molecules as signals, including not only peptides, but also large proteins, individual amino acids, nucleotides, steroids and other lipids.

Even dissolved gases are used as signals. Nitric oxide (NO) plays a role in mediating male erections (Viagra functions by stimulating NO release).

Some of these molecules are attached to the surface of the signaling cell; others are secreted through the plasma membrane or released by exocytosis.

Cell Surface Receptors

Any given cell of a multicellular organism is exposed to a constant stream of signals. At any time, hundreds of different chemical signals may be in the environment surrounding the cell. However, each cell responds only to certain signals and ignores the rest (figure 7.2), like a person following the conversation of one or two individuals in a noisy, crowded room. How does a cell “choose” which signals to respond to? Located on or within the cell are **receptor proteins**, each with a three-dimensional shape that fits the shape of a specific signal molecule. When a signal molecule approaches a receptor protein of the right shape, the two can bind. This binding induces a change in the receptor protein’s shape, ultimately producing a response in the cell. Hence, a given cell responds to the signal molecules that fit the particular set of receptor proteins it possesses, and ignores those for which it lacks receptors.

The Hunt for Receptor Proteins

The characterization of receptor proteins has presented a very difficult technical problem, because of their relative scarcity in the cell. Because these proteins may constitute less than 0.01% of the total mass of protein in a cell, purifying them is analogous to searching for a particular grain of sand in a sand dune! However, two recent techniques have enabled cell biologists to make rapid progress in this area.

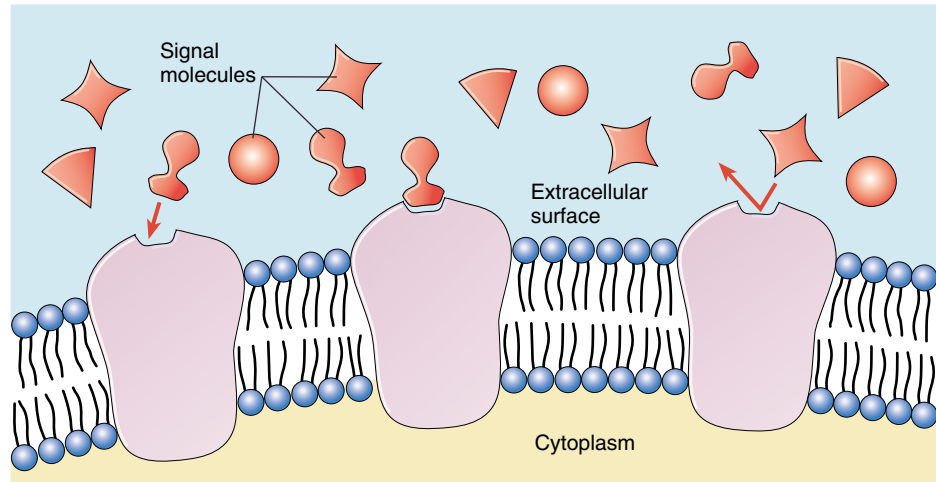


FIGURE 7.2

Cell surface receptors recognize only specific molecules. Signal molecules will bind only to those cells displaying receptor proteins with a shape into which they can fit snugly.

Monoclonal antibodies. The first method uses *monoclonal antibodies*. An antibody is an immune system protein that, like a receptor, binds specifically to another molecule. Each individual immune system cell can make only one particular type of antibody, which can bind to only one specific target molecule. Thus, a cell-line derived from a single immune system cell (a clone) makes one specific antibody (a *monoclonal* antibody). Monoclonal antibodies that bind to particular receptor proteins can be used to isolate those proteins from the thousands of others in the cell.

Gene analysis. The study of mutants and isolation of gene sequences has had a tremendous impact on the field of receptor analysis. In chapter 19 we will present a detailed account of how this is done. These advances make it feasible to identify and isolate the many genes that encode for various receptor proteins.

Remarkably, these techniques have revealed that the enormous number of receptor proteins can be grouped into just a handful of “families” containing many related receptors. Later in this chapter we will meet some of the members of these receptor families.

Cells in a multicellular organism communicate with others by releasing signal molecules that bind to receptor proteins on the surface of the other cells. Recent advances in protein isolation have yielded a wealth of information about the structure and function of these proteins.

Types of Cell Signaling

Cells communicate through any of four basic mechanisms, depending primarily on the distance between the signaling and responding cells (figure 7.3). In addition to using these four basic mechanisms, some cells actually send signals to themselves, secreting signals that bind to specific receptors on their own plasma membranes. This process, called **autocrine signaling**, is thought to play an important role in reinforcing developmental changes.

Direct Contact

As we saw in chapter 6, the surface of a eukaryotic cell is a thick layer of proteins, carbohydrates, and lipids attached to and extending outward from the plasma membrane. When cells are very close to one another, some of the molecules on the cells' plasma membranes may bind together in specific ways. Many of the important interactions between cells in early development occur by means of direct contact between cell surfaces (figure 7.3a). We'll examine contact-dependent interactions more closely later in this chapter.

Paracrine Signaling

Signal molecules released by cells can diffuse through the extracellular fluid to other cells. If those molecules are taken up by neighboring cells, destroyed by extracellular enzymes, or quickly removed from the extracellular fluid in some other way, their influence is restricted to cells in the immediate vicinity of the releasing cell. Signals with such short-lived, local effects are called **paracrine** signals (figure 7.3b). Like direct contact, paracrine signaling plays an important role in early development, coordinating the activities of clusters of neighboring cells.

Endocrine Signaling

If a released signal molecule remains in the extracellular fluid, it may enter the organism's circulatory system and travel widely throughout the body. These longer lived signal molecules, which may affect cells very distant from the releasing cell, are called **hormones**, and this type of intercellular communication is known as **endocrine** signaling (figure 7.3c). Chapter 58 discusses endocrine signaling in detail. Both animals and plants use this signaling mechanism extensively.

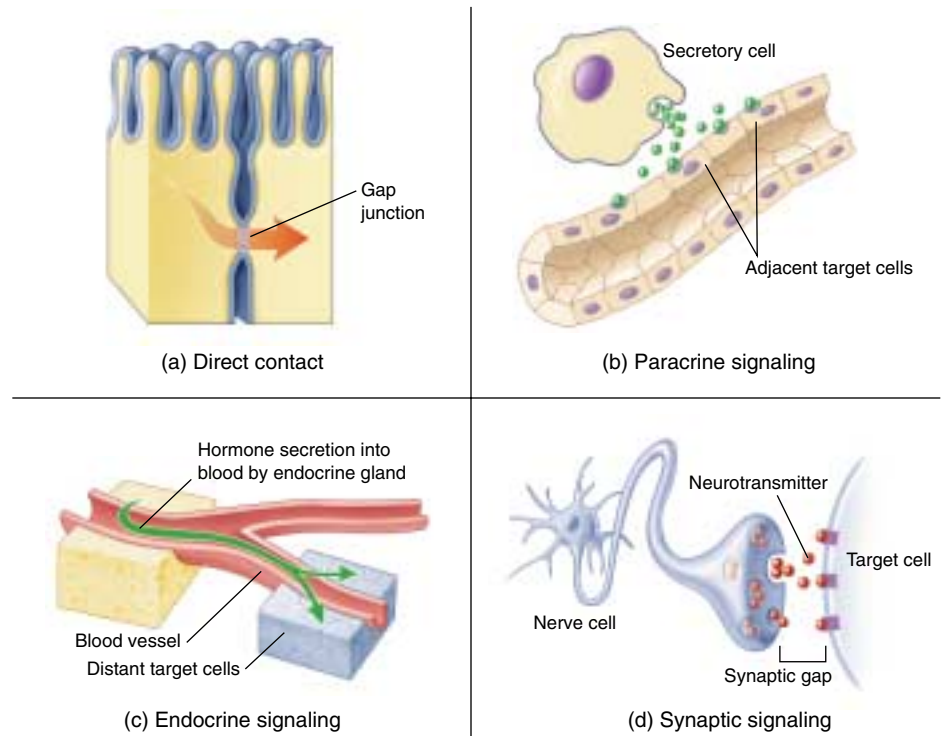


FIGURE 7.3

Four kinds of cell signaling. Cells communicate in several ways. (a) Two cells in *direct contact* with each other may send signals across gap junctions. (b) In *paracrine signaling*, secretions from one cell have an effect only on cells in the immediate area. (c) In *endocrine signaling*, hormones are released into the circulatory system, which carries them to the target cells. (d) Chemical *synaptic signaling* involves transmission of signal molecules, called neurotransmitters, from a neuron over a small synaptic gap to the target cell.

Synaptic Signaling

In animals, the cells of the nervous system provide rapid communication with distant cells. Their signal molecules, **neurotransmitters**, do not travel to the distant cells through the circulatory system like hormones do. Rather, the long, fiberlike extensions of nerve cells release neurotransmitters from their tips very close to the target cells (figure 7.3d). The narrow gap between the two cells is called a **chemical synapse**. While paracrine signals move through the fluid between cells, neurotransmitters cross the synapse and persist only briefly. We will examine synaptic signaling more fully in chapter 54.

Adjacent cells can signal others by direct contact, while nearby cells that are not touching can communicate through paracrine signals. Two other systems mediate communication over longer distances: in endocrine signaling the blood carries hormones to distant cells, and in synaptic signaling nerve cells secrete neurotransmitters from long cellular extensions close to the responding cells.

7.2 Proteins in the cell and on its surface receive signals from other cells.

Intracellular Receptors

All cell signaling pathways share certain common elements, including a chemical signal that passes from one cell to another and a receptor that receives the signal in or on the target cell. We've looked at the sorts of signals that pass from one cell to another. Now let's consider the nature of the receptors that receive the signals. Table 7.1 summarizes the types of receptors we will discuss in this chapter.

Many cell signals are lipid-soluble or very small molecules that can readily pass across the plasma membrane of the target cell and into the cell, where they interact with a receptor. Some bind to protein receptors located in the cytoplasm; others pass across the nuclear membrane as well and bind to receptors within the nucleus. These **intracellular receptors** (figure 7.4) may trigger a variety of responses in the cell, depending on the receptor.

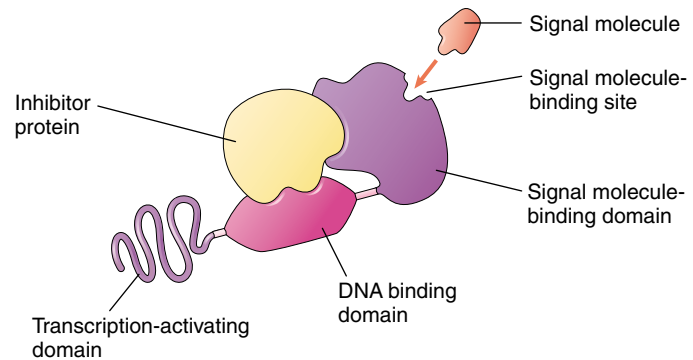


FIGURE 7.4

Basic structure of a gene-regulating intracellular receptor.

These receptors are located within the cell and function in the reception of signals such as steroid hormones, vitamin D, and thyroid hormone.

Table 7.1 Cell Communicating Mechanisms

Mechanism	Structure	Function	Example
INTRACELLULAR RECEPTORS	No extracellular signal-binding site	Receives signals from lipid-soluble or noncharged, nonpolar small molecules	Receptors for NO, steroid hormone, vitamin D, and thyroid hormone
CELL SURFACE RECEPTORS			
Chemically gated ion channels	Multipass transmembrane protein forming a central pore	Molecular “gates” triggered chemically to open or close	Neurons
Enzymic receptors	Single-pass transmembrane protein	Binds signal extracellularly, catalyzes response intracellularly	Phosphorylation of protein kinases
G-protein-linked receptors	Seven-pass transmembrane protein with cytoplasmic binding site for G protein	Binding of signal to receptor causes GTP to bind a G protein; G protein, with attached GTP, detaches to deliver the signal inside the cell	Peptide hormones, rod cells in the eyes
PHYSICAL CONTACT WITH OTHER CELLS			
Surface markers	Variable; integral proteins or glycolipids in cell membrane	Identify the cell	MHC complexes, blood groups, antibodies
Tight junctions	Tightly bound, leakproof, fibrous protein “belt” that surrounds cell	Organizing junction: holds cells together such that material passes <i>through</i> but not <i>between</i> the cells	Junctions between epithelial cells in the gut
Desmosomes	Intermediate filaments of cytoskeleton linked to adjoining cells through cadherins	Anchoring junction: “buttons” cells together	Epithelium
Adherens junctions	Transmembrane fibrous proteins	Anchoring junction: “roots” extracellular matrix to cytoskeleton	Tissues with high mechanical stress, such as the skin
Gap junctions	Six transmembrane connexon proteins creating a “pipe” that connects cells	Communicating junction: allows passage of small molecules from cell to cell in a tissue	Excitable tissue such as heart muscle
Plasmodesmata	Cytoplasmic connections between gaps in adjoining plant cell walls	Communicating junction between plant cells	Plant tissues

Receptors That Act as Gene Regulators

Some intracellular receptors act as regulators of gene transcription. Among them are the receptors for steroid hormones, such as cortisol, estrogen, and progesterone, as well as the receptors for a number of other small, lipid-soluble signal molecules, such as vitamin D and thyroid hormone. All of these receptors have similar structures; the genes that code for them may well be the evolutionary descendants of a single ancestral gene. Because of their structural similarities, they are all part of the *intracellular receptor superfamily*.

Each of these receptors has a binding site for DNA. In its inactive state, the receptor typically cannot bind DNA because an inhibitor protein occupies the binding site. When the signal molecule binds to another site on the receptor, the inhibitor is released and the DNA binding site is exposed (figure 7.5). The receptor then binds to a specific nucleotide sequence on the DNA, which activates (or, in a few instances, suppresses) a particular gene, usually located adjacent to the regulatory site.

The lipid-soluble signal molecules that intracellular receptors recognize tend to persist in the blood far longer than water-soluble signals. Most water-soluble hormones break down within minutes, and neurotransmitters within seconds or even milliseconds. A steroid hormone like cortisol or estrogen, on the other hand, persists for hours.

The target cell's response to a lipid-soluble cell signal can vary enormously, depending on the nature of the cell. This is true even when different target cells have the same intracellular receptor, for two reasons: First, the binding site for the receptor on the target DNA differs from one cell type to another, so that different genes are affected when the signal-receptor complex binds to the DNA, and second, most eukaryotic genes have complex controls. We will discuss them in detail in chapter 16, but for now it is sufficient to note that several different regulatory proteins are usually involved in reading a eukaryotic gene. Thus the intracellular receptor interacts with different signals in different tissues. Depending on the cell-specific controls operating in different tissues, the effect the intracellular receptor produces when it binds with DNA will vary.

Receptors That Act as Enzymes

Other intracellular receptors act as enzymes. A very interesting example is the receptor for the signal molecule, **nitric oxide (NO)**. A small gas molecule, NO diffuses readily out of the cells where it is produced and passes directly into neighboring cells, where it binds to the enzyme guanylyl cyclase. Binding of NO activates the enzyme, enabling it to catalyze the synthesis of cyclic guanosine monophosphate (GMP), an intracellular messenger molecule that produces cell-specific responses such as the relaxation of smooth muscle cells.

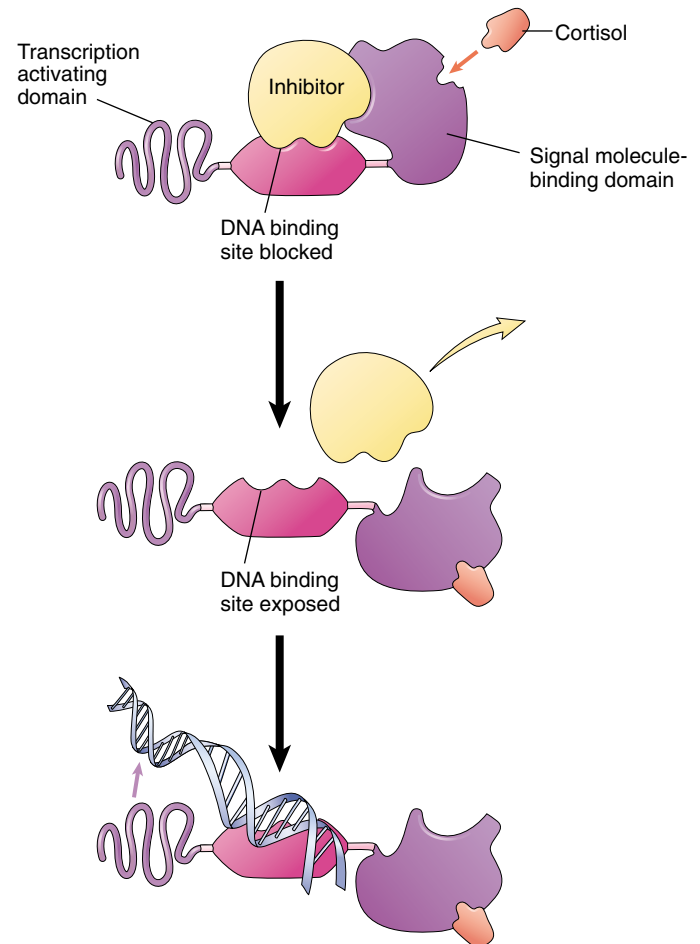


FIGURE 7.5

How intracellular receptors regulate gene transcription. In this model, the binding of the steroid hormone cortisol to a DNA regulatory protein causes it to alter its shape. The inhibitor is released, exposing the DNA binding site of the regulatory protein. The DNA binds to the site, positioning a specific nucleotide sequence over the transcription activating domain of the receptor and initiating transcription.

NO has only recently been recognized as a signal molecule in vertebrates. Already, however, a wide variety of roles have been documented. For example, when the brain sends a nerve signal relaxing the smooth muscle cells lining the walls of vertebrate blood vessels, the signal molecule acetylcholine released by the nerve near the muscle does not interact with the muscle cell directly. Instead, it causes nearby epithelial cells to produce NO, which then causes the smooth muscle to relax, allowing the vessel to expand and thereby increase blood flow.

Many target cells possess intracellular receptors, which are activated by substances that pass through the plasma membrane.

Cell Surface Receptors

Most signal molecules are water-soluble, including neurotransmitters, peptide hormones, and the many proteins that multicellular organisms employ as “growth factors” during development. Water-soluble signals cannot diffuse through cell membranes. Therefore, to trigger responses in cells, they must bind to receptor proteins on the surface of the cell. These **cell surface receptors** (figure 7.6) convert the extracellular signal to an intracellular one, responding to the binding of the signal molecule by producing a change within the cell’s cytoplasm. Most of a cell’s receptors are cell surface receptors, and almost all of them belong to one of three receptor superfamilies: chemically gated ion channels, enzymic receptors, and G-protein-linked receptors.

Chemically Gated Ion Channels

Chemically gated ion channels are receptor proteins that ions pass through. The receptor proteins that bind many neurotransmitters have the same basic structure (figure 7.6*a*). Each is a “multi-pass” transmembrane protein, meaning that the chain of amino acids threads back and forth

across the plasma membrane several times. In the center of the protein is a pore that connects the extracellular fluid with the cytoplasm. The pore is big enough for ions to pass through, so the protein functions as an **ion channel**. The channel is said to be chemically gated because it opens when a chemical (the neurotransmitter) binds to it. The type of ion (sodium, potassium, calcium, chloride, for example) that flows across the membrane when a chemically gated ion channel opens depends on the specific three-dimensional structure of the channel.

Enzymic Receptors

Many cell surface receptors either act as enzymes or are directly linked to enzymes (figure 7.6*b*). When a signal molecule binds to the receptor, it activates the enzyme. In almost all cases, these enzymes are **protein kinases**, enzymes that add phosphate groups to proteins. Most enzymic receptors have the same general structure. Each is a single-pass transmembrane protein (the amino acid chain passes through the plasma membrane only once); the portion that binds the signal molecule lies outside the cell, and the portion that carries out the enzyme activity is exposed to the cytoplasm.

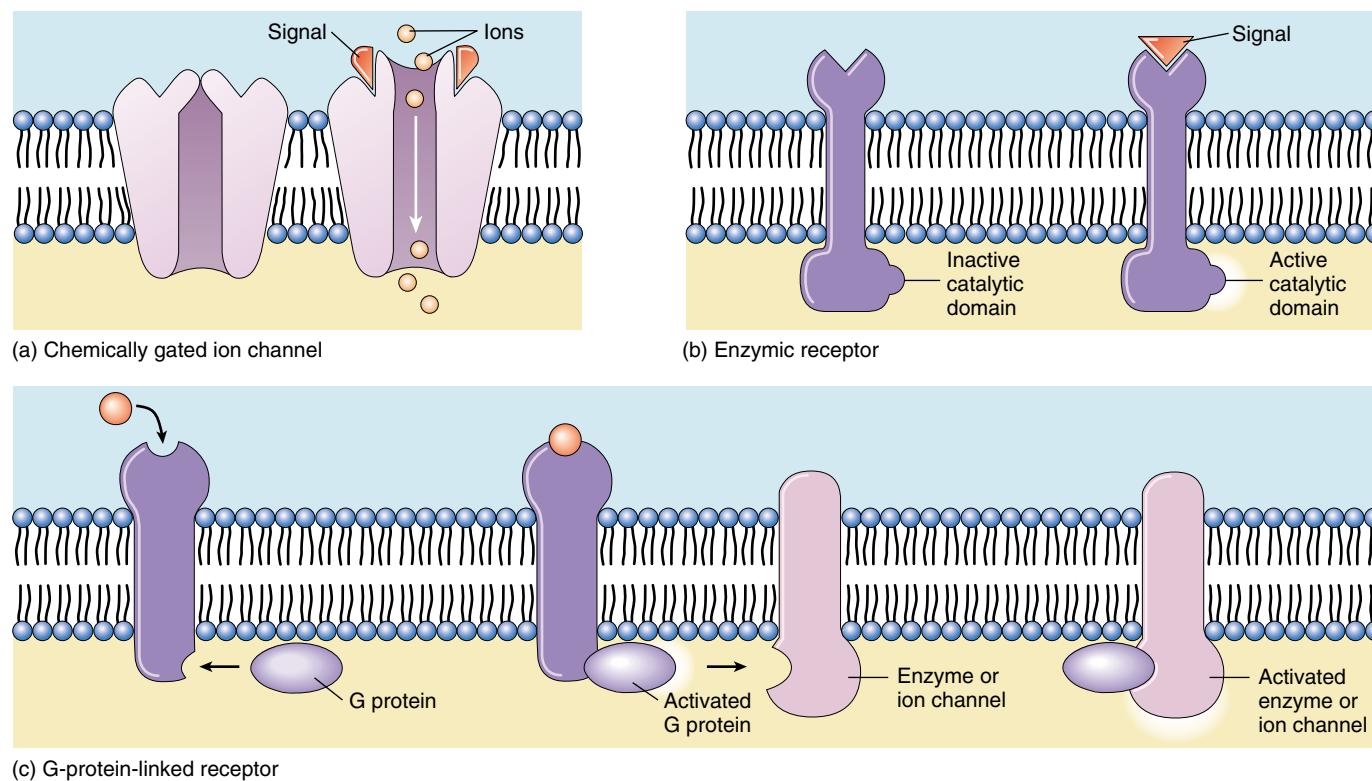


FIGURE 7.6

Cell surface receptors. (a) Chemically gated ion channels are multi-pass transmembrane proteins that form a pore in the cell membrane. This pore is opened or closed by chemical signals. (b) Enzymic receptors are single-pass transmembrane proteins that bind the signal on the extracellular surface. A catalytic region on their cytoplasmic portion then initiates enzymatic activity inside the cell. (c) G-protein-linked receptors bind to the signal outside the cell and to G proteins inside the cell. The G protein then activates an enzyme or ion channel, mediating the passage of a signal from the cell’s surface to its interior.

G-Protein-Linked Receptors

A third class of cell surface receptors acts indirectly on enzymes or ion channels in the plasma membrane with the aid of an assisting protein, called a *guanosine triphosphate* (GTP)-*binding protein*, or **G protein** (figure 7.6c). Receptors in this category use G proteins to mediate passage of the signal from the membrane surface into the cell interior.

How G-Protein-Linked Receptors Work. G proteins are mediators that initiate a diffusible signal in the cytoplasm. They form a transient link between the receptor on the cell surface and the signal pathway within the cytoplasm. Importantly, this signal has a relatively short life span whose active age is determined by GTP. When a signal arrives, it finds the G protein nestled into the G-protein-linked receptor on the cytoplasmic side of the plasma membrane. Once the signal molecule binds to the receptor, the G-protein-linked receptor changes shape. This change in receptor shape twists the G protein, causing it to bind GTP. The G protein can now diffuse away from the receptor. The “activated” complex of a G protein with attached GTP is then free to initiate a number of events. However, this activation is short-lived, because GTP has a relatively short life span (seconds to minutes). This elegant arrangement allows the G proteins to activate numerous pathways, but only in a transient manner. In order for a pathway to “stay on,” there must be a continuous source of incoming extracellular signals. When the rate of external signal drops off, the pathway shuts down.

The Largest Family of Cell Surface Receptors. Scientists have identified more than 100 different G-protein-linked receptors, more than any other kind of cell surface receptor. They mediate an incredible range of cell signals, including peptide hormones, neurotransmitters, fatty acids, and amino acids. Despite this great variation in specificity, however, all G-protein-linked receptors whose amino acid sequences are known have a similar structure. They are almost certainly closely related in an evolutionary sense, arising from a single ancestral sequence. Each of these G-protein-linked receptors is a seven-pass transmembrane protein (figure 7.7)—a single polypeptide chain that threads back and forth across the lipid bilayer seven times, creating a channel through the membrane.

Evolutionary Origin of G-Protein-Linked Receptors. As research revealed the structure of G-protein-linked receptors, an interesting pattern emerged: the same seven-pass structural motif is seen again and again, in sensory receptors such as the light-activated rhodopsin protein in the vertebrate eye, in the light-activated bacteriorhodopsin proton pump that plays a key role in bacterial photosynthesis, in the receptor that recognizes the yeast mating factor

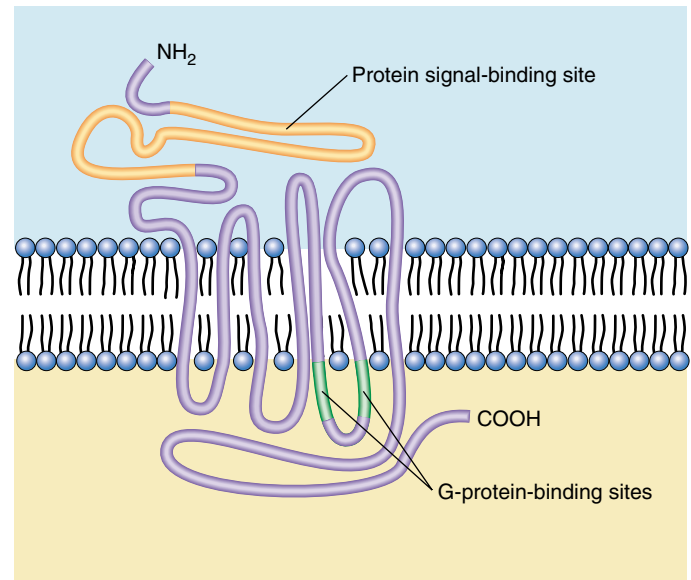


FIGURE 7.7
The G-protein-linked receptor is a seven-pass transmembrane protein.

protein discussed earlier, and in many other sensory receptors. Vertebrate rhodopsin is in fact a G-protein-linked receptor and utilizes a G protein. Bacteriorhodopsin is not. The occurrence of the seven-pass structural motif in both, and in so many other G-protein-linked receptors, suggests that this motif is a very ancient one, and that G-protein-linked receptors may have evolved from sensory receptors of single-celled ancestors.

Discovery of G Proteins. Martin Rodbell of the National Institute of Environmental Health Sciences and Alfred Gilman of the University of Texas Southwestern Medical Center received the 1994 Nobel Prize for Medicine or Physiology for their work on G proteins. Rodbell and Gilman’s work has proven to have significant ramifications. G proteins are involved in the mechanism employed by over half of all medicines in use today. Studying G proteins will vastly expand our understanding of how these medicines work. Furthermore, the investigation of G proteins should help elucidate how cells communicate in general and how they contribute to the overall physiology of organisms. As Gilman says, G proteins are “involved in everything from sex in yeast to cognition in humans.”

Most receptors are located on the surface of the plasma membrane. Chemically gated ion channels open or close when signal molecules bind to the channel, allowing specific ions to diffuse through. Enzyme receptors typically activate intracellular proteins by phosphorylation. G-protein-linked receptors activate an intermediary protein, which then effects the intracellular change.

7.3 Follow the journey of information into the cell.

Initiating the Intracellular Signal

Some enzymic receptors and most G-protein-linked receptors carry the signal molecule's message into the target cell by utilizing other substances to relay the message within the cytoplasm. These other substances, small molecules or ions commonly called **second messengers** or intracellular mediators, alter the behavior of particular proteins by binding to them and changing their shape. The two most widely used second messengers are cyclic adenosine monophosphate (cAMP) and calcium.

cAMP

All animal cells studied thus far use **cAMP** as a second messenger (chapter 56 discusses cAMP in detail). To see how cAMP typically works as a messenger, let's examine what happens when the hormone epinephrine binds to a particular type of G-protein-linked receptor called the β -adrenergic receptor (figure 7.8). When epinephrine binds with this receptor, it activates a G protein, which then stimulates the enzyme **adenylyl cyclase** to produce large amounts of cAMP within the cell (figure 7.9a). The cAMP then binds to and activates the enzyme α -kinase, which adds phosphates to specific proteins in the cell. The effect this phosphorylation has on cell function depends on the identity of the cell and the proteins that are phosphorylated. In muscle cells, for example, the α -kinase phosphorylates and thereby activates enzymes that stimulate the breakdown of glycogen into glucose and inhibit the synthesis of glycogen from glucose. Glucose is then more available to the muscle cells for metabolism.

Calcium

Calcium (Ca^{++}) ions serve even more widely than cAMP as second messengers. Ca^{++} levels inside the cytoplasm of a cell are normally very low (less than 10^{-7} M), while outside the cell and in the endoplasmic reticulum Ca^{++} levels are quite high (about 10^{-3} M). Chemically gated calcium channels in the endoplasmic reticulum membrane act as switches; when they open, Ca^{++} rushes into the cytoplasm and triggers proteins sensitive to Ca^{++} to initiate a variety of

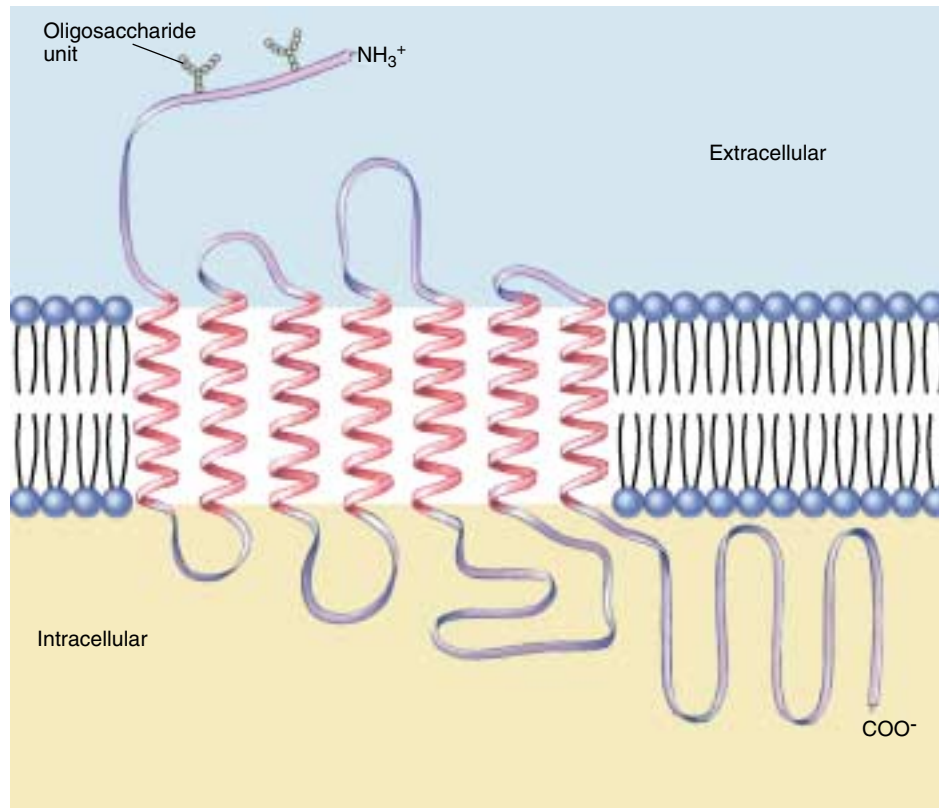


FIGURE 7.8

Structure of the β -adrenergic receptor. The receptor is a G-protein-linked molecule which, when it binds to an extracellular signal molecule, stimulates voluminous production of cAMP inside the cell, which then effects the cellular change.

activities. For example, the efflux of Ca^{++} from the endoplasmic reticulum causes skeletal muscle cells to contract and some endocrine cells to secrete hormones.

The gated Ca^{++} channels are opened by a G-protein-linked receptor. In response to signals from other cells, the receptor activates its G protein, which in turn activates the enzyme, *phospholipase C*. This enzyme catalyzes the production of *inositol trisphosphate* (IP_3) from phospholipids in the plasma membrane. The IP_3 molecules diffuse through the cytoplasm to the endoplasmic reticulum and bind to the Ca^{++} channels. This opens the channels and allows Ca^{++} to flow from the endoplasmic reticulum into the cytoplasm (figure 7.9b).

Ca^{++} initiates some cellular responses by binding to *calmodulin*, a 148-amino acid cytoplasmic protein that contains four binding sites for Ca^{++} (figure 7.10). When four Ca^{++} ions are bound to calmodulin, the calmodulin/ Ca^{++} complex binds to other proteins, and activates them.

Cyclic AMP and Ca^{++} often behave as second messengers, intracellular substances that relay messages from receptors to target proteins.

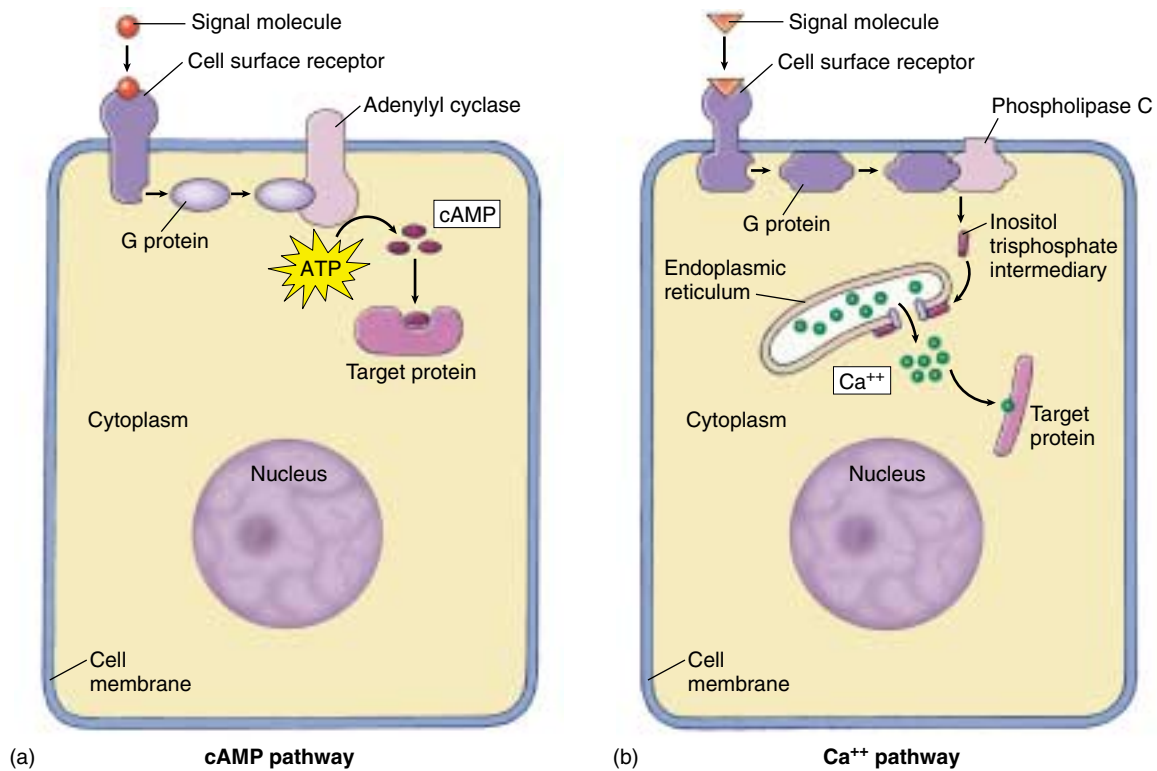


FIGURE 7.9

How second messengers work. (a) The cyclic AMP (cAMP) pathway. An extracellular receptor binds to a signal molecule and, through a G protein, activates the membrane-bound enzyme, adenylyl cyclase. This enzyme catalyzes the synthesis of cAMP, which binds to the target protein to initiate the cellular change. (b) The calcium (Ca⁺⁺) pathway. An extracellular receptor binds to another signal molecule and, through another G protein, activates the enzyme phospholipase C. This enzyme stimulates the production of inositol trisphosphate, which binds to and opens calcium channels in the membrane of the endoplasmic reticulum. Ca⁺⁺ is released into the cytoplasm, effecting a change in the cell.

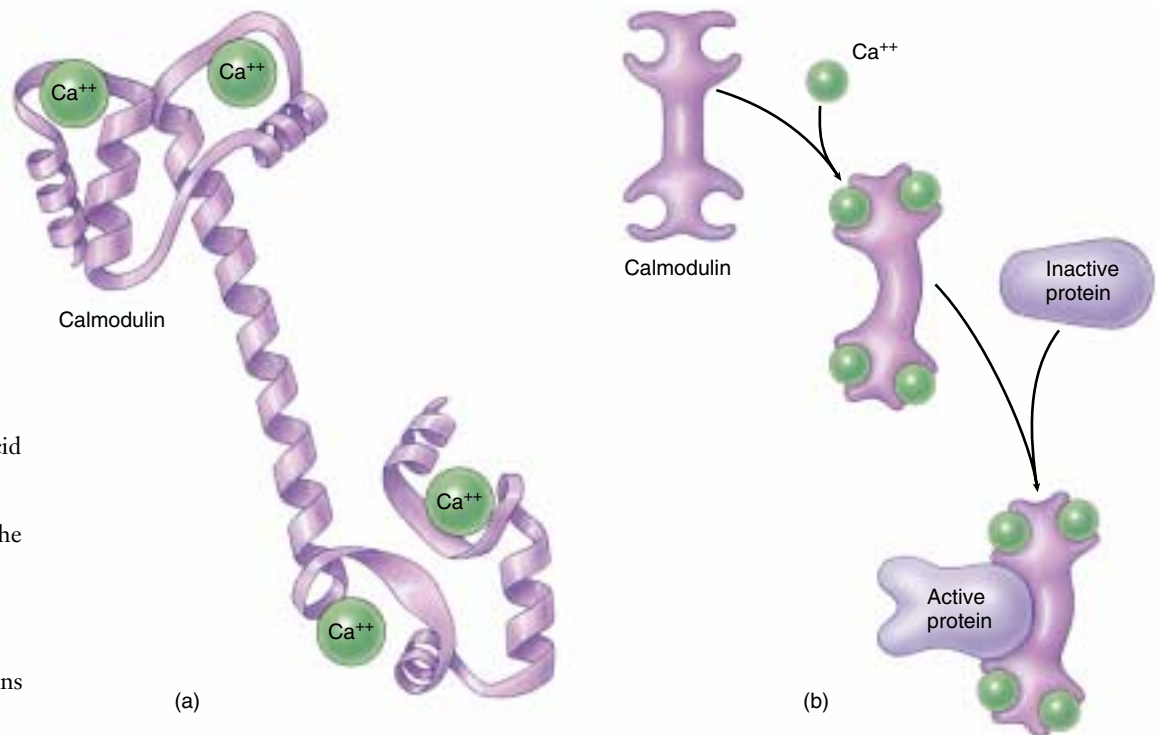


FIGURE 7.10

Calmodulin. (a) Calmodulin is a protein containing 148 amino acid residues that mediates Ca⁺⁺ function. (b) When four Ca⁺⁺ are bound to the calmodulin molecule, it undergoes a conformational change that allows it to bind to other cytoplasmic proteins and effect cellular responses.

Amplifying the Signal: Protein Kinase Cascades

Both enzyme-linked and G-protein-linked receptors receive signals at the surface of the cell, but as we've seen, the target cell's response rarely takes place there. In most cases the signals are relayed to the cytoplasm or the nucleus by second messengers, which influence the activity of one or more enzymes or genes and so alter the behavior of the cell. But most signaling molecules are found in such low concentrations that their diffusion across the cytoplasm would take a great deal of time unless the signal is amplified. Therefore, most enzyme-linked and G-protein-linked

receptors use a chain of other protein messengers to amplify the signal as it is being relayed to the nucleus.

How is the signal amplified? Imagine a relay race where, at the end of each stage, the finishing runner tags five new runners to start the next stage. The number of runners would increase dramatically as the race progresses: 1, then 5, 25, 125, and so on. The same sort of process takes place as a signal is passed from the cell surface to the cytoplasm or nucleus. First the receptor activates a stage-one protein, almost always by phosphorylating it. The receptor either adds a phosphate group directly, or, it activates a G protein that goes on to activate a second protein that does the phosphorylation. Once activated, each of these stage-one proteins in turn activates a large number of stage-two pro-

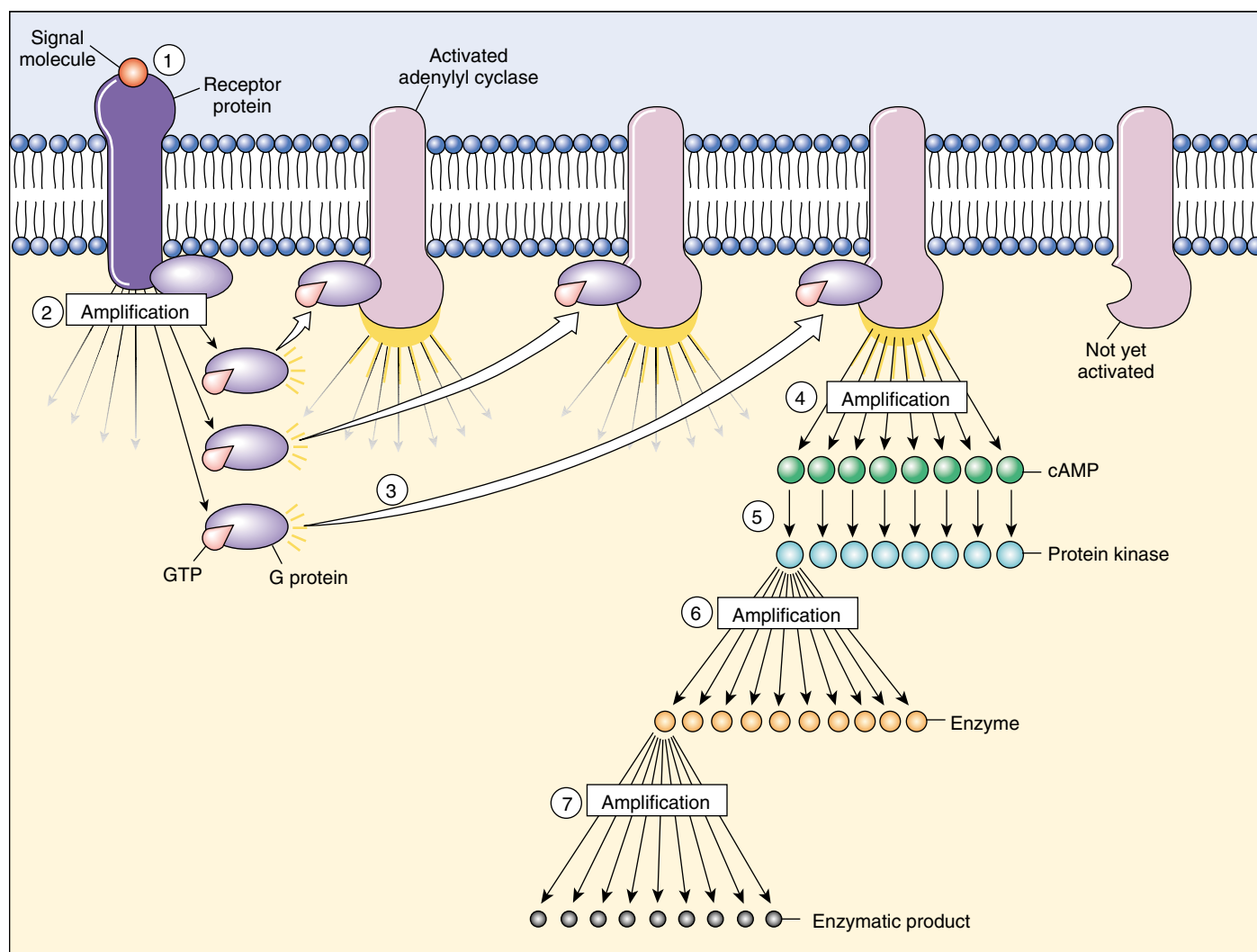
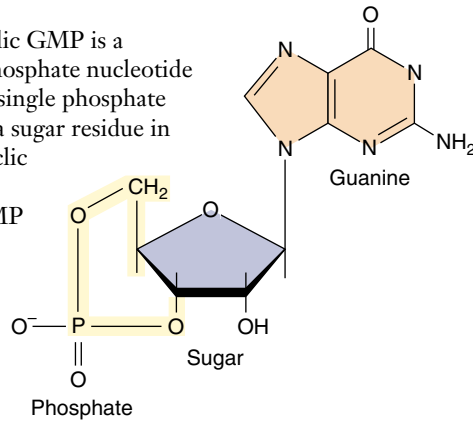


FIGURE 7.11

Signal amplification. Amplification at many steps of the cell-signaling process can ultimately produce a large response by the cell. One cell surface receptor (1), for example, may activate many G protein molecules (2), each of which activates a molecule of adenylyl cyclase (3), yielding an enormous production of cAMP (4). Each cAMP molecule in turn will activate a protein kinase (5), which can phosphorylate and thereby activate several copies of a specific enzyme (6). Each of *those* enzymes can then catalyze many chemical reactions (7). Starting with 10^{-10} M of signaling molecule, one cell surface receptor can trigger the production of 10^{-6} M of one of the products, an amplification of four orders of magnitude.

FIGURE 7.12

Cyclic GMP. Cyclic GMP is a guanosine monophosphate nucleotide molecule with the single phosphate group attached to a sugar residue in two places (this cyclic part is shown in yellow). Cyclic GMP is an important second messenger linking G proteins to signal transduction pathways within the cytoplasm.



teins; then each of them activates a large number of stage-three proteins, and so on (figure 7.11). A single cell surface receptor can thus stimulate a cascade of protein kinases to amplify the signal.

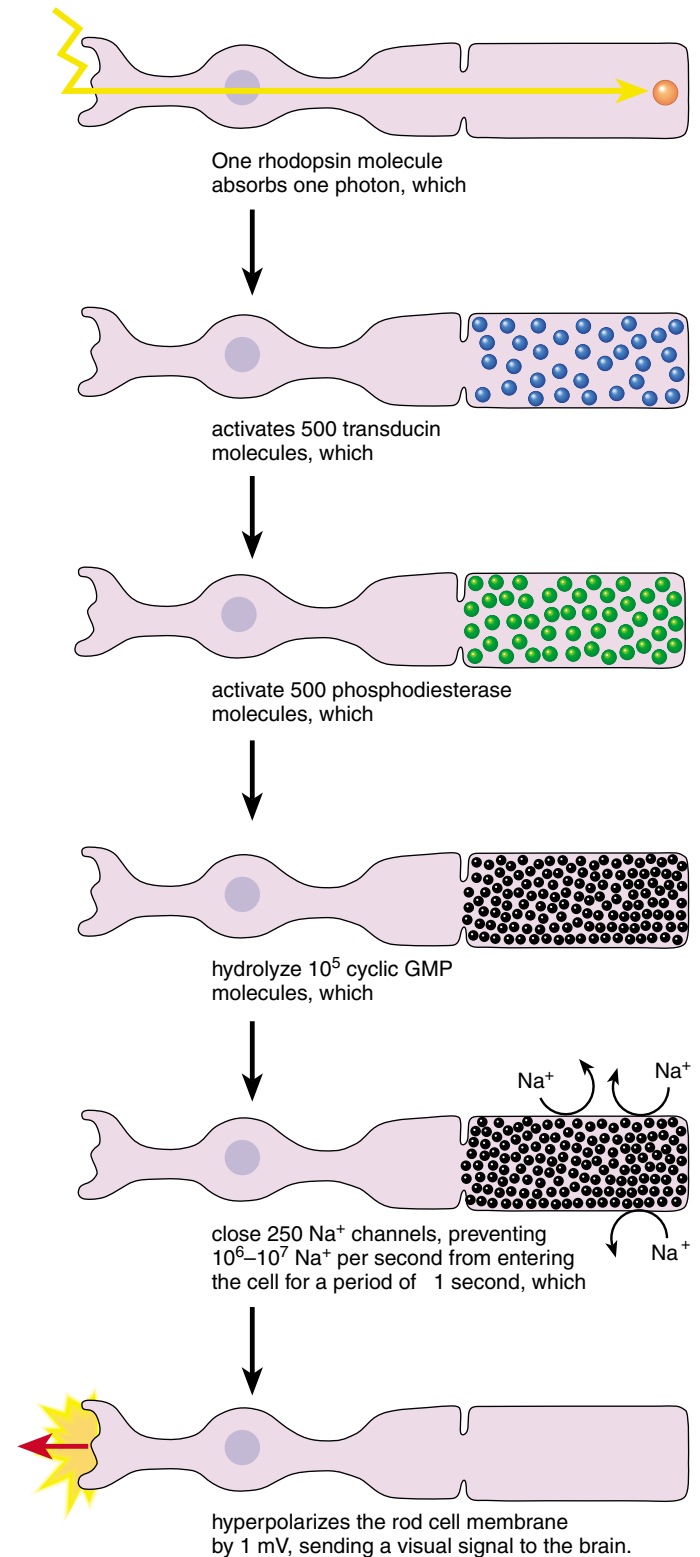
The Vision Amplification Cascade

Let's trace a protein amplification cascade to see exactly how one works. In vision, a single light-activated rhodopsin (a G-protein-linked receptor) activates hundreds of molecules of the G protein transducin in the first stage of the relay. In the second stage, each transducin causes an enzyme to modify thousands of molecules of a special inside-the-cell messenger called cyclic GMP (figure 7.12). (We will discuss cyclic GMP in more detail later.) In about 1 second, a single rhodopsin signal passing through this two-step cascade splits more than 10^5 (100,000) cyclic GMP molecules (figure 7.13)! The rod cells of humans are sufficiently sensitive to detect brief flashes of 5 photons.

The Cell Division Amplification Cascade

The amplification of signals traveling from the plasma membrane to the nucleus can be even more complex than the process we've just described. Cell division, for example, is controlled by a receptor that acts as a protein kinase. The receptor responds to growth-promoting signals by phosphorylating an intracellular protein called ras, which then activates a series of interacting phosphorylation cascades, some with five or more stages. If the ras protein becomes hyperactive for any reason, the cell acts as if it is being constantly stimulated to divide. Ras proteins were first discovered in cancer cells. A mutation of the gene that encodes ras had caused it to become hyperactive, resulting in unrestrained cell proliferation. Almost one-third of human cancers have such a mutation in a *ras* gene.

A small number of surface receptors can generate a vast intracellular response, as each stage of the pathway amplifies the next.

**FIGURE 7.13**

The role of signal amplification in vision. In this vertebrate rod cell (the cells of the eye responsible for interpreting light and dark), *one* single rhodopsin pigment molecule, when excited by a photon, ultimately yields *100,000* split cGMP molecules, which will then effect a change in the membrane of the rod cell, which will be interpreted by the organism as a visual event.

7.4 Cell surface proteins mediate cell-cell interactions.

The Expression of Cell Identity

With the exception of a few primitive types of organisms, the hallmark of multicellular life is the development of highly specialized groups of cells called **tissues**, such as blood and muscle. Remarkably, each cell within a tissue performs the functions of that tissue and no other, even though all cells of the body are derived from a single fertilized cell and contain the same genetic information. How do cells sense where they are, and how do they “know” which type of tissue they belong to?

Tissue-Specific Identity Markers

As it develops, each animal cell type acquires a unique set of cell surface molecules. These molecules serve as markers proclaiming the cells’ tissue-specific identity. Other cells that make direct physical contact with them “read” the markers.

Glycolipids. Most tissue-specific cell surface markers are glycolipids, lipids with carbohydrate heads. The glycolipids on the surface of red blood cells are also responsible for the differences among A, B, and O blood types. As the cells in a tissue divide and differentiate, the population of cell surface glycolipids changes dramatically.

MHC Proteins. The immune system uses other cell surface markers to distinguish between “self” and “nonself” cells. All of the cells of a given individual, for example, have the same “self” markers, called *major histocompatibility complex* (MHC) *proteins*. Because practically every individual makes a different set of MHC proteins, they serve as distinctive identity tags for each individual. The MHC proteins and other self-identifying markers are single-pass proteins anchored in the plasma membrane, and many of them are members of a large superfamily of receptors, the immunoglobulins (figure 7.14). Cells of the immune system continually inspect the other cells they encounter in the body, triggering the destruction of cells that display foreign or “nonself” identity markers.

The immune systems of vertebrates, described in detail in chapter 57, shows an exceptional ability to distinguish self from nonself. However, other vertebrates and even some simple animals like sponges are able to make this distinction to some degree, even though they lack a complex immune system.

Every cell contains a specific array of marker proteins on its surface. These markers identify each type of cell in a very precise way.

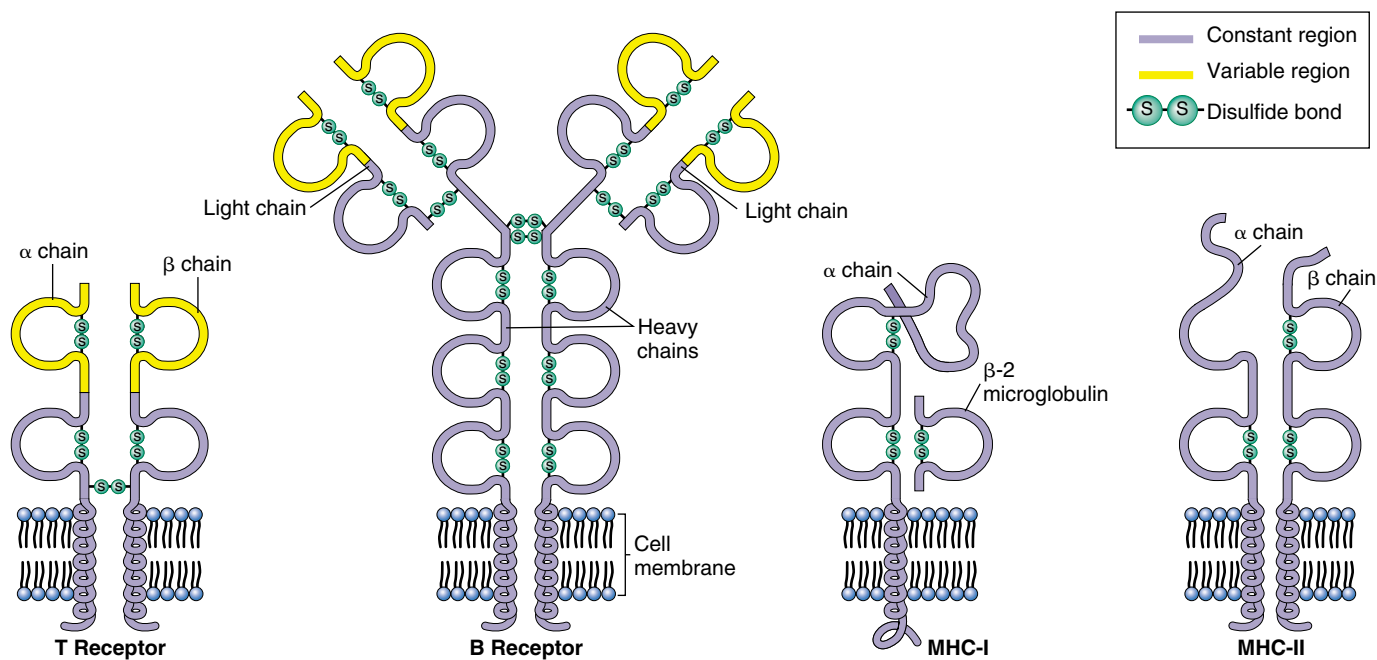


FIGURE 7.14

Structure of the immunoglobulin family of cell surface marker proteins. T and B cell receptors help mediate the immune response in organisms by recognizing and binding to foreign cell markers. MHC antigens label cells as “self,” so that the immune system attacks only invading entities, such as bacteria, viruses, and usually even the cells of transplanted organs!

Intercellular Adhesion

Not all physical contacts between cells in a multicellular organism are fleeting touches. In fact, most cells are in physical contact with other cells at all times, usually as members of organized tissues such as those in the lungs, heart, or gut. These cells and the mass of other cells clustered around them form long-lasting or permanent connections with each other called **cell junctions** (figure 7.15). The nature of the physical connections between the cells of

a tissue in large measure determines what the tissue is like. Indeed, a tissue's proper functioning often depends critically upon how the individual cells are arranged within it. Just as a house cannot maintain its structure without nails and cement, so a tissue cannot maintain its characteristic architecture without the appropriate cell junctions.

Cells attach themselves to one another with long-lasting bonds called cell junctions.

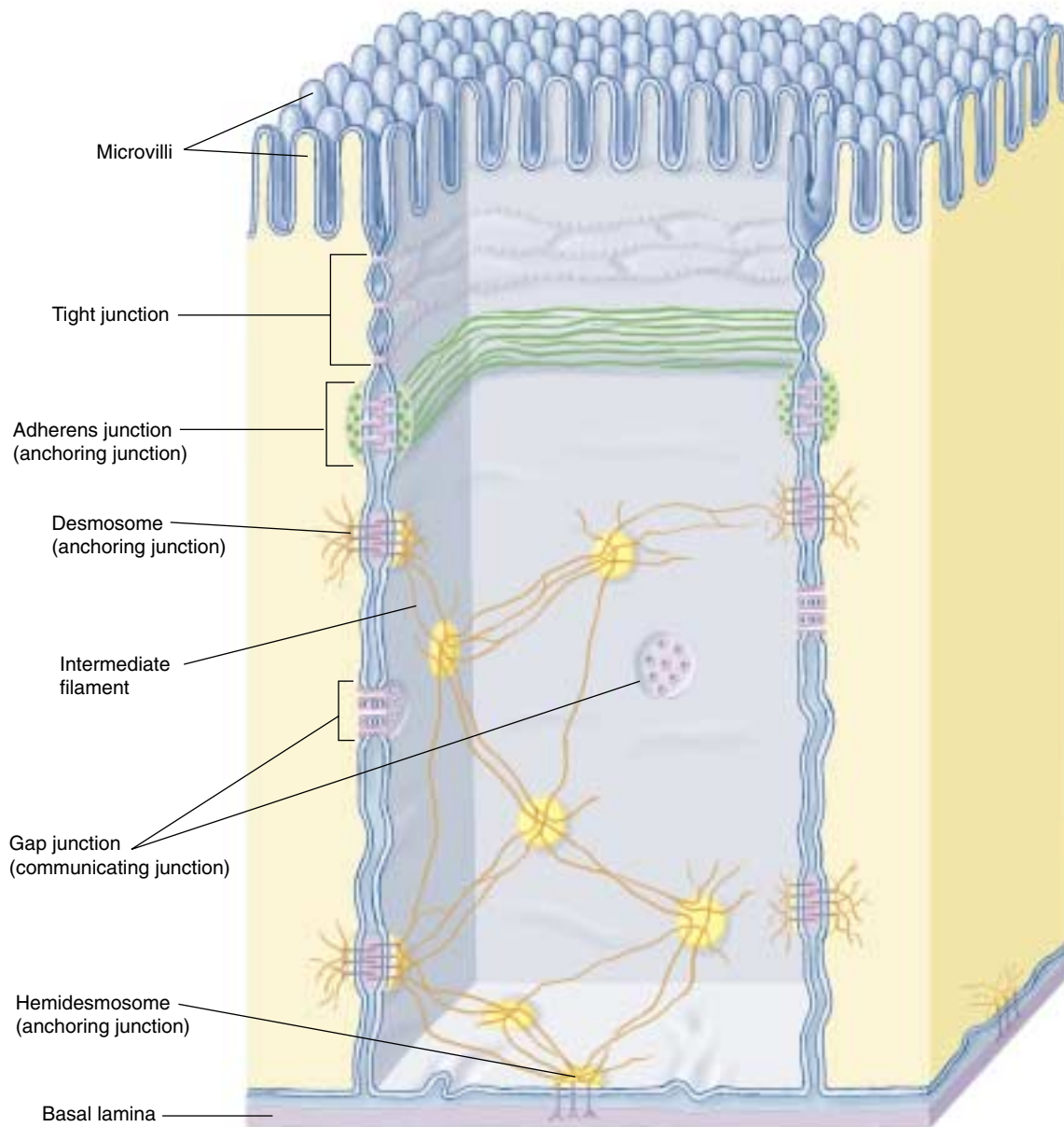


FIGURE 7.15

A summary of cell junction types. Gut epithelial cells are used here to illustrate the comparative structures and locations of common cell junctions.

Tight Junctions

Cell junctions are divided into three categories, based upon the functions they serve (figure 7.16): tight junctions, anchoring junctions, and communicating junctions.

Sometimes called occluding junctions, tight junctions connect the plasma membranes of adjacent cells in a sheet, preventing small molecules from leaking between the cells and through the sheet (figure 7.17). This allows the sheet of cells to act as a wall within the organ, keeping molecules on one side or the other.

Creating Sheets of Cells

The cells that line an animal's digestive tract are organized in a sheet only one cell thick. One surface of the sheet faces the inside of the tract and the other faces the extracellular space where blood vessels are located. Tight junctions encircle each cell in the sheet, like a belt cinched around a pair of pants. The junctions between neighboring cells are so securely attached that there is no space between them for leakage. Hence, nutrients absorbed from the food in the digestive tract must pass directly through the cells in the sheet to enter the blood.

Partitioning the Sheet

The tight junctions between the cells lining the digestive tract also partition the plasma membranes of these cells into separate compartments. Transport proteins in the membrane facing the inside of the tract carry nutrients from that side to the cytoplasm of the cells. Other proteins, located in the membrane on the opposite side of the cells, transport those nutrients from the cytoplasm to the extracellular fluid, where they can enter the blood. For the sheet to absorb nutrients properly, these proteins must remain in the correct locations within the fluid membrane. Tight junctions effectively segregate the proteins on opposite sides of the sheet, preventing them from drifting within the membrane from one side of the sheet to the other. When tight junctions are experimentally disrupted, just this sort of migration occurs.

Tight junctions connect the plasma membranes of adjacent cells into sheets.

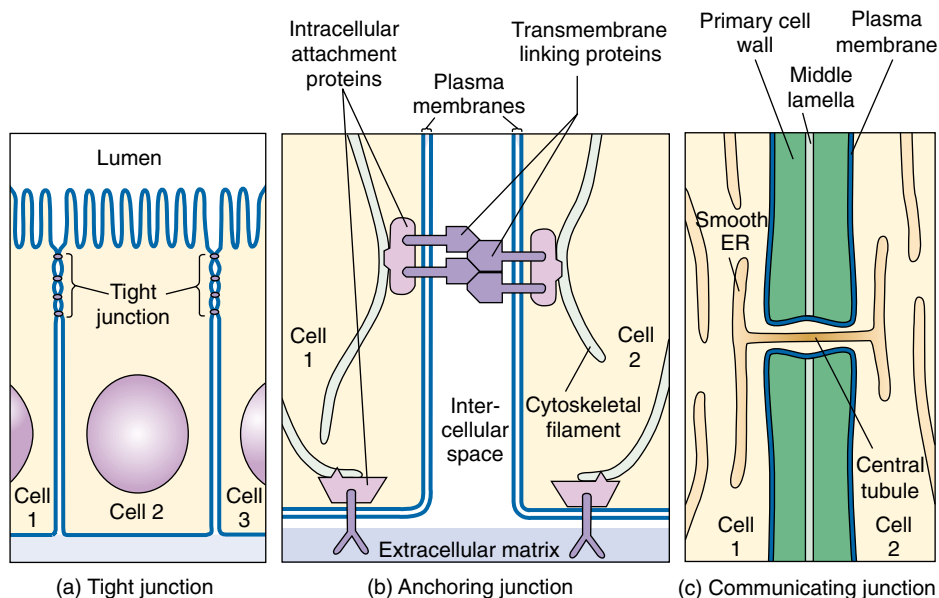


FIGURE 7.16

The three types of cell junctions. These three models represent current thinking on how the structures of the three major types of cell junctions facilitate their function: (a) tight junction; (b) anchoring junction; (c) communicating junction.

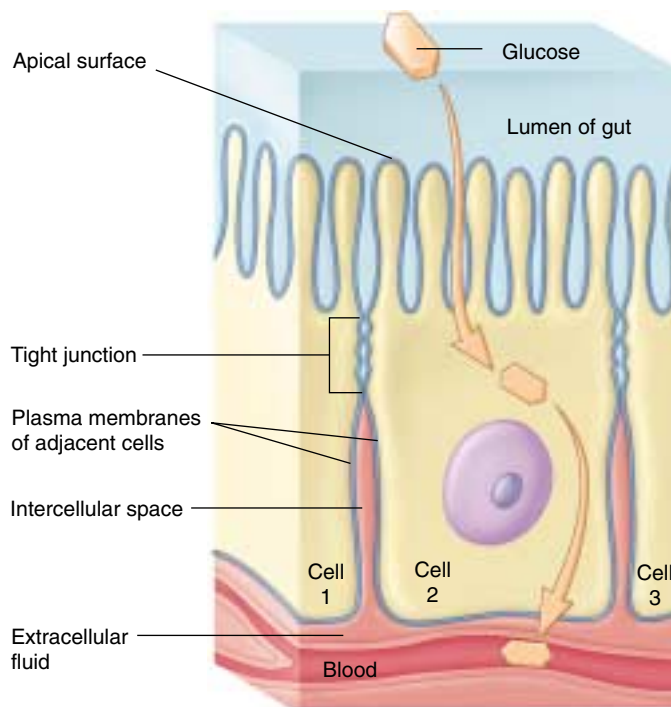


FIGURE 7.17

Tight junctions. Encircling the cell like a tight belt, these intercellular contacts ensure that materials move through the cells rather than between them.

Anchoring Junctions

Anchoring junctions mechanically attach the cytoskeleton of a cell to the cytoskeletons of other cells or to the extracellular matrix. They are commonest in tissues subject to mechanical stress, such as muscle and skin epithelium.

Cadherin and Intermediate Filaments: Desmosomes

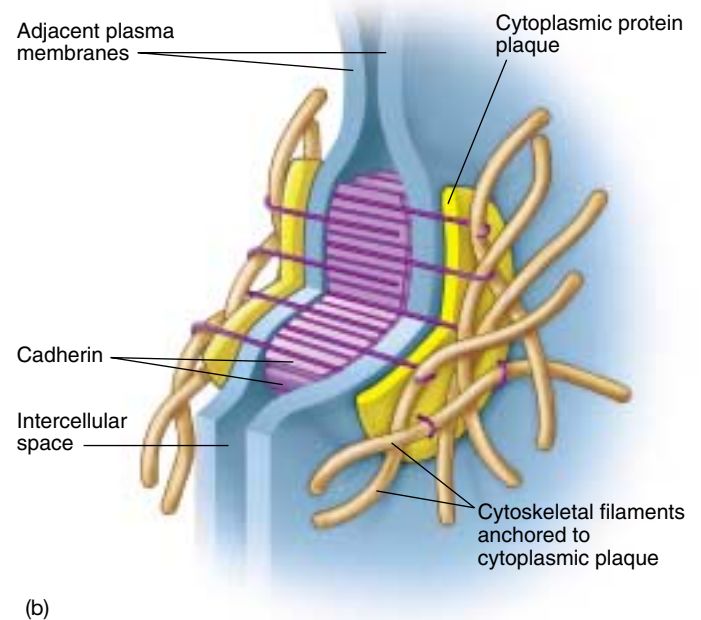
Anchoring junctions called **desmosomes** connect the cytoskeletons of adjacent cells (figure 7.18), while hemidesmosomes anchor epithelial cells to a basement membrane. Proteins called cadherins, most of which are

single-pass transmembrane glycoproteins, create the critical link. A variety of attachment proteins link the short cytoplasmic end of a cadherin to the intermediate filaments in the cytoskeleton. The other end of the cadherin molecule projects outward from the plasma membrane, joining directly with a cadherin protruding from an adjacent cell in a firm handshake binding the cells together.

Connections between proteins tethered to the intermediate filaments are much more secure than connections between free-floating membrane proteins. Proteins are suspended within the membrane by relatively weak interactions between the nonpolar portions of the protein and the membrane lipids. It would not take much force to pull an untethered protein completely out of the membrane, as if pulling an unanchored raft out of the water.



(a)



(b)

FIGURE 7.18

Desmosomes. (a) Desmosomes anchor adjacent cells to each other. (b) Cadherin proteins create the adhering link between adjoining cells.

Cadherin and Actin Filaments

Cadherins can also connect the actin frameworks of cells in cadherin-mediated junctions (figure 7.19). When they do, they form less stable links between cells than when they connect intermediate filaments. Many kinds of actin-linking cadherins occur in different tissues, as well as in the same tissue at different times. During vertebrate development, the migration of neurons in the embryo is associated with changes in the type of cadherin expressed on their plasma membranes. This suggests that gene-controlled changes in cadherin expression may provide the migrating cells with a “roadmap” to their destination.

Integrin-Mediated Links

Anchoring junctions called **adherens junctions** are another type of junction that connects the actin filaments of one cell with those of neighboring cells or with the extracellular matrix (figure 7.20). The linking proteins in these junctions are members of a large superfamily of cell surface receptors called integrins. Each integrin is a trans-membrane protein composed of two different glycoprotein subunits that extend outward from the plasma membrane. Together, these subunits bind a protein component of the extracellular matrix, like two hands clasping a pole. There appear to be many different kinds of integrin (cell biologists have identified 20), each with a slightly different shaped “hand.” The exact component of the matrix that a given cell binds to depends on which combination of integrins that cell has in its plasma membrane.

Anchoring junctions attach the cytoskeleton of a cell to the matrix surrounding the cell, or to the cytoskeleton of another cell.

FIGURE 7.20

An integrin-mediated junction. These adherens junctions link the actin filaments inside cells to their neighbors and to the extracellular matrix.

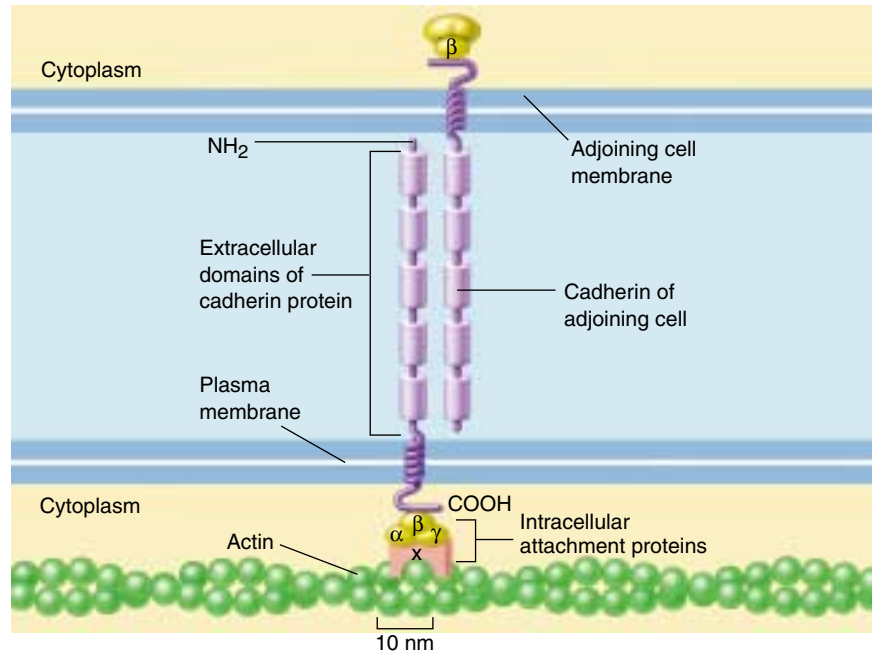
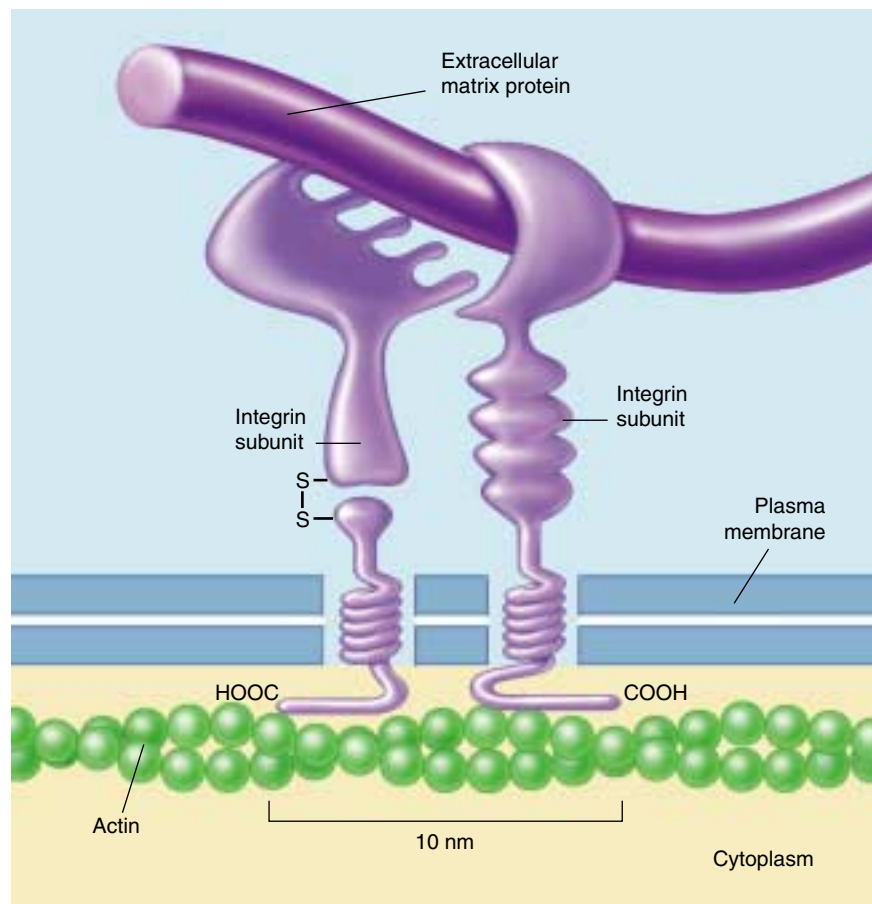


FIGURE 7.19

A cadherin-mediated junction. The cadherin molecule is anchored to actin in the cytoskeleton and passes through the membrane to interact with the cadherin of an adjoining cell.



Communicating Junctions

Many cells communicate with adjacent cells through direct connections, called **communicating junctions**. In these junctions, a chemical signal passes directly from one cell to an adjacent one. Communicating junctions establish direct physical connections that link the cytoplasms of two cells together, permitting small molecules or ions to pass from one to the other. In animals, these direct communication channels between cells are called gap junctions. In plants, they are called plasmodesmata.

Gap Junctions in Animals

Communicating junctions called gap junctions are composed of structures called connexons, complexes of six identical transmembrane proteins (figure 7.21). The proteins in a connexon are arranged in a circle to create a channel through the plasma membrane that protrudes several nanometers from the cell surface. A gap junction forms when the connexons of two cells align perfectly, creating an open channel spanning the plasma membranes of both cells. Gap junctions provide passageways large enough to permit small substances, such as simple sugars and amino acids, to pass from the cytoplasm of one cell to that of the next, yet small enough to prevent the passage of larger molecules such as proteins. The connexons hold the plasma membranes of the paired cells about 4 nanometers apart, in marked contrast to the more-or-less direct contact between the lipid bilayers in a tight junction.

Gap junction channels are dynamic structures that can open or close in response to a variety of factors, including Ca^{++} and H^{+} ions. This gating serves at least one important function. When a cell is damaged, its plasma membrane often becomes leaky. Ions in high concentrations outside the cell, such as Ca^{++} , flow into the damaged cell and shut its gap junction channels. This isolates the cell and so prevents the damage from spreading to other cells.

Plasmodesmata in Plants

In plants, cell walls separate every cell from all others. Cell-cell junctions occur only at holes or gaps in the walls, where the plasma membranes of adjacent cells can come into contact with each other. Cytoplasmic connections that form across the touching plasma membranes are called plasmodesmata (figure 7.22). The majority of living cells within a higher plant are connected with their neighbors by these junctions. Plasmodesmata function much like gap junctions in animal cells, although their structure is more complex. Unlike gap junctions, plasmodesmata are lined with plasma membrane and contain a central tubule that connects the endoplasmic reticulum of the two cells.

Communicating junctions permit the controlled passage of small molecules or ions between cells.

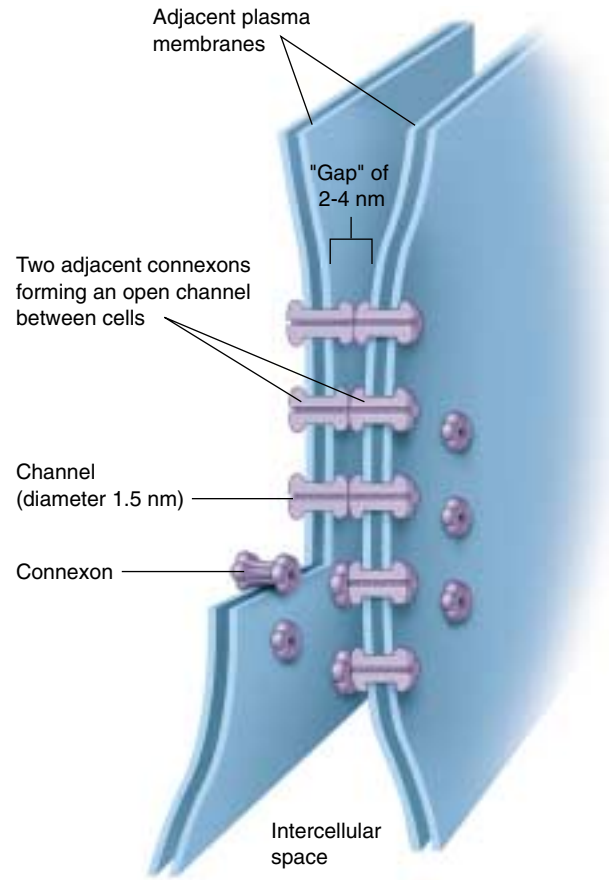


FIGURE 7.21

Gap junctions. Connexons in gap junctions create passageways that connect the cytoplasms of adjoining cells. Gap junctions readily allow the passage of small molecules and ions required for rapid communication (such as in heart tissue), but do not allow the passage of larger molecules like proteins.

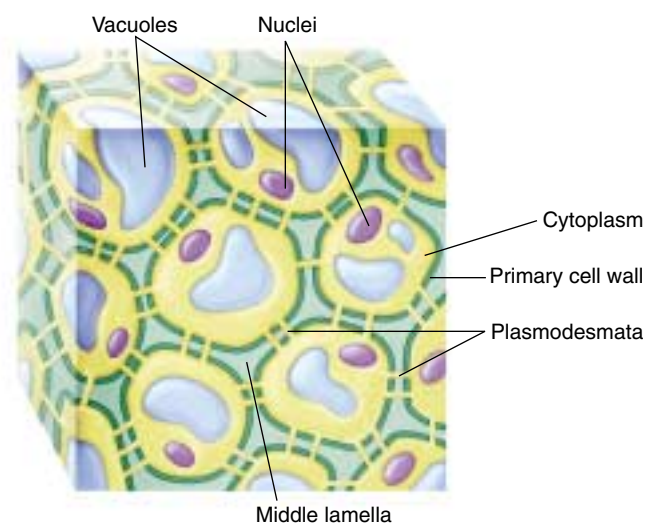


FIGURE 7.22

Plasmodesmata. Plant cells can communicate through specialized openings in their cell walls, called plasmodesmata, where the cytoplasms of adjoining cells are connected.



Summary

Questions

Media Resources

7.1 Cells signal one another with chemicals.

- Cell signaling is accomplished through the recognition of signal molecules by target cells.

- What determines which signal molecules in the extracellular environment a cell will respond to?
- How do paracrine, endocrine, and synaptic signaling differ?



- Cell Interactions



- Student Research: Retrograde Messengers between Nerve Cells
- Student Research: Vertebrate Limb formation

7.2 Proteins in the cell and on its surface receive signals from other cells.

- The binding of a signal molecule to an intracellular receptor usually initiates transcription of specific regions of DNA, ultimately resulting in the production of specific proteins.
- Cell surface receptors bind to specific molecules in the extracellular fluid. In some cases, this binding causes the receptor to enzymatically alter other (usually internal) proteins, typically through phosphorylation.
- G proteins behave as intracellular shuttles, moving from an activated receptor to other areas in the cell.

- Describe two of the ways in which intracellular receptors control cell activities.
- What structural features are characteristic of chemically gated ion channels, and how are these features related to the function of the channels?
- What are G proteins? How do they participate in cellular responses mediated by G-protein-linked receptors?

7.3 Follow the journey of information into the cell.

- There are usually several amplifying steps between the binding of a signal molecule to a cell surface receptor and the response of the cell. These steps often involve phosphorylation by protein kinases.

- How does the binding of a single signal molecule to a cell surface receptor result in an amplified response within the target cell?



- Exploration: Cell-Cell Interactions



- Scientists on Science: G Proteins

7.4 Cell surface proteins mediate cell-cell interactions.

- Tight junctions and desmosomes enable cells to adhere in tight, leakproof sheets, holding the cells together such that materials cannot pass between them.
- Gap junctions (in animals) and plasmodesmata (in plants) permit small substances to pass directly from cell to cell through special passageways.

- What are the functions of tight junctions? What are the functions of desmosomes and adherens junctions, and what proteins are involved in these junctions?
- What are the molecular components that make up gap junctions? What sorts of substances can pass through gap junctions?
- Where are plasmodesmata found? What cellular constituents are found in plasmodesmata?

How Do Proteins Help Chlorophyll Carry Out Photosynthesis?

Much public attention in recent years has been focused on high-profile science—headline-creating advances in the Human Genome Project, genetic engineering, and the battle against AIDS and cancer. Meanwhile, great advances have been made more quietly in other areas of biology. Among the greatest of these achievements has been the unmasking in the last decade of the underlying mechanism of photosynthesis.

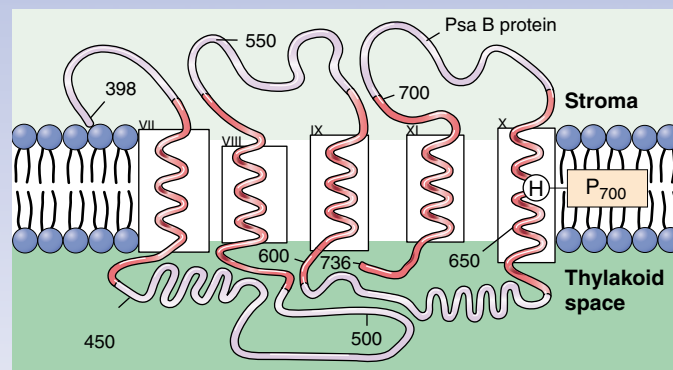
In photosynthesis, photons of light are absorbed by chlorophyll molecules, causing them to donate a high-energy electron that is put to work making NADPH and pumping protons to produce ATP.

When researchers looked at the light-absorbing chlorophylls that carry out photosynthesis more closely, they found the chlorophylls to be arranged in clusters called photosystems, supported by proteins and accessory pigments. Within a photosystem, hundreds of chlorophyll molecules act like antennae, absorbing light and passing the energy they capture inward to a single chlorophyll molecule that acts as the reaction center. This chlorophyll acts as the primary electron donor of photosynthesis. Once it releases a light-energized electron, the complex series of chemical events we call photosynthesis begins, and, like a falling row of dominos, is difficult to stop.

Plants possess two kinds of photosystems that work together to harvest light energy. One of them, called photosystem I, is similar to a simpler photosystem found in photosynthetic bacteria, and is thought to have evolved from it.

Photosystem I has been the subject of intense research. In its reaction center, a pair of chlorophyll molecules act as the trap for photon energy, passing an excited electron on to an acceptor molecule outside the reaction center. This moves the photon energy away from the chlorophylls, and is the key conversion of light energy to chemical energy, the very heart of photosynthesis.

Because the pair of chlorophyll molecules in the reaction center of photosystem I absorb light at a wavelength of 700 nm, they are together given the name P₇₀₀. The P₇₀₀ dimer is positioned within the photosystem by two related proteins that act as scaffolds. These proteins, discovered less



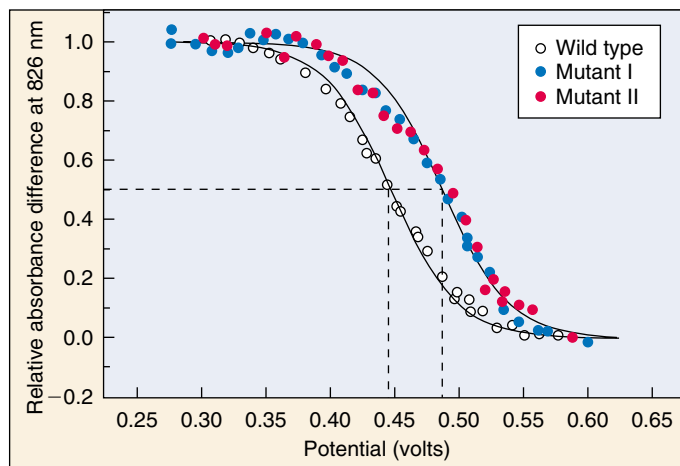
The proposed antenna complex of the PsaB protein. Position 656 is a histidine (H) in the tenth pass (helix X) of the PsaB protein across the thylakoid membrane within chloroplasts. This histidine is where the PsaB protein makes contact with a P₇₀₀ chlorophyll molecule.

than 10 years ago, turn out to play a pivotal role in the photosynthetic process. Passing back and forth across the internal chloroplast membranes 11 times, they form a molecular frame that positions P₇₀₀ to accept energy from other chlorophyll molecules of the photosystem, and to donate a photo-excited electron to an acceptor molecule outside the photosystem.

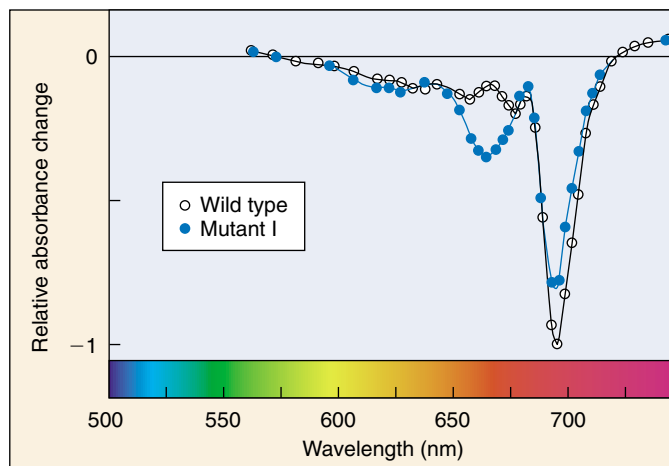
Recent research suggests that the role of these scaffold proteins, called PsaA and PsaB, is far more active than the passive support provided by a scaffold. Analysis of highly purified photosystems carried out in 1995 revealed that the distribution of electric charge over the two halves of the P₇₀₀ dimer is highly asymmetric—one chlorophyll molecule exhibits a far greater charge density than the other. Because the two chlorophyll molecules of P₇₀₀ are themselves identical, this suggests that the PsaA and PsaB proteins are actively modulating the physicochemical properties of the chlorophyll.

How can a protein pull off this physical-chemical sleight-of-hand? Just what are these proteins *doing* to the chlorophyll molecules? To look more closely at what is going on, you have to first figure out what part of the protein to look at. One way to get a handle on this problem is to compare the amino acid sequences of PsaA and PsaB with that of the bacterial photosystem from which they are thought to have evolved. It is likely that such an important part of the sequence would have been conserved and will be found in all three.

Several sequences are indeed conserved, but most of them prove not to interact directly with chlorophyll. One, however, is a promising candidate. A single amino acid in the helix X domain (that is, the tenth pass of the PsaB protein across the membrane), dubbed His-656, is conserved in all sequences, and is positioned right where the PsaB protein touches the P₇₀₀ chlorophyll (see above). This amino acid, a histidine, has become the focus of recent efforts to clarify how proteins help chlorophyll carry out photosynthesis.



(a)



(b)

Effect of altering position 656. (a) When P_{700} interacts with normal and mutant forms of PsaB, the midpoint potentials are 447 ± 6 mV in the wild type and 487 ± 6 in the mutants, the mutant value being about 40 mV higher. (b) The bleaching band (dip in the absorbance) of P_{700} is shifted to the blue (left) and exhibits a new bleaching band at 667 nm when interacting with mutant forms of PsaB.

The Experiment

To determine the importance of His-656, and more generally of the helix X domain of the PsaB protein, Professor Andrew Webber of Arizona State University, working with his research team and the group of Professor Wolfgang Lubitz at Technische Universität Berlin, has created site-directed mutations of His-656 in the photosynthetic protist *Chlamydomonas reinhardtii*. *C. reinhardtii* is widely used to study photosynthesis because of the ease with which lab experiments can be done.

Webber and his collaborators set out to change the amino acid located at position 656 of PsaB, and then to look and see what effect the change had on photosynthesis. If His-656 indeed plays a critical role in modifying the P_{700} chlorophylls, then a change at that position to a different amino acid should have profound effects.

Creating PsaB Proteins Mutant at Position 656. The first and key step in Webber's experimental approach was to genetically alter the chloroplast of *C. reinhardtii*, introducing a mutation of the PsaB gene at the His-656 position. To do this, the team employed site-specific mutagenesis to construct mutant plasmids pHN(B656) and pHS(B656), inserting a gene carrying either the Ser or Asn amino acids in place of His. The two mutation-carrying plasmids were then cloned into *C. reinhardtii*, cells carrying the mutant plasmids isolated, and presence of the mutated gene directly confirmed by sequencing the DNA.

Characterizing the Effects of 656 Mutations. Once researchers confirmed that the *C. reinhardtii* chloroplast DNA now contained the mutant forms of the PsaB gene, they proceeded to test the function of the mutated PsaB protein in coordinating P_{700} , examining interior thylakoid membranes isolated from the chloroplasts. To do this, the researchers measured the oxidation midpoint potentials of the P_{700} complexes, an indication of how tightly the chlorophyll molecules are holding onto their electrons.

The researchers further characterized the P_{700} complexes by measuring the changes in absorbance of the mutants versus the wild type to see if the mutations altered the spectral properties of the P_{700} chlorophylls.

The Results

The results of the examination of the oxidation midpoint potentials revealed that the influence of the PsaB protein on P_{700} had been profoundly altered by the mutations. The midpoint potential of P_{700} in the wild type was determined to be 447 ± 6 mV, while the midpoint potential had increased to 487 ± 6 mV in both the PsaB mutant I, HN(B656), and the PsaB mutant II, HS(B656) (see graph a). This increase in the oxidation midpoint potential by approximately 40 mV indicates that the mutations to the His residue significantly altered the redox property of P_{700} and, therefore, that His-656 is closely interacting with one of the chlorophyll molecules of the P_{700} dimer.

These results and this conclusion are further supported by changes observed in the spectral properties of the mutants and wild type (see graph b). There is a reduction and a slight shift in the 696 nm bleaching band (dip in absorbance) in PsaB mutant I toward the blue end of the spectrum and a new bleaching band appearing at 667 nm, both changes in the spectral properties of chlorophyll induced by the mutational changes in the PsaB protein.

Ultimately, the researchers conclude that the His-656 of PsaB directly coordinates the central magnesium atom of one of the two chlorophyll molecules of P_{700} . Their results are consistent with a model of photosystem I in which the first six spans of PsaB constitute an antenna domain for receiving energy from other chlorophylls and the last five membrane spans interact with the P_{700} reaction complex.



To explore this experiment further, go to the Virtual Lab at www.mhhe.com/raven6/vlab3.mhtml

8

Energy and Metabolism

Concept Outline

8.1 The laws of thermodynamics describe how energy changes.

The Flow of Energy in Living Things. Potential energy is present in the valence electrons of atoms, and so can be transferred from one molecule to another.

The Laws of Thermodynamics. Energy is never lost but as it is transferred, more and more of it dissipates as heat, a disordered form of energy.

Free Energy. In a chemical reaction, the energy released or supplied is the difference in bond energies between reactants and products, corrected for disorder.

Activation Energy. To start a chemical reaction, an input of energy is required to destabilize existing chemical bonds.

8.2 Enzymes are biological catalysts.

Enzymes. Globular proteins called enzymes catalyze chemical reactions within cells.

How Enzymes Work. Enzymes have sites on their surface shaped to fit their substrates snugly, forcing chemically reactive groups close enough to facilitate a reaction.

Enzymes Take Many Forms. Some enzymes are associated in complex groups; others are not even proteins.

Factors Affecting Enzyme Activity. Each enzyme works most efficiently at its optimal temperature and pH. Metal ions or other substances often help enzymes carry out their catalysis.

8.3 ATP is the energy currency of life.

What Is ATP? Cells store and release energy from the phosphate bonds of ATP, the energy currency of the cell.

8.4 Metabolism is the chemical life of a cell.

Biochemical Pathways: The Organizational Units of Metabolism. Biochemical pathways are the organizational units of metabolism.

The Evolution of Metabolism. The major metabolic processes evolved over a long period, building on what came before.



FIGURE 8.1

Lion at lunch. Energy that this lion extracts from its meal of giraffe will be used to power its roar, fuel its running, and build a bigger lion.

Life can be viewed as a constant flow of energy, channeled by organisms to do the work of living. Each of the significant properties by which we define life—order, growth, reproduction, responsiveness, and internal regulation—requires a constant supply of energy (figure 8.1). Deprived of a source of energy, life stops. Therefore, a comprehensive study of life would be impossible without discussing bioenergetics, the analysis of how energy powers the activities of living systems. In this chapter, we will focus on energy—on what it is and how organisms capture, store, and use it.

8.1 The laws of thermodynamics describe how energy changes.

The Flow of Energy in Living Things

Energy is defined as the capacity to do work. It can be considered to exist in two states. **Kinetic energy** is the energy of motion (figure 8.2). Moving objects perform work by causing other matter to move. **Potential energy** is stored energy. Objects that are not actively moving but have the capacity to do so possess potential energy. A boulder perched on a hilltop has potential energy; as it begins to roll downhill, some of its potential energy is converted into kinetic energy. Much of the work that living organisms carry out involves transforming potential energy to kinetic energy.

Energy can take many forms: mechanical energy, heat, sound, electric current, light, or radioactive radiation. Because it can exist in so many forms, there are many ways to measure energy. The most convenient is in terms of heat, because all other forms of energy can be converted into heat. In fact, the study of energy is called **thermodynamics**, meaning heat changes. The unit of heat most commonly employed in biology is the **kilocalorie** (kcal). One kilocalorie is equal to 1000 calories (cal), and one calorie is the heat required to raise the temperature of one gram of water one degree Celsius ($^{\circ}\text{C}$). (It is important not to confuse calories with a term related to diets and nutrition, the Calorie with a capital C, which is actually another term for kilocalorie.) Another energy unit, often used in physics, is the **joule**; one joule equals 0.239 cal.

Oxidation-Reduction

Energy flows into the biological world from the sun, which shines a constant beam of light on the earth. It is estimated that the sun provides the earth with more than 13×10^{23} calories per year, or 40 million billion calories per second! Plants, algae, and certain kinds of bacteria capture a fraction of this energy through photosynthesis. In photosynthesis, energy garnered from sunlight is used to combine small molecules (water and carbon dioxide) into more complex molecules (sugars). The energy is stored as potential energy in the covalent bonds between atoms in the sugar molecules. Recall from chapter 2 that an atom consists of a central nucleus surrounded by one or more orbiting electrons, and a covalent bond forms when two atomic nuclei share valence electrons. Breaking such a bond requires energy to pull the nuclei apart. Indeed, the strength of a covalent bond is measured by the amount of energy required to break it. For example, it takes 98.8 kcal to break one mole (6.023×10^{23}) of carbon-hydrogen (C—H) bonds.

During a chemical reaction, the energy stored in chemical bonds may transfer to new bonds. In some of these reactions, electrons actually pass from one atom or molecule to another. When an atom or molecule loses an electron, it is said to be oxidized, and the process by which this occurs is called **oxidation**. The name reflects the fact that in biological systems oxygen, which attracts electrons strongly, is the most common electron accep-

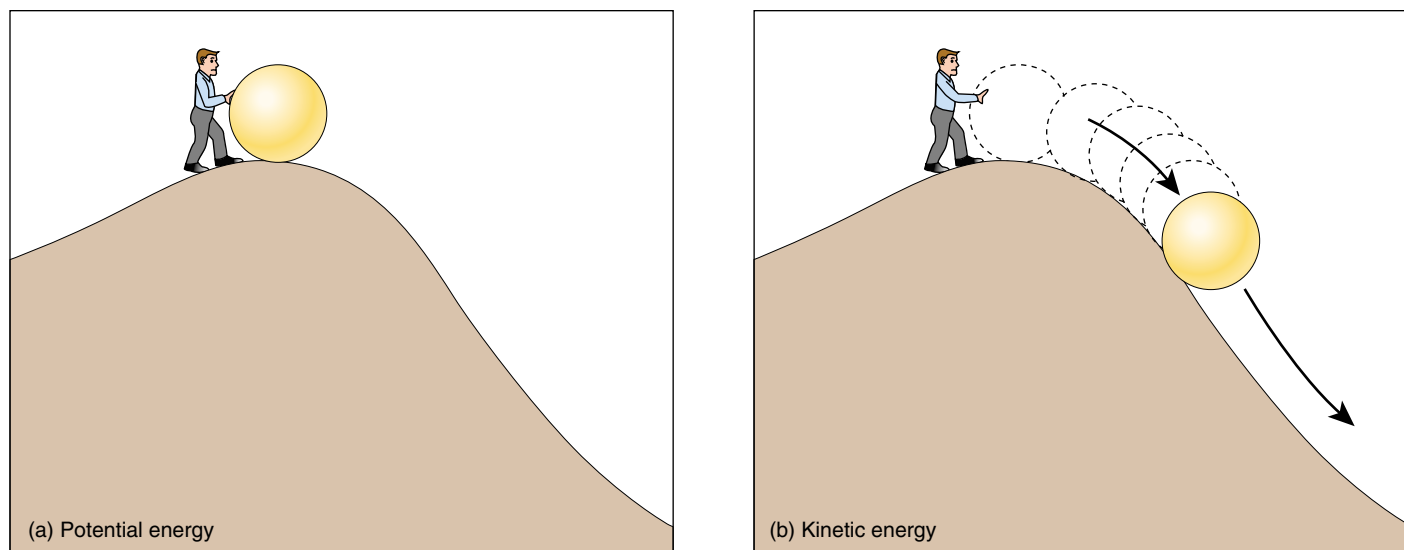


FIGURE 8.2

Potential and kinetic energy. (a) Objects that have the capacity to move but are not moving have potential energy. The energy required to move the ball up the hill is stored as potential energy. (b) Objects that are in motion have kinetic energy. The stored energy is released as kinetic energy as the ball rolls down the hill.

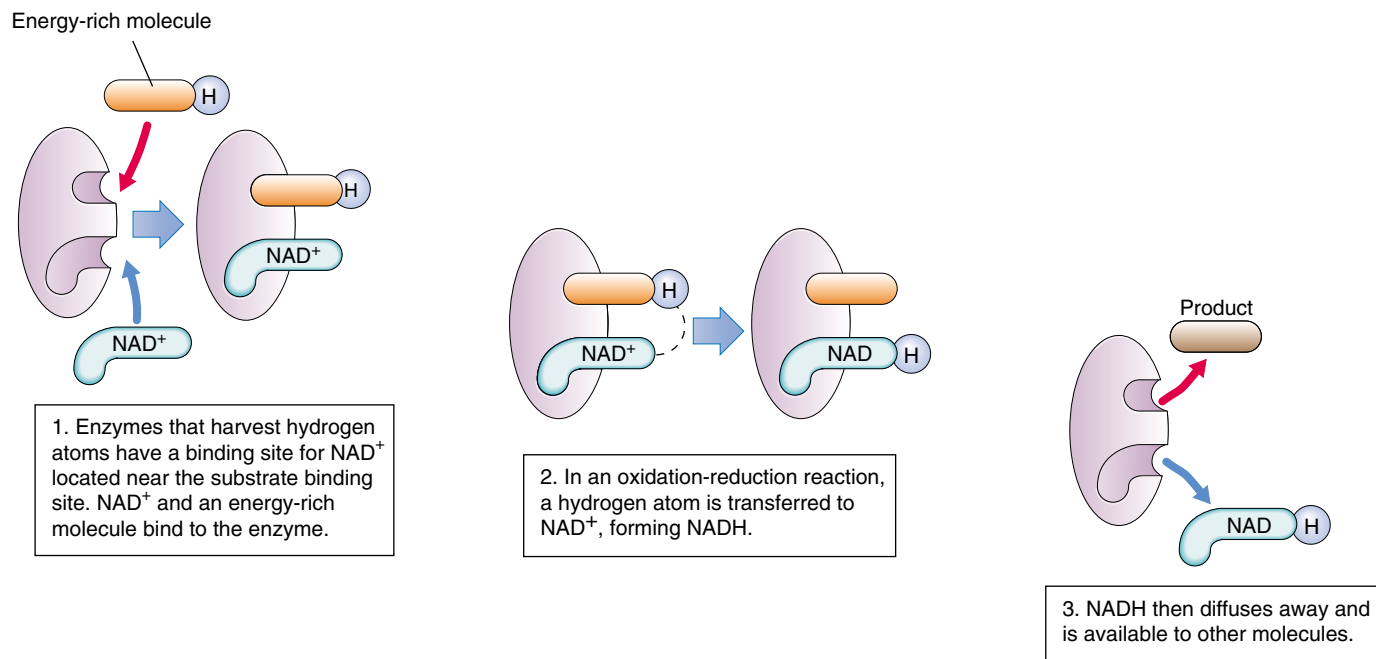


FIGURE 8.3

An oxidation-reduction reaction. Cells use a chemical called NAD^+ to carry out oxidation-reduction reactions. Energetic electrons are often paired with a proton as a hydrogen atom. Molecules that gain energetic electrons are said to be reduced, while ones that lose energetic electrons are said to be oxidized. NAD^+ oxidizes energy-rich molecules by acquiring their hydrogens (in the figure, this proceeds 1→2→3) and then reduces other molecules by giving the hydrogens to them (in the figure, this proceeds 3→2→1).

tor. Conversely, when an atom or molecule gains an electron, it is said to be reduced, and the process is called **reduction**. Oxidation and reduction always take place together, because every electron that is lost by an atom through oxidation is gained by some other atom through reduction. Therefore, chemical reactions of this sort are called **oxidation-reduction (redox) reactions** (figure 8.3). Energy is transferred from one molecule to another via redox reactions. The reduced form of a molecule thus has a higher level of energy than the oxidized form (figure 8.4).

Oxidation-reduction reactions play a key role in the flow of energy through biological systems because the electrons that pass from one atom to another carry energy with them. The amount of energy an electron possesses depends on how far it is from the nucleus and how strongly the nucleus attracts it. Light (and other forms of energy) can add energy to an electron and boost it to a higher energy level. When this electron departs from one atom (oxidation) and moves to another (reduction), the electron's added energy is transferred with it, and the electron orbits the second atom's nucleus at the higher energy level. The added energy is stored as potential chemical energy that the atom can later release when the electron returns to its original energy level.

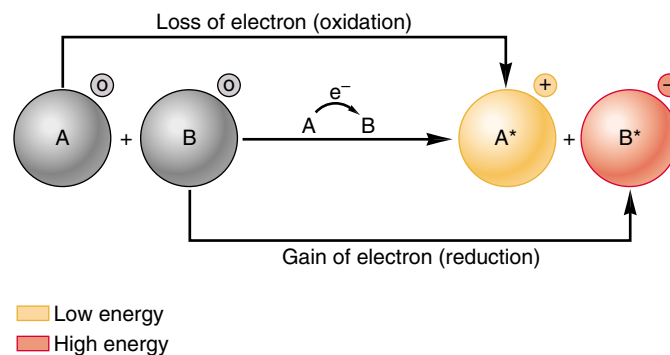


FIGURE 8.4

Redox reactions. Oxidation is the loss of an electron; reduction is the gain of an electron. In this example, the charges of molecules A and B are shown in small circles to the upper right of each molecule. Molecule A loses energy as it loses an electron, while molecule B gains energy as it gains an electron.

Energy is the capacity to do work, either actively (kinetic energy) or stored for later use (potential energy). Energy is transferred with electrons. Oxidation is the loss of an electron; reduction is the gain of one.

The Laws of Thermodynamics

Running, thinking, singing, reading these words—all activities of living organisms involve changes in energy. A set of universal laws we call the Laws of Thermodynamics govern all energy changes in the universe, from nuclear reactions to the buzzing of a bee.

The First Law of Thermodynamics

The first of these universal laws, the **First Law of Thermodynamics**, concerns the amount of energy in the universe. It states that energy cannot be created or destroyed; it can only change from one form to another (from potential to kinetic, for example). The total amount of energy in the universe remains constant.

The lion eating a giraffe in figure 8.1 is in the process of acquiring energy. Rather than creating new energy or capturing the energy in sunlight, the lion is merely transferring some of the potential energy stored in the giraffe's tissues to its own body (just as the giraffe obtained the potential energy stored in the plants it ate while it was alive). Within any living organism, this chemical potential energy can be shifted to other molecules and stored in different chemical bonds, or it can convert into other forms, such as kinetic energy, light, or electricity. During each conversion, some of the energy dissipates into the environment as **heat**, a measure of the random motions of molecules (and, hence, a measure of one form of kinetic energy). Energy continuously flows through the biological world in one direction, with new energy from the sun constantly entering the system to replace the energy dissipated as heat.

Heat can be harnessed to do work only when there is a heat gradient, that is, a temperature difference between two areas (this is how a steam engine functions). Cells are too small to maintain significant internal temperature differences, so heat energy is incapable of doing the work of cells. Thus, although the total amount of energy in the universe remains constant, the energy available to do work decreases, as progressively more of it dissipates as heat.

The Second Law of Thermodynamics

The **Second Law of Thermodynamics** concerns this transformation of potential energy into heat, or random molecular motion. It states that the disorder (more formally called *en-*



FIGURE 8.5

Entropy in action. As time elapses, a child's room becomes more disorganized. It takes effort to clean it up.

trophy) in the universe is continuously increasing. Put simply, disorder is more likely than order. For example, it is much more likely that a column of bricks will tumble over than that a pile of bricks will arrange themselves spontaneously to form a column. In general, energy transformations proceed spontaneously to convert matter from a more ordered, less stable form, to a less ordered, more stable form (figure 8.5).

Entropy

Entropy is a measure of the disorder of a system, so the Second Law of Thermodynamics can also be stated simply as “entropy increases.” When the universe formed, it held all the potential energy it will ever have. It has become progressively more disordered ever since, with every energy exchange increasing the amount of entropy.

The First Law of Thermodynamics states that energy cannot be created or destroyed; it can only undergo conversion from one form to another. The **Second Law of Thermodynamics** states that disorder (entropy) in the universe is increasing. As energy is used, more and more of it is converted to heat, the energy of random molecular motion.

Free Energy

It takes energy to break the chemical bonds that hold the atoms in a molecule together. Heat energy, because it increases atomic motion, makes it easier for the atoms to pull apart. Both chemical bonding and heat have a significant influence on a molecule, the former reducing disorder and the latter increasing it. The net effect, the amount of energy actually available to break and subsequently form other chemical bonds, is called the **free energy** of that molecule. In a more general sense, free energy is defined as the energy available to do work in any system. In a molecule within a cell, where pressure and volume usually do not change, the free energy is denoted by the symbol G (for “Gibbs’ free energy,” which limits the system being considered to the cell). G is equal to the energy contained in a molecule’s chemical bonds (called *enthalpy* and designated H) minus the energy unavailable because of disorder (called *entropy* and given the symbol S) times the absolute temperature, T , in degrees Kelvin ($K = ^\circ C + 273$):

$$G = H - TS$$

Chemical reactions break some bonds in the reactants and form new bonds in the products. Consequently, reactions can produce changes in free energy. When a chemical reaction occurs under conditions of constant temperature, pressure, and volume—as do most biological reactions—the change in free energy (ΔG) is simply:

$$\Delta G = \Delta H - T \Delta S$$

The change in free energy, or ΔG , is a fundamental property of chemical reactions.

In some reactions, the ΔG is positive. This means that the products of the reaction contain *more* free energy than the reactants; the bond energy (H) is higher or the disorder (S) in the system is lower. Such reactions do not proceed spontaneously because they require an input of energy. Any reaction that requires an input of energy is said to be **endergonic** (“inward energy”).

In other reactions, the ΔG is negative. The products of the reaction contain less free energy than the reactants; either the bond energy is lower or the disorder is higher, or both. Such reactions tend to proceed spontaneously. Any chemical reaction will tend to proceed spontaneously if the difference in disorder ($T \Delta S$) is *greater* than the difference in bond energies between reactants and products (ΔH). Note that spontaneous does not mean the same thing as instantaneous. A spontaneous reaction may proceed very slowly. These reactions release the excess free energy as heat and are thus said to be exergonic (“outward energy”). Figure 8.6 sums up these reactions.

Free energy is the energy available to do work. Within cells, the change in free energy (ΔG) is the difference in bond energies between reactants and products (ΔH), minus any change in the degree of disorder of the system ($T \Delta S$). Any reaction whose products contain less free energy than the reactants (ΔG is negative) will tend to proceed spontaneously.

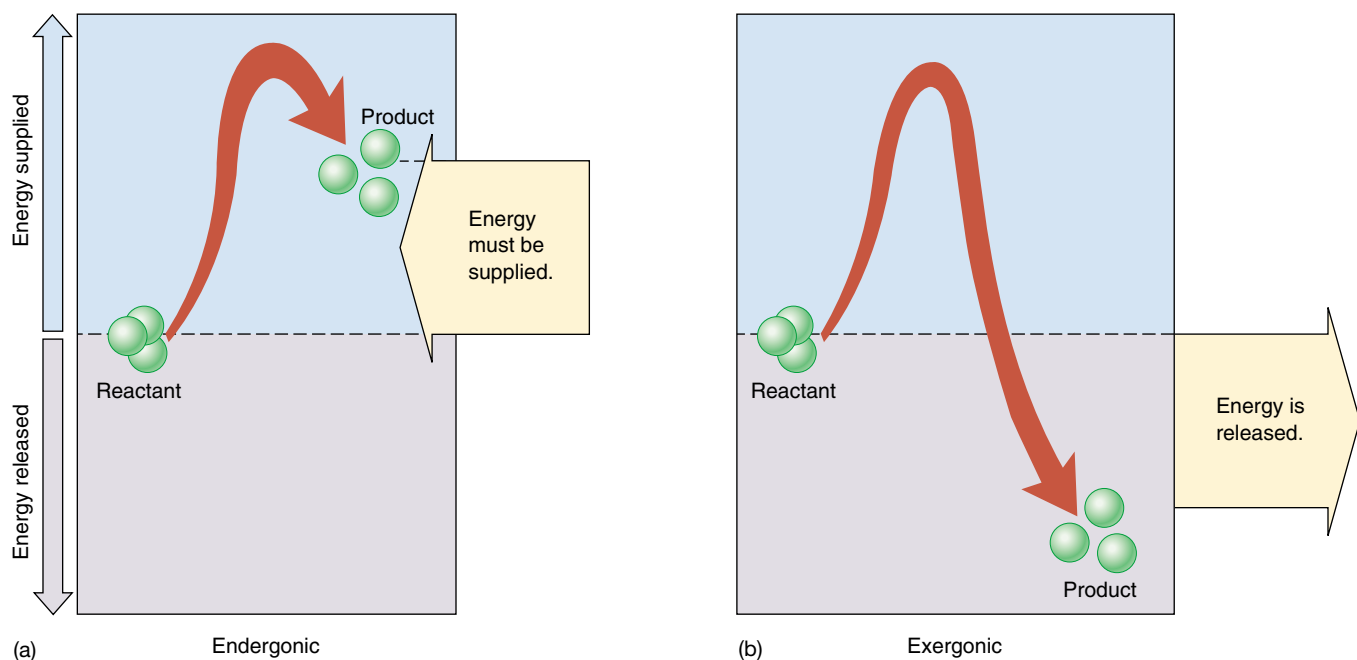


FIGURE 8.6

Energy in chemical reactions. (a) In an endergonic reaction, the products of the reaction contain more energy than the reactants, and the extra energy must be supplied for the reaction to proceed. (b) In an exergonic reaction, the products contain less energy than the reactants, and the excess energy is released.

Activation Energy

If all chemical reactions that release free energy tend to occur spontaneously, why haven't all such reactions already occurred? One reason they haven't is that most reactions require an input of energy to get started. Before it is possible to form new chemical bonds, even bonds that contain less energy, it is first necessary to break the existing bonds, and that takes energy. The extra energy required to destabilize existing chemical bonds and initiate a chemical reaction is called **activation energy** (figure 8.7a).

The rate of an exergonic reaction depends on the activation energy required for the reaction to begin. Reactions with larger activation energies tend to proceed more slowly because fewer molecules succeed in overcoming the initial energy hurdle. Activation energies are not constant, however. Stressing particular chemical bonds can make them easier to break. The process of influencing chemical bonds in a way that lowers the activation energy needed to initiate a reaction is called **catalysis**, and substances that accomplish this are known as catalysts (figure 8.7b).

Catalysts cannot violate the basic laws of thermodynamics; they cannot, for example, make an endergonic reaction proceed spontaneously. By reducing the activation energy, a catalyst accelerates both the forward and the reverse reactions by exactly the same amount. Hence, it does not alter the proportion of reactant ultimately converted into product.

To grasp this, imagine a bowling ball resting in a shallow depression on the side of a hill. Only a narrow rim of dirt below the ball prevents it from rolling down the hill. Now imagine digging away that rim of dirt. If you remove enough dirt from below the ball, it will start to roll down the hill—but removing dirt from below the ball will *never* cause the ball to roll UP the hill! Removing the lip of dirt simply allows the ball to move freely; gravity determines the direction it then travels. Lowering the resistance to the ball's movement will promote the movement dictated by its position on the hill.

Similarly, the direction in which a chemical reaction proceeds is determined solely by the difference in free energy. Like digging away the soil below the bowling ball on the hill, catalysts reduce the energy barrier preventing the reaction from proceeding. Catalysts don't favor endergonic reactions any more than digging makes the hypothetical bowling ball roll uphill. Only exergonic reactions can proceed spontaneously, and catalysts cannot change that. What catalysts *can* do is make a reaction proceed much faster.

The rate of a reaction depends on the activation energy necessary to initiate it. Catalysts reduce the activation energy and so increase the rates of reactions, although they do not change the final proportions of reactants and products.

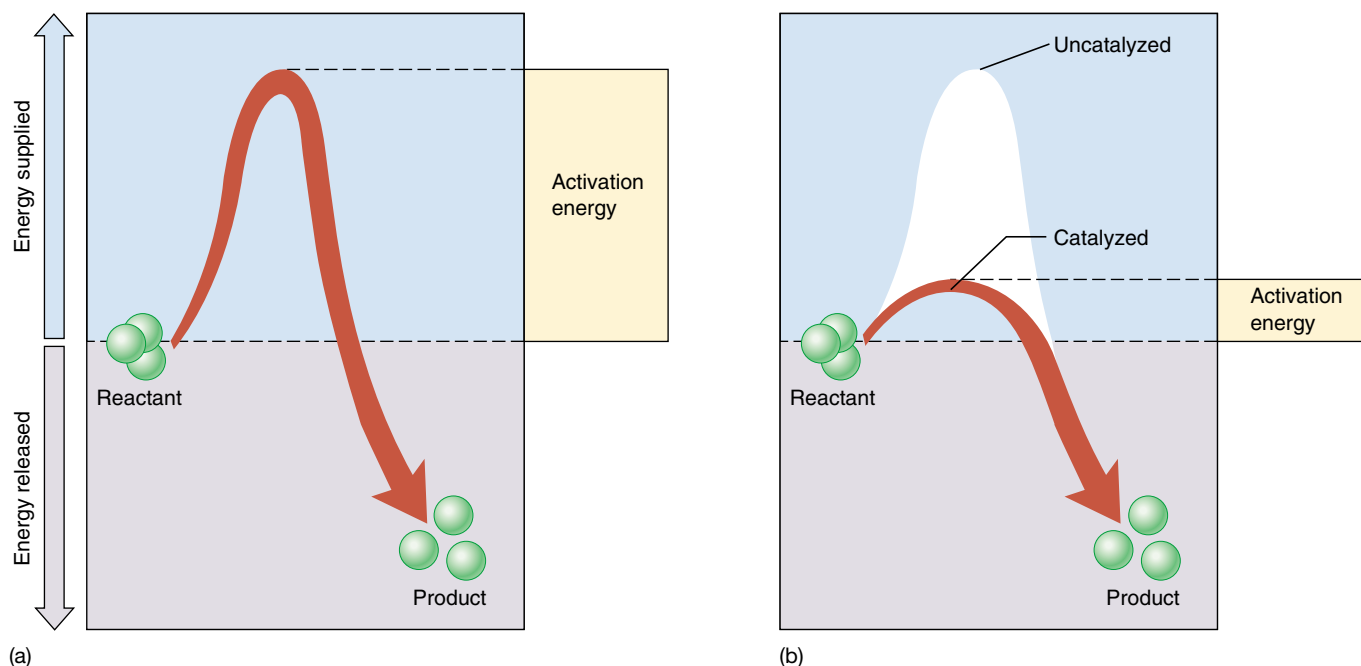


FIGURE 8.7

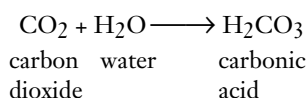
Activation energy and catalysis. (a) Exergonic reactions do not necessarily proceed rapidly because energy must be supplied to destabilize existing chemical bonds. This extra energy is the activation energy for the reaction. (b) Catalysts accelerate particular reactions by lowering the amount of activation energy required to initiate the reaction.

8.2 Enzymes are biological catalysts.

Enzymes

The chemical reactions within living organisms are regulated by controlling the points at which catalysis takes place. Life itself is, therefore, regulated by catalysts. The agents that carry out most of the catalysis in living organisms are proteins called **enzymes**. (There is increasing evidence that some types of biological catalysis are carried out by RNA molecules.) The unique three-dimensional shape of an enzyme enables it to stabilize a temporary association between **substrates**, the molecules that will undergo the reaction. By bringing two substrates together in the correct orientation, or by stressing particular chemical bonds of a substrate, an enzyme lowers the activation energy required for new bonds to form. The reaction thus proceeds much more quickly than it would without the enzyme. Because the enzyme itself is not changed or consumed in the reaction, only a small amount of an enzyme is needed, and it can be used over and over.

As an example of how an enzyme works, let's consider the reaction of carbon dioxide and water to form carbonic acid. This important enzyme-catalyzed reaction occurs in vertebrate red blood cells:



This reaction may proceed in either direction, but because it has a large activation energy, the reaction is very slow in the absence of an enzyme: perhaps 200 molecules of carbonic acid form in an hour in a cell. Reactions that proceed this slowly are of little use to a cell. Cells overcome this problem by employing an enzyme within their cytoplasm called *carbonic anhydrase* (enzyme names usually end in “-ase”). Under the same conditions, but in the presence of carbonic anhydrase, an estimated 600,000 molecules of carbonic acid form every *second*! Thus, the enzyme increases the reaction rate more than 10 million times.

Thousands of different kinds of enzymes are known, each catalyzing one or a few specific chemical reactions. By facilitating particular chemical reactions, the enzymes in a cell determine the course of metabolism—the collection of all chemical reactions—in that cell. Different types of cells contain different sets of enzymes, and this difference contributes to structural and functional variations among cell types. The chemical reactions taking place within a red blood cell differ from those that occur within a nerve cell, in part because the cytoplasm and membranes of red blood cells and nerve cells contain different arrays of enzymes.

Cells use proteins called enzymes as catalysts to lower activation energies.

Catalysis: A Closer Look at Carbonic Anhydrase

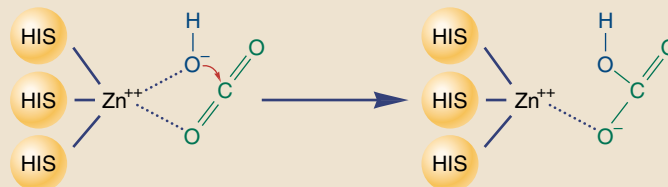
One of the most rapidly acting enzymes in the human body is carbonic anhydrase, which plays a key role in blood by converting dissolved CO_2 into carbonic acid, which dissociates into bicarbonate and hydrogen ions:



Fully 70% of the CO_2 transported by the blood is transported as bicarbonate ion. This reaction is exergonic, but its energy of activation is significant, so that little conversion to bicarbonate occurs spontaneously. In the presence of the enzyme carbonic anhydrase, however, the rate of the reaction accelerates by a factor of more than 10 million!

How does carbonic anhydrase catalyze this reaction so effectively? The active site

FIGURE 8.A



of the enzyme is a deep cleft traversing the enzyme, as if it had been cut with the blade of an ax. Deep within the cleft, some 1.5 nm from the surface, are located three histidines, their imidazole (nitrogen ring) groups all pointed at the same place in the center of the cleft. Together they hold a zinc ion firmly in position. This zinc ion will be the cutting blade of the catalytic process.

Here is how the zinc catalyzes the reaction. Immediately adjacent to the position of the zinc atom in the cleft are a group of amino acids that recognize and bind carbon dioxide. The zinc atom interacts with this carbon dioxide molecule, orienting it in the plane of the cleft. Meanwhile, water bound

to the zinc is rapidly converted to hydroxide ion. This hydroxide ion is now precisely positioned to attack the carbon dioxide. When it does so, HCO_3^- is formed—and the enzyme is unchanged (figure 8.A).

Carbonic anhydrase is an effective catalyst because it brings its two substrates into close proximity and optimizes their orientation for reaction. Other enzymes use other mechanisms. Many, for example, use charged amino acids to polarize substrates or electronegative amino acids to stress particular bonds. Whatever the details of the reaction, however, the precise positioning of substrates achieved by the particular shape of the enzyme always plays a key role.

How Enzymes Work

Most enzymes are globular proteins with one or more pockets or clefts on their surface called **active sites** (figure 8.8). Substrates bind to the enzyme at these active sites, forming an **enzyme-substrate complex**. For catalysis to occur within the complex, a substrate molecule must fit precisely into an active site. When that happens, amino acid side groups of the enzyme end up in close proximity to certain bonds of the substrate. These side groups interact chemically with the substrate, usually stressing or distorting a particular bond and consequently lowering the activation energy needed to break the bond. The substrate, now a product, then dissociates from the enzyme.

Proteins are not rigid. The binding of a substrate induces the enzyme to adjust its shape slightly, leading to a better *induced fit* between enzyme and substrate (figure 8.9). This interaction may also facilitate the binding of other substrates; in such cases, the substrate itself “activates” the enzyme to receive other substrates.

Enzymes typically catalyze only one or a few similar chemical reactions because they are specific in their choice of substrates. This specificity is due to the active site of the enzyme, which is shaped so that only a certain substrate molecule will fit into it.

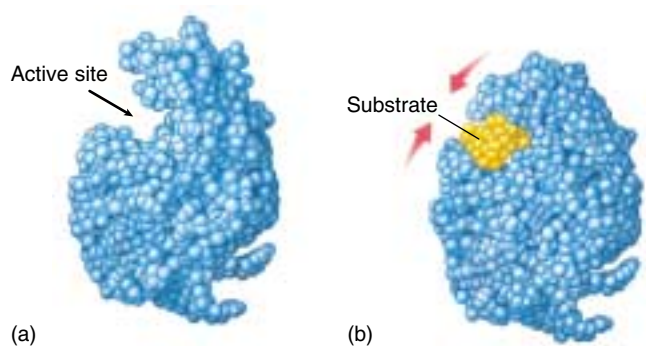


FIGURE 8.8

How the enzyme lysozyme works. (a) A groove runs through lysozyme that fits the shape of the polysaccharide (a chain of sugars) that makes up bacterial cell walls. (b) When such a chain of sugars, indicated in yellow, slides into the groove, its entry induces the protein to alter its shape slightly and embrace the substrate more intimately. This induced fit positions a glutamic acid residue in the protein next to the bond between two adjacent sugars, and the glutamic acid “steals” an electron from the bond, causing it to break.

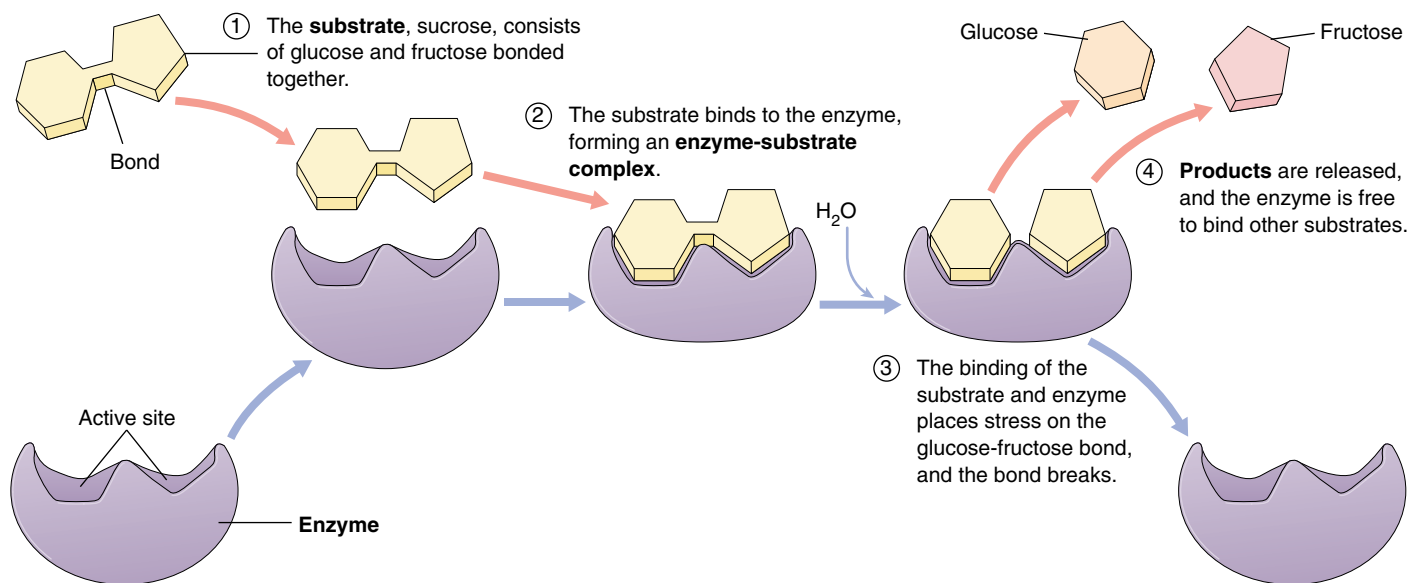


FIGURE 8.9

The catalytic cycle of an enzyme. Enzymes increase the speed with which chemical reactions occur, but they are not altered themselves as they do so. In the reaction illustrated here, the enzyme sucrase is splitting the sugar sucrose (present in most candy) into two simpler sugars: glucose and fructose. (1) First, the sucrose substrate binds to the active site of the enzyme, fitting into a depression in the enzyme surface. (2) The binding of sucrose to the active site forms an enzyme-substrate complex and induces the sucrase molecule to alter its shape, fitting more tightly around the sucrose. (3) Amino acid residues in the active site, now in close proximity to the bond between the glucose and fructose components of sucrose, break the bond. (4) The enzyme releases the resulting glucose and fructose fragments, the products of the reaction, and is then ready to bind another molecule of sucrose and run through the catalytic cycle once again. This cycle is often summarized by the equation: $E + S \leftrightarrow [ES] \leftrightarrow E + P$, where E = enzyme, S = substrate, ES = enzyme-substrate complex, and P = products.

Enzymes Take Many Forms

While many enzymes are suspended in the cytoplasm of cells, free to move about and not attached to any structure, other enzymes function as integral parts of cell structures and organelles.

Multienzyme Complexes

Often in cells the several enzymes catalyzing the different steps of a sequence of reactions are loosely associated with one another in non-covalently bonded assemblies called *multienzyme complexes*. The bacterial pyruvate dehydrogenase multienzyme complex seen in figure 8.10 contains enzymes that carry out three sequential reactions in oxidative metabolism. Each complex has multiple copies of each of the three enzymes—60 protein subunits in all. The many subunits work in concert, like a tiny factory.

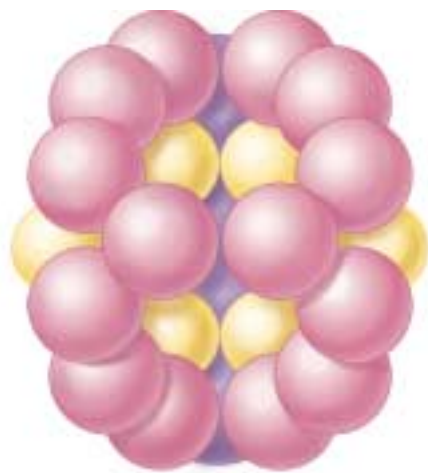
Multienzyme complexes offer significant advantages in catalytic efficiency:

1. The rate of any enzyme reaction is limited by the frequency with which the enzyme collides with its substrate. If a series of sequential reactions occurs within a multienzyme complex, the product of one reaction can be delivered to the next enzyme without releasing it to diffuse away.
2. Because the reacting substrate never leaves the complex during its passage through the series of reactions, the possibility of unwanted side reactions is eliminated.
3. All of the reactions that take place within the multienzyme complex can be controlled as a unit.

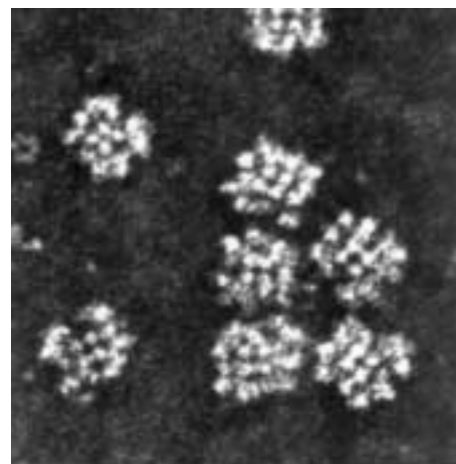
In addition to pyruvate dehydrogenase, which controls entry to the Krebs cycle, several other key processes in the cell are catalyzed by multienzyme complexes. One well-studied system is the fatty acid synthetase complex that catalyzes the synthesis of fatty acids from two-carbon precursors. There are seven different enzymes in this multienzyme complex, and the reaction intermediates remain associated with the complex for the entire series of reactions.

Not All Biological Catalysts Are Proteins

Until a few years ago, most biology textbooks contained statements such as “Enzymes are the catalysts of biological systems.” We can no longer make that statement without qualification. As discussed in chapter 4, Tom Cech and his colleagues at the University of Colorado reported in 1981 that certain reactions involving RNA molecules appear to



(a)



(b)

FIGURE 8.10

The enzyme pyruvate dehydrogenase. The enzyme (model, *a*) that carries out the oxidation of pyruvate is one of the most complex enzymes known—it has 60 subunits, many of which can be seen in the electron micrograph (*b*) (200,000 $\times$).

be catalyzed in cells by RNA itself, rather than by enzymes. This initial observation has been corroborated by additional examples of RNA catalysis in the last few years. Like enzymes, these RNA catalysts, which are loosely called “ribozymes,” greatly accelerate the rate of particular biochemical reactions and show extraordinary specificity with respect to the substrates on which they act.

There appear to be at least two sorts of ribozymes. Those that carry out *intramolecular* catalysis have folded structures and catalyze reactions on themselves. Those that carry out *intermolecular* catalysis act on other molecules without themselves being changed in the process. Many important cellular reactions involve small RNA molecules, including reactions that chip out unnecessary sections from RNA copies of genes, that prepare ribosomes for protein synthesis, and that facilitate the replication of DNA within mitochondria. In all of these cases, the possibility of RNA catalysis is being actively investigated. It seems likely, particularly in the complex process of photosynthesis, that both enzymes and RNA play important catalytic roles.

The ability of RNA, an informational molecule, to act as a catalyst has stirred great excitement among biologists, as it appears to provide a potential answer to the “chicken-and-egg” riddle posed by the spontaneous origin of life hypothesis discussed in chapter 3. Which came first, the protein or the nucleic acid? It now seems at least possible that RNA may have evolved first and catalyzed the formation of the first proteins.

Not all biological catalysts float free in the cytoplasm. Some are part of other structures, and some are not even proteins.

Factors Affecting Enzyme Activity

The rate of an enzyme-catalyzed reaction is affected by the concentration of substrate, and of the enzyme that works on it. In addition, any chemical or physical factor that alters the enzyme's three-dimensional shape—such as temperature, pH, salt concentration, and the binding of specific regulatory molecules—can affect the enzyme's ability to catalyze the reaction.

Temperature

Increasing the temperature of an uncatalyzed reaction will increase its rate because the additional heat represents an increase in random molecular movement. The rate of an enzyme-catalyzed reaction also increases with temperature, but only up to a point called the *temperature optimum* (figure 8.11a). Below this temperature, the hydrogen bonds and hydrophobic interactions that determine the enzyme's shape are not flexible enough to permit the induced fit that is optimum for catalysis. Above the temperature optimum, these forces are too weak to maintain the enzyme's shape against the increased random movement of the atoms in the enzyme. At these higher temperatures, the enzyme denatures, as we described in chapter 3. Most human enzymes have temperature optima between 35°C and 40°C, a range that includes normal body temperature. Bacteria that live in hot springs have more stable enzymes (that is, enzymes held together more strongly), so the temperature optima for those enzymes can be 70°C or higher.

pH

Ionic interactions between oppositely charged amino acid residues, such as glutamic acid (–) and lysine (+), also hold enzymes together. These interactions are sensitive to the hydrogen ion concentration of the fluid the enzyme is dissolved in, because changing that concentration shifts the balance between positively and negatively charged amino acid residues. For this reason, most enzymes have a **pH optimum** that usually ranges from pH 6 to 8. Those enzymes able to function in very acid environments are proteins that maintain their three-dimensional shape even in the presence of high levels of hydrogen ion. The enzyme pepsin, for example, digests proteins in the stomach at pH 2, a very acidic level (figure 8.11b).

Inhibitors and Activators

Enzyme activity is sensitive to the presence of specific substances that bind to the enzyme and cause changes in its shape. Through these substances, a cell is able to regulate which enzymes are active and which are inactive at a particular time. This allows the cell to increase its efficiency and to control changes in its characteristics during develop-

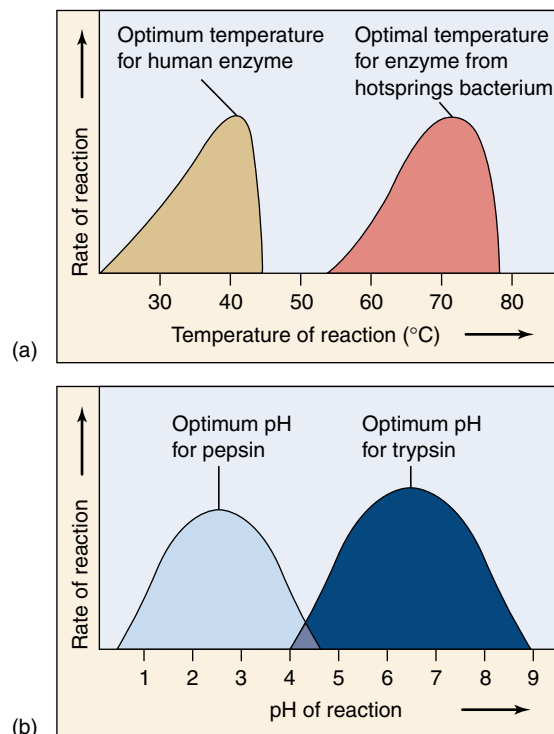


FIGURE 8.11

Enzymes are sensitive to their environment. The activity of an enzyme is influenced by both (a) temperature and (b) pH. Most human enzymes, such as the protein-degrading enzyme trypsin, work best at temperatures of about 40°C and within a pH range of 6 to 8.

ment. A substance that binds to an enzyme and *decreases* its activity is called an **inhibitor**. Very often, the end product of a biochemical pathway acts as an inhibitor of an early reaction in the pathway, a process called *feedback inhibition* (to be discussed later).

Enzyme inhibition occurs in two ways: **competitive inhibitors** compete with the substrate for the same binding site, displacing a percentage of substrate molecules from the enzymes; **noncompetitive inhibitors** bind to the enzyme in a location other than the active site, changing the shape of the enzyme and making it unable to bind to the substrate (figure 8.12). Most noncompetitive inhibitors bind to a specific portion of the enzyme called an **allosteric site** (Greek *allos*, “other” + *steros*, “form”). These sites serve as chemical on/off switches; the binding of a substance to the site can switch the enzyme between its active and inactive configurations. A substance that binds to an allosteric site and reduces enzyme activity is called an **allosteric inhibitor** (figure 8.12b). Alternatively, **activators** bind to allosteric sites and keep the enzymes in their active configurations, thereby *increasing* enzyme activity.

Enzyme Cofactors

Enzyme function is often assisted by additional chemical components known as **cofactors**. For example, the active sites of many enzymes contain metal ions that help draw electrons away from substrate molecules. The enzyme carboxypeptidase digests proteins by employing a zinc ion (Zn^{++}) in its active site to remove electrons from the bonds joining amino acids. Other elements, such as molybdenum and manganese, are also used as cofactors. Like zinc, these substances are required in the diet in small amounts. When the cofactor is a nonprotein organic molecule, it is called a **coenzyme**. Many vitamins are parts of coenzymes.

In numerous oxidation-reduction reactions that are catalyzed by enzymes, the electrons pass in pairs from the active site of the enzyme to a coenzyme that serves as the electron acceptor. The coenzyme then transfers the electrons to a different enzyme, which releases them (and the energy they bear) to the substrates in another reaction. Often, the electrons pair with protons (H^+) as hydrogen atoms. In this way, coenzymes shuttle energy in the form of hydrogen atoms from one enzyme to another in a cell.

One of the most important coenzymes is the hydrogen acceptor **nicotinamide adenine dinucleotide (NAD^+)** (figure 8.13). The NAD^+ molecule is composed of two nucleotides bound together. As you may recall from chapter 3, a nucleotide is a five-carbon sugar with one or more phosphate groups attached to one end and an organic base attached to the other end. The two nucleotides that make up NAD^+ , nicotinamide monophosphate (NMP) and adenine monophosphate (AMP), are joined head-to-head by their phosphate groups. The two nucleotides serve different functions in the NAD^+ molecule: AMP acts as the core, providing a shape recognized by many enzymes; NMP is the active part of the molecule, contributing a site that is readily reduced (that is, easily accepts electrons).

When NAD^+ acquires an electron and a hydrogen atom (actually, two electrons and a proton) from the active site of an enzyme, it is reduced to NADH . The NADH molecule now carries the two energetic electrons and the proton. The oxidation of energy-containing molecules, which provides energy to cells, involves stripping electrons from those molecules and donating them to NAD^+ . As we'll see, much of the energy of NADH is transferred to another molecule.

Enzymes have an optimum temperature and pH, at which the enzyme functions most effectively. Inhibitors decrease enzyme activity, while activators increase it. The activity of enzymes is often facilitated by cofactors, which can be metal ions or other substances. Cofactors that are nonprotein organic molecules are called **coenzymes**.

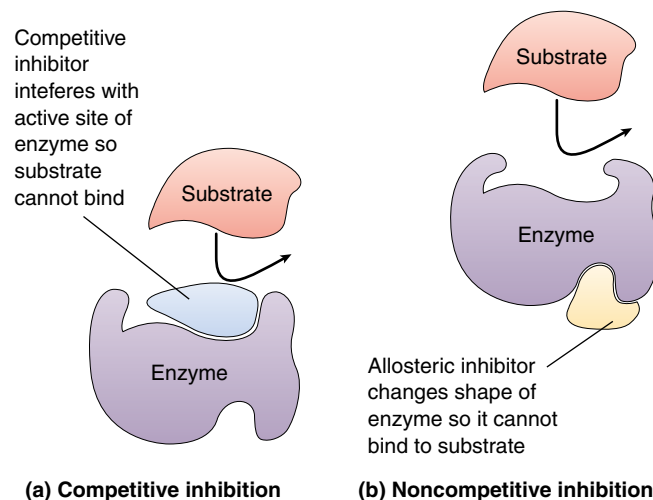


FIGURE 8.12

How enzymes can be inhibited. (a) In competitive inhibition, the inhibitor interferes with the active site of the enzyme. (b) In noncompetitive inhibition, the inhibitor binds to the enzyme at a place away from the active site, effecting a conformational change in the enzyme so that it can no longer bind to its substrate.

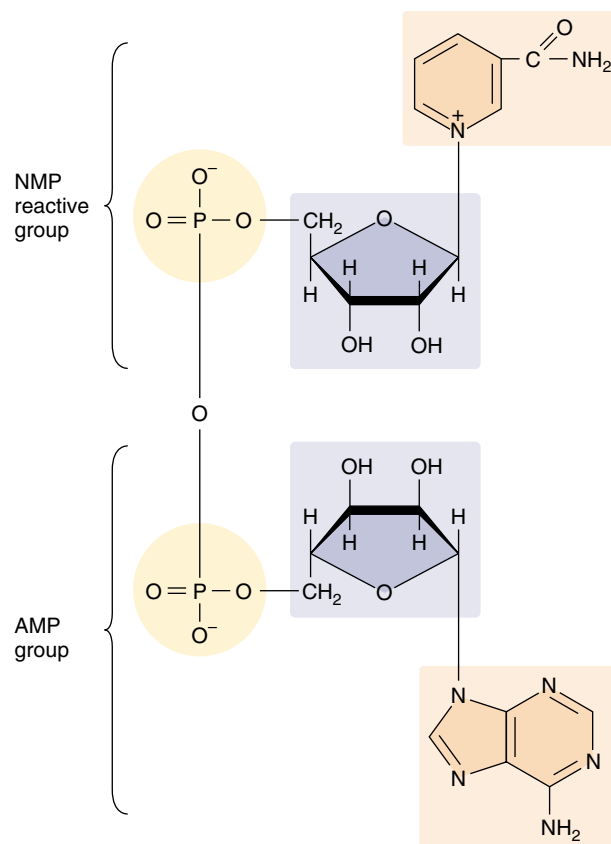


FIGURE 8.13

The chemical structure of nicotinamide adenine dinucleotide (NAD^+). This key cofactor is composed of two nucleotides, NMP and AMP, attached head-to-head.

8.3 ATP is the energy currency of life.

What Is ATP?

The chief energy currency all cells use is a molecule called **adenosine triphosphate (ATP)**. Cells use their supply of ATP to power almost every energy-requiring process they carry out, from making sugars, to supplying activation energy for chemical reactions, to actively transporting substances across membranes, to moving through their environment and growing.

Structure of the ATP Molecule

Each ATP molecule is a nucleotide composed of three smaller components (figure 8.14). The first component is a five-carbon sugar, ribose, which serves as the backbone to which the other two subunits are attached. The second component is adenine, an organic molecule composed of two carbon-nitrogen rings. Each of the nitrogen atoms in the ring has an unshared pair of electrons and weakly attracts hydrogen ions. Adenine, therefore, acts chemically as a base and is usually referred to as a nitrogenous base (it is one of the four nitrogenous bases found in DNA and RNA). The third component of ATP is a triphosphate group (a chain of three phosphates).

How ATP Stores Energy

The key to how ATP stores energy lies in its triphosphate group. Phosphate groups are highly negatively charged, so they repel one another strongly. Because of the electrostatic repulsion between the charged phosphate groups, the two covalent bonds joining the phosphates are unstable. The ATP molecule is often referred to as a “coiled spring,” the phosphates straining away from one another.

The unstable bonds holding the phosphates together in the ATP molecule have a low activation energy and are easily broken. When they break, they can transfer a considerable amount of energy. In most reactions involving ATP, only the outermost high-energy phosphate bond is hydrolyzed, cleaving off the phosphate group on the end. When this happens, ATP becomes **adenosine diphosphate (ADP)**, and energy equal to 7.3 kcal/mole is released under standard conditions. The liberated phosphate group usually attaches temporarily to some intermediate molecule. When that molecule is dephosphorylated, the phosphate group is released as inorganic phosphate (P_i).

How ATP Powers Energy-Requiring Reactions

Cells use ATP to drive endergonic reactions. Such reactions do not proceed spontaneously, because their products possess more free energy than their reactants. However, if the cleavage of ATP's terminal high-energy bond releases

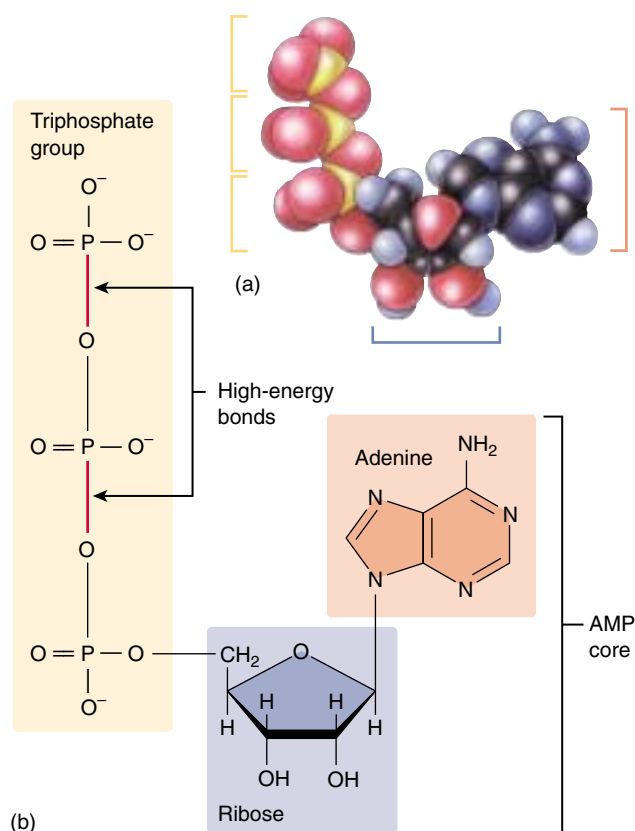


FIGURE 8.14

The ATP molecule. (a) The model and (b) structural diagram both show that like NAD^+ , ATP has a core of AMP. In ATP the reactive group added to the end of the AMP phosphate group is not another nucleotide but rather a chain of two additional phosphate groups. The bonds connecting these two phosphate groups to each other and to AMP are energy-storing bonds.

more energy than the other reaction consumes, the overall energy change of the two coupled reactions will be exergonic (energy releasing) and they will both proceed. Because almost all endergonic reactions require less energy than is released by the cleavage of ATP, ATP can provide most of the energy a cell needs.

The same feature that makes ATP an effective energy donor—the instability of its phosphate bonds—precludes it from being a good long-term energy storage molecule. Fats and carbohydrates serve that function better. Most cells do not maintain large stockpiles of ATP. Instead, they typically have only a few seconds' supply of ATP at any given time, and they continually produce more from ADP and P_i .

The instability of its phosphate bonds makes ATP an excellent energy donor.

8.4 Metabolism is the chemical life of a cell.

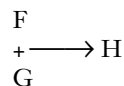
Biochemical Pathways: The Organizational Units of Metabolism

This living chemistry, the total of all chemical reactions carried out by an organism, is called **metabolism** (Greek *metabole*, “change”). Those reactions that expend energy to make or transform chemical bonds are called *anabolic* reactions, or **anabolism**. Reactions that harvest energy when chemical bonds are broken are called *catabolic* reactions, or **catabolism**.

Organisms contain thousands of different kinds of enzymes that catalyze a bewildering variety of reactions. Many of these reactions in a cell occur in sequences called **biochemical pathways**. In such pathways, the product of one reaction becomes the substrate for the next (figure 8.15). Biochemical pathways are the organizational units of metabolism, the elements an organism controls to achieve coherent metabolic activity. Most sequential enzyme steps in biochemical pathways take place in specific compartments of the cell; the steps of the citric acid cycle (chapter 9), for example, occur inside mitochondria. By determining where many of the enzymes that catalyze these steps are located, we can “map out” a model of metabolic processes in the cell.

How Biochemical Pathways Evolved

In the earliest cells, the first biochemical processes probably involved energy-rich molecules scavenged from the environment. Most of the molecules necessary for these processes are thought to have existed in the “organic soup” of the early oceans. The first catalyzed reactions are thought to have been simple, one-step reactions that brought these molecules together in various combinations. Eventually, the energy-rich molecules became depleted in the external environment, and only organisms that had evolved some means of making those molecules from other substances in the environment could survive. Thus, a hypothetical reaction,



where two energy-rich molecules (F and G) react to produce compound H and release energy, became more complex when the supply of F in the environment ran out. A new reaction was added in which the depleted molecule, F, is made from another molecule, E, which was also present in the environment:

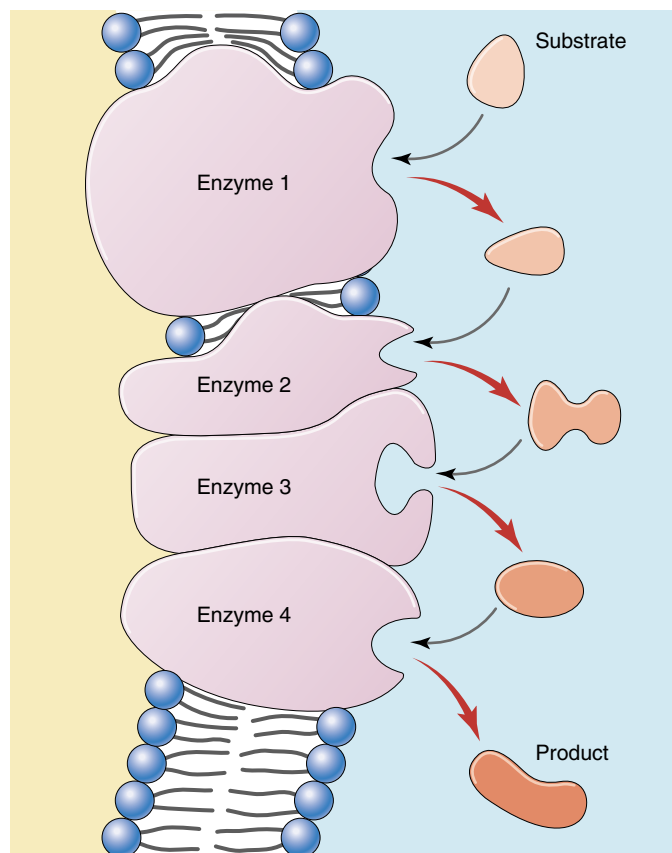
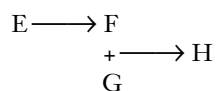
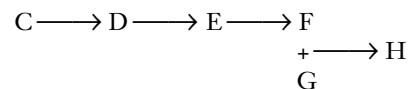


FIGURE 8.15

A biochemical pathway. The original substrate is acted on by enzyme 1, changing the substrate to a new form recognized by enzyme 2. Each enzyme in the pathway acts on the product of the previous stage.

When the supply of E in turn became depleted, organisms that were able to make it from some other available precursor, D, survived. When D became depleted, those organisms in turn were replaced by ones able to synthesize D from another molecule, C:



This hypothetical biochemical pathway would have evolved slowly through time, with the final reactions in the pathway evolving first and earlier reactions evolving later. Looking at the pathway now, we would say that the organism, starting with compound C, is able to synthesize H by means of a series of steps. This is how the biochemical pathways within organisms are thought to have evolved—not all at once, but one step at a time, backward.

How Biochemical Pathways Are Regulated

For a biochemical pathway to operate efficiently, its activity must be coordinated and regulated by the cell. Not only is it unnecessary to synthesize a compound when plenty is already present, doing so would waste energy and raw materials that could be put to use elsewhere. It is, therefore, advantageous for a cell to temporarily shut down biochemical pathways when their products are not needed.

The regulation of simple biochemical pathways often depends on an elegant feedback mechanism: the end product of the pathway binds to an allosteric site on the enzyme that catalyzes the first reaction in the pathway. In the hypothetical pathway we just described, the enzyme catalyzing the reaction $C \longrightarrow D$ would possess an allosteric site for H, the end product of the pathway. As the pathway churned out its product and the amount of H in the cell increased, it would become increasingly likely that one of the H molecules would encounter the allosteric site on the $C \longrightarrow D$ enzyme. If the product H functioned as an allosteric inhibitor of the enzyme, its binding to the enzyme would essentially shut down the reaction $C \longrightarrow D$. Shutting down this reaction, the first reaction in the pathway, effectively shuts down the whole pathway. Hence, as the cell produces increasing quantities of the product H, it automatically inhibits its ability to produce more. This mode of regulation is called **feedback inhibition** (figure 8.16).

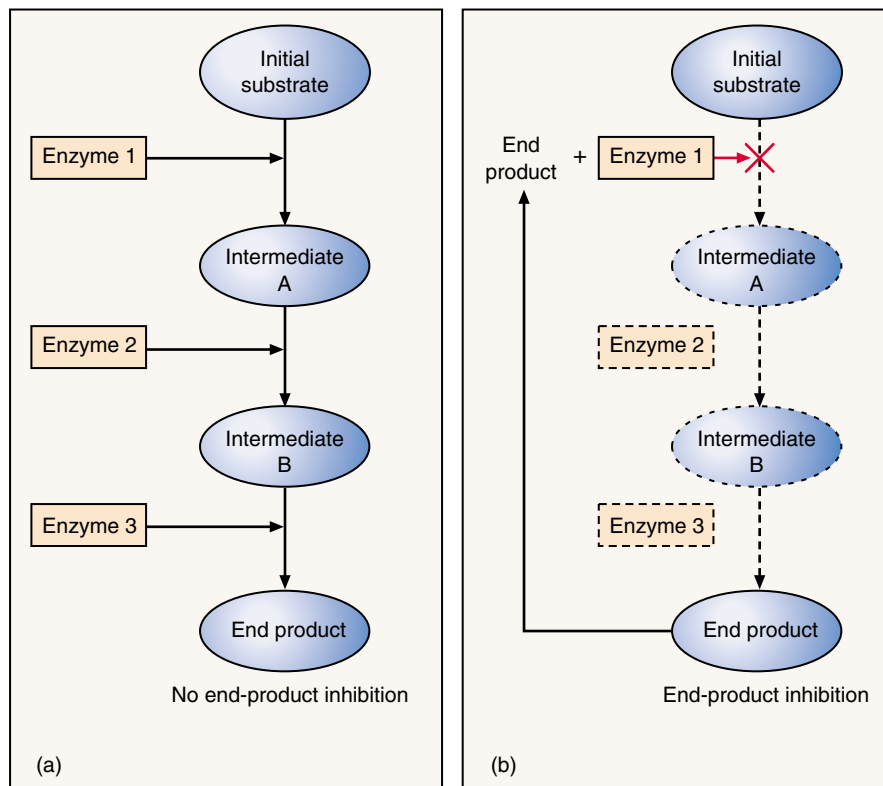


FIGURE 8.16

Feedback inhibition. (a) A biochemical pathway with no feedback inhibition. (b) A biochemical pathway in which the final end product becomes the allosteric effector for the first enzyme in the pathway. In other words, the formation of the pathway's final end product stops the pathway.

A biochemical pathway is an organized series of reactions, often regulated as a unit.

A Vocabulary of Metabolism

activation energy The energy required to destabilize chemical bonds and to initiate a chemical reaction.

catalysis Acceleration of the rate of a chemical reaction by lowering the activation energy.

coenzyme A nonprotein organic molecule that plays an accessory role in enzyme-catalyzed reactions, often by acting as a donor or acceptor of electrons. NAD^+ is a coenzyme.

endergonic reaction A chemical reaction to which energy from an outside source must be added before the reaction proceeds; the opposite of an exergonic reaction.

entropy A measure of the randomness or disorder of a system. In cells, it is a measure of how much energy has become so dispersed (usually as evenly distributed heat) that it is no longer available to do work.

exergonic reaction. An energy-yielding chemical reaction. Exergonic reactions tend to proceed spontaneously, although activation energy is required to initiate them.

free energy Energy available to do work.

kilocalorie 1000 calories. A calorie is the heat required to raise the temperature of 1 gram of water by 1°C .

metabolism The sum of all chemical processes occurring within a living cell or organism.

oxidation The loss of an electron by an atom or molecule. It occurs simultaneously with reduction of some other atom or molecule because an electron that is lost by one is gained by another.

reduction The gain of an electron by an atom or molecule. Oxidation-reduction reactions are an important means of energy transfer within living systems.

substrate A molecule on which an enzyme acts; the initial reactant in an enzyme-catalyzed reaction.

The Evolution of Metabolism

Metabolism has changed a great deal as life on earth has evolved. This has been particularly true of the reactions organisms use to capture energy from the sun to build organic molecules (anabolism), and then break down organic molecules to obtain energy (catabolism). These processes, the subject of the next two chapters, evolved in concert with each other.

Degradation

The most primitive forms of life are thought to have obtained chemical energy by degrading, or breaking down, organic molecules that were abiotically produced.

The first major event in the evolution of metabolism was the origin of the ability to harness chemical bond energy. At an early stage, organisms began to store this energy in the bonds of ATP, an energy carrier used by all organisms today.

Glycolysis

The second major event in the evolution of metabolism was glycolysis, the initial breakdown of glucose. As proteins evolved diverse catalytic functions, it became possible to capture a larger fraction of the chemical bond energy in organic molecules by breaking chemical bonds in a series of steps. For example, the progressive breakdown of the six-carbon sugar glucose into three-carbon molecules is performed in a series of 10 steps that results in the net production of two ATP molecules. The energy for the synthesis of ATP is obtained by breaking chemical bonds and forming new ones with less bond energy, the energy difference being channeled into ATP production. This biochemical pathway is called glycolysis.

Glycolysis undoubtedly evolved early in the history of life on earth, since this biochemical pathway has been retained by all living organisms. It is a chemical process that does not appear to have changed for well over 3 billion years.

Anaerobic Photosynthesis

The third major event in the evolution of metabolism was anaerobic photosynthesis. Early in the history of life, some organisms evolved a different way of generating ATP, called photosynthesis. Instead of obtaining energy for ATP synthesis by reshuffling chemical bonds, as in glycolysis, these organisms developed the ability to use light to pump protons out of their cells, and to use the resulting proton gradient to power the production of ATP, a process called chemiosmosis.

Photosynthesis evolved in the absence of oxygen and works well without it. Dissolved H_2S , present in the oceans beneath an atmosphere free of oxygen gas, served as a ready source of hydrogen atoms for building organic molecules. Free sulfur was produced as a by-product of this reaction.

Nitrogen Fixation

Nitrogen fixation was the fourth major step in the evolution of metabolism. Proteins and nucleic acids cannot be synthesized from the products of photosynthesis because both of these biologically critical molecules contain nitrogen. Obtaining nitrogen atoms from N_2 gas, a process called *nitrogen fixation*, requires the breaking of an $\text{N}\equiv\text{N}$ triple bond. This important reaction evolved in the hydrogen-rich atmosphere of the early earth, an atmosphere in which no oxygen was present. Oxygen acts as a poison to nitrogen fixation, which today occurs only in oxygen-free environments, or in oxygen-free compartments within certain bacteria.

Oxygen-Forming Photosynthesis

The substitution of H_2O for H_2S in photosynthesis was the fifth major event in the history of metabolism. Oxygen-forming photosynthesis employs H_2O rather than H_2S as a source of hydrogen atoms and their associated electrons. Because it garners its hydrogen atoms from reduced oxygen rather than from reduced sulfur, it generates oxygen gas rather than free sulfur.

More than 2 billion years ago, small cells capable of carrying out this oxygen-forming photosynthesis, such as cyanobacteria, became the dominant forms of life on earth. Oxygen gas began to accumulate in the atmosphere. This was the beginning of a great transition that changed conditions on earth permanently. Our atmosphere is now 20.9% oxygen, every molecule of which is derived from an oxygen-forming photosynthetic reaction.

Aerobic Respiration

Aerobic respiration is the sixth and final event in the history of metabolism. This cellular process harvests energy by stripping energetic electrons from organic molecules. Aerobic respiration employs the same kind of proton pumps as photosynthesis, and is thought to have evolved as a modification of the basic photosynthetic machinery. However, the hydrogens and their associated electrons are not obtained from H_2S or H_2O , as in photosynthesis, but rather from the breakdown of organic molecules.

Biologists think that the ability to carry out photosynthesis without H_2S first evolved among purple nonsulfur bacteria, which obtain their hydrogens from organic compounds instead. It was perhaps inevitable that among the descendants of these respiring photosynthetic bacteria, some would eventually do without photosynthesis entirely, subsisting only on the energy and hydrogens derived from the breakdown of organic molecules. The mitochondria within all eukaryotic cells are thought to be their descendants.

Six major innovations highlight the evolution of metabolism as we know it today.



Summary

Questions

Media Resources

8.1 The laws of thermodynamics describe how energy changes.

- Energy is the capacity to bring about change, to provide motion against a force, or to do work.
- Kinetic energy is actively engaged in doing work, while potential energy has the capacity to do so.
- An oxidation-reduction (redox) reaction is one in which an electron is taken from one atom or molecule (oxidation) and donated to another (reduction).
- The First Law of Thermodynamics states that the amount of energy in the universe is constant; energy is neither lost nor created.
- The Second Law of Thermodynamics states that disorder in the universe (entropy) tends to increase.
- Any chemical reaction whose products contain less free energy than the original reactants can proceed spontaneously. However, the difference in free energy does not determine the rate of the reaction.
- The rate of a reaction depends on the amount of activation energy required to break existing bonds.
- Catalysis is the process of lowering activation energies by stressing chemical bonds.

1. What is the difference between anabolism and catabolism?
2. Define oxidation and reduction. Why must these two reactions always occur in concert?
3. State the First and Second Laws of Thermodynamics.
4. What is heat? What is entropy? What is free energy?
5. What is the difference between an exergonic and an endergonic reaction? Which type of reaction tends to proceed spontaneously?
6. Define activation energy. How does a catalyst affect the final proportion of reactant converted into product?



- Energy Conversion
- Catalysis



- Thermodynamics
- Coupled Reactions

8.2 Enzymes are biological catalysts.

- Enzymes are the major catalysts of cells; they affect the rate of a reaction but not the ultimate balance between reactants and products.
- Cells contain many different enzymes, each of which catalyzes a specific reaction.
- The specificity of an enzyme is due to its active site, which fits only one or a few types of substrate molecules.

7. How are the rates of enzyme-catalyzed reactions affected by temperature? What is the molecular basis for the effect on reaction rate?
8. What is the difference between the active site and an allosteric site on an enzyme?



- Exploration: Thermodynamics
- Exploration: Kinetics



- Enzymes

8.3 ATP is the energy currency of life.

- Cells obtain energy from photosynthesis and the oxidation of organic molecules and use it to manufacture ATP from ADP and phosphate.
- The energy stored in ATP is then used to drive endergonic reactions.

9. What part of the ATP molecule contains the bond that is employed to provide energy for most of the endergonic reactions in cells?



- ATP

8.4 Metabolism is the chemical life of a cell.

- Generally, the final reactions of a biochemical pathway evolved first; preceding reactions in the pathway were added later, one step at a time.

10. What is a biochemical pathway? How does feedback inhibition regulate the activity of a biochemical pathway?



- Feedback Inhibition

9

How Cells Harvest Energy

Concept Outline

9.1 Cells harvest the energy in chemical bonds.

Using Chemical Energy to Drive Metabolism. The energy in C—H, C—O, and other chemical bonds can be captured and used to fuel the synthesis of ATP.

9.2 Cellular respiration oxidizes food molecules.

An Overview of Glucose Catabolism. The chemical energy in sugar is harvested by both substrate-level phosphorylation and by aerobic respiration.

Stage One: Glycolysis. The 10 reactions of glycolysis capture energy from glucose by reshuffling the bonds.

Stage Two: The Oxidation of Pyruvate. Pyruvate, the product of glycolysis, is oxidized to acetyl-CoA.

Stage Three: The Krebs Cycle. In a series of reactions, electrons are stripped from acetyl-CoA.

Harvesting Energy by Extracting Electrons. The respiration of glucose is a series of oxidation-reduction reactions which involve stripping electrons from glucose and using the energy of these electrons to power the synthesis of ATP.

Stage Four: The Electron Transport Chain. The electrons harvested from glucose pass through a chain of membrane proteins that use the energy to pump protons, driving the synthesis of ATP.

Summarizing Aerobic Respiration. The oxidation of glucose by aerobic respiration in eukaryotes produces up to three dozen ATP molecules, over half the energy in the chemical bonds of glucose.

Regulating Aerobic Respiration. High levels of ATP tend to shut down cellular respiration by feedback-inhibiting key reactions.

9.3 Catabolism of proteins and fats can yield considerable energy.

Glucose Is Not the Only Source of Energy. Proteins and fats are dismantled and the products fed into cellular respiration.

9.4 Cells can metabolize food without oxygen.

Fermentation. Fermentation allows continued metabolism in the absence of oxygen by donating the electrons harvested in glycolysis to organic molecules.



FIGURE 9.1
Harvesting chemical energy. Organisms such as these harvest mice depend on the energy stored in the chemical bonds of the food they eat to power their life processes.

Life is driven by energy. All the activities organisms carry out—the swimming of bacteria, the purring of a cat, your reading of these words—use energy. In this chapter, we will discuss the processes all cells use to derive chemical energy from organic molecules and to convert that energy to ATP. We will consider photosynthesis, which uses light energy rather than chemical energy, in detail in chapter 10. We examine the conversion of chemical energy to ATP first because all organisms, both photosynthesizers and the organisms that feed on them (like the field mice in figure 9.1), are capable of harvesting energy from chemical bonds. As you will see, though, this process and photosynthesis have much in common.

9.1 Cells harvest the energy in chemical bonds.

Using Chemical Energy to Drive Metabolism

Plants, algae, and some bacteria harvest the energy of sunlight through photosynthesis, converting radiant energy into chemical energy. These organisms, along with a few others that use chemical energy in a similar way, are called **autotrophs** (“self-feeders”). All other organisms live on the energy autotrophs produce and are called **heterotrophs** (“fed by others”). At least 95% of the kinds of organisms on earth—all animals and fungi, and most protists and bacteria—are heterotrophs.

Where is the chemical energy in food, and how do heterotrophs harvest it to carry out the many tasks of living (figure 9.2)? Most foods contain a variety of carbohydrates, proteins, and fats, all rich in energy-laden chemical bonds. Carbohydrates and fats, for example, possess many carbon-hydrogen (C—H), as well as carbon-oxygen (C—O) bonds. The job of extracting energy from this complex organic mixture is tackled in stages. First, enzymes break the large molecules down into smaller ones, a process called **digestion**. Then, other enzymes dismantle these fragments a little at a time, harvesting energy from C—H and other chemical bonds at each stage. This process is called **catabolism**.

While you obtain energy from many of the constituents of food, it is traditional to focus first on the catabolism of carbohydrates. We will follow the six-carbon sugar, glucose ($C_6H_{12}O_6$), as its chemical bonds are progressively harvested for energy. Later, we will come back and examine the catabolism of proteins and fats.

Cellular Respiration

The energy in a chemical bond can be visualized as potential energy borne by the electrons that make up the covalent bond. Cells harvest this energy by putting the electrons to work, often to produce ATP, the energy currency of the cell. Afterward, the energy-depleted electron (associated with a proton as a hydrogen atom) is donated to some other molecule. When oxygen gas (O_2) accepts the hydrogen atom, water forms, and the process is called **aerobic respiration**. When an inorganic molecule other than oxygen accepts the hydrogen, the process is called **anaerobic respiration**. When an organic molecule accepts the hydrogen atom, the process is called **fermentation**.

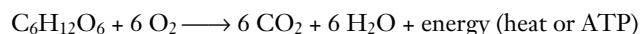
Chemically, there is little difference between the catabolism of carbohydrates in a cell and the burning of wood in a



FIGURE 9.2

Start every day with a good breakfast. The carbohydrates, proteins, and fats in this fish contain energy that the bear’s cells can use to power their daily activities.

fireplace. In both instances, the reactants are carbohydrates and oxygen, and the products are carbon dioxide, water, and energy:



The change in free energy in this reaction is -720 kilocalories (-3012 kilojoules) per mole of glucose under the conditions found within a cell (the traditional value of -686 kilocalories, or -2870 kJ, per mole refers to standard conditions—room temperature, one atmosphere of pressure, etc.). This change in free energy results largely from the breaking of the six C—H bonds in the glucose molecule. The negative sign indicates that the products possess *less* free energy than the reactants. The same amount of energy is released whether glucose is catabolized or burned, but when it is burned most of the energy is released as heat. This heat cannot be used to perform work in cells. The key to a cell’s ability to harvest useful energy from the catabolism of food molecules such as glucose is its conversion of a portion of the energy into a more useful form. Cells do this by using some of the energy to drive the production of ATP, a molecule that can power cellular activities.

The ATP Molecule

Adenosine triphosphate (ATP) is the energy currency of the cell, the molecule that transfers the energy captured during respiration to the many sites that use energy in the cell. How is ATP able to transfer energy so readily? Recall from chapter 8 that ATP is composed of a sugar (ribose) bound to an organic base (adenine) and a chain of three phosphate groups. As shown in figure 9.3, each phosphate group is negatively charged. Because like charges repel each other, the linked phosphate groups push against the bond that holds them together. Like a cocked mousetrap, the linked phosphates store the energy of their electrostatic repulsion. Transferring a phosphate group to another molecule relaxes the electrostatic spring of ATP, at the same time cocking the spring of the molecule that is phosphorylated. This molecule can then use the energy to undergo some change that requires work.

How Cells Use ATP

Cells use ATP to do most of those activities that require work. One of the most obvious is movement. Some bacteria swim about, propelling themselves through the water by rapidly spinning a long, tail-like flagellum, much as a ship moves by spinning a propeller. During your development as an embryo, many of your cells moved about, crawling over one another to reach new positions. Movement also occurs within cells. Tiny fibers within muscle cells pull against one another when muscles contract. Mitochondria pass a meter or more along the narrow nerve cells that connect your feet with your spine. Chromosomes are pulled by microtubules during cell division. All of these movements by cells require the expenditure of ATP energy.

A second major way cells use ATP is to *drive endergonic reactions*. Many of the synthetic activities of the cell are endergonic, because building molecules takes energy. The chemical bonds of the products of these reactions contain more energy, or are more organized, than the reactants. The reaction can't proceed until that extra energy is supplied to the reaction. It is ATP that provides this needed energy.

How ATP Drives Endergonic Reactions

How does ATP drive an endergonic reaction? The enzyme that catalyzes the endergonic reaction has *two* binding sites on its surface, one for the reactant and another for ATP. The ATP site splits the ATP molecule, liberating over 7 kcal (30 kJ) of chemical energy. This energy pushes the reactant at the second site “uphill,” driving the endergonic reaction. (In a similar way, you can make water in a swimming pool leap straight up in the air, despite the fact that gravity prevents water from rising spontaneously—just jump in the pool! The energy you add going in more than compensates for the force of gravity holding the water back.)

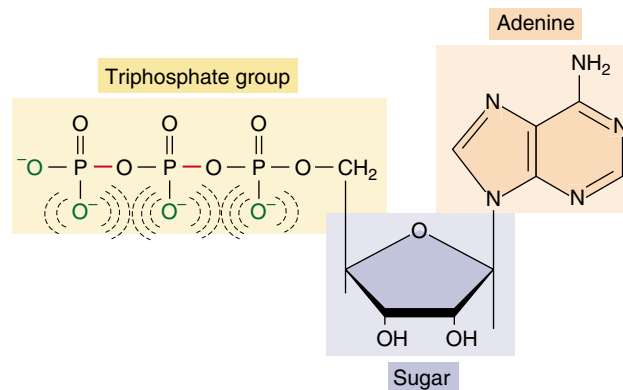


FIGURE 9.3

Structure of the ATP molecule. ATP is composed of an organic base and a chain of phosphates attached to opposite ends of a five-carbon sugar. Notice that the charged regions of the phosphate chain are close to one another. These like charges tend to repel one another, giving the bonds that hold them together a particularly high energy transfer potential.

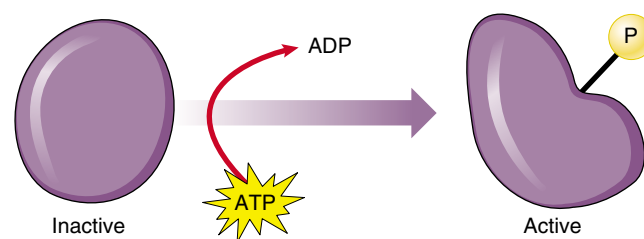


FIGURE 9.4

How ATP drives an endergonic reaction. In many cases, a phosphate group split from ATP activates a protein, catalyzing an endergonic process.

When the splitting of ATP molecules drives an energy-requiring reaction in a cell, the two parts of the reaction—ATP-splitting and endergonic—take place in concert. In some cases, the two parts both occur on the surface of the same enzyme; they are physically linked, or “coupled,” like two legs walking. In other cases, a high-energy phosphate from ATP attaches to the protein catalyzing the endergonic process, activating it (figure 9.4). Coupling energy-requiring reactions to the splitting of ATP in this way is one of the key tools cells use to manage energy.

The catabolism of glucose into carbon dioxide and water in living organisms releases about 720 kcal (3012 kJ) of energy per mole of glucose. This energy is captured in ATP, which stores the energy by linking charged phosphate groups near one another. When the phosphate bonds in ATP are hydrolyzed, energy is released and available to do work.

9.2 Cellular respiration oxidizes food molecules.

An Overview of Glucose Catabolism

Cells are able to make ATP from the catabolism of organic molecules in two different ways.

- 1. Substrate-level phosphorylation.** In the first, called **substrate-level phosphorylation**, ATP is formed by transferring a phosphate group directly to ADP from a phosphate-bearing intermediate (figure 9.5). During glycolysis, discussed below, the chemical bonds of glucose are shifted around in reactions that provide the energy required to form ATP.
- 2. Aerobic respiration.** In the second, called **aerobic respiration**, ATP forms as electrons are harvested, transferred along the electron transport chain, and eventually donated to oxygen gas. Eukaryotes produce the majority of their ATP from glucose in this way.

In most organisms, these two processes are combined. To harvest energy to make ATP from the sugar glucose in the presence of oxygen, the cell carries out a complex series of enzyme-catalyzed reactions that occur in four stages: the first stage captures energy by substrate-level phosphorylation through glycolysis, the following three stages carry out aerobic respiration by oxidizing the end product of glycolysis.

Glycolysis

Stage One: Glycolysis. The first stage of extracting energy from glucose is a 10-reaction biochemical pathway called glycolysis that produces ATP by substrate-level phosphorylation. The enzymes that catalyze the glycolytic reactions are in the cytoplasm of the cell, not bound to any membrane or organelle. Two ATP molecules are used up early in the pathway, and four ATP molecules are formed by substrate-level phosphorylation. This yields a net of two ATP molecules for each molecule of glucose catabolized. In addition, four electrons are harvested as NADH that can be used to form ATP by aerobic respiration. Still, the total yield of ATP is small. When the glycolytic process is completed, the two molecules of pyruvate that are formed still contain most of the energy the original glucose molecule held.

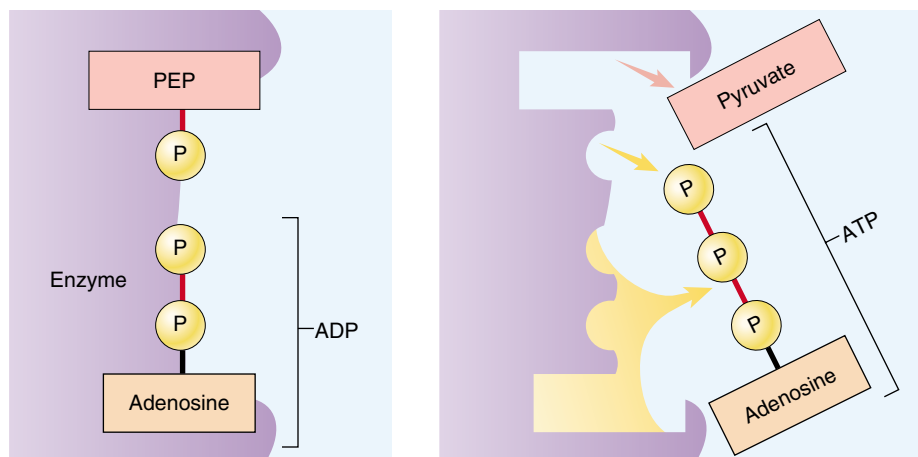


FIGURE 9.5

Substrate-level phosphorylation. Some molecules, such as phosphoenolpyruvate (PEP), possess a high-energy phosphate bond similar to the bonds in ATP. When PEP's phosphate group is transferred enzymatically to ADP, the energy in the bond is conserved and ATP is created.

Aerobic Respiration

Stage Two: Pyruvate Oxidation. In the second stage, pyruvate, the end product from glycolysis, is converted into carbon dioxide and a two-carbon molecule called acetyl-CoA. For each molecule of pyruvate converted, one molecule of NAD^+ is reduced to NADH.

Stage Three: The Krebs Cycle. The third stage introduces this acetyl-CoA into a cycle of nine reactions called the Krebs cycle, named after the British biochemist, Sir Hans Krebs, who discovered it. (The Krebs cycle is also called the citric acid cycle, for the citric acid, or citrate, formed in its first step, and less commonly, the tricarboxylic acid cycle, because citrate has three carboxyl groups.) In the Krebs cycle, two more ATP molecules are extracted by substrate-level phosphorylation, and a large number of electrons are removed by the reduction of NAD^+ to NADH.

Stage Four: Electron Transport Chain. In the fourth stage, the energetic electrons carried by NADH are employed to drive the synthesis of a large amount of ATP by the electron transport chain.

Pyruvate oxidation, the reactions of the Krebs cycle, and ATP production by electron transport chains occur within many forms of bacteria and inside the mitochondria of all eukaryotes. Recall from chapter 5 that mitochondria are thought to have evolved from bacteria. Although plants and algae can produce ATP by photosynthesis, they also produce ATP by aerobic respiration, just as animals and other nonphotosynthetic eukaryotes do. Figure 9.6 provides an overview of aerobic respiration.

Anaerobic Respiration

In the presence of oxygen, cells can respire aerobically, using oxygen to accept the electrons harvested from food molecules. In the absence of oxygen to accept the electrons, some organisms can still respire anaerobically, using inorganic molecules to accept the electrons. For example, many bacteria use sulfur, nitrate, or other inorganic compounds as the electron acceptor in place of oxygen.

Methanogens. Among the heterotrophs that practice anaerobic respiration are primitive archaeobacteria such as the thermophiles discussed in chapter 4. Some of these, called methanogens, use CO_2 as the electron acceptor, reducing CO_2 to CH_4 (methane) with the hydrogens derived from organic molecules produced by other organisms.

Sulfur Bacteria. Evidence of a second anaerobic respiratory process among primitive bacteria is seen in a group of rocks about 2.7 billion years old, known as the Woman River iron formation. Organic material in these rocks is enriched for the light isotope of sulfur, ^{32}S , rela-

tive to the heavier isotope ^{34}S . No known geochemical process produces such enrichment, but biological sulfur reduction does, in a process still carried out today by certain primitive bacteria. In this sulfate respiration, the bacteria derive energy from the reduction of inorganic sulfates (SO_4) to H_2S . The hydrogen atoms are obtained from organic molecules other organisms produce. These bacteria thus do the same thing methanogens do, but they use SO_4 as the oxidizing (that is, electron-accepting) agent in place of CO_2 .

The sulfate reducers set the stage for the evolution of photosynthesis, creating an environment rich in H_2S . As discussed in chapter 8, the first form of photosynthesis obtained hydrogens from H_2S using the energy of sunlight.

In aerobic respiration, the cell harvests energy from glucose molecules in a sequence of four major pathways: glycolysis, pyruvate oxidation, the Krebs cycle, and the electron transport chain. Oxygen is the final electron acceptor. Anaerobic respiration donates the harvested electrons to other inorganic compounds.

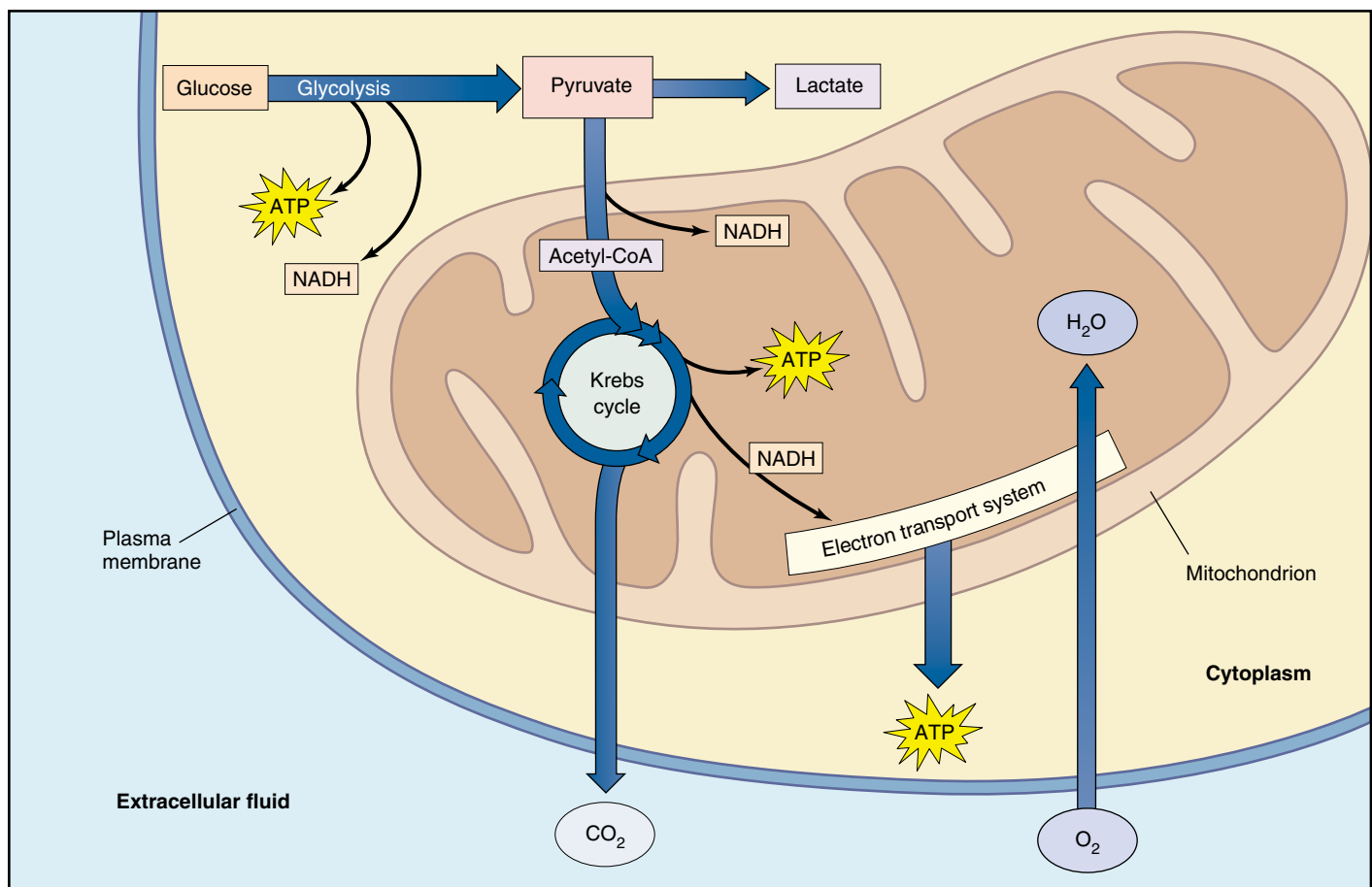


FIGURE 9.6
An overview of aerobic respiration.

Stage One: Glycolysis

The metabolism of primitive organisms focused on glucose. Glucose molecules can be dismantled in many ways, but primitive organisms evolved a glucose-catabolizing process that releases enough free energy to drive the synthesis of ATP in coupled reactions. This process, called glycolysis, occurs in the cytoplasm and involves a sequence of 10 reactions that convert glucose into 2 three-carbon molecules of pyruvate (figure 9.7). For each molecule of glucose that passes through this transformation, the cell nets two ATP molecules by substrate-level phosphorylation.

Priming

The first half of glycolysis consists of five sequential reactions that convert one molecule of glucose into two molecules of the three-carbon compound, glyceraldehyde 3-phosphate (G3P). These reactions demand the expenditure of ATP, so they are an energy-requiring process.

Step A: Glucose priming. Three reactions “prime” glucose by changing it into a compound that can be cleaved readily into 2 three-carbon phosphorylated molecules. Two of these reactions require the cleavage of ATP, so this step requires the cell to use two ATP molecules.

Step B: Cleavage and rearrangement. In the first of the remaining pair of reactions, the six-carbon product of step A is split into 2 three-carbon molecules. One is G3P, and the other is then converted to G3P by the second reaction (figure 9.8).

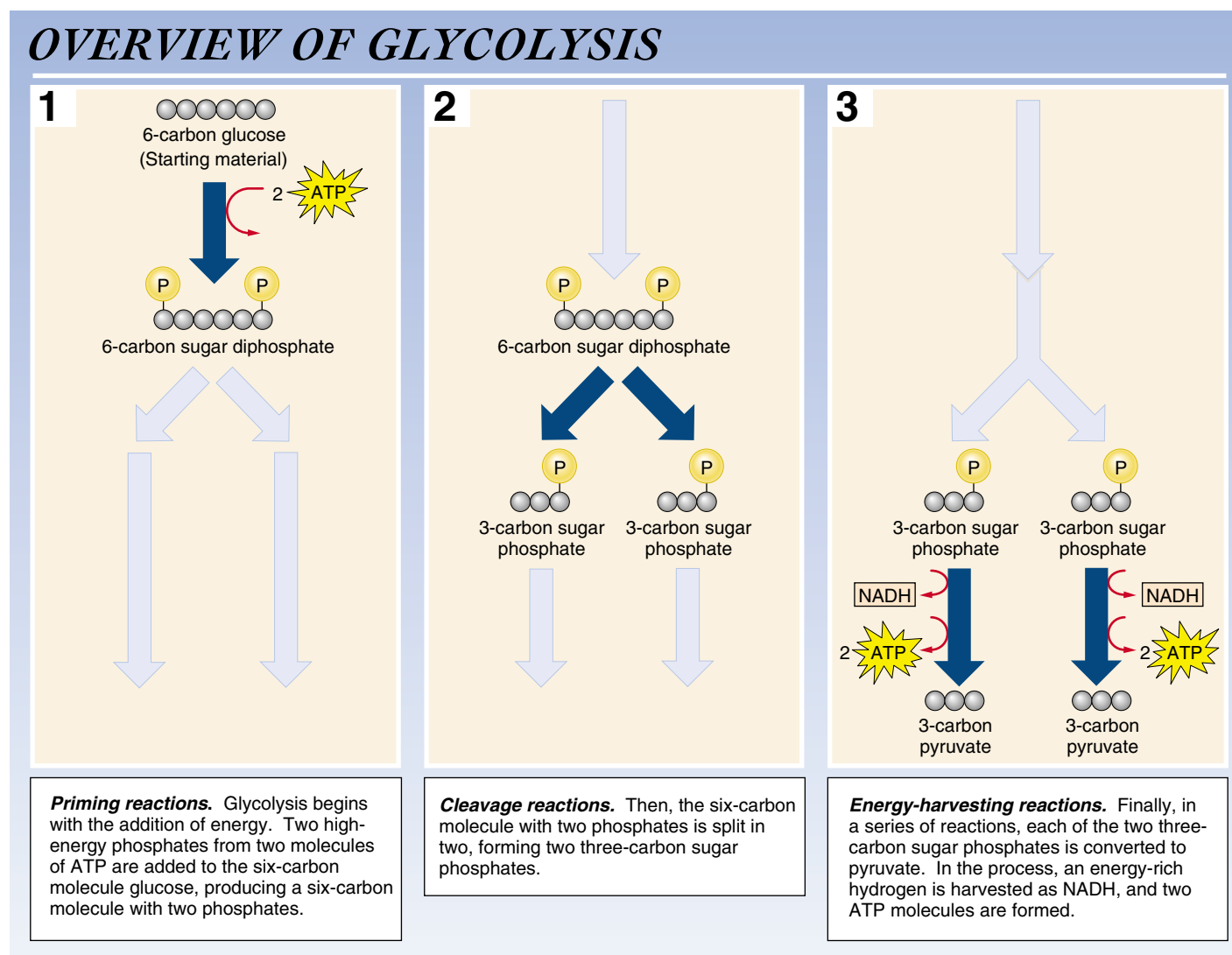


FIGURE 9.7
How glycolysis works.

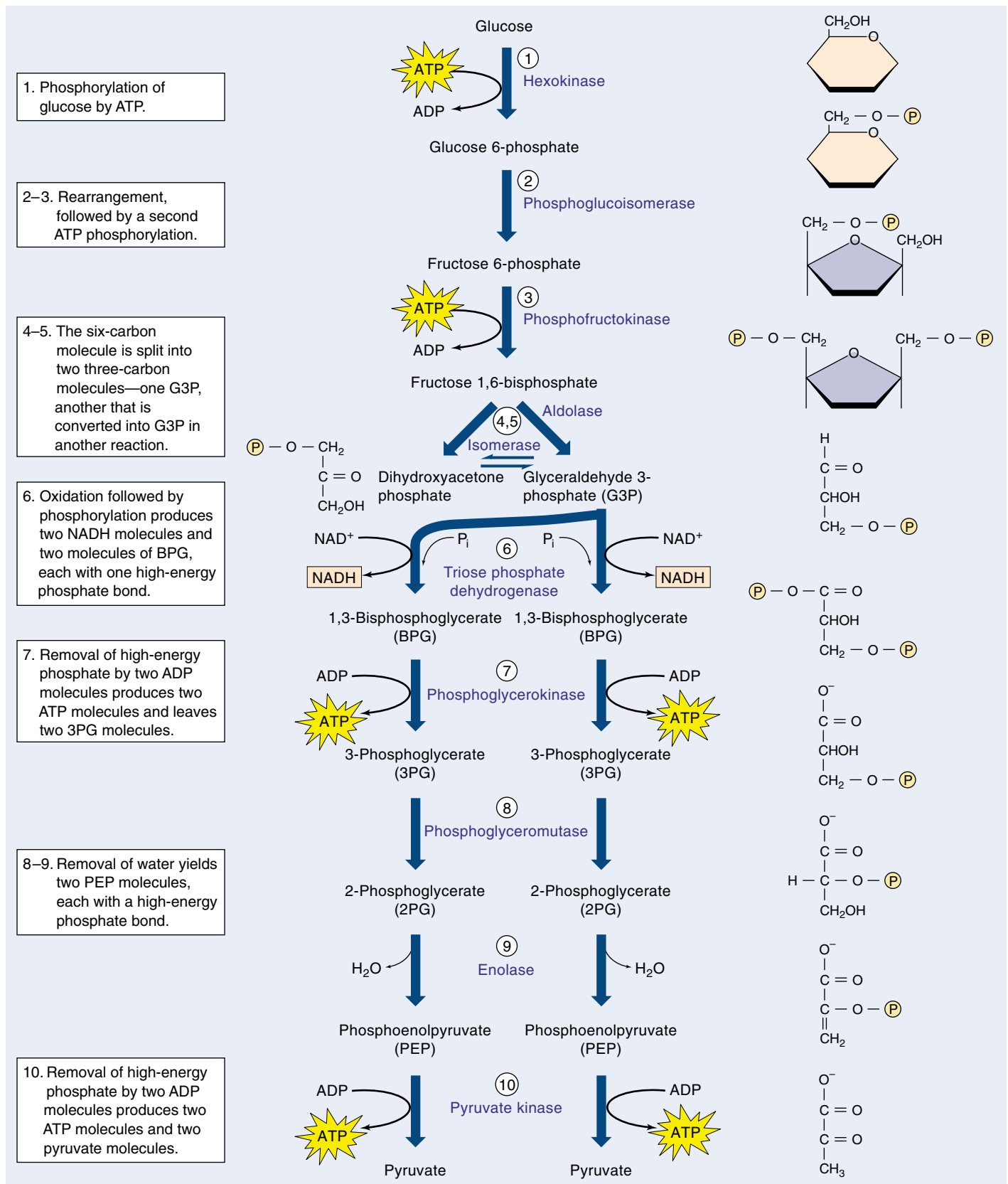


FIGURE 9.8

The glycolytic pathway. The first five reactions convert a molecule of glucose into two molecules of G3P. The second five reactions convert G3P into pyruvate.

Substrate-Level Phosphorylation

In the second half of glycolysis, five more reactions convert G3P into pyruvate in an energy-yielding process that generates ATP. Overall, then, glycolysis is a series of 10 enzyme-catalyzed reactions in which some ATP is invested in order to produce more.

Step C: Oxidation. Two electrons and one proton are transferred from G3P to NAD^+ , forming NADH. Note that NAD^+ is an ion, and that both electrons in the new covalent bond come from G3P.

Step D: ATP generation. Four reactions convert G3P into another three-carbon molecule, pyruvate. This process generates two ATP molecules (see figure 9.5).

Because each glucose molecule is split into two G3P molecules, the overall reaction sequence yields two molecules of ATP, as well as two molecules of NADH and two of pyruvate:

$$\begin{array}{r} 4 \text{ ATP (2 ATP for each of the 2 G3P molecules in step D)} \\ -2 \text{ ATP (used in the two reactions in step A)} \\ \hline 2 \text{ ATP} \end{array}$$

Under the nonstandard conditions within a cell, each ATP molecule produced represents the capture of about 12 kcal (50 kJ) of energy per mole of glucose, rather than the 7.3 traditionally quoted for standard conditions. This means glycolysis harvests about 24 kcal/mole (100 kJ/mole). This is not a great deal of energy. The total energy content of the chemical bonds of glucose is 686 kcal (2870 kJ) per mole, so glycolysis harvests only 3.5% of the chemical energy of glucose.

Although far from ideal in terms of the amount of energy it releases, glycolysis does generate ATP. For more than a billion years during the anaerobic first stages of life on earth, it was the primary way heterotrophic organisms generated ATP from organic molecules. Like many biochemical pathways, glycolysis is believed to have evolved backward, with the last steps in the process being the most ancient. Thus, the second half of glycolysis, the ATP-yielding breakdown of G3P, may have been the original process early heterotrophs used to generate ATP. The synthesis of G3P from glucose would have appeared later, perhaps when alternative sources of G3P were depleted.

All Cells Use Glycolysis

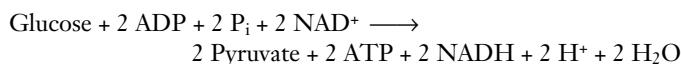
The glycolytic reaction sequence is thought to have been among the earliest of all biochemical processes to evolve. It uses no molecular oxygen and occurs readily in an anaerobic environment. All of its reactions occur free in the cytoplasm; none is associated with any organelle or membrane structure. Every living creature is capable of carrying out

glycolysis. Most present-day organisms, however, can extract considerably more energy from glucose through aerobic respiration.

Why does glycolysis take place even now, since its energy yield in the absence of oxygen is comparatively so paltry? The answer is that evolution is an incremental process: change occurs by improving on past successes. In catabolic metabolism, glycolysis satisfied the one essential evolutionary criterion: it was an improvement. Cells that could not carry out glycolysis were at a competitive disadvantage, and only cells capable of glycolysis survived the early competition of life. Later improvements in catabolic metabolism built on this success. Glycolysis was not discarded during the course of evolution; rather, it served as the starting point for the further extraction of chemical energy. Metabolism evolved as one layer of reactions added to another, just as successive layers of paint cover the walls of an old building. Nearly every present-day organism carries out glycolysis as a metabolic memory of its evolutionary past.

Closing the Metabolic Circle: The Regeneration of NAD^+

Inspect for a moment the net reaction of the glycolytic sequence:



You can see that three changes occur in glycolysis: (1) glucose is converted into two molecules of pyruvate; (2) two molecules of ADP are converted into ATP via substrate level phosphorylation; and (3) two molecules of NAD^+ are reduced to NADH.

The Need to Recycle NADH

As long as food molecules that can be converted into glucose are available, a cell can continually churn out ATP to drive its activities. In doing so, however, it accumulates NADH and depletes the pool of NAD^+ molecules. A cell does not contain a large amount of NAD^+ , and for glycolysis to continue, NADH must be recycled into NAD^+ . Some other molecule than NAD^+ must ultimately accept the hydrogen atom taken from G3P and be reduced. Two molecules can carry out this key task (figure 9.9):

1. Aerobic respiration. Oxygen is an excellent electron acceptor. Through a series of electron transfers, the hydrogen atom taken from G3P can be donated to oxygen, forming water. This is what happens in the cells of eukaryotes in the presence of oxygen. Because air is rich in oxygen, this process is also referred to as aerobic metabolism.

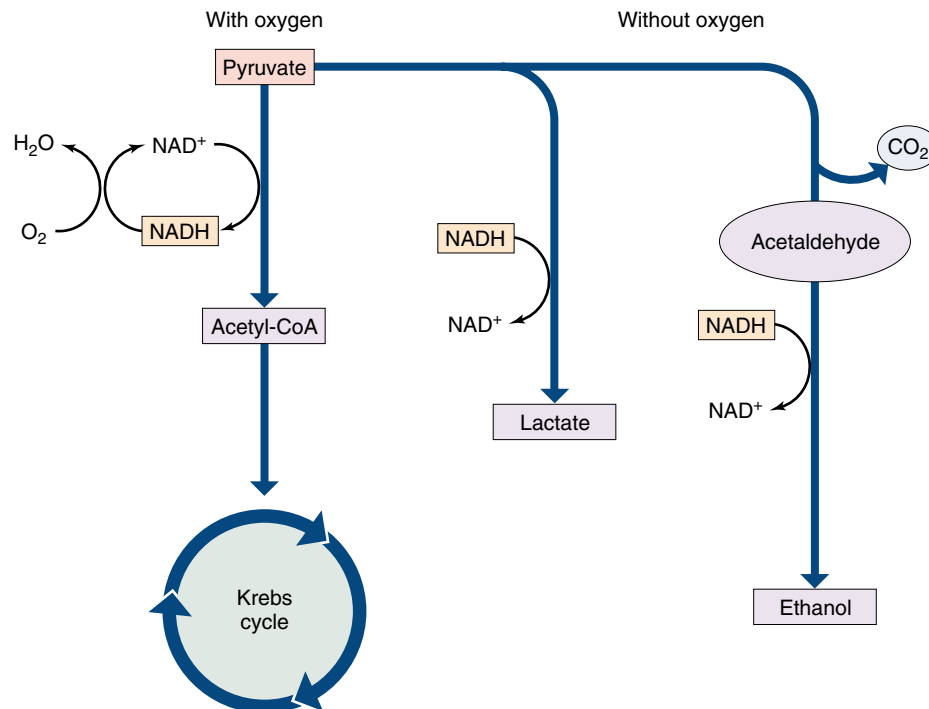


FIGURE 9.9

What happens to pyruvate, the product of glycolysis? In the presence of oxygen, pyruvate is oxidized to acetyl-CoA, which enters the Krebs cycle. In the absence of oxygen, pyruvate is instead reduced, accepting the electrons extracted during glycolysis and carried by NADH. When pyruvate is reduced directly, as in muscle cells, the product is lactate. When CO₂ is first removed from pyruvate and the product, acetaldehyde, is then reduced, as in yeast cells, the product is ethanol.

2. Fermentation. When oxygen is unavailable, an organic molecule can accept the hydrogen atom instead. Such fermentation plays an important role in the metabolism of most organisms (figure 9.10), even those capable of aerobic respiration.

The fate of the pyruvate that is produced by glycolysis depends upon which of these two processes takes place. The aerobic respiration path starts with the oxidation of pyruvate to a molecule called acetyl-CoA, which is then further oxidized in a series of reactions called the Krebs cycle. The fermentation path, by contrast, involves the reduction of all or part of pyruvate. We will start by examining aerobic respiration, then look briefly at fermentation.

Glycolysis generates a small amount of ATP by reshuffling the bonds of glucose molecules. In glycolysis, two molecules of NAD⁺ are reduced to NADH. NAD⁺ must be regenerated for glycolysis to continue unabated.



FIGURE 9.10

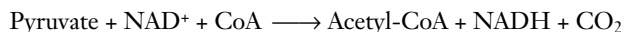
Fermentation. The conversion of pyruvate to ethanol takes place naturally in grapes left to ferment on vines, as well as in fermentation vats of crushed grapes. Yeasts carry out the process, but when their conversion increases the ethanol concentration to about 12%, the toxic effects of the alcohol kill the yeast cells. What is left is wine.

Stage Two: The Oxidation of Pyruvate

In the presence of oxygen, the oxidation of glucose that begins in glycolysis continues where glycolysis leaves off—with pyruvate. In eukaryotic organisms, the extraction of additional energy from pyruvate takes place exclusively inside mitochondria. The cell harvests pyruvate's considerable energy in two steps: first, by oxidizing pyruvate to form acetyl-CoA, and then by oxidizing acetyl-CoA in the Krebs cycle.

Producing Acetyl-CoA

Pyruvate is oxidized in a single “decarboxylation” reaction that cleaves off one of pyruvate's three carbons. This carbon then departs as CO_2 (figure 9.11, *top*). This reaction produces a two-carbon fragment called an acetyl group, as well as a pair of electrons and their associated hydrogen, which reduce NAD^+ to NADH. The reaction is complex, involving three intermediate stages, and is catalyzed within mitochondria by a *multienzyme complex*. As chapter 8 noted, such a complex organizes a series of enzymatic steps so that the chemical intermediates do not diffuse away or undergo other reactions. Within the complex, component polypeptides pass the substrates from one enzyme to the next, without releasing them. *Pyruvate dehydrogenase*, the complex of enzymes that removes CO_2 from pyruvate, is one of the largest enzymes known: it contains 60 subunits! In the course of the reaction, the acetyl group removed from pyruvate combines with a cofactor called coenzyme A (CoA), forming a compound known as **acetyl-CoA**:



This reaction produces a molecule of NADH, which is later used to produce ATP. Of far greater significance than the reduction of NAD^+ to NADH, however, is the production of acetyl-CoA (figure 9.11, *bottom*). Acetyl-CoA is important because so many different metabolic processes generate it. Not only does the oxidation of pyruvate, an intermediate in carbohydrate catabolism, produce it, but the metabolic breakdown of proteins, fats, and other lipids also generate acetyl-CoA. Indeed, almost all molecules catabolized for energy are converted into acetyl-CoA. Acetyl-CoA is then channeled into fat synthesis or into ATP production, depending on the organism's energy requirements. Acetyl-CoA is a key point of focus for the many catabolic processes of the eukaryotic cell.

Using Acetyl-CoA

Although the cell forms acetyl-CoA in many ways, only a limited number of processes use acetyl-CoA. Most of it is either directed toward energy storage (lipid synthesis, for example) or oxidized in the Krebs cycle to produce ATP. Which of these two options is taken depends on the level of ATP in the cell. When ATP levels are high, the oxida-

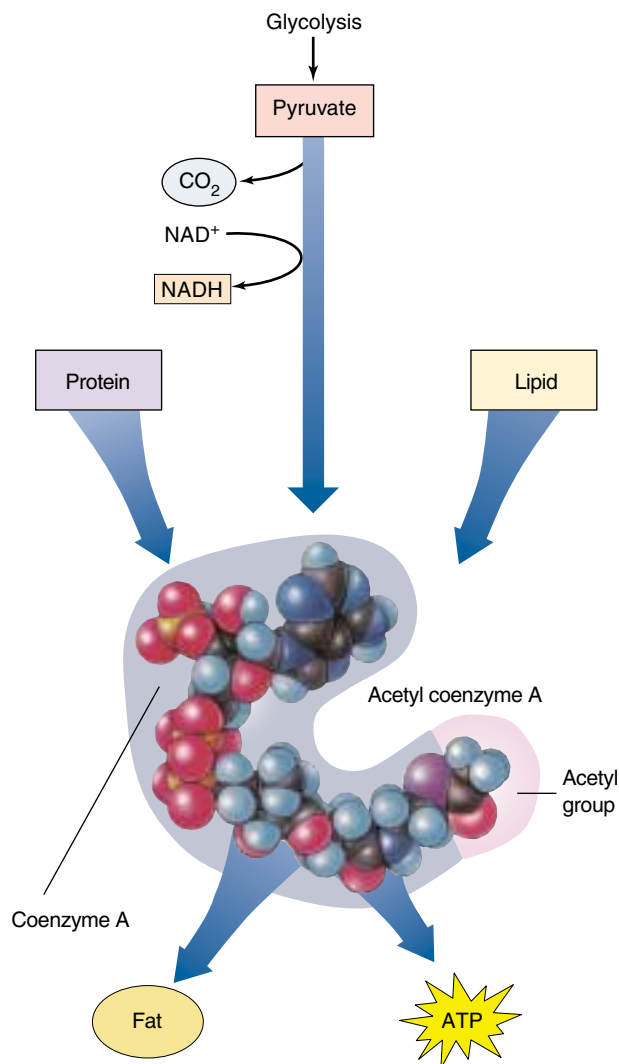


FIGURE 9.11

The oxidation of pyruvate. This complex reaction involves the reduction of NAD^+ to NADH and is thus a significant source of metabolic energy. Its product, acetyl-CoA, is the starting material for the Krebs cycle. Almost all molecules that are catabolized for energy are converted into acetyl-CoA, which is then channeled into fat synthesis or into ATP production.

tive pathway is inhibited, and acetyl-CoA is channeled into fatty acid synthesis. This explains why many animals (humans included) develop fat reserves when they consume more food than their bodies require. Alternatively, when ATP levels are low, the oxidative pathway is stimulated, and acetyl-CoA flows into energy-producing oxidative metabolism.

In the second energy-harvesting stage of glucose catabolism, pyruvate is decarboxylated, yielding acetyl-CoA, NADH, and CO_2 . This process occurs within the mitochondrion.

Stage Three: The Krebs Cycle

After glycolysis catabolizes glucose to produce pyruvate, and pyruvate is oxidized to form acetyl-CoA, the third stage of extracting energy from glucose begins. In this third stage, acetyl-CoA is oxidized in a series of nine reactions called the Krebs cycle. These reactions occur in the matrix of mitochondria. In this cycle, the two-carbon acetyl group of acetyl-CoA combines with a four-carbon molecule called oxaloacetate (figure 9.12). The resulting six-carbon molecule then goes through a sequence of electron-yielding oxidation reactions, during which two CO_2 molecules split off, restoring oxaloacetate. The oxaloacetate is then recycled to bind to another acetyl group. In each turn of the cycle, a new acetyl group replaces the two CO_2 molecules lost, and

more electrons are extracted to drive proton pumps that generate ATP.

Overview of the Krebs Cycle

The nine reactions of the Krebs cycle occur in two steps:

Step A: Priming. Three reactions prepare the six-carbon molecule for energy extraction. First, acetyl-CoA joins the cycle, and then chemical groups are rearranged.

Step B: Energy extraction. Four of the six reactions in this step are oxidations in which electrons are removed, and one generates an ATP equivalent directly by substrate-level phosphorylation.

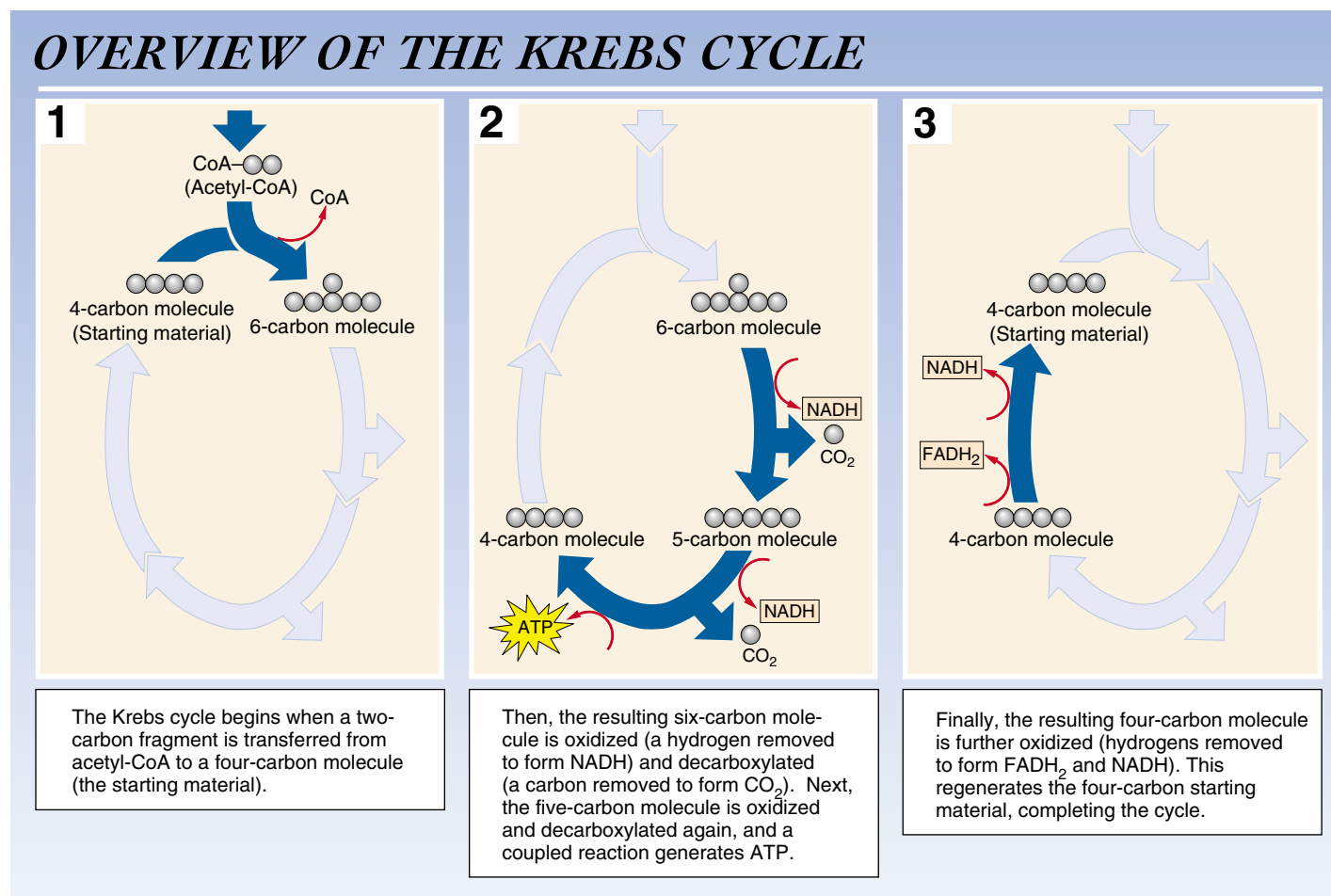


FIGURE 9.12
How the Krebs cycle works.

The Reactions of the Krebs Cycle

The **Krebs cycle** consists of nine sequential reactions that cells use to extract energetic electrons and drive the synthesis of ATP (figure 9.13). A two-carbon group from acetyl-CoA enters the cycle at the beginning, and two CO₂ molecules and several electrons are given off during the cycle.

Reaction 1: Condensation. The two-carbon group from acetyl-CoA joins with a four-carbon molecule, oxaloacetate, to form a six-carbon molecule, citrate. This condensation reaction is irreversible, committing the two-carbon acetyl group to the Krebs cycle. The reaction is inhibited when the cell's ATP concentration is high and stimulated when it is low. Hence, when the cell possesses ample amounts of ATP, the Krebs cycle shuts down and acetyl-CoA is channeled into fat synthesis.

Reactions 2 and 3: Isomerization. Before the oxidation reactions can begin, the hydroxyl (—OH) group of citrate must be repositioned. This is done in two steps: first, a water molecule is removed from one carbon; then, water is added to a different carbon. As a result, an —H group and an —OH group change positions. The product is an isomer of citrate called isocitrate.

Reaction 4: The First Oxidation. In the first energy-yielding step of the cycle, isocitrate undergoes an oxidative decarboxylation reaction. First, isocitrate is oxidized, yielding a pair of electrons that reduce a molecule of NAD⁺ to NADH. Then the oxidized intermediate is decarboxylated; the central carbon atom splits off to form CO₂, yielding a five-carbon molecule called α -ketoglutarate.

Reaction 5: The Second Oxidation. Next, α -ketoglutarate is decarboxylated by a multienzyme complex similar to pyruvate dehydrogenase. The succinyl group left after the removal of CO₂ joins to coenzyme A, forming succinyl-CoA. In the process, two electrons are extracted, and they reduce another molecule of NAD⁺ to NADH.

Reaction 6: Substrate-Level Phosphorylation. The linkage between the four-carbon succinyl group and CoA is a high-energy bond. In a coupled reaction similar to those that take place in glycolysis, this bond is cleaved, and the

energy released drives the phosphorylation of guanosine diphosphate (GDP), forming guanosine triphosphate (GTP). GTP is readily converted into ATP, and the four-carbon fragment that remains is called succinate.

Reaction 7: The Third Oxidation. Next, succinate is oxidized to fumarate. The free energy change in this reaction is not large enough to reduce NAD⁺. Instead, flavin adenine dinucleotide (FAD) is the electron acceptor. Unlike NAD⁺, FAD is not free to diffuse within the mitochondrion; it is an integral part of the inner mitochondrial membrane. Its reduced form, FADH₂, contributes electrons to the electron transport chain in the membrane.

Reactions 8 and 9: Regeneration of Oxaloacetate. In the final two reactions of the cycle, a water molecule is added to fumarate, forming malate. Malate is then oxidized, yielding a four-carbon molecule of oxaloacetate and two electrons that reduce a molecule of NAD⁺ to NADH. Oxaloacetate, the molecule that began the cycle, is now free to combine with another two-carbon acetyl group from acetyl-CoA and reinitiate the cycle.

The Products of the Krebs Cycle

In the process of aerobic respiration, glucose is entirely consumed. The six-carbon glucose molecule is first cleaved into a pair of three-carbon pyruvate molecules during glycolysis. One of the carbons of each pyruvate is then lost as CO₂ in the conversion of pyruvate to acetyl-CoA; two other carbons are lost as CO₂ during the oxidations of the Krebs cycle. All that is left to mark the passing of the glucose molecule into six CO₂ molecules is its energy, some of which is preserved in four ATP molecules and in the reduced state of 12 electron carriers. Ten of these carriers are NADH molecules; the other two are FADH₂.

The Krebs cycle generates two ATP molecules per molecule of glucose, the same number generated by glycolysis. More importantly, the Krebs cycle and the oxidation of pyruvate harvest many energized electrons, which can be directed to the electron transport chain to drive the synthesis of much more ATP.

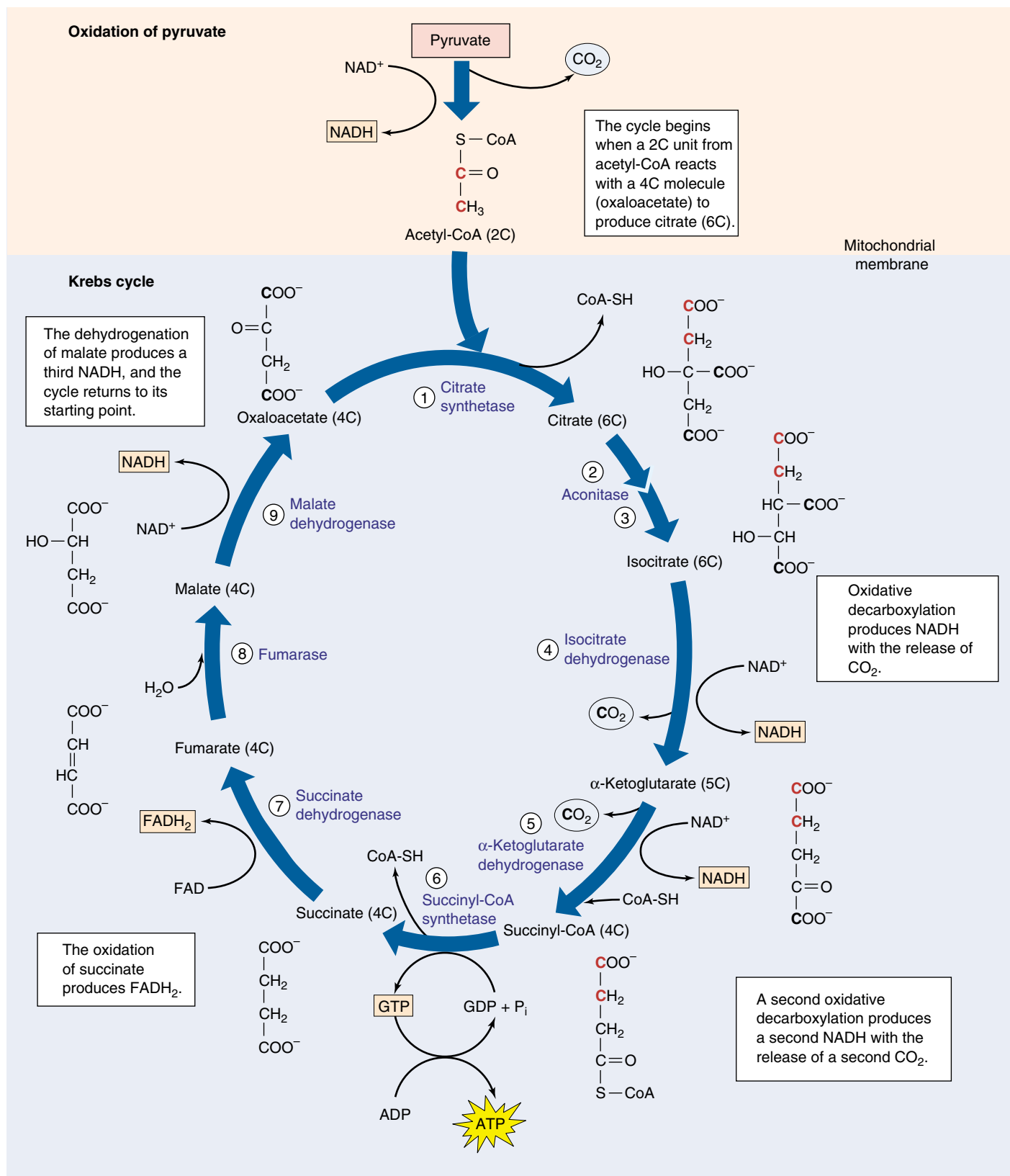


FIGURE 9.13

The Krebs cycle. This series of reactions takes place within the matrix of the mitochondrion. For the complete breakdown of a molecule of glucose, the two molecules of acetyl-CoA produced by glycolysis and pyruvate oxidation will each have to make a trip around the Krebs cycle. Follow the different carbons through the cycle, and notice the changes that occur in the carbon skeletons of the molecules as they proceed through the cycle.

Harvesting Energy by Extracting Electrons

To understand how cells direct some of the energy released during glucose catabolism into ATP production, we need to take a closer look at the electrons in the C—H bonds of the glucose molecule. We stated in chapter 8 that when an electron is removed from one atom and donated to another, the electron's potential energy of position is also transferred. In this process, the atom that receives the electron is reduced. We spoke of reduction in an all-or-none fashion, as if it involved the complete transfer of an electron from one atom to another. Often this is just what happens. However, sometimes a reduction simply changes the *degree of sharing* within a covalent bond. Let us now revisit that discussion and consider what happens when the transfer of electrons is incomplete.

A Closer Look at Oxidation-Reduction

The catabolism of glucose is an oxidation-reduction reaction. The covalent electrons in the C—H bonds of glucose are shared approximately equally between the C and H atoms because carbon and hydrogen nuclei have about the same affinity for valence electrons (that is, they exhibit similar *electronegativity*). However, when the carbon atoms of glucose react with oxygen to form carbon dioxide, the electrons in the new covalent bonds take a different position. Instead of being shared equally, the electrons that were associated with the carbon atoms in glucose shift far toward the oxygen atom in CO_2 because oxygen is very electronegative. Since these electrons are pulled farther from the carbon atoms, the carbon atoms of glucose have been oxidized (loss of electrons) and the oxygen atoms reduced (gain of electrons). Similarly, when the hydrogen atoms of glucose combine with oxygen atoms to form water, the oxygen atoms draw the shared electrons strongly toward them; again, oxygen is reduced and glucose is oxidized. In this reaction, oxygen is an oxidizing (electron-attracting) agent because it oxidizes the atoms of glucose.

Releasing Energy

The key to understanding the oxidation of glucose is to focus on the energy of the shared electrons. In a covalent bond, energy must be added to remove an electron from an atom, just as energy must be used to roll a boulder up a hill. The more electronegative the atom, the steeper the energy hill that must be climbed to pull an electron away from it. However, energy is released when an electron is shifted away from a less electronegative atom and *closer* to a more electronegative atom, just as energy is released when a boulder is allowed to roll down a hill. In the catabolism of glucose, energy is released when glucose is oxidized, as electrons relocate closer to oxygen (figure 9.14).

Glucose is an energy-rich food because it has an abundance of C—H bonds. Viewed in terms of oxidation-reduction, glucose possesses a wealth of electrons held far from their atoms, all with the potential to move closer toward oxygen. In oxidative respiration, energy is released not simply because the hydrogen atoms of the C—H bonds are transferred from glucose to oxygen, but because the positions of the valence electrons shift. This shift releases energy that can be used to make ATP.

Harvesting the Energy in Stages

It is generally true that the larger the release of energy in any single step, the more of that energy is released as heat (random molecular motion) and the less there is available to be channeled into more useful paths. In the combustion of gasoline, the same amount of energy is released whether all of the gasoline in a car's gas tank explodes at once, or whether the gasoline burns in a series of very small explosions inside the cylinders. By releasing the energy in gasoline a little at a time, the harvesting efficiency is greater and

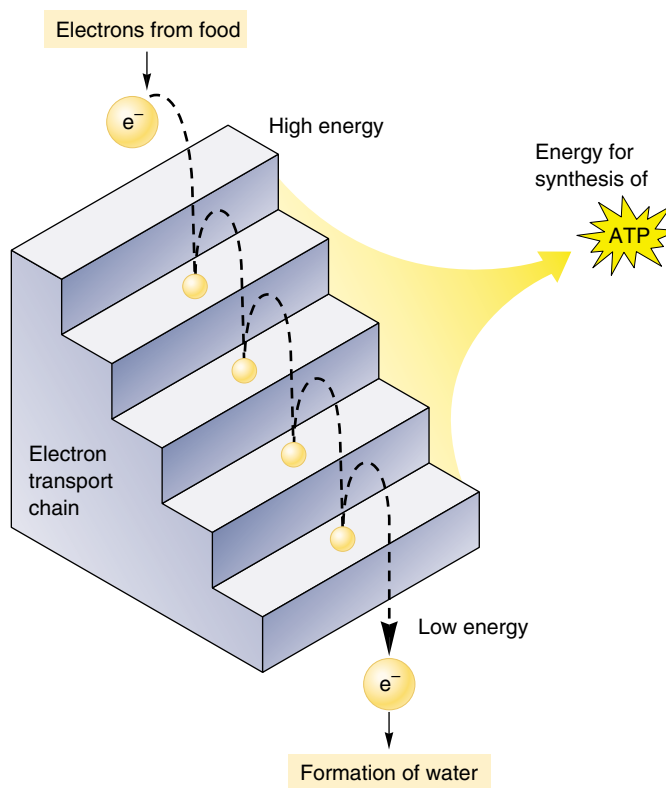
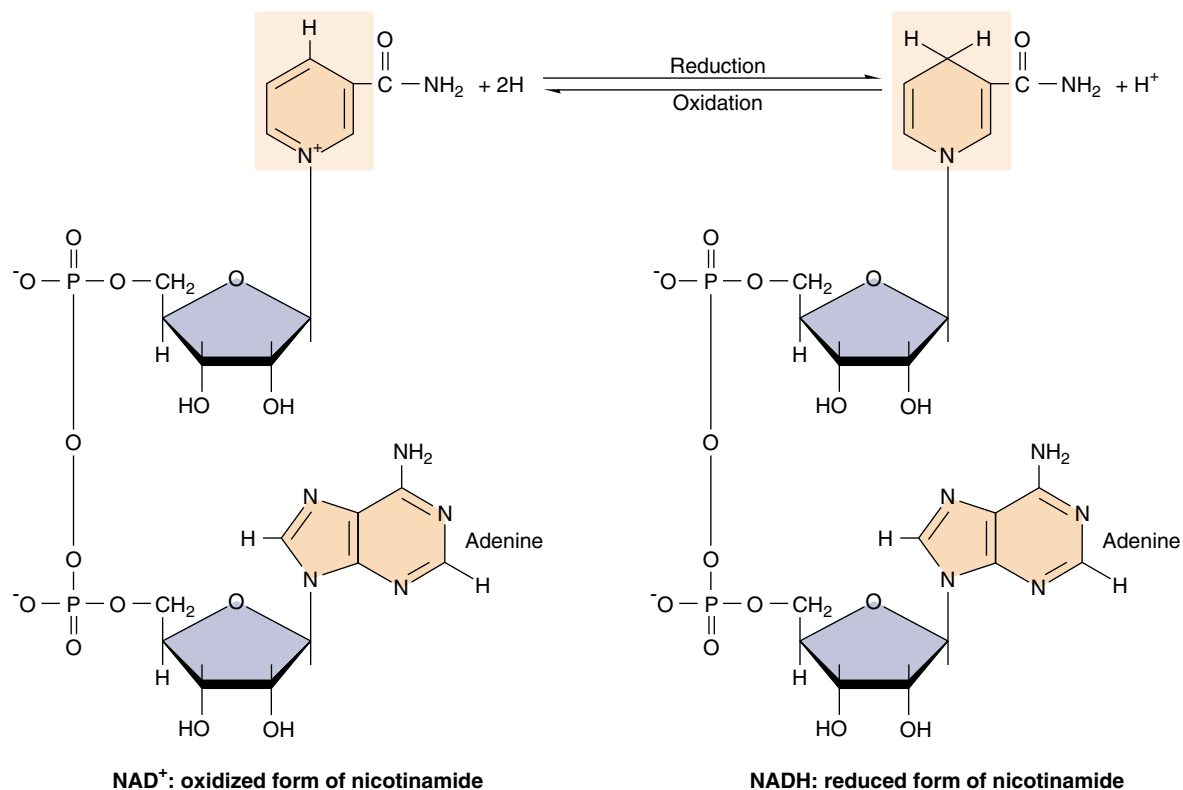


FIGURE 9.14

How electron transport works. This diagram shows how ATP is generated when electrons transfer from one energy level to another. Rather than releasing a single explosive burst of energy, electrons “fall” to lower and lower energy levels in steps, releasing stored energy with each fall as they tumble to the lowest (most electronegative) electron acceptor.

FIGURE 9.15
NAD⁺ and NADH.

This dinucleotide serves as an “electron shuttle” during cellular respiration. NAD⁺ accepts electrons from catabolized macromolecules and is reduced to NADH.



more of the energy can be used to push the pistons and move the car.

The same principle applies to the oxidation of glucose inside a cell. If all of the hydrogens were transferred to oxygen in one explosive step, releasing all of the free energy at once, the cell would recover very little of that energy in a useful form. Instead, cells burn their fuel much as a car does, a little at a time. The six hydrogens in the C—H bonds of glucose are stripped off in stages in the series of enzyme-catalyzed reactions collectively referred to as glycolysis and the Krebs cycle. We have had a great deal to say about these reactions already in this chapter. Recall that the hydrogens are removed by transferring them to a coenzyme carrier, NAD⁺ (figure 9.15). Discussed in chapter 8, NAD⁺ is a very versatile electron acceptor, shuttling energy-bearing electrons throughout the cell. In harvesting the energy of glucose, NAD⁺ acts as the primary electron acceptor.

Following the Electrons

As you examine these reactions, try not to become confused by the changes in electrical charge. Always *follow the electrons*. Enzymes extract two hydrogens—that is, two electrons and two protons—from glucose and transfer both electrons and one of the protons to NAD⁺. The other proton is released as a hydrogen ion, H⁺, into the surrounding solution. This transfer converts NAD⁺ into NADH; that is,

two negative electrons and one positive proton are added to one positively charged NAD⁺ to form NADH, which is electrically neutral.

Energy captured by NADH is *not* harvested all at once. Instead of being transferred directly to oxygen, the two electrons carried by NADH are passed along the **electron transport chain** if oxygen is present. This chain consists of a series of molecules, mostly proteins, embedded within the inner membranes of mitochondria. NADH delivers electrons to the top of the electron transport chain and oxygen captures them at the bottom. The oxygen then joins with hydrogen ions to form water. At each step in the chain, the electrons move to a slightly more electronegative carrier, and their positions shift slightly. Thus, the electrons move *down* an energy gradient. The entire process releases a total of 53 kcal/mole (222 kJ/mole) under standard conditions. The transfer of electrons along this chain allows the energy to be extracted gradually. In the next section, we will discuss how this energy is put to work to drive the production of ATP.

The catabolism of glucose involves a series of oxidation-reduction reactions that release energy by repositioning electrons closer to oxygen atoms. Energy is thus harvested from glucose molecules in gradual steps, using NAD⁺ as an electron carrier.

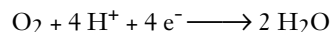
Stage Four: The Electron Transport Chain

The NADH and FADH₂ molecules formed during the first three stages of aerobic respiration each contain a pair of electrons that were gained when NAD⁺ and FAD were reduced. The NADH molecules carry their electrons to the inner mitochondrial membrane, where they transfer the electrons to a series of membrane-associated proteins collectively called the electron transport chain.

Moving Electrons through the Electron Transport Chain

The first of the proteins to receive the electrons is a complex, membrane-embedded enzyme called **NADH dehydrogenase**. A carrier called ubiquinone then passes the electrons to a protein-cytochrome complex called the *bc₁* complex. This complex, along with others in the chain, operates as a proton pump, driving a proton out across the membrane. Cytochromes are respiratory proteins that contain heme groups, complex carbon rings with many alternating single and double bonds and an iron atom in the center.

The electron is then carried by another carrier, *cytochrome c*, to the cytochrome oxidase complex. This complex uses four such electrons to reduce a molecule of oxygen, each oxygen then combines with two hydrogen ions to form water:



This series of membrane-associated electron carriers is collectively called the electron transport chain (figure 9.16).

NADH contributes its electrons to the first protein of the electron transport chain, NADH dehydrogenase. FADH₂, which is always attached to the inner mitochondrial membrane, feeds its electrons into the electron transport chain later, to ubiquinone.

It is the availability of a plentiful electron acceptor (often oxygen) that makes oxidative respiration possible. As we'll see in chapter 10, the electron transport chain used in aerobic respiration is similar to, and may well have evolved from, the chain employed in aerobic photosynthesis.

The electron transport chain is a series of five membrane-associated proteins. Electrons delivered by NADH and FADH₂ are passed from protein to protein along the chain, like a baton in a relay race.

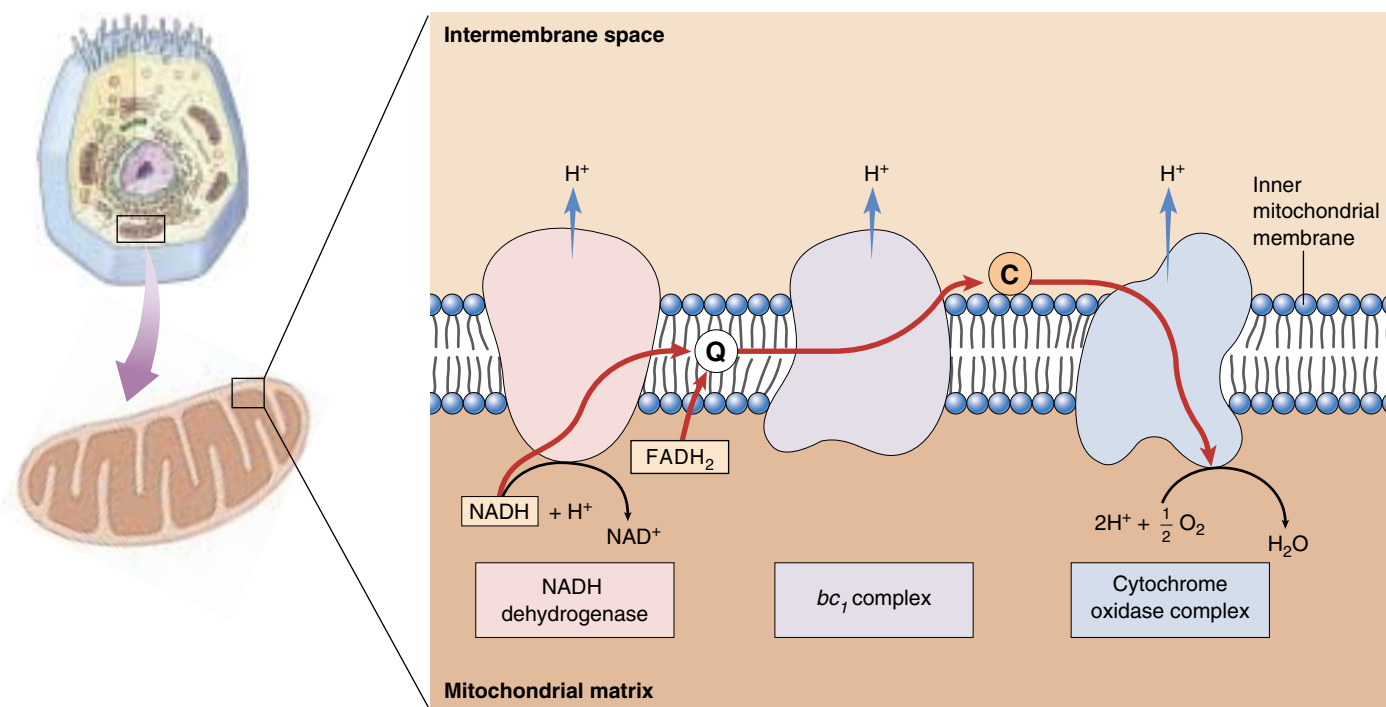


FIGURE 9.16

The electron transport chain. High-energy electrons harvested from catabolized molecules are transported (red arrows) by mobile electron carriers (ubiquinone, marked Q, and cytochrome *c*, marked C) along a chain of membrane proteins. Three proteins use portions of the electrons' energy to pump protons (blue arrows) out of the matrix and into the intermembrane space. The electrons are finally donated to oxygen to form water.

Building an Electrochemical Gradient

In eukaryotes, aerobic metabolism takes place within the mitochondria present in virtually all cells. The internal compartment, or matrix, of a mitochondrion contains the enzymes that carry out the reactions of the Krebs cycle. As the electrons harvested by oxidative respiration are passed along the electron transport chain, the energy they release transports protons out of the matrix and into the outer compartment, sometimes called the intermembrane space. Three transmembrane proteins in the inner mitochondrial membrane (see figure 9.16) actually accomplish the transport. The flow of excited electrons induces a change in the shape of these pump proteins, which causes them to transport protons across the membrane. The electrons contributed by NADH activate all three of these proton pumps, while those contributed by FADH_2 activate only two.

Producing ATP: Chemiosmosis

As the proton concentration in the intermembrane space rises above that in the matrix, the matrix becomes slightly negative in charge. This internal negativity attracts the positively charged protons and induces them to reenter the matrix. The higher outer concentration tends to drive protons back in by diffusion; since membranes are relatively impermeable to ions, most of the protons that reenter the matrix pass through special proton channels in the inner mitochondrial membrane. When the protons pass through, these channels synthesize ATP from $\text{ADP} + \text{P}_i$ within the matrix. The ATP is then transported by facilitated diffusion out of the mitochondrion and into the cell's cytoplasm. Because the chemical formation of ATP is driven by a diffusion force similar to osmosis, this process is referred to as **chemiosmosis** (figure 9.17).

Thus, the electron transport chain uses electrons harvested in aerobic respiration to pump a large number of protons across the inner mitochondrial membrane. Their subsequent reentry into the mitochondrial matrix drives the synthesis of ATP by chemiosmosis. Figure 9.18 summarizes the overall process.

The electrons harvested from glucose are pumped out of the mitochondrial matrix by the electron transport chain. The return of the protons into the matrix generates ATP.

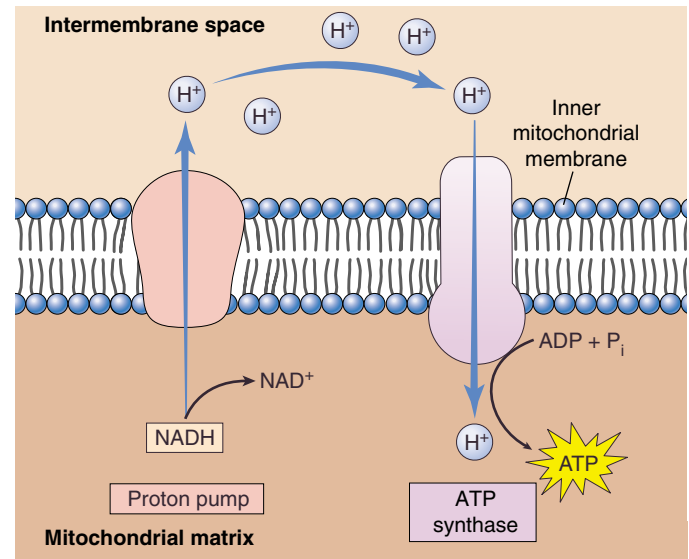


FIGURE 9.17

Chemiosmosis. NADH transports high-energy electrons harvested from the catabolism of macromolecules to “proton pumps” that use the energy to pump protons out of the mitochondrial matrix. As a result, the concentration of protons in the intermembrane space rises, inducing protons to diffuse back into the matrix. Many of the protons pass through special channels that couple the reentry of protons to the production of ATP.

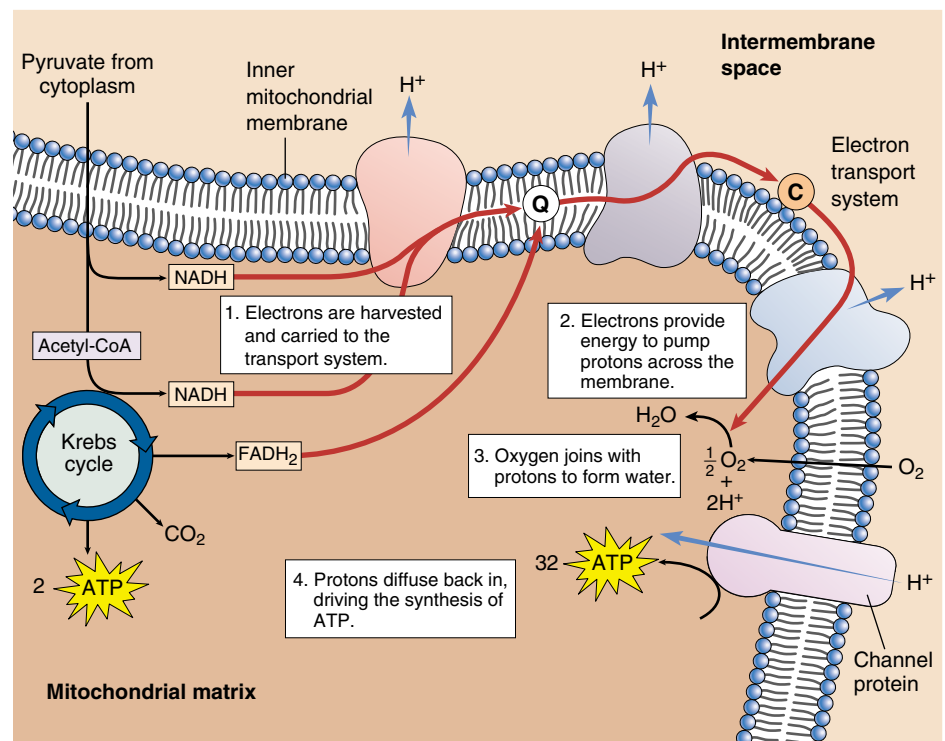


FIGURE 9.18

ATP generation during the Krebs cycle and electron transport chain. This process begins with pyruvate, the product of glycolysis, and ends with the synthesis of ATP.

Summarizing Aerobic Respiration

How much metabolic energy does a cell actually gain from the electrons harvested from a molecule of glucose, using the electron transport chain to produce ATP by chemiosmosis?

Theoretical Yield

The chemiosmotic model suggests that one ATP molecule is generated for each proton pump activated by the electron transport chain. Since the electrons from NADH activate three pumps and those from FADH_2 activate two, we would expect each molecule of NADH and FADH_2 to generate three and two ATP molecules, respectively. However, because eukaryotic cells carry out glycolysis in their cytoplasm and the Krebs cycle within their mitochondria, they must transport the two molecules of NADH produced during glycolysis across the mitochondrial membranes, which requires one ATP per molecule of NADH. Thus, the net ATP production is decreased by two. Therefore, the overall ATP production resulting from aerobic respiration *theoretically* should be 4 (from substrate-level phosphorylation during glycolysis) + 30 (3 from each of 10 molecules of NADH) + 4 (2 from each of 2 molecules of FADH_2) – 2 (for transport of glycolytic NADH) = 36 molecules of ATP (figure 9.19).

Actual Yield

The amount of ATP *actually* produced in a eukaryotic cell during aerobic respiration is somewhat lower than 36, for two reasons. First, the inner mitochondrial membrane is somewhat “leaky” to protons, allowing some of them to reenter the matrix without passing through ATP-generating channels. Second, mitochondria often use the proton gradient generated by chemiosmosis for purposes other than ATP synthesis (such as transporting pyruvate into the matrix). Consequently, the actual measured values of ATP generated by NADH and FADH_2 are closer to 2.5 for each NADH and 1.5 for each FADH_2 . With these corrections, the overall harvest of ATP from a molecule of glucose in a eukaryotic cell is closer to 4 (from substrate-level phosphorylation) + 25 (2.5 from each of 10 molecules of NADH) + 3 (1.5 from each of 2 molecules of FADH_2) – 2 (transport of glycolytic NADH) = 30 molecules of ATP.

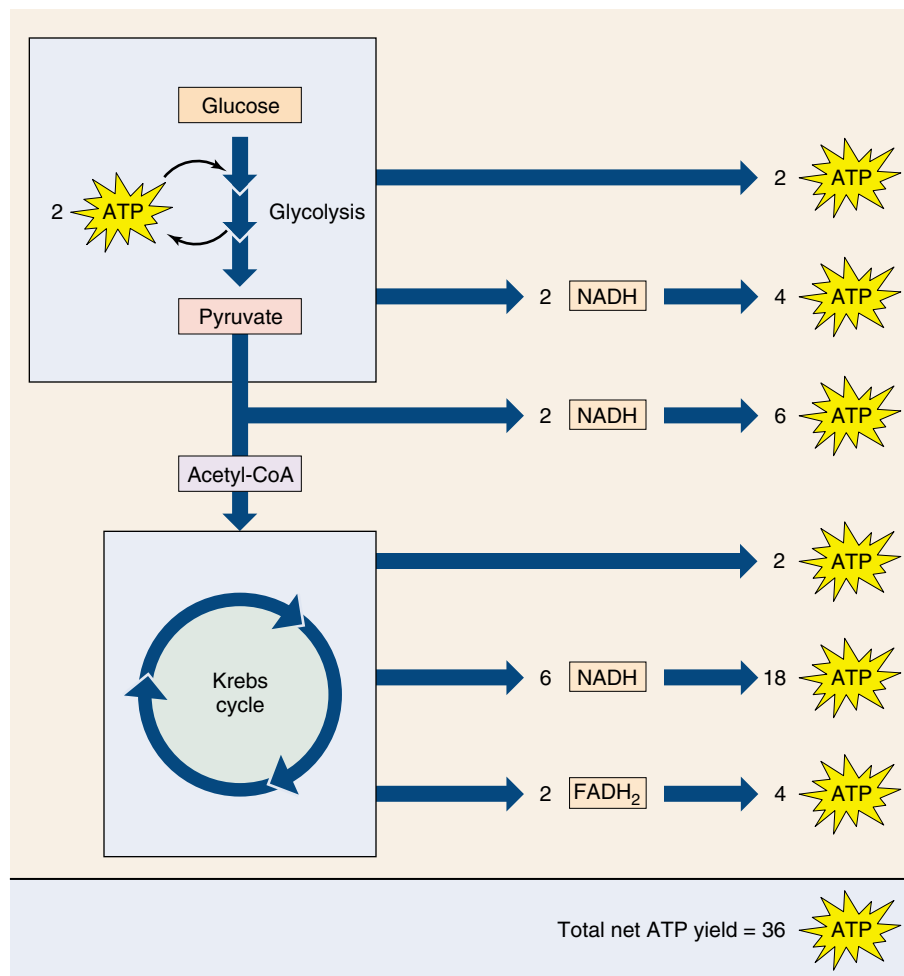


FIGURE 9.19

Theoretical ATP yield. The theoretical yield of ATP harvested from glucose by aerobic respiration totals 36 molecules.

The catabolism of glucose by aerobic respiration, in contrast to glycolysis, is quite efficient. Aerobic respiration in a eukaryotic cell harvests about $7.3 \times 30 \div 686 = 32\%$ of the energy available in glucose. (By comparison, a typical car converts only about 25% of the energy in gasoline into useful energy.) The efficiency of oxidative respiration at harvesting energy establishes a natural limit on the maximum length of food chains.

The high efficiency of aerobic respiration was one of the key factors that fostered the evolution of heterotrophs. With this mechanism for producing ATP, it became feasible for nonphotosynthetic organisms to derive metabolic energy exclusively from the oxidative breakdown of other organisms. As long as some organisms captured energy by photosynthesis, others could exist solely by feeding on them.

Oxidative respiration produces approximately 30 molecules of ATP from each molecule of glucose in eukaryotic cells. This represents more than half of the energy in the chemical bonds of glucose.

Regulating Aerobic Respiration

When cells possess plentiful amounts of ATP, the key reactions of glycolysis, the Krebs cycle, and fatty acid breakdown are inhibited, slowing ATP production. The regulation of these biochemical pathways by the level of ATP is an example of feedback inhibition. Conversely, when ATP levels in the cell are low, ADP levels are high; and ADP activates enzymes in the pathways of carbohydrate catabolism to stimulate the production of more ATP.

Control of glucose catabolism occurs at two key points of the catabolic pathway (figure 9.20). The control point in glycolysis is the enzyme phosphofructokinase, which catalyzes reaction 3, the conversion of fructose phosphate to fructose biphosphate. This is the first reaction of glycolysis that is not readily reversible, committing the substrate to the glycolytic sequence. High levels of ADP relative to ATP (implying a need to convert more ADP to ATP) stimulate phosphofructokinase, committing more sugar to the catabolic pathway; so do low levels of citrate (implying the Krebs cycle is not running at full tilt and needs more input). The main control point in the oxidation of pyruvate occurs at the committing step in the Krebs cycle with the enzyme pyruvate decarboxylase. It is inhibited by high levels of NADH (implying no more is needed).

Another control point in the Krebs cycle is the enzyme citrate synthetase, which catalyzes the first reaction, the conversion of oxaloacetate and acetyl-CoA into citrate. High levels of ATP inhibit citrate synthetase (as well as pyruvate decarboxylase and two other Krebs cycle enzymes), shutting down the catabolic pathway.

Relative levels of ADP and ATP regulate the catabolism of glucose at key committing reactions.

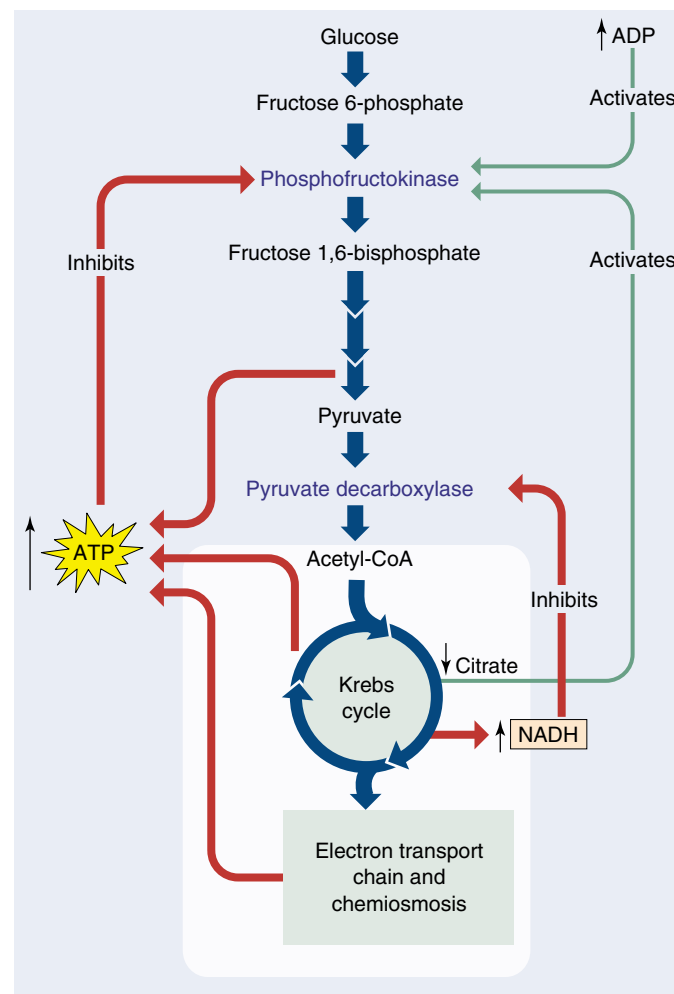


FIGURE 9.20
Control of glucose catabolism. The relative levels of ADP and ATP control the catabolic pathway at two key points: the committing reactions of glycolysis and the Krebs cycle.

A Vocabulary of ATP Generation

aerobic respiration The portion of cellular respiration that requires oxygen as an electron acceptor; it includes pyruvate oxidation, the Krebs cycle, and the electron transport chain.

anaerobic respiration Cellular respiration in which inorganic electron acceptors other than oxygen are used; it includes glycolysis.

cellular respiration The oxidation of organic molecules to produce ATP in

which the final electron acceptor is organic; it includes aerobic and anaerobic respiration.

chemiosmosis The passage of high-energy electrons along the electron transport chain, which is coupled to the pumping of protons across a membrane and the return of protons to the original side of the membrane through ATP-generating channels.

fermentation Alternative ATP-producing pathway performed by some cells in the absence of oxygen, in which the final electron acceptor is an organic molecule.

maximum efficiency The maximum number of ATP molecules generated by

oxidizing a substance, relative to the free energy of that substance; in organisms, the actual efficiency is usually less than the maximum.

oxidation The loss of an electron. In cellular respiration, high-energy electrons are stripped from food molecules, oxidizing them.

photosynthesis The chemiosmotic generation of ATP and complex organic molecules powered by the energy derived from light.

substrate-level phosphorylation The generation of ATP by the direct transfer of a phosphate group to ADP from another phosphorylated molecule.

9.3 Catabolism of proteins and fats can yield considerable energy.

Glucose Is Not the Only Source of Energy

Thus far we have discussed oxidative respiration of glucose, which organisms obtain from the digestion of carbohydrates or from photosynthesis. Other organic molecules than glucose, particularly proteins and fats, are also important sources of energy (figure 9.21).

Cellular Respiration of Protein

Proteins are first broken down into their individual amino acids. The nitrogen-containing side group (the amino group) is then removed from each amino acid in a process called **deamination**. A series of reactions convert the carbon chain that remains into a molecule that takes part in glycolysis or the Krebs cycle. For example, alanine is converted into pyruvate, glutamate into α -ketoglutarate (figure 9.22), and aspartate into oxaloacetate. The reactions of glycolysis and the Krebs cycle then extract the high-energy electrons from these molecules and put them to work making ATP.

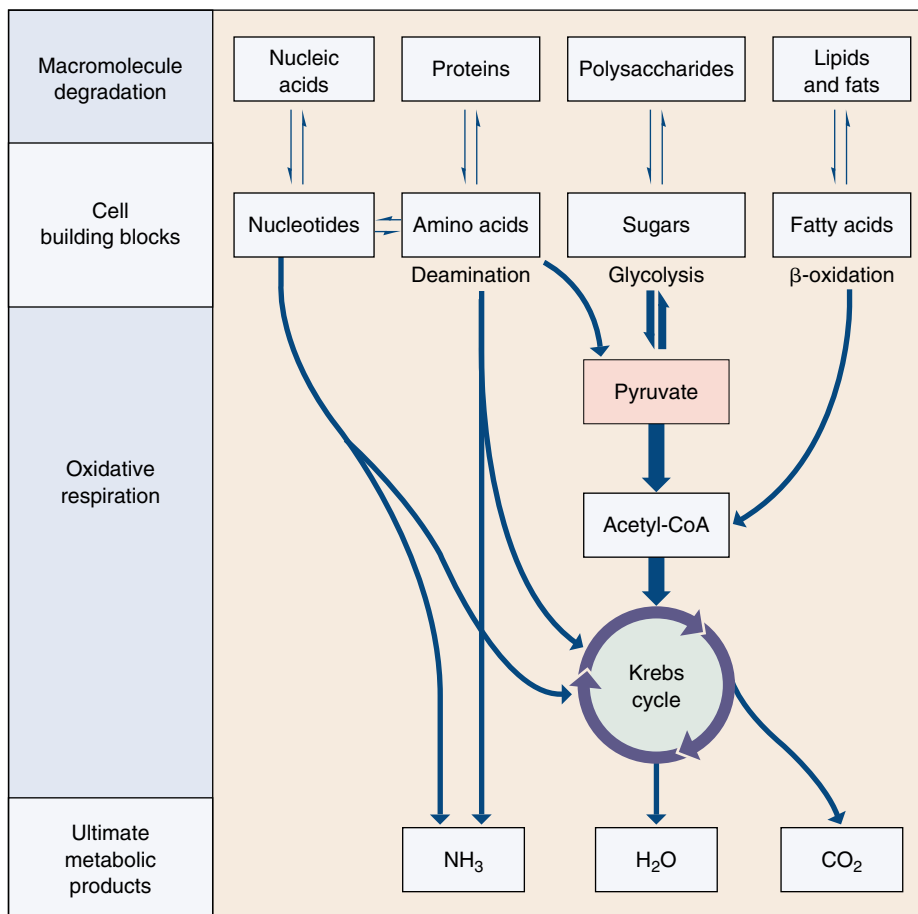
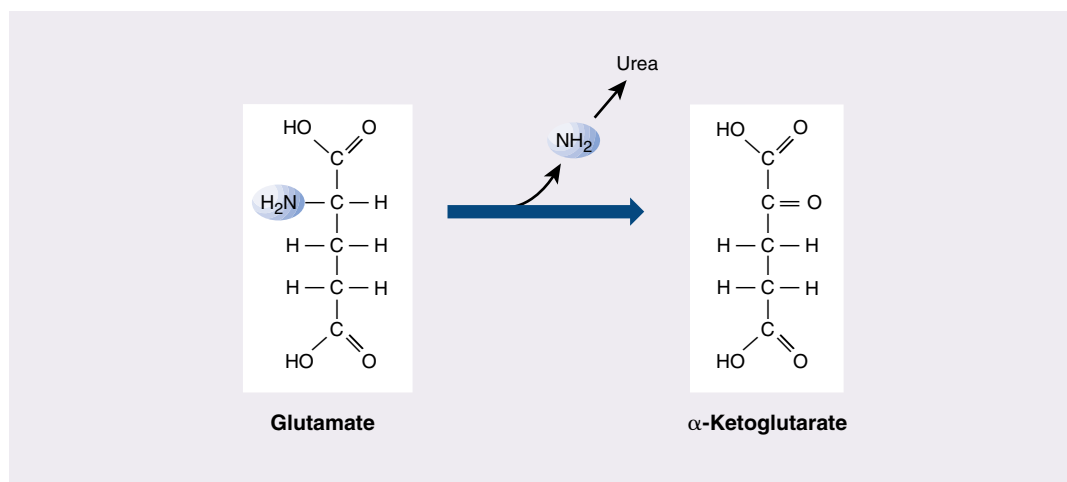


FIGURE 9.21

How cells extract chemical energy. All eukaryotes and many prokaryotes extract energy from organic molecules by oxidizing them. The first stage of this process, breaking down macromolecules into their constituent parts, yields little energy. The second stage, oxidative or aerobic respiration, extracts energy, primarily in the form of high-energy electrons, and produces water and carbon dioxide.

FIGURE 9.22

Deamination. After proteins are broken down into their amino acid constituents, the amino groups are removed from the amino acids to form molecules that participate in glycolysis and the Krebs cycle. For example, the amino acid glutamate becomes α -ketoglutarate, a Krebs cycle molecule, when it loses its amino group.



Cellular Respiration of Fat

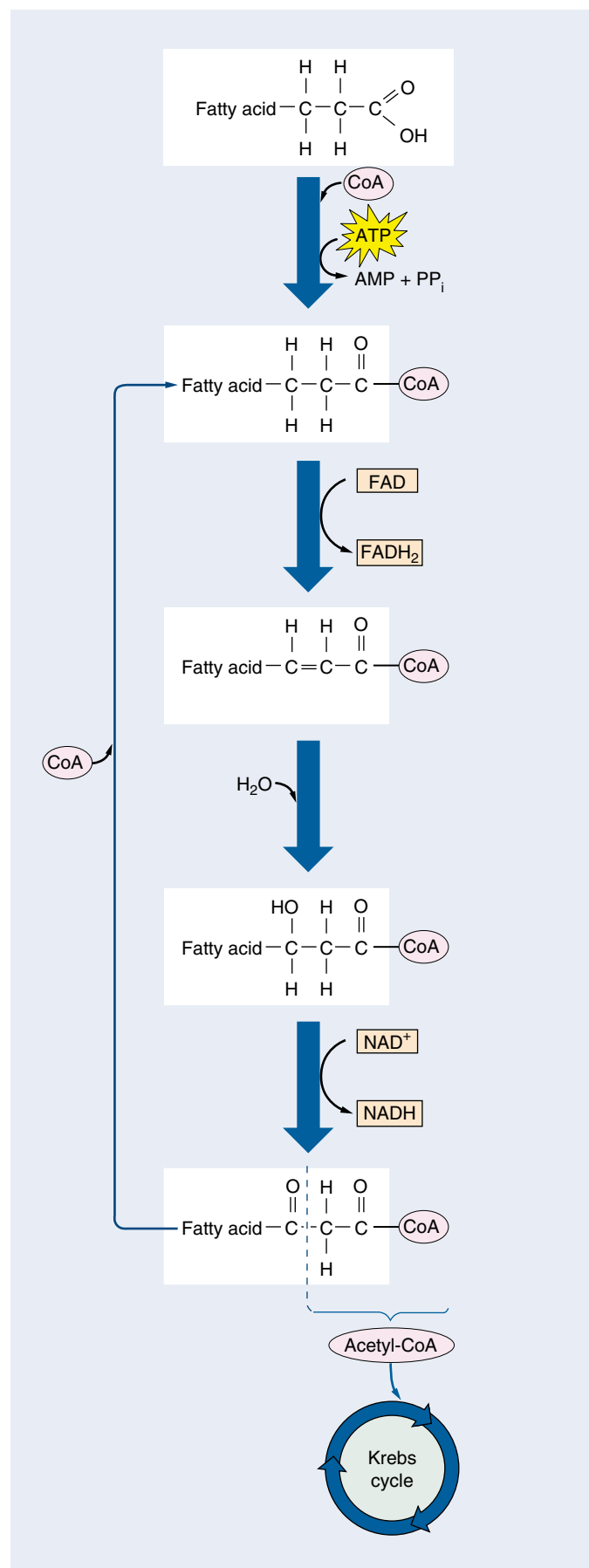
Fats are broken down into fatty acids plus glycerol. The tails of fatty acids typically have 16 or more —CH_2 links, and the many hydrogen atoms in these long tails provide a rich harvest of energy. Fatty acids are oxidized in the matrix of the mitochondrion. Enzymes there remove the two-carbon acetyl groups from the end of each fatty acid tail until the entire fatty acid is converted into acetyl groups (figure 9.23). Each acetyl group then combines with coenzyme A to form acetyl-CoA. This process is known as β -oxidation.

How much ATP does the catabolism of fatty acids produce? Let's compare a hypothetical six-carbon fatty acid with the six-carbon glucose molecule, which we've said yields about 30 molecules of ATP in a eukaryotic cell. Two rounds of β -oxidation would convert the fatty acid into three molecules of acetyl-CoA. Each round requires one molecule of ATP to prime the process, but it also produces one molecule of NADH and one of FADH_2 . These molecules together yield four molecules of ATP (assuming 2.5 ATPs per NADH and 1.5 ATPs per FADH_2). The oxidation of each acetyl-CoA in the Krebs cycle ultimately produces an additional 10 molecules of ATP. Overall, then, the ATP yield of a six-carbon fatty acid would be approximately 8 (from two rounds of β -oxidation) + 30 (from oxidizing the three acetyl-CoAs) = 36 molecules of ATP. Therefore, the respiration of a six-carbon fatty acid yields 20% more ATP than the respiration of glucose. Moreover, a fatty acid of that size would weigh less than two-thirds as much as glucose, so a gram of fatty acid contains more than twice as many kilocalories as a gram of glucose. That is why fat is a storage molecule for excess energy in many types of animals. If excess energy were stored instead as carbohydrate, as it is in plants, animal bodies would be much bulkier.

Proteins, fats, and other organic molecules are also metabolized for energy. The amino acids of proteins are first deaminated, while fats undergo a process called β -oxidation.

FIGURE 9.23

β -oxidation. Through a series of reactions known as β -oxidation, the last two carbons in a fatty acid tail combine with coenzyme A to form acetyl-CoA, which enters the Krebs cycle. The fatty acid, now two carbons shorter, enters the pathway again and keeps reentering until all its carbons have been used to form acetyl-CoA molecules. Each round of β -oxidation uses one molecule of ATP and generates one molecule each of FADH_2 and NADH, not including the molecules generated from the Krebs cycle.



Metabolic Efficiency and the Length of Food Chains

It has been estimated that a heterotroph limited to glycolysis captures only 3.5% of the energy in the food it consumes. Hence, if such a heterotroph preserves 3.5% of the energy in the autotrophs it consumes, then any other heterotrophs that consume the first heterotroph will capture through glycolysis 3.5% of the energy in it, or 0.12% of the energy available in the original autotrophs. A very large base of autotrophs would thus be needed to support a small number of heterotrophs.

When organisms became able to extract energy from organic molecules by oxidative metabolism, this constraint became far less severe, because the efficiency of oxidative respiration is estimated to be about 52 to 63%. This increased efficiency results in the transmission of much more energy from one trophic level to another than does glycolysis. (A trophic level is a step in the movement of energy through an ecosystem.) The efficiency of oxidative metabolism has made possible the evolution of food chains, in which autotrophs are consumed by heterotrophs, which are consumed by other heterotrophs, and so on. You will read more about food chains in chapter 28.

Even with oxidative metabolism, approximately two-thirds of the available energy is lost at each trophic level, and that puts a limit on how long a food chain can be. Most food chains, like the one illustrated in figure 9.A, involve only three or rarely four trophic levels. Too much energy is lost at each transfer to allow chains to be much longer than that. For example, it would be impossible for a large human population to subsist by eating lions captured from the Serengeti Plain of Africa; the amount of grass available there would not support enough zebras and other herbivores to maintain the number of lions needed to feed the human population. Thus, the ecological complexity of our world is fixed in a fundamental way by the chemistry of oxidative respiration.



Stage 1: Photosynthesizers



Stage 2: Herbivores



Stage 3: Carnivore



Stage 4: Scavengers



Stage 5: Refuse utilizers

FIGURE 9.A

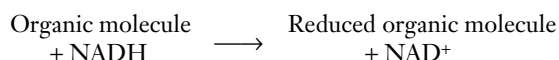
A food chain in the savannas, or open grasslands, of East Africa. Stage 1: *Photosynthesizer*. The grass under these palm trees grows actively during the hot, rainy season, capturing the energy of the sun and storing it in molecules of glucose, which are then converted into starch and stored in the grass. Stage 2: *Herbivore*. These large antelopes, known as wildebeests, consume the grass and transfer some of its stored energy into their own bodies. Stage 3: *Carnivore*. The lion feeds on wildebeests and other animals, capturing part of their stored energy and storing it in its own body. Stage 4: *Scavenger*. This hyena and the vultures occupy the same stage in the food chain as the lion. They are also consuming the body of the dead wildebeest, which has been abandoned by the lion. Stage 5: *Refuse utilizer*. These butterflies, mostly *Precis octavia*, are feeding on the material left in the hyena's dung after the food the hyena consumed had passed through its digestive tract. At each of these four levels, only about a third or less of the energy present is used by the recipient.

9.4 Cells can metabolize food without oxygen.

Fermentation

In the absence of oxygen, aerobic metabolism cannot occur, and cells must rely exclusively on glycolysis to produce ATP. Under these conditions, the hydrogen atoms generated by glycolysis are donated to organic molecules in a process called **fermentation**.

Bacteria carry out more than a dozen kinds of fermentations, all using some form of organic molecule to accept the hydrogen atom from NADH and thus recycle NAD^+ :



Often the reduced organic compound is an organic acid—such as acetic acid, butyric acid, propionic acid, or lactic acid—or an alcohol.

Ethanol Fermentation

Eukaryotic cells are capable of only a few types of fermentation. In one type, which occurs in single-celled fungi called yeast, the molecule that accepts hydrogen from NADH is pyruvate, the end product of glycolysis itself. Yeast enzymes remove a terminal CO_2 group from pyruvate through decarboxylation, producing a two-carbon molecule called acetaldehyde. The CO_2 released causes bread made with yeast to rise, while bread made without yeast (unleavened bread) does not. The acetaldehyde accepts a hydrogen atom from NADH, producing NAD^+ and **ethanol** (ethyl alcohol). This particular type of fermentation is of great interest to humans, since it is the source of the ethanol in wine and beer (figure 9.24). Ethanol is a by-product of fermentation that is actually toxic to yeast; as it approaches a concentration of about 12%, it begins to kill the yeast. That explains why naturally fermented wine contains only about 12% ethanol.

Lactic Acid Fermentation

Most animal cells regenerate NAD^+ without decarboxylation. Muscle cells, for example, use an enzyme called lactate dehydrogenase to transfer a hydrogen atom from NADH back to the pyruvate that is produced by glycolysis. This reaction converts pyruvate into lactic acid and regenerates NAD^+ from NADH. It therefore closes the metabolic circle, allowing glycolysis to continue as long as glucose is available. Circulating blood removes excess lactate (the ionized form of lactic acid) from muscles, but when removal cannot keep pace with production, the accumulating lactic acid interferes with muscle function and contributes to muscle fatigue.

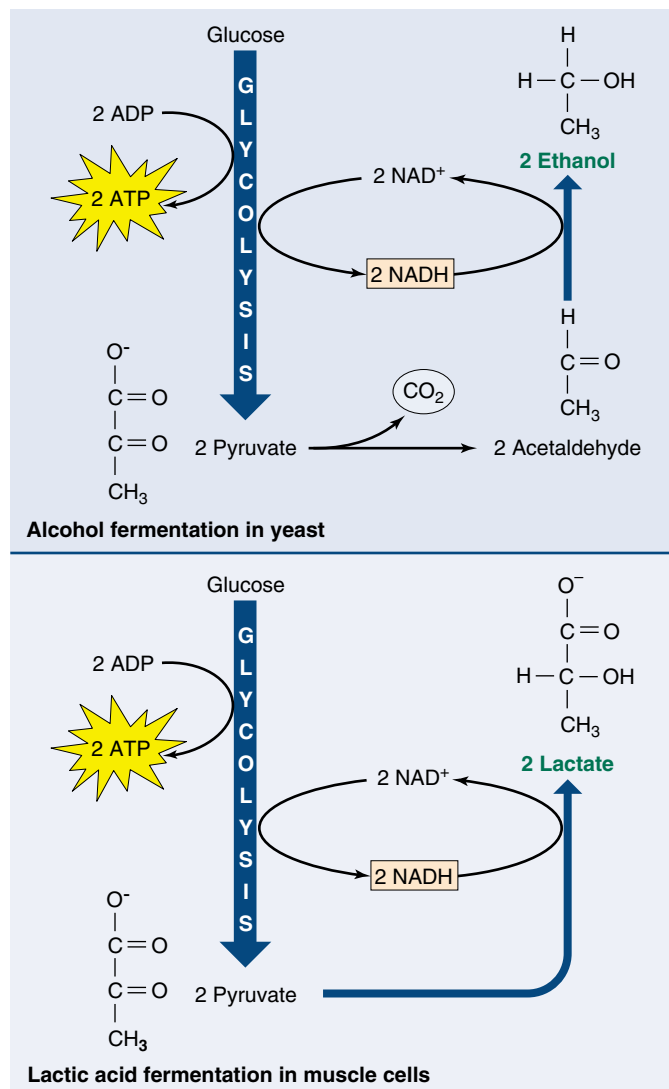


FIGURE 9.24

How wine is made. Yeasts carry out the conversion of pyruvate to ethanol. This takes place naturally in grapes left to ferment on vines, as well as in fermentation vats containing crushed grapes. When the ethanol concentration reaches about 12%, its toxic effects kill the yeast; what remains is wine. Muscle cells convert pyruvate into lactate, which is less toxic than ethanol. However, lactate is still toxic enough to produce a painful sensation in muscles during heavy exercise, when oxygen in the muscles is depleted.

In fermentation, which occurs in the absence of oxygen, the electrons that result from the glycolytic breakdown of glucose are donated to an organic molecule, regenerating NAD^+ from NADH.



Summary

Questions

Media Resources

9.1 Cells harvest the energy in chemical bonds.

- The reactions of cellular respiration are oxidation-reduction (redox) reactions. Those that require a net input of free energy are coupled to the cleavage of ATP, which releases free energy.
- The mechanics of cellular respiration are often dictated by electron behavior, which is in turn influenced by the presence of electron acceptors. Some atoms, such as oxygen, are very electronegative and thus behave as good oxidizing agents.

1. What is the difference between an autotroph and a heterotroph? How does each obtain energy?
2. What is the difference between digestion and catabolism? Which provides more energy?

9.2 Cellular respiration oxidizes food molecules.

- In eukaryotic cells, the oxidative respiration of pyruvate takes place within the matrix of mitochondria.
- The electrons generated in the process are passed along the electron transport chain, a sequence of electron carriers in the inner mitochondrial membrane.
- Some of the energy released by passage of electrons along the electron transport chain is used to pump protons out of the mitochondrial matrix. The reentry of protons into the matrix is coupled to the production of ATP. This process is called chemiosmosis. The ATP then leaves the mitochondrion by facilitated diffusion.

3. Where in a eukaryotic cell does glycolysis occur? What is the net production of ATP during glycolysis, and why is it different from the number of ATP molecules synthesized during glycolysis?
4. By what two mechanisms can the NADH that results from glycolysis be converted back into NAD⁺?
5. What is the theoretical maximum number of ATP molecules produced during the oxidation of a glucose molecule by these processes? Why is the actual number of ATP molecules produced usually lower than the theoretical maximum?



- Exploration: Oxidative Respiration
- Art Activity: Aerobic Cellular Respiration
- Art Activity: Organization of Cristae



- Electron Transport and ATP



- Glycolysis
- Krebs Cycle
- Electron Transport

9.3 Catabolism of proteins and fats can yield considerable energy.

- The catabolism of fatty acids begins with β -oxidation and provides more energy than the catabolism of carbohydrates.

6. How is acetyl-CoA produced during the aerobic oxidation of carbohydrates, and what happens to it? How is it produced during the aerobic oxidation of fatty acids, and what happens to it then?

9.4 Cells can metabolize food without oxygen.

- Fermentation is an anaerobic process that uses an organic molecule instead of oxygen as a final electron acceptor.
- It occurs in bacteria as well as eukaryotic cells, including yeast and the muscle cells of animals.

7. How do the amounts of ATP produced by the aerobic oxidation of glucose and fatty acids compare? Which type of substance contains more energy on a per-weight basis?



- Fermentation

10

Photosynthesis

Concept Outline

10.1 What is photosynthesis?

The Chloroplast as a Photosynthetic Machine. The highly organized system of membranes in chloroplasts is essential to the functioning of photosynthesis.

10.2 Learning about photosynthesis: An experimental journey.

The Role of Soil and Water. The added mass of a growing plant comes mostly from photosynthesis. In plants, water supplies the electrons used to reduce carbon dioxide.

Discovery of the Light-Independent Reactions. Photosynthesis is a two-stage process. Only the first stage directly requires light.

The Role of Light. The oxygen released during green plant photosynthesis comes from water, and carbon atoms from carbon dioxide are incorporated into organic molecules.

The Role of Reducing Power. Electrons released from the splitting of water reduce NADP^+ ; ATP and NADPH are then used to reduce CO_2 and form simple sugars.

10.3 Pigments capture energy from sunlight.

The Biophysics of Light. The energy in sunlight occurs in “packets” called photons, which are absorbed by pigments.

Chlorophylls and Carotenoids. Photosynthetic pigments absorb light and harvest its energy.

Organizing Pigments into Photosystems. A photosystem uses light energy to eject an energized electron.

How Photosystems Convert Light to Chemical Energy. Some bacteria rely on a single photosystem to produce ATP. Plants use two photosystems in series to generate enough energy to reduce NADP^+ and generate ATP.

How the Two Photosystems of Plants Work Together. Photosystems II and I drive the synthesis of the ATP and NADPH needed to form organic molecules.

10.4 Cells use the energy and reducing power captured by the light reactions to make organic molecules.

The Calvin Cycle. ATP and NADPH are used to build organic molecules, a process reversed in mitochondria.

Reactions of the Calvin Cycle. Ribulose biphosphate binds CO_2 in the process of carbon fixation.

Photorespiration. The enzyme that catalyzes carbon fixation also affects CO_2 release.



FIGURE 10.1

Capturing energy. These sunflower plants, growing vigorously in the August sun, are capturing light energy for conversion into chemical energy through photosynthesis.

Life on earth would be impossible without photosynthesis. Every oxygen atom in the air we breathe was once part of a water molecule, liberated by photosynthesis. The energy released by the burning of coal, firewood, gasoline, and natural gas, and by our bodies' burning of all the food we eat—all, directly or indirectly, has been captured from sunlight by photosynthesis. It is vitally important that we understand photosynthesis. Research may enable us to improve crop yields and land use, important goals in an increasingly crowded world. In the previous chapter we described how cells extract chemical energy from food molecules and use that energy to power their activities. In this chapter, we will examine photosynthesis, the process by which organisms capture energy from sunlight and use it to build food molecules rich in chemical energy (figure 10.1).

10.1 What is photosynthesis?

The Chloroplast as a Photosynthetic Machine

Life is powered by sunshine. The energy used by most living cells comes ultimately from the sun, captured by plants, algae, and bacteria through the process of photosynthesis. The diversity of life is only possible because our planet is awash in energy streaming earthward from the sun. Each day, the radiant energy that reaches the earth equals about 1 million Hiroshima-sized atomic bombs. Photosynthesis captures about 1% of this huge supply of energy, using it to provide the energy that drives all life.

The Photosynthetic Process: A Summary

Photosynthesis occurs in many kinds of bacteria and algae, and in the leaves and sometimes the stems of green plants. Figure 10.2 describes the levels of organization in a plant leaf. Recall from chapter 5 that the cells of plant leaves contain organelles called chloroplasts that actually carry out the photosynthetic process. No other structure in a plant cell is able to carry out photosynthesis. Photosynthe-

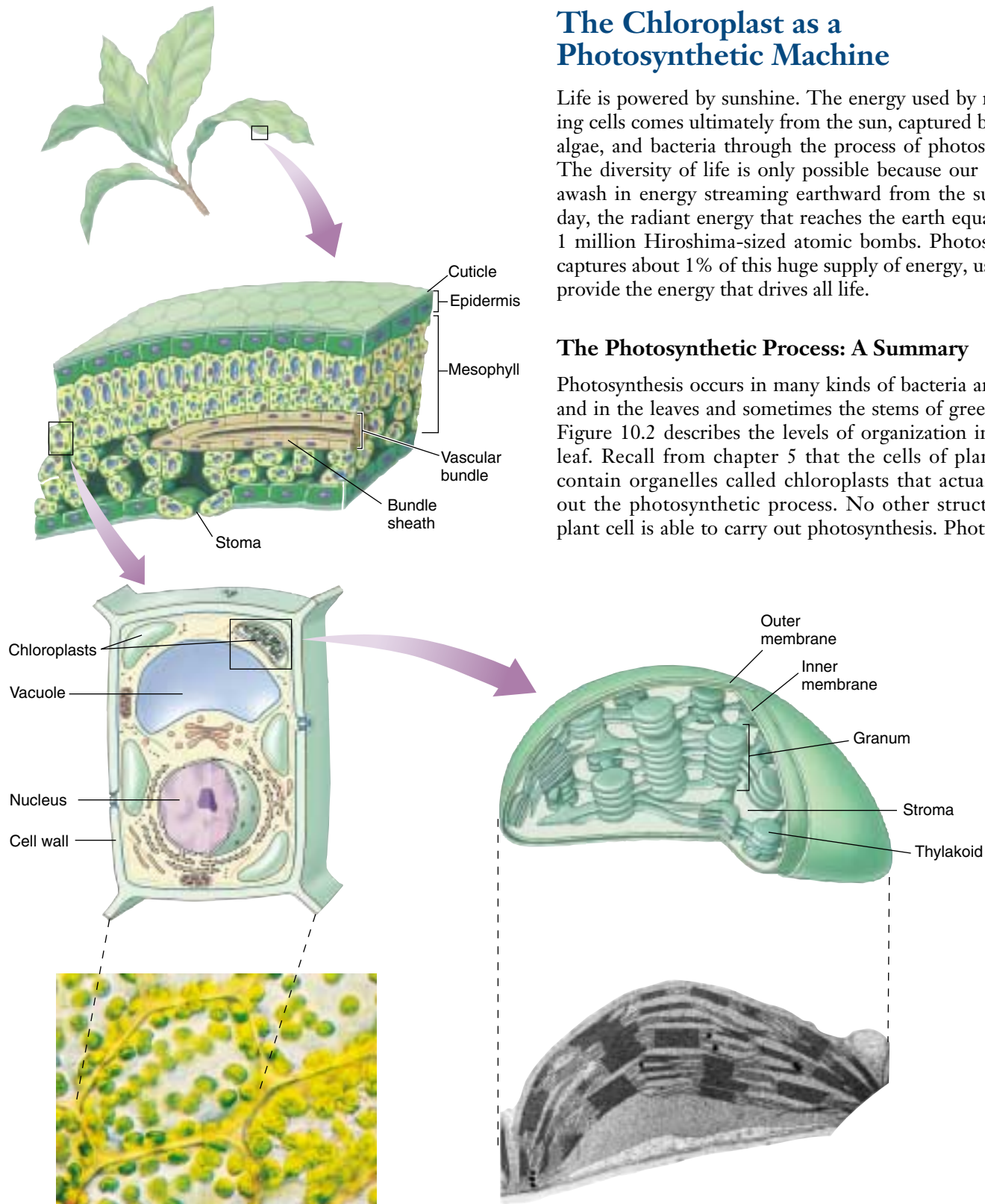


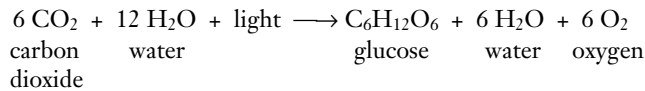
FIGURE 10.2

Journey into a leaf. A plant leaf possesses a thick layer of cells (the mesophyll) rich in chloroplasts. The flattened thylakoids in the chloroplast are stacked into columns called grana (singular, granum). The light reactions take place on the thylakoid

sis takes place in three stages: (1) capturing energy from sunlight; (2) using the energy to make ATP and reducing power in the form of a compound called NADPH; and (3) using the ATP and NADPH to power the synthesis of organic molecules from CO₂ in the air (carbon fixation).

The first two stages take place in the presence of light and are commonly called the **light reactions**. The third stage, the formation of organic molecules from atmospheric CO₂, is called the **Calvin cycle**. As long as ATP and NADPH are available, the Calvin cycle may occur in the absence of light.

The following simple equation summarizes the overall process of photosynthesis:



Inside the Chloroplast

The internal membranes of chloroplasts are organized into sacs called *thylakoids*, and often numerous thylakoids are stacked on one another in columns called *grana*. The thylakoid membranes house the photosynthetic pigments for capturing light energy and the machinery to make ATP. Surrounding the thylakoid membrane system is a semiliquid substance called *stroma*. The stroma houses the enzymes needed to assemble carbon molecules. In the mem-

branes of thylakoids, photosynthetic pigments are clustered together to form a **photosystem**.

Each pigment molecule within the photosystem is capable of capturing *photons*, which are packets of energy. A lattice of proteins holds the pigments in close contact with one another. When light of a proper wavelength strikes a pigment molecule in the photosystem, the resulting excitation passes from one chlorophyll molecule to another. The excited electron is not transferred physically—it is the *energy* that passes from one molecule to another. A crude analogy to this form of energy transfer is the initial “break” in a game of pool. If the cue ball squarely hits the point of the triangular array of 15 pool balls, the two balls at the far corners of the triangle fly off, but none of the central balls move. The energy passes through the central balls to the most distant ones.

Eventually the energy arrives at a key chlorophyll molecule that is touching a membrane-bound protein. The energy is transferred as an excited electron to that protein, which passes it on to a series of other membrane proteins that put the energy to work making ATP and NADPH and building organic molecules. The photosystem thus acts as a large antenna, gathering the light harvested by many individual pigment molecules.

The reactions of photosynthesis take place within thylakoid membranes within chloroplasts in leaf cells.

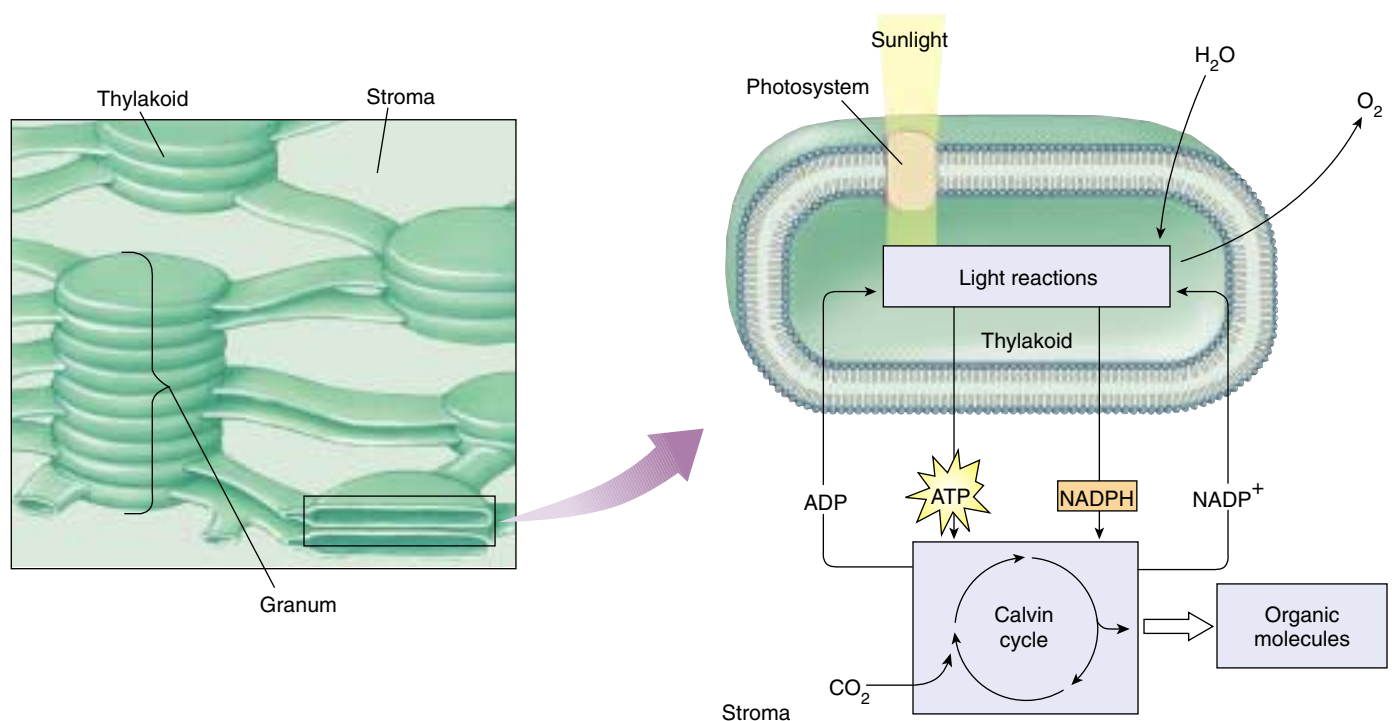


FIGURE 10.2 (continued)

membrane and generate the ATP and NADPH that fuel the Calvin cycle. The fluid interior matrix of a chloroplast, the stroma, contains the enzymes that carry out the Calvin cycle.

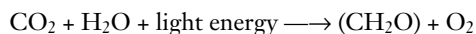
10.2 Learning about photosynthesis: An experimental journey.

The Role of Soil and Water

The story of how we learned about photosynthesis is one of the most interesting in science and serves as a good introduction to this complex process. The story starts over 300 years ago, with a simple but carefully designed experiment by a Belgian doctor, Jan Baptista van Helmont (1577–1644). From the time of the Greeks, plants were thought to obtain their food from the soil, literally sucking it up with their roots; van Helmont thought of a simple way to test the idea. He planted a small willow tree in a pot of soil after weighing the tree and the soil. The tree grew in the pot for several years, during which time van Helmont added only water. At the end of five years, the tree was much larger: its weight had increased by 74.4 kilograms. However, *all of this added mass could not have come from the soil*, because the soil in the pot weighed only 57 grams less than it had five years earlier! With this experiment, van Helmont demonstrated that the substance of the plant was not produced only from the soil. He incorrectly concluded that mainly the water he had been adding accounted for the plant's increased mass.

A hundred years passed before the story became clearer. The key clue was provided by the English scientist Joseph Priestly, in his pioneering studies of the properties of air. On the 17th of August, 1771, Priestly “accidentally hit upon a method of restoring air that had been injured by the burning of candles.” He “put a [living] sprig of mint into air in which a wax candle had burnt out and found that, on the 27th of the same month, another candle could be burned in this same air.” Somehow, the vegetation seemed to have restored the air! Priestly found that while a mouse could not breathe candle-exhausted air, air “restored” by vegetation was not “at all inconvenient to a mouse.” The key clue was that living vegetation *adds something to the air*.

How does vegetation “restore” air? Twenty-five years later, Dutch physician Jan Ingenhousz solved the puzzle. Working over several years, Ingenhousz reproduced and significantly extended Priestly's results, demonstrating that air was restored only in the presence of sunlight, and only by a plant's green leaves, not by its roots. He proposed that the green parts of the plant carry out a process (which we now call photosynthesis) that uses sunlight to split carbon dioxide (CO_2) into carbon and oxygen. He suggested that the oxygen was released as O_2 gas into the air, while the carbon atom combined with water to form carbohydrates. His proposal was a good guess, even though the later step was subsequently modified. Chemists later found that the proportions of carbon, oxygen, and hydrogen atoms in carbohydrates are indeed about one atom of carbon per molecule of water (as the term *carbohydrate* indicates). A Swiss botanist found in 1804 that water was a necessary reactant. By the end of that century the overall reaction for photosynthesis could be written as:



It turns out, however, that there's more to it than that. When researchers began to examine the process in more detail in the last century, the role of light proved to be unexpectedly complex.

Van Helmont showed that soil did not add mass to a growing plant. Priestly and Ingenhousz and others then worked out the basic chemical reaction.

Discovery of the Light-Independent Reactions

Ingenhousz's early equation for photosynthesis includes one factor we have not discussed: light energy. What role does light play in photosynthesis? At the beginning of the previous century, the English plant physiologist F. F. Blackman began to address the question of the role of light in photosynthesis. In 1905, he came to the startling conclusion that photosynthesis is in fact a two-stage process, only one of which uses light directly.

Blackman measured the effects of different light intensities, CO_2 concentrations, and temperatures on photosynthesis. As long as light intensity was relatively low, he found photosynthesis could be accelerated by increasing the amount of light, but not by increasing the temperature or CO_2 concentration (figure 10.3). At high light intensities, however, an increase in temperature or CO_2 concentration greatly accelerated photosynthesis. Blackman concluded that photosynthesis consists of an initial set of what he called “light” reactions, that are largely independent of temperature, and a second set of “dark” reactions, that seemed to be independent of light but limited by CO_2 . Do not be confused by Blackman's labels—the so-called “dark” reactions occur in the light (in fact, they require the products of the light reactions); their name simply indicates that light is not directly involved in those reactions.

Blackman found that increased temperature increases the rate of the dark carbon-reducing reactions, but only up to about 35°C . Higher temperatures caused the rate to fall off rapidly. Because 35°C is the temperature at which many plant enzymes begin to be denatured (the hydrogen bonds that hold an enzyme in its particular catalytic shape begin to be disrupted), Blackman concluded that enzymes must carry out the dark reactions.

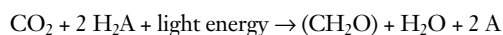
Blackman showed that capturing photosynthetic energy requires sunlight, while building organic molecules does not.

The Role of Light

The role of light in the so-called light and dark reactions was worked out in the 1930s by C. B. van Niel, then a graduate student at Stanford University studying photosynthesis in bacteria. One of the types of bacteria he was studying, the purple sulfur bacteria, does not release oxygen during photosynthesis; instead, they convert hydrogen sulfide (H_2S) into globules of pure elemental sulfur that accumulate inside themselves. The process that van Niel observed was

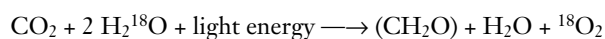


The striking parallel between this equation and Ingenhousz's equation led van Niel to propose that the generalized process of photosynthesis is in fact

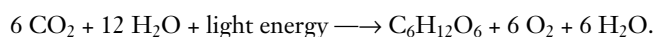


In this equation, the substance H_2A serves as an electron donor. In photosynthesis performed by green plants, H_2A is water, while among purple sulfur bacteria, H_2A is hydrogen sulfide. The product, A, comes from the splitting of H_2A . Therefore, the O_2 produced during green plant photosynthesis results from splitting water, not carbon dioxide.

When isotopes came into common use in biology in the early 1950s, it became possible to test van Niel's revolutionary proposal. Investigators examined photosynthesis in green plants supplied with ^{18}O water; they found that the ^{18}O label ended up in oxygen gas rather than in carbohydrate, just as van Niel had predicted:



In algae and green plants, the carbohydrate typically produced by photosynthesis is the sugar glucose, which has six carbons. The complete balanced equation for photosynthesis in these organisms thus becomes

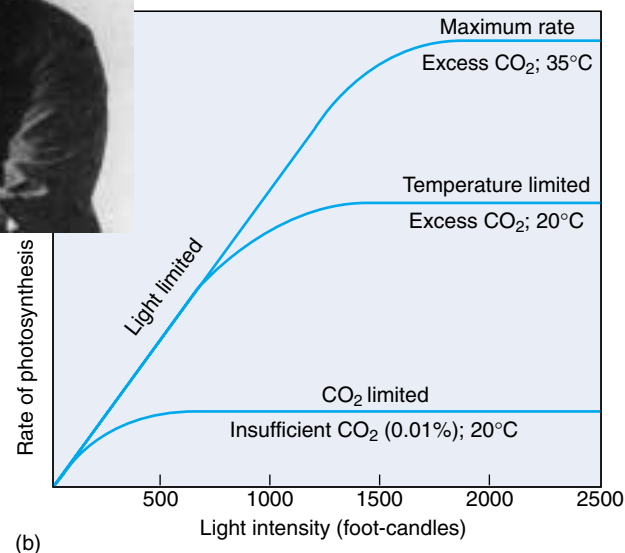


We now know that the first stage of photosynthesis, the light reactions, uses the energy of light to reduce NADP (an electron carrier molecule) to NADPH and to manufacture ATP. The NADPH and ATP from the first stage of photosynthesis are then used in the second stage, the Calvin cycle, to reduce the carbon in carbon dioxide and form a simple sugar whose carbon skeleton can be used to synthesize other organic molecules.

Van Niel discovered that photosynthesis splits water molecules, incorporating the carbon atoms of carbon dioxide gas and the hydrogen atoms of water into organic molecules and leaving oxygen gas.



(a)



(b)

FIGURE 10.3

Discovery of the dark reactions. (a) Blackman measured photosynthesis rates under differing light intensities, CO_2 concentrations, and temperatures. (b) As this graph shows, light is the limiting factor at low light intensities, while temperature and CO_2 concentration are the limiting factors at higher light intensities.

The Role of Reducing Power

In his pioneering work on the light reactions, van Niel had further proposed that the reducing power (H^+) generated by the splitting of water was used to convert CO_2 into organic matter in a process he called **carbon fixation**. Was he right?

In the 1950s Robin Hill demonstrated that van Niel was indeed right, and that light energy could be used to generate reducing power. Chloroplasts isolated from leaf cells were able to reduce a dye and release oxygen in response to light. Later experiments showed that the electrons released from water were transferred to NADP $^+$. Arnon and coworkers showed that illuminated chloroplasts deprived of CO_2 accumulate ATP. If CO_2 is then introduced, neither ATP nor NADPH accumulate, and the CO_2 is assimilated into organic molecules. These experiments are important for three reasons. First, they firmly demonstrate that photosynthesis occurs only within chloroplasts. Second, they show that the light-dependent reactions use light energy to reduce NADP $^+$ and to manufacture ATP. Thirdly, they confirm that the ATP and NADPH from this early stage of photosynthesis are then used in the later light-independent reactions to reduce carbon dioxide, forming simple sugars.

Hill showed that plants can use light energy to generate reducing power. The incorporation of carbon dioxide into organic molecules in the light-independent reactions is called carbon fixation.

10.3 Pigments capture energy from sunlight.

The Biophysics of Light

Where is the energy in light? What is there in sunlight that a plant can use to reduce carbon dioxide? This is the mystery of photosynthesis, the one factor fundamentally different from processes such as respiration. To answer these questions, we will need to consider the physical nature of light itself. James Clerk Maxwell had theorized that light was an electromagnetic wave—that is, that light moved through the air as oscillating electric and magnetic fields. Proof of this came in a curious experiment carried out in a laboratory in Germany in 1887. A young physicist, Heinrich Hertz, was attempting to verify a highly mathematical theory that predicted the existence of electromagnetic waves. To see whether such waves existed, Hertz designed a clever experiment. On one side of a room he constructed a powerful spark generator that consisted of two large, shiny metal spheres standing near each other on tall, slender rods. When a very high static electrical charge was built up on one sphere, sparks would jump across to the other sphere.

After constructing this device, Hertz set out to investigate whether the sparking would create invisible electromagnetic waves, so-called radio waves, as predicted by the mathematical theory. On the other side of the room, he placed the world's first radio receiver, a thin metal hoop on an insulating stand. There was a small gap at the bottom of the hoop, so that the hoop did not quite form a complete circle. When Hertz turned on the spark generator across the room, he saw tiny sparks passing across the gap in the hoop! This was the first demonstration of radio waves. But Hertz noted another curious phenomenon. When UV light was shining across the gap on the hoop, the sparks were produced more readily. This unexpected facilitation, called the photoelectric effect, puzzled investigators for many years.

The photoelectric effect was finally explained using a concept proposed by Max Planck in 1901. Planck developed an equation that predicted the blackbody radiation curve based upon the assumption that light and other forms of radiation behaved as units of energy called photons. In 1905 Albert Einstein explained the photoelectric effect utilizing the photon concept. Ultraviolet light has photons of sufficient energy that when they fell on the loop, electrons were ejected from the metal surface. The photons had transferred their energy to the electrons, literally blasting them from the ends of the hoop and thus facilitating the

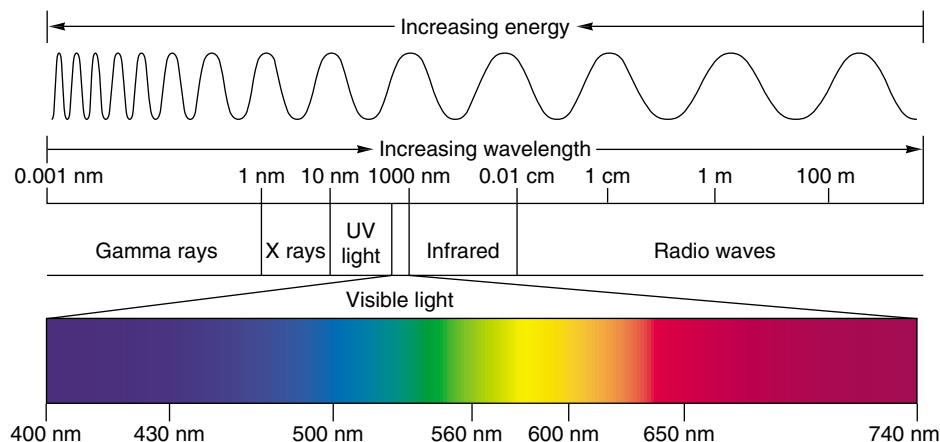


FIGURE 10.4

The electromagnetic spectrum. Light is a form of electromagnetic energy conveniently thought of as a wave. The shorter the wavelength of light, the greater its energy. Visible light represents only a small part of the electromagnetic spectrum between 400 and 740 nanometers.

passage of the electric spark induced by the radio waves. Visible wavelengths of light were unable to remove the electrons because their photons did not have enough energy to free the electrons from the metal surface at the ends of the hoop.

The Energy in Photons

Photons do not all possess the same amount of energy (figure 10.4). Instead, the energy content of a photon is inversely proportional to the wavelength of the light: short-wavelength light contains photons of higher energy than long-wavelength light. X rays, which contain a great deal of energy, have very short wavelengths—much shorter than visible light, making them ideal for high-resolution microscopes.

Hertz had noted that the strength of the photoelectric effect depends on the wavelength of light; short wavelengths are much more effective than long ones in producing the photoelectric effect. Einstein's theory of the photoelectric effect provides an explanation: sunlight contains photons of many different energy levels, only some of which our eyes perceive as visible light. The highest energy photons, at the short-wavelength end of the electromagnetic spectrum (see figure 10.4), are gamma rays, with wavelengths of less than 1 nanometer; the lowest energy photons, with wavelengths of up to thousands of meters, are radio waves. Within the visible portion of the spectrum, violet light has the shortest wavelength and the most energetic photons, and red light has the longest wavelength and the least energetic photons.

Ultraviolet Light

The sunlight that reaches the earth's surface contains a significant amount of ultraviolet (UV) light, which, because of its shorter wavelength, possesses considerably more energy than visible light. UV light is thought to have been an important source of energy on the primitive earth when life originated. Today's atmosphere contains ozone (derived from oxygen gas), which absorbs most of the UV photons in sunlight, but a considerable amount of UV light still manages to penetrate the atmosphere. This UV light is a potent force in disrupting the bonds of DNA, causing mutations that can lead to skin cancer. As we will describe in a later chapter, loss of atmospheric ozone due to human activities threatens to cause an enormous jump in the incidence of human skin cancers throughout the world.

Absorption Spectra and Pigments

How does a molecule “capture” the energy of light? A photon can be envisioned as a very fast-moving packet of energy. When it strikes a molecule, its energy is either lost as heat or absorbed by the electrons of the molecule, boosting those electrons into higher energy levels. Whether or not the photon's energy is absorbed depends on how much energy it carries (defined by its wavelength) and on the chemical nature of the molecule it hits. As we saw in chapter 2, electrons occupy discrete energy levels in their orbits around atomic nuclei. To boost an electron into a different energy level requires just the right amount of energy, just as reaching the next rung on a ladder requires you to raise your foot just the right distance. A specific atom can, therefore, absorb only certain photons of light—namely, those that correspond to the atom's available electron energy levels. As a result, each molecule has a characteristic **absorption spectrum**, the range and efficiency of photons it is capable of absorbing.

Molecules that are good absorbers of light in the visible range are called **pigments**. Organisms have evolved a variety of different pigments, but there are only two general types used in green plant photosynthesis: carotenoids and chlorophylls. Chlorophylls absorb photons within narrow energy ranges. Two kinds of chlorophyll in plants, chlorophylls *a* and *b*, preferentially absorb violet-blue and red light (figure 10.5). Neither of these pigments absorbs photons with wavelengths between about 500 and 600 nanometers, and light of these wavelengths is, therefore, reflected by plants. When these photons are subsequently

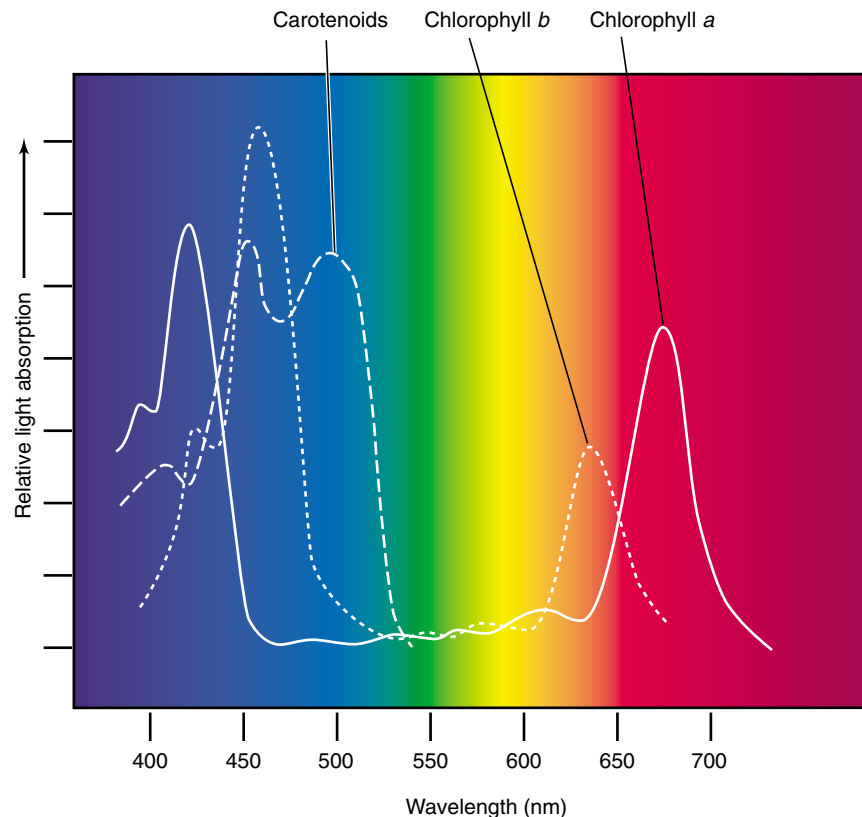


FIGURE 10.5

The absorption spectrum of chlorophyll. The peaks represent wavelengths of sunlight that the two common forms of photosynthetic pigment, chlorophyll *a* (solid line) and chlorophyll *b* (dashed line), strongly absorb. These pigments absorb predominately violet-blue and red light in two narrow bands of the spectrum and reflect the green light in the middle of the spectrum. Carotenoids (not shown here) absorb mostly blue and green light and reflect orange and yellow light.

absorbed by the pigment in our eyes, we perceive them as green.

Chlorophyll *a* is the main photosynthetic pigment and is the only pigment that can act directly to convert light energy to chemical energy. However, chlorophyll *b*, acting as an **accessory** or secondary light-absorbing pigment, complements and adds to the light absorption of chlorophyll *a*. Chlorophyll *b* has an absorption spectrum shifted toward the green wavelengths. Therefore, chlorophyll *b* can absorb photons chlorophyll *a* cannot. Chlorophyll *b* therefore greatly increases the proportion of the photons in sunlight that plants can harvest. An important group of accessory pigments, the carotenoids, assist in photosynthesis by capturing energy from light of wavelengths that are not efficiently absorbed by either chlorophyll.

In photosynthesis, photons of light are absorbed by pigments; the wavelength of light absorbed depends upon the specific pigment.

Chlorophylls and Carotenoids

Chlorophylls absorb photons by means of an excitation process analogous to the photoelectric effect. These pigments contain a complex ring structure, called a porphyrin ring, with alternating single and double bonds. At the center of the ring is a magnesium atom. Photons absorbed by the pigment molecule excite electrons in the ring, which are then channeled away through the alternating carbon-bond system. Several small side groups attached to the outside of the ring alter the absorption properties of the molecule in different kinds of chlorophyll (figure 10.6). The precise absorption spectrum is also influenced by the local microenvironment created by the association of chlorophyll with specific proteins.

Once Ingenhousz demonstrated that only the green parts of plants can “restore” air, researchers suspected chlorophyll was the primary pigment that plants employ to absorb light in photosynthesis. Experiments conducted in the 1800s clearly verified this suspicion. One such experiment, performed by T. W. Englemann in 1882 (figure 10.7), serves as a particularly elegant example, simple in design and clear in outcome. Englemann set out to characterize the **action spectrum** of photosynthesis, that is, the relative effectiveness of different wavelengths of light in promoting photosynthesis. He carried out the entire experiment utilizing a single slide mounted on a microscope. To obtain different

wavelengths of light, he placed a prism under his microscope, splitting the light that illuminated the slide into a spectrum of colors. He then arranged a filament of green algal cells across the spectrum, so that different parts of the filament were illuminated with different wavelengths, and allowed the algae to carry out photosynthesis. To assess how fast photosynthesis was proceeding, Englemann chose to monitor the rate of oxygen production. Lacking a mass spectrometer and other modern instruments, he added aerotactic (oxygen-seeking) bacteria to the slide; he knew they would gather along the filament at locations where oxygen was being produced. He found that the bacteria accumulated in areas illuminated by red and violet light, the two colors most strongly absorbed by chlorophyll.

All plants, algae, and cyanobacteria use chlorophyll *a* as their primary pigments. It is reasonable to ask why these photosynthetic organisms do not use a pigment like retinal (the pigment in our eyes), which has a broad absorption spectrum that covers the range of 500 to 600 nanometers. The most likely hypothesis involves *photoefficiency*. Although retinal absorbs a broad range of wavelengths, it does so with relatively low efficiency. Chlorophyll, in contrast, absorbs in only two narrow bands, but does so with high efficiency. Therefore, plants and most other photosynthetic organisms achieve far higher overall photon capture rates with chlorophyll than with other pigments.

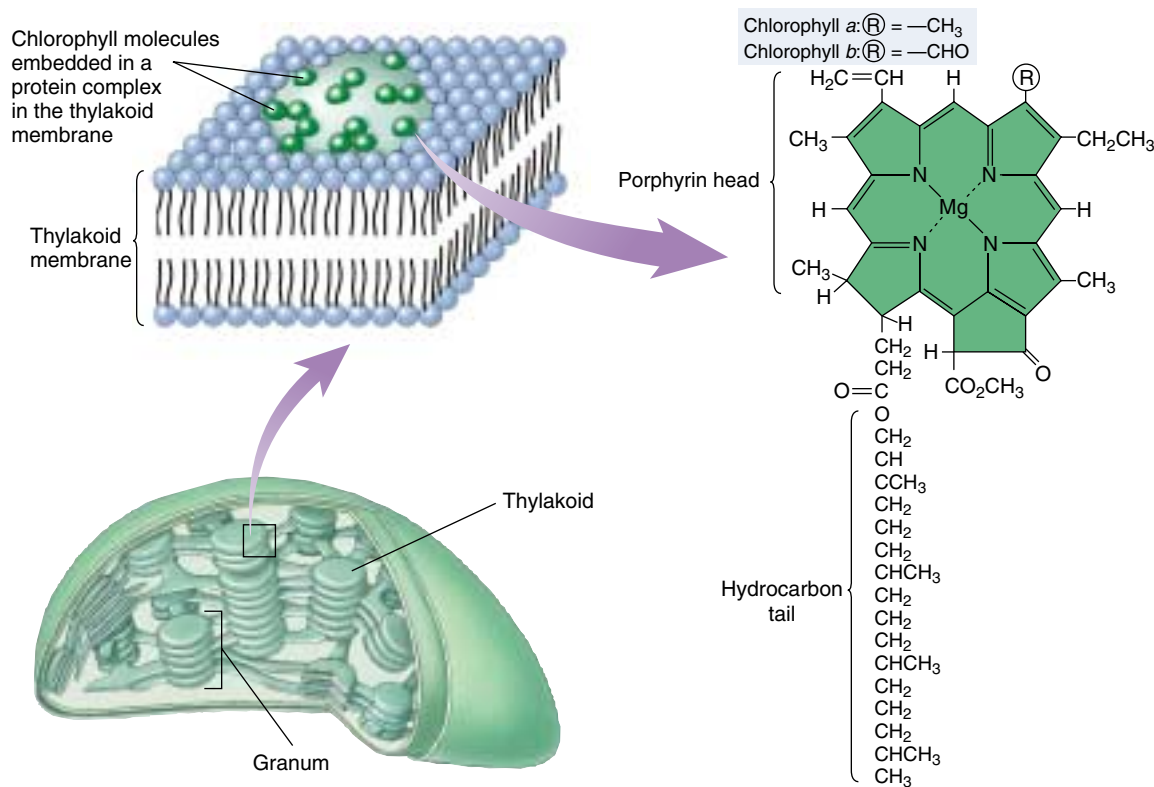


FIGURE 10.6
Chlorophyll.

Chlorophyll molecules consist of a porphyrin head and a hydrocarbon tail that anchors the pigment molecule to hydrophobic regions of proteins embedded within the membranes of thylakoids. The only difference between the two chlorophyll molecules is the substitution of a —CHO (aldehyde) group in chlorophyll *b* for a —CH₃ (methyl) group in chlorophyll *a*.

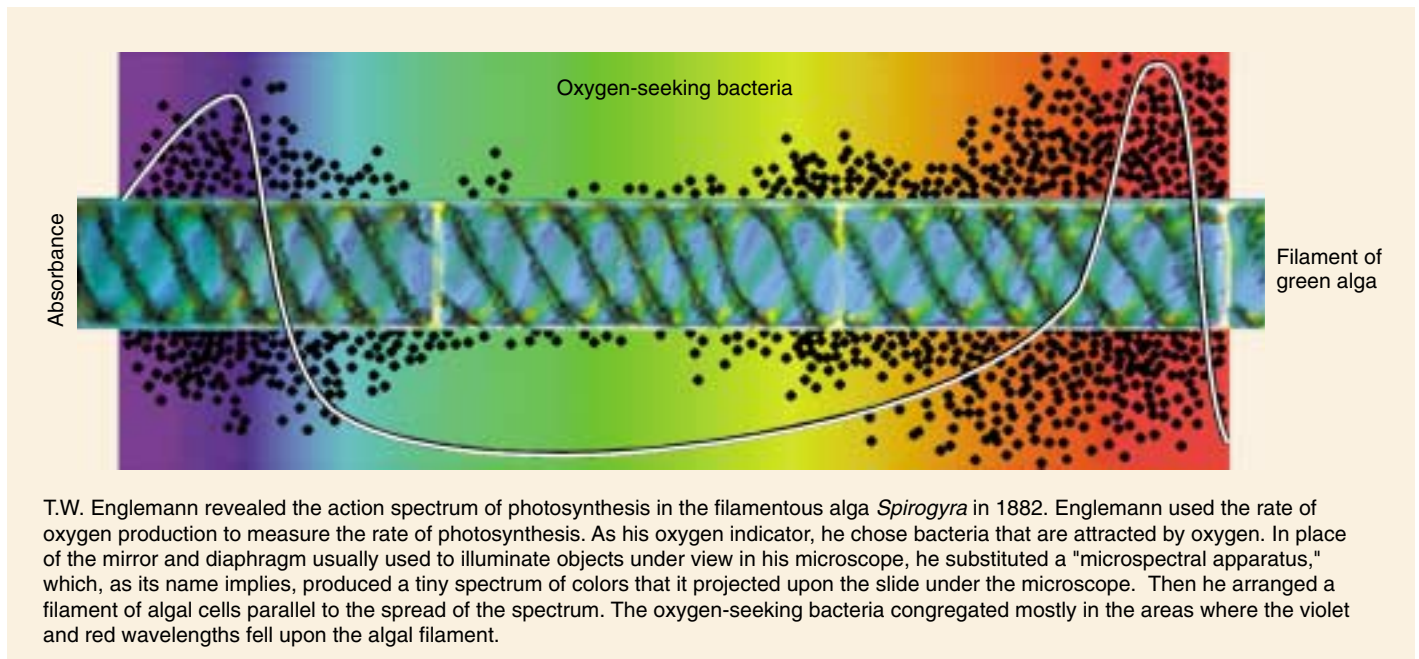


FIGURE 10.7

Constructing an action spectrum for photosynthesis. As you can see, the action spectrum for photosynthesis that Englemann revealed in his experiment parallels the absorption spectrum of chlorophyll (see figure 10.5).



FIGURE 10.8

Fall colors are produced by carotenoids and other accessory pigments. During the spring and summer, chlorophyll in leaves masks the presence of carotenoids and other accessory pigments. When cool fall temperatures cause leaves to cease manufacturing chlorophyll, the chlorophyll is no longer present to reflect green light, and the leaves reflect the orange and yellow light that carotenoids and other pigments do not absorb.

Carotenoids consist of carbon rings linked to chains with alternating single and double bonds. They can absorb photons with a wide range of energies, although they are not always highly efficient in transferring this energy. Carotenoids assist in photosynthesis by capturing energy from light of wavelengths that are not efficiently absorbed by chlorophylls (figure 10.8; see figure 10.5).

A typical carotenoid is β -carotene, whose two carbon rings are connected by a chain of 18 carbon atoms with alternating single and double bonds. Splitting a molecule of

β -carotene into equal halves produces two molecules of vitamin A. Oxidation of vitamin A produces retinal, the pigment used in vertebrate vision. This explains why carrots, which are rich in β -carotene, enhance vision.

A pigment is a molecule that absorbs light. The wavelengths absorbed by a particular pigment depend on the available energy levels to which light-excited electrons can be boosted in the pigment.

Organizing Pigments into Photosystems

The light reactions of photosynthesis occur in membranes. In bacteria like those studied by van Niel, the plasma membrane itself is the photosynthetic membrane. In plants and algae, by contrast, photosynthesis is carried out by organelles that are the evolutionary descendants of photosynthetic bacteria, chloroplasts—the photosynthetic membranes exist *within* the chloroplasts. The light reactions take place in four stages:

1. **Primary photoevent.** A photon of light is captured by a pigment. The result of this primary photoevent is the excitation of an electron within the pigment.
2. **Charge separation.** This excitation energy is transferred to a specialized chlorophyll pigment termed a reaction center, which reacts by transferring an energetic electron to an acceptor molecule, thus initiating electron transport.
3. **Electron transport.** The excited electron is shuttled along a series of electron-carrier molecules embedded within the photosynthetic membrane. Several of them react by transporting protons across the membrane, generating a gradient of proton concentration. Its arrival at the pump induces the transport of a proton across the membrane. The electron is then passed to an acceptor.
4. **Chemiosmosis.** The protons that accumulate on one side of the membrane now flow back across the membrane through specific protein complexes where chemiosmotic synthesis of ATP takes place, just as it does in aerobic respiration.

Discovery of Photosystems

One way to study how pigments absorb light is to measure the dependence of the output of photosynthesis on the intensity of illumination—that is, how much photosynthesis is produced by how much light. When experiments of this sort are done on plants, they show that the output of photosynthesis increases linearly at low intensities but lessens at higher intensities, finally saturating at high-intensity light (figure 10.9). Saturation occurs because all of the light-absorbing capacity of the plant is in use; additional light doesn't increase the output because there is nothing to absorb the added photons.

It is tempting to think that at saturation, all of a plant's pigment molecules are in use. In 1932 plant physiologists Robert Emerson and William Arnold set out to test this hypothesis in an organism where they could measure both the number of chlorophyll molecules and the output of photosynthesis. In their experiment, they measured the oxygen yield of photosynthesis when *Chlorella* (unicellular green algae) were exposed to very brief light flashes lasting

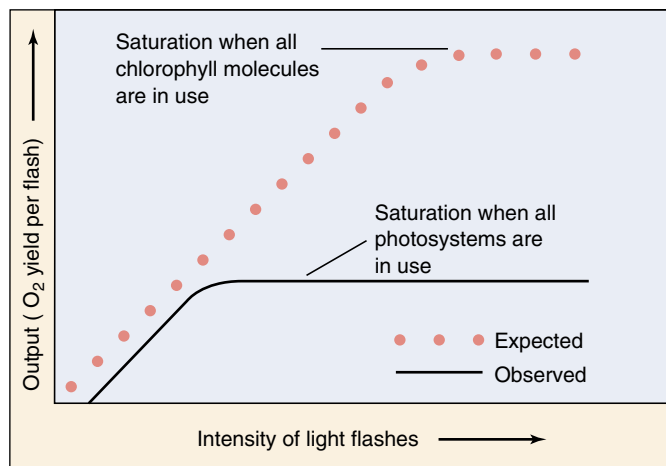


FIGURE 10.9

Emerson and Arnold's experiment. When photosynthetic saturation is achieved, further increases in intensity cause no increase in output.

only a few microseconds. Assuming the hypothesis of pigment saturation to be correct, they expected to find that as they increased the intensity of the flashes, the yield per flash would increase, until each chlorophyll molecule absorbed a photon, which would then be used in the light reactions, producing a molecule of O_2 .

Unexpectedly, this is not what happened. Instead, saturation was achieved much earlier, with only one molecule of O_2 per 2500 chlorophyll molecules! This led Emerson and Arnold to conclude that light is absorbed not by independent pigment molecules, but rather by clusters of chlorophyll and accessory pigment molecules which have come to be called *photosystems*. Light is absorbed by any one of the hundreds of pigment molecules in a photosystem, which transfer their excitation energy to one with a lower energy level than the others. This **reaction center** of the photosystem acts as an energy sink, trapping the excitation energy. It was the saturation of these reaction centers, not individual molecules, that was observed by Emerson and Arnold.

Architecture of a Photosystem

In chloroplasts and all but the most primitive bacteria, light is captured by such photosystems. Each photosystem is a network of chlorophyll *a* molecules, accessory pigments, and associated proteins held within a protein matrix on the surface of the photosynthetic membrane. Like a magnifying glass focusing light on a precise point, a photosystem channels the excitation energy gathered by any one of its pigment molecules to a specific molecule, the reaction center chlorophyll. This molecule then passes the energy out

of the photosystem so it can be put to work driving the synthesis of ATP and organic molecules.

A photosystem thus consists of two closely linked components: (1) an *antenna complex* of hundreds of pigment molecules that gather photons and feed the captured light energy to the reaction center; and (2) a *reaction center*, consisting of one or more chlorophyll *a* molecules in a matrix of protein, that passes the energy out of the photosystem.

The Antenna Complex. The antenna complex captures photons from sunlight (figure 10.10). In chloroplasts, the antenna complex is a web of chlorophyll molecules linked together and held tightly on the thylakoid membrane by a matrix of proteins. Varying amounts of carotenoid accessory pigments may also be present. The protein matrix serves as a sort of scaffold, holding individual pigment molecules in orientations that are optimal for energy transfer. The excitation energy resulting from the absorption of a photon passes from one pigment molecule to an adjacent molecule on its way to the reaction center. After the transfer, the excited electron in each molecule returns to the low-energy level it had before the photon was absorbed. Consequently, it is energy, not the excited electrons themselves, that passes from one pigment molecule to the next. The antenna complex funnels the energy from many electrons to the reaction center.

The Reaction Center. The reaction center is a transmembrane protein-pigment complex. In the reaction center of purple photosynthetic bacteria, which is simpler than in chloroplasts but better understood, a pair of chlorophyll *a* molecules acts as a trap for photon energy, passing an excited electron to an acceptor precisely positioned as its neighbor. Note that here the excited electron itself is transferred, not just the energy as we saw in pigment-pigment transfers. This allows the photon excitation to move away from the chlorophylls and is the key conversion of light to chemical energy.

Figure 10.11 shows the transfer of energy from the reaction center to the primary electron acceptor. By energizing an electron of the reaction center chlorophyll, light creates a strong electron donor where none existed before. The chlorophyll transfers the energized electron to the primary acceptor, a molecule of quinone, reducing the quinone and converting it to a strong electron donor. A weak electron donor then donates a low-energy electron to the chlorophyll, restoring it to its original condition. In plant chloroplasts, water serves as the electron donor.

Photosystems contain pigments that capture photon energy from light. The pigments transfer the energy to reaction centers. There, the energy excites electrons, which are channeled away to do chemical work.

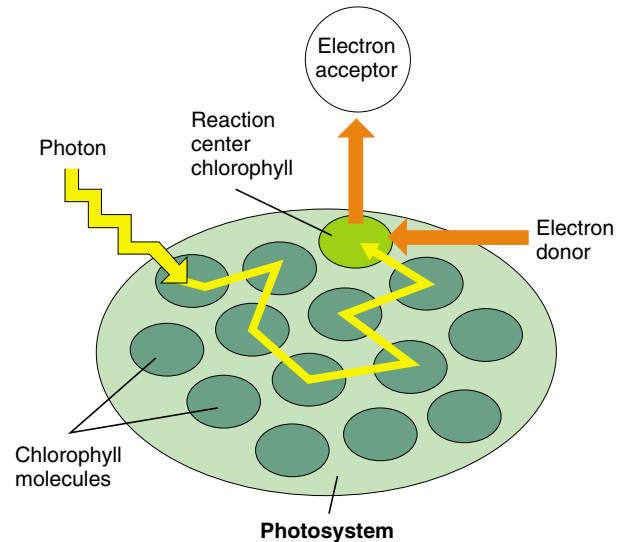


FIGURE 10.10

How the antenna complex works. When light of the proper wavelength strikes any pigment molecule within a photosystem, the light is absorbed by that pigment molecule. The excitation energy is then transferred from one molecule to another within the cluster of pigment molecules until it encounters the reaction center chlorophyll *a*. When excitation energy reaches the reaction center chlorophyll, electron transfer is initiated.

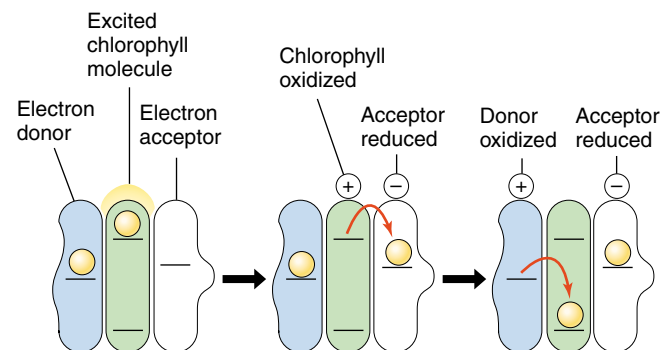


FIGURE 10.11

Converting light to chemical energy. The reaction center chlorophyll donates a light-energized electron to the primary electron acceptor, reducing it. The oxidized chlorophyll then fills its electron "hole" by oxidizing a donor molecule.

How Photosystems Convert Light to Chemical Energy

Bacteria Use a Single Photosystem

Photosynthetic pigment arrays are thought to have evolved more than 3 billion years ago in bacteria similar to the sulfur bacteria studied by van Niel.

1. Electron is joined with a proton to make hydrogen.

In these bacteria, the absorption of a photon of light at a peak absorption of 870 nanometers (near infrared, not visible to the human eye) by the photosystem results in the transmission of an energetic electron along an electron transport chain, eventually combining with a proton to form a hydrogen atom. In the sulfur bacteria, the proton is extracted from hydrogen sulfide, leaving elemental sulfur as a by-product. In bacteria that evolved later, as well as in plants and algae, the proton comes from water, producing oxygen as a by-product.

2. Electron is recycled to chlorophyll. The ejection of an electron from the bacterial reaction center leaves it short one electron. Before the photosystem of the sulfur bacteria can function again, an electron must be returned. These bacteria channel the electron back to the pigment through an electron transport system similar to the one described in chapter 9; the electron's passage drives a proton pump that promotes the chemiosmotic synthesis of ATP. One molecule of ATP is produced for every three electrons that follow this path. Viewed overall (figure 10.12), the path of the electron is thus a circle. Chemists therefore call the electron transfer process leading to ATP formation **cyclic photophosphorylation**. Note, however, that the electron that left the P_{870} reaction center was a high-energy electron, boosted by the absorption of a photon of light, while the electron that returns has only as much energy as it had before the photon was absorbed. The difference in the energy of that electron is the photosynthetic payoff, the energy that drives the proton pump.

For more than a billion years, cyclic photophosphorylation was the only form of photosynthetic light reaction that organisms used. However, its major limitation is that it is geared only toward energy production, not toward biosynthesis. Most photosynthetic organisms incorporate atmospheric carbon dioxide into carbohydrates. Because the carbohydrate molecules are more reduced (have more hydrogen atoms) than carbon dioxide, a source of reducing power (that is, hydrogens) must be provided. Cyclic photophosphorylation does not do this. The hydrogen atoms extracted from H_2S are used as a source of protons, and are not available to join to carbon. Thus bacteria that are restricted to this process must scavenge hydrogens from other sources, an inefficient undertaking.

Why Plants Use Two Photosystems

After the sulfur bacteria appeared, other kinds of bacteria evolved an improved version of the photosystem that overcame the limitation of cyclic photophosphorylation in a neat and simple way: a second, more powerful photosystem using another arrangement of chlorophyll *a* was combined with the original.

In this second photosystem, called **photosystem II**, molecules of chlorophyll *a* are arranged with a different geometry, so that more shorter wavelength, higher energy photons are absorbed than in the ancestral photosystem, which is called **photosystem I**. As in the ancestral photosystem, energy is transmitted from one pigment molecule to another within the antenna complex of these photosystems until it reaches the reaction center, a particular pigment molecule positioned near a strong membrane-bound electron acceptor. In photosystem II, the absorption peak (that is, the wavelength of light most strongly absorbed) of the pigments is approximately 680 nanometers; therefore, the reaction center pigment is called P_{680} . The absorption peak of photosystem I pigments in plants is 700 nanometers, so its reaction center pigment is called P_{700} . Working together, the two photosystems carry out a noncyclic electron transfer.

When the rate of photosynthesis is measured using two light beams of different wavelengths (one red and the

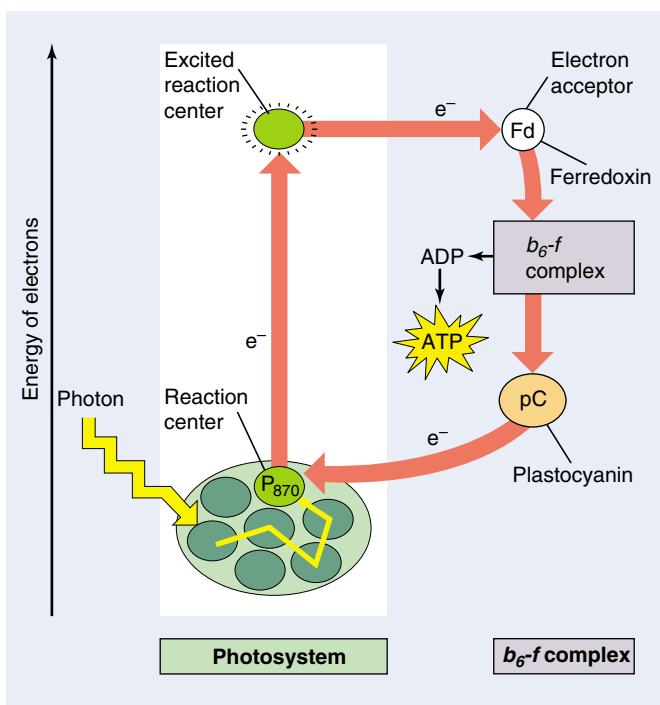


FIGURE 10.12

The path of an electron in purple sulfur bacteria. When a light-energized electron is ejected from the photosystem reaction center (P_{870}), it passes in a circle, eventually returning to the photosystem from which it was ejected.

other far-red), the rate was greater than the sum of the rates using individual beams of red and far-red light (figure 10.13). This surprising result, called the enhancement effect, can be explained by a mechanism involving two photosystems acting in series (that is, one after the other), one of which absorbs preferentially in the red, the other in the far-red.

The use of two photosystems solves the problem of obtaining reducing power in a simple and direct way, by harnessing the energy of two photosystems. The scheme shown in figure 10.14, called a *Z diagram*, illustrates the two electron-energizing steps, one catalyzed by each photosystem. The electrons originate from water, which holds onto its electrons very tightly (redox potential = +820 mV), and end up in NADPH, which holds its electrons much more loosely (redox potential = -320 mV).

In sulfur bacteria, excited electrons ejected from the reaction center travel a circular path, driving a proton pump and then returning to their original photosystem. Plants employ two photosystems in series, which generates power to reduce NADP^+ to NADPH with enough left over to make ATP.

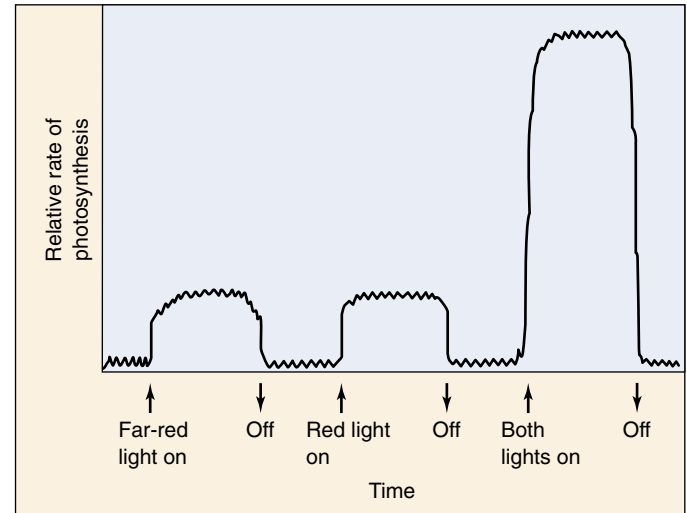


FIGURE 10.13

The “enhancement effect.” The rate of photosynthesis when red and far-red light are provided together is greater than the sum of the rates when each wavelength is provided individually. This result baffled researchers in the 1950s. Today it provides the key evidence that photosynthesis is carried out by two photochemical systems with slightly different wavelength optima.

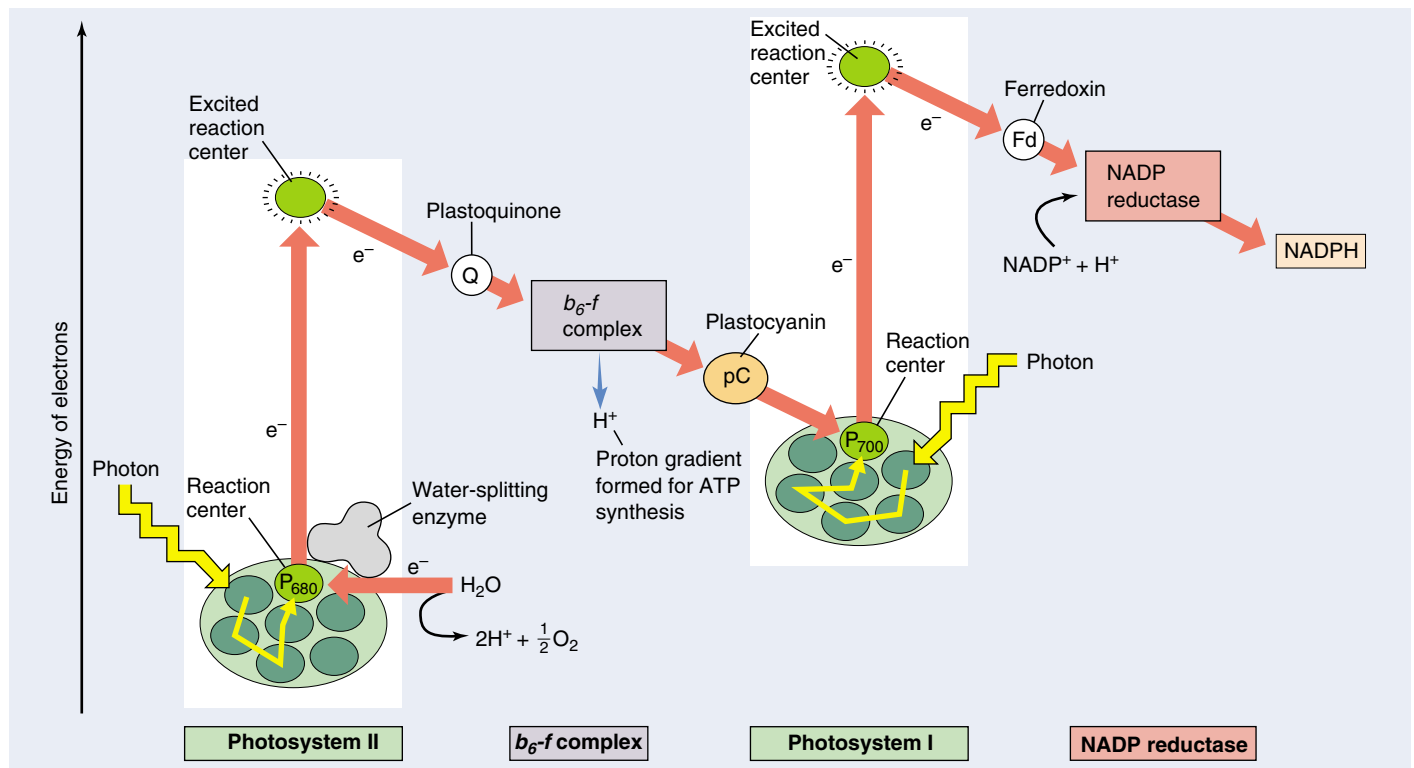


FIGURE 10.14

A Z diagram of photosystems I and II. Two photosystems work sequentially. First, a photon of light ejects a high-energy electron from photosystem II; that electron is used to pump a proton across the membrane, contributing chemiosmotically to the production of a molecule of ATP. The ejected electron then passes along a chain of cytochromes to photosystem I. When photosystem I absorbs a photon of light, it ejects a high-energy electron used to drive the formation of NADPH.

How the Two Photosystems of Plants Work Together

Plants use the two photosystems discussed earlier in series, first one and then the other, to produce both ATP and NADPH. This two-stage process is called **noncyclic photophosphorylation**, because the path of the electrons is not a circle—the electrons ejected from the photosystems do not return to it, but rather end up in NADPH. The photosystems are replenished instead with electrons obtained by splitting water. Photosystem II acts first. High-energy electrons generated by photosystem II are used to synthesize ATP and then passed to photosystem I to drive the production of NADPH. For every pair of electrons obtained from water, one molecule of NADPH and slightly more than one molecule of ATP are produced.

Photosystem II

The reaction center of photosystem II, called P_{680} , closely resembles the reaction center of purple bacteria. It consists of more than 10 transmembrane protein subunits. The light-harvesting antenna complex consists of some 250 molecules of chlorophyll *a* and accessory pigments bound to several protein chains. In photosystem II, the oxygen atoms of two water molecules bind to a cluster of manganese atoms which are embedded within an enzyme and bound to the reaction center. In a way that is poorly understood, this enzyme splits water, removing electrons one at a time to fill the holes left in the reaction center by departure of light-energized electrons. As soon as four electrons have been removed from the two water molecules, O_2 is released.

The Path to Photosystem I

The primary electron acceptor for the light-energized electrons leaving photosystem II is a quinone molecule, as it was in the bacterial photosystem described earlier. The reduced quinone which results (*plastoquinone*, symbolized Q) is a strong electron donor; it passes the excited electron to a proton pump called the *b₆-f complex* embedded within the thylakoid membrane (figure 10.15). This complex closely resembles the *bc₁* complex in the respiratory electron trans-

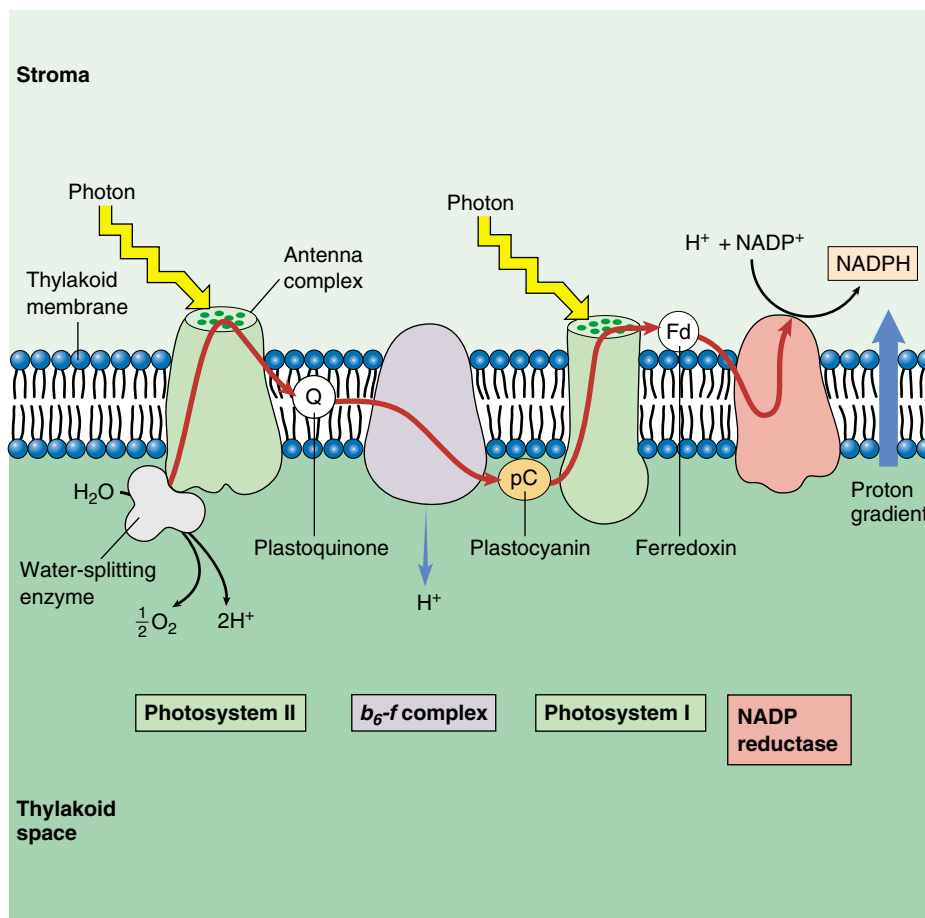


FIGURE 10.15

The photosynthetic electron transport system. When a photon of light strikes a pigment molecule in photosystem II, it excites an electron. This electron is coupled to a proton stripped from water by an enzyme and is passed along a chain of membrane-bound cytochrome electron carriers (*red arrow*). When water is split, oxygen is released from the cell, and the hydrogen ions remain in the thylakoid space. At the proton pump (*b₆-f complex*), the energy supplied by the photon is used to transport a proton across the membrane into the thylakoid. The concentration of hydrogen ions within the thylakoid thus increases further. When photosystem I absorbs another photon of light, its pigment passes a second high-energy electron to a reduction complex, which generates NADPH.

port chain of mitochondria discussed in chapter 9. Arrival of the energetic electron causes the *b₆-f complex* to pump a proton into the thylakoid space. A small copper-containing protein called *plastocyanin* (symbolized pC) then carries the electron to photosystem I.

Making ATP: Chemiosmosis

Each thylakoid is a closed compartment into which protons are pumped from the stroma by the *b₆-f complex*. The splitting of water also produces added protons that contribute to the gradient. The thylakoid membrane is impermeable to protons, so protons cross back out almost exclusively via the channels provided by *ATP synthases*. These channels protrude like knobs on the external surface of the thylakoid membrane. As protons pass out of

the thylakoid through the ATP synthase channel, ADP is phosphorylated to ATP and released into the stroma, the fluid matrix inside the chloroplast (figure 10.16). The stroma contains the enzymes that catalyze the reactions of carbon fixation.

Photosystem I

The reaction center of photosystem I, called P_{700} , is a transmembrane complex consisting of at least 13 protein subunits. Energy is fed to it by an antenna complex consisting of 130 chlorophyll *a* and accessory pigment molecules. Photosystem I accepts an electron from plastocyanin into the hole created by the exit of a light-energized electron. This arriving electron has by no means lost all of its light-excited energy; almost half remains. Thus, the absorption of a photon of light energy by photosystem I boosts the electron leaving the reaction center to a very high energy level. Unlike photosystem II and the bacterial photosystem, photosystem I does not rely on quinones as electron acceptors. Instead, it passes electrons to an iron-sulfur protein called *ferredoxin* (Fd).

Making NADPH

Photosystem I passes electrons to ferredoxin on the stromal side of the membrane (outside the thylakoid). The reduced ferredoxin carries a very-high-potential electron. Two of them, from two molecules of reduced ferredoxin, are then donated to a molecule of $NADP^+$ to form NADPH. The reaction is catalyzed by the membrane-bound enzyme *NADP reductase*. Because the reaction occurs on the stromal side of the membrane and involves the uptake of a proton in forming NADPH, it contributes further to the proton gradient established during photosynthetic electron transport.

Making More ATP

The passage of an electron from water to NADPH in the noncyclic photophosphorylation described previously generates one molecule of NADPH and slightly more than one molecule of ATP. However, as you will learn later in this chapter, building organic molecules takes more energy than that—it takes one-and-a-half ATP molecules per NADPH molecule to fix carbon. To produce the extra ATP, many

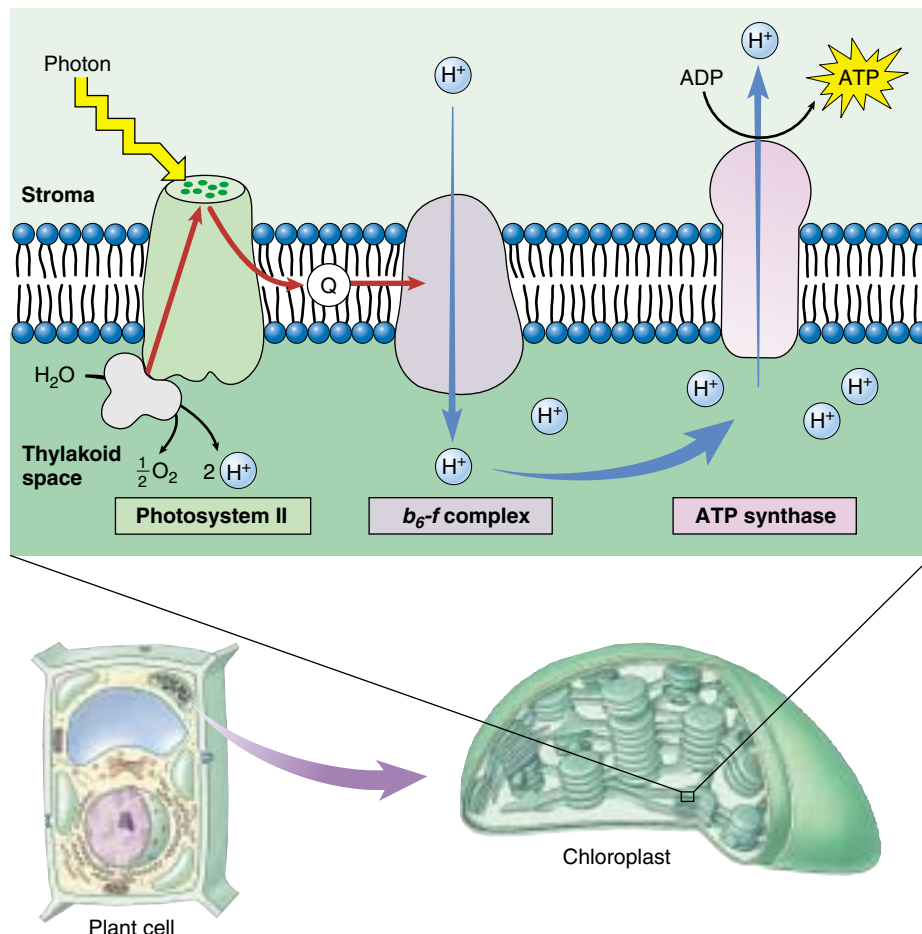


FIGURE 10.16

Chemiosmosis in a chloroplast. The b_6-f complex embedded in the thylakoid membrane pumps protons into the interior of the thylakoid. ATP is produced on the outside surface of the membrane (stroma side), as protons diffuse back out of the thylakoid through ATP synthase channels.

plant species are capable of short-circuiting photosystem I, switching photosynthesis into a *cyclic photophosphorylation* mode, so that the light-excited electron leaving photosystem I is used to make ATP instead of NADPH. The energetic electron is simply passed back to the b_6-f complex rather than passing on to $NADP^+$. The b_6-f complex pumps out a proton, adding to the proton gradient driving the chemiosmotic synthesis of ATP. The relative proportions of cyclic and noncyclic photophosphorylation in these plants determines the relative amounts of ATP and NADPH available for building organic molecules.

The electrons that photosynthesis strips from water molecules provide the energy to form ATP and NADPH. The residual oxygen atoms of the water molecules combine to form oxygen gas.

10.4 Cells use the energy and reducing power captured by the light reactions to make organic molecules.

The Calvin Cycle

Photosynthesis is a way of making organic molecules from carbon dioxide (CO_2). These organic molecules contain many C—H bonds and are highly reduced compared with CO_2 . To build organic molecules, cells use raw materials provided by the light reactions:

1. **Energy.** ATP (provided by cyclic and noncyclic photophosphorylation) drives the endergonic reactions.
2. **Reducing power.** NADPH (provided by photosystem I) provides a source of hydrogens and the energetic electrons needed to bind them to carbon atoms. Much of the light energy captured in photosynthesis ends up invested in the energy-rich C—H bonds of sugars.

Carbon Fixation

The key step in the Calvin cycle—the event that makes the reduction of CO_2 possible—is the attachment of CO_2 to a very special organic molecule. Photosynthetic cells produce this molecule by reassembling the bonds of two intermediates in glycolysis, fructose 6-phosphate and glyceraldehyde 3-phosphate, to form the energy-rich five-carbon sugar, *ribulose 1,5-bisphosphate (RuBP)*, and a four-carbon sugar.

CO_2 binds to RuBP in the key process called **carbon fixation**, forming two three-carbon molecules of *phosphoglycerate (PGA)* (figure 10.17). The enzyme that carries out this reaction, *ribulose biphosphate carboxylase/oxygenase* (usually abbreviated *rubisco*) is a very large four-subunit enzyme present in the chloroplast stroma. This enzyme works very sluggishly, processing only about three molecules of RuBP per second (a typical enzyme processes about 1000 substrate molecules per second). Because it works so slowly, many molecules of rubisco are needed. In a typical leaf, over 50% of all the protein is rubisco. It is thought to be the most abundant protein on earth.

Discovering the Calvin Cycle

Nearly 100 years ago, Blackman concluded that, because of its temperature dependence, photosynthesis might involve enzyme-catalyzed reactions. These reactions form a cycle of enzyme-catalyzed steps similar to the Krebs cycle. This cycle of reactions is called the **Calvin cycle**, after its discoverer, Melvin Calvin of the University of California, Berkeley. Because the cycle begins when CO_2 binds RuBP to form PGA, and PGA contains three carbon atoms, this process is also called **C_3 photosynthesis**.

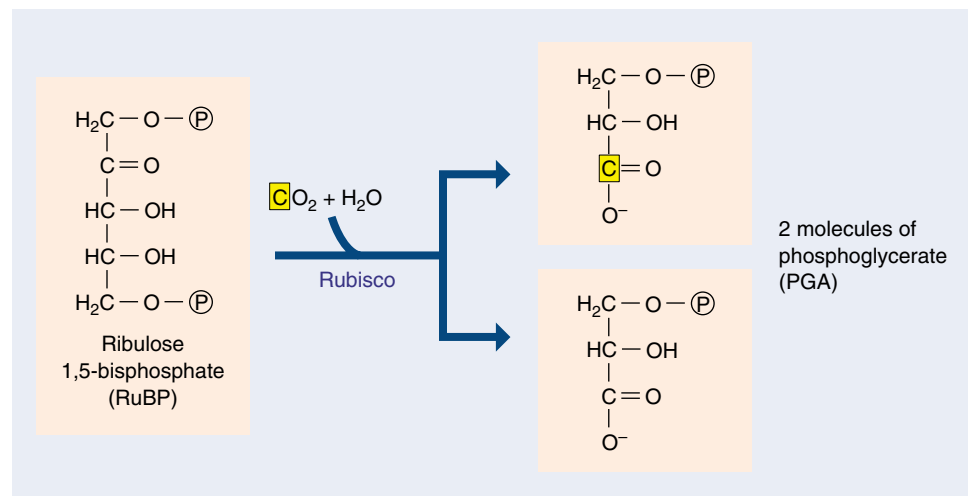


FIGURE 10.17

The key step in the Calvin cycle.

Melvin Calvin and his coworkers at the University of California worked out the first step of what later became known as the Calvin cycle. They exposed photosynthesizing algae to radioactive carbon dioxide ($^{14}\text{CO}_2$). By following the fate of a radioactive carbon atom, they found that it first binds to a molecule of ribulose 1,5-bisphosphate (RuBP), then immediately splits, forming two molecules of phosphoglycerate (PGA). One of these PGAs contains the radioactive carbon atom. In 1948, workers isolated the enzyme responsible for this remarkable carbon-fixing reaction: rubisco.

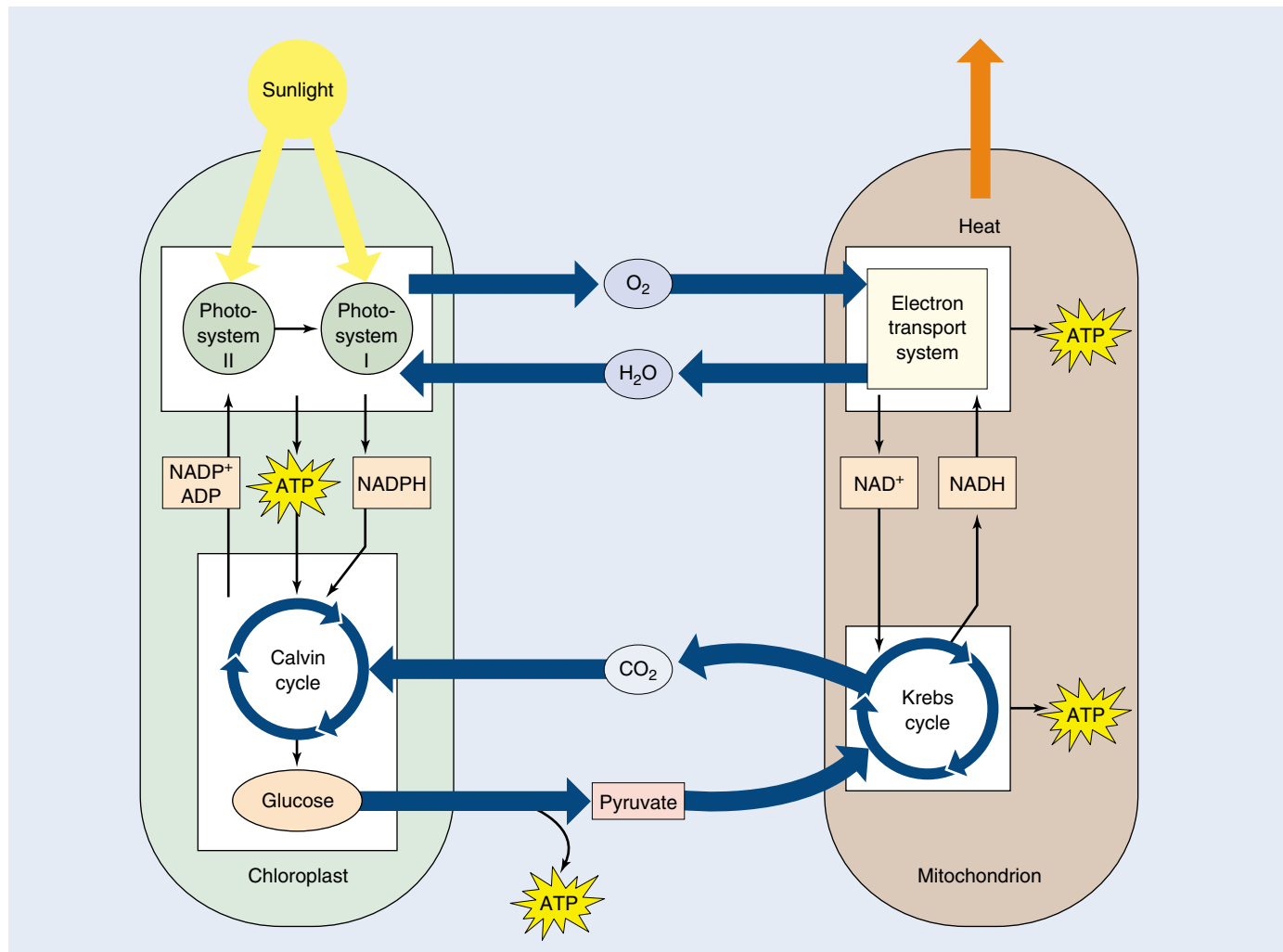


FIGURE 10.18

Chloroplasts and mitochondria: Completing an energy cycle. Water and oxygen gas cycle between chloroplasts and mitochondria within a plant cell, as do glucose and CO_2 . Cells with chloroplasts require an outside source of CO_2 and water and generate glucose and oxygen. Cells without chloroplasts, such as animal cells, require an outside source of glucose and oxygen and generate CO_2 and water.

The Energy Cycle

The energy-capturing metabolisms of the chloroplasts studied in this chapter and the mitochondria studied in the previous chapter are intimately related. Photosynthesis uses the products of respiration as starting substrates, and respiration uses the products of photosynthesis as its starting substrates (figure 10.18). The Calvin cycle even uses part of the ancient glycolytic pathway, run in reverse, to produce glucose. And, the principal proteins involved in electron transport in plants are related to those in mitochondria, and in many cases are actually the same.

Photosynthesis is but one aspect of plant biology, although it is an important one. In chapters 37 through 43,

we will examine plants in more detail. We have treated photosynthesis here, in a section devoted to cell biology, because photosynthesis arose long before plants did, and all organisms depend directly or indirectly on photosynthesis for the energy that powers their lives.

Chloroplasts put ATP and NADPH to work building carbon-based molecules, a process that essentially reverses the breakdown of such molecules that occurs in mitochondria. Taken together, chloroplasts and mitochondria carry out a cycle in which energy enters from the sun and leaves as heat and work.

THE CALVIN CYCLE

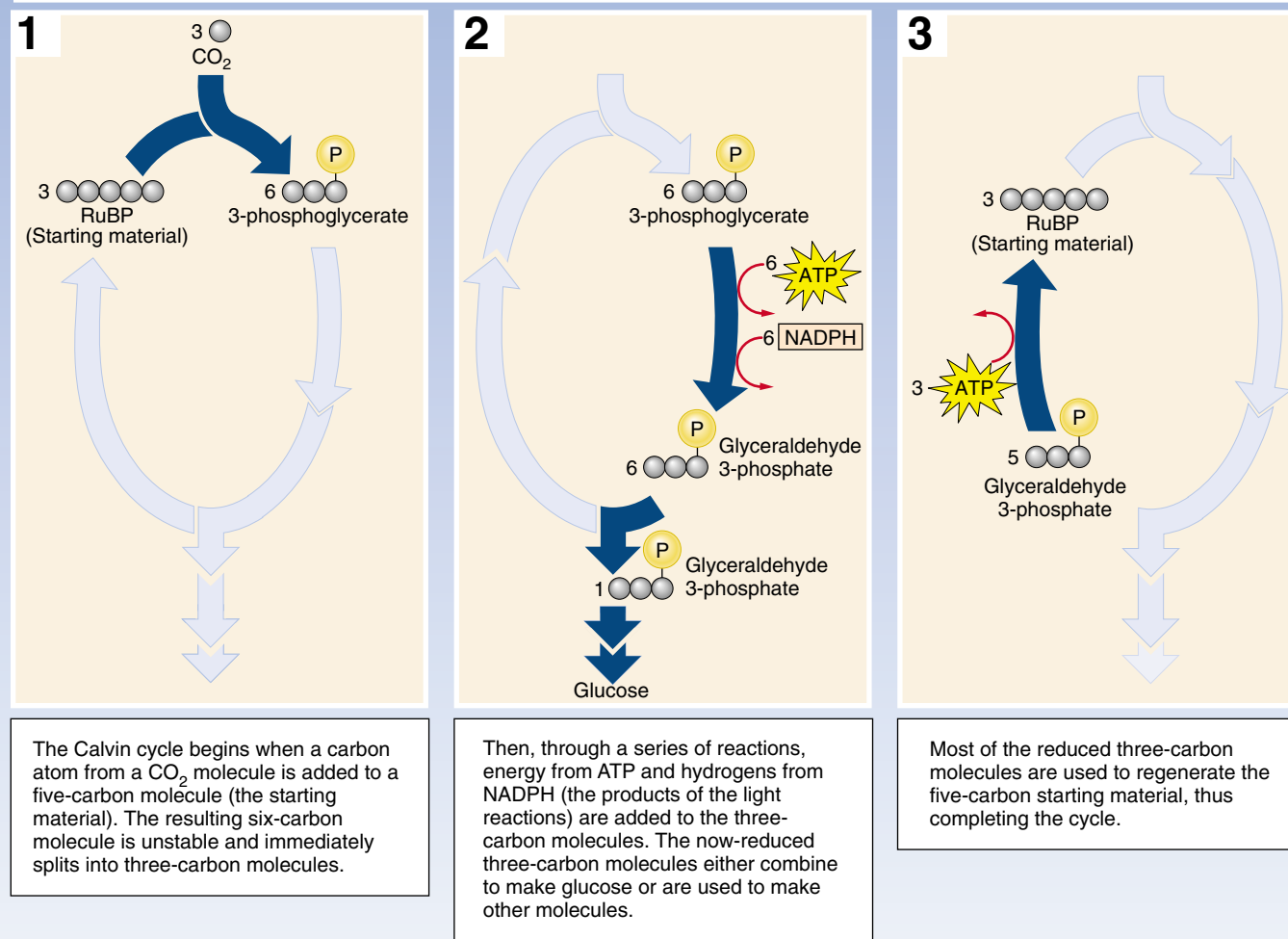
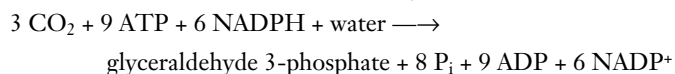


FIGURE 10.19
How the Calvin cycle works.

Reactions of the Calvin Cycle

In a series of reactions (figure 10.19), three molecules of CO₂ are fixed by rubisco to produce six molecules of PGA (containing $6 \times 3 = 18$ carbon atoms in all, three from CO₂ and 15 from RuBP). The 18 carbon atoms then undergo a cycle of reactions that regenerates the three molecules of RuBP used in the initial step (containing $3 \times 5 = 15$ carbon atoms). This leaves one molecule of glyceraldehyde 3-phosphate (three carbon atoms) as the net gain.

The net equation of the Calvin cycle is:



With three full turns of the cycle, three molecules of carbon dioxide enter, a molecule of glyceraldehyde 3-phosphate (G3P) is produced, and three molecules of RuBP are regenerated (figure 10.20).

We now know that light is required *indirectly* for different segments of the CO₂ reduction reactions. Five of the Calvin cycle enzymes—including rubisco—are light activated; that is, they become functional or operate more efficiently in the presence of light. Light also promotes transport of three-carbon intermediates across chloroplast membranes that are required for Calvin cycle reactions. And finally, light promotes the influx of Mg²⁺ into the chloroplast stroma, which further activates the enzyme rubisco.

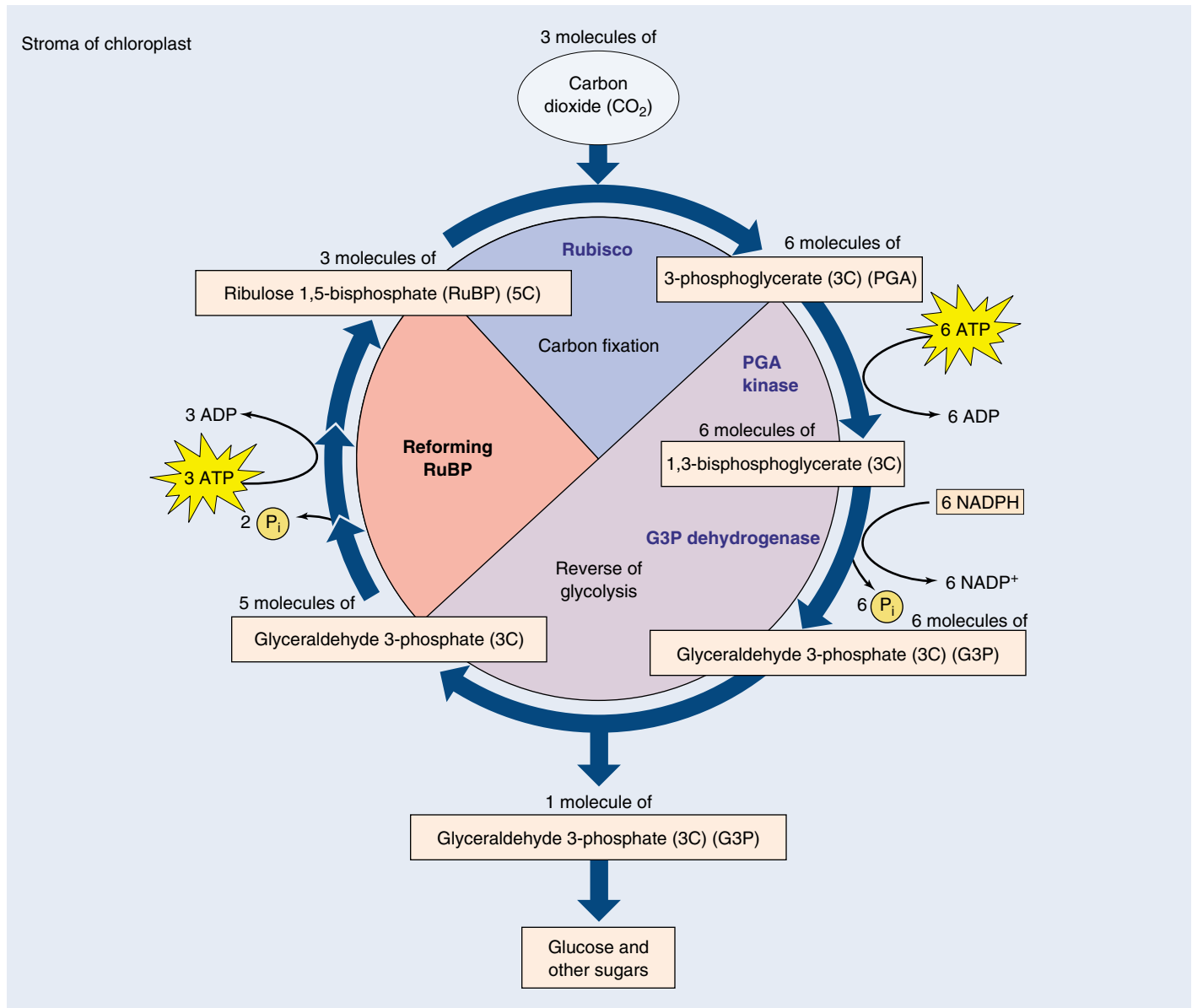


FIGURE 10.20

The Calvin cycle. For every three molecules of CO_2 that enter the cycle, one molecule of the three-carbon compound, glyceraldehyde 3-phosphate (G3P), is produced. Notice that the process requires energy stored in ATP and NADPH, which are generated by the light reactions. This process occurs in the stroma of the chloroplast.

Output of the Calvin Cycle

The glyceraldehyde 3-phosphate that is the product of the Calvin cycle is a three-carbon sugar that is a key intermediate in glycolysis. Much of it is exported from the chloroplast to the cytoplasm of the cell, where the reversal of several reactions in glycolysis allows it to be converted to fructose 6-phosphate and glucose 1-phosphate, and from that to sucrose, a major transport sugar in plants (sucrose, common table sugar, is a disaccharide made of fructose and glucose).

In times of intensive photosynthesis, glyceraldehyde 3-phosphate levels in the stroma of the chloroplast rise. As a

consequence, some glyceraldehyde 3-phosphate in the chloroplast is converted to glucose 1-phosphate, in an analogous set of reactions to those done in the cytoplasm, by reversing several reactions similar to those of glycolysis. The glucose 1-phosphate is then combined into an insoluble polymer, forming long chains of starch stored as bulky starch grains in chloroplasts.

Plants incorporate carbon dioxide into sugars by means of a cycle of reactions called the Calvin cycle, which is driven by the ATP and NADPH produced in the light reactions which are consumed as CO_2 is reduced to G3P.

Photorespiration

Evolution does not necessarily result in optimum solutions. Rather, it favors workable solutions that can be derived from others that already exist. Photosynthesis is no exception. Rubisco, the enzyme that catalyzes the key carbon-fixing reaction of photosynthesis, provides a decidedly sub-optimal solution. This enzyme has a second enzyme activity that interferes with the Calvin cycle, *oxidizing* ribulose 1,5-bisphosphate. In this process, called **photorespiration**, O_2 is incorporated into ribulose 1,5-bisphosphate, which undergoes additional reactions that actually release CO_2 . Hence, photorespiration releases CO_2 —essentially undoing the Calvin cycle which reduces CO_2 to carbohydrate.

The carboxylation and oxidation of ribulose 1,5-bisphosphate are catalyzed at the same active site on rubisco, and compete with each other. Under normal conditions at $25^\circ C$, the rate of the carboxylation reaction is four times that of the oxidation reaction, meaning that 20% of photosynthetically fixed carbon is lost to photorespiration. This loss rises substantially as temperature increases, because the rate of the oxidation reaction increases with temperature far faster than the carboxylation reaction rate.

Plants that fix carbon using only C_3 photosynthesis (the Calvin cycle) are called C_3 plants. In C_3 photosynthesis, ribulose 1,5-bisphosphate is carboxylated to form a three-carbon compound via the activity of rubisco. Other plants use **C_4 photosynthesis**, in which phosphoenolpyruvate, or PEP, is carboxylated to form a four-carbon compound using the enzyme PEP carboxylase. This enzyme has no oxidation activity, and thus no photorespiration. Furthermore, PEP carboxylase has a much greater affinity for CO_2 than does rubisco. In the C_4 pathway, the four-carbon compound undergoes further modification, only to be decarboxylated. The CO_2 which is released is then captured by rubisco and drawn into the Calvin cycle. Because an organic compound is donating the CO_2 , the effective concentration of CO_2 relative to O_2 is increased, and photorespiration is minimized.

The loss of fixed carbon as a result of photorespiration is not trivial. C_3 plants lose between 25 and 50% of their photosynthetically fixed carbon in this way. The rate depends largely upon the temperature. In tropical climates, especially those in which the temperature is often above $28^\circ C$, the problem is severe, and it has a major impact on tropical agriculture.

The C_4 Pathway

Plants that adapted to these warmer environments have evolved two principal ways that use the C_4 pathway to deal with this problem. In one approach, plants conduct C_4 photosynthesis in the mesophyll cells and the Calvin cycle in the bundle sheath cells. This creates high local levels of CO_2 to favor the carboxylation reaction of rubisco. These plants are called C_4 plants and include corn,

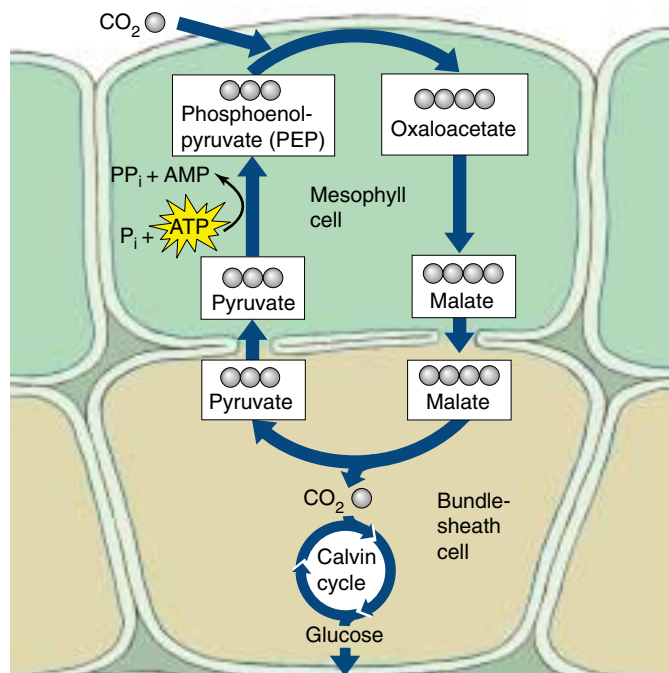


FIGURE 10.21

Carbon fixation in C_4 plants. This process is called the C_4 pathway because the starting material, oxaloacetate, is a molecule containing four carbons.

sugarcane, sorghum, and a number of other grasses. In the C_4 pathway, the three-carbon metabolite phosphoenolpyruvate is carboxylated to form the four-carbon molecule oxaloacetate, which is the first product of CO_2 fixation (figure 10.21). In C_4 plants, oxaloacetate is in turn converted into the intermediate malate, which is transported to an adjacent bundle-sheath cell. Inside the bundle-sheath cell, malate is decarboxylated to produce pyruvate, releasing CO_2 . Because bundle-sheath cells are impermeable to CO_2 , the CO_2 is retained within them in high concentrations. Pyruvate returns to the mesophyll cell, where two of the high-energy bonds in an ATP molecule are split to convert the pyruvate back into phosphoenolpyruvate, thus completing the cycle.

The enzymes that carry out the Calvin cycle in a C_4 plant are located within the bundle-sheath cells, where the increased CO_2 concentration decreases photorespiration. Because each CO_2 molecule is transported into the bundle-sheath cells at a cost of two high-energy ATP bonds, and because six carbons must be fixed to form a molecule of glucose, 12 additional molecules of ATP are required to form a molecule of glucose. In C_4 photosynthesis, the energetic cost of forming glucose is almost twice that of C_3 photosynthesis: 30 molecules of ATP versus 18. Nevertheless, C_4 photosynthesis is advantageous in a hot climate: photorespiration would otherwise remove more than half of the carbon fixed.

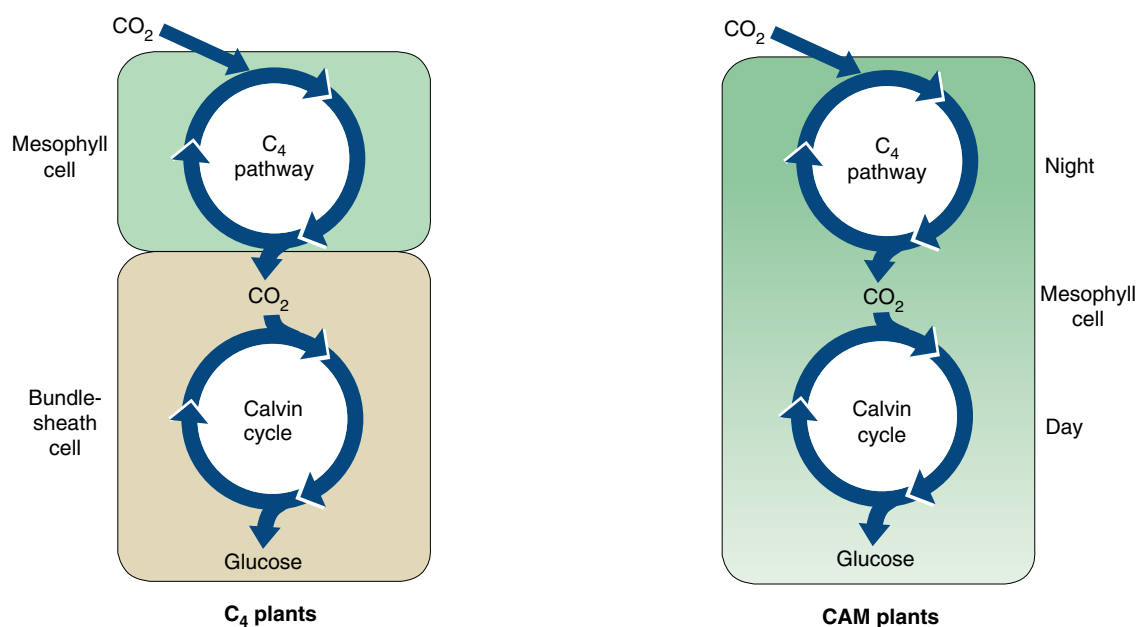


FIGURE 10.22

A comparison of C_4 and CAM plants. Both C_4 and CAM plants utilize the C_4 and the C_3 pathways. In C_4 plants, the pathways are separated spatially: the C_4 pathway takes place in the mesophyll cells and the C_3 pathway in the bundle-sheath cells. In CAM plants, the two pathways are separated temporally: the C_4 pathway is utilized at night and the C_3 pathway during the day.

The Crassulacean Acid Pathway

A second strategy to decrease photorespiration in hot regions has been adopted by many succulent (water-storing) plants such as cacti, pineapples, and some members of about two dozen other plant groups. This mode of initial carbon fixation is called **crassulacean acid metabolism (CAM)**, after the plant family Crassulaceae (the stonecrops or hens-and-chicks), in which it was first discovered. In these plants, the stomata (singular, stoma), specialized openings in the leaves of all plants through which CO_2 enters and water vapor is lost, open during the night and close during the day. This pattern of stomatal opening and closing is the reverse of that in most plants. CAM plants open stomata at night and initially fix CO_2 into organic compounds using the C_4 pathway. These organic compounds accumulate throughout the night and

are decarboxylated during the day to yield high levels of CO_2 . In the day, these high levels of CO_2 drive the Calvin cycle and minimize photorespiration. Like C_4 plants, CAM plants use both C_4 and C_3 pathways. They differ from C_4 plants in that they use the C_4 pathway at night and the C_3 pathway during the day *within the same cells*. In C_4 plants, the two pathways take place in different cells (figure 10.22).

Photorespiration results in decreased yields of photosynthesis. C_4 and CAM plants circumvent this problem through modifications of leaf architecture and photosynthetic chemistry that locally increase CO_2 concentrations. C_4 plants isolate CO_2 production spatially, CAM plants temporally.



Summary

Questions

Media Resources

10.1 What is photosynthesis?

- Light is used by plants, algae, and some bacteria, in a process called photosynthesis, to convert atmospheric carbon (CO₂) into carbohydrate.

1. Where do the oxygen atoms in the O₂ produced during photosynthesis come from?



- Art Activity: Chloroplast Structure



- Chloroplast

10.2 Learning about photosynthesis: An experimental journey.

- A series of simple experiments demonstrated that plants capture energy from light and use it to convert the carbon atoms of CO₂ and the hydrogen atoms of water into organic molecules.

2. How did van Helmont determine that plants do not obtain their food from the soil?



- Energy Conversion
- Photosynthesis



- Light Dependent Reactions
- Light Independent Reactions
- Light and Pigmentation

10.3 Pigments capture energy from sunlight.

- Light consists of energy packets called photons; the shorter the wavelength of light, the more its energy. When photons are absorbed by a pigment, electrons in the pigment are boosted to a higher energy level.
- Photosynthesis channels photon excitation energy into a single pigment molecule. In bacteria, that molecule then donates an electron to an electron transport chain, which drives a proton pump and ultimately returns the electron to the pigment.
- Plants employ two photosystems. Light is first absorbed by photosystem II and passed to photosystem I, driving a proton pump and bringing about the chemiosmotic synthesis of ATP.
- When the electron arrives at photosystem I, another photon of light is absorbed, and energized electrons are channeled to a primary electron acceptor, which reduces NADP⁺ to NADPH. Use of NADPH rather than NADH allows plants and algae to keep the processes of photosynthesis and oxidative respiration separate from each other.

3. How is the energy of light captured by a pigment molecule? Why does light reflected by the pigment chlorophyll appear green?



- Art Activity: Electromagnetic Spectrum

4. What is the function of the reaction center chlorophyll? What is the function of the primary electron acceptor?

5. Explain how photosynthesis in the sulfur bacteria is a cyclic process. What is its energy yield in terms of ATP molecules synthesized per electron?

6. How do the two photosystems in plants and algae work? Which stage generates ATP and which generates NADPH?

10.4 Cells use the energy and reducing power captured by the light reactions to make organic molecules.

- The ATP and reducing power produced by the light reactions are used to fix carbon in a series of reactions called the Calvin cycle.
- RuBP carboxylase, the enzyme that fixes carbon in the Calvin cycle, also carries out an oxidative reaction that uses the products of photosynthesis, a process called photorespiration.
- Many tropical plants inhibit photorespiration by expending ATP to increase the intracellular concentration of CO₂. This process, called the C₄ pathway, nearly doubles the energetic cost of synthesizing glucose.

7. In a C₃ plant, where do the light reactions occur? Where does the Calvin cycle occur?

8. What is photorespiration? What advantage do C₄ plants have over C₃ plants with respect to photorespiration? What disadvantage do C₄ plants have that limits their distribution primarily to warm regions of the earth?



- The Calvin Cycle



- Photorespiration

Reproduction and Heredity

Why Do Some Genes Maintain More Than One Common Allele in a Population?

When Mendel did his crosses of pea plants, he knew what a pea plant was supposed to look like: a small plant with green leaves, purple flowers, and smooth seeds. But if all pea plants were like that, he would never have been able to sort out the rules of heredity—in a cross of green peas with green peas, there would have been no visible differences to reveal the 3:1 pattern of gene segregation. The variant alleles that Mendel employed in his studies—yellow leaves, white flowers, wrinkled seeds—were rare “accidents” maintained in seed collections for their novelty. In nature, such unusual kinds of peas had never been encountered by Mendel.

By the time Mendel’s work was rediscovered in 1900, Darwin had provided a ready explanation of why alternative alleles seemed to be rare in natural populations. Natural selection was simply scouring the population, cleansing it in each generation of less fit alternatives. While recombination can complicate the process in interesting ways among sexual organisms like peas, asexual organisms like bacteria were predicted to be very sensitive to the effects of selection. Left to do its work, natural selection should crown as winner in bacterial population the best allele of each gene, producing a uniform population.

Why do populations contain variants at all? In 1932 the famous geneticist Herman Muller formulated what has come to be called the “classical model,” explaining gene variation in natural populations of asexual organisms as a temporary, transient condition, new variations arising by random mutation only to be established or eliminated by selection. Except for the brief periods when populations are undergoing this periodic cleansing, they should remain genetically uniform.

The removal of variants was proposed to be a very straightforward process. During the periodic cleansing periods envisioned by Muller, his classical model operates under a “competitive exclusion” principle first proposed by Gause: whenever a new variant appears, it is weighed in the balance by natural selection, and either wins or loses. There are no ties. One version of the gene becomes universal in the population, and the other is eliminated.



These bacterial cells are dividing. As the population grows, gene variants arise by mutation. Do the new variants persist, or are they eliminated by natural selection?

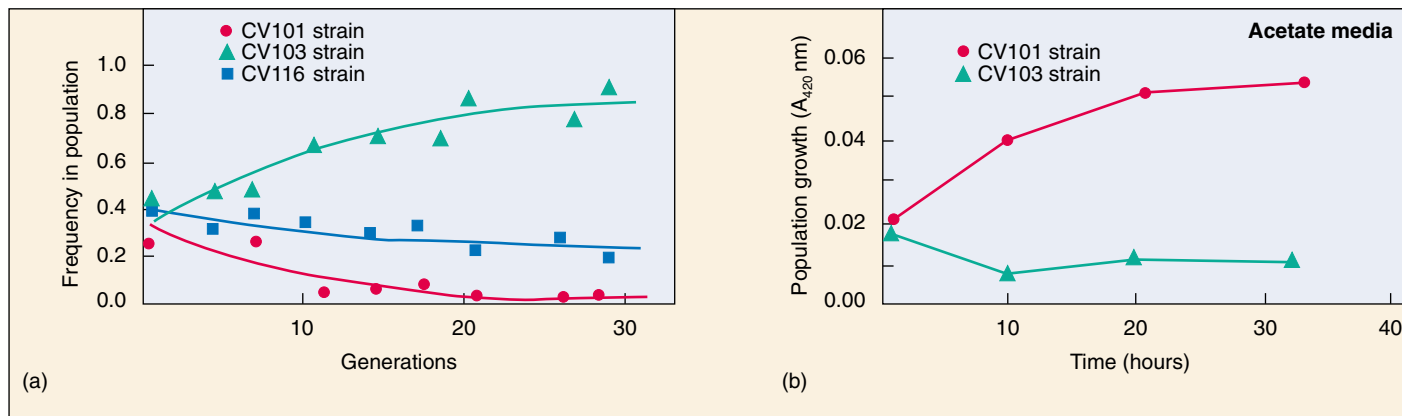
Muller’s classical model thus makes a very straightforward prediction: in nature, most populations of asexual organisms should be genetically uniform most of the time. However, this is not at all what is observed. Natural populations of most species, including asexual ones like bacteria, appear to have lots of common variants—they are said to be “polymorphic.”

So where are all of these variants coming from? Variation in the environment, either spatial or temporal, can be used to explain how some polymorphisms arise. Selection favors one form at a particular place and time, a different form at a different place or time. In a nutshell, varying selection can encourage polymorphism.

Is that all there is to it? Is it really impossible for more than one variant to become common in a population, if the population lives in a constant uniform environment, an environment that does not vary from one place to another or from one time to another? Theory says so.

Biologists that study microbial communities have begun to report that bacteria are not aware of Muller’s theory. Bacterial cultures started from a single cell living in simple unstructured environments rapidly become polymorphic.

There is a way to reconcile theory and experiment. Perhaps the variant individuals in the population are interacting with one another. Muller’s theory assumes that every individual undergoes an independent trial by selection. But what if that’s not so? What if different kinds of individuals help each other out? Stable coexistence of variants in a population might be possible if interactions between them contribute to the welfare of both (what a biologist calls mutualism) or favors one (what a biologist calls commensalism). In essence, cooperation would be counterbalancing the effects of competition.



Maintaining stable polymorphism. (a) Three new strains emerge in culture and are maintained. (b) Two strains are grown on media containing acetate. The strain CV103 was found to excrete acetate, while the strain CV101 was found to thrive in media with acetate as the sole source of carbon. Population growth is measured by an increase in the turbidity of the liquid medium; turbidity is measured as an increase in light absorbance at a wavelength of 420 nm (A_{420} nm).

The Experiment

To investigate this intriguing possibility, Julian Adams and co-workers at the University of Michigan set out to see if polymorphism for metabolic abilities would develop spontaneously in bacteria growing in a uniform environment.

For a bacterial subject they chose *Escherichia coli* (*E. coli*), a widely studied bacterium whose growth under laboratory conditions is well understood. Cultures of *Escherichia coli* can be maintained in chemostat culture for many hundreds of generations. A chemostat is a large container holding liquid culture medium. A little bit of the liquid is continuously removed, and an equal amount of fresh culture medium added to replace what leaves. The growth of the *E. coli* culture is limited by the amount of glucose remaining in the culture medium to feed the growing cells.

Researchers inoculated a glucose-limited chemostat culture media with the *E. coli* strain JA122, and maintained the continuous culture for 773 generations. A sample was taken from the chemostat after 773 generations and analyzed for the presence of new strains of *E. coli*. Any variation among the cells in the sample would indicate that polymorphism had arisen.

To detect metabolic variation within the sample of growing cells, Adams's team analyzed the rate of glucose uptake and the concentration of acetate, among other variables. By examining such biochemical parameters, the researchers could determine if the different strains were filling different metabolic "niches"—that is, using the metabolic environment in different ways. Metabolic niches were characterized by looking at the normal products of aerobic fermentation, acetate and glycerol, which appear in the growth medium as a by-product of *E. coli* metabolism.

To further classify the strains, batch cultures containing two strains were established to analyze interactions between the two groups.

The Results

Three distinct variants were detected in the 773-generation *E. coli*, each being maintained at stable levels in the continuously growing culture. Clearly polymorphism *can* appear within an initially uniform bacterial population growing in a simple homogeneous environment.

When mixed together and allowed to compete, one strain does not drive the other two to extinction, as theory had predicted. Instead, the three new strains, CV101, CV103, and CV116, all persist (see graph *a* above).

The three strains were then analyzed to see how they differed. CV103 exhibited the highest rate of glucose uptake and produced the most acetate (an end product of glucose aerobic fermentation). Is this difference important? To see, the CV103 strain was co-cultured with CV101. They maintained stable growth levels, which indicated that the contribution of the third strain, CV116, was not required to maintain their growth.

What is the difference between CV101 and CV103? CV101 could grow in culture filtrate of CV103 but in the reverse situation, CV103 could not grow. This indicates that CV103 secretes a substance upon which CV101 can grow. Is CV101 utilizing the acetate produced by CV103 as its carbon source?

To test this possibility, CV101 and CV103 were grown together in media with acetate as the only carbon source. The results from this experiment are shown in graph *b* above and indicate that CV101 thrives on an acetate carbon source, while CV103 does not and requires an additional carbon source such as glucose.

These results indicate that two of the strains are maintained in polymorphism at stable levels because they have evolved different adaptations that allow them to coexist by filling different niches. One strain (CV101) is maintained in the population because it is able to use a metabolic by-product released by another strain (CV103).



To explore this experiment further, go to the Virtual Lab at www.mhhe.com/raven6/vlab4.mhtml

11

How Cells Divide

Concept Outline

11.1 Bacteria divide far more simply than do eukaryotes.

Cell Division in Prokaryotes. Bacterial cells divide by splitting in two.

11.2 Chromosomes are highly ordered structures.

Discovery of Chromosomes. All eukaryotic cells contain chromosomes, but different organisms possess differing numbers of chromosomes.

The Structure of Eukaryotic Chromosomes. Proteins play an important role in packaging DNA in chromosomes.

11.3 Mitosis is a key phase of the cell cycle.

Phases of the Cell Cycle. The cell cycle consists of three growth phases, a nuclear division phase, and a cytoplasmic division stage.

Interphase: Preparing for Mitosis. In interphase, the cell grows, replicates its DNA, and prepares for cell division.

Mitosis. In prophase, the chromosomes condense and microtubules attach sister chromosomes to opposite poles of the cell. In metaphase, chromosomes align along the center of the cell. In anaphase, the chromosomes separate; in telophase the spindle dissipates and the nuclear envelope reforms.

Cytokinesis. In cytokinesis, the cytoplasm separates into two roughly equal halves.

11.4 The cell cycle is carefully controlled.

General Strategy of Cell Cycle Control. At three points in the cell cycle, feedback from the cell determines whether the cycle will continue.

Molecular Mechanisms of Cell Cycle Control. Special proteins regulate the “checkpoints” of the cell cycle.

Cancer and the Control of Cell Proliferation. Cancer results from damage to genes encoding proteins that regulate the cell division cycle.



FIGURE 11.1

Cell division in bacteria. It's hard to imagine fecal coliform bacteria as beautiful, but here is *Escherichia coli*, inhabitant of the large intestine and the biotechnology lab, spectacularly caught in the act of fission.

All species of organisms—bacteria, alligators, the weeds in a lawn—grow and reproduce. From the smallest of creatures to the largest, all species produce offspring like themselves and pass on the hereditary information that makes them what they are. In this chapter, we begin our consideration of heredity with an examination of how cells reproduce (figure 11.1). The mechanism of cell reproduction and its biological consequences have changed significantly during the evolution of life on earth.

11.1 Bacteria divide far more simply than do eukaryotes.

Cell Division in Prokaryotes

In bacteria, which are prokaryotes and lack a nucleus, cell division consists of a simple procedure called **binary fission** (literally, “splitting in half”), in which the cell divides into two equal or nearly equal halves (figure 11.2). The genetic information, or *genome*, replicates early in the life of the cell. It exists as a single, circular, double-stranded DNA molecule. Fitting this DNA circle into the bacterial cell is a remarkable feat of packaging—fully stretched out, the DNA of a bacterium like *Escherichia coli* is about 500 times longer than the cell itself.

The DNA circle is attached at one point to the cytoplasmic surface of the bacterial cell’s plasma membrane. At a specific site on the DNA molecule called the *replication origin*, a battery of more than 22 different proteins begins the process of copying the DNA (figure 11.3). When these enzymes have proceeded all the way around the circle of DNA, the cell possesses two copies of the genome. These “daughter” genomes are attached side-by-side to the plasma membrane.

The growth of a bacterial cell to about twice its initial size induces the onset of cell division. A wealth of recent evidence suggests that the two daughter chromosomes are actively partitioned during this process. As this process proceeds, the cell lays down new plasma membrane and cell wall materials in the zone between the attachment sites of the two daughter genomes. A new plasma membrane grows between the genomes; eventually, it reaches all the way into the center of the cell, dividing it in two. Because the membrane forms between the two genomes, each new cell is assured of retaining one of the genomes. Finally, a new cell wall forms around the new membrane.

The evolution of the eukaryotes introduced several additional factors into the process of cell division. Eukaryotic



FIGURE 11.2
Fission (40,000 $\times$). Bacteria divide by a process of simple cell fission. Note the newly formed plasma membrane between the two daughter cells.

cells are much larger than bacteria, and their genomes contain much more DNA. Eukaryotic DNA is contained in a number of linear chromosomes, whose organization is much more complex than that of the single, circular DNA molecules in bacteria. In chromosomes, DNA forms a complex with packaging proteins called histones and is wound into tightly condensed coils.

Bacteria divide by binary fission. Fission begins in the middle of the cell. An active partitioning process ensures that one genome will end up in each daughter cell.

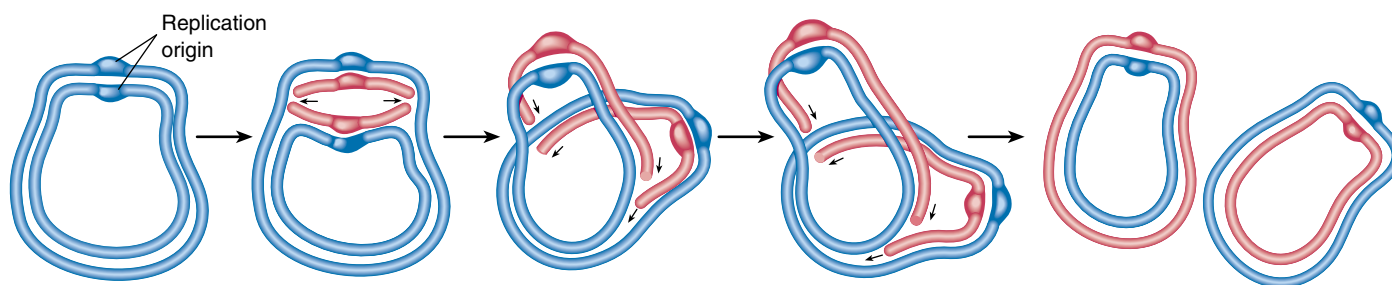


FIGURE 11.3
How bacterial DNA replicates. The replication of the circular DNA molecule (*blue*) that constitutes the genome of a bacterium begins at a single site, called the replication origin. The replication enzymes move out in both directions from that site and make copies (*red*) of each strand in the DNA duplex. When the enzymes meet on the far side of the molecule, replication is complete.

11.2 Chromosomes are highly ordered structures.

Discovery of Chromosomes

Chromosomes were first observed by the German embryologist Walther Fleming in 1882, while he was examining the rapidly dividing cells of salamander larvae. When Fleming looked at the cells through what would now be a rather primitive light microscope, he saw minute threads within their nuclei that appeared to be dividing lengthwise. Fleming called their division **mitosis**, based on the Greek word *mitos*, meaning “thread.”

Chromosome Number

Since their initial discovery, chromosomes have been found in the cells of all eukaryotes examined. Their number may vary enormously from one species to another. A few kinds of organisms—such as the Australian ant *Myrmecia*, the plant *Haplopappus gracilis*, a relative of the sunflower that grows in North American deserts; and the fungus *Penicillium*—have only 1 pair of chromosomes, while some ferns have more than 500 pairs (table 11.1). Most eukaryotes have between 10 and 50 chromosomes in their body cells.

Human cells each have 46 chromosomes, consisting of 23 nearly identical pairs (figure 11.4). Each of these 46 chromosomes contains hundreds or thousands of genes that play important roles in determining how a person’s body develops and functions. For this reason, possession of all the chromosomes is essential to survival. Humans missing even one chromosome, a condition called monosomy, do not survive embryonic development in most cases. Nor does the human embryo develop properly with an extra copy of any one chromosome, a condition called trisomy. For all but a few of the smallest chromosomes,



FIGURE 11.4
Human chromosomes. This photograph (950×) shows human chromosomes as they appear immediately before nuclear division. Each DNA molecule has already replicated, forming identical copies held together by a constriction called the centromere.

trisomy is fatal, and even in those few cases, serious problems result. Individuals with an extra copy of the very small chromosome 21, for example, develop more slowly than normal and are mentally retarded, a condition called Down syndrome.

All eukaryotic cells store their hereditary information in chromosomes, but different kinds of organisms utilize very different numbers of chromosomes to store this information.

Table 11.1 Chromosome Number in Selected Eukaryotes

Group	Total Number of Chromosomes	Group	Total Number of Chromosomes	Group	Total Number of Chromosomes
FUNGI		PLANTS		VERTEBRATES	
<i>Neurospora</i> (haploid)	7	<i>Haplopappus gracilis</i>	2	Opossum	22
<i>Saccharomyces</i> (a yeast)	16	Garden pea	14	Frog	26
INSECTS		Corn	20	Mouse	40
Mosquito	6	Bread wheat	42	Human	46
<i>Drosophila</i>	8	Sugarcane	80	Chimpanzee	48
Honeybee	32	Horsetail	216	Horse	64
Silkworm	56	Adder’s tongue fern	1262	Chicken	78
				Dog	78

The Structure of Eukaryotic Chromosomes

In the century since discovery of chromosomes, we have learned a great deal about their structure and composition.

Composition of Chromatin

Chromosomes are composed of **chromatin**, a complex of DNA and protein; most are about 40% DNA and 60% protein. A significant amount of RNA is also associated with chromosomes because chromosomes are the sites of RNA synthesis. The DNA of a chromosome is one very long, double-stranded fiber that extends unbroken through the entire length of the chromosome. A typical human chromosome contains about 140 million (1.4×10^8) nucleotides in its DNA. The amount of information one chromosome contains would fill about 280 printed books of 1000 pages each, if each nucleotide corresponded to a “word” and each page had about 500 words on it. Further-

more, if the strand of DNA from a single chromosome were laid out in a straight line, it would be about 5 centimeters (2 inches) long. Fitting such a strand into a nucleus is like cramming a string the length of a football field into a baseball—and that’s only 1 of 46 chromosomes! In the cell, however, the DNA is coiled, allowing it to fit into a much smaller space than would otherwise be possible.

Chromosome Coiling

How can this long DNA fiber coil so tightly? If we gently disrupt a eukaryotic nucleus and examine the DNA with an electron microscope, we find that it resembles a string of beads (figure 11.5). Every 200 nucleotides, the DNA duplex is coiled around a core of eight histone proteins, forming a complex known as a **nucleosome**. Unlike most proteins, which have an overall negative charge, histones are positively charged, due to an abundance of the basic amino acids arginine and lysine. They are thus strongly attracted to the negatively charged phosphate groups of the

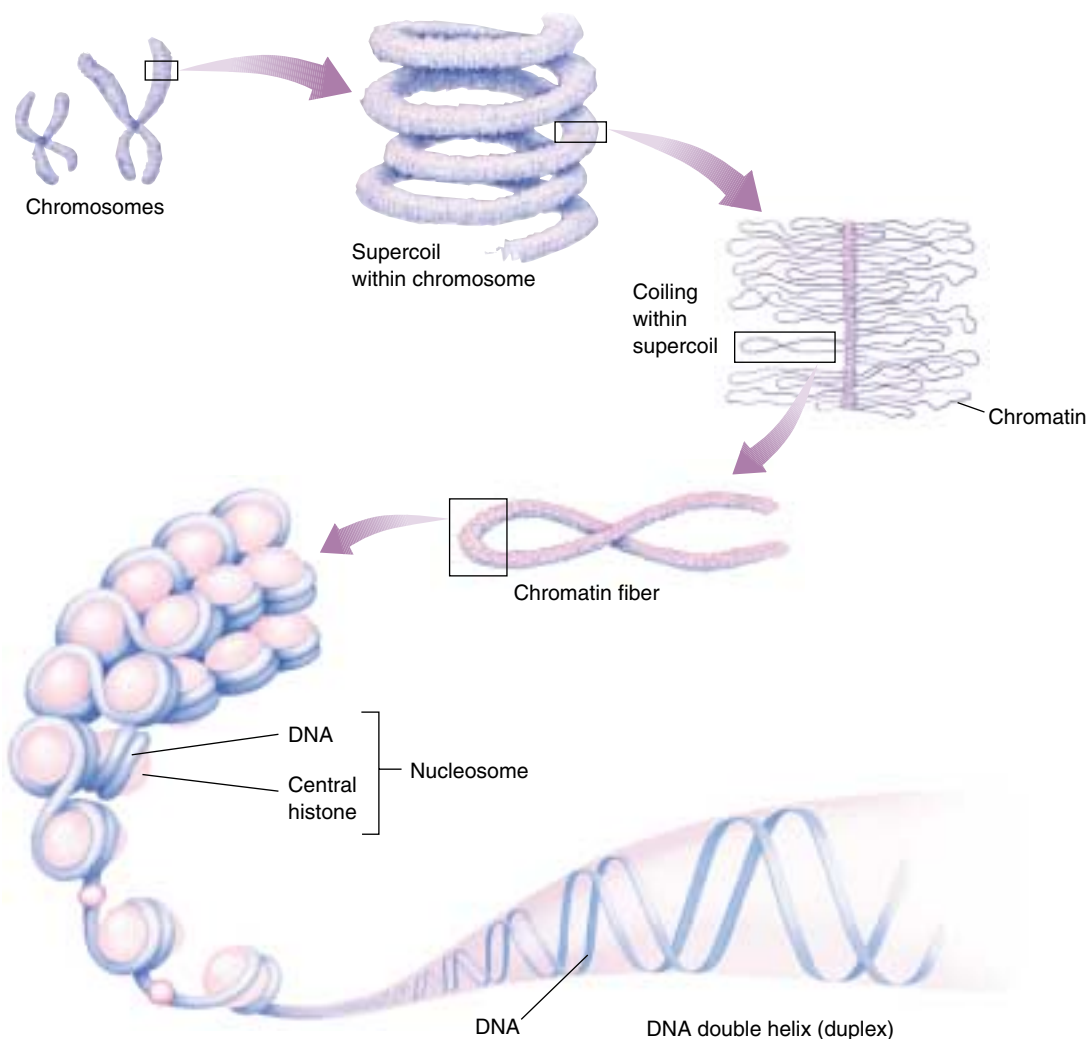


FIGURE 11.5
Levels of eukaryotic chromosomal organization.

Nucleotides assemble into long double strands of DNA molecules. These strands require further packaging to fit into the cell nucleus. The DNA duplex is tightly bound to and wound around proteins called *histones*. The DNA-wrapped histones are called *nucleosomes*. The nucleosomes then coalesce into *chromatin* fibers, ultimately coiling around into *supercoils* that make up the form of DNA recognized as a *chromosome*.

DNA. The histone cores thus act as “magnetic forms” that promote and guide the coiling of the DNA. Further coiling occurs when the string of nucleosomes wraps up into higher order coils called supercoils.

Highly condensed portions of the chromatin are called **heterochromatin**. Some of these portions remain permanently condensed, so that their DNA is never expressed. The remainder of the chromosome, called **euchromatin**, is condensed only during cell division, when compact packaging facilitates the movement of the chromosomes. At all other times, euchromatin is present in an open configuration, and its genes can be expressed. The way chromatin is packaged when the cell is not dividing is not well understood beyond the level of nucleosomes and is a topic of intensive research.

Chromosome Karyotypes

Chromosomes may differ widely in appearance. They vary in size, staining properties, the location of the *centromere* (a constriction found on all chromosomes), the relative length of the two arms on either side of the centromere, and the positions of constricted regions along the arms. The particular array of chromosomes that an individual possesses is called its **karyotype** (figure 11.6). Karyotypes show marked differences among species and sometimes even among individuals of the same species.

To examine a human karyotype, investigators collect a cell sample from blood, amniotic fluid, or other tissue and add chemicals that induce the cells in the sample to divide. Later, they add other chemicals to stop cell division at a stage when the chromosomes are most condensed and thus most easily distinguished from one another. The cells are then broken open and their contents, including the chromosomes, spread out and stained. To facilitate the examination of the karyotype, the chromosomes are usually photographed, and the outlines of the chromosomes are cut out of the photograph and arranged in order (see figure 11.6).

How Many Chromosomes Are in a Cell?

With the exception of the **gametes** (eggs or sperm) and a few specialized tissues, every cell in a human body is **diploid** ($2n$). This means that the cell contains two nearly identical copies of each of the 23 types of chromosomes, for a total of 46 chromosomes. The **haploid** ($1n$) gametes contain only one copy of each of the 23 chromosome types, while certain tissues have unusual numbers of chromosomes—many liver cells, for example, have two nuclei, while mature red blood cells have no nuclei at all. The two copies of each chromosome in body cells are called **homologous chromosomes**, or **homologues** (Greek *homologia*, “agreement”). Before cell division, each homologue replicates, producing two identical **sister chromatids** joined at the **centromere**, a condensed area found on all eukaryotic chromosomes (figure 11.7). Hence, as cell division begins, a

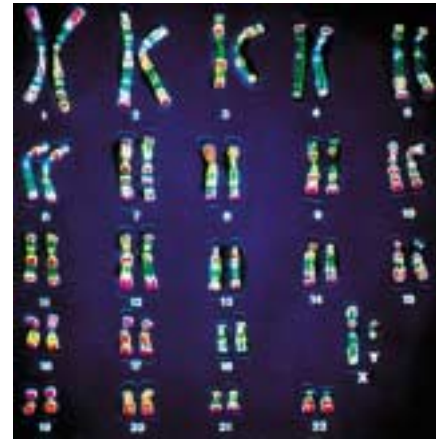


FIGURE 11.6

A human karyotype. The individual chromosomes that make up the 23 pairs differ widely in size and in centromere position. In this preparation, the chromosomes have been specifically stained to indicate further differences in their composition and to distinguish them clearly from one another.

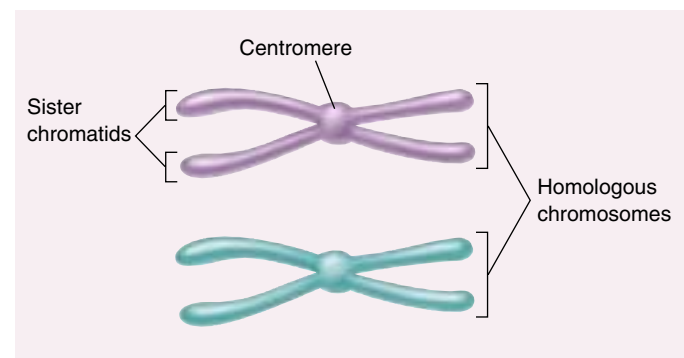


FIGURE 11.7

The difference between homologous chromosomes and sister chromatids. Homologous chromosomes are a pair of the same chromosome—say, chromosome number 16. Sister chromatids are the two replicas of a single chromosome held together by the centromeres after DNA replication.

human body cell contains a total of 46 replicated chromosomes, each composed of two sister chromatids joined by one centromere. The cell thus contains 46 centromeres and 92 chromatids (2 sister chromatids for each of 2 homologues for each of 23 chromosomes). The cell is said to contain 46 chromosomes rather than 92 because, by convention, the number of chromosomes is obtained by counting centromeres.

Eukaryotic genomes are larger and more complex than those of bacteria. Eukaryotic DNA is packaged tightly into chromosomes, enabling it to fit inside cells. Haploid cells contain one set of chromosomes, while diploid cells contain two sets.

11.3 Mitosis is a key phase of the cell cycle.

Phases of the Cell Cycle

The increased size and more complex organization of eukaryotic genomes over those of bacteria required radical changes in the process by which the two replicas of the genome are partitioned into the daughter cells during cell division. This division process is diagrammed as a **cell cycle**, consisting of five phases (figure 11.8).

The Five Phases

G₁ is the primary growth phase of the cell. For many organisms, this encompasses the major portion of the cell's life span. **S** is the phase in which the cell synthesizes a replica of the genome. **G₂** is the second growth phase, in which preparations are made for genomic separation. During this phase, mitochondria and other organelles replicate, chromosomes condense, and microtubules begin to assemble at a spindle. **G₁**, **S**, and **G₂** together constitute **interphase**, the portion of the cell cycle between cell divisions.

M is the phase of the cell cycle in which the microtubular apparatus assembles, binds to the chromosomes, and moves the sister chromatids apart. Called **mitosis**, this process is the essential step in the separation of the two daughter genomes. We will discuss mitosis as it occurs in animals and plants, where the process does not vary much (it is somewhat different among fungi and some protists). Although mitosis is a continuous process, it is traditionally subdivided into four stages: prophase, metaphase, anaphase, and telophase.

C is the phase of the cell cycle when the cytoplasm divides, creating two daughter cells. This phase is called **cytokinesis**. In animal cells, the microtubule spindle helps position a contracting ring of actin that constricts like a drawstring to pinch the cell in two. In cells with a cell wall, such as plant cells, a plate forms between the dividing cells.

Duration of the Cell Cycle

The time it takes to complete a cell cycle varies greatly among organisms. Cells in growing embryos can complete their cell cycle in under 20 minutes; the shortest known animal nuclear division cycles occur in fruit fly embryos (8 minutes). Cells such as these simply divide their nuclei as quickly as they can replicate their DNA, without cell growth. Half of the cycle is taken up by **S**, half by **M**, and essentially none by **G₁** or **G₂**. Because mature cells require time to grow, most of their cycles are much longer than those of embryonic tissue. Typically, a dividing mammalian cell completes its cell cycle in about

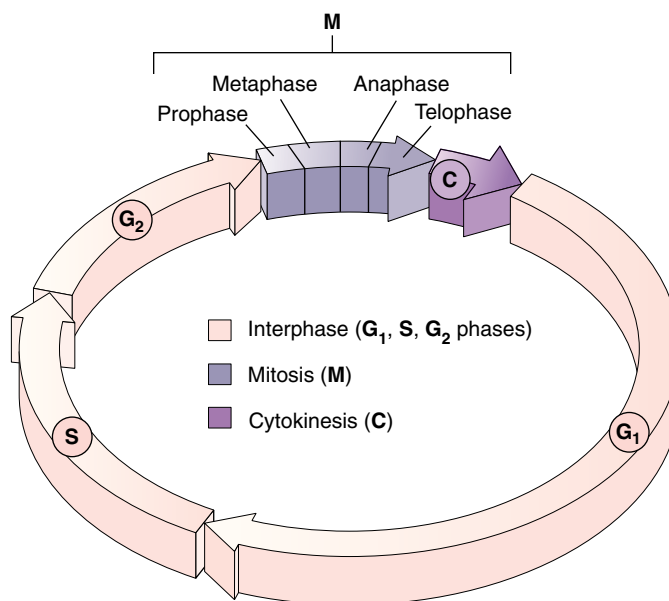


FIGURE 11.8

The cell cycle. Each wedge represents one hour of the 22-hour cell cycle in human cells growing in culture. **G₁** represents the primary growth phase of the cell cycle, **S** the phase during which a replica of the genome is synthesized, and **G₂** the second growth phase.

24 hours, but some cells, like certain cells in the human liver, have cell cycles lasting more than a year. During the cycle, growth occurs throughout the **G₁** and **G₂** phases (referred to as “gap” phases, as they separate **S** from **M**), as well as during the **S** phase. The **M** phase takes only about an hour, a small fraction of the entire cycle.

Most of the variation in the length of the cell cycle from one organism or tissue to the next occurs in the **G₁** phase. Cells often pause in **G₁** before DNA replication and enter a resting state called **G₀ phase**; they may remain in this phase for days to years before resuming cell division. At any given time, most of the cells in an animal's body are in **G₀** phase. Some, such as muscle and nerve cells, remain there permanently; others, such as liver cells, can resume **G₁** phase in response to factors released during injury.

Most eukaryotic cells repeat a process of growth and division referred to as the cell cycle. The cycle can vary in length from a few minutes to several years.

Interphase: Preparing for Mitosis

The events that occur during interphase, made up of the G_1 , S, and G_2 phases, are very important for the successful completion of mitosis. During G_1 , cells undergo the major portion of their growth. During the S phase, each chromosome replicates to produce two sister chromatids, which remain attached to each other at the **centromere**. The centromere is a point of constriction on the chromosome, containing a specific DNA sequence to which is bound a disk of protein called a **kinetochore**. This disk functions as an attachment site for fibers that assist in cell division (figure 11.9). Each chromosome's centromere is located at a characteristic site.

The cell grows throughout interphase. The G_1 and G_2 segments of interphase are periods of active growth, when proteins are synthesized and cell organelles produced. The cell's DNA replicates only during the S phase of the cell cycle.

After the chromosomes have replicated in S phase, they remain fully extended and uncoiled. This makes them invisible under the light microscope. In G_2 phase, they begin the long process of **condensation**, coiling ever more tightly. Special *motor proteins* are involved in the rapid final condensation of the chromosomes that occurs early in mitosis. Also during G_2 phase, the cells begin to assemble the machinery they will later use to move the chromosomes to opposite poles of the cell. In animal cells, a pair of microtubule-organizing centers called **centrioles** replicate. All eukaryotic cells undertake an extensive synthesis of *tubulin*, the protein of which microtubules are formed.

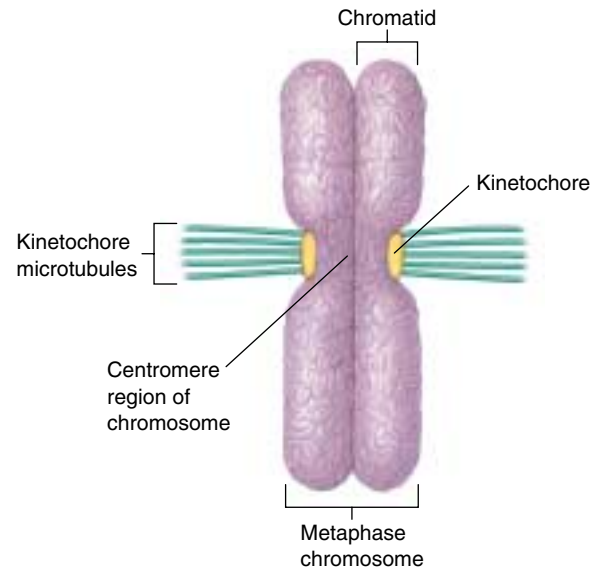


FIGURE 11.9

Kinetochores. In a metaphase chromosome, kinetochore microtubules are anchored to proteins at the centromere.

Interphase is that portion of the cell cycle in which the chromosomes are invisible under the light microscope because they are not yet condensed. It includes the G_1 , S, and G_2 phases. In the G_2 phase, the cell mobilizes its resources for cell division.

A Vocabulary of Cell Division

binary fission Asexual reproduction of a cell by division into two equal or nearly equal parts. Bacteria divide by binary fission.

centromere A constricted region of a chromosome about 220 nucleotides in length, composed of highly repeated DNA sequences (satellite DNA). During mitosis, the centromere joins the two sister chromatids and is the site to which the kinetochores are attached.

chromatid One of the two copies of a replicated chromosome, joined by a single centromere to the other strand.

chromatin The complex of DNA and proteins of which eukaryotic chromosomes are composed.

chromosome The structure within cells that contains the genes. In eukaryotes, it consists of a single linear DNA molecule associated with proteins. The DNA is replicated during S phase, and the replicas separated during M phase.

cytokinesis Division of the cytoplasm of a cell after nuclear division.

euchromatin The portion of a chromosome that is extended except during cell division, and from which RNA is transcribed.

heterochromatin The portion of a chromosome that remains permanently condensed and, therefore, is not transcribed into RNA. Most centromere regions are heterochromatic.

homologues Homologous chromosomes; in diploid cells, one of a pair of chromosomes that carry equivalent genes.

kinetochore A disk of protein bound to the centromere and attached to microtubules during mitosis, linking each chromatid to the spindle apparatus.

microtubule A hollow cylinder, about 25 nanometers in diameter, composed of subunits of the protein tubulin. Microtubules lengthen by the addition of tubulin subunits to their end(s) and shorten by the removal of subunits.

mitosis Nuclear division in which replicated chromosomes separate to form two genetically identical daughter nuclei. When accompanied by cytokinesis, it produces two identical daughter cells.

nucleosome The basic packaging unit of eukaryotic chromosomes, in which the DNA molecule is wound around a cluster of histone proteins. Chromatin is composed of long strings of nucleosomes that resemble beads on a string.

Mitosis

Prophase: Formation of the Mitotic Apparatus

When the chromosome condensation initiated in G₂ phase reaches the point at which individual condensed chromosomes first become visible with the light microscope, the first stage of mitosis, **prophase**, has begun. The condensation process continues throughout prophase; consequently, some chromosomes that start prophase as minute threads appear quite bulky before its conclusion. Ribosomal RNA synthesis ceases when the portion of the chromosome bearing the rRNA genes is condensed.

Assembling the Spindle Apparatus. The assembly of the microtubular apparatus that will later separate the sister chromatids also continues during prophase. In animal cells, the two centriole pairs formed during G₂ phase begin to move apart early in prophase, forming between them an axis of microtubules referred to as spindle fibers. By the time the centrioles reach the opposite poles of the cell, they have established a bridge of microtubules called the spindle apparatus between them. In plant cells, a similar bridge of microtubular fibers forms between opposite poles of the cell, although centrioles are absent in plant cells.

During the formation of the spindle apparatus, the nuclear envelope breaks down and the endoplasmic reticulum reabsorbs its components. At this point, then, the microtubular spindle fibers extend completely across the cell, from one pole to the other. Their orientation determines the plane in which the cell will subsequently divide, through the center of the cell at right angles to the spindle apparatus.

In animal cell mitosis, the centrioles extend a radial array of microtubules toward the plasma membrane when they reach the poles of the cell. This arrangement of microtubules is called an **aster**. Although the aster's function is not fully understood, it probably braces the centrioles against the membrane and stiffens the point of microtubular attachment during the retraction of the spindle. Plant cells, which have rigid cell walls, do not form asters.

Linking Sister Chromatids to Opposite Poles. Each chromosome possesses two kinetochores, one attached to the centromere region of each sister chromatid (see figure 11.9). As prophase continues, a second group of microtubules appears to grow from the poles of the cell toward the centromeres. These microtubules connect the kinetochores on each pair of sister chromatids to the two poles of the spindle. Because microtubules extending from the two poles attach to opposite sides of the centromere, they attach one sister chromatid to one pole and the other sister chromatid to the other pole. This arrangement is absolutely critical to the process of mitosis; any mistakes in microtubule positioning can be disas-

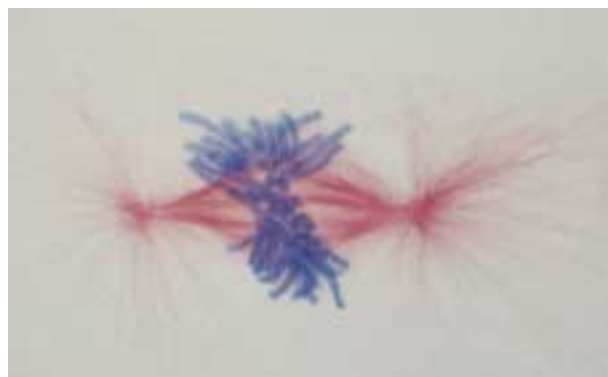
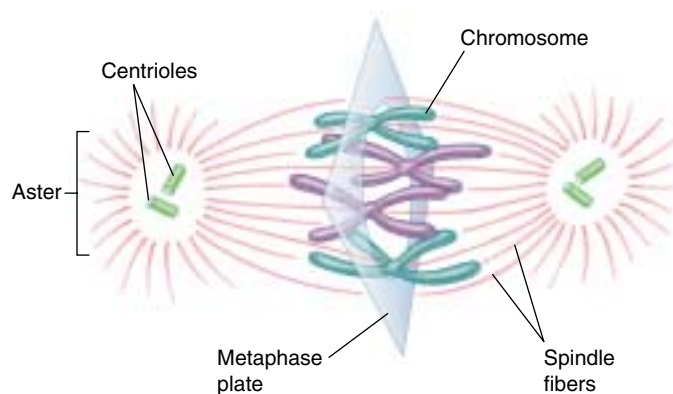


FIGURE 11.10

Metaphase. In metaphase, the chromosomes array themselves in a circle around the spindle midpoint.

trous. The attachment of the two sides of a centromere to the same pole, for example, leads to a failure of the sister chromatids to separate, so that they end up in the same daughter cell.

Metaphase: Alignment of the Centromeres

The second stage of mitosis, **metaphase**, is the phase where the chromosomes align in the center of the cell. When viewed with a light microscope, the chromosomes appear to array themselves in a circle along the inner circumference of the cell, as the equator girdles the earth (figure 11.10). An imaginary plane perpendicular to the axis of the spindle that passes through this circle is called the *metaphase plate*. The metaphase plate is not an actual structure, but rather an indication of the future axis of cell division. Positioned by the microtubules attached to the kinetochores of their centromeres, all of the chromosomes line up on the metaphase plate (figure 11.11). At this point, which marks the end of metaphase, their centromeres are neatly arrayed in a circle, equidistant from the two poles of the cell, with microtubules extending back towards the opposite poles of the cell in an arrangement called a spindle because of its shape.

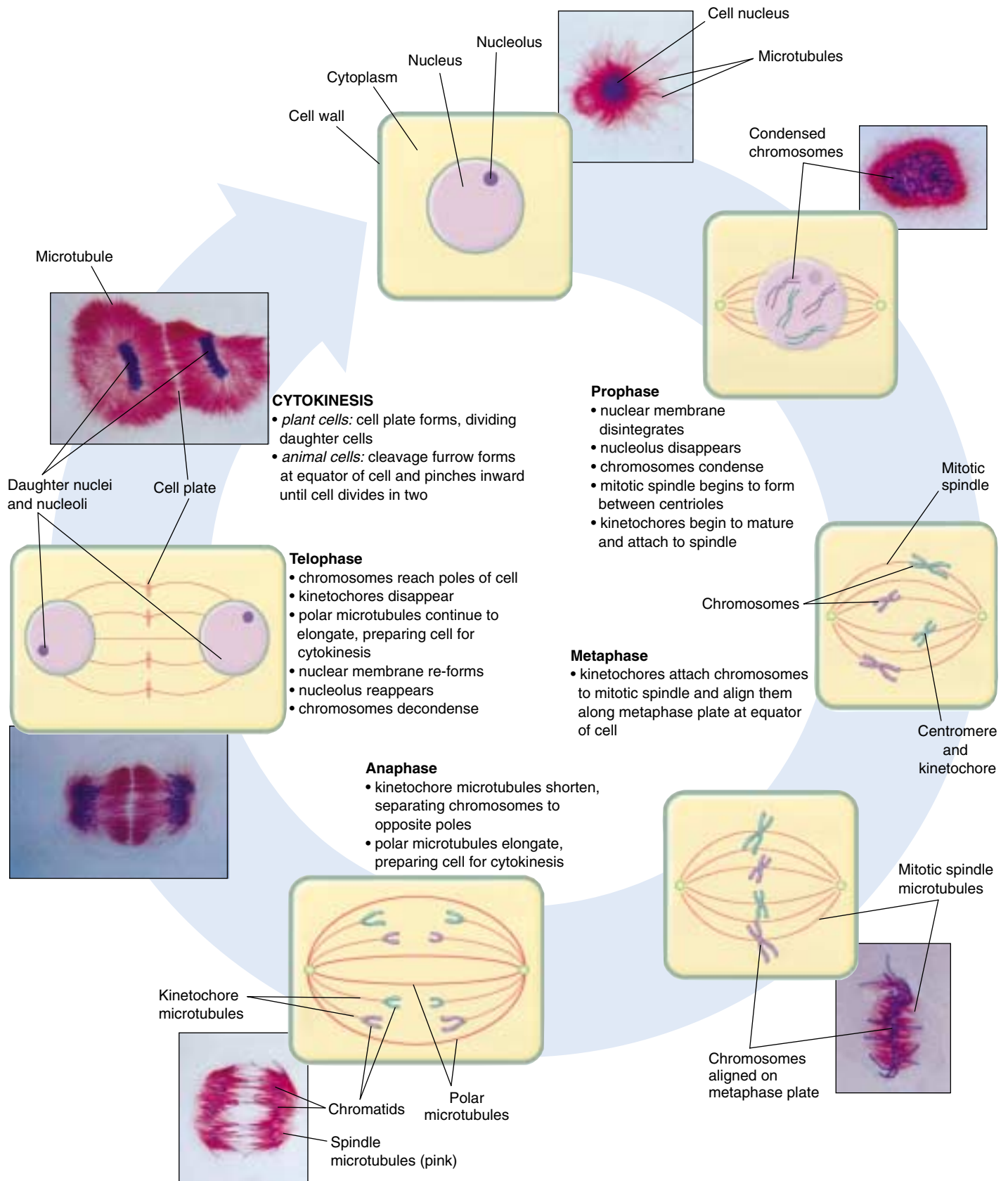


FIGURE 11.11

Mitosis and cytokinesis. Mitosis (separation of the two genomes) occurs in four stages—prophase, metaphase, anaphase, and telophase—and is followed by cytokinesis (division into two separate cells). In this depiction, the chromosomes of the African blood lily, *Haemantbus katherinae*, are stained blue, and microtubules are stained red.

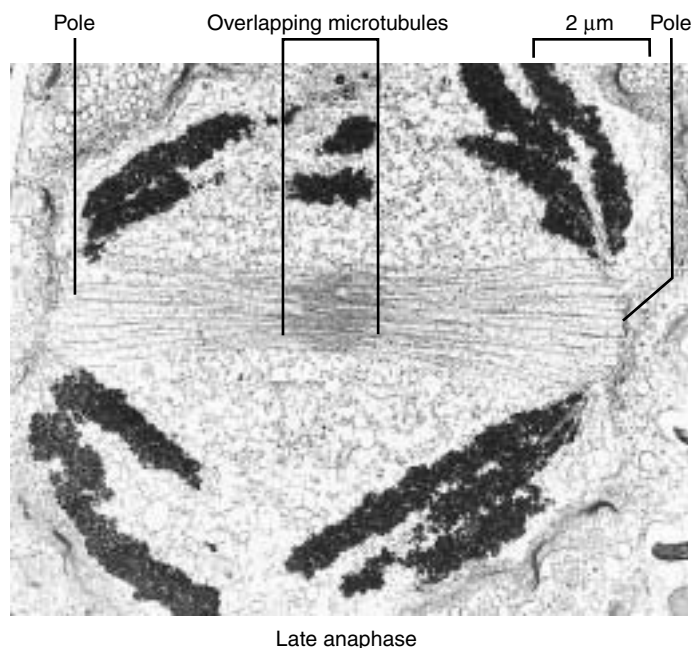
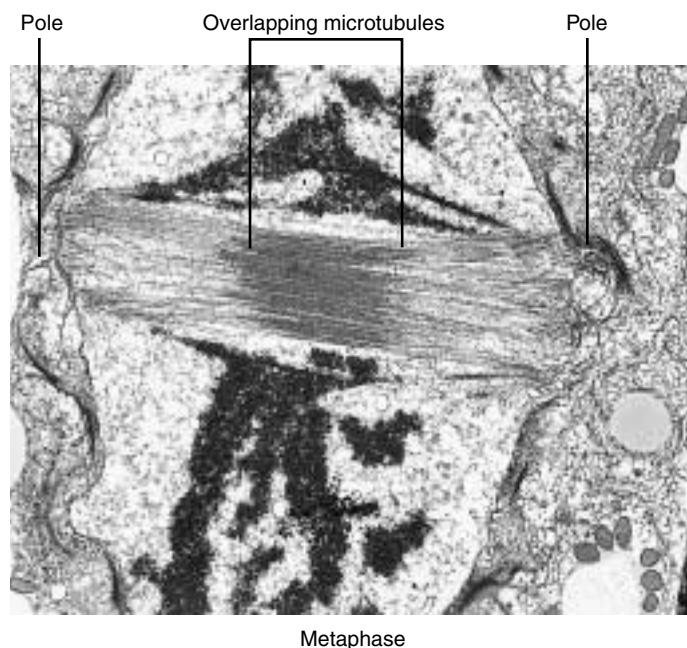


FIGURE 11.12

Microtubules slide past each other as the chromosomes separate. In these electron micrographs of dividing diatoms, the overlap of the microtubules lessens markedly during spindle elongation as the cell passes from metaphase to anaphase.

Anaphase and Telophase: Separation of the Chromatids and Reformation of the Nuclei

Of all the stages of mitosis, **anaphase** is the shortest and the most beautiful to watch. It starts when the centromeres divide. Each centromere splits in two, freeing the two sister chromatids from each other. The centromeres of all the chromosomes separate simultaneously, but the mechanism that achieves this synchrony is not known.

Freed from each other, the sister chromatids are pulled rapidly toward the poles to which their kinetochores are attached. In the process, two forms of movement take place simultaneously, each driven by microtubules.

First, *the poles move apart* as microtubular spindle fibers physically anchored to opposite poles slide past each other, away from the center of the cell (figure 11.12). Because another group of microtubules attach the chromosomes to the poles, the chromosomes move apart, too. If a flexible membrane surrounds the cell, it becomes visibly elongated.

Second, *the centromeres move toward the poles* as the microtubules that connect them to the poles shorten. This shortening process is not a contraction; the microtubules do not get any thicker. Instead, tubulin subunits are removed from the kinetochore ends of the microtubules by the organizing center. As more subunits are removed, the chromatid-bearing microtubules are progressively disassembled, and the chromatids are pulled ever closer to the poles of the cell.

When the sister chromatids separate in anaphase, the accurate partitioning of the replicated genome—the essential element of mitosis—is complete. In **telophase**, the spindle apparatus disassembles, as the microtubules are broken down into tubulin monomers that can be used to construct the cytoskeletons of the daughter cells. A nuclear envelope forms around each set of sister chromatids, which can now be called chromosomes because each has its own centromere. The chromosomes soon begin to uncoil into the more extended form that permits gene expression. One of the early group of genes expressed are the rRNA genes, resulting in the reappearance of the nucleolus.

During prophase, microtubules attach the centromeres joining pairs of sister chromatids to opposite poles of the spindle apparatus. During metaphase, each chromosome is drawn to a ring along the inner circumference of the cell by the microtubules extending from the centromere to the two poles of the spindle apparatus. During anaphase, the poles of the cell are pushed apart by microtubular sliding, and the sister chromatids are drawn to opposite poles by the shortening of the microtubules attached to them. During telophase, the spindle is disassembled, nuclear envelopes are reestablished, and the normal expression of genes present in the chromosomes is reinitiated.

Cytokinesis

Mitosis is complete at the end of telophase. The eukaryotic cell has partitioned its replicated genome into two nuclei positioned at opposite ends of the cell. While mitosis is going on, the cytoplasmic organelles, including mitochondria and chloroplasts (if present), were reassorted to areas that will separate and become the daughter cells. The replication of organelles takes place before cytokinesis, often in the S or G₂ phase. Cell division is still not complete at the end of mitosis, however, because the division of the cell proper has not yet begun. The phase of the cell cycle when the cell actually divides is called **cytokinesis**. It generally involves the cleavage of the cell into roughly equal halves.

Cytokinesis in Animal Cells

In animal cells and the cells of all other eukaryotes that lack cell walls, cytokinesis is achieved by means of a constricting belt of actin filaments. As these filaments slide past one another, the diameter of the belt decreases, pinching the cell and creating a *cleavage furrow* around the cell's circumference (figure 11.13*a*). As constriction proceeds, the furrow deepens until it eventually slices all the way into the center of the cell. At this point, the cell is divided in two (figure 11.13*b*).

Cytokinesis in Plant Cells

Plant cells possess a cell wall far too rigid to be squeezed in two by actin filaments. Instead, these cells assemble membrane components in their interior, at right angles to the spindle apparatus (figure 11.14). This expanding membrane partition, called a **cell plate**, continues to grow outward until it reaches the interior surface of the plasma membrane and fuses with it, effectively dividing the cell in two. Cellulose is then laid down on the new membranes, creating two new cell walls. The space between the daughter cells becomes impregnated with pectins and is called a **middle lamella**.

Cytokinesis in Fungi and Protists

In fungi and some groups of protists, the nuclear membrane does not dissolve and, as a result, all the events of mitosis occurs entirely *within* the nucleus. Only after mitosis is complete in these organisms does the nucleus then divide into two daughter nuclei, and one nucleus goes to each daughter cell during cytokinesis. This separate nuclear division phase of the cell cycle does not occur in plants, animals, or most protists.

After cytokinesis in any eukaryotic cell, the two daughter cells contain all of the components of a complete cell. While mitosis ensures that both daughter cells contain a full complement of chromosomes, no similar mechanism ensures that organelles such as mitochondria and chloro-

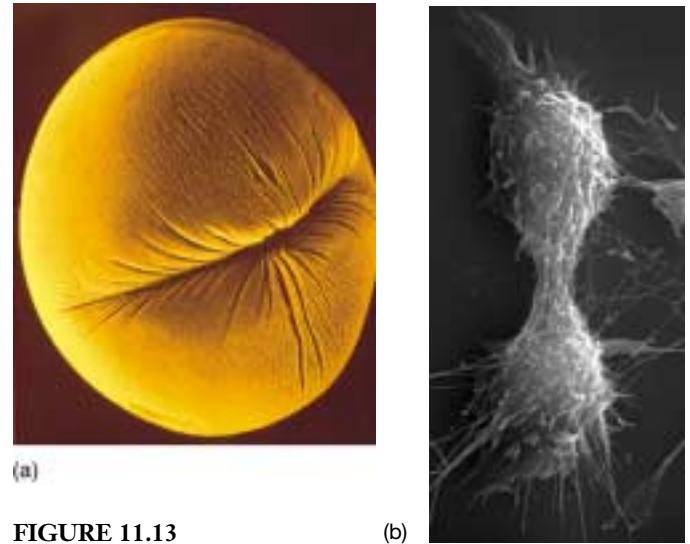


FIGURE 11.13

Cytokinesis in animal cells.

(*a*) A cleavage furrow forms around a dividing sea urchin egg (30 $\times$). (*b*) The completion of cytokinesis in an animal cell. The two daughter cells are still joined by a thin band of cytoplasm occupied largely by microtubules.

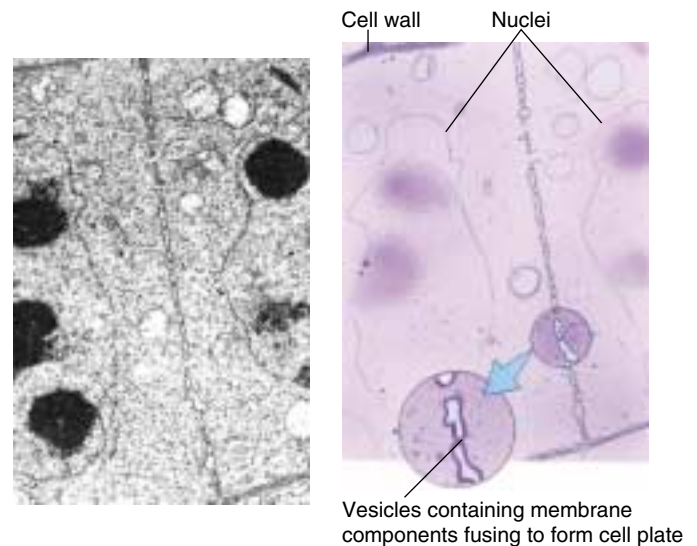


FIGURE 11.14

Cytokinesis in plant cells. In this photograph and companion drawing, a cell plate is forming between daughter nuclei. Once the plate is complete, there will be two cells.

plasts are distributed equally between the daughter cells. However, as long as some of each organelle are present in each cell, the organelles can replicate to reach the number appropriate for that cell.

Cytokinesis is the physical division of the cytoplasm of a eukaryotic cell into two daughter cells.

11.4 The cell cycle is carefully controlled.

General Strategy of Cell Cycle Control

The events of the cell cycle are coordinated in much the same way in all eukaryotes. The control system human cells utilize first evolved among the protists over a billion years ago; today, it operates in essentially the same way in fungi as it does in humans.

The goal of controlling any cyclic process is to adjust the duration of the cycle to allow sufficient time for all events to occur. In principle, a variety of methods can achieve this goal. For example, an internal “clock” can be employed to allow adequate time for each phase of the cycle to be completed. This is how many organisms control their daily activity cycles. The disadvantage of using such a clock to control the cell cycle is that it is not very flexible. One way to achieve a more flexible and sensitive regulation of a cycle is simply to let the completion of each phase of the cycle trigger the beginning of the next phase, as a runner passing a baton starts the next leg in a relay race. Until recently, biologists thought this type of mechanism controlled the cell division cycle. However, we now know that eukaryotic cells employ a separate, centralized controller to regulate the process: at critical points in the cell cycle, further progress depends upon a central set of “go/no-go” switches that are regulated by feedback from the cell.

This mechanism is the same one engineers use to control many processes. For example, the furnace that heats a home in the winter typically goes through a daily heating cycle. When the daily cycle reaches the morning “turn on” checkpoint, sensors report whether the house temperature is below the set point (for example, 70°F). If it is, the thermostat triggers the furnace, which warms the house. If the house is already at least that warm, the thermostat does not start up the furnace. Similarly, the cell cycle has key checkpoints where feedback signals from the cell about its size and the condition of its chromosomes can either trigger subsequent phases of the cycle, or delay them to allow more time for the current phase to be completed.

Architecture of the Control System

Three principal checkpoints control the cell cycle in eukaryotes (figure 11.15):

Cell growth is assessed at the G_1 checkpoint. Located near the end of G_1 , just before entry into S phase, this checkpoint makes the key decision of whether the cell should divide, delay division, or enter a resting stage (figure 11.16). In yeasts, where researchers first studied this checkpoint, it is called START. If conditions are favorable for division, the cell begins to copy its DNA,

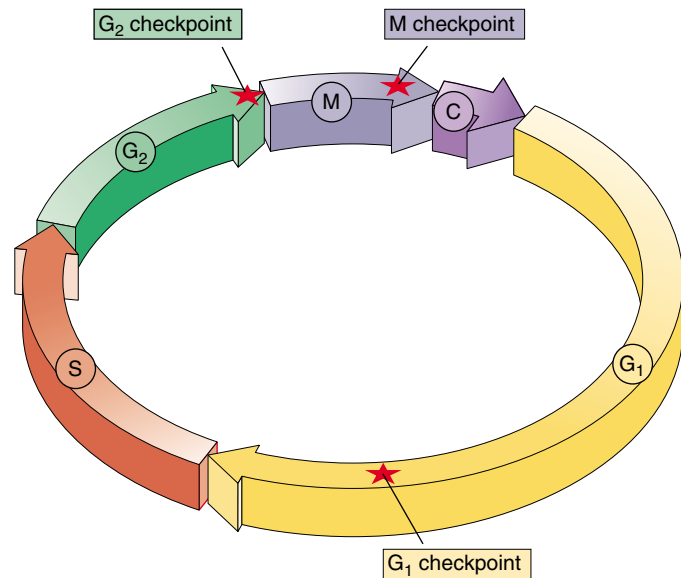


FIGURE 11.15

Control of the cell cycle. Cells use a centralized control system to check whether proper conditions have been achieved before passing three key “checkpoints” in the cell cycle.



FIGURE 11.16

The G_1 checkpoint. Feedback from the cell determines whether the cell cycle will proceed to the S phase, pause, or withdraw into G_0 for an extended rest period.

initiating S phase. The G_1 checkpoint is where the more complex eukaryotes typically arrest the cell cycle if environmental conditions make cell division impossible, or if the cell passes into G_0 for an extended period.

The success of DNA replication is assessed at the G_2 checkpoint. The second checkpoint, which occurs at the end of G_2 , triggers the start of M phase. If this checkpoint is passed, the cell initiates the many molecular processes that signal the beginning of mitosis.

Mitosis is assessed at the M checkpoint. Occurring at metaphase, the third checkpoint triggers the exit from mitosis and cytokinesis and the beginning of G_1 .

The cell cycle is controlled at three checkpoints.

Molecular Mechanisms of Cell Cycle Control

Exactly how does a cell achieve central control of the division cycle? The basic mechanism is quite simple. A set of proteins sensitive to the condition of the cell interact at the checkpoints to trigger the next events in the cycle. Two key types of proteins participate in this interaction: cyclin-dependent protein kinases and cyclins (figure 11.17).

The Cyclin Control System

Cyclin-dependent protein kinases (Cdks) are enzymes that phosphorylate (add phosphate groups to) the serine and threonine amino acids of key cellular enzymes and other proteins. At the G₂ checkpoint, for example, Cdks phosphorylate histones, nuclear membrane filaments, and the microtubule-associated proteins that form the mitotic spindle. Phosphorylation of these components of the cell division machinery initiates activities that carry the cycle past the checkpoint into mitosis.

Cyclins are proteins that bind to Cdks, enabling the Cdks to function as enzymes. Cyclins are so named because they are destroyed and resynthesized during each turn of the cell cycle (figure 11.18). Different cyclins regulate the G₁ and G₂ cell cycle checkpoints.

The G₂ Checkpoint. During G₂, the cell gradually accumulates G₂ cyclin (also called mitotic cyclin). This cyclin binds to Cdk to form a complex called MPF (mitosis-promoting factor). At first, MPF is not active in carrying the cycle past the G₂ checkpoint. But eventually, other cellular enzymes phosphorylate and so activate a few molecules of MPF. These activated MPFs in turn increase the activity of the enzymes that phosphorylate MPF, setting up a positive feedback that leads to a very rapid increase in the cellular concentration of activated MPF. When the level of activated MPF exceeds the threshold necessary to trigger mitosis, G₂ phase ends.

MPF sows the seeds of its own destruction. The length of time the cell spends in M phase is determined by the activity of MPF, for one of its many functions is to activate proteins that destroy cyclin. As mitosis proceeds to the end of metaphase, Cdk levels stay relatively constant, but increasing amounts of G₂ cyclin are degraded, causing progressively less MPF to be available and so initiating the events that end mitosis. After mitosis, the gradual accumulation of new cyclin starts the next turn of the cell cycle.

The G₁ Checkpoint. The G₁ checkpoint is thought to be regulated in a similar fashion. In unicellular eukaryotes such as yeasts, the main factor triggering DNA replication is cell size. Yeast cells grow and divide as rapidly as possible, and they make the START decision by comparing the volume of cytoplasm to the size of the genome. As a

FIGURE 11.17

A complex of two proteins triggers passage through cell cycle checkpoints. Cdk is a protein kinase that activates numerous cell proteins by phosphorylating them. Cyclin is a regulatory protein required to activate Cdk; in other words, Cdk does not function unless cyclin is bound to it.

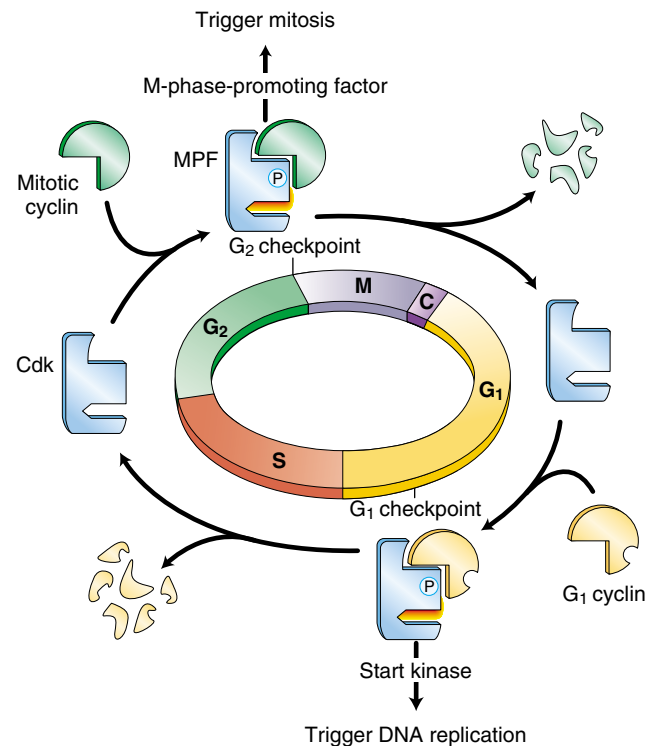
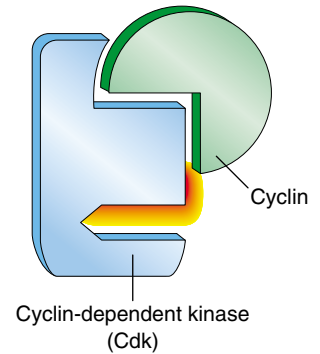


FIGURE 11.18

How cell cycle control works. As the cell cycle passes through the G₁ and G₂ checkpoints, Cdk becomes associated with different cyclins and, as a result, activates different cellular processes. At the completion of each phase, the cyclins are degraded, bringing Cdk activity to a halt until the next set of cyclins appears.

cell grows, its cytoplasm increases in size, while the amount of DNA remains constant. Eventually a threshold ratio is reached that promotes the production of cyclins and thus triggers the next round of DNA replication and cell division.

Controlling the Cell Cycle in Multicellular Eukaryotes

The cells of multicellular eukaryotes are not free to make individual decisions about cell division, as yeast cells are. The body’s organization cannot be maintained without severely limiting cell proliferation, so that only certain cells divide, and only at appropriate times. The way that cells inhibit individual growth of other cells is apparent in mammalian cells growing in tissue culture: a single layer of cells expands over a culture plate until the growing border of cells comes into contact with neighboring cells, and then the cells stop dividing. If a sector of cells is cleared away, neighboring cells rapidly refill that sector and then stop dividing again. How are cells able to sense the density of the cell culture around them? Each growing cell apparently binds minute amounts of positive regulatory signals called **growth factors**, proteins that stimulate cell division (such as MPF). When neighboring cells have used up what little growth factor is present, not enough is left to trigger cell division in any one cell.

Growth Factors and the Cell Cycle

As you may recall from chapter 7 (cell-cell interactions), growth factors work by triggering intracellular signaling systems. Fibroblasts, for example, possess numerous receptors on their plasma membranes for one of the first growth

factors to be identified: platelet-derived growth factor (PDGF). When PDGF binds to a membrane receptor, it initiates an amplifying chain of internal cell signals that stimulates cell division. PDGF was discovered when investigators found that fibroblasts would grow and divide in tissue culture only if the growth medium contained blood serum (the liquid that remains after blood clots); blood plasma (blood from which the cells have been removed without clotting) would not work. The researchers hypothesized that platelets in the blood clots were releasing into the serum one or more factors required for fibroblast growth. Eventually, they isolated such a factor and named it PDGF. Growth factors such as PDGF override cellular controls that otherwise inhibit cell division. When a tissue is injured, a blood clot forms and the release of PDGF triggers neighboring cells to divide, helping to heal the wound. Only a tiny amount of PDGF (approximately 10⁻¹⁰ M) is required to stimulate cell division.

Characteristics of Growth Factors. Over 50 different proteins that function as growth factors have been isolated (table 11.2 lists a few), and more undoubtedly exist. A specific cell surface receptor “recognizes” each growth factor, its shape fitting that growth factor precisely. When the growth factor binds with its receptor, the receptor reacts by triggering events within the cell (figure 11.19). The cellular selectivity of a particular growth factor depends upon which target cells bear its unique receptor. Some growth

Table 11.2 Growth Factors of Mammalian Cells		
Growth Factor	Range of Specificity	Effects
Epidermal growth factor (EGF)	Broad	Stimulates cell proliferation in many tissues; plays a key role in regulating embryonic development
Erythropoietin	Narrow	Required for proliferation of red blood cell precursors and their maturation into erythrocytes (red blood cells)
Fibroblast growth factor (FGF)	Broad	Initiates the proliferation of many cell types; inhibits maturation of many types of stem cells; acts as a signal in embryonic development
Insulin-like growth factor	Broad	Stimulates metabolism of many cell types; potentiates the effects of other growth factors in promoting cell proliferation
Interleukin-2	Narrow	Triggers the division of activated T lymphocytes during the immune response
Mitosis-promoting factor (MPF)	Broad	Regulates entrance of the cell cycle into the M phase
Nerve growth factor (NGF)	Narrow	Stimulates the growth of neuron processes during neural development
Platelet-derived growth factor (PDGF)	Broad	Promotes the proliferation of many connective tissues and some neuroglial cells
Transforming growth factor β (TGF-β)	Broad	Accentuates or inhibits the responses of many cell types to other growth factors; often plays an important role in cell differentiation

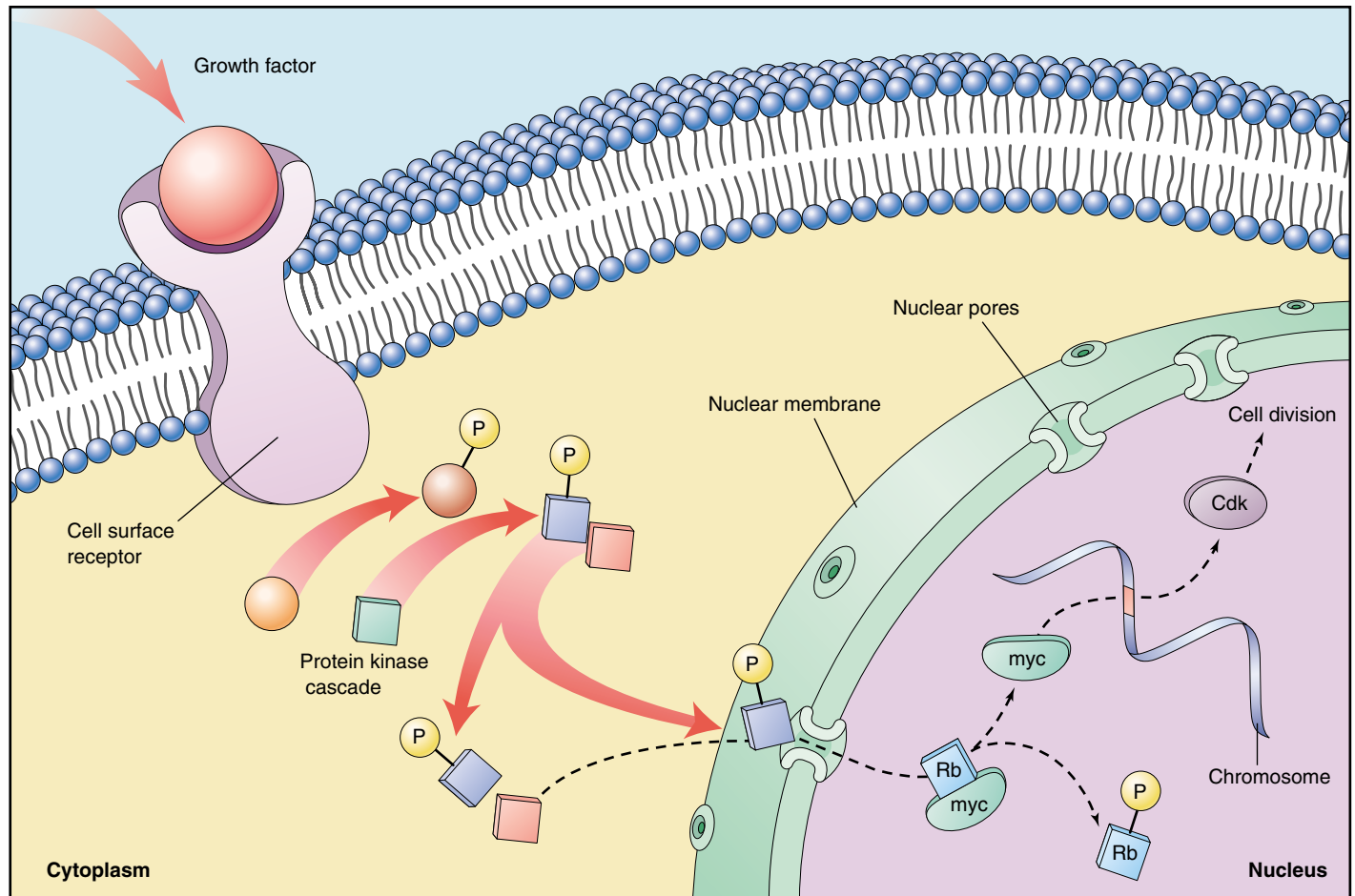


FIGURE 11.19

The cell proliferation-signaling pathway. Binding of a growth factor sets in motion a cascading intracellular signaling pathway (described in chapter 7), which activates nuclear regulatory proteins that trigger cell division. In this example, when the nuclear protein Rb is phosphorylated, another nuclear protein (myc) is released and is then able to stimulate the production of Cdk proteins.

factors, like PDGF and epidermal growth factor (EGF), affect a broad range of cell types, while others affect only specific types. For example, nerve growth factor (NGF) promotes the growth of certain classes of neurons, and erythropoietin triggers cell division in red blood cell precursors. Most animal cells need a combination of several different growth factors to overcome the various controls that inhibit cell division.

The G_0 Phase. If cells are deprived of appropriate growth factors, they stop at the G_1 checkpoint of the cell cycle. With their growth and division arrested, they remain in the G_0 phase, as we discussed earlier. This nongrowing state is distinct from the interphase stages of the cell cycle, G_1 , S, and G_2 .

It is the ability to enter G_0 that accounts for the incredible diversity seen in the length of the cell cycle among different tissues. Epithelial cells lining the gut divide more than twice a day, constantly renewing the lining of the digestive tract. By contrast, liver cells divide only once every year or two, spending most of their time in G_0 phase. Mature neurons and muscle cells usually never leave G_0 .

Two groups of proteins, cyclins and Cdks, interact to regulate the cell cycle. Cells also receive protein signals called growth factors that affect cell division.

Cancer and the Control of Cell Proliferation

The unrestrained, uncontrolled growth of cells, called cancer, is addressed more fully in chapter 18. However, cancer certainly deserves mention in a chapter on cell division, as it is essentially a disease of cell division—a failure of cell division *control*. Recent work has identified one of the culprits. Working independently, cancer scientists have repeatedly identified what has proven to be the same gene! Officially dubbed *p53* (researchers italicize the gene symbol to differentiate it from the protein), this gene plays a key role in the G₁ checkpoint of cell division. The gene's product, the p53 protein, monitors the integrity of DNA, checking that it is undamaged. If the p53 protein detects damaged DNA, it halts cell division and stimulates the activity of special enzymes to repair the damage. Once the DNA has been repaired, *p53* allows cell division to continue. In cases where the DNA is irreparable, *p53* then directs the cell to kill itself, activating an apoptosis

(cell suicide) program (see chapter 17 for a discussion of apoptosis).

By halting division in damaged cells, *p53* prevents the development of many mutated cells, and it is therefore considered a tumor-suppressor gene (even though its activities are not limited to cancer prevention). Scientists have found that *p53* is entirely absent or damaged beyond use in the majority of cancerous cells they have examined! It is precisely because *p53* is nonfunctional that these cancer cells are able to repeatedly undergo cell division without being halted at the G₁ checkpoint (figure 11.20). To test this, scientists administered healthy p53 protein to rapidly dividing cancer cells in a petri dish: the cells soon ceased dividing and died.

Scientists at Johns Hopkins University School of Medicine have further reported that cigarette smoke causes mutations in the *p53* gene. This study, published in 1995, reinforced the strong link between smoking and cancer described in chapter 18.

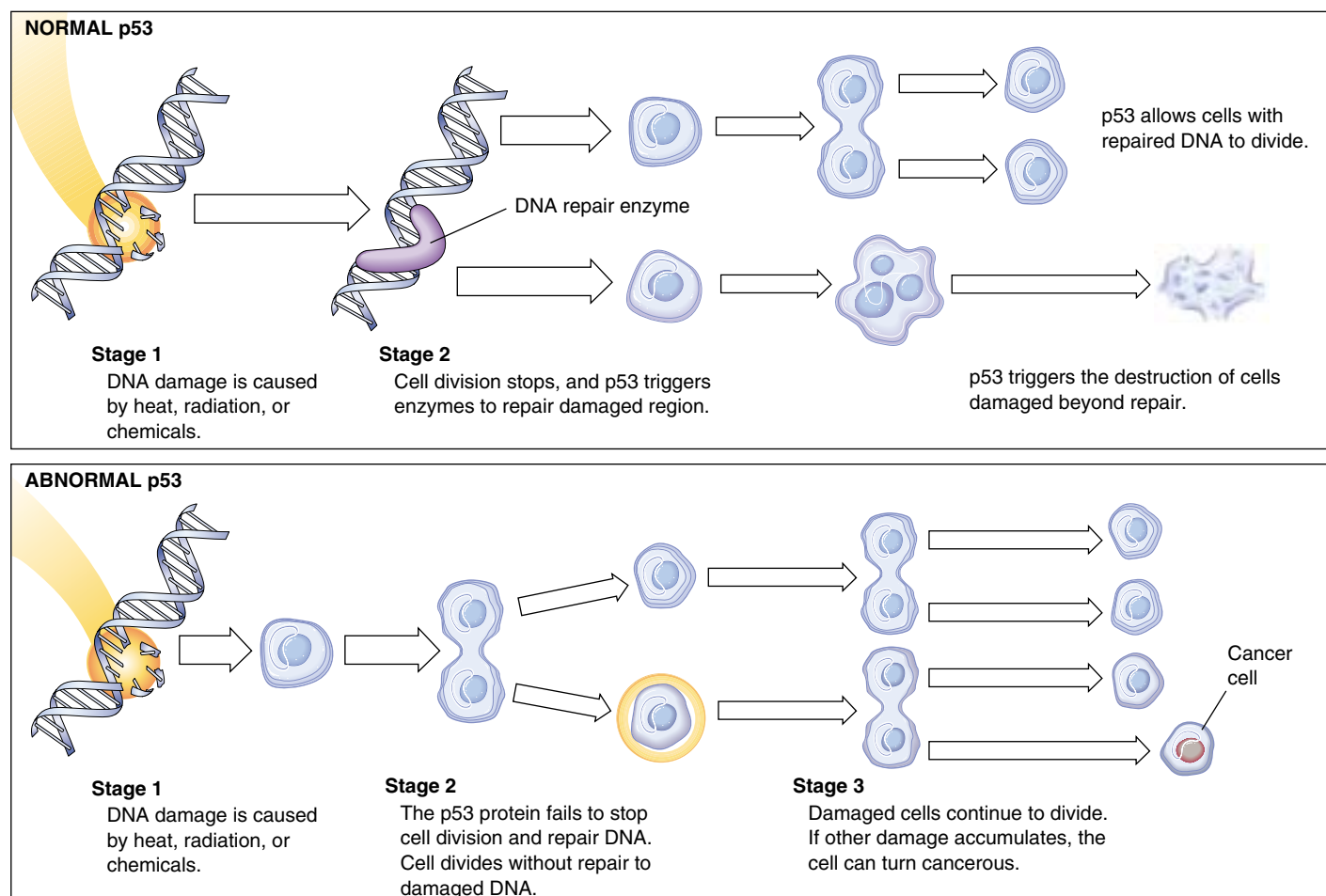


FIGURE 11.20

Cell division and p53 protein. Normal p53 protein monitors DNA, destroying cells with irreparable damage to their DNA. Abnormal p53 protein fails to stop cell division and repair DNA. As damaged cells proliferate, cancer develops.

Growth Factors and Cancer

How do growth factors influence the cell cycle? As you have seen, there are two different approaches, one positive and the other negative.

Proto-oncogenes. PDGF and many other growth factors utilize the positive approach, stimulating cell division. They trigger passage through the G₁ checkpoint by aiding the formation of cyclins and so activating genes that promote cell division. Genes that normally stimulate cell division are sometimes called *proto-oncogenes* because mutations that cause them to be overexpressed or hyperactive convert them into oncogenes (Greek *onco*, “cancer”), leading to the excessive cell proliferation that is characteristic of cancer. Even a single mutation (creating a heterozygote) can lead to cancer if the other cancer-preventing genes are nonfunctional. Geneticists, using Mendel’s terms, call such mutations of proto-oncogenes *dominant*.

Some 30 different proto-oncogenes are known. Some act very quickly after stimulation by growth factors. Among the most intensively studied of these are *myc*, *fos*, and *jun*, all of which cause unrestrained cell growth and division when they are overexpressed. In a normal cell, the *myc* proto-oncogene appears to be important in regulating the G₁ checkpoint. Cells in which *myc* expression is prevented will not divide, even in the presence of growth factors. A critical activity of *myc* and other genes in this group of immediately responding proto-oncogenes is to stimulate a second group of “delayed response” genes, including those that produce cyclins and Cdk proteins (figure 11.21).

Tumor-suppressor Genes. Other growth factors utilize a negative approach to cell cycle control. They block passage through the G₁ checkpoint by preventing cyclins from binding to Cdk, thus inhibiting cell division. Genes that normally inhibit cell division are called tumor-suppressor genes. When mutated, they can also lead to unrestrained cell division, but only if both copies of the gene are mutant. Hence, these cancer-causing mutations are *recessive*.

The most thoroughly understood of the tumor-suppressor genes is the retinoblastoma (*Rb*) gene. This gene was originally cloned from children with a rare form of eye cancer inherited as a recessive trait, implying that the normal gene product was a cancer suppressor that helped keep cell division in check. The *Rb* gene encodes a protein present in ample amounts within the nucleus. This protein interacts with many key regulatory proteins of the cell cycle, but how it does so depends upon its state of phosphorylation. In G₀ phase, the Rb protein is dephosphorylated. In this state, it binds to and ties up a set of regulatory proteins, like *myc* and *fos*, needed for cell proliferation, blocking their action and so inhibiting cell division (see figure 11.19). When phosphorylated, the Rb protein releases its captive regulatory proteins, freeing

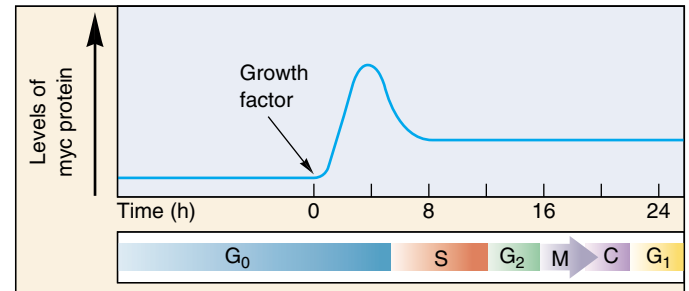


FIGURE 11.21

The role of *myc* in triggering cell division. The addition of a growth factor leads to transcription of the *myc* gene and rapidly increasing levels of the *myc* protein. This causes G₀ cells to enter the S phase and begin proliferating.

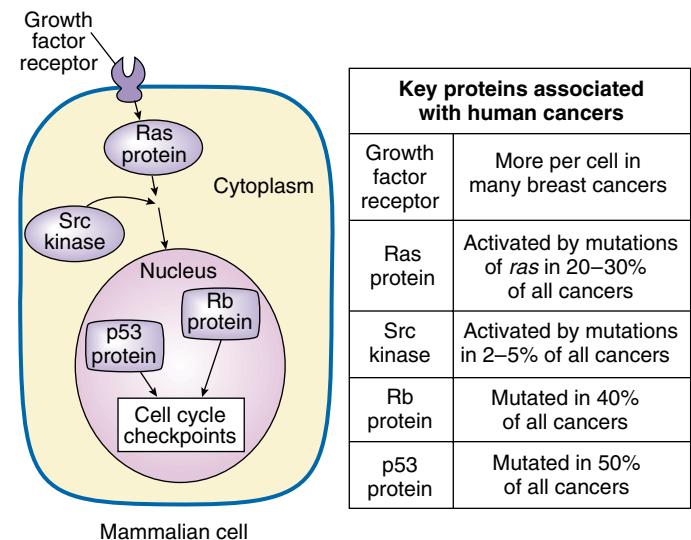


FIGURE 11.22

Mutations cause cancer. Mutations in genes encoding key components of the cell division-signaling pathway are responsible for many cancers. Among them are proto-oncogenes encoding growth factor receptors, such as ras protein, and kinase enzymes, such as *src*, that aid ras function. Mutations that disrupt tumor-suppressor proteins, such as Rb and p53, also foster cancer development.

them to act and so promoting cell division. Growth factors lessen the inhibition the Rb protein imposes by activating kinases that phosphorylate it. Free of Rb protein inhibition, cells begin to produce cyclins and Cdk, pass the G₁ checkpoint, and proceed through the cell cycle. Figure 11.22 summarizes the types of genes that can cause cancer when mutated.

The progress of mitosis is regulated by the interaction of two key classes of proteins, cyclin-dependent protein kinases and cyclins. Some growth factors accelerate the cell cycle by promoting cyclins and Cdks, others suppress it by inhibiting their action.



Summary

Questions

Media Resources

11.1 Bacteria divide far more simply than do eukaryotes.

- Bacterial cells divide by simple binary fission.
- The two replicated circular DNA molecules attach to the plasma membrane at different points, and fission is initiated between those points.

1. How is the genome replicated prior to binary fission in a bacterial cell?



• Cell Division Introduction



• Prokaryotes
• Scientists on Science: Ribozymes

11.2 Chromosomes are highly ordered structures.

- Eukaryotic DNA forms a complex with histones and other proteins and is packaged into chromosomes.
- In eukaryotic cells, DNA replication is completed during the S phase of the cell cycle, and during the G₂ phase the cell makes its final preparation for mitosis.
- Along with G₁, these two phases constitute the portion of the cell cycle called interphase, which alternates with mitosis and cytokinesis.

2. What are nucleosomes composed of, and how do they participate in the coiling of DNA?



• Chromosomes

3. What are the differences between heterochromatin and euchromatin?

4. What is a karyotype? How are chromosomes distinguished from one another in a karyotype?

11.3 Mitosis is a key phase of the cell cycle.

- The first stage of mitosis is prophase, during which the mitotic spindle apparatus forms.
- In the second stage of mitosis, metaphase, the chromosomes are arranged in a circle around the periphery of the cell.
- At the beginning of the third stage of mitosis, anaphase, the centromeres joining each pair of sister chromatids separate, freeing the sister chromatids from each other.
- After the chromatids physically separate, they are pulled to opposite poles of the cell by the microtubules attached to their centromeres.
- In the fourth and final stage of mitosis, telophase, the mitotic apparatus is disassembled, the nuclear envelope re-forms, and the chromosomes uncoil.
- When mitosis is complete, the cell divides in two, so that the two sets of chromosomes separated by mitosis end up in different daughter cells.

5. Which phases of the cell cycle is generally the longest in the cells of a mature eukaryote?



• Art Activity: Mitosis Overview
• Art Activity: Plant Cell Mitosis

6. What happens to the chromosomes during S phase?



• Mitosis

7. What changes with respect to ribosomal RNA occur during prophase?



• Mitosis

8. What event signals the initiation of metaphase?

9. What molecular mechanism seems to be responsible for the movement of the poles during anaphase?



• Student Research: Nuclear Division in *Drosophila*

10. Describe three events that occur during telophase.

11. How is cytokinesis in animal cells different from that in plant cells?

11.4 The cell cycle is carefully controlled.

- The cell cycle is regulated by two types of proteins, cyclins and cyclin-dependent protein kinases, which permit progress past key “checkpoints” in the cell cycle only if the cell is ready to proceed further.
- Failures of cell cycle regulation can lead to uncontrolled cell growth and lie at the root of cancer.

12. What aspects of the cell cycle are controlled by the G₁, G₂, and M checkpoints? How are cyclins and cyclin-dependent protein kinases involved in cell cycle regulation at checkpoints?



• Exploration: Regulating the cell cycle

12

Sexual Reproduction and Meiosis

Concept Outline

12.1 Meiosis produces haploid cells from diploid cells.

Discovery of Reduction Division. Sexual reproduction does not increase chromosome number because gamete production by meiosis involves a decrease in chromosome number. Individuals produced from sexual reproduction inherit chromosomes from two parents.

12.2 Meiosis has three unique features.

Unique Features of Meiosis. Three unique features of meiosis are synapsis, homologous recombination, and reduction division.

12.3 The sequence of events during meiosis involves two nuclear divisions.

Prophase I. Homologous chromosomes pair intimately, and undergo crossing over that locks them together.

Metaphase I. Spindle microtubules align the chromosomes in the central plane of the cell.

Completing Meiosis. The second meiotic division is like a mitotic division, but has a very different outcome.

12.4 The evolutionary origin of sex is a puzzle.

Why Sex? Sex may have evolved as a mechanism to repair DNA, or perhaps as a means for contagious elements to spread. Sexual reproduction increases genetic variability by shuffling combinations of genes.

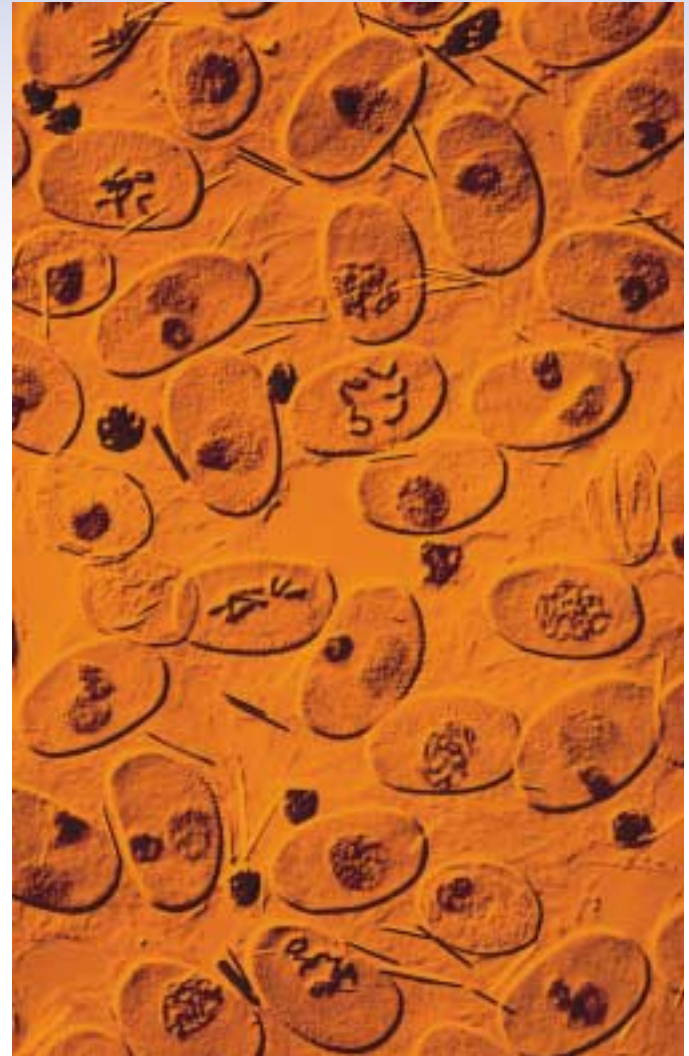


FIGURE 12.1

Plant cells undergoing meiosis (600 $\times$). This preparation of pollen cells of a spiderwort, *Tradescantia*, was made by freezing the cells and then fracturing them. It shows several stages of meiosis.

Most animals and plants reproduce sexually. Gametes of opposite sex unite to form a cell that, dividing repeatedly by mitosis, eventually gives rise to an adult body with some 100 trillion cells. The gametes that give rise to the initial cell are the products of a special form of cell division called meiosis (figure 12.1), the subject of this chapter. Far more intricate than mitosis, the details of meiosis are not as well understood. The basic process, however, is clear. Also clear are the profound consequences of sexual reproduction: it plays a key role in generating the tremendous genetic diversity that is the raw material of evolution.

12.1 Meiosis produces haploid cells from diploid cells.

Discovery of Reduction Division

Only a few years after Walther Fleming's discovery of chromosomes in 1882, Belgian cytologist Pierre-Joseph van Beneden was surprised to find different numbers of chromosomes in different types of cells in the roundworm *Ascaris*. Specifically, he observed that the **gametes** (eggs and sperm) each contained two chromosomes, while the **somatic** (nonreproductive) cells of embryos and mature individuals each contained four.

Fertilization

From his observations, van Beneden proposed in 1887 that an egg and a sperm, each containing half the complement of chromosomes found in other cells, fuse to produce a single cell called a **zygote**. The zygote, like all of the somatic cells ultimately derived from it, contains two copies of each chromosome. The fusion of gametes to form a new cell is called **fertilization**, or **syngamy**.

Reduction Division

It was clear even to early investigators that gamete formation must involve some mechanism that reduces the number of chromosomes to half the number found in other cells. If it did not, the chromosome number would double with each fertilization, and after only a few generations, the

number of chromosomes in each cell would become impossibly large. For example, in just 10 generations, the 46 chromosomes present in human cells would increase to over 47,000 (46×2^{10}).

The number of chromosomes does not explode in this way because of a special reduction division that occurs during gamete formation, producing cells with half the normal number of chromosomes. The subsequent fusion of two of these cells ensures a consistent chromosome number from one generation to the next. This reduction division process, known as **meiosis**, is the subject of this chapter.

The Sexual Life Cycle

Meiosis and fertilization together constitute a cycle of reproduction. Two sets of chromosomes are present in the somatic cells of adult individuals, making them **diploid** cells (Greek *diploos*, “double” + *eidos*, “form”), but only one set is present in the gametes, which are thus **haploid** (Greek *haploos*, “single” + *plōion*, “vessel”). Reproduction that involves this alternation of meiosis and fertilization is called **sexual reproduction**. Its outstanding characteristic is that offspring inherit chromosomes from *two* parents (figure 12.2). You, for example, inherited 23 chromosomes from your mother, contributed by the egg fertilized at your conception, and 23 from your father, contributed by the sperm that fertilized that egg.

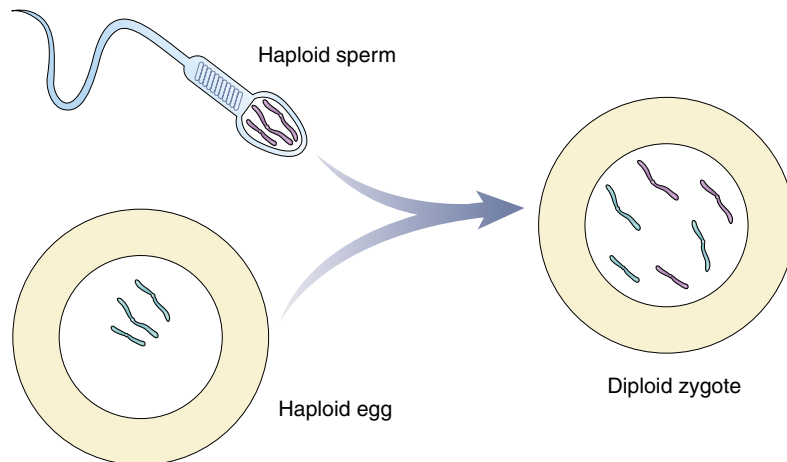


FIGURE 12.2

Diploid cells carry chromosomes from two parents. A diploid cell contains two versions of each chromosome, one contributed by the haploid egg of the mother, the other by the haploid sperm of the father.

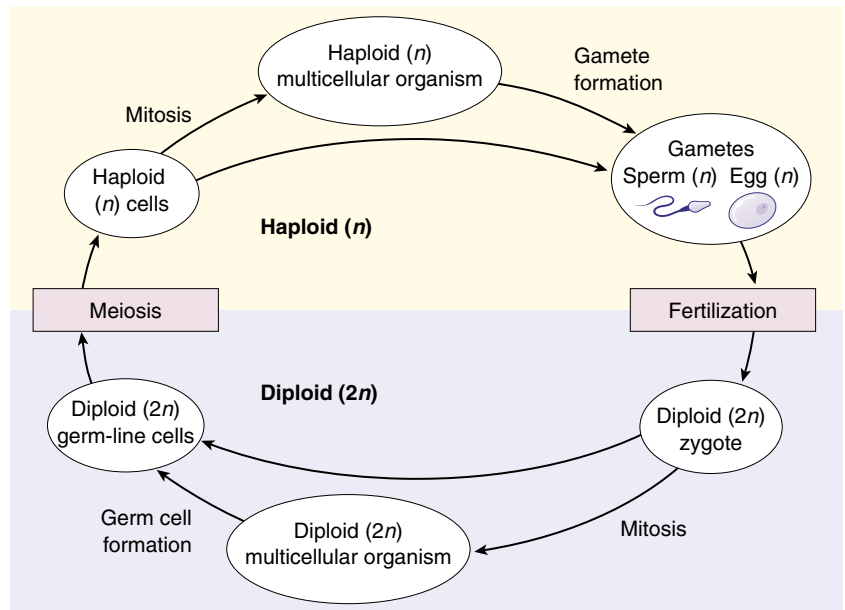


FIGURE 12.3
Alternation of generations. In sexual reproduction, haploid cells or organisms alternate with diploid cells or organisms.

Somatic Tissues. The life cycles of all sexually reproducing organisms follow the same basic pattern of alternation between the diploid and haploid chromosome numbers (figures 12.3 and 12.4). After fertilization, the resulting zygote begins to divide by mitosis. This single diploid cell eventually gives rise to all of the cells in the adult. These cells are called somatic cells, from the Latin word for “body.” Except when rare accidents occur, or in special variation-creating situations such as occur in the immune system, every one of the adult’s somatic cells is genetically identical to the zygote.

In unicellular eukaryotic organisms, including most protists, individual cells function as gametes, fusing with other gamete cells. The zygote may undergo mitosis, or it may divide immediately by meiosis to give rise to haploid individuals. In plants, the haploid cells that meiosis produces divide by mitosis, forming a multicellular haploid phase. Certain cells of this haploid phase eventually differentiate into eggs or sperm.

Germ-Line Tissues. In animals, the cells that will eventually undergo meiosis to produce gametes are set aside from somatic cells early in the course of development. These cells are often referred to as germ-line cells. Both the somatic cells and the gamete-producing germ-line cells are diploid, but while somatic cells undergo mitosis to form genetically identical, diploid daughter cells, gamete-producing germ-line cells undergo meiosis, producing haploid gametes.

Meiosis is a process of cell division in which the number of chromosomes in certain cells is halved during gamete formation. In the sexual life cycle, there is an alternation of diploid and haploid generations.

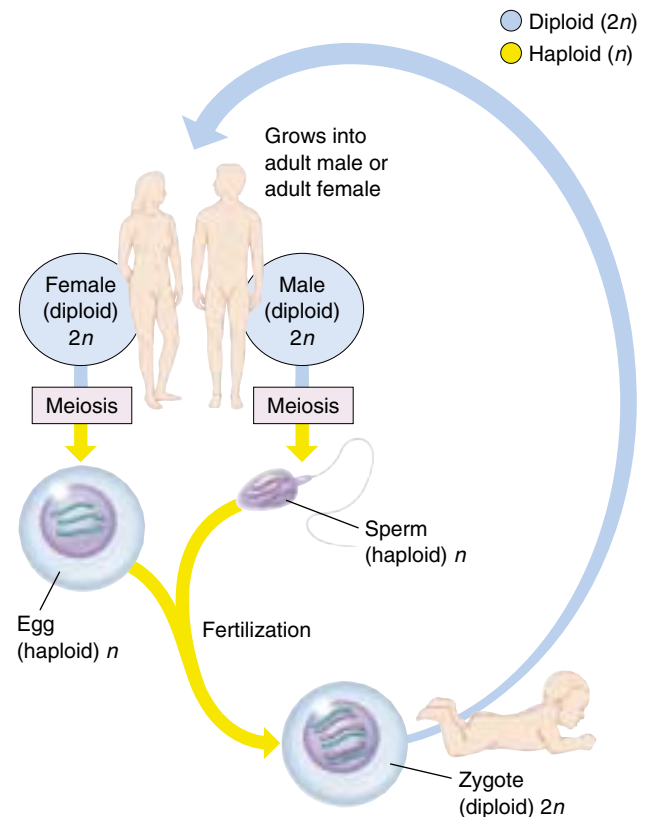


FIGURE 12.4
The sexual life cycle. In animals, the completion of meiosis is followed soon by fertilization. Thus, the vast majority of the life cycle is spent in the diploid stage.

12.2 Meiosis has three unique features.

Unique Features of Meiosis

The mechanism of cell division varies in important details in different organisms. This is particularly true of chromosomal separation mechanisms, which differ substantially in protists and fungi from the process in plants and animals that we will describe here. Meiosis in a diploid organism consists of two rounds of division, mitosis of one. Although meiosis and mitosis have much in common, meiosis has three unique features: synapsis, homologous recombination, and reduction division.

Synapsis

The first unique feature of meiosis happens early during the first nuclear division. Following chromosome replication, *homologous chromosomes*, or *homologues* (see chapter 11), *pair all along their length*. The process of forming these complexes of homologous chromosomes is called **synapsis**

Homologous Recombination

The second unique feature of meiosis is that *genetic exchange occurs between the homologous chromosomes* while they are thus physically joined (figure 12.5a). The exchange process that occurs between paired chromosomes is called **crossing over**. Chromosomes are then drawn together along the equatorial plane of the dividing cell; subsequently, homologues are pulled by microtubules toward opposite poles of the cell. When this process is complete, the cluster of chromosomes at each pole contains one of the two homologues of each chromosome. Each pole is haploid, containing half the number of chromosomes present in the original diploid cell. Sister chromatids do not separate from each other in the first nuclear division, so each homologue is still composed of two chromatids.

Reduction Division

The third unique feature of meiosis is that *the chromosomes do not replicate between the two nuclear divisions*, so that at the end of meiosis, each cell contains only half the original complement of chromosomes (figure 12.5b). In most respects, the second meiotic division is identical to a normal mitotic division. However, because of the crossing over that occurred during the first division, the sister chromatids in meiosis II are not identical to each other.

Meiosis is a continuous process, but it is most easily studied when we divide it into arbitrary stages. The stages of meiosis are traditionally called meiosis I and meiosis II. Like mitosis, each stage is subdivided further into prophase, metaphase, anaphase, and telophase (figure 12.6). In meiosis, however, prophase I is more complex than in mitosis.

In meiosis, homologous chromosomes become intimately associated and do not replicate between the two nuclear divisions.

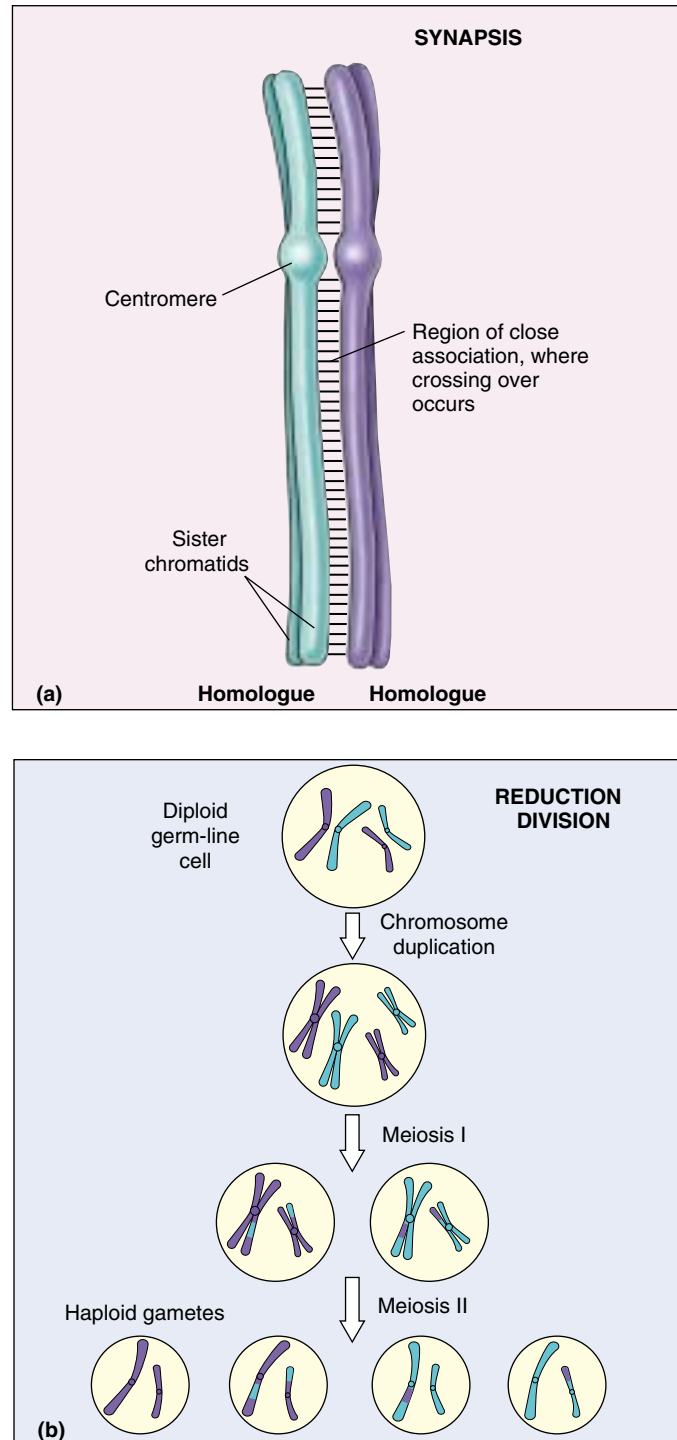


FIGURE 12.5

Unique features of meiosis. (a) Synapsis draws homologous chromosomes together, creating a situation where the two chromosomes can physically exchange parts, a process called crossing over. (b) Reduction division, by omitting a chromosome duplication before meiosis II, produces haploid gametes, thus ensuring that chromosome number remains stable during the reproduction cycle.

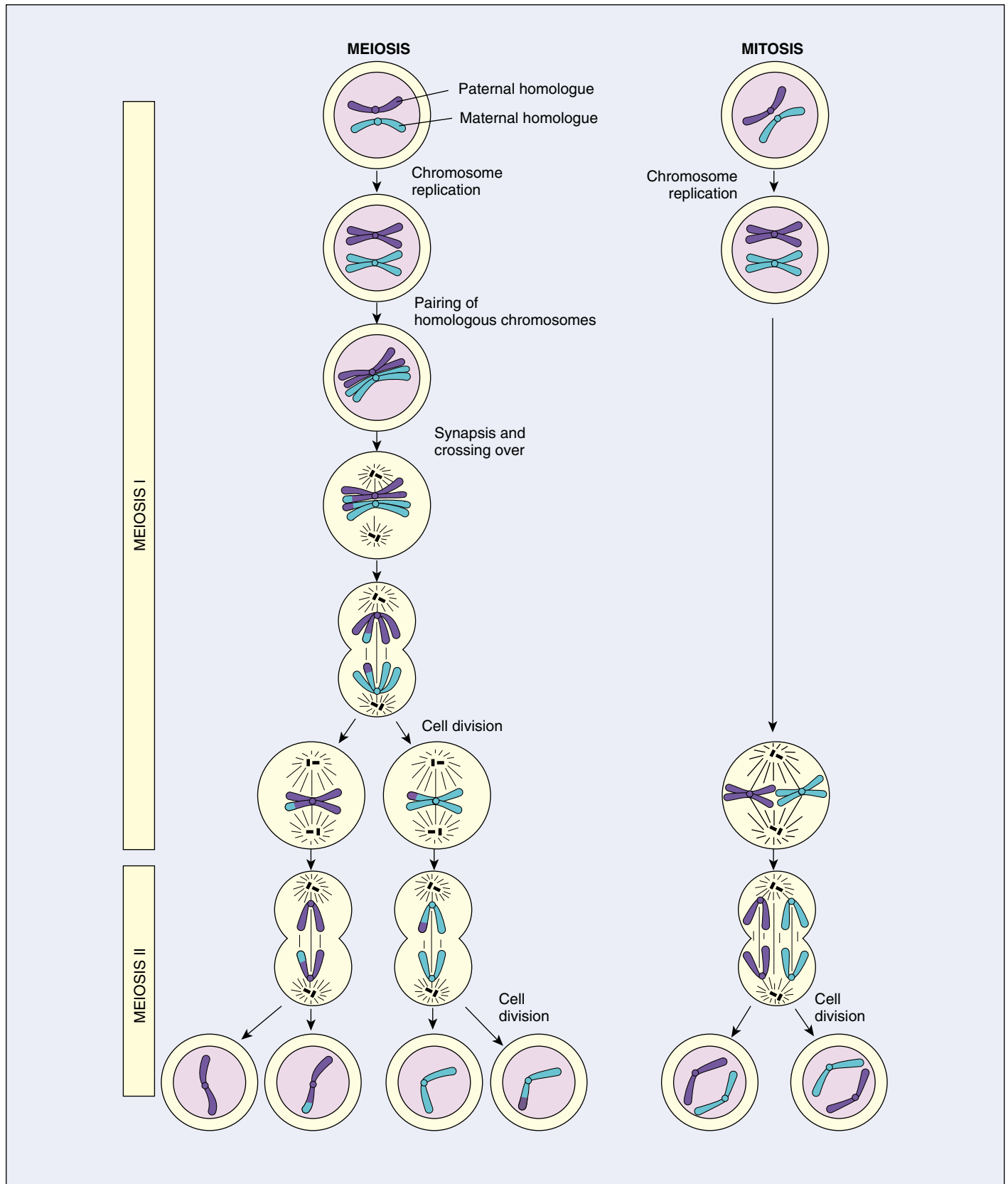


FIGURE 12.6

A comparison of meiosis and mitosis. Meiosis involves two nuclear divisions with no DNA replication between them. It thus produces four daughter cells, each with half the original number of chromosomes. Crossing over occurs in prophase I of meiosis. Mitosis involves a single nuclear division after DNA replication. It thus produces two daughter cells, each containing the original number of chromosomes.

12.3 The sequence of events during meiosis involves two nuclear divisions.

Prophase I

In prophase I of meiosis, the DNA coils tighter, and individual chromosomes first become visible under the light microscope as a matrix of fine threads. Because the DNA has already replicated before the onset of meiosis, each of these threads actually consists of two sister chromatids joined at their centromeres. In prophase I, homologous chromosomes become closely associated in synapsis, exchange segments by crossing over, and then separate.

An Overview

Prophase I is traditionally divided into five sequential stages: leptotene, zygotene, pachytene, diplotene, and diakinesis.

Leptotene. Chromosomes condense tightly.

Zygotene. A lattice of protein is laid down between the homologous chromosomes in the process of synapsis, forming a structure called a *synaptonemal complex* (figure 12.7).

Pachytene. Pachytene begins when synapsis is complete (just after the synaptonemal complex forms; figure 12.8), and lasts for days. This complex, about 100 nm across, holds the two replicated chromosomes in precise register, keeping each gene directly across from its partner on the homologous chromosome, like the teeth of a

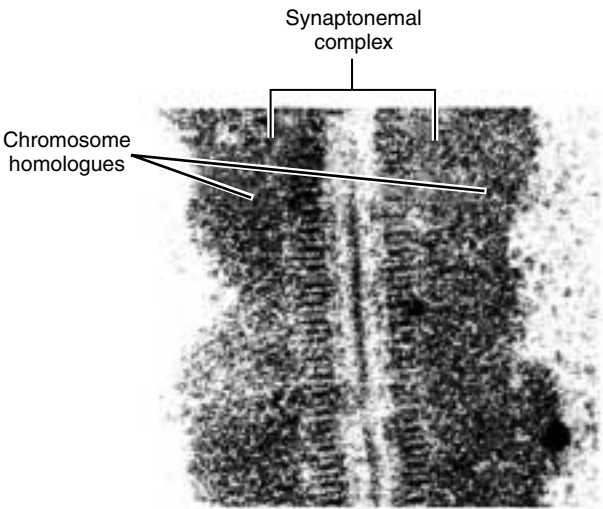


FIGURE 12.7
Structure of the synaptonemal complex. A portion of the synaptonemal complex of the ascomycete *Neotiella rutilans*, a cup fungus.

zipper. Within the synaptonemal complex, the DNA duplexes unwind at certain sites, and single strands of DNA form base-pairs with complementary strands *on the other homologue*. The synaptonemal complex thus provides the structural framework that enables crossing over between the homologous chromosomes. As you

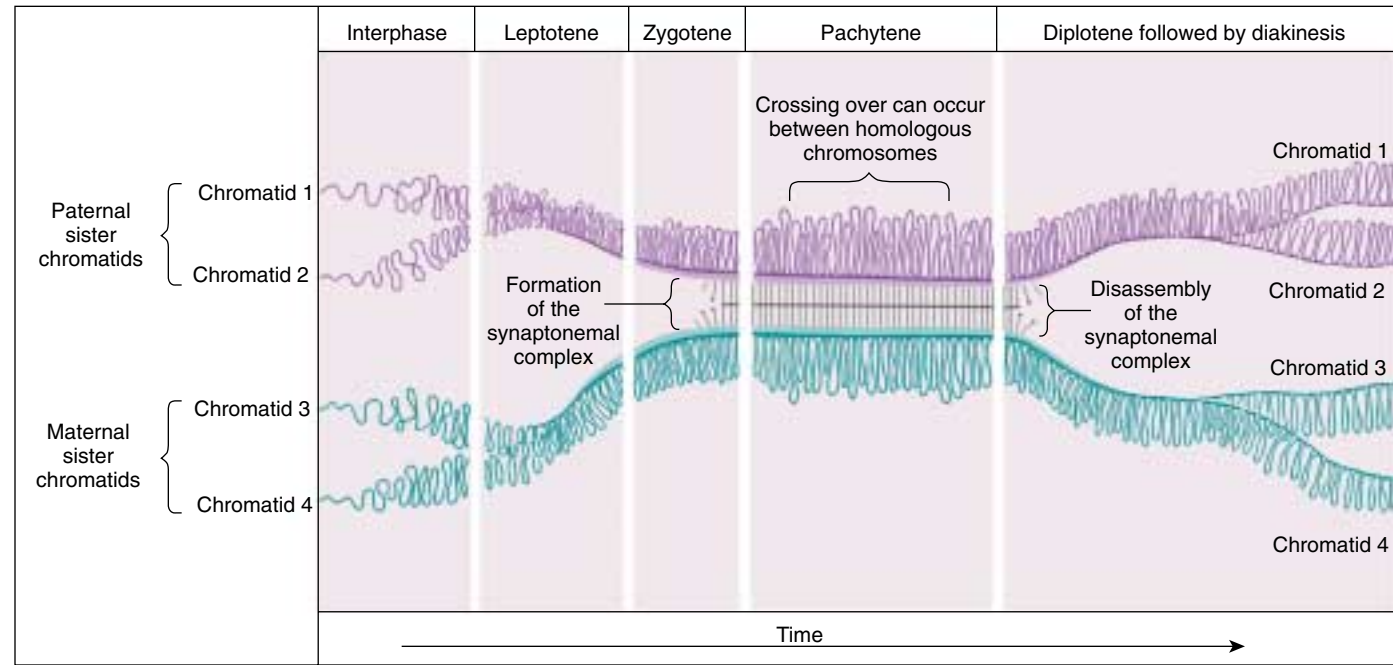


FIGURE 12.8
Time course of prophase I. The five stages of prophase I represent stages in the formation and subsequent disassembly of the synaptonemal complex, the protein lattice that holds homologous chromosomes together during synapsis.

will see, this has a key impact on how the homologues separate later in meiosis.

Diplotene. At the beginning of diplotene, the protein lattice of the synaptonemal complex disassembles. Diplotene is a period of intense cell growth. During this period the chromosomes decondense and become very active in transcription.

Diakinesis. At the beginning of diakinesis, the transition into metaphase, transcription ceases and the chromosomes recondense.



FIGURE 12.9

Chiasmata. This micrograph shows two distinct crossovers, or chiasmata.

Synapsis

During prophase, the ends of the chromatids attach to the nuclear envelope at specific sites. The sites the homologues attach to are adjacent, so that the members of each homologous pair of chromosomes are brought close together. They then line up side by side, apparently guided by heterochromatin sequences, in the process called synapsis.

Crossing Over

Within the synaptonemal complex, recombination is thought to be carried out during pachytene by very large protein assemblies called **recombination nodules**. A nodule's diameter is about 90 nm, spanning the central element of the synaptonemal complex. Spaced along the synaptonemal complex, these recombination nodules act as large multienzyme "recombination machines," each nodule bringing about a recombination event. The details of the crossing over process are not well understood, but involve a complex series of events in which DNA segments are exchanged between nonsister or sister chromatids. In humans, an average of two or three such crossover events occur per chromosome pair.

When crossing over is complete, the synaptonemal complex breaks down, and the homologous chromosomes are released from the nuclear envelope and begin to move away from each other. At this point, there are four chromatids for each type of chromosome (two homologous chromosomes, each of which consists of two sister chromatids). The four chromatids do not separate completely, however, because they are held together in two ways: (1) the two sister chromatids of each homologue, recently created by DNA replication, are held near by their common centromeres; and (2) the paired homologues are held together at the points where crossing over occurred within the synaptonemal complex.

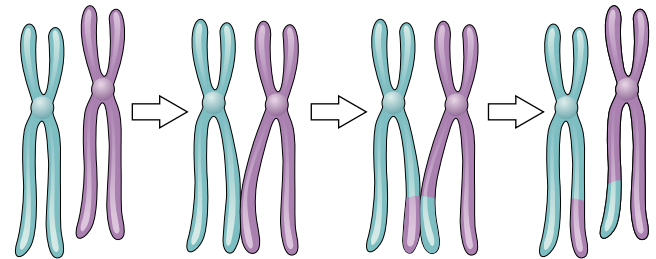


FIGURE 12.10

The results of crossing over. During crossing over, nonsister (shown above) or sister chromatids may exchange segments.

Chiasma Formation

Evidence of crossing over can often be seen under the light microscope as an X-shaped structure known as a **chiasma** (Greek, "cross"; plural, **chiasmata**; figure 12.9). The presence of a chiasma indicates that two chromatids (one from each homologue) have exchanged parts (figure 12.10). Like small rings moving down two strands of rope, the chiasmata move to the end of the chromosome arm as the homologous chromosomes separate.

Synapsis is the close pairing of homologous chromosomes that takes place early in prophase I of meiosis. Crossing over occurs between the paired DNA strands, creating the chromosomal configurations known as chiasmata. The two homologues are locked together by these exchanges and they do not disengage readily.

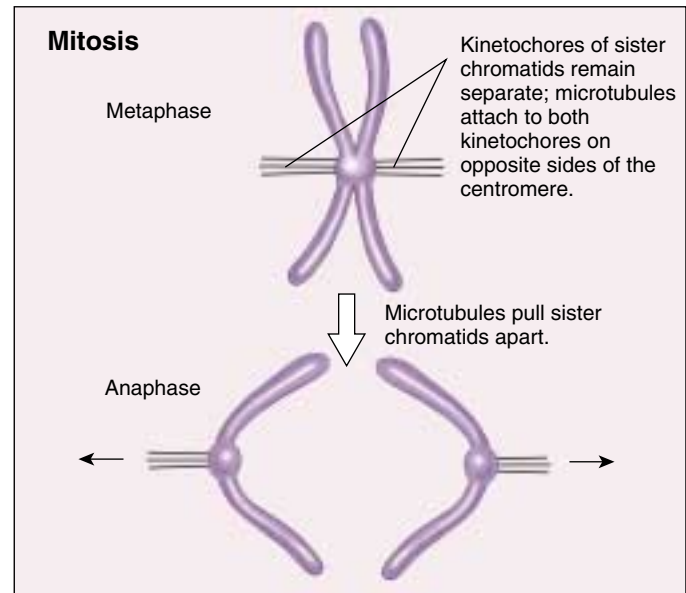
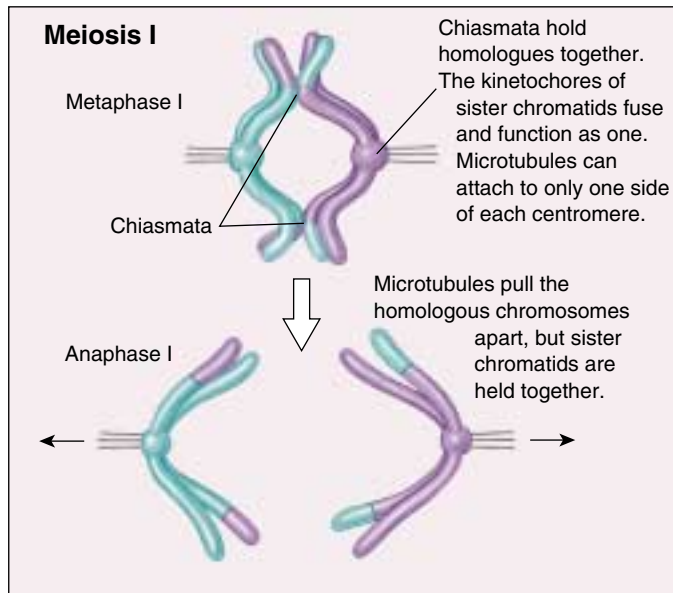


FIGURE 12.11

Chiasmata created by crossing over have a key impact on how chromosomes align in metaphase I. In the first meiotic division, the chiasmata hold one sister chromatid to the other sister chromatid; consequently, the spindle microtubules can bind to only one side of each centromere, and the homologous chromosomes are drawn to opposite poles. In mitosis, microtubules attach to *both* sides of each centromere; when the microtubules shorten, the sister chromatids are split and drawn to opposite poles.

Metaphase I

By metaphase I, the second stage of meiosis I, the nuclear envelope has dispersed and the microtubules form a spindle, just as in mitosis. During diakinesis of prophase I, the chiasmata move down the paired chromosomes from their original points of crossing over, eventually reaching the ends of the chromosomes. At this point, they are called terminal chiasmata. Terminal chiasmata hold the homologous chromosomes together in metaphase I, so that only one side of each centromere faces outward from the complex; the other side is turned inward toward the other homologue (figure 12.11). Consequently, spindle microtubules are able to attach to kinetochore proteins only on the outside of each centromere, and the centromeres of the two homologues attach to microtubules originating from opposite poles. This one-sided attachment is in marked contrast to the attachment in mitosis, when kinetochores on *both* sides of a centromere bind to microtubules.

Each joined pair of homologues then lines up on the metaphase plate. The orientation of each pair on the spindle axis is random: either the maternal or the paternal homologue may orient toward a given pole (figure 12.12). Figure 12.13 illustrates the alignment of chromosomes during metaphase I.

Chiasmata play an important role in aligning the chromosomes on the metaphase plate.

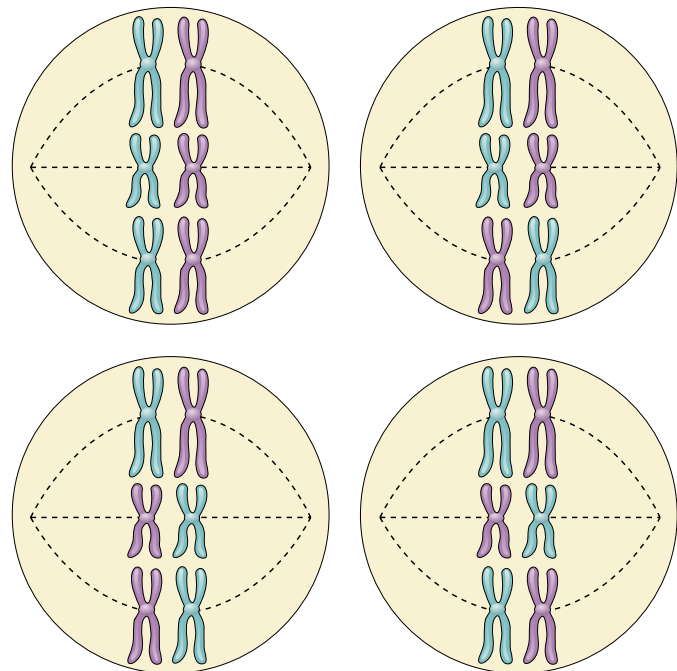
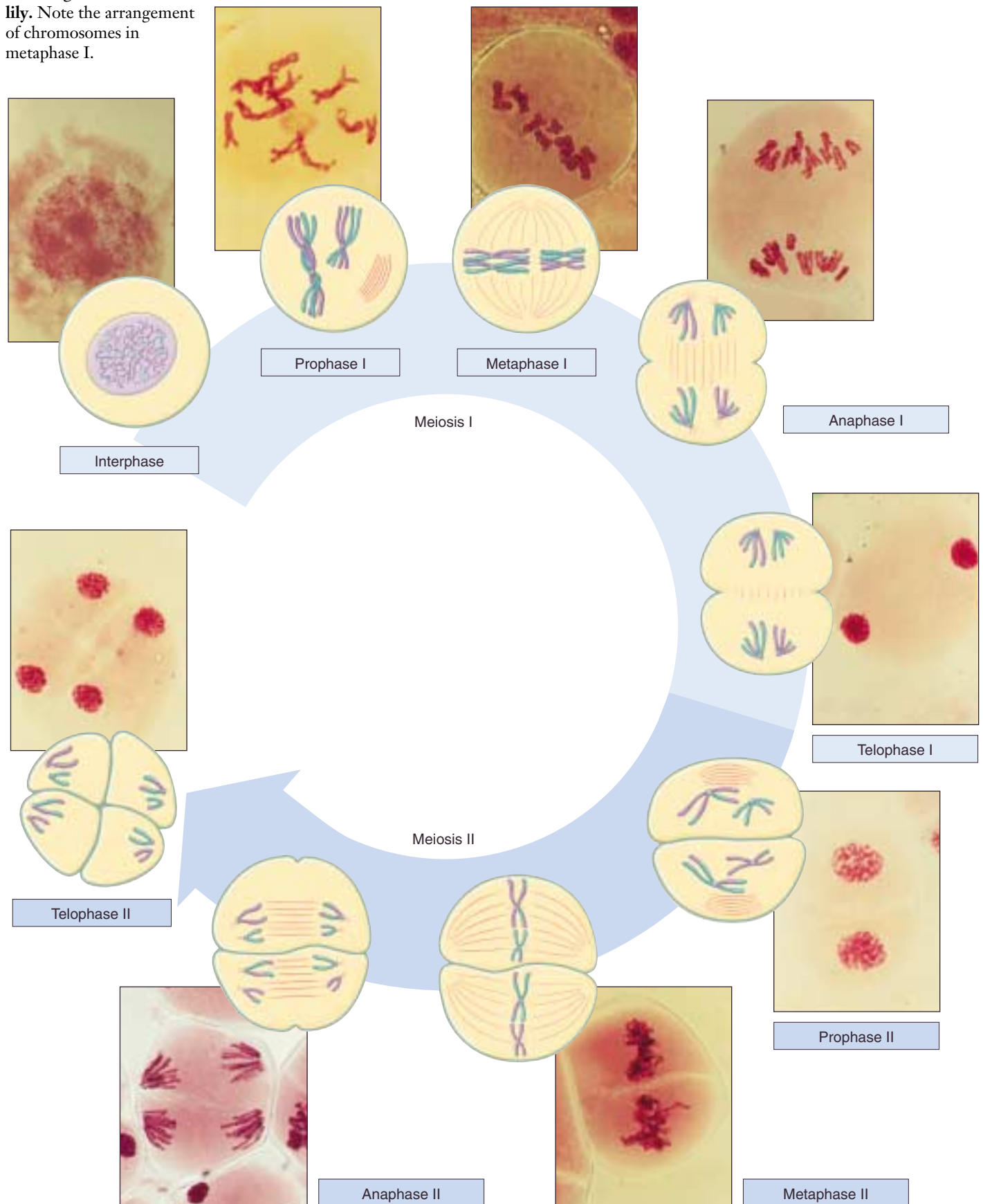


FIGURE 12.12

Random orientation of chromosomes on the metaphase plate. The number of possible chromosome orientations equals 2 raised to the power of the number of chromosome pairs. In this hypothetical cell with three chromosome pairs, eight (2^3) possible orientations exist, four of them illustrated here. Each orientation produces gametes with different combinations of parental chromosomes.

FIGURE 12.13
The stages of meiosis in a lily. Note the arrangement of chromosomes in metaphase I.



Completing Meiosis

After the long duration of prophase and metaphase, which together make up 90% or more of the time meiosis I takes, meiosis I rapidly concludes. Anaphase I and telophase I proceed quickly, followed—without an intervening period of DNA synthesis—by the second meiotic division.

Anaphase I

In anaphase I, the microtubules of the spindle fibers begin to shorten. As they shorten, they break the chiasmata and pull the centromeres toward the poles, dragging the chromosomes along with them. Because the microtubules are attached to kinetochores on only one side of each centromere, the individual centromeres are not pulled apart to form two daughter centromeres, as they are in mitosis. Instead, the entire centromere moves to one pole, taking both sister chromatids with it. When the spindle fibers have fully contracted, each pole has a complete haploid set of chromosomes consisting of one member of each homologous pair. Because of the random orientation of homologous chromosomes on the metaphase plate, a pole may receive either the maternal or the paternal homologue from each chromosome pair. As a result, the genes on different chromosomes assort independently; that is, meiosis I results in the **independent assortment** of maternal and paternal chromosomes into the gametes.

Telophase I

By the beginning of telophase I, the chromosomes have segregated into two clusters, one at each pole of the cell. Now the nuclear membrane re-forms around each daughter nucleus. Because each chromosome within a daughter nucleus replicated before meiosis I began, each now contains two sister chromatids attached by a common centromere. Importantly, *the sister chromatids are no longer identical*, because of the crossing over that occurred in prophase I (figure 12.14). Cytokinesis may or may not occur after telophase I. The second meiotic division, meiosis II, occurs after an interval of variable length.

The Second Meiotic Division

After a typically brief interphase, in which no DNA synthesis occurs, the second meiotic division begins.

Meiosis II resembles a normal mitotic division. Prophase II, metaphase II, anaphase II, and telophase II follow in quick succession.

Prophase II. At the two poles of the cell the clusters of chromosomes enter a brief prophase II, each nuclear envelope breaking down as a new spindle forms.

Metaphase II. In metaphase II, spindle fibers bind to both sides of the centromeres.



FIGURE 12.14

After meiosis I, sister chromatids are not identical. So-called “harlequin” chromosomes, each containing one fluorescent DNA strand, illustrate the reciprocal exchange of genetic material during meiosis I between sister chromatids.

Anaphase II. The spindle fibers contract, splitting the centromeres and moving the sister chromatids to opposite poles.

Telophase II. Finally, the nuclear envelope re-forms around the four sets of daughter chromosomes.

The final result of this division is four cells containing haploid sets of chromosomes (figure 12.15). No two are alike, because of the crossing over in prophase I. Nuclear envelopes then form around each haploid set of chromosomes. The cells that contain these haploid nuclei may develop directly into gametes, as they do in animals. Alternatively, they may themselves divide mitotically, as they do in plants, fungi, and many protists, eventually producing greater numbers of gametes or, as in the case of some plants and insects, adult individuals of varying ploidy.

During meiosis I, homologous chromosomes move toward opposite poles in anaphase I, and individual chromosomes cluster at the two poles in telophase I. At the end of meiosis II, each of the four haploid cells contains one copy of every chromosome in the set, rather than two. Because of crossing over, no two cells are the same. These haploid cells may develop directly into gametes, as in animals, or they may divide by mitosis, as in plants, fungi, and many protists.

MEIOSIS

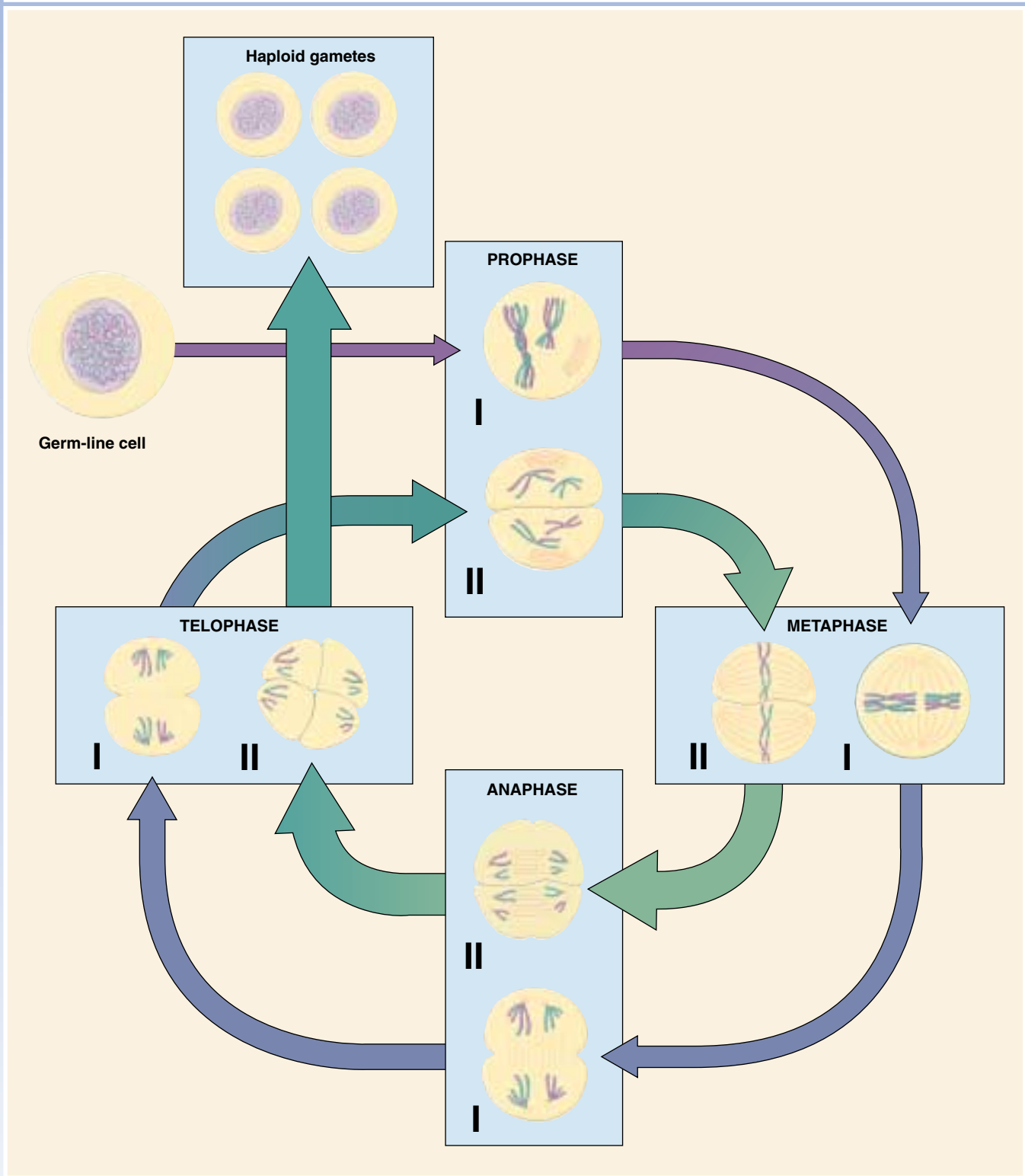


FIGURE 12.15

How meiosis works. Meiosis consists of two rounds of cell division and produces four haploid cells.

12.3 The evolutionary origin of sex is a puzzle.

Why Sex?

Not all reproduction is sexual. In **asexual reproduction**, an individual inherits all of its chromosomes from a single parent and is, therefore, genetically identical to its parent. Bacterial cells reproduce asexually, undergoing binary fission to produce two daughter cells containing the same genetic information. Most protists reproduce asexually except under conditions of stress; then they switch to sexual reproduction. Among plants, asexual reproduction is common, and many other multicellular organisms are also capable of reproducing asexually. In animals, asexual reproduction often involves the budding off of a localized mass of cells, which grows by mitosis to form a new individual.

Even when meiosis and the production of gametes occur, there may still be reproduction without sex. The development of an adult from an unfertilized egg, called **parthenogenesis**, is a common form of reproduction in arthropods. Among bees, for example, fertilized eggs develop into diploid females, but unfertilized eggs develop into haploid males. Parthenogenesis even occurs among the vertebrates. Some lizards, fishes, and amphibians are capable of reproducing in this way; their unfertilized eggs undergo a mitotic nuclear division without cell cleavage to produce a diploid cell, which then develops into an adult.

Recombination Can Be Destructive

If reproduction can occur without sex, why does sex occur at all? This question has generated considerable discussion, particularly among evolutionary biologists. Sex is of great evolutionary advantage for populations or species, which benefit from the variability generated in meiosis by random orientation of chromosomes and by crossing over. However, evolution occurs because of changes at the level of *individual* survival and reproduction, rather than at the population level, and no obvious advantage accrues to the progeny of an individual that engages in sexual reproduction. In fact, recombination is as destructive as well as a constructive process in evolution. The segregation of chromosomes during meiosis tends to disrupt advantageous combinations of genes more often than it creates new, better adapted combinations; as a result, some of the diverse progeny produced by sexual reproduction will not be as well adapted as their parents were. In fact, the more complex the adaptation of an individual organism, the less likely that recombination will improve it, and the more likely that recombination will disrupt it. It is, therefore, a puzzle to know what a well-adapted individual gains from participating in sexual reproduction, as *all* of its progeny could maintain its successful gene combinations if that individual simply reproduced asexually.

The Origin and Maintenance of Sex

There is no consensus among evolutionary biologists regarding the evolutionary origin or maintenance of sex. Conflicting hypotheses abound. Alternative hypotheses seem to be correct to varying degrees in different organisms.

The DNA Repair Hypothesis. If recombination is often detrimental to an individual's progeny, then what benefit promoted the evolution of sexual reproduction? Although the answer to this question is unknown, we can gain some insight by examining the protists. Meiotic recombination is often absent among the protists, which typically undergo sexual reproduction only occasionally. Often the fusion of two haploid cells occurs only under stress, creating a diploid zygote.

Why do some protists form a diploid cell in response to stress? Several geneticists have suggested that this occurs because only a diploid cell can effectively repair certain kinds of chromosome damage, particularly double-strand breaks in DNA. Both radiation and chemical events within cells can induce such breaks. As organisms became larger and longer-lived, it must have become increasingly important for them to be able to repair such damage. The synaptonemal complex, which in early stages of meiosis precisely aligns pairs of homologous chromosomes, may well have evolved originally as a mechanism for repairing double-strand damage to DNA, using the undamaged homologous chromosome as a template to repair the damaged chromosome. A transient diploid phase would have provided an opportunity for such repair. In yeast, mutations that inactivate the repair system for double-strand breaks of the chromosomes also prevent crossing over, suggesting a common mechanism for both synapsis and repair processes.

The Contagion Hypothesis. An unusual and interesting alternative hypothesis for the origin of sex is that it arose as a secondary consequence of the infection of eukaryotes by mobile genetic elements. Suppose a replicating transposable element were to infect a eukaryotic lineage. If it possessed genes promoting fusion with uninfected cells and synapsis, the transposable element could readily copy itself onto homologous chromosomes. It would rapidly spread by infection through the population, until all members contained it. The bizarre mating type "alleles" found in many fungi are very nicely explained by this hypothesis. Each of several mating types is in fact not an allele but an "idiomorph." Idiomorphs are genes occupying homologous positions on the chromosome but having such dissimilar sequences that they cannot be of homologous origin. These idiomorph genes may simply be the relics of several ancient infections by transposable elements.

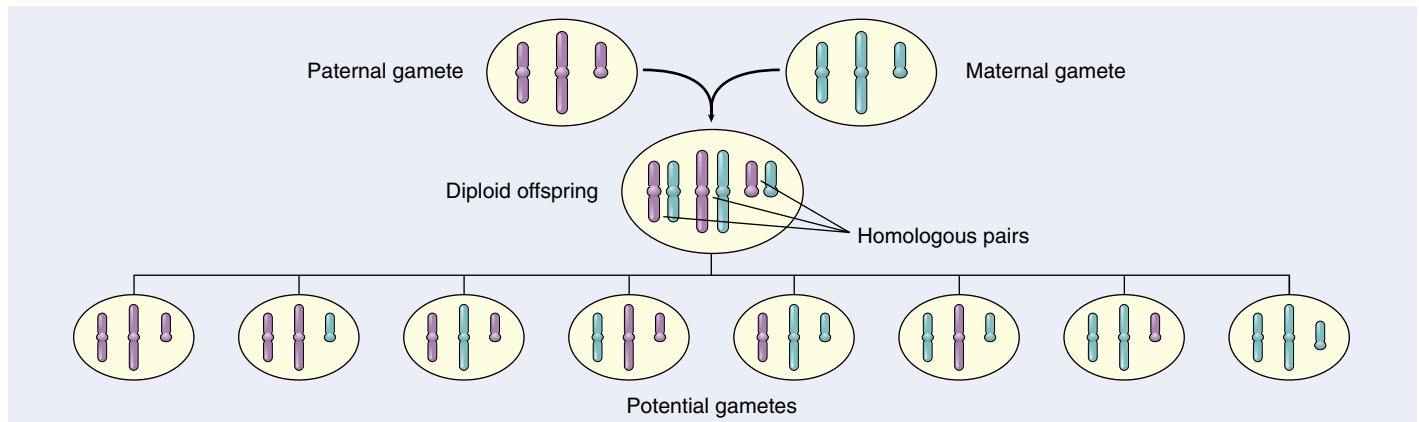


FIGURE 12.16

Independent assortment increases genetic variability. Independent assortment contributes new gene combinations to the next generation because the orientation of chromosomes on the metaphase plate is random. In the cells shown above with three chromosome pairs, eight different gametes can result, each with different combinations of parental chromosomes.

The Red Queen Hypothesis. One evolutionary advantage of sex may be that it allows populations to “store” recessive alleles that are currently bad but have promise for reuse at some time in the future. Because populations are constrained by a changing physical and biological environment, selection is constantly acting against such alleles, but in sexual species can never get rid of those sheltered in heterozygotes. The evolution of most sexual species, most of the time, thus manages to keep pace with ever-changing physical and biological constraints. This “treadmill evolution” is sometimes called the “Red Queen hypothesis,” after the Queen of Hearts in Lewis Carroll’s *Through the Looking Glass*, who tells Alice, “Now, here, you see, it takes all the running you can do, to keep in the same place.”

Miller’s Ratchet. The geneticist Herman Miller pointed out in 1965 that asexual populations incorporate a kind of mutational ratchet mechanism—once harmful mutations arise, asexual populations have no way of eliminating them, and they accumulate over time, like turning a ratchet. Sexual populations, on the other hand, can employ recombination to generate individuals carrying fewer mutations, which selection can then favor. Sex may just be a way to keep the mutational load down.

The Evolutionary Consequences of Sex

While our knowledge of how sex evolved is sketchy, it is abundantly clear that sexual reproduction has an enormous impact on how species evolve today, because of its ability to rapidly generate new genetic combinations. Independent assortment (figure 12.16), crossing over, and random fertilization each help generate genetic diversity.

Whatever the forces that led to sexual reproduction, its evolutionary consequences have been profound. No genetic

process generates diversity more quickly; and, as you will see in later chapters, genetic diversity is the raw material of evolution, the fuel that drives it and determines its potential directions. In many cases, the pace of evolution appears to increase as the level of genetic diversity increases. Programs for selecting larger stature in domesticated animals such as cattle and sheep, for example, proceed rapidly at first, but then slow as the existing genetic combinations are exhausted; further progress must then await the generation of new gene combinations. Racehorse breeding provides a graphic example: thoroughbred racehorses are all descendants of a small initial number of individuals, and selection for speed has accomplished all it can with this limited amount of genetic variability—the winning times in major races ceased to improve decades ago.

Paradoxically, the evolutionary process is thus both revolutionary and conservative. It is revolutionary in that the pace of evolutionary change is quickened by genetic recombination, much of which results from sexual reproduction. It is conservative in that evolutionary change is not always favored by selection, which may instead preserve existing combinations of genes. These conservative pressures appear to be greatest in some asexually reproducing organisms that do not move around freely and that live in especially demanding habitats. In vertebrates, on the other hand, the evolutionary premium appears to have been on versatility, and sexual reproduction is the predominant mode of reproduction by an overwhelming margin.

The close association between homologous chromosomes that occurs during meiosis may have evolved as mechanisms to repair chromosomal damage, although several alternative mechanisms have also been proposed.



Summary

Questions

Media Resources

12.1 Meiosis produces haploid cells from diploid cells.

- Meiosis is a special form of nuclear division that produces the gametes of the sexual cycle. It involves two chromosome separations but only one chromosome replication.

1. What are the cellular products of meiosis called, and are they haploid or diploid? What is the cellular product of syngamy called, and is it haploid or diploid?

12.2 Meiosis has three unique features.

- The three unique features of meiosis are synapsis, homologous recombination, and reduction division.

2. What three unique features distinguish meiosis from mitosis?

12.3 The sequence of events during meiosis involves two nuclear divisions.

- The crossing over that occurs between homologues during synapsis is an essential element of meiosis.
- Because crossing over binds the homologues together, only one side of each homologue is accessible to the spindle fibers. Hence, the spindle fibers separate the paired homologues rather than the sister chromatids.
- At the end of meiosis I, one homologue of each chromosome type is present at each of the two poles of the dividing nucleus. The homologues still consist of two chromatids, which may differ from each other as a result of crossing over that occurred during synapsis.
- No further DNA replication occurs before the second nuclear division, which is essentially a mitotic division occurring at each of the two poles.
- The sister chromatids of each chromosome are separated, resulting in the formation of four daughter nuclei, each with half the number of chromosomes that were present before meiosis.
- Cytokinesis typically but not always occurs at this point. When it does, each daughter nucleus has one copy of every chromosome.

3. What are synaptonemal complexes? How do they participate in crossing over? At what stage during meiosis are they formed?

4. How many chromatids are present for each type of chromosome at the completion of crossing over? What two structures hold the chromatids together at this stage?

5. How is the attachment of spindle microtubules to centromeres in metaphase I of meiosis different from that which occurs in metaphase of mitosis? What effect does this difference have on the movement of chromosomes during anaphase I?

6. What mechanism is responsible for the independent assortment of chromosomes?



• Art Activity: Meiosis I



• Meiosis



• Meiosis

12.4 The evolutionary origin of sex is a puzzle.

- In asexual reproduction, mitosis produces offspring genetically identical to the parent.
- Meiosis is thought to have evolved initially as a mechanism to repair double-strand breaks in DNA, in which the broken chromosome is paired with its homologue while it is being repaired.
- The evolutionary significance of meiosis is that it generates large amounts of recombination, rapidly reshuffling gene combinations, producing variability upon which evolutionary processes can act.

7. What is one of the current scientific explanations for the evolution of synapsis?

8. By what three mechanisms does sexual reproduction increase genetic variability? How does this increase in genetic variability affect the evolution of species?



• Evolution of Sex
• Review of Cell Division

13

Patterns of Inheritance

Concept Outline

13.1 Mendel solved the mystery of heredity.

Early Ideas about Heredity: The Road to Mendel.

Before Mendel, the mechanism of inheritance was not known.

Mendel and the Garden Pea. Mendel experimented with heredity in edible peas counted his results.

What Mendel Found. Mendel found that alternative traits for a character segregated among second-generation progeny in the ratio 3:1. Mendel proposed that information for a trait rather than the trait itself is inherited.

How Mendel Interpreted His Results. Mendel found that one alternative of a character could mask the other in heterozygotes, but both could subsequently be expressed in homozygotes of future generations.

Mendelian Inheritance Is Not Always Easy to Analyze. A variety of factors can influence the Mendelian segregation of alleles.

13.2 Human genetics follows Mendelian principles.

Most Gene Disorders Are Rare. Tay-Sachs disease is due to a recessive allele.

Multiple Alleles: The ABO Blood Groups. The human ABO blood groups are determined by three *I* gene alleles.

Patterns of Inheritance Can Be Deduced from Pedigrees. Hemophilia is sex-linked.

Gene Disorders Can Be Due to Simple Alterations of Proteins. Sickle cell anemia is caused by a single amino acid change.

Some Defects May Soon Be Curable. Cystic fibrosis may soon be cured by gene replacement therapy.

13.3 Genes are on chromosomes.

Chromosomes: The Vehicles of Mendelian Inheritance. Mendelian segregation reflects the random assortment of chromosomes in meiosis.

Genetic Recombination. Crossover frequency reflect the physical distance between genes.

Human Chromosomes. Humans possess 23 pairs of chromosomes, one of them determining the sex.

Human Abnormalities Due to Alterations in Chromosome Number. Loss or addition of chromosomes has serious consequences.

Genetic Counseling. Some gene defects can be detected early in pregnancy.



FIGURE 13.1

Human beings are extremely diverse in appearance. The differences between us are partly inherited and partly the result of environmental factors we encounter in our lives.

Every living creature is a product of the long evolutionary history of life on earth. While all organisms share this history, only humans wonder about the processes that led to their origin. We are still far from understanding everything about our origins, but we have learned a great deal. Like a partially completed jigsaw puzzle, the boundaries have fallen into place, and much of the internal structure is becoming apparent. In this chapter, we will discuss one piece of the puzzle—the enigma of heredity. Why do groups of people from different parts of the world often differ in appearance (figure 13.1)? Why do the members of a family tend to resemble one another more than they resemble members of other families?

13.1 Mendel solved the mystery of heredity.

Early Ideas about Heredity: The Road to Mendel

As far back as written records go, patterns of resemblance among the members of particular families have been noted and commented on (figure 13.2). Some familial features are unusual, such as the protruding lower lip of the European royal family Hapsburg, evident in pictures and descriptions of family members from the thirteenth century onward. Other characteristics, like the occurrence of redheaded children within families of redheaded parents, are more common (figure 13.3). Inherited features, the building blocks of evolution, will be our concern in this chapter.

Classical Assumption 1: Constancy of Species

Two concepts provided the basis for most of the thinking about heredity before the twentieth century. The first is that *heredity occurs within species*. For a very long time people believed that it was possible to obtain bizarre composite animals by breeding (crossing) widely different species. The minotaur of Cretan mythology, a creature with the body of a bull and the torso and head of a man, is one example. The giraffe was thought to be another; its scientific name, *Giraffa camelopardalis*, suggests the belief that it was the result of a cross between a camel and a leopard. From the Middle Ages onward, however, people discovered that such extreme crosses were not possible and that variation and heredity occur mainly within the boundaries of a particular species. Species were thought to have been maintained without significant change from the time of their creation.

Classical Assumption 2: Direct Transmission of Traits

The second early concept related to heredity is that *traits are transmitted directly*. When variation is inherited by offspring from their parents, *what* is transmitted? The ancient Greeks suggested that the parents' body parts were transmitted directly to their offspring. Hippocrates called this type of reproductive material *gonos*, meaning "seed." Hence, a characteristic such as a misshapen limb was the result of material that came from the misshapen limb of a parent. Information from each part of the body was supposedly passed along independently of the information from the other parts, and the child was formed after the hereditary material from all parts of the parents' bodies had come together.

This idea was predominant until fairly recently. For example, in 1868, Charles Darwin proposed that all cells and tissues excrete microscopic granules, or "gemmules," that



FIGURE 13.2

Heredity is responsible for family resemblance. Family resemblances are often strong—a visual manifestation of the mechanism of heredity. This is the Johnson family, the wife and daughters of one of the authors. While each daughter is different, all clearly resemble their mother.



FIGURE 13.3

Red hair is inherited. Many different traits are inherited in human families. This redhead is exhibiting one of these traits.

are passed to offspring, guiding the growth of the corresponding part in the developing embryo. Most similar theories of the direct transmission of hereditary material assumed that the male and female contributions *blend* in the offspring. Thus, parents with red and brown hair would produce children with reddish brown hair, and tall and short parents would produce children of intermediate height.

Koelreuter Demonstrates Hybridization between Species

Taken together, however, these two concepts lead to a paradox. If no variation enters a species from outside, and if the variation within each species blends in every generation, then all members of a species should soon have the same appearance. Obviously, this does not happen. Individuals within most species differ widely from each other, and they differ in characteristics that are transmitted from generation to generation.

How could this paradox be resolved? Actually, the resolution had been provided long before Darwin, in the work of the German botanist Josef Koelreuter. In 1760, Koelreuter carried out successful **hybridizations** of plant species, crossing different strains of tobacco and obtaining fertile offspring. The hybrids differed in appearance from both parent strains. When individuals within the hybrid generation were crossed, their offspring were highly variable. Some of these offspring resembled plants of the hybrid generation (their parents), but a few resembled the original strains (their grandparents).

The Classical Assumptions Fail

Koelreuter's work represents the beginning of modern genetics, the first clues pointing to the modern theory of heredity. Koelreuter's experiments provided an important clue about how heredity works: the traits he was studying could be masked in one generation, only to reappear in the next. This pattern contradicts the theory of direct transmission. How could a trait that is transmitted directly disappear and then reappear? Nor were the traits of Koelreuter's plants blended. A contemporary account stated that the traits reappeared in the third generation "fully restored to all their original powers and properties."

It is worth repeating that the offspring in Koelreuter's crosses were not identical to one another. Some resembled the hybrid generation, while others did not. The alternative



FIGURE 13.4
The garden pea, *Pisum sativum*. Easy to cultivate and able to produce many distinctive varieties, the garden pea was a popular experimental subject in investigations of heredity as long as a century before Gregor Mendel's experiments.

forms of the characters Koelreuter was studying were distributed among the offspring. Referring to a heritable feature as a **character**, a modern geneticist would say the alternative forms of each character were **segregating** among the progeny of a mating, meaning that some offspring exhibited one alternative form of a character (for example, hairy leaves), while other offspring from the same mating exhibited a different alternative (smooth leaves). This segregation of alternative forms of a character, or **traits**, provided the clue that led Gregor Mendel to his understanding of the nature of heredity.

Knight Studies Heredity in Peas

Over the next hundred years, other investigators elaborated on Koelreuter's work. Prominent among them were English gentleman farmers trying to improve varieties of agricultural plants. In one such series of experiments, carried out in the 1790s, T. A. Knight crossed two true-breeding varieties (varieties that remain uniform from one generation to the next) of the garden pea, *Pisum sativum* (figure 13.4). One of these varieties had purple flowers, and the other had white flowers. All of the progeny of the cross had purple flowers. Among the offspring of

these hybrids, however, were some plants with purple flowers and others, less common, with white flowers. Just as in Koelreuter's earlier studies, a trait from one of the parents disappeared in one generation only to reappear in the next.

In these deceptively simple results were the makings of a scientific revolution. Nevertheless, another century passed before the process of gene segregation was fully appreciated. Why did it take so long? One reason was that early workers did not quantify their results. A numerical record of results proved to be crucial to understanding the process. Knight and later experimenters who carried out other crosses with pea plants noted that some traits had a "stronger tendency" to appear than others, but they did not record the numbers of the different classes of progeny. Science was young then, and it was not obvious that the numbers were important.

Early geneticists demonstrated that some forms of an inherited character (1) can disappear in one generation only to reappear unchanged in future generations; (2) segregate among the offspring of a cross; and (3) are more likely to be represented than their alternatives.

Mendel and the Garden Pea

The first quantitative studies of inheritance were carried out by Gregor Mendel, an Austrian monk (figure 13.5). Born in 1822 to peasant parents, Mendel was educated in a monastery and went on to study science and mathematics at the University of Vienna, where he failed his examinations for a teaching certificate. He returned to the monastery and spent the rest of his life there, eventually becoming abbot. In the garden of the monastery (figure 13.6), Mendel initiated a series of experiments on plant hybridization. The results of these experiments would ultimately change our views of heredity irrevocably.

Why Mendel Chose the Garden Pea

For his experiments, Mendel chose the garden pea, the same plant Knight and many others had studied earlier. The choice was a good one for several reasons. First, many earlier investigators had produced hybrid peas by crossing different varieties. Mendel knew that he could expect to observe segregation of traits among the offspring. Second, a large number of true-breeding varieties of peas were available. Mendel initially examined 32. Then, for further study, he selected lines that differed with respect to seven easily distinguishable traits, such as round versus wrinkled seeds and purple versus white flowers, a character that Knight had studied. Third, pea plants are small and easy to grow, and they have a relatively short generation time. Thus, one can conduct experiments involving numerous plants, grow several generations in a single year, and obtain results relatively quickly.

A fourth advantage of studying peas is that the sexual organs of the pea are enclosed within the flower (figure 13.7). The flowers of peas, like those of many flowering plants, contain both male and female sex organs. Furthermore, the gametes produced by the male and female parts of the same flower, unlike those of many flowering plants, can fuse to form viable offspring. Fertilization takes place automatically within an individual flower if it is not disturbed, resulting in offspring that are the progeny from a single individual. Therefore, one can either let individual flowers engage in **self-fertilization**, or remove the flower's male parts before fertilization and introduce pollen from a strain with a different trait, thus performing *cross-pollination* which results in **cross-fertilization**.

FIGURE 13.6

The garden where Mendel carried out his plant-breeding experiments. Gregor Mendel did his key scientific experiments in this small garden in a monastery.

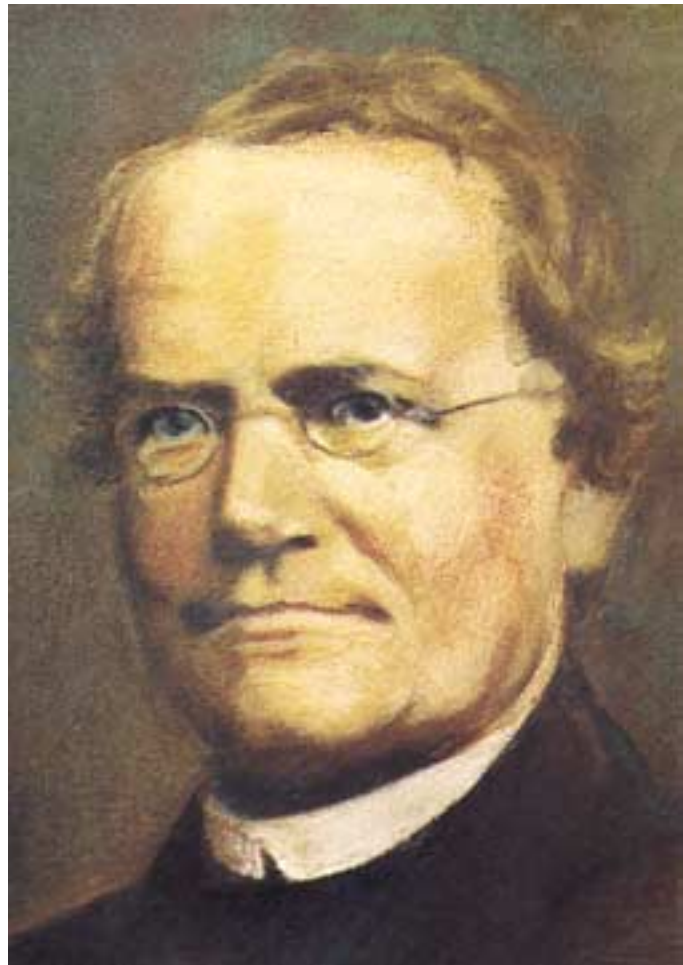


FIGURE 13.5

Gregor Johann Mendel. Cultivating his plants in the garden of a monastery in Brunn, Austria (now Brno, Czech Republic), Mendel studied how differences among varieties of peas were inherited when the varieties were crossed. Similar experiments had been done before, but Mendel was the first to quantify the results and appreciate their significance.



Mendel's Experimental Design

Mendel was careful to focus on only a few specific differences between the plants he was using and to ignore the countless other differences he must have seen. He also had the insight to realize that the differences he selected to analyze must be comparable. For example, he appreciated that trying to study the inheritance of round seeds versus tall height would be useless.

Mendel usually conducted his experiments in three stages:

1. First, he allowed pea plants of a given variety to produce progeny by self-fertilization for several generations. Mendel thus was able to assure himself that the traits he was studying were indeed constant, transmitted unchanged from generation to generation. Pea plants with white flowers, for example, when crossed with each other, produced only offspring with white flowers, regardless of the number of generations.
2. Mendel then performed crosses between varieties exhibiting alternative forms of characters. For example, he removed the male parts from the flower of a plant that produced white flowers and fertilized it with pollen from a purple-flowered plant. He also carried out the reciprocal cross, using pollen from a white-flowered individual to fertilize a flower on a pea plant that produced purple flowers (figure 13.8).
3. Finally, Mendel permitted the hybrid offspring produced by these crosses to self-pollinate for several generations. By doing so, he allowed the alternative forms of a character to segregate among the progeny. This was the same experimental design that Knight and others had used much earlier. But Mendel went an important step farther: he counted the numbers of offspring exhibiting each trait in each succeeding generation. No one had ever done that before. The quantitative results Mendel obtained proved to be of supreme importance in revealing the process of heredity.

Mendel's experiments with the garden pea involved crosses between true-breeding varieties, followed by a generation or more of inbreeding.

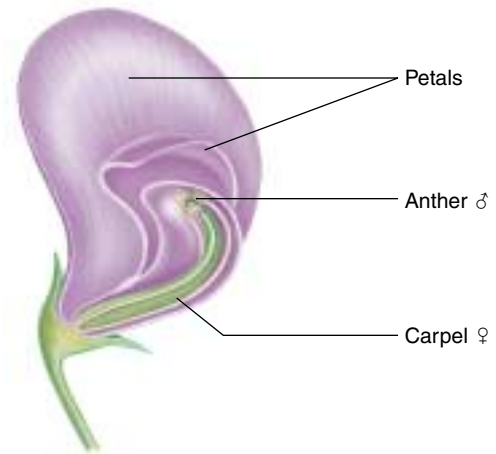


FIGURE 13.7

Structure of the pea flower (longitudinal section). In a pea plant flower, the petals enclose the male anther (containing pollen grains, which give rise to haploid sperm) and the female carpel (containing ovules, which give rise to haploid eggs). This ensures that self-fertilization will take place unless the flower is disturbed.

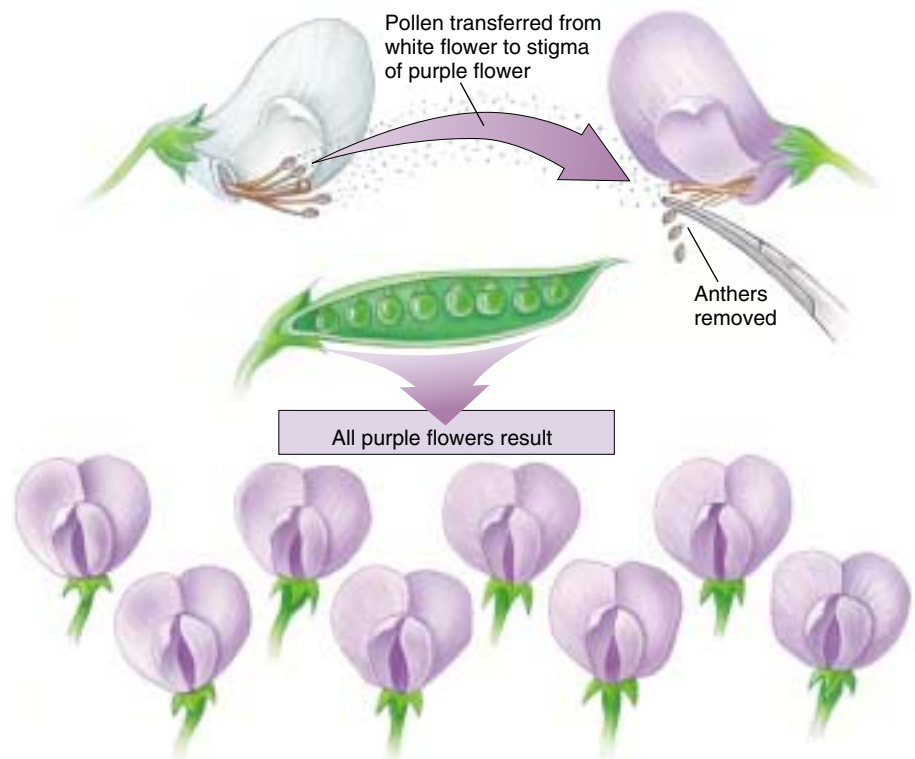


FIGURE 13.8

How Mendel conducted his experiments. Mendel pushed aside the petals of a white flower and collected pollen from the anthers. He then placed that pollen onto the stigma (part of the carpel) of a purple flower whose anthers had been removed, causing cross-fertilization to take place. All the seeds in the pod that resulted from this pollination were hybrids of the white-flowered male parent and the purple-flowered female parent. After planting these seeds, Mendel observed the pea plants they produced. All of the progeny of this cross had purple flowers.

What Mendel Found

The seven characters Mendel studied in his experiments possessed several variants that differed from one another in ways that were easy to recognize and score (figure 13.9). We will examine in detail Mendel’s crosses with flower color. His experiments with other characters were similar, and they produced similar results.

The F₁ Generation

When Mendel crossed two contrasting varieties of peas, such as white-flowered and purple-flowered plants, the hybrid offspring he obtained did not have flowers of intermediate color, as the theory of blending inheritance would predict. Instead, in every case the flower color of the offspring resembled one of their parents. It is customary to refer to these offspring as the **first filial** (*filius* is

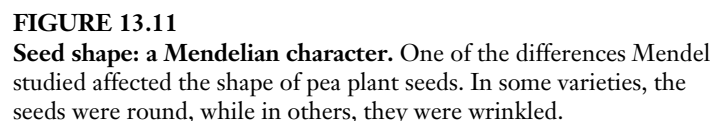
Character	Dominant vs. recessive trait	F ₂ generation		Ratio
		Dominant form	Recessive form	
Flower color	 X  Purple White	705	224	3.15:1
Seed color	 X  Yellow Green	6022	2001	3.01:1
Seed shape	 X  Round Wrinkled	5474	1850	2.96:1
Pod color	 X  Green Yellow	428	152	2.82:1
Pod shape	 X  Inflated Constricted	882	299	2.95:1
Flower position	 X  Axial Terminal	651	207	3.14:1
Plant height	 X  Tall Dwarf	787	277	2.84:1

FIGURE 13.9
Mendel’s experimental results. This table illustrates the seven characters Mendel studied in his crosses of the garden pea and presents the data he obtained from these crosses. Each pair of traits appeared in the F₂ generation in very close to a 3:1 ratio.

Mendel referred to the trait expressed in the F₁ plants as **dominant** and to the alternative form that was not expressed in the F₁ plants as **recessive**. For each of the seven pairs of contrasting traits that Mendel examined, one of the pair proved to be dominant and the other recessive.

After allowing individual F₁ plants to mature and self-pollinate, Mendel collected and planted the seeds from each plant to see what the offspring in the **second filial**, or **F₂**, generation would look like. He found, just as Knight had earlier, that some F₂ plants exhibited white flowers, the recessive trait. Hidden in the F₁ generation, the recessive form reappeared among some F₂ individuals.

Believing the proportions of the F_2 types would provide some clue about the mechanism of heredity, Mendel counted the numbers of each type among the F_2 progeny (figure 13.10). In the cross between the purple-flowered F_1 plants, he counted a total of 929 F_2 individuals (see figure 13.9). Of these, 705 (75.9%) had purple flowers and 224 (24.1%) had white flowers. Approximately $\frac{1}{4}$ of the F_2 individuals exhibited the recessive form of the character. Mendel obtained the same numerical result with the other six characters he examined: $\frac{1}{4}$ of the F_2 individuals exhibited the dominant trait, and $\frac{1}{4}$ displayed the recessive trait. In other words, the dominant:recessive ratio among the F_2 plants was always close to 3:1. Mendel carried out similar experiments with other traits, such as wrinkled versus round seeds (figure 13.11), and obtained the same result.



A Disguised 1:2:1 Ratio

Mendel went on to examine how the F_2 plants passed traits on to subsequent generations. He found that the recessive $\frac{1}{4}$ were always true-breeding. In the cross of white-flowered with purple-flowered plants, for example, the white-flowered F_2 individuals reliably produced white-flowered offspring when they were allowed to self-fertilize. By contrast, only $\frac{1}{3}$ of the dominant purple-flowered F_2 individuals ($\frac{1}{4}$ of all F_2 offspring) proved true-breeding, while $\frac{2}{3}$ were not. This last class of plants produced dominant and recessive individuals in the third filial (F_3) generation in a 3:1 ratio. This result suggested that, for the entire sample, the 3:1 ratio that Mendel observed in the F_2 generation was really a disguised 1:2:1 ratio: $\frac{1}{4}$ pure-breeding dominant individuals, $\frac{2}{4}$ not-pure-breeding dominant individuals, and $\frac{1}{4}$ pure-breeding recessive individuals (figure 13.12).

Mendel's Model of Heredity

From his experiments, Mendel was able to understand four things about the nature of heredity. *First*, the plants he crossed did not produce progeny of intermediate appearance, as a theory of blending inheritance would have predicted. Instead, different plants inherited each alternative intact, as a discrete characteristic that either was or was not visible in a particular generation. *Second*, Mendel learned that for each pair of alternative forms of a character, one alternative was not expressed in the F_1 hybrids, although it reappeared in some F_2 individuals. *The trait that “disappeared” must therefore be latent (present but not expressed) in the F_1 individuals.* *Third*, the pairs of alternative traits examined segregated among the progeny of a particular cross, some individuals exhibiting one trait, some the other. *Fourth*, these alternative traits were expressed in the F_2 generation in the ratio of $\frac{3}{4}$ dominant to $\frac{1}{4}$ recessive. This characteristic 3:1 segregation is often referred to as the **Mendelian ratio**.

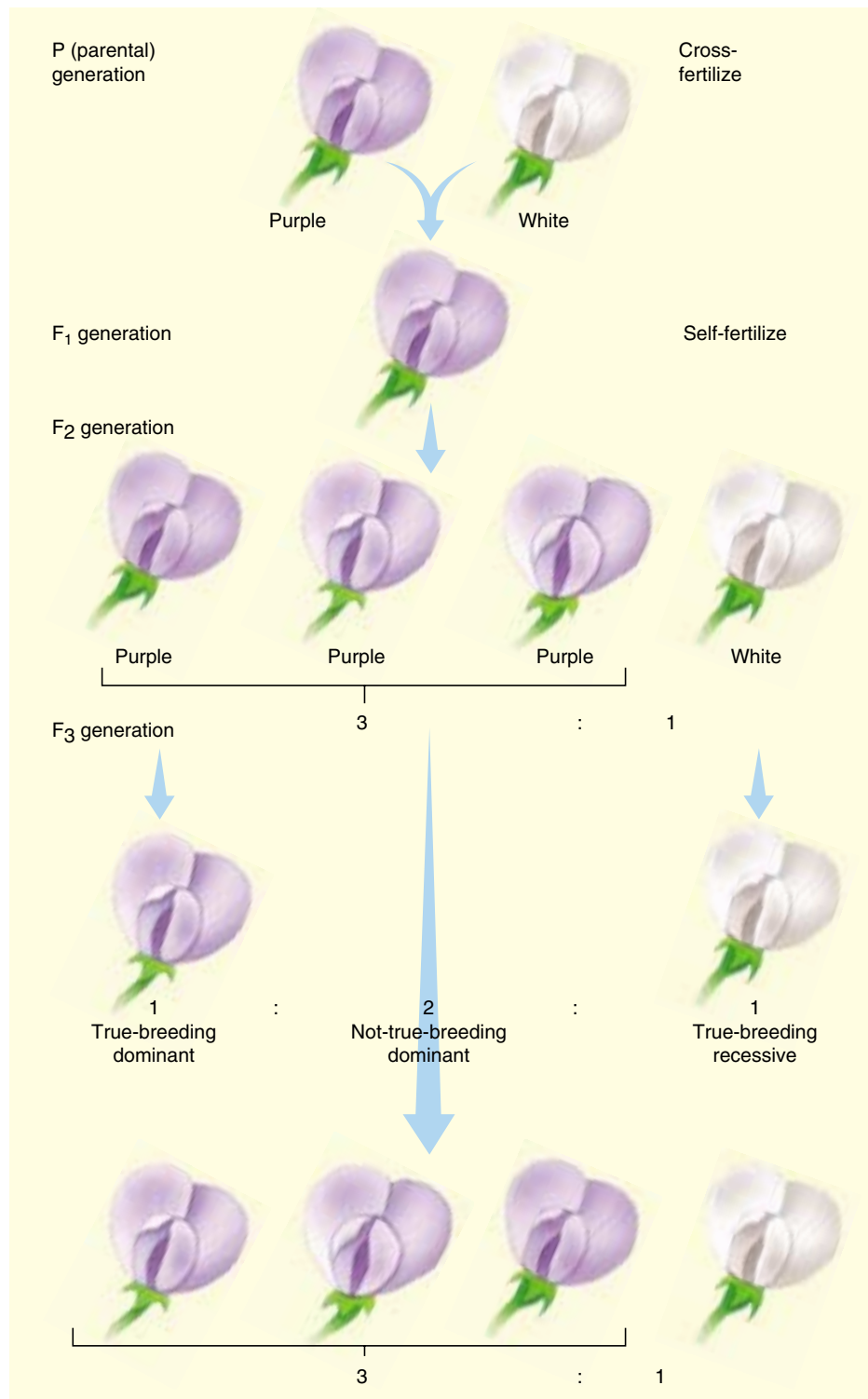


FIGURE 13.12

The F_2 generation is a disguised 1:2:1 ratio. By allowing the F_2 generation to self-fertilize, Mendel found from the offspring (F_3) that the ratio of F_2 plants was one true-breeding dominant, two not-true-breeding dominant, and one true-breeding recessive.